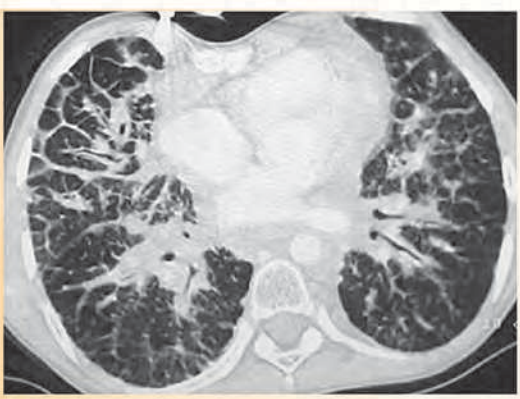
Clinico Radiological Series

IMAGING OF INTERSTITIAL LUNG DISEASES



Editors
Ashu Seith Bhalla
Manisha Jana

Clinical Content Editors
Randeep Guleria
GC Khilnani

Co-Editor
Priyanka Naranje

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Dedicated to

The concept of multidisciplinary teams working together to care for patients and to find solutions

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Preface

Interstitial lung disease, the very phrase sends many a shivers down the spines of young radiologists. And truly so, interstitial lung diseases (ILDs) in itself remain an enigma; only to be decoded with an in-depth knowledge of the pathophysiology, imaging and clinical manifestations. The range of ILDs is wide, and with advances in knowledge, everyday a new entity is being added to the list! A thorough knowledge of the newer classification and terminology is crucial in order to identify, and describes the imaging abnormalities. In addition, there are several diseases that can simulate ILDs radiologically; and hence preference of the term, *Diffuse Lung Diseases (DLDs)* by many. Even a trained eye may miss significant yet subtle findings unless a systematic approach is adopted for reporting.

Pulmonary medicine is a rapidly expanding medical specialty which relies heavily on imaging. Practicing physicians need to be aware of the basics of imaging, as that will affect the management of a critically ill patient. Similarly, radiologists in practice need to be aware of the imaging patterns and their differential diagnoses. This book is not intended to be a detailed text on ILDs, but we hope that it will help the busy radiologists and physicians in their daily practice. The pattern approach at the end of each section is aimed at this.

We wish to thank all the contributors, for their efforts in compiling the text. We sincerely thank Professor Randeep Guleria and Professor GC Khilnani for guiding us in this new endeavor. We are also obliged to Professor Sushil K Kabra and Professor Rakesh Lodha for their valuable contributions on ILD in children.

We would also like to extend our appreciation to Shri Jitendar P Vij (Group Chairman), Mr Ankit Vij (Group President), and all the staff of M/s Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, India, for their efforts and inputs enabling timely publication of the book.

We welcome any feedback on enhancing the content of the book, which can be incorporated in the subsequent editions.

Ashu Seith Bhalla
Manisha Jana

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Section 1

General

- Terminology and Classification
- Optimizing the Imaging Modalities
- Interstitial Lung Diseases: Pathologic Diagnosis
- Clinical Diagnosis including Pulmonary Function Test

1

Terminology and Classification

Devasenathipathy Kandasamy, Surabhi Vyas

INTRODUCTION

- Interstitial lung disease (ILD) is a heterogeneous group of disorders affecting the pulmonary interstitium comprising of interalveolar connective tissue, peribronchovascular and perilymphatic tissues including the alveolar epithelium, capillary endothelium and basement membranes.
- The term ILD is often used interchangeably with diffuse lung diseases (DLDs), however neither all DLD (e.g. diffuse alveolar hemorrhage) involve the interstitium nor all ILDs are diffuse [e.g. idiopathic pulmonary fibrosis (IPF)].
- In this book we will cover in subsequent chapters, the various conditions included under ILD as well as DLD which can mimic ILD.

CLASSIFICATION

- Interstitial lung diseases (ILDs) can be broadly classified into idiopathic and those secondary to a known factor like connective tissue diseases, infections, drug or irritant exposure related.
- The American Thoracic Society–European Respiratory Society classification of 2002 had described individual diseases in the idiopathic interstitial pneumonias (IIPs) and their diagnostic approach. This joint statement underwent a revision in 2013, with the multidisciplinary approach taking cognizance of the clinical, imaging and histopathological findings.[1,2]
- In the 2013 classification, the major entities in IIPs have been classified as:
  - Chronic fibrosing IIPs.
  - Smoking-related IIPs (respiratory bronchiolitis–associated interstitial lung disease [RB-ILD] and desquamative interstitial pneumonia).
  - Acute or subacute IIPs (cryptogenic organizing pneumonia and acute interstitial pneumonia).
  - Rare IIPs (idiopathic pleuroparenchymal fibroelastosis, lymphoid interstitial pneumonia) and unclassifiable IIPs.[3]

We have categorized the contents of this book according to various ILDs in the following manner for ease of discussion:

- Idiopathic interstitial pneumonia
  - Usual interstitial pneumonia (UIP)
  - Nonspecific interstitial pneumonia (NSIP)
  - Desquamative interstitial pneumonia (DIP)
  - Respiratory bronchiolitis-ILD (RB-ILD)
  - Acute interstitial pneumonia (AIP)
  - Lymphocytic interstitial pneumonia (LIP)
  - Cryptogenic organizing pneumonia (COP)
  - Lymphangioleiomyomatosis
  - Eosinophilic granuloma
  - Sarcoidosis
- Connective tissue diseases
  - Scleroderma
  - Dermatomyositis
  - Systemic lupus erythematosus
  - Sjögren's syndrome
  - Ankylosing spondylitis
  - Mixed connective tissue disorder
- Miscellaneous
  - Infectious causes
    - Atypical pneumonia
    - *Pneumocystis jirovecii* pneumonia
  - Malignant causes
    - Pulmonary adenocarcinoma in situ
    - Lymphangitic carcinomatosis
  - Caused by inhalational agents
    - Organic—animal/bird/farm antigens
    - Inorganic—silica, asbestos, etc.
  - Drug-induced
    - Antiarrhythmic drugs
    - Anticancer drugs
    - Antidepressants
    - Radiation
    - Anti-inflammatory drugs.

SALIENT ANATOMY

- The organization of structures in the subsegmental level consists of bronchovascular bundles and secondary pulmonary lobule.
- Bronchovascular bundle consists of bronchi, arteries and associated connective tissue (central interstitium) **(Fig. 1.1)**.
- On CT, vessels are seen as solid linear or circular structures whereas bronchi are seen as parallel linear or ring shaped structures.

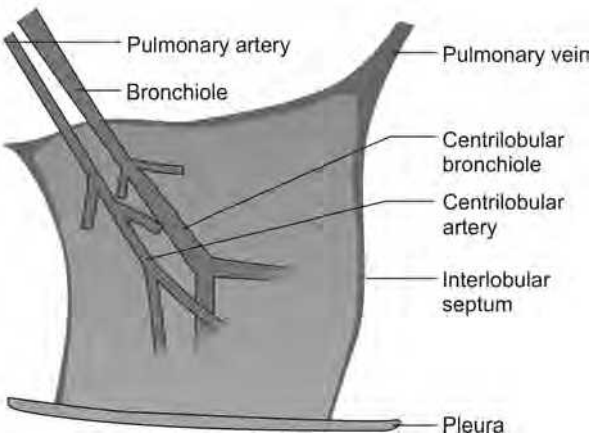


Fig. 1.1: Schematic diagram of secondary pulmonary lobule showing the bronchiole and pulmonary arterial branch entering in the center and the pulmonary veins are draining from the periphery of the lobule. The central bronchovascular interstitium is seen in the center and the peripheral interstitium is seen in the periphery of the lobule

- Secondary pulmonary lobule is a polyhedral structure (around 2 cm) with bronchovascular structures in the center and perilobular veins (peripheral interstitium) in the periphery.
- Lymphatics are seen in central as well as in the periphery of the lobule
- Each secondary pulmonary lobule is supplies by a terminal bronchiole and a centrilobular pulmonary artery.
- Centrilobular arteries can be seen on CT images whereas the bronchiole is not usually seen as it is too thin for the current resolution of scanners.
- The terminal bronchiole is thin, does not have cilia and has muscular walls which makes them a favored location for bronchospasm leading to air-trapping.
- Terminal bronchioles give rise to multiple respiratory bronchioles which lead to alveolar ducts.
- There are fine septations seen with in the secondary pulmonary lobule which forms the framework of acini, alveolar ducts, alveolar sac, etc.
- These intralobular interstitial structures are not seen routinely on CT scans
- Pleuropulmonary interface forms the boundary for the lungs and it is usually smooth and devoid of irregularities.

PATTERNS AND TERMINOLOGIES

Reticular Pattern

- This is one of the commonly seen abnormalities in ILDs.
- It manifests as linear opacities on high-resolution computed tomography (HRCT) which is usually as a result of thickening of the peripheral interstitium especially interlobular septum.

- However, there can be associated central interstitial thickening as well.
- There are various causes which can result in reticular pattern such as fibrosis, volume overload and malignant infiltration.[4]
- The reticular pattern can be *smooth, nodular or irregular*.
- *Smooth reticular opacities* are seen as smooth linear opacities without any irregularity or nodularity **(Fig. 1.2)**.
- It can be seen in edema or by infiltration of the interlobular septum
- *Nodular reticular opacities* are usually because of interstitial thickening with associated micronodules making the interface nodular **(Fig. 1.3)**.
- It can be seen in sarcoidosis, pneumoconiosis, lymphangitis carcinomatosis etc.
- *Irregular reticular pattern* is as a result of extensive interstitial fibrosis leading to irregular thickening of septae and distortion of the pulmonary lobule framework **(Fig. 1.4)**.
- Abnormality of the centrilobular interstitium can also be seen in the form of irregular thickening and tortuosity.
- This pattern can be seen in a variety of fibrosing ILDs such as UIP, NSIP, sarcoidosis, etc.

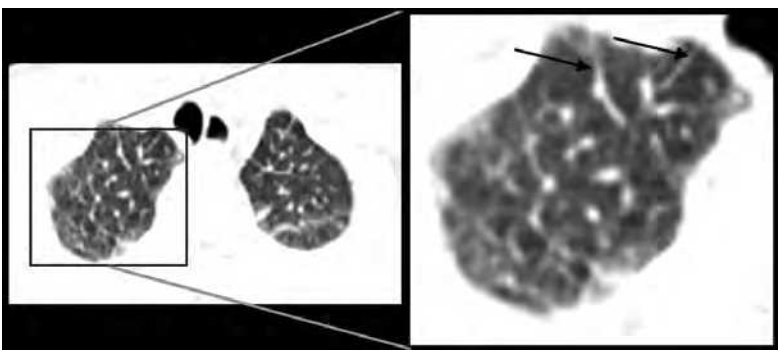


Fig. 1.2: Axial CT scan image in lung window in a patient with fluid overload features showing smooth interlobular septal thickening (arrows) involving bilateral lungs

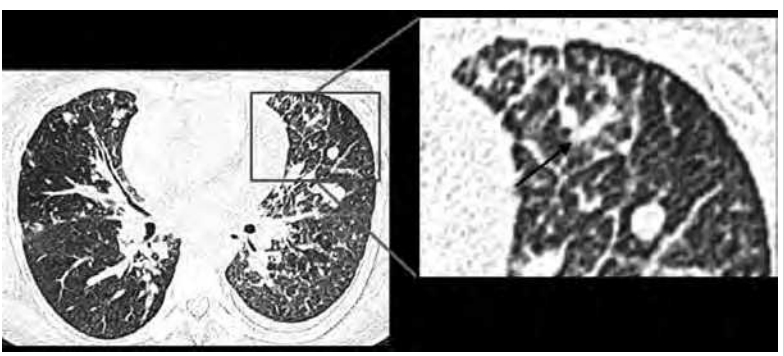


Fig. 1.3: Axial CT scan image of a patient with left lung carcinoma showing nodular interlobular septal thickening (arrow) in the left lung suggestive of lymphangitic carcinomatosis. Multiple nodules suggestive of metastases are also noted in both lungs

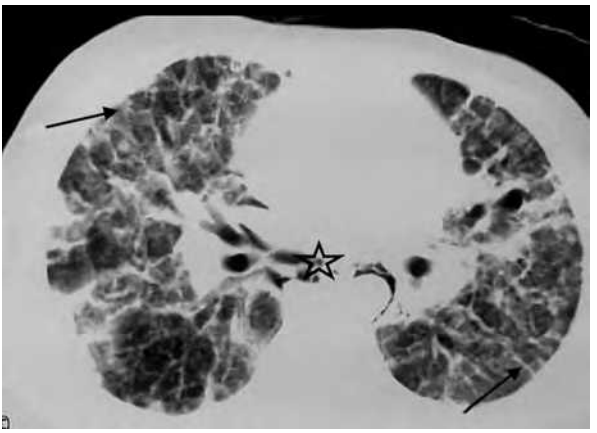


Fig. 1.4: Axial CT scan image showing irregular thickening (arrows) of the interlobular septa in both lungs with associated architectural distortion. Pneumomediastinum (star) is also noted in this image

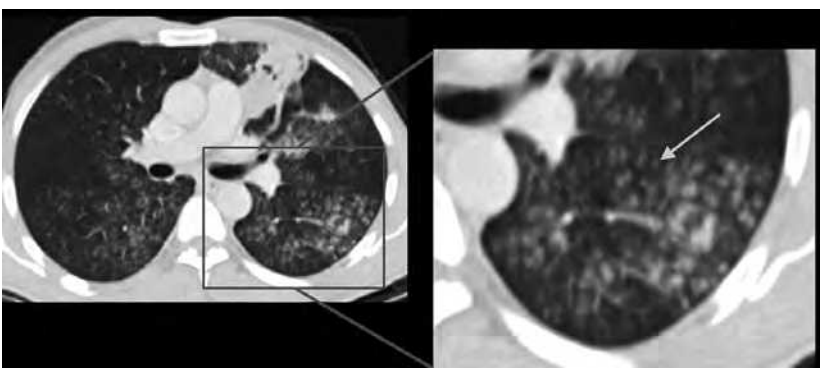


Fig. 1.5: Axial CT scan image showing multiple tiny nodules (arrow) seen in both lungs which are seen separate from pleura and fissure suggestive of centrilobular nodules

Nodular Pattern

- This pattern consists of discrete nodules of varying sizes (micronodules <3 mm, macronodules 3 mm to 1 cm) distributed in the pulmonary lobule.
- There are various causes such as granulomatous diseases, pneumoconiosis, metastases, etc.
- There are three types of nodules based on the distribution in relation to the structures in secondary pulmonary lobule.
- They can be *centrilobular, perilymphatic and random.*
- The *centrilobular nodules* are seen in relation to the centrilobular interstitium and they are separated from the peripheral interstitium by a distance **(Fig. 1.5)**.
- On HRCT, they are usually seen roughly in the centre of the secondary pulmonary lobule and not along the pleura or fissures.

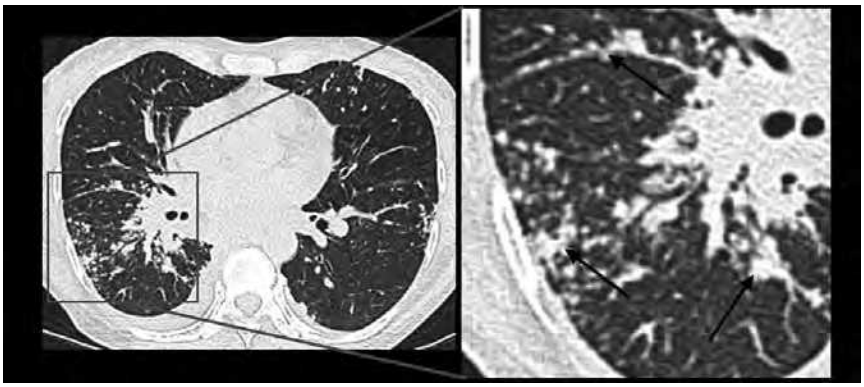


Fig. 1.6: Axial CT scan image of a patient diagnosed with sarcoidosis showing multiple tiny discrete nodules seen in the perilymphatic distribution (arrows). The fissure, pleura and the bronchovascular interstitium appear nodular

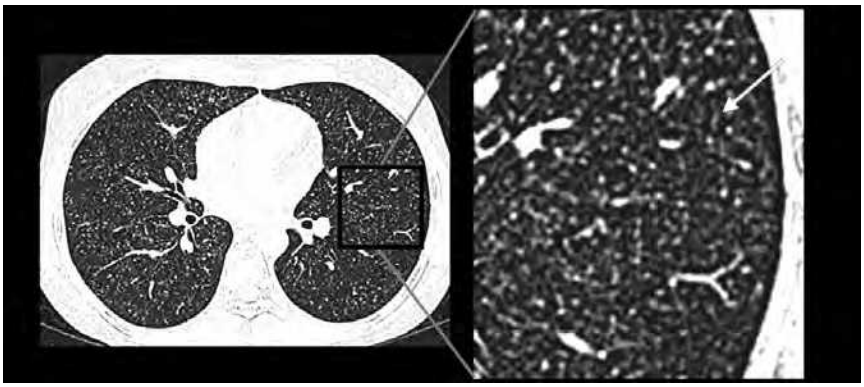


Fig. 1.7: Axial CT scan image of a patient diagnosed with miliary TB showing multiple tiny discrete randomly distributed nodules (arrow) in bilateral lungs

- The diseases primarily affecting the terminal part of the bronchioles such as hypersensitivity pneumonitis and RB-ILD produce this pattern.
- They can be sometimes associated by surrounding ground glass opacity
- *Perilymphatic nodules* are seen in along the peripheral interstitium which means the nodules are seen along the pleura and fissures on HRCT **(Fig. 1.6)**.
- Diseases which involve the lymphatics such as sarcoidosis and other granulomatous processes can manifest with this pattern.
- *Random nodules* will not follow any particular pattern and in general the distribution is uniform in the secondary pulmonary lobule **(Fig. 1.7)**.
- They can be seen in relation to the central interstitium or the peripheral interstitium.
- Diseases with hematogenous spread such as miliary TB and metastases can follow this pattern.

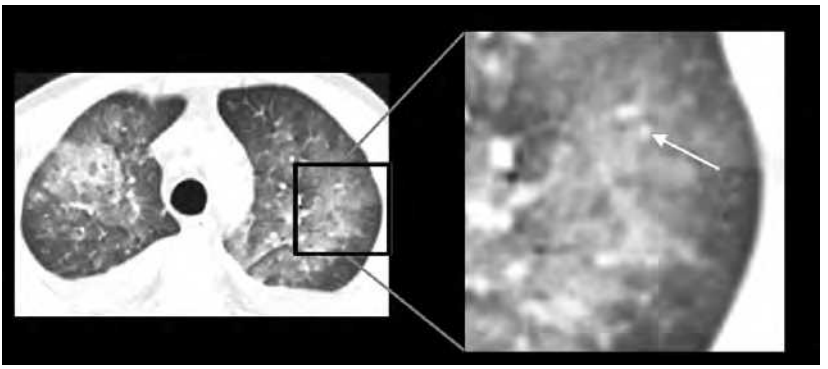


Fig. 1.8: Axial CT scan showing mild increase in the attenuation of bilateral lungs which is not seen to efface the pulmonary vessels (arrow) suggestive of ground glass opacity

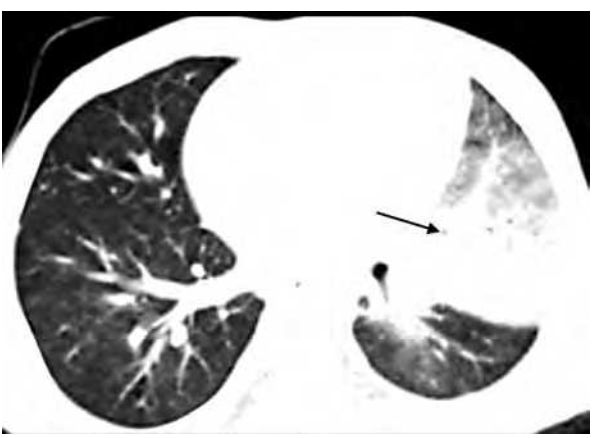


Fig. 1.9: Axial CT scan of a patient presented with acute fever and LRTI symptoms showing increased attenuation of left upper lobe with effacement of pulmonary vessels and patent bronchi (air bronchogram) (arrow) suggestive of consolidation

Ground-glass Opacities and Consolidation Pattern

- Both are abnormal increase in lung density with (*consolidation*) or without (*ground glass*) effacement of pulmonary vessels on CT **(Figs 1.8 and 1.9).**
- This pattern results from the opacification of alveolar spaces and/or interstitium.
- The type of manifestation will depend on the extent of involvement.
- Sometimes ground glass opacity can be associated with reticular pattern in which case it is called as *crazy paving pattern* **(Fig. 1.10)**.
- Alveolar pattern can be seen in a variety of acute, subacute or chronic disease processes such as AIP, ARDS, alveolar hemorrhage, pneumocystis pneumonia, bronchoalveolar carcinoma, etc.

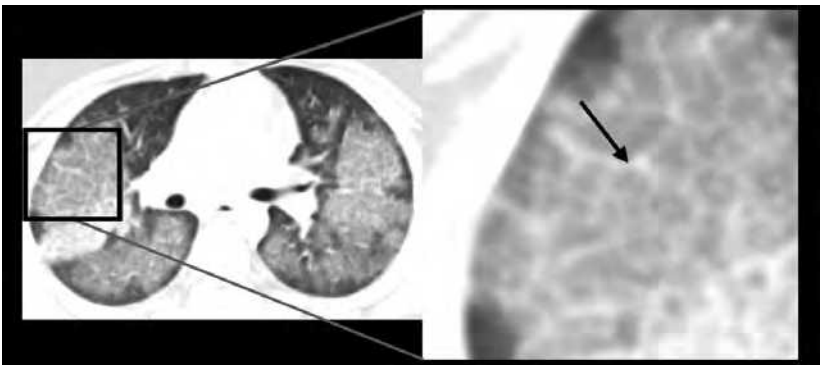


Fig. 1.10: Axial CT scan image showing ground-glass opacity in bilateral lungs with associated interlobular septal thickening (arrow) suggestive of crazy paving appearance



Fig. 1.11: Axial CT scan image showing patchy hyperlucent areas (arrows) in bilateral lungs suggestive of mosaic attenuation pattern

- *Mosaic attenuation* is another pattern which is commonly encountered presenting as patchy areas of increased or decreased lucencies leading to heterogeneous appearance of lung parenchyma **(Fig. 1.11)**.
- This pattern can be as a result of abnormality in small airway, peripheral pulmonary vasculature or interstitium.[5]
- When there is abnormality in the small airway (terminal or respiratory bronchioles) the air gets trapped in distally and there will not be a normal outflow of air during expiration.
- This leads to patchy areas of hyperlucency and it is a challenge to recognise the abnormal area (hyper or hypolucent area). There can be associated pulmonary oligemia in the affected areas because of hypoxia due to air trapping.
- By performing expiratory scans the abnormal areas showing air trapping can be readily identified. There will be normal outflow of air in the normal

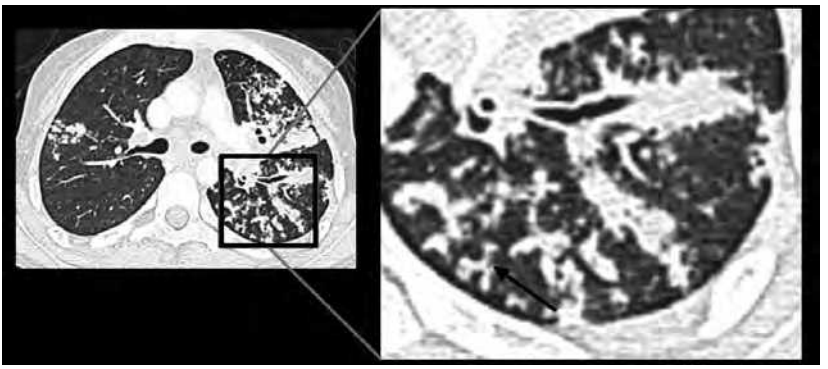


Fig. 1.12: Axial CT scan image of a patient diagnosed with pulmonary tuberculosis showing multiple centrilobular nodules (arrow) in bilateral lungs giving an appearance of tree in bud

lung during expiration whereas there will be air trapping in the affected areas which leads to accentuation of differential lucency.

- Whenever the heterogeneity is more pronounced in expiratory scans air trapping due to small airway disease.
- Mosaic attenuation can also occur as a result of mosaic perfusion which occurs in patients with pulmonary hypertension of any cause such as chronic thromboembolic pulmonary hypertension and primary pulmonary hypertension.[6]
- In these patients there will be focal pulmonary oligemia in the affected areas however there will not be air trapping in expiratory scans.
- Sometimes mosaic attenuation can be due to patchy ground glass infiltration wherein there will not be air trapping or pulmonary oligemia.
- *Tree in bud pattern* is a manifestation of endobronchial spread of infection such as TB or aspiration **(Fig. 1.12)**.
- It is characterized by linear branching structures (opacified distal bronchioles) with nodular opacities (acinar opacification) at the tip.

Cystic Pattern

- It is characterized by the presence of complete lucency which may or may not have walls.
- Cystic lesions in lung are usually formed by the abnormal dilatation of the bronchus/bronchioles or due to abnormal distention of air spaces.
- There are various patterns of cystic lesions of lung such as *cluster of grapes, string of pearls, honeycombing and random cysts*.
- *Cluster of grapes* pattern is usually due to the abnormal dilatation of airway involving a lobe of segment which is seen in cystic bronchiectasis **(Fig. 1.13)**.
- They usually have thick walls and may contain mucus, pus, debris or even fungal ball.
- Air fluid level in these cystic lesions often denote secondary infection.

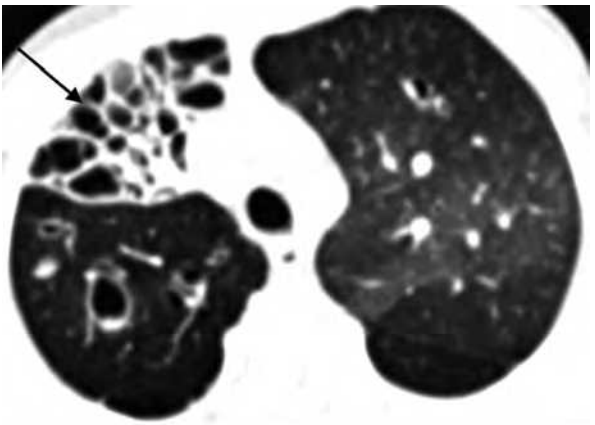


Fig. 1.13: Axial CT scan image showing clustering of cystic lesions (arrow) predominantly in the right upper lobe associated with volume loss suggestive of cystic bronchiectasis

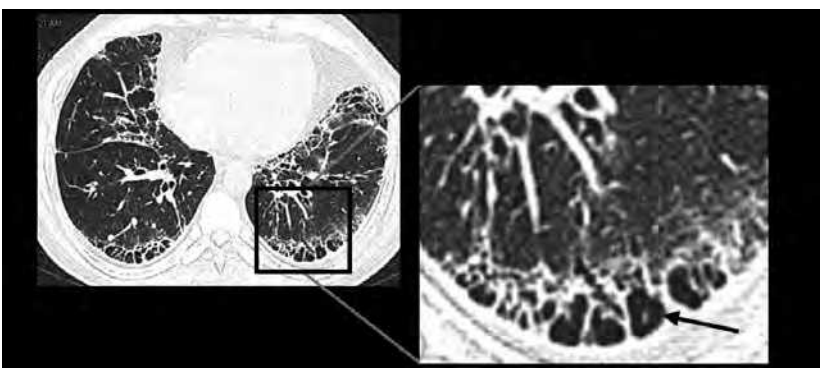


Fig. 1.14: Axial CT scan showing multiple thin walled tiny cysts (arrow) which are arranged in multiple rows in the periphery of bilateral lungs suggestive of honeycombing

- *String of pearls* pattern is characterized by the presence of multiple cysts arranged in a single row along the pleura (subpleural blebs).
- They may have a thin wall and when they are more than 1 cm in size they are called as bullae.
- Paraseptal emphysema is a typical example for this type of pattern.
- *Honeycombing* as the name suggests consists of relatively smaller thick walled cysts arranges in multiple layers **(Fig. 1.14)**.
- They are usually seen in fibrosing ILDs in which case there can be associated irregular reticular pattern, lung volume loss and tractional bronchiectasis.
- The architecture of the secondary pulmonary lobules gets distorted due to extensive fibrosis.
- Some disease processes such as centriacinar emphysema, pulmonary LCH and lymphangioleiomyomatosis can present with randomly distributed cysts **(Fig. 1.15)**.

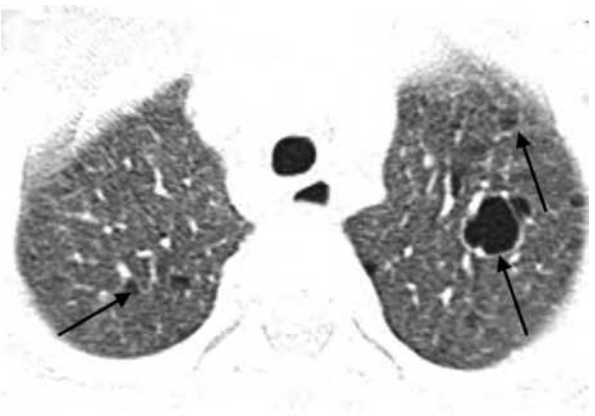


Fig. 1.15: Axial CT scan image of a patient diagnosed with lymphangioleiomyomatosis showing multiple randomly distributed thin walled cysts of varying sizes seen in bilateral lungs. Note that the background lung is normal

- Wall of the cysts, their craniocaudal distribution and other associated findings will dictate the differential diagnosis.

It is important to recognize various patterns of lung changes occurring in ILDs, to classify them accurately. The differential diagnosis depends on the pattern of abnormality thereby directing patients to an appropriate treatment protocol.

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2

Optimizing the Imaging Modalities

Devasenathipathy Kandasamy, Ankur Goyal

INTRODUCTION

- As the clinical findings are usually not accurate in the diagnosis and classification of interstitial lung diseases (ILDs), imaging plays an important role
- Imaging is now being considered as an integral part of management
- Earlier, radiographs were the only imaging modality available. However, the introduction of computed tomography (CT) had changed the entire perspective of imaging.
- Currently, CT scan has become the modality of choice for the evaluation of ILD
- In this chapter, the basics of imaging modalities, principles of imaging and recent advances will be discussed.

IMAGING MODALITIES

Chest Radiographs

- Chest radiograph posteroanterior (PA) view in suspended inspiration is the standard view used for the evaluation of ILDs
- Lateral views are not recommended
- Radiographs are not sensitive in detecting the abnormalities associated with ILDs
- By the time, the abnormalities become apparent on imaging, significant changes would have happened in the lung parenchyma
- It can show volume loss in the lung fields, changes in the major airways and lung parenchyma **(Fig. 2.1)**.
- Most of the ILDs result in fibrosis eventually which causes volume loss and ectasia of airways which can be seen on radiographs
- Thickening of interlobar septum and centrilobular interstitium is seen as reticulonodular opacification of lung fields, especially in the periphery of the lungs **(Fig. 2.1)**.

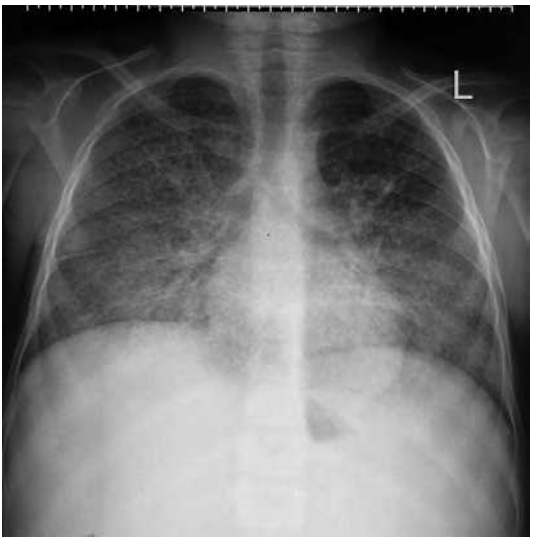
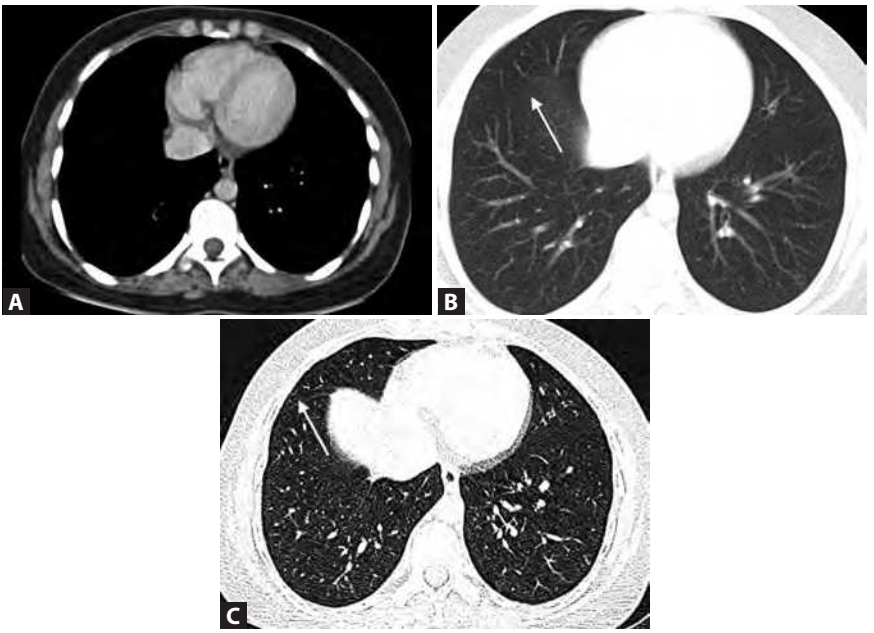


Fig. 2.1: Chest radiograph posteroanterior (PA) view showing reticulonodular opacities with associated volume loss in bilateral lung fields suggestive of interstitial lung disease

- There can be reduction in the normal lucency of lungs (ground glass) because of diffuse infiltration of lung parenchyma
- Sometimes changes seen on radiographs can be focal as seen in radiation induced changes
- Radiographs can also show the severity and distribution of abnormalities which can be helpful in the differential diagnosis
- In diseases like sarcoidosis and asbestosis features on radiographs can give valuable clues to the diagnosis.

Computed Tomography

- Computed tomography (CT) has become the modality of choice in most of the chest diseases
- Conventional CT has excellent spatial resolution in terms of localizing the compartment of the abnormality and its characterization
- The reconstruction kernel used in conventional CT is aimed at optimizing the signal to noise ratio (SNR)
- However, the changes in ILDs are often subtle and may not be seen on conventional CT scans
- This led to the development of an improvised technique called high-resolution computed tomography (CT) (HRCT) in 1985 by Naidich et al.
- The HRCT provides exquisite details about the pulmonary architecture and can also detect subtle ground glass opacities
- The HRCT is performed by using thin collimation during acquisition of data and then high resolution reconstruction algorithm is used to generate the image **(Figs 2.2A to C)**.



Figs 2.2A to C: Axial contrast-enhanced computed tomography (CECT) scan image (A) in mediastinal window setting showing the mediastinal and chest wall structures. CT scan image in lung window (B) reconstructed using soft tissue kernel showing the lung parenchyma. Note that the fissures are not visualized directly and the location of fissure is indirectly denoted by the avascular region (arrow). Thin slice (~1 mm) HRCT image (C) of the same patient which is reconstructed using bone kernel and smaller FOV showing the lung parenchyma in much better resolution than image B. Also note that the right major fissure is directly shown as a thin linear structure (arrow) owing to the higher spatial resolution

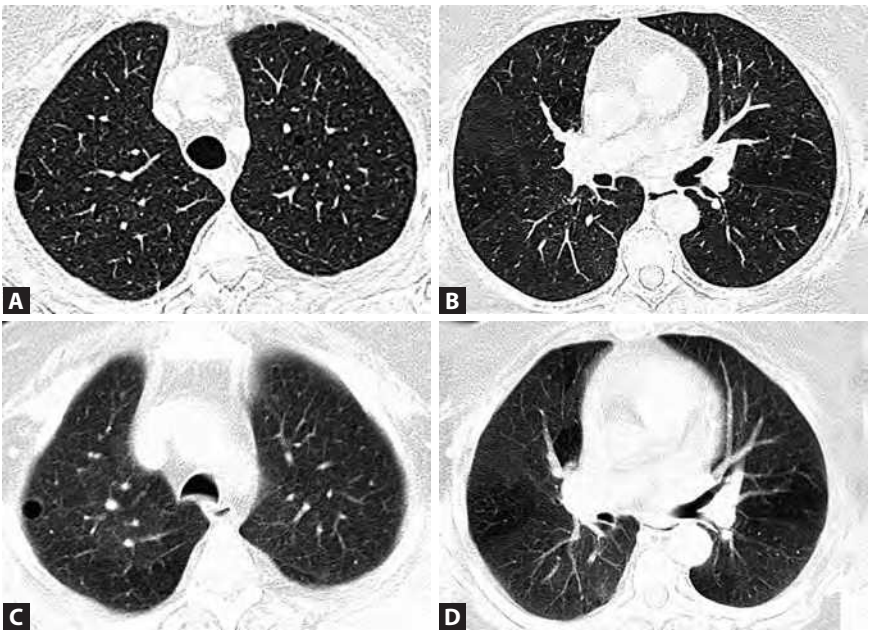
- The reconstruction field of view (FOV) should be adjusted in such a way to include only the lung fields. It is important because unnecessary increase in FOV will compromise the resolution
- Sometimes if it is needed, images of single lung can also be reconstructed instead of entire thorax to increase the spatial resolution
- Thin collimation as well as high resolution reconstruction increases the spatial resolution of the image in both z and x-y axes **(Figs 2.2A to C)**.
- Before the introduction of multidetector computed tomography (MDCT) the method used for performing HRCT was *axial incremental method*.
- In this method between two thin slices a gap of 10–20 mm is given so as to reduce the radiation load and time of examination
- The gap between the slices will not usually compromise the diagnostic quality because the abnormality in ILDs is usually diffuse and hence not missed
- The advantage of this method is reduction in radiation dose and the disadvantage is that the multiplanar reconstruction is not possible
- With the advent of MDCT, *volumetric method* has been used to perform HRCT

- In this method, the entire volume of lung is radiated and then high resolution images are reconstructed
- This method is widely used as it is fast and the volume data can be reconstructed in any plane. However, the radiation dose is more than the incremental method
- The characteristics and parameters used to acquire HRCT images are shown in the **Table 2.1**

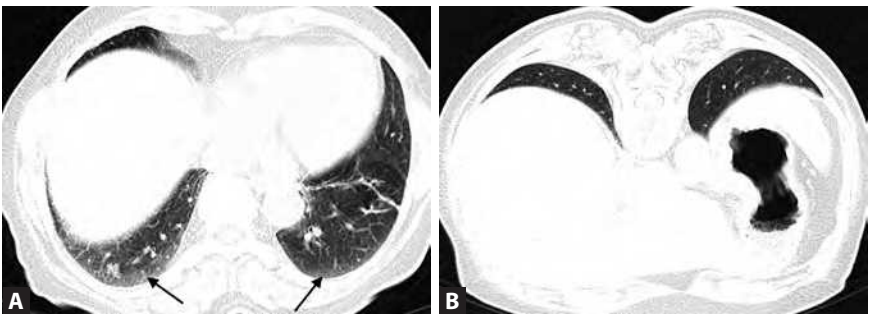
Table 2.1: Characteristics and parameters used to acquire HRCT images

Parameter	*Recommended Value*
Section thickness	0.625–1.5 mm
kVp	120
mAs	100–240 (tube current modulation recommended)
Gantry rotation time	Less than 1 sec
Matrix size	512 x 512 or largest available
Reconstruction algorithm	High spatial frequency (high resolution algorithm such as bone algorithm)
Slice increment in axial incremental HRCT	10–20 mm
Viewing window setting	Lung window

- Images are acquired with patient in supine position with suspended full inspiration
- Occasionally, it may be necessary to do scans in prone position or in expiratory phase, in which case incremental method of acquisition may be preferred to reduce radiation **(Figs 2.3 and 2.4)**
- The exposure factors should be optimized to reduce radiation
- Tube current modulation and iterative reconstruction may be used if available to reduce radiation
- There is no benefit of administering IV contrast for the evaluation of parenchyma in ILDs. Rather subtle findings can be obscured by contrast. Hence, IV contrast should not be used
- Low dose HRCT is usually performed with mAs around 50 which is as good as routine HRCT in most aspects
- Disadvantage of low dose HRCT is that the noise and streak artifacts are more prominent and subtle ground glass opacities can be missed. Hence, it is recommended only for following-up of patients with known ILD
- The acquisition of images can be either craniocaudal or caudocranial. Sometimes caudocranial acquisition can be used to reduce the artifacts close to the diaphragm.
- This approach is suited in diseases where the abnormalities are in lower lobes which may otherwise be affected by motion artifacts if the patient is not able to hold the breath adequately
- The evaluation of emphysema on CT can be qualitative, semiquantitative or quantitative

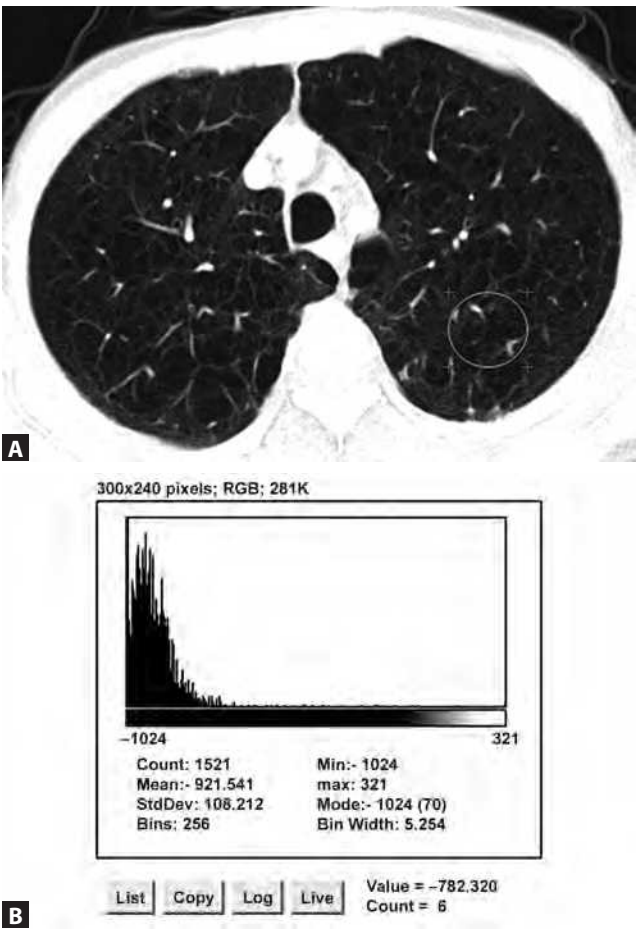


Figs 2.3A to D: Axial CT scan image in inspiratory phase (A and B) and in expiratory phase (C and D) showing features of air trapping. In the inspiratory phase, there is subtle mosaic attenuation seen in both lungs which has become more pronounced on expiratory phase. The phase of respiration on CT image is evidenced by the tracheal shape. The above findings are characteristic of air trapping in small air way disease



Figs 2.4A and B: Axial CT scan of lower chest in supine (A) and prone (B) position showing dependent opacities in bilateral lower lobes (arrows) on supine scan which got resolved in the prone scan. This highlights the utility of prone scan in differentiating dependent atelectasis from ground glass opacity

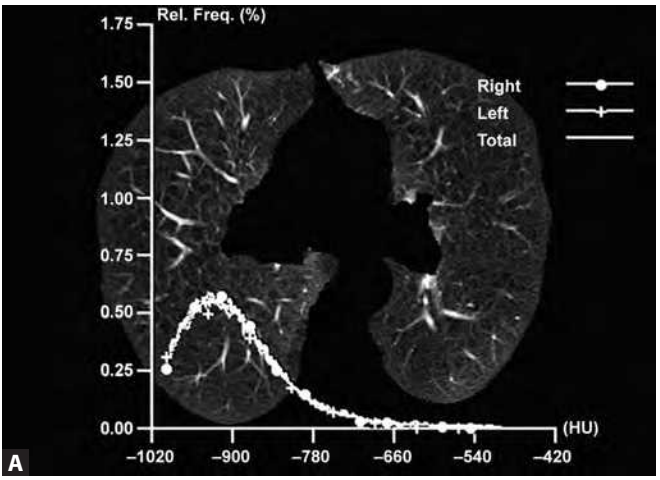
- Qualitative evaluation is done using visual analysis of the pathology and the type of emphysema
- There are various methods and softwares to perform semiquantitative and quantitative analysis. Various metrics are evaluated and proposed

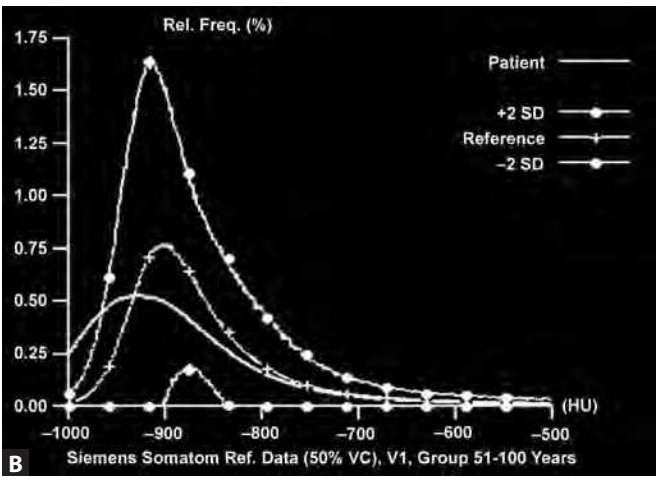


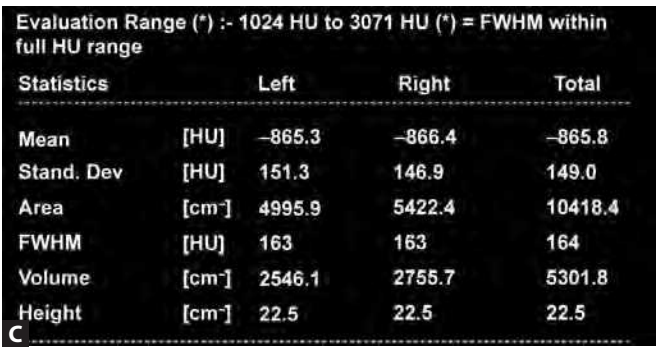
Figs 2.5A and B: CT densitometry evaluation in a patient with emphysema showing quantitative parameters on histogram analysis of region of interest (ROI). A ROI is drawn in the left lung (A) and the ROI analysis of the CT attenuation numbers within the ROI are represented in image B. The total number of pixels in the ROI are 1521 and around 48% of pixels are showing HU values less than -950 suggestive of extensive emphysema

for emphysema evaluation. Commonly used parameter is the percentage of voxels having HU values less than -950. It is obtained by performing histogram analysis on the lung region of interest (ROI) **(Figs 2.5A and B)**. Another method is percentile method and it is expressed as the HU value below which 15% of all voxels are distributed. It is known as 15th percentile cut off attenuation or Perc15.

- *Quantitative evaluation* of lung parenchymal pathology can be done using various softwares (e.g. Pulmo CT, Siemens, Erlangen, Germany) **(Figs 2.6A to C)**. Automatic recognition and segmentation of lung parenchyma using predefined HU values can be done. The airway,






Evaluation Range (*) :- 1024 HU to 3071 HU (*) = FWHM within full HU range

Statistics		Left	Right	Total
Mean	[HU]	−865.3	−866.4	−865.8
Stand. Dev	[HU]	151.3	146.9	149.0
Area	[cm⁻]	4995.9	5422.4	10418.4
FWHM	[HU]	163	163	164
Volume	[cm⁻]	2546.1	2755.7	5301.8
Height	[cm⁻]	22.5	22.5	22.5

C

Figs 2.6A to C: Quantitative evaluation of lung parenchyma using Pulmo CT software (A to C) showing histogram of the pixel values (A) and based on which a curve is generated. Image B showing the curve corresponding to the patient along with two standard deviations and a reference standard. The quantitative parameters of right and left lungs separately are summarized (C). Various parameters such as mean lung density, standard deviation, area and volume are generated automatically by the software

pulmonary vessels and other nonparenchymal structures are excluded from the calculation. Histogram with the frequency distribution of various pixels, HU values are shown and curves are generated. Attenuation distribution, percentage of pixel values, mean lung density can also be calculated and displayed using color coded images

- Recent advancements in dual energy CT have the potential to provide information regarding the lung perfusion. With material decomposition using dual energy CT, iodine in the contrast material can be mapped which may be valuable in characterization of ILDs.

Magnetic Resonance Imaging

- Magnetic resonance imaging (MRI) once considered as a unsuitable modality for evaluation of lung parenchyma, has been used successfully in the recent years
- The challenges of performing MRI for lung are low proton density in lung which affects SNR and susceptibility artifacts produced by air-lung interface
- With the recent advances in MRI such as newer sophisticated sequences, parallel imaging, etc. lung MRI has become a reality and can be performed in variety of conditions such as cystic fibrosis, pulmonary embolism, pneumonia and ILD
- It is especially useful in children, young patients who are in need of repeated imaging and in pregnant patients
- MRI can visualize and recognize the morphological changes and patterns in ILD
- It can assess the activity of disease process and the changes in perfusion and contrast enhancement
- Interstitial and air-space abnormalities increase the proton density and hence they are seen as hyperintense areas on T2-weighted sequences **(Fig. 2.7)**.
- Honeycombing and reticular opacities can be visualized on T1-weighted GRE images with and without contrast
- Signal intensity on T2-weighted images and the contrast kinetics on post-contrast images has been shown to correlate with the disease activity
- The basic sequences used for lung MRI are fast T1- and T2-weighted sequences, steady state free precession and postcontrast fat saturated 3D GRE sequence
- Advances in *functional MRI* has opened avenues for the evaluation of chronic obstructive pulmonary disease (COPD) for which multiple repeat imaging may be necessary. Although CT is the commonly used modality for their evaluation, MRI can be an alternative to reduce radiation burden. Dynamic contrast enhanced MRI (DCE-MRI) can be used to assess the severity of COPD. The principle is based on the fact that the ventilation abnormality in COPD will lead to perfusion abnormality which can be detected on DCE MRI. Another novel method to evaluate the extent of damage in COPD is by performing T1 mapping which basically evaluates the proton density in the lung.

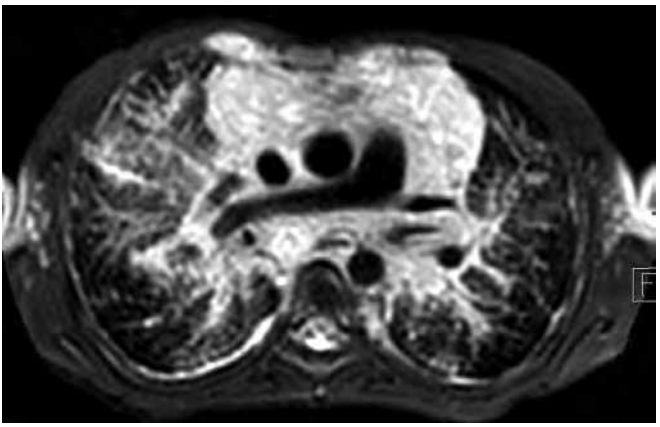


Fig. 2.7: Axial T2 W image showing extensive hyperintense lesions seen in the mediastinum and in the lung interstitium in a patient with pulmonary lymphangiectasia

CONCLUSION

The HRCT is currently the imaging modality of choice and will remain so in the near future. It has performed well in showing the anatomical information in great detail since its introduction three decades ago. In the recent years, MRI has been shown to be useful in many clinical situations especially in children and pregnant patients. With continued improvement in the sequences and technique the role of lung MRI will increase significantly in the future.

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3

Interstitial Lung Diseases: Pathologic Diagnosis

Deepali Jain

INTRODUCTION

- Group of diffuse pulmonary parenchymal diseases
- Classified together because of similar clinical, radiologic, physiologic, and/or pathologic manifestations
- Surgical lung biopsy is ideal for diagnosis of interstitial lung disease (ILD) to differentiate between the patterns of interstitial pneumonias. Biopsy includes pleura, interstitium, alveolar space and airway, and vessels
- Airway-centered or perilymphatic processes can be diagnosed by transbronchial biopsy
- Newly available transbronchial cryobiopsy produces larger tissue fragments with very little artefact
- Bronchoalveolar lavage (BAL) has limited role in diagnosis.
  Interstitial lung disorders—major categories[1,2] classification (*See* Chapter 1).

STAINING AND EVALUATION OF SECTIONS

- Hematoxylin and Eosin (H and E) is usually sufficient
- Gram, Ziehl-Neelsen, silver methenamine, Periodic acid-Schiff (PAS)—for tuberculosis (TB) and fungal infection
- Elastic Van Gieson (EVG) stains—for vasculitic process.

HISTOPATHOLOGIC PATTERNS ON SURGICAL LUNG BIOPSY

Six histopathologic patterns are described:

1. Acute injury
2. Fibrosis
3. Cellular infiltrates
4. Alveolar filling
5. Nodules
6. Minimal changes.

Acute Injury

- Prototype is acute interstitial pneumonia (*diffuse alveolar damage—DAD pattern*)
- Histopathology variable, highly dependent on timing of biopsy
- Hyaline membranes most prominent in initial 2–4 days of injury **(Fig. 3.1)**
- Late stage shows inflammatory cells and alveolar filling pattern of organizing pneumonia type; no or tiny remnants of hyaline membranes
- Clue of DAD in late stage is fibrin thrombi in pulmonary artery branches
- Other patterns of acute lung injury without hyaline membrane
  - Acute eosinophilic pneumonia
  - Acute fibrinous and organizing pneumonia **(Fig. 3.2)**
  - Diffuse alveolar hemorrhage **(Figs 3.3 and 3.4)**.

Fibrosis

- Prototype is idiopathic pulmonary fibrosis (*usual interstitial pneumonia—UIP pattern*)
- Surgical lung biopsy remains the gold standard for diagnosis of UIP, and sampling of at least two sites is recommended[3]

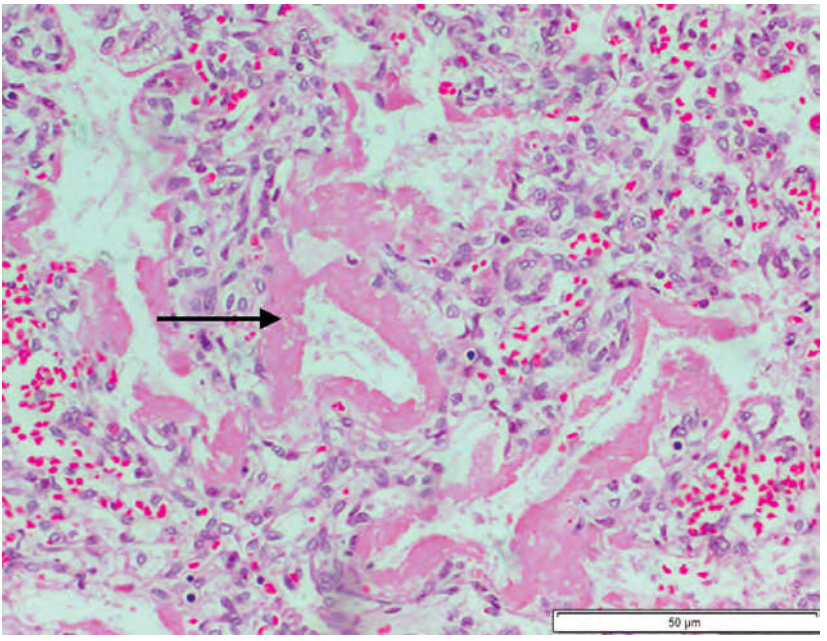


Fig. 3.1: Photomicrograph shows hyaline membrane (arrow) lining alveolar walls in a case of AIP (H and E x 400)

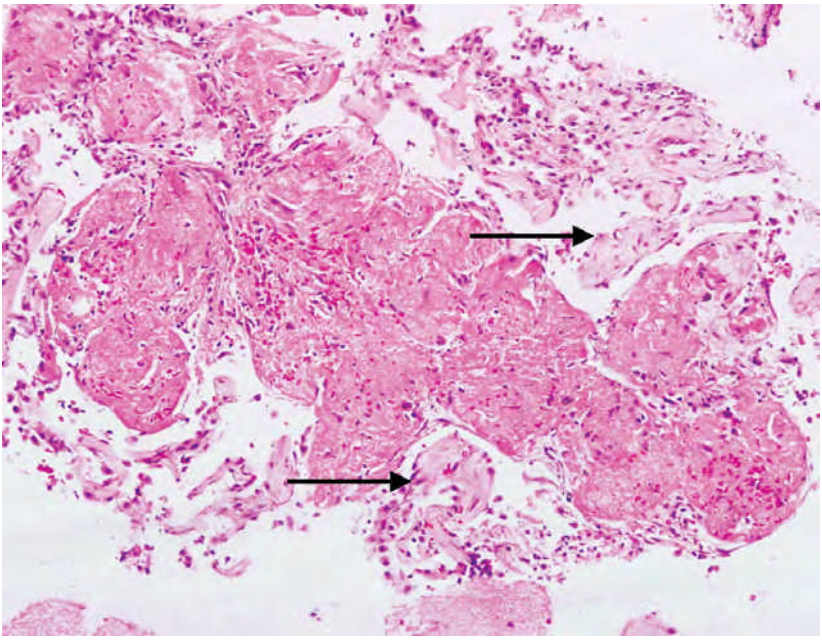


Fig. 3.2: Acute fibrinous and organizing pneumonia (AFOP) with fibrinous exudate within alveolar lumina and evidence of organization (arrows) at other parts of biopsy (H and E x 40)

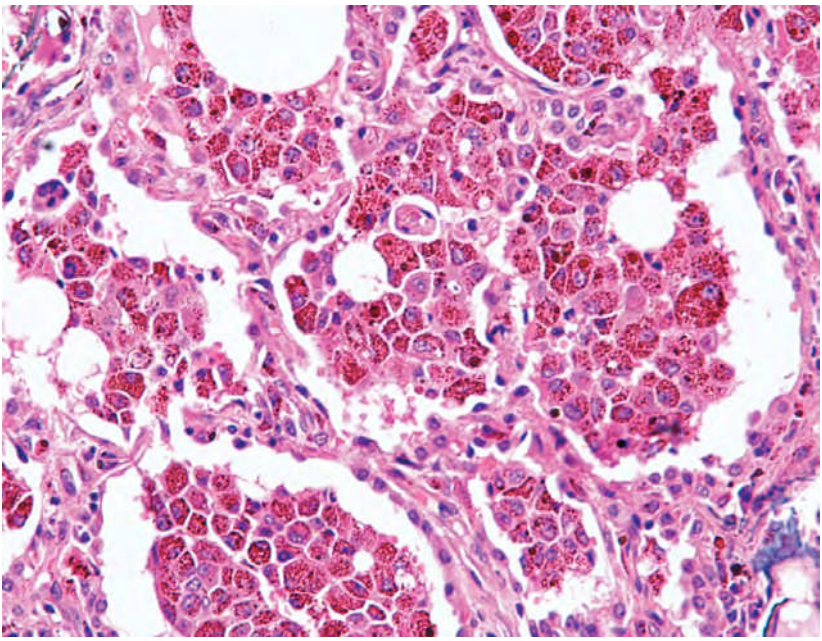


Fig. 3.3: Diffuse alveolar hemorrhage (H and E x 400)

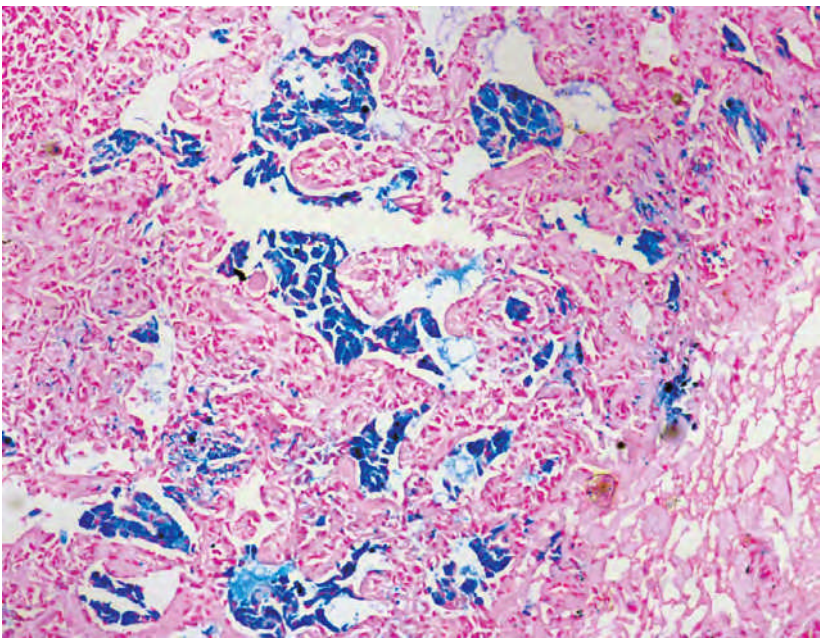


Fig. 3.4: Diffuse alveolar hemorrhage (DAH) shows hemosiderin laden macrophages. Prussian blue stain

- Solid areas of well-established fibrosis that alternate with zones of normal lung (spatial or geometric heterogeneity) **(Fig. 3.5)**
- Foci of immature fibroblastic proliferation (fibroblastic foci) and old fibrosis (temporal heterogeneity) **(Fig. 3.6)**
- Other diseases with UIP pattern—rheumatic ILD and chronic hypersensitivity pneumonitis.

Definite UIP (all four of these following):

1. Marked fibrosis or architectural distortion with or without honeycombing in a predominantly subpleural or paraseptal distribution
2. Patchy involvement of lung parenchyma by fibrosis
3. Presence of fibroblast foci
4. Absence of features against a diagnosis of UIP suggesting an alternate diagnosis, that is, "not UIP."

Not UIP (*any of these six criteria*): (1) Hyaline membranes; (2) Organizing pneumonia; (3) Granulomas; (4) Marked interstitial inflammatory infiltrate away from honeycombing; (5) Predominant airway-centered changes; or (6) Other features suggestive of an alternate diagnosis.[4]

Cellular Infiltrates

- *Prototype:* Hypersensitivity pneumonitis (HP)

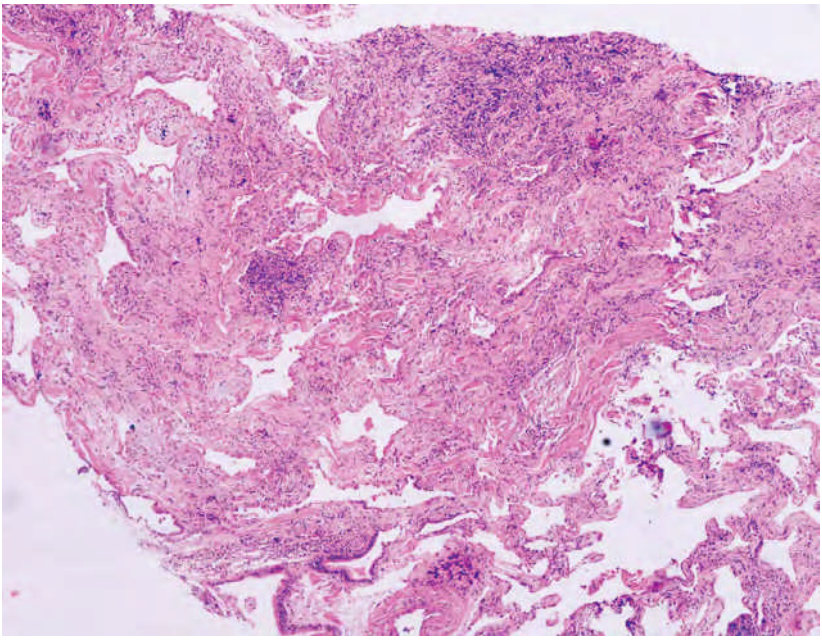


Fig. 3.5: Usual interstitial pneumonia (UIP) with spatial heterogeneity; normal appearing lung at right lower corner whereas fibrosis in upper part (H and E x 40)

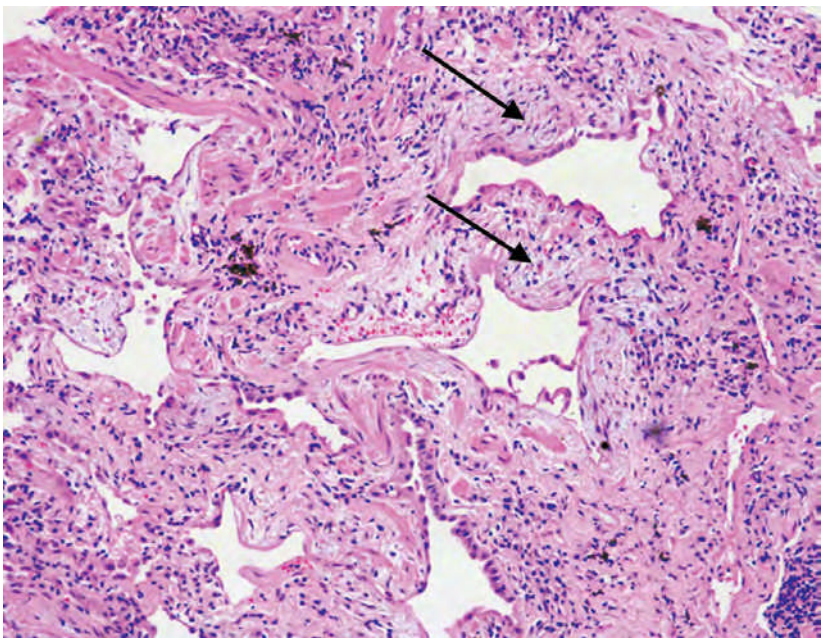


Fig. 3.6: Fibroblast foci (arrows) in UIP (H and E x 400)

- Immunologically-mediated pulmonary disease due to a reaction to inhaled organic dusts or exposure to an inciting antigen such as fungi, thermophilic bacteria or animal protein
- *A triad of features seen:*
  - Diffuse lymphoplasmacytic interstitial infiltrate with bronchiolocentric accentuation **(Fig. 3.7)**
  - Poorly formed non-necrotizing granulomas or Schaumann bodies **(Fig. 3.8)**
  - Foci of organizing pneumonia **(Fig. 3.9)**
- *Nonspecific interstitial pneumonia (NSIP):*
  - Now accepted as a specific clinicopathologic entity
  - Idiopathic NSIP is a distinct clinical-radiologic-pathologic entity
  - Patchy distribution of uniform interstitial thickening by a cellular or a fibrosing process
  - Lymphoid follicles, enlarged air spaces, interstitial inflammation, focal organizing pneumonia, pleural fibrosis and vascular medial thickening.

Alveolar Filling

Alveoli in the biopsy specimen filled with cells or noncellular material.

Prototype: Organizing pneumonia (OP)

- Causes may be aspiration, bacterial, viral and fungal infections, distal to obstruction, inhalation of toxic compounds, drug reaction

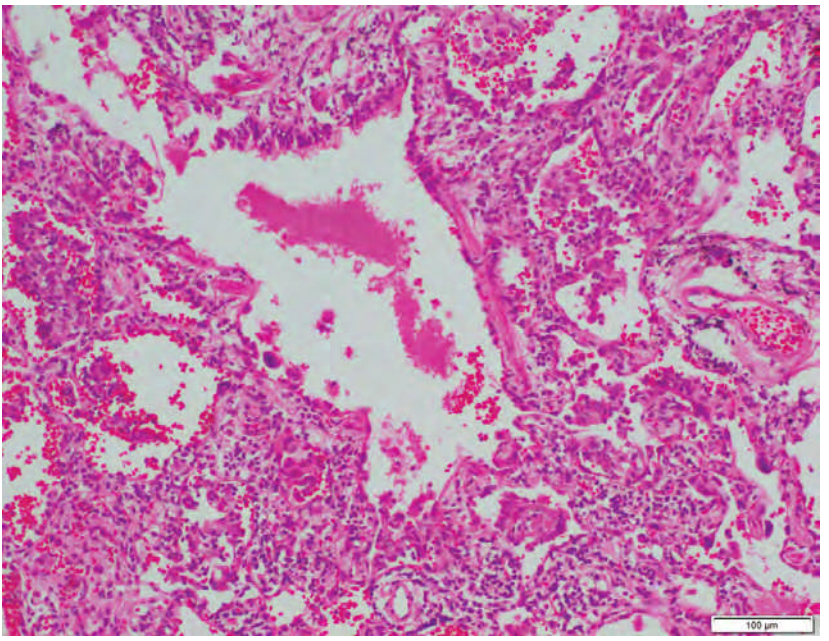


Fig. 3.7: Peribronchial lymphoplasmacytic inflammation in hypersensitivity pneumonitis (HP) (H and E x 400)

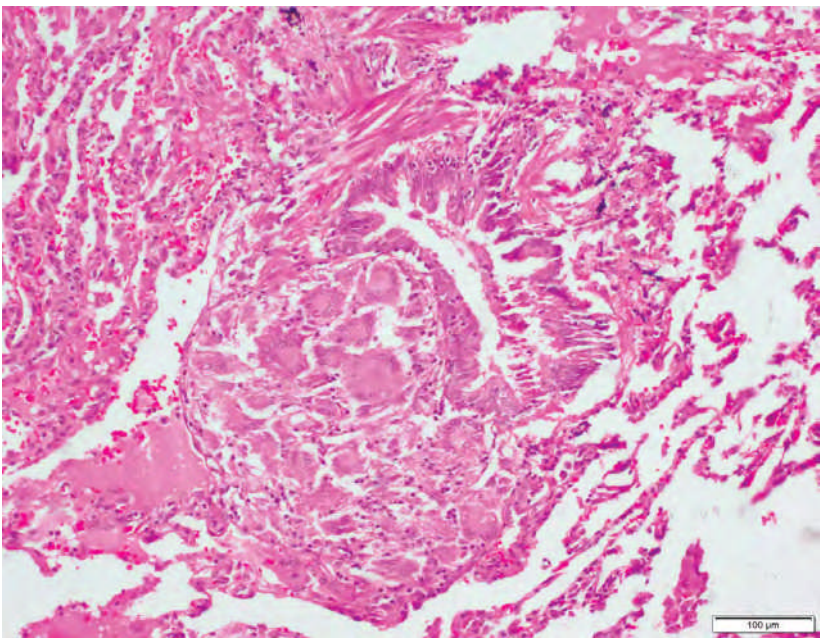


Fig. 3.8: Peribronchial ill-defined collection of giant cells in HP (H and E x 400)

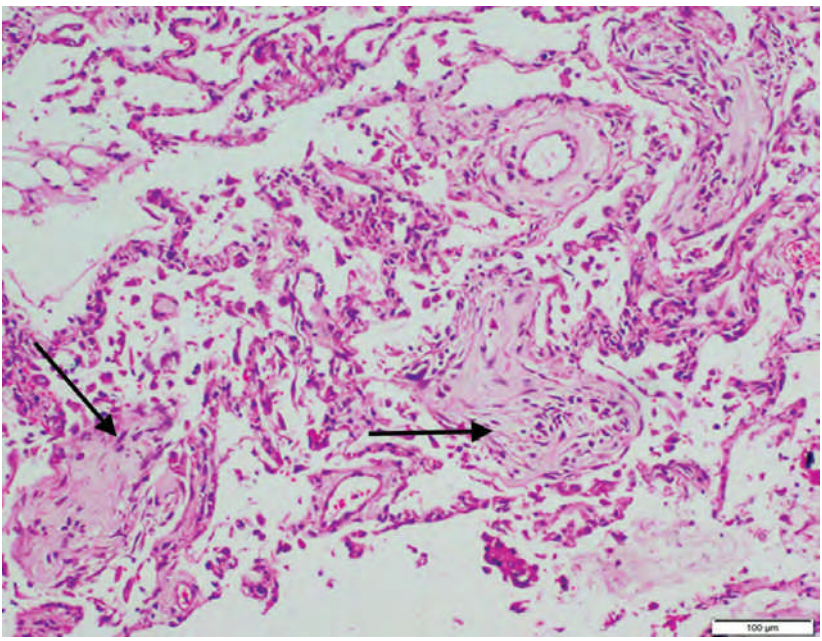


Fig. 3.9: Intra-alveolar polypoid plugs of fibrous tissue
(Masson bodies; arrows) in organizing pneumonia (OP) (H and E x 400)

- Cryptogenic OP (previously referred as BOOP) is a clinicopathological diagnosis of exclusion
- Intraluminal plugs are patchy and are composed of fibroblasts and myofibroblasts in a loose connective tissue **(*See* Fig. 3.9)**
- Associated mild interstitial infiltrate of lymphocytes, plasma cells, and histiocytes.

Other alveolar filling patterns:
Diffuse alveolar hemorrhage (DAH) **(*See* Figs 3.3 and 3.4)**, acute bronchopneumonia, pulmonary alveolar proteinosis **(Figs 3.10A and B)**, chronic eosinophilic pneumonia, desquamative interstitial pneumonia (DIP) **(Fig. 3.11)**, respiratory bronchiolitis-associated interstitial lung disease (RBILD).

Respiratory Bronchiolitis-associated Interstitial Lung Disease

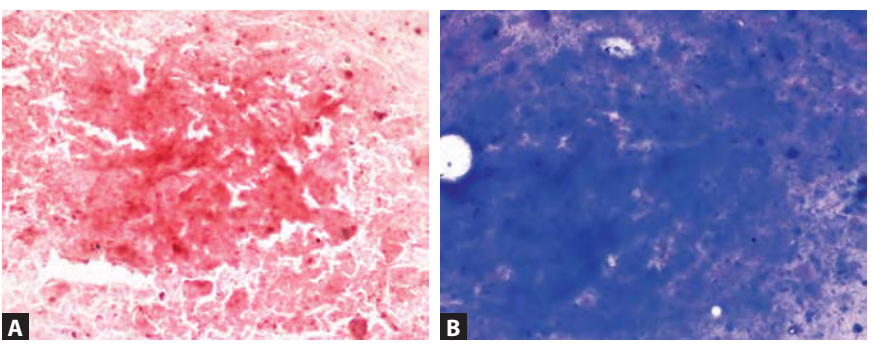
- Alveolar macrophage accumulation in centrilobular alveolar spaces
- Mild fibrosis
- Patchy distribution.

Desquamative Interstitial Pneumonia

- Diffuse distribution and large numbers of macrophages filling the alveolar spaces throughout
- Characteristic macrophage of DIP has abundant pale eosinophilic cytoplasm with finely granular light-brown pigment (so-called smokers macrophages)
- Intra-alveolar eosinophils, giant cells and lymphoid follicles within the lung interstitium (not seen in RBILD).

Lymphocytic Interstitial Pneumonia

- Diffuse, dense, interstitial lymphoplasma cellular infiltration **(Fig. 3.12)**
- Small non-necrotizing granulomas and giant cells.



Figs 3.10A and B: Acellular proteinaceous material in pulmonary alveolar proteinosis (PAP)
(10:Papanicolaou, 11:MGG stains)

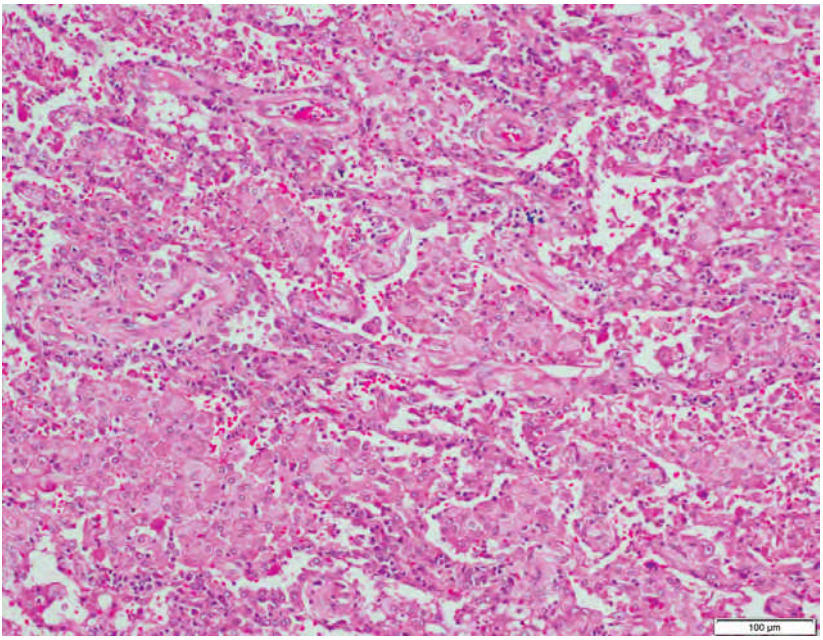


Fig. 3.11: Alveolar macrophages filling lumina of alveolar spaces in desquamative interstitial pneumonia (H and E x 400)

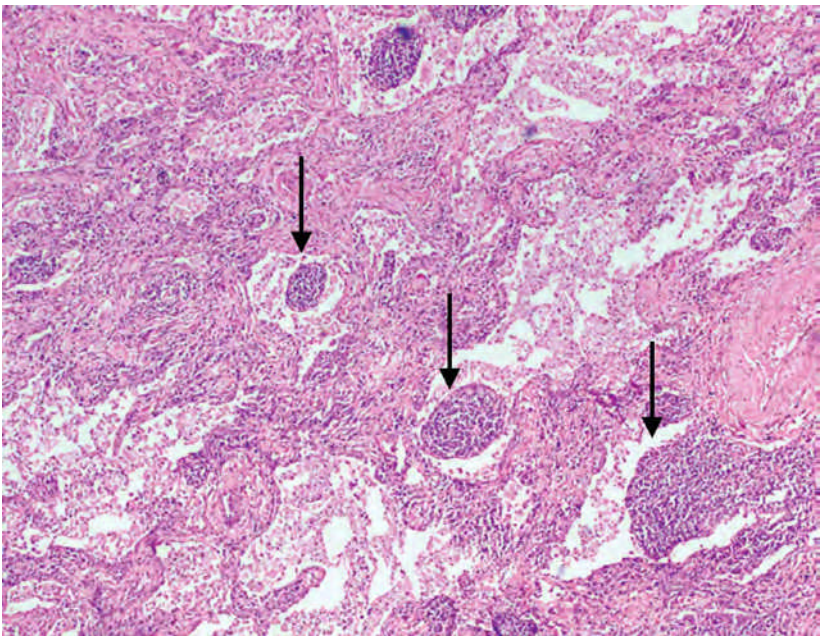


Fig. 3.12: Presence of lymphoid aggregates (arrows) in interstitium (H and E x 100)

Idiopathic Pleuroparenchymal Fibroelastosis

- Pleuroparenchymal fibroelastosis is a newly recognized entity in the 2013 ATS/ERS revision/update of the idiopathic interstitial pneumonia (IIP) classification
- Upper lobe fibrosis elastotic in nature.

Nodules

Three types of nodules are described:

1. Granulomas
2. Nodules in silicate disease
3. Nodular Langerhans cell histiocytosis (LCH).

Sarcoidosis

- Conglomerates of non-necrotizing granulomas distributed in lymphatic route **(Fig. 3.13A)**
- Distinctive Inclusions within giant cells—hematoxyphilic Schaumann body, lucent oxalate crystals and asteroid bodies **(Fig. 3.13B)**.

Nodules in Silicate Disease

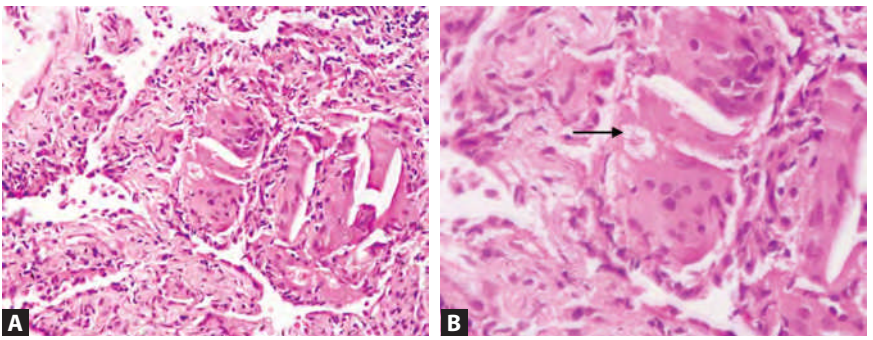
- Fibrotic nodules with admixed polarizable silicates
- Similar to sarcoidosis, pneumoconioses will have a lymphatic pattern (disease occurring along bronchovascular bundles, interlobular septa and pleura).

Nodular Langerhans Cell Histiocytosis

Nodules of fibroblasts, collagen, Langerhans cells and pigmented alveolar macrophages; immunoreactivity for S-100 protein and the CD1a antigen.

Minimal Changes

Includes vascular or lymphatic pathology and cystic lesions such as lymphangioleiomyomatosis.



Figs 3.13A and B: Granuloma in a case of sarcoidosis. Asteroid body (arrow) (H and E x 400)

PATHOLOGY REPORT CONTENT

- Adequacy of the specimen
- Pattern and severity
- IIP classification.

TRANSBRONCHIAL LUNG BIOPSY

Conditions which can be diagnosed on transbronchial lung biopsy (TBLB)[5]:

- Acute injury
- Cellular infiltrates
- Granulomatous disorders
- Lepidic pattern adenocarcinoma which on radiology mimics pneumonic consolidation or ILD
- Lymphangitic spread of metastatic carcinoma
- Alveolar proteinosis
- Alveolar microlithiasis.

BRONCHOALVEOLAR LAVAGE

Cellular Analysis in Interstitial Lung Disease[6]

- Bronchoalveolar lavage (BAL) cytologic examination may be useful in the diagnostic work-up of patients who lack usual interstitial pneumonia (UIP) pattern on high-resolution computed tomography (HRCT) imaging
- Although BAL differential leukocyte count (DLC) helps in narrowing down the different types of ILD, it is not very specific. However, in an appropriate clinico radiological context abnormal findings on BAL may support a specific diagnosis
- A normal BAL DLC does not exclude histologic abnormalities in the lung parenchyma due to ILD **(Fig. 3.14)**
- BAL cellular analysis is neither a prognostic nor a predictive marker in the diagnosis of ILD.

Technique, Transport, Processing, and Analysis

- Site of BAL may be chosen on the basis of radiologic abnormality present rather than performing a traditional BAL from right middle lobe or lingual lobe
- During BAL, the total instilled volume of normal saline should be 100–300 mL
- For satisfactory sampling of distal airspaces, the total retrieval volume in pooled aliquots should be at least 30% of the total instilled volume
- A minimum of 5 mL of a pooled BAL sample is needed for BAL DLC although 10–20 mL is ideal
- In addition to DLC, BAL should also be sent for microbiology, virology, and/or malignant cell cytology testing if clinically required
- BAL DLC with more than 15% lymphocytes signifies a lymphocytic cellular pattern. Similarly, greater than 3% neutrophils, more than 1% eosinophils,

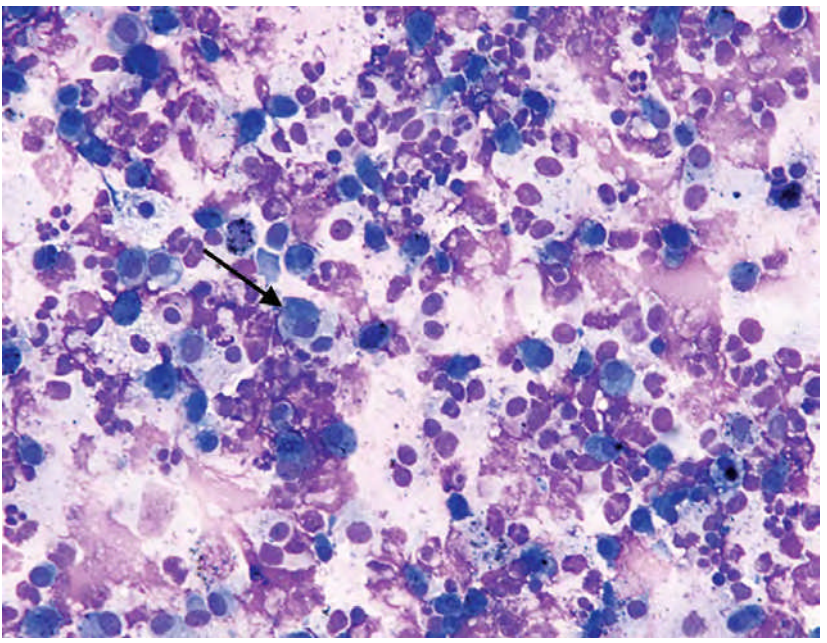


Fig. 3.14: Normal bronchoalveolar lavage pattern with predominantly alveolar macrophages (arrow) (MGG x 200)

and more than 0.5% mast cells indicate neutrophilic, eosinophilic and mastocytosis cellular patterns respectively

- A predominance of smokers macrophages without increase in other cell types is suggestive of smoking-related ILD such as desquamative interstitial pneumonia (DIP), respiratory bronchiolitis interstitial lung disease (RBILD), and Langerhans cell histiocytosis (LCH).

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4

Clinical Diagnosis including Pulmonary Function Test

Vijay Hadda

INTRODUCTION

- Interstitial lung diseases (ILDs) include a heterogeneous group of more than 200 lung diseases that share common clinical picture.
- ILDs involve parenchyma of the lung including the alveoli, the alveolar epithelium, the capillary endothelium, and the spaces between those structures—as well as the perivascular and lymphatic tissues.
- Sometimes inflammation and fibrosis extend to the airways (like in respiratory bronchiolitis and chronic hypersensitivity pneumonitis) or airways may be the predominant component involved (like bronchiolitis obliterans).
- Therefore, these disorders are sometimes called *diffuse parenchymal lung disorders* to make the point that the interstitium is not the only compartment of the lung affected.

CLASSIFICATION

Various classification schemes for ILDs have been proposed from time to time. However, currently, the classification proposed by the American Thoracic Society (ATS) and European Respiratory Society (ETS) published in 2001 and revised in 2013 (*See* Chapter 1) is accepted globally.

CLINICAL FEATURES

- Usually present with symptoms of exertional breathlessness, nonproductive cough with bilateral infiltrates on chest radiograph.
- Diagnosis is usually based upon comprehensive history and clinical examination, pulmonary function tests, chest imaging and occasionally on pathological examination of tissue obtained from lung biopsy.
- A pattern-based approach to HRCT can help in making a specific diagnosis in many cases or differentiate between diseases with similar presentations or at least, narrowing the differential diagnosis.

- In overlapping cases, lung biopsy might be necessary, and a multidisciplinary approach (integrating clinical presentation and laboratory data, chest radiology, and lung pathology) is considered the best approach to formulate a definitive diagnosis.

History

- The clinical presentation of ILD is often nonspecific.
- Symptoms such as dyspnea on exertion, cough and an abnormal chest radiograph mimics number of diseases.
- Therefore, a comprehensive patient history and clinical examination are crucial for diagnosis. In history, specific attention should be given to four main questions.
  1. Onset of symptoms
  2. Progression of the symptoms
  3. Current or past exposure to a known etiological agent responsible for ILD
  4. Severity of the disease.

Onset of Symptoms

Based on the onset of disease ILDs can be subdivided into four categories:

1. *Acute presentation (days to weeks)*: Acute interstitial pneumonia (AIP), acute eosinophilic pneumonia, acute hypersensitivity pneumonitis, alveolar hemorrhage syndromes, some drug related ILDs, acute immunological pneumonias complicating SLE or polymyositis.
2. *Subacute presentation (weeks to months)*: Sarcoidosis, drug induced ILDs, cryptogenic organizing pneumonias (COP).
3. *Chronic presentation (months to years)*: Most ILDs have chronic presentation idiopathic pulmonary fibrosis (IPF), sarcoidosis, connective tissue related ILDs, Langerhans cell histiocytosis (PLCH), pneumoconiosis.
4. *Episodic (symptomatic phase followed by asymptomatic phase)*: Include eosinophilic pneumonia, hypersensitivity pneumonitis, COP, vasculitides, alveolar hemorrhage syndromes, and Churg-Strauss syndrome.

Symptoms of Interstitial Lung Disease

Pulmonary symptoms

- *Dyspnea*
  - Dyspnea is the most common presenting symptom. It is usually insidious in onset and progressive.
  - In elderly or disabled patients with limited mobility, dyspnea may be a late symptom. It is often attributed to aging and deconditioning.
  - Acute worsening of dyspnea in known patients of cystic ILD may indicate the development of a spontaneous pneumothorax; especially if acute chest pain is present.
  - Degree of dyspnea is related to severity of disease and prognosis, however some ILDs have minimal dyspnea even with extensive parenchymal involvement such as hypersensitivity pneumonitis, sarcoidosis.

Table 4.1: Modified Medical Research Council Dyspnea scale

Grade	*Description of the grade*
0	No breathlessness on activity, except strenuous
1	Breathlessness while hurrying on level or walking up slight hills
2	Walks at slower pace than the peers or has to stop after some distance while walking at own pace due to breathlessness
3	Stops due to breathlessness after 100 yards or after few minutes of walking
4	Too breathless to leave home or while dressing/undressing

- Modified Medical Research Council (mMRC) dyspnea scale is used to assess the severity of dyspnea presence due to various respiratory diseases. The table below summarizes mMRC dyspnea scale **(Table 4.1)**.

- *Cough*
  - Cough is the second most common symptom which is often disturbing and bothersome.
  - It is predominantly nonproductive and is particularly seen in airway centered ILDs such as sarcoidosis, HP, organizing pneumonia or respiratory bronchiolitis.
- *Wheezing*
  - Wheezing as such is infrequent among patients with ILD.
  - Like cough, wheezing is also associated with airway centered ILDs including Churg-Strauss syndrome, HP, eosinophilic pneumonias or airway-stenotic sarcoidosis.
- *Chest pain*
  - Chest tightness or discomfort is not unusual however frank pain is an uncommon clinical presentation.
  - When present pleuritic chest pain may indicate development of pneumothorax in cystic ILDs; or serositis in ILDs associated with connective tissue disorders or drug-induced or asbestos-related diseases.
  - Substernal chest discomfort is often described by patients with sarcoidosis.
- *Hemoptysis*
  - Hemoptysis is rare in patients with ILD.
  - Frank hemoptysis or blood streaked sputum can be seen in patients with diffuse alveolar hemorrhage (DAH) syndrome, LAM, tuberous sclerosis or granulomatous vasculitides.
  - Complications such as pulmonary infections or pulmonary embolism can also presents with varying degree of hemoptysis.

Extrapulmonary Manifestations

- Extrapulmonary symptoms associated with various ILDs provide a clue to the underlying diagnosis. Therefore, a thorough systemic review should be done in all patients with ILD.

Table 4.2: Extrapulmonary features and differential diagnoses

Extrapulmonary feature	*Differential diagnosis*
Cutaneous thickness, Raynaud's phenomenon, digital ulceration, dysphagia or regurgitation	Systemic sclerosis
Malar rash, photosensitivity, hair loss, arthralgia, arthritis	Systemic lupus erythematosus
Erythema nodosum, lupus pernio, scleritis	Sarcoidosis
Heliotrope rash, Gottron's papule (popular lesions on knuckles), muscle weakness	Dermatomyositis
Arthralgias, arthritis (morning stiffness, joint erythema and swelling), and joint deformities	Connective tissue disease
Dryness of eyes and mouth	Sjögren's syndrome

- Common extrapulmonary features and their differential diagnoses are summarized in **Table 4.2**.

Past Medical History

History of past medical illness is important and may suggest the etiology of ILD. Some examples are summarized below.

History suggestive of:
- Connective tissue disorder—connective disease associated ILD
- Allergic rhinitis and asthma—chronic eosinophilic pneumonia
- Nasal polyposis and asthma—Churg-Strauss syndrome
- Acute or chronic renal failure—pulmonary-renal syndromes, connective tissue disorders
- HIV—lymphocytic interstitial pneumonia.

Occupational and Exposure History

- A detailed occupational history both past and current, with specific attention on type of work/duties (like plumbing, metal working, stone cutting or mining) and exposures (like asbestos, silica, hard metals or beryllium) should be sought in all patients with ILD. The degree of exposure, duration, the latency of exposure and use of protective devices should also be elicited.
- Smoking is considered as a risk factor for IPF, Goodpasture syndrome and is strongly linked with PLCH, DIP (desquamating interstitial pneumonia) and RB-ILD (respiratory bronchiolitis-ILD).
- Radiation induced lung injury is related to the cumulative radiation dose and volume of lung involved.

Family History

- Family history should be elicited in all patients of ILD.
- Several inborn errors of metabolism such as Gaucher disease and Niemann-Pick (autosomal recessive) and Hermansky–Pudlak syndrome, Birt–Hogg–

Dubé syndrome, and neurofibromatosis type I (autosomal dominant) are rarely associated with ILD.

- Familial lung fibrosis has been reported to be associated with mutations in the surfactant protein C gene, surfactant protein A2 gene, telomerase reverse transcriptase (TERT), telomerase RNA component (TERC), and the promoter of a mucin gene (MUC5B). Familial clustering has been identified in sarcoidosis.

Drug History

- A detailed review of drugs current or used in the recent or distant past should be taken.
- Drug induced ILDs can present as acute pneumonitis or chronic fibrotic lung disease.
- Drugs commonly implicated for development of ILD e.g. Nitrofurantoin (used for UTI), NSAIDs, Amiodarone (anti arrhythmic drug).
- Less common drugs including chemotherapeutic agents, immune-modulating drugs for inflammatory bowel disease or Rheumatoid arthritis.

Physical Examination

- The physical examination of patients with ILD is usually nonspecific, except when extrapulmonary findings suggest a particular systemic disease.
- The most common examination findings are tachypnea and bibasilar end-inspiratory dry crackles or velcro rales.
- Crackles are common in ILDs associated with inflammation like IPF and in patients with connective tissue disease–associated ILD.
- Crackles are less frequent in granulomatous lung diseases like HP, and rare in sarcoidosis.
- Crackles may be present in the absence of radiographic abnormalities on the chest radiograph.
- Wheezing and scattered late inspiratory high-pitched rhonchi (inspiratory squeaks) are observed in patients with airway centered disease such as sarcoidosis, HP and bronchiolitis.
- Clubbing of digits which indicates advanced fibrotic disease is common in patients with idiopathic or familial pulmonary fibrosis. It is rare in other ILDs like sarcoidosis, HP and PLCH.
- Signs of pulmonary hypertension and right heart failure include lower extremity pitting edema, a loud P2 component of second heart sound and right ventricular parasternal heave.

A detailed physical examination may reveal findings which are specific to some disorders associated with ILD **(Table 4.3)**.

Laboratory Testing

- There is no laboratory test that is specific to ILD, but may strongly support the specific diagnosis.

Table 4.3: Important physical findings and conditions associated with ILD

Physical findings	*Associated disease*
Cutaneous thickening, digital ulcerations, Raynaud's phenomenon, calcinosis cutis, telangiectasias	Scleroderma
Keratoconjunctivitis sicca, xerostomia, salivary gland enlargement	Sjögrens syndrome
Heliotrope rash, Gottron's papules, mechanic's hands, myositis, muscle weakness	Dermatomyositis
Erythema nodosum, lupus pernio, uveitis, parotid enlargement, lymphadenopathy, hepatosplenomegaly	Sarcoidosis
Malar rash, photosensitivity skin reaction, or hair loss	Systemic lupus erythematosus (SLE)
Arthritis, subcutaneous nodules, joint deformity	Rheumatoid arthritis
Skin rash, neuropathy, cutaneous vasculitis, uveitis, scleritis	Systemic vasculitis
Cutaneous neurofibromas or café-au-lait spots	Neurofibromatosis
Adenoma sebaceum, periungual fibromas, and Shagreen patch	Tuberous sclerosis

- The routine laboratory evaluation typically includes biochemical tests to evaluate hepatic and renal function; hematologic tests with differential blood count to check for evidence of anemia, polycythemia, leukocytosis, or eosinophilia; urinalysis for active urinary sediments.
- Anemia may be due to underlying chronic disease, gastrointestinal blood loss as in case of inflammatory bowel disease or hemolytic process in connective tissue disorders.
- Blood eosinophilia is seen in patients with systemic vasculitis, chronic eosinophilic pneumonia (CEP), Churg-Strauss syndrome or drug reaction.
- Pulmonary renal syndromes and connective tissue disorders present with deranged renal functions, active urinary sediments (proteinuria and glomerular hematuria).
- Elevated liver enzymes (especially alkaline phosphatase) and hypercalcemia suggest diagnosis of sarcoidosis.

Additional advanced laboratory testing which is often required to support the specific diagnosis is summarized in **Table 4.4**.

Pulmonary Function Testing

- All patients of suspected ILD should undergo a comprehensive pulmonary function testing (PFTs) including spirometry, lung volumes, diffusing capacity.
- Pulmonary function testing does not support a specific ILD diagnosis, however helps to determine severity of the disease, disease progression, response to therapy and prognosis.

Spirometry and Lung Volumes (Table 4.5)

- On spirometry evaluation most of the ILDs have a restrictive defect due to decreased compliance and increased recoil of the lung parenchyma.

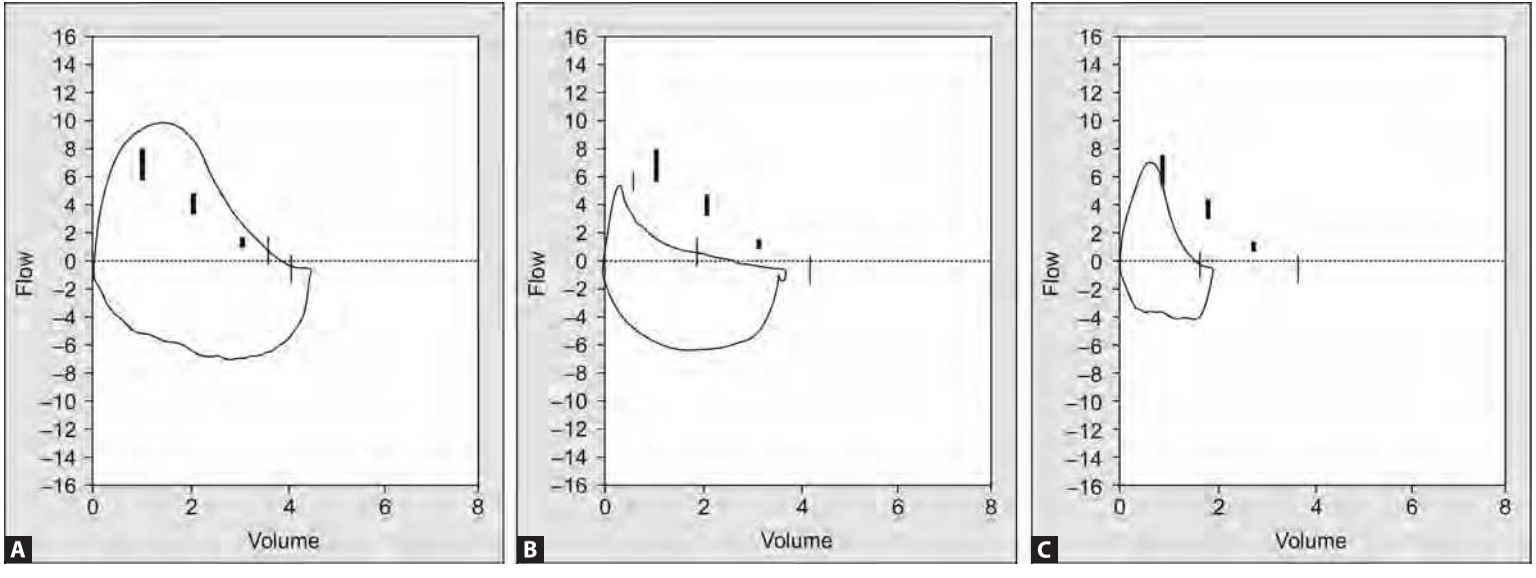
Table 4.4: Advanced laboratory investigations done during work-up of ILD

Laboratory test	*Indication*
ANA; rheumatoid factor; ANA differentiation, including Jo-1 or ScL-70 antibodies	Suspected CTD or idiopathic ILD for which CTD cannot be excluded
Creatine kinase activity, anti-Jo-1 antibodies, aldolase	Suspected myositis
Hypersensitivity precipitin panel (Specific serum IgG antibodies)	Exposure to antigens that cause HP (Valid only within an appropriate clinical context)
c-ANCA, p-ANCA	Suspected vasculitis (c ANCA suggest Wegner's granulomatosis and p ANCA microscopic polyangiitis/Churg-Strauss syndrome)
Antiglomerular basement membrane antibody	Goodpasture syndrome
Serum angiotensin-converting enzyme activity, serum soluble interleukin 2 receptor	Sarcoidosis
Anti-SS-A (Ro) Anti-SS-B (La)	Sjögren syndrome
Antiribonuclear protein (U1Rnp)	Mixed connective tissue disease

Table 4.5: Typical example of normal, restrictive and obstructive changes on spirometry

	Normal		*Restrictive pattern*		*Obstructive pattern*	
Parameters	*Observed*	*% predicted*	*Observed*	*% predicted*	*Observed*	*% predicted*
FVC	4.42 Lt	111	1.88 Lt	52	3.81 Lt	89
FEV1	3.76 Lt	112	1.65 Lt	56	2.06 Lt	59
FEV1/FVC	85.6		87.7		54.2	

- Useful parameters on spirometry include forced vital capacity (FVC), forced expiratory volume in one second (FEV1) and FEV1/FVC ratio.
- Typical finding is decreased FVC with normal to reduced FEV1, thus, the FEV1/FVC ratio is usually normal or increased.
- Lung volume measurements will show reductions in total lung capacity (TLC), functional residual capacity (FRC), and residual volume (RV). The reductions in lung volumes become more pronounced with disease progression as lung becomes stiffer.
- The presence of obstruction suggests either presence of an airway-centered ILD (like sarcoidosis, LAM, HP, PLCH, constrictive bronchiolitis or concomitant obstructive lung disease (combined COPD and ILD) **(Figs 4.1A to C)**.



Figs 4.1A to C: Representative spirograms showing various patterns: Normal (A), obstructive (B) and restrictive (C) pattern. The corresponding spirometry values for **normal** (Fig. A) - FEV1/FVC = 85.6%; FVC = 4.42 lt (111% of predicted) and FEV1 = 3.76 lt (112% of predicted); **obstructive** (Fig. B) - FEV1/FVC = 54.2%; FVC = 3.89 lt (89% of predicted) and FEV1 = 2.06 lt (59% of predicted); and **restrictive** (Fig. C) - FEV1/FVC = 87.7%; FVC = 1.88 lt (52% of predicted) and FEV1 = 1.65 lt (56% of predicted)

Diffusing Capacity

- Diffusion impairment, usually measured as diffusion capacity of carbon mono-oxide (DLCO), although nonspecific, is often the earliest physiologic abnormality in patients with ILD.
- The decrease in DLCO is due to effacement of the alveolar capillary units and interstitium of lung by fibrotic tissue or inflammatory cells.
- A disproportionate decrease in diffusion capacity compared to the restrictive ventilatory defect should prompt suspicion of associated emphysema or pulmonary vascular disease (pulmonary hypertension, pulmonary embolism).
- DLCO reduction does not correlate well with disease severity (unless values is less than 35 percent of predicted), however, serial assessments of DLCO have been used to assess disease progression or regression.

Exercise Testing

- Patients with early ILD have normal resting arterial blood gases. However, as disease progresses it leads to hypoxemia (secondary to ventilation perfusion mismatch) and respiratory alkalosis.
- During early in the course of ILD, hypoxemia can be unmasked by exercise testing (cardiopulmonary exercise testing, 6 minute walk test, or ambulatory oximetry).
- Full cardiopulmonary exercise testing is not always necessary for every patient with ILD. However, in patients with significant symptoms with minimal or unclear radiographic abnormality a normal maximal cardiopulmonary exercise test effectively excludes significant ILD.
- Drop in oxygen saturation, failure to decrease dead space ventilation appropriately with exercise (i.e. a high dead space/tidal volume ratio) and an excessive increase in respiratory rate provide useful information about physiologic abnormalities and extent of disease.
- *6-minute walk testing* is useful for assessment of exercise capacity in terms of distance walked, heart rate response, and to identify alternative cause of exercise limitation (like musculoskeletal and cardiovascular diseases).
- Serial assessment of resting and exercise gas exchange is one of the methods used to follow ILD activity and responsiveness to treatment.

Role of Bronchoscopy

In evaluation of patient with ILD, fiberoptic bronchoscopy is used to obtain fluid from the most distal airways and lung tissue cytological and histopathological examination respectively.

Bronchoalveolar Lavage

- Bronchoalveolar lavage (BAL) has limited role in the diagnosis of ILD. The most important reason of obtaining BAL fluid in ILD is to rule out infection causing diffuse lung disease or secondary to immunosuppressive treatment.

Table 4.6: Differential diagnosis based on BAL fluid analysis

BAL fluid finding	*Clinical condition*
More than 25% eosinophils	Eosinophilic lung disease
Lymphocytosis	Granulomatous ILDs (Sarcoidosis) Hypersensitivity pneumonitis Drug reaction Cellular NSIP (lymphocytes > 50%)
High CD4+/CD8+ > 3.5 times	Sarcoidosis
Presence of foamy macrophages and mixed pattern of increased cells characteristic; decreased CD4:CD8 ratio	Organizing pneumonia
Hemosiderin laden macrophages	Diffuse alveolar hemorrhage
Type II pneumocyte hyperplasia	Diffuse alveolar damage
CD1a+ cells	Pulmonary Langerhan cell histiocytosis
Positive lymphocyte proliferation assay	Chronic Berylliosis
Asbestos bodies/ferruginous bodies	Asbestosis
Milky BAL fluid, foamy macrophages and lipoproteinaceous material positive for periodic acid–Schiff stain	Alveolar proteinosis
Dust particles on polarized microscopy	Silicosis

- Appearance of BAL fluid is itself occasionally diagnostic, e.g. progressively hemorrhagic BAL fluid in alveolar hemorrhage syndrome, milky white BAL fluid in alveolar proteinosis.
- In selected diseases cellular analysis of BAL fluid may be useful in making a specific diagnosis among various types of ILD **(Table 4.6)**.

Lung Biopsy

Lung biopsy is considered as the gold standard for the diagnosis of ILD. In appropriate clinical setting, however, HRCT may establish the diagnosis of ILD with acceptable certainty.

- HRCT may be diagnostic in cases of IPF, sarcoidosis, HP, lymphangitis carcinomatosis and PLCH.
- Lung biopsy may be required n situation where clinical evaluation including laboratory testing, pulmonary functions and HRCT are unable to establish a confident diagnosis of a given type and stage of ILD.
- Various methods of obtaining lung biopsy include bronchoscopic technique [transbronchial lung biopsy (TBLB) and core biopsy] or by surgical technique [video assisted thoracoscopic surgery (VATS) or open surgical lung biopsy].
- TBLB has high yield for the diagnosis of sarcoidosis. Other ILD where TBLB can be helpful in making diagnosis include PLCH, alveolar proteinosis, lipoid pneumonia, drug induced pneumonitis and eosinophilic pneumonia.

- The utility of TBLB for most of the idiopathic interstitial pneumonias is low and surgical biopsy is often required for accurate diagnosis.
- A new bronchoscopic technique (transbronchial cryobiopsy) using flexible cryoprobes to obtain TBLB is minimally invasive procedure with high diagnostic yield as compared to conventional TBLB even in patients with idiopathic interstitial pneumonias.

Even after lung biopsy a multidisciplinary approach with inputs from pulmonologist, radiologist and pathologist is required to reach final diagnosis in many patients. **Table 4.7** summarizes the characteristic histopathological features of various ILDs.

Table 4.7: Characteristic histopathological features of various ILD

Type of ILD	*Histopathological features*
Idiopathic pulmonary fibrosis (IPF)	*Nonuniform distribution (temporal heterogeneity)*: Areas of fibrosis and fibroblastic foci with honey combing with adjacent normal areas of lung. Changes are more marked at periphery and lower lobes.
Nonspecific interstitial pneumonia (NSIP)	*Uniform distribution (temporal homogeneity)*: Honeycombing and fibroblastic foci are rare. Three patterns are seen based on degree of inflammation and fibrosis. *Cellular*: predominant interstitial inflammation; *Fibrotic*: predominant fibrosis; and *Mixed*: combination of inflammation and fibrosis.
Cryptogenic organizing pneumonia (COP)	No honeycombing or fibroblastic foci. Varying degree of interstitial inflammation. Characteristic fibroblastic intraluminal plugs (Masson's bodies) involving alveolar ducts and alveolar spaces. Airspaces also contain foamy macrophages.
Acute interstitial pneumonia (AIP)	Diffuse alveolar damage. Hyaline membranes (histological landmark), marked hyperplasia of type II pneumocytes with fibroblastic proliferation. Squamous metaplasia and arteriolar thrombi.
Respiratory bronchiolitis-ILD (RB-ILD)	Nonuniform and bronchiolocentric distribution of inflammation. Peribronchiolar fibrosis seen. Respiratory bronchioles, alveolar ducts, and peribronchiolar alveolar spaces contain pigmented macrophages (fine granular cytoplasm) stains with Perl's stain.
Desquamative interstitial pneumonia (DIP)	Uniform involvement of lung and lacks bronchiocentric distribution. There is accumulation of macrophages in distal airspaces eosinophilic cytoplasm stains weakly with Perl's stain.
Hypersensitivity pneumonitis (HP)	Triad of patchy nonspecific interstitial inflammation, non-necrotizing poorly formed granulomas and foci of bronchiolitis obliterans.

Contd…

Contd…

Type of ILD	*Histopathological features*
Lymphangioleiomyomatosis (LAM)	Multiple air filled cysts. Proliferation of immature smooth muscle cells in interstitium, around bronchioles, blood vessels, lymphatics and walls of cystic spaces. These cells are positive for Melan A and HMB-45.
Pulmonary Langerhans cell histiocytosis (PLCH)	Nodular aggregates of Langerhans cells. Cells are positive for CD1a and S-100 immunostains.

SUMMARY

- Clinical symptoms of ILD are nonspecific. Dyspnea and dry cough are the common symptoms.
- Disease onset, progression and associated clinical conditions are important while making diagnosis of ILD.
- Pulmonary functions tests are useful to suspect ILD early and may be useful for follow-up. However, there is no specific pattern which can differentiate one ILD from other.
- Lung biopsy is the gold standard for diagnosis, however, in typical cases HRCT is sufficient for diagnosis.
- For definite diagnosis of ILD a multidisciplinary approach consisting pulmonologist, radiologist and pathologist is required.

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Section 2

Idiopathic Interstitial Pneumonias

- Imaging in Idiopathic Interstitial Pneumonias
- Clinical Diagnosis and Management Issues

5

Imaging in Idiopathic Interstitial Pneumonias

Ankur Goyal

The term 'idiopathic interstitial pneumonias (IIPs)' refers to interstitial lung diseases (ILDs) after exclusion of known causes (connective tissue diseases, drug-induced, etc.). Connective tissue diseases and hypersensitivity pneumonitis are important differentials which can produce similar morphologic pattern of involvement. Diagnosis involves a multidisciplinary approach with integration of clinical, imaging and pathologic findings. Exposure history to cigarette smoke, medications, and laboratory evaluation for an underlying connective tissue disease are important clinical information required to refine the differential diagnoses.

The recent classification system is 2013 Updated American Thoracic Society–European Respiratory Society (ATS–ERS) classification of the IIPs.[1] This update was designed to be a supplement to the 2002 classification, which defined best diagnostic approach to the key diseases. It also replaced histology as the conventional 'gold standard' and laid emphasis on an integrated approach. Assessment of disease behavior and molecular and genetic features are also important.

Classification categories include ***(Table 5.1)****:*

- Chronic fibrosing IIPs
- Smoking related IIPs
- Acute/subacute
- Rare IIPs
- Unclassifiable IIPs—where final diagnosis is not achieved.

IIPs can also be classified depending on the clinical course and management **(Table 5.2)**.

STEP 1: IMAGING MODALITY AND TECHNIQUE

- Chest radiographs are usually the first imaging investigation in patients suspected to have lung disease **(Figs 5.1A and B)**. They may also serve

Table 5.1: Updated American Thoracic Society–European Respiratory Society (ATS-ERS) classification of the Idiopathic Interstitial Pneumonias (IIPs)

Category	*Clinical-radiologic-pathologic diagnosis*	*Associated radiologic and/or pathologic-morphologic patterns*
Chronic fibrosing IP	Idiopathic pulmonary fibrosis	Usual interstitial pneumonia
	Idiopathic non-specific interstitial pneumonia	Non-specific interstitial pneumonia
Smoking related IP	Respiratory bronchiolitis-Interstitial lung disease	Respiratory bronchiolitis
	Desquamative interstitial pneumonia	Desquamative interstitial pneumonia
Acute/ subacute	Cryptogenic organizing pneumonia	Organizing pneumonia
	Acute interstitial pneumonia	Diffuse alveolar damage
Rare IP	Lymphoid interstitial pneumonia	
	Idiopathic pleuroparenchymal fibroelastosis	

Table 5.2: Classification of IIPs based on clinical course and management

Clinical course	*Entity*	*Aim of treatment*	*Follow-up needed*
Reversible	RB-ILD	Avoid cause (Smoking)	Up to 6 months
Reversible but risk of progression	Cellular NSIP, DIP, COP	Response achievement immediately and then long-term therapy	Long-term observation for early detection of active disease
Stable with residual disease	Fibrotic NSIP	Prevent deterioration	Long-term observation
Progressive disease with potential stabilization	Fibrotic NSIP	Stabilize	Long -term observation
Progressive despite treatment	IPF	Delay progression	Long-term observation and palliation

to detect lung disease in relatively asymptomatic individuals and are an inexpensive tool for follow-up of patients. In addition to identifying the disease pattern, radiographs also determine the distribution and enable quick assessment of lung volumes and evaluation of complications.

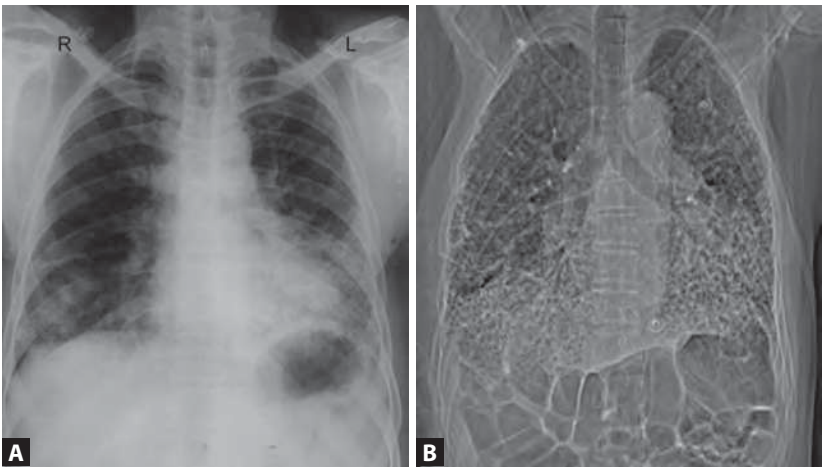
- The IIPs are characterized by typical morphological patterns on computed tomography (CT) and thus making CT (including high-resolution

reconstructions HRCT) the imaging investigation of choice. Obtaining high quality thin-section CT reconstructions along with multiplanar reformats is essential and imaging should always be done in full inspiration.

- It is important to understand the evolution of IIPs on serial imaging, especially in cases where diagnosis is not certain.[2]
- Volumetric CT acquisition is indispensable, especially for the first-time evaluation as well as whenever there is considerable change in the clinical profile or radiographic appearance.
- Noncontiguous sequential imaging may miss patchy/subtle disease, cannot give good delineation of disease distribution (since multiplanar reformats are not feasible), and may not be able to differentiate honeycombing from traction bronchiectasis.
- Because of inherent good contrast, chest CT can usually be performed with reduced radiation dose techniques.
- Limited prone CT, especially of the lung bases, may be helpful to differentiate dependent opacities from early interstitial lung disease.
- Additional expiratory phase imaging is pivotal to confidently detect air trapping, which may suggest an alternative diagnosis. This may be performed in noncontiguous fashion and at an even lower radiation dose. Some authors advocate acquiring random interspersed CT sections from all the three lung zones.
- Use of standard terminology and determining the distribution of disease in both axial and craniocaudal orientations are essential components of a systematic approach in CT evaluation.
- Imaging helps in diagnosis by identifying the pattern of involvement and serves as a marker for severity and activity assessment in follow-up.

STEP 2: IS IT INTERSTITIAL LUNG DISEASE?

- The foremost step in imaging evaluation is to determine whether there is focal or diffuse lung disease. Are the findings bilateral, symmetrical and in similar distribution?
- Interstitial lung diseases (ILDs) are characterized by reticulonodular lesions on radiograph **(Figs 5.1A and B)** and septal thickening, traction bronchiectasis and architectural distortion on CT with volume loss in later stages. These findings are characteristic of chronic fibrosing ILD.
- Acute/subacute IIPs, especially in acute phase may demonstrate ground glass and consolidations.
- Smoking-related IIPs occur at a younger age, fibrosis occurs late and have better prognosis.
- Important to remember that radiological patterns of IIPs like UIP and NSIP may be found in non-IIP diseases like connective tissue diseases and hypersensitivity pneumonitis.
- Presence of mosaic attenuation, perilymphatic/centrilobular nodules, sparing of the lung bases, cysts, pleural thickening, effusion, etc. may point towards a non-IIP diagnosis **(Table 5.3)**.



Figs 5.1A and B: (A) Chest radiograph in a patient of IIP shows bilateral lower zone reticular lesions; (B) Scout CT topogram shows typical features of UIP pattern. There is extensive honeycombing in both bibasilar regions with reticular abnormalities and traction bronchiectasis. Note is made of right apical pleural thickening as well

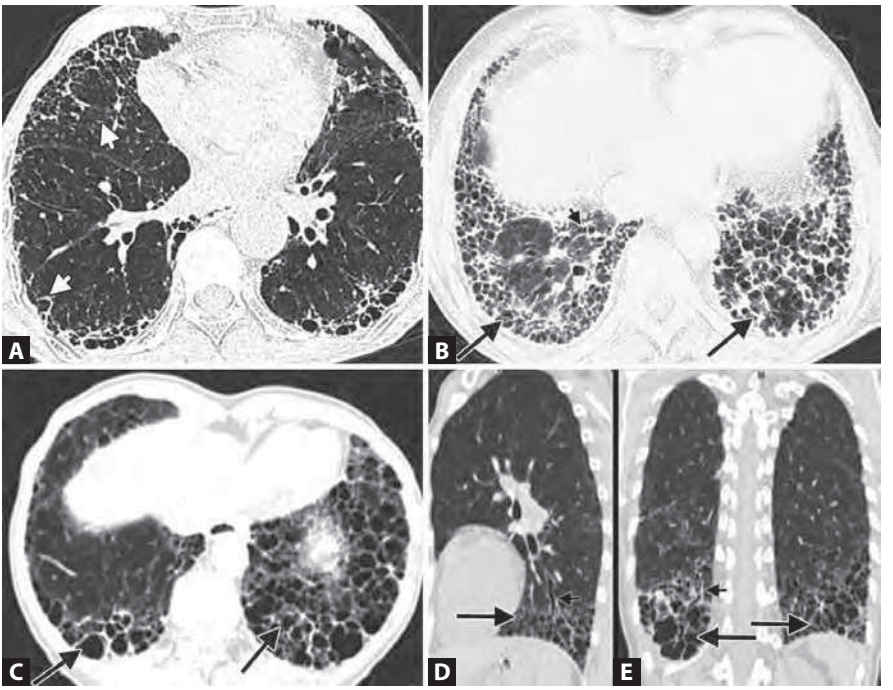
Table 5.3: Findings atypical for IIP and associated alternative diagnosis

Cysts	Pulmonary LCH, LM
Perilymphatic nodules	Sarcoidosis, lymphangitis carcinomatosis, lymphoma
Centrilobular nodules	Hypersensitivity pneumonitis, infection, aspiration, bronchiolitis
Air trapping	Hypersensitivity pneumonitis, obliterative bronchiolitis

Abbreviations: LCH, Langerhans cell histiocytosis; LM, lymphangioleiomyomatosis

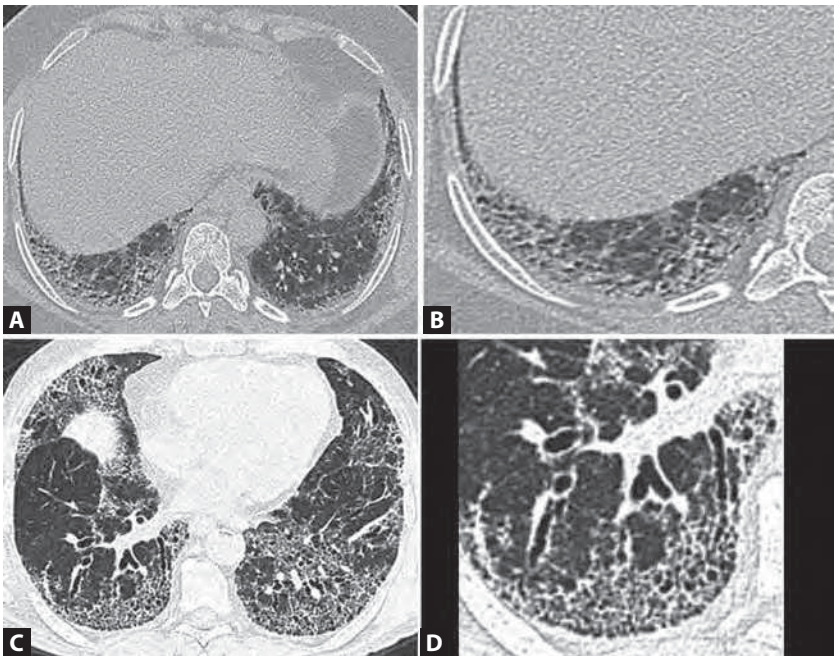
STEP 3: IS IT USUAL INTERSTITIAL PNEUMONIA?

- The most important task is to identify radiological pattern of usual interstitial pneumonia (UIP) because this pattern has poorer prognosis and requires different pharmacological management.
- There are three levels of certainty in CT diagnosis of UIP.
- Presence of any of the following imaging findings is inconsistent with UIP pattern:
  - Predominance of findings in upper or middle portions of lungs
  - Peribronchovascular predominance
  - Extensive ground glass abnormalities (greater than reticular lesions)
  - Profuse micronodules (bilateral, in upper lobes)
  - Presence of discrete cysts (multiple, bilateral and away from the areas of honeycombing)
  - Diffuse mosaic attenuation due to air trapping (bilateral, in three or more lobes)
  - Segmental/lobar consolidation



Figs 5.2A to E: CT features of definite UIP. There is presence of reticular abnormalities (small solid white arrows in (A), honeycombing (long black arrows) in subpleural basal distribution along with traction bronchiectasis (small black arrows) in axial (A to C) HRCT, sagittal (D) and coronal (E) MPR lung window images. No GGO is seen. (E) Coronal and sagittal MPR images depict craniocaudal distribution with basal predominance of honeycombing and traction bronchiectasis

- The CT diagnosis of definite UIP pattern can be made if all of the following features are fulfilled **(Figs 5.2A to E)**:
  - Subpleural basal predominance of findings
  - Reticular lesions
  - Honeycombing (+/– traction bronchiectasis)
  - Absence of findings which are inconsistent with UIP (see above)
- If honeycombing is not seen, but other three conditions are fulfilled, then UIP pattern is possible **(Figs 5.3A to D)**. Thus, confident diagnosis of UIP requires presence of honeycombing.
- In cases of possible UIP, surgical lung biopsy is indicated only if the clinical profile is not typical; in concordant clinical profile there is no need of sampling.
- Sometimes histopathology may show UIP pattern even though CT demonstrates neither classic nor possible UIP.
- Patients with classic UIP pattern have shorter survival than those with possible UIP. The latter, in turn have poorer survival than those with indeterminate CT features.

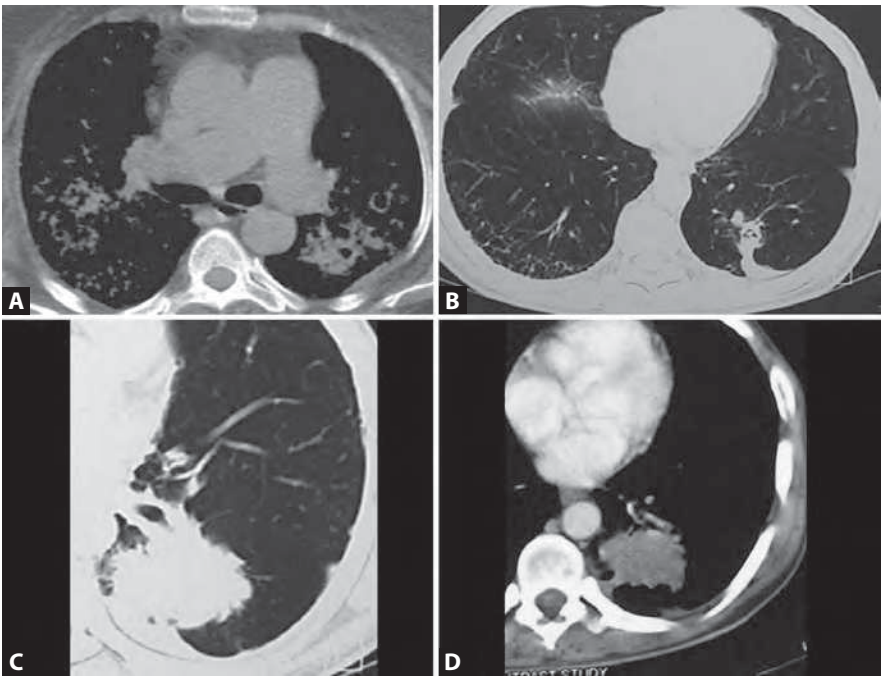


Figs 5.3A to D: Possible UIP. CT not diagnostic but multidisciplinary approach (including clinical and histological findings) confirmed UIP in axial HRCT images (lung window) of different patients. (A) Axial CT image shows reticular abnormalities with traction bronchiectasis with basal subpleural predominance but no honeycombing; (B) Zoomed image of (A) showing no honeycombing; (C) Axial HRCT image depicts extensive reticular lesions with interstitial septal thickening and traction bronchiectasis. Honeycombing not seen; (D) Zoomed image of (C) showing septal thickening and traction bronchiectasis

CHRONIC FIBROSING IDIOPATHIC INTERSTITIAL PNEUMONIA

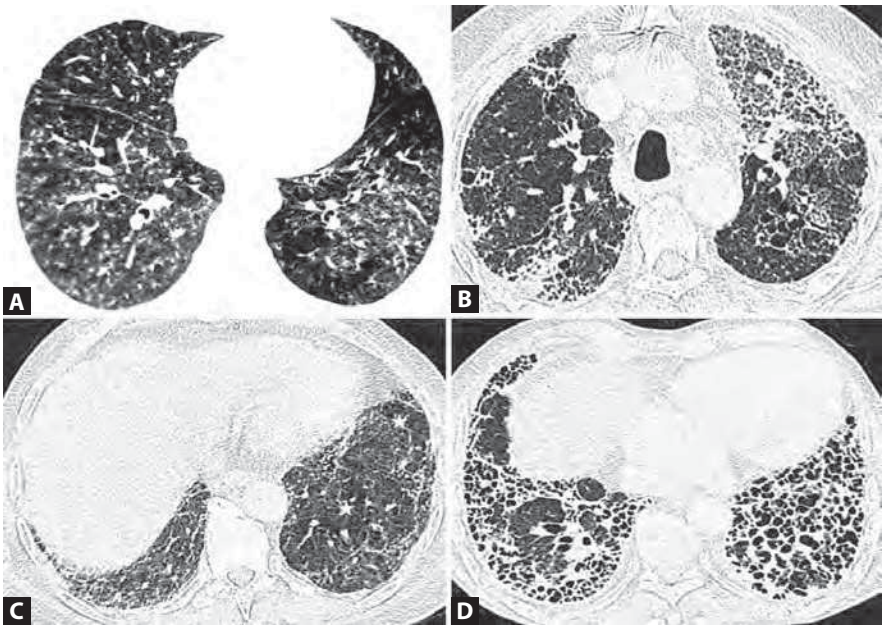
Idiopathic Pulmonary Fibrosis (IPF)

- Chronic progressive fibrosing disease of unknown cause
- Affects older adults (> 50 years)
- Poor prognosis with median survival less than 5 years
- Radiological and pathological pattern of UIP.
- Radiographs show bilateral reticular pattern, lower zone predominance, progressive honeycombing and loss of lung volume in later stages.
- Peripheral areas are most affected, although distribution is often patchy and asymmetric in one-third of cases.
- Areas of fibrosis may show linear/nodular foci of calcification.
- Ground-glass opacification (GGO) may be seen but less extensive than reticular abnormalities.
- Mild mediastinal lymph node (LN) enlargement is seen in 70% patients.
- Diagnosis of IPF is based on the following **(Figs 5.2 and 5.3)**:
  - Exclusion of known causes of ILD
  - Presence of classic UIP pattern on high-resolution computer tomography (HRCT). There is no need of sampling in such patients



Figs 5.4A to D: Complications of ILD /IIP. (A) Axial mediastinal window shows pulmonary arterial dilatation in a patient with UIP pattern, suggesting pulmonary hypertension; (B and C) Axial CT images lung window of a different patient shows subpleural basal reticular lesions along with an irregular spiculated mass in left lower lobe, biopsy from which revealed squamous cell carcinoma. (C) Zoomed image of left lower lobe at a cranial section to (B) shows lobulated neoplastic mass; (D) Corresponding mediastinal window

- Specific combinations of HRCT and surgical lung biopsy pattern in patients undergoing sampling
- Coexisting emphysema is common and can make the diagnosis difficult and associated with poorer prognosis.
- On follow-up, there is increase in fibrosis with involvement of upper lobes as well. GGOs may improve with treatment but progression to reticulation and honeycombing is usually relentless.
- Complications include pulmonary hypertension, accelerated phase and development of lung cancer **(Figs 5.4A to D)**.
- Differential diagnoses for UIP pattern of IPF include:
  - *Asbestosis:* Exposure history, presence of pleural plaques and thickening, subpleural dot-like or branching lesions help in differentiating from IPF.
  - Connective tissue diseases
  - C*hronic hypersensitivity pneumonitis*: Differentiating features include presence of ill-defined micronodules, areas of air-trapping and sparing of the lung bases in chronic HP **(Figs 5.5A to D)**.
  - *End-stage sarcoidosis*: Presence of peribronchovascular nodules and larger cysts in areas of honeycombing.



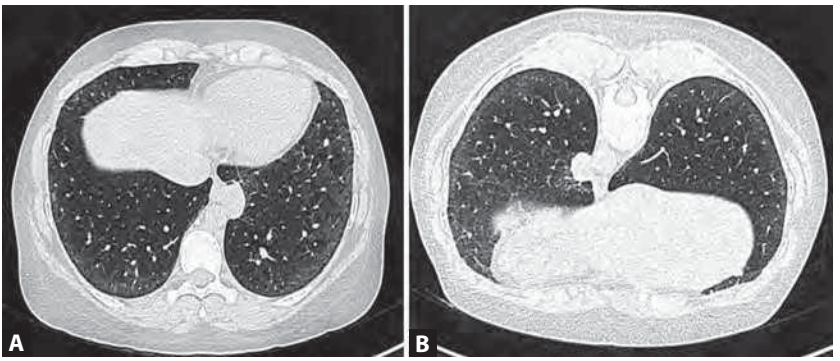
Figs 5.5A to D: IPF (D) vs chronic hypersensitivity pneumonitis (HP) (A to C). (A) Axial HRCT image shows multiple ill-defined micronodules in peribronchovascular distribution along with GGO and air trapping in a patient with subacute HP; (B) and (C) Axial HRCT images in a patient of chronic HP demonstrate reticular lesions and septal thickening in both upper lobes with relative sparing of the lung bases (C) and patchy areas of air trapping (asterisks) in (C); (D) Axial HRCT image in a patient of IPF for comparison showing classical subpleural basal honeycombing

- *Chronic interstitial pneumonia associated with coal worker's pneumoconiosis and silicosis:* Fibrosis is randomly distributed and there is less traction bronchiectasis.

- About 30–60% cases of IPF may show atypical CT features (similar to NSIP, chronic HP or sarcoidosis) and thus diagnosis of IPF cannot be excluded on the basis of CT alone.
- Older age (>65 yrs) and profound reticular lesions are also predictive of histopathologic confirmation of IPF.

Idiopathic Nonspecific Interstitial Pneumonia (NSIP)

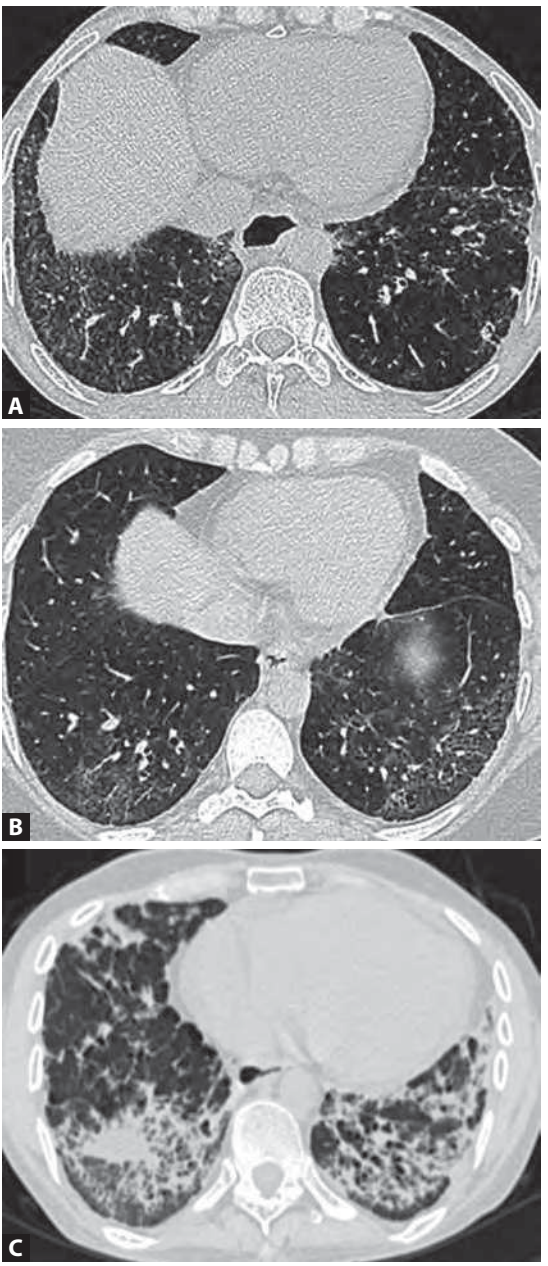
- Less common but distinct entity with variable prognosis, however better than IPF and longer survival.
- No longer a provisional diagnosis, rather a unique clinicopathological disease.
- Age of presentation is younger (40–50 years) and symptoms are milder relative to IPF.
- Surgical lung biopsy often required to secure the diagnosis.



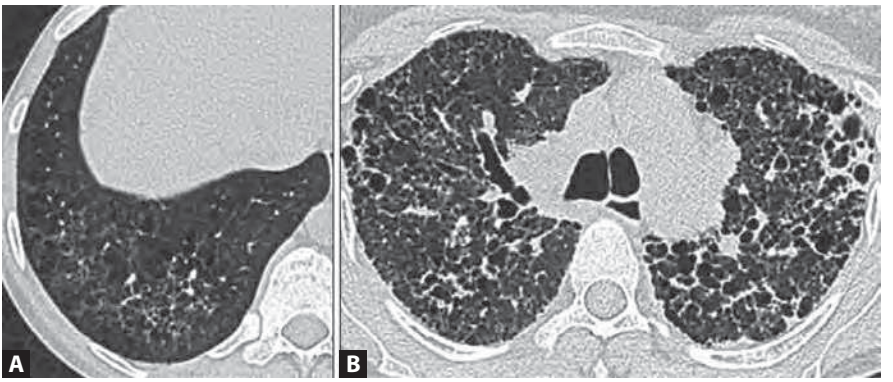
Figs 5.6A and B: Cellular NSIP pattern. (A) Axial HRCT image shows peripheral basal GGO with immediate subpleural sparing; (B) Prone scan shows persistence of the abnormality, further confirming early ILD, likely cellular NSIP

- Radiologic and histologic nonspecific interstitial pneumonia (NSIP) pattern is commonly associated with an underlying cause (connective tissue diseases, hypersensitivity pneumonitis and drug toxicity).
- CT features vary as per the proportion of underlying inflammation (cellular component) and fibrosis. Former is characterized by GGO and consolidation **(Figs 5.6A and B)** while latter shows predominant reticulation and traction bronchiectasis **(Figs 5.7A to C)**.
- Most common features include reticular pattern, traction bronchiectasis and lobar volume loss. Mild mediastinal lymphadenopathy is also usually seen.
- Ground-glass opacities are present in half of the cases. Consolidations are less common.
- Honeycombing is characteristically absent or inconspicuous **(Figs 5.8A and B)**.
- Lower lobes are predominantly involved.
- Peribronchovascular distribution or subpleural sparing (in cases of peripheral distribution) is a characteristic feature of idiopathic NSIP and seen in up to 64% cases.
- The extent of GGOs decreases with time but reticular abnormality persists and honeycombing may increase on follow-up.
- Differentiation of fibrotic NSIP from UIP pattern **(Figs 5.9A to F)** is difficult and may be indicated by
  - Peribronchovascular distribution or subpleural sparing in former
  - More extensive honeycombing in latter

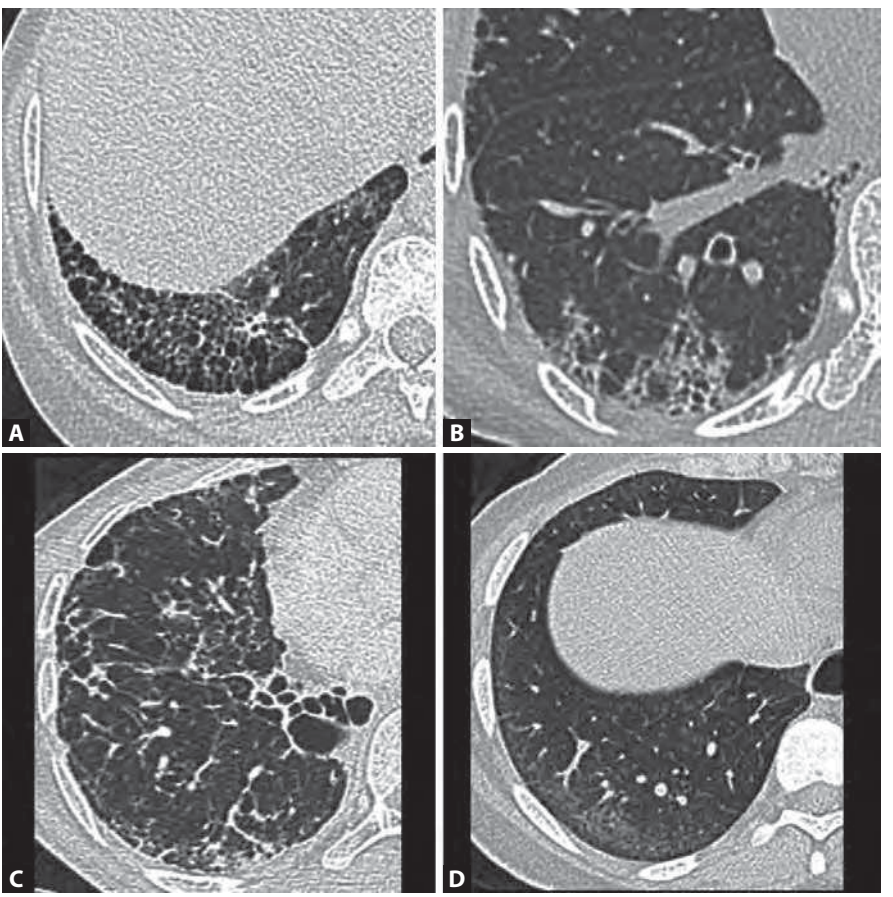
  Differences between UIP and NSIP pattern are summarized in **Table 5.4**.
- Distinction between NSIP and UIP is more difficult in smokers with emphysema
- NSIP pattern in HP, connective tissue diseases and drug-toxicity may be indistinguishable from idiopathic NSIP. Clinical and CT features which may help in this scenario are as follows:
  - Older age of presentation, male sex and history of smoking point towards IPF rather than NSIP.



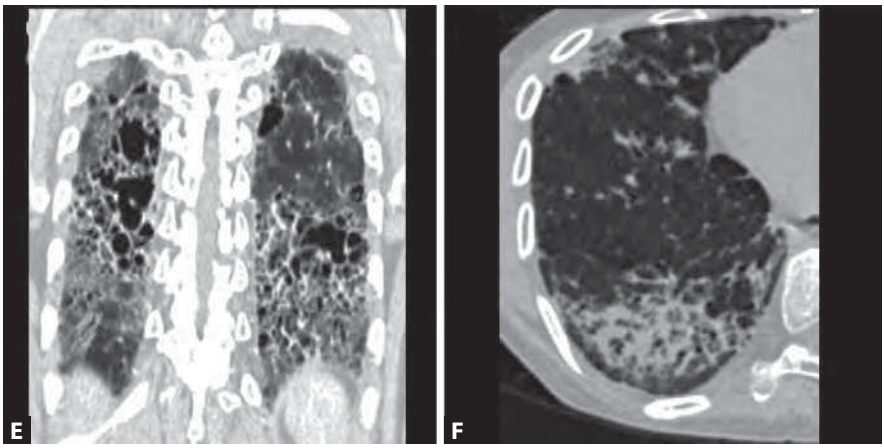
Figs 5.7A to C: NSIP–fibrotic and mixed patterns in different patients. (A) Axial HRCT image in a patient of systemic sclerosis shows reticular lesions with no definite subpleural predominance—fibrotic NSIP; (B) Axial HRCT image in a different patient shows reticular abnormalities, with GGO and traction bronchiectasis in peribronchovascular and subpleural distribution, indicative of mixed NSIP; (C) Axial HRCT image in a patient of NSIP pattern who presented with clinical deterioration, shows that in addition to reticular lesions and traction bronchiectasis, there is peribronchovascular consolidation with immediate subpleural sparing (Mixed cellular and fibrotic pattern)



Figs 5.8A and B: Atypical imaging features of NSIP. (A) Axial HRCT section shows presence of honeycombing without significant GGO; however the distribution is peribronchovascular; (B) Axial HRCT image depicts subpleural honeycombing without GGO; however the disease is more prominent in upper lobes



Figs 5.9A to D



Figs 5.9E and F

Figs 5.9A to F: UIP (A, C and E) vs NSIP (B, D and F). Left panel shows UIP pattern while right panel demonstrates findings in NSIP pattern. (A) Axial HRCT image depicts reticular lesions, traction bronchiectasis and honeycombing in basal subpleural distribution; (B) Comparative HRCT image in NSIP shows traction bronchiectasis in peribronchovascular distribution with absence of honeycombing; (C) Axial HRCT image in another patient with of IPF depicts early subpleural honeycombing and patchy heterogeneous involvement with no GGO. (D) Comparative axial HRCT image in NSIP depicts GGO predominant homogeneous abnormality with sparing of immediate subpleural region; (E) Coronal MPR lung window shows that upper lobes are also involved with disease progression in UIP. (F) Comparative HRCT image in acute exacerbation of NSIP demonstrates lower lobe predominant disease with peribronchovascular interstitial thickening, GGO and immediate subpleural sparing

Table 5.4: UIP pattern vs idiopathic NSIP

UIP	*NSIP*
Older age of presentation (> 50 years)	40–50 years
Honeycombing is diagnostic	Honeycombing absent
GGO less common and less extensive than reticular abnormalities	GGO predominance favors NSIP
Peripheral distribution	Peribronchovascular/peripheral distribution
Subpleural areas involved	Immediate subpleural sparing in cases of peripheral distribution
Can progressively involve upper lobes	Lower lobe predominance is almost always there
Patchy heterogeneous involvement	Homogeneous involvement

- Young age, centrilobular nodules, air trapping and sparing of the lung bases, lymphocytosis in bronchoalveolar lavage fluid favour HP.
- Young age, female sex, involvement of skin/musculoskeletal system and positive serology indicate underlying vasculitis and connective tissue disease. CT evidence of dilated esophagus (in systemic sclerosis), cysts, pleural thickening/effusion, pericardial effusion, mixed patterns (organizing pneumonia and NSIP) or disproportionate pulmonary arterial enlargement further points towards CTD.
- Cryptogenic organizing pneumonia and desquamative interstitial pneumonia are other differentials.

SMOKING-RELATED IDIOPATHIC INTERSTITIAL PNEUMONIAS

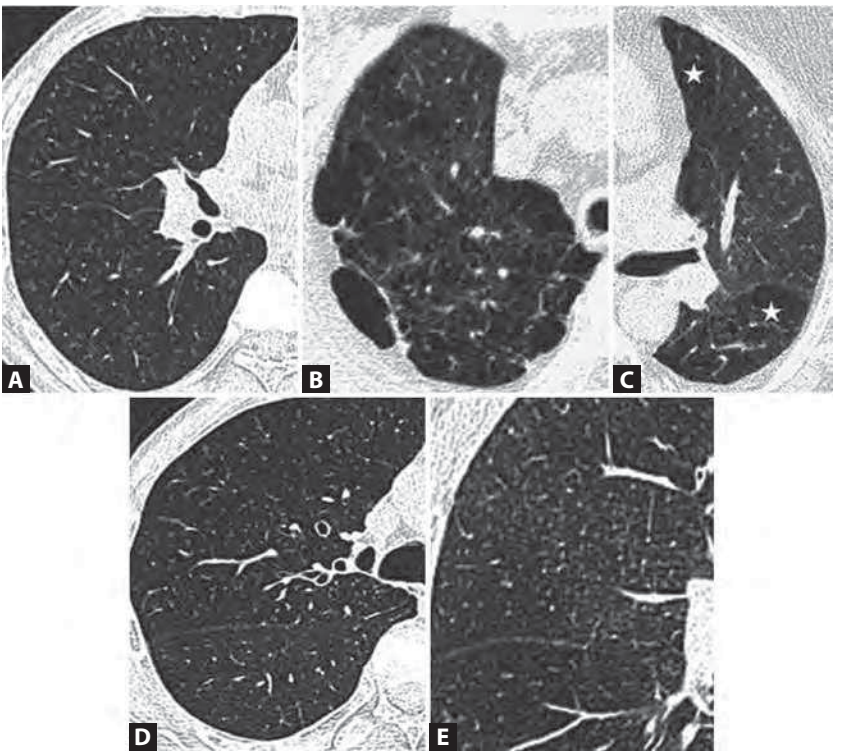
These occur in younger patients (30–40 years) and have good prognosis with cessation of smoking.

Respiratory Bronchiolitis-associated Interstitial Lung Disease

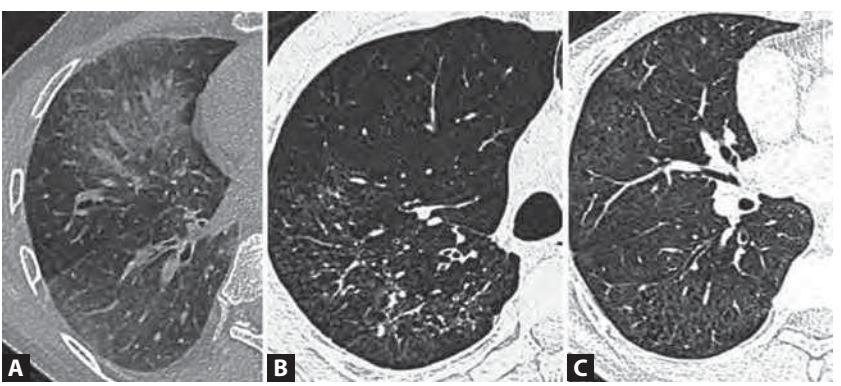
- Respiratory bronchiolitis is the most common type of smoking-related lung injury and usually asymptomatic.
- When symptomatic or associated with physiological abnormalities, it is termed respiratory bronchiolitis (RB-ILD).
- BAL fluid shows pigment-laden macrophages.
- Typical CT findings include poorly-defined centrilobular nodules (especially in upper lobes), patchy areas of GGO, lobular areas of air trapping (more prominent on expiratory scan) and mild interlobular septal thickening **(Figs 5.10A to E)**.
- Ancillary findings are emphysematous changes in upper lobes and bronchial wall thickening.
- Fibrosis is rare.
- Differential diagnoses:
  - *Subacute HP:* Patients are nonsmokers and BAL is characterized by lymphocytosis. Centrilobular nodules are more diffuse and profuse.
  - *Desquamative interstitial pneumonia:* GGO in RB-ILD is patchy and less extensive, while it is profuse in DIP. Centrilobular nodules are sparse in DIP. RB-ILD is upper lobe predominant while DIP predominates in lower lobes. Both the entities may co-exist as well.
  - NSIP
- Centrilobular nodules and GGO may regress with cessation of smoking but reticulation, bronchiectasis and emphysema usually persist.

Desquamative Interstitial Pneumonia

- Severer and more extensive end of spectrum of smoking-related IIP
- GGO is the dominant pattern and is always seen **(Figs 5.11A to C)**. Majority cases have lower zone predominance, with subpleural distribution in half the cases and random in one-fourth. Remaining one-fourth have uniform



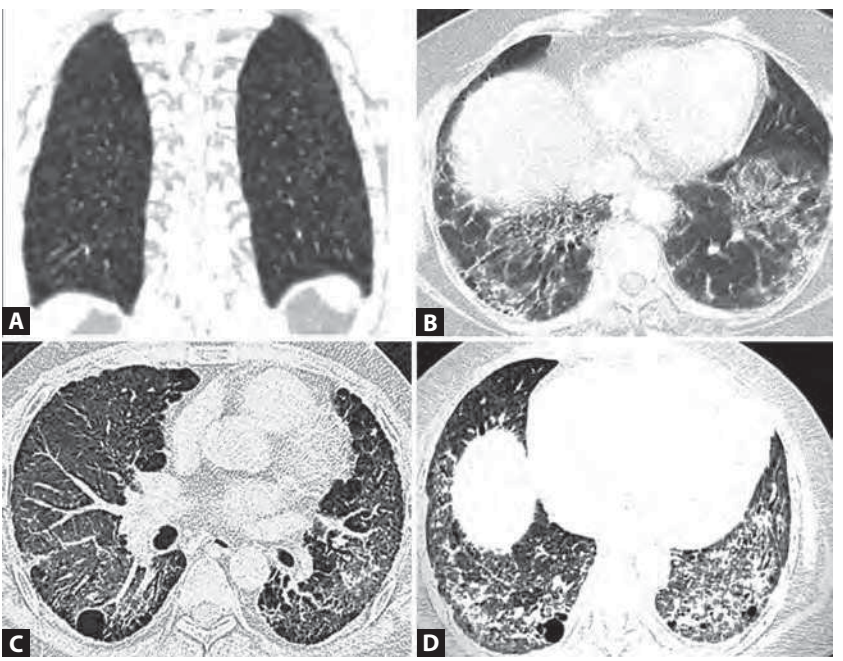
Figs 5.10A to E: Smoking related IIPs: RB-ILD. (A) Axial HRCT image depicts patchy ill-defined centrilobular nodules in right lung. Concomitant findings like subpleural bullae and centrilobular emphysema are seen in (B); (C) Axial HRCT image depicts patchy areas of air trapping (asterisks); (D) Axial HRCT image shows associated bronchial wall thickening.;(E) Axial HRCT zoomed image shows patchy ill-defined GGO and centrilobular nodules without any fibrosis



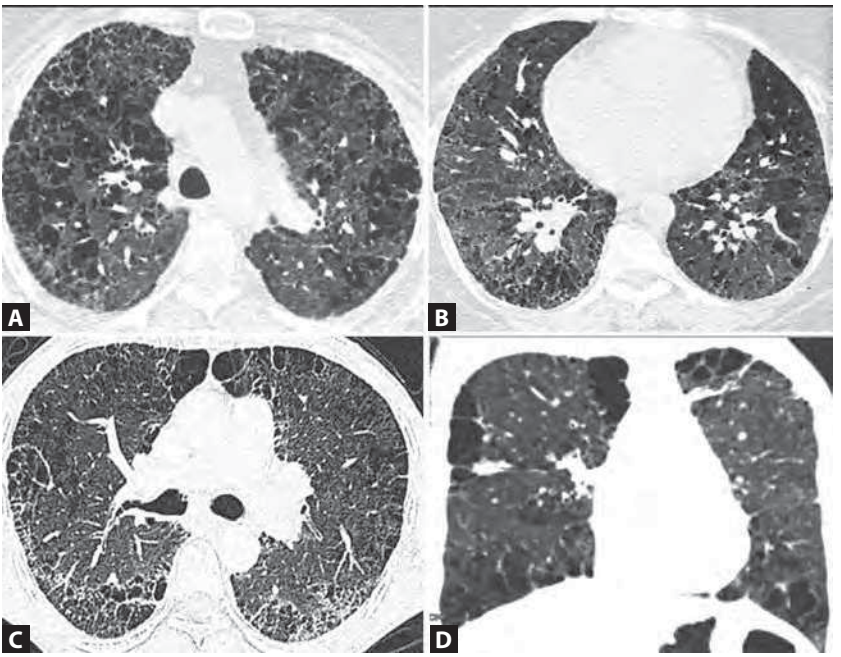
Figs 5.11A to C: Smoking-related IIPs: DIP. (A) Axial CT image shows GGO predominant disease in right middle and lower lobes; (B) Axial HRCT image depicts associated reticulation in lower lobe; (C) Axial HRCT image demonstrates subtle GGO, reticular abnormalities and tiny cystic change in middle and lower lobes

diffuse distribution. Numerous cystic airspaces may be seen within the areas of GGO and are distinct from honeycombing.

- Basal reticulation and irregular linear opacities are usually accompanying. Peripheral honeycombing is seen in less than one-third cases.
- GG opacification is partially or completely reversible with treatment. Eventually patients develop irreversible fibrosis, frequently fibrotic NSIP.
- DIP may be seen in nonsmokers, as a continuation of childhood disease, especially due to surfactant gene mutations.
- Differential diagnoses:
  - *RB-ILD:* It may coexist with DIP as part of smoking-related lung injury.
  - *NSIP:* DIP can be distinguished from NSIP by the absence of fibrosis (traction bronchiectasis and volume loss) **(Figs 5.12A to D)**. History of smoking and pigmented macrophages on BAL also suggest DIP
  - *HP:* Sparing of lung bases
- 'Airspace enlargement with fibrosis (AEF)' is a smoking-related phenomenon, not regarded as a distinct IIP. It is characterized by more prominent reticulation and fibrosis along with emphysema.



Figs 5.12A to D: DIP (A and C) vs NSIP (B and D). (A) Coronal MPR lung window image in a patient with DIP shows ill-defined GGO and centrilobular nodules diffusely scattered in both lungs without fibrosis; (B) Compared to NSIP where axial HRCT image depicts GGO and fibrosis in peribronchovascular distribution; (C) Axial HRCT image in DIP depicts paraseptal emphysema and diffuse uniform bilateral GGO along with few reticular lesions; (D) Axial HRCT image in active NSIP may look similar but demonstrates peribronchovascular interstitial thickening, GGO, traction bronchiectasis and lower lobe predominance



Figs 5.13A to D: Combined pulmonary fibrosis and emphysema (CPFE). (A) Axial HRCT image depicts multiple layers of cystic change in both upper lobes; (B) Axial HRCT image at a lower level shows peribronchovascular interstitial thickening, traction bronchiectasis and honeycombing; (C) Axial HRCT image in a different patient shows reticulation, interstitial thickening and early honeycombing in lower lobes along with paraseptal emphysema; (D) Coronal MPR lung window shows subpleural bullae and paraseptal emphysematous changes

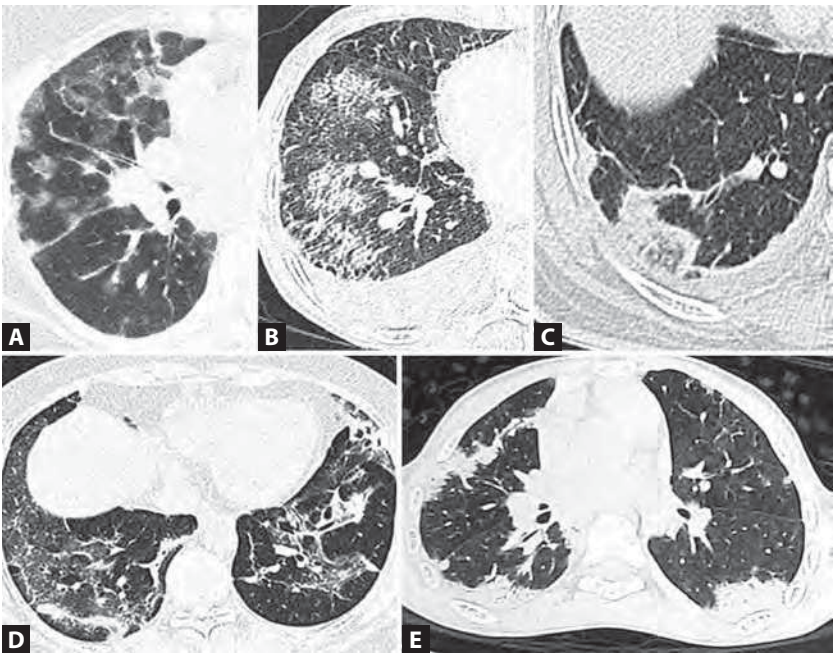
Combined Pulmonary Fibrosis and Emphysema

- Multiple patterns may coexist in smokers like LCH, RB, DIP, chronic fibrosis (UIP/NSIP) and emphysema.
- While emphysema predominates in upper lobes, fibrosis occurs in lower lobes **(Figs 5.13A to D)**.
- Combined pulmonary fibrosis and emphysema (CPFE) refers to a distinct set of patients and not a separate IIP.
- While AEF is usually an incidental finding on CT, CPFE is always asymptomatic.
- There are poor prognosis and increased risk of pulmonary hypertension.

ACUTE/SUBACUTE IDIOPATHIC INTERSTITIAL PNEUMONIAS

Cryptogenic Organizing Pneumonia

- Patients present with 4–6 weeks history of flu-like features.
- Organizing pneumonia is the generic term while COP is the idiopathic form.



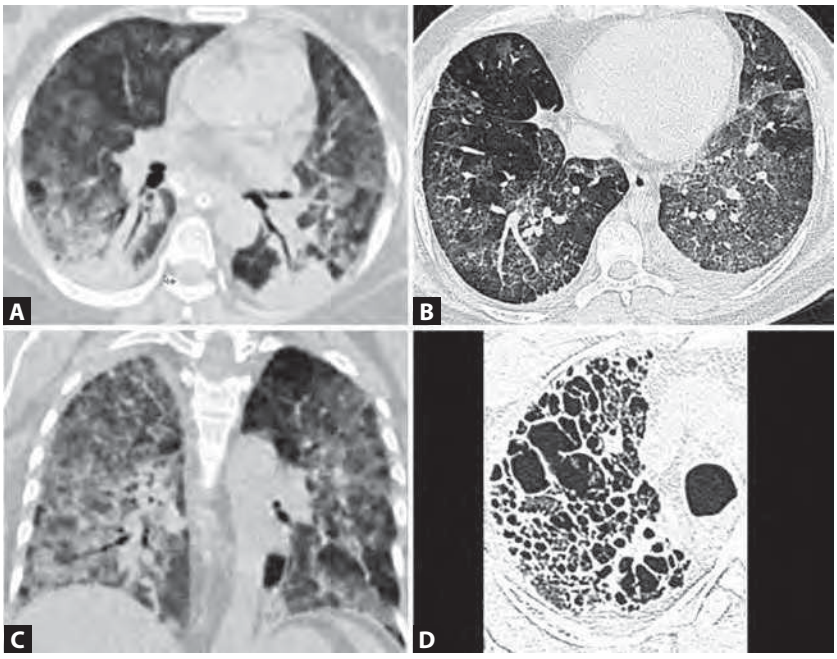
Figs 5.14A to E: Cryptogenic Organizing pneumonia (COP). (A) Axial CT image lung window depicts patchy areas of GGO in lobular configuration in peribronchovascular and subpleural regions; (B) Axial HRCT image in a different patient shows peribronchovascular consolidation and GGO. Right pleural effusion is also seen; (C) Axial HRCT image in different patient shows pleural based consolidation with central GGO (atoll sign); (D) Axial HRCT image demonstrates patchy areas of GGO bilaterally with consolidations in subpleural band-like configuration and peribronchovascular distribution; (E) Axial CT image lung window shows subpleural and bronchiolocentric consolidations with some having band-like configuration

- CT findings include patchy unilateral or bilateral consolidations which may contain air bronchograms with bronchial dilatation. Typical presentation is subacute consolidations persisting despite antibiotic therapy.
- Consolidations have subpleural or peribronchovascular predilection **(Figs 5.14A to E)**. Basal predominace has also been reported.
- Randomly distributed GGOs usually co-exist in majority of the cases. Occasionally GGOs are the dominant manifestation.
- Perilobular opacities or increased attenuation **(Figs 5.14A to E)** is seen in 57% cases
- Reverse halo or atoll sign is seen in 20% cases and refers to central ground glass and surrounding denser consolidation **(Figs 5.14A to E)**. Though initially described to be characteristic for COP, atoll sign may be seen in infections (TB, fungal), granulomatous diseases (sarcoidosis, Wegener's granulomatosis).

- Less common findings include band-like opacities **(Figs 5.14A to E)**, large nodules or mass-like areas of consolidation, ill-defined micronodules, satellite nodules around the consolidations.
- Reticulation may be seen but seldom the dominant feature.
- Mediastinal LN enlargement may be seen.
- Small effusions can also occur.
- On follow-up, consolidations tend to migrate and show fleeting behavior, even without treatment. COP responds well to treatment, though relapse rate is high. Areas of consolidation may be replaced by GGO and reticulation surrounding bronchiectatic airways and thus may resemble NSIP.
- Differential diagnoses:
  - Infection and aspiration are important differentials. Septic emboli may also look similar on CT.
  - Chronic eosinophilic pneumonia is also characterized by subpleural consolidations but has upper lobe predominance. Presence of nodules is distinctly rare in eosinophilic pneumonia.
  - Vasculitis and sarcoidosis may occasionally mimic COP.
  - Mucinous adenocarcinoma, metastases and lymphoma may come in differentials when the manifestation is in the form of mass-like consolidations.

Acute Interstitial Pneumonia

- Presents with rapidly progressive hypoxia with high mortality.
- Pathology shows diffuse alveolar damage similar to acute respiratory distress syndrome (ARDS).
- Common CT findings include GGO, bronchial dilatation and architectural distortion **(Figs 5.15A to D)**.
- Bilateral patchy areas of ground-glass attenuation with focal sparing of lung lobules, producing a geographic appearance.
- Basal predominance but occasionally diffuse or upper lobe preponderance. Dependent areas of lungs most often involved.
- Consolidations are common.
- CT based scoring systems in the acute phase aid in prognostication as well as are independent predictors of mortality.
- Late organizing stage depicts traction bronchiectasis, distortion of bronchovascular bundles and honeycombing.
- The disease has poor prognosis and thus studies on longitudinal CT findings are rare. Subjects who survive show resolution of GGO and consolidations. Residual findings include regions of hypoattenuation, cysts, reticular abnormalities and architectural distortion affecting the nondependent portions of the lungs.
- Differential diagnoses:
  - *ARDS:* CT features are similar but ARDS is secondary to an inciting cause. Bilateral symmetrical distribution with lower lobe predominance is commoner in AIP.



Figs 5.15A to D: AIP. (A) Axial CT image shows bilateral diffuse GGO and patchy consolidations in the dependent regions. There is peribronchial consolidation as well; (B) Axial HRCT image in a different patient shows GGO and septal thickening predominantly in dependent regions; (C) Coronal MPR lung window image depicts widespread GGO both lungs with septal thickening; (D) Axial HRCT image shows sequelae of AIP in the form of diffuse honeycombing and architectural distortion

- *COP:* Septal thickening, reticular lesions and traction bronchiectasis are more prevalent in AIP than COP.
- Widespread infection (*Pneumocystis jirovecii* pneumonia) shows peribronchovascular predominance of GGO and septal thickening.
- Pulmonary edema and acute eosinophilic pneumonia demonstrate prominent interlobular septal thickening and pleural effusions.
- *Diffuse alveolar hemorrhage:* Widely distributed traction bronchiectasis is more characteristic of AIP than other acute lung diseases.
- Acute HP is characterized by profuse centrilobular nodules and mosaic attenuation.
- *Alveolar proteinosis:* Scattered geographic areas of septal thickening and GGO.

Acute Exacerbation of Idiopathic Interstitial Pneumonias (Accelerated Phase)

- Any chronic IIP may show acute respiratory worsening (clinically simulating pulmonary embolism).
- Histologically, this is characterized by acute or organizing diffuse alveolar damage.

- Commonly seen in patients of IPF with new extensive rapidly progressive GGO and sometimes consolidation.
- Background IIP pattern (UIP most commonly but NSIP may also be seen) can often be identified.
- Three types of distribution may be seen: peripheral, multifocal or diffuse.
- Diagnostic differentials include superimposed infection and heart failure.

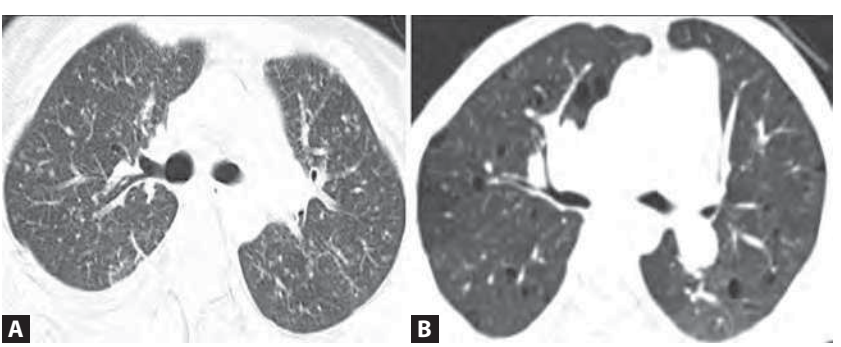
RARE IDIOPATHIC INTERSTITIAL PNEUMONIAS

Lymphoid Interstitial Pneumonia

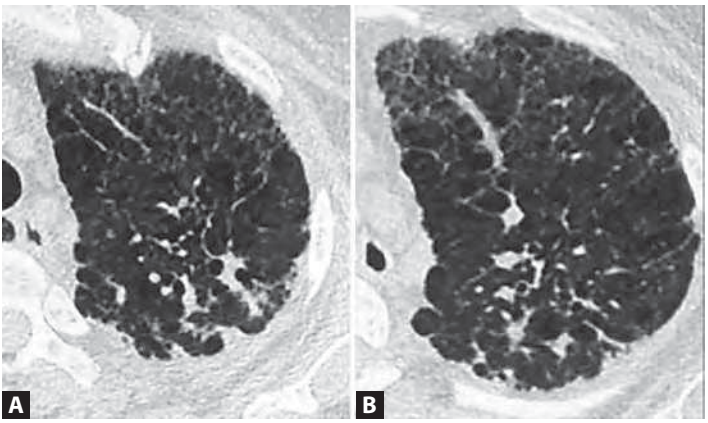
- It is a benign lymphoproliferative condition
- Most cases are associated with AIDS and Sjögren syndrome and truly idiopathic LIP is rare. Cellular NSIP is sometimes wrongly labelled as LIP.
- CT demonstrates GGO, varying-size nodules, interlobular septal thickening, thickened peribronchovascular interstitium with predominant involvement of lower lobe **(Fig. 5.16A)**.
- Thin walled cysts (1 mm to 3 cm in diameter)—discrete or clustered in peribronchovascular distribution are characteristic **(Fig. 5.16B)**.
- On follow-up, GGO resolve but cysts persist.
- Differential diagnoses:
  - *Cellular NSIP:* Presence of cysts is helpful in differentiating.
  - *Pulmonary Langerhans cell histiocytosis (LCH):* Presence of Bizzare-shaped cavitating nodules with sparing of CP angles favors LCH while lower lobe predominance with GGO and reticulation suggests LIP.
  - *Lymphangioleiomyomatosis (LM):* GGO and reticulation are found in LIP and unusual in LM.

Idiopathic Pleuroparenchymal Fibroelastosis

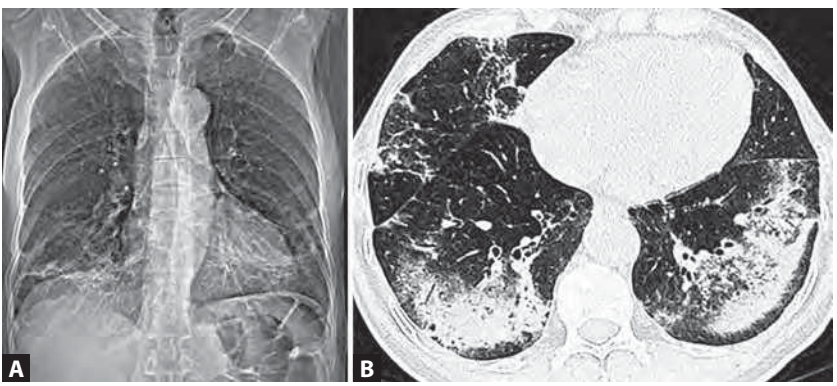
- Predominantly involves the pleura and subpleural parenchyma, especially of the upper lobes.
- Progression of disease occurs in 60% and mortality rate is 40%.



Figs 5.16A and B: Lymphoid interstitial pneumonia. (A) Axial HRCT image depicts ill-defined nodules in peribronchovascular distribution; (B) Few interspersed well-defined thin-walled cysts are also seen in similar distribution in lower sections



Figs 5.17A and B: Idiopathic pleuroparenchymal fibroelastosis. (A and B) Axial HRCT images depict apical irregular pleural thickening, subpleural reticular lesions, architectural distortion and traction bronchiectasis

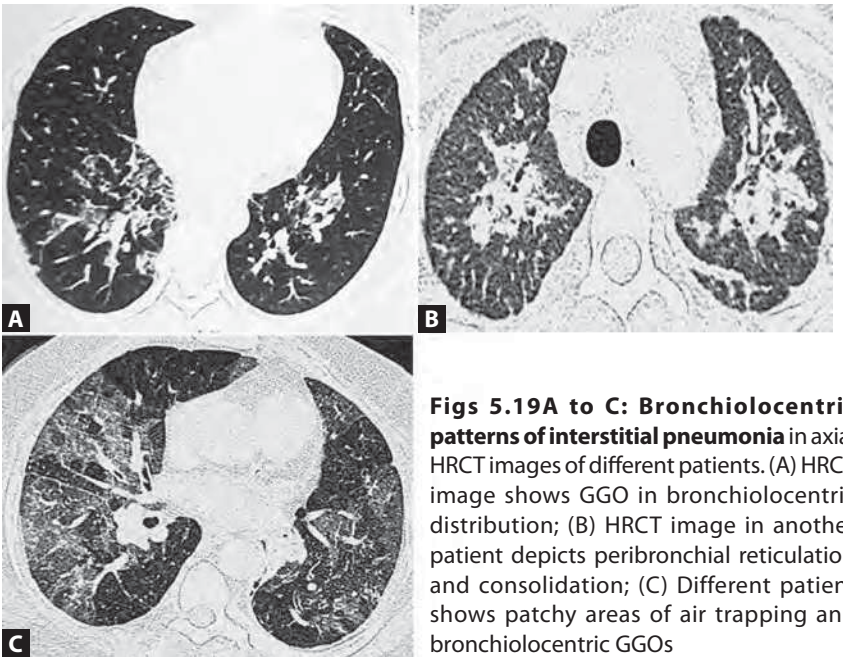


Figs 5.18A and B: Acute fibrinous and organizing pneumonia. (A) CT topogram image depicts bilateral lower zone consolidations; (B) Axial HRCT image of same patient shows bibasilar dense consolidations

- CT shows irregular pleural thickening, pleural tags in the upper lobes along with fibrotic changes in the adjacent lung parenchyma **(Figs 5.17A and B)**.
- Significant upper lobe volume loss also occurs with traction bronchiectasis and architectural distortion.
- Dense subpleural consolidations and pneumothorax may also occur.
- Differential diagnoses include connective tissue diseases (ankylosing spondylitis), end stage sarcoidosis and chronic HP.

Acute Fibrinous and Organizing Pneumonia (AFOP)

- Principal CT findings include bilateral basal consolidations **(Figs 5.18A and B)**.
- Differentials include eosinophilic pneumonia, COP, AIP, connective tissue diseases, HP.



Figs 5.19A to C: Bronchiolocentric patterns of interstitial pneumonia in axial HRCT images of different patients. (A) HRCT image shows GGO in bronchiolocentric distribution; (B) HRCT image in another patient depicts peribronchial reticulation and consolidation; (C) Different patient shows patchy areas of air trapping and bronchiolocentric GGOs

Interstitial Pneumonias with a Bronchiolocentric Distribution

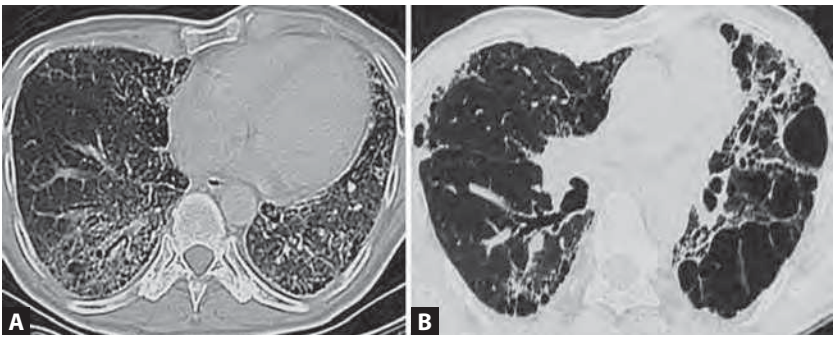
- These are characterized by bronchiolocentric inflammation and fibrosis.
- There is peribronchovascular GGO, reticular abnormalities and traction bronchiectasis **(Figs 5.19A to C)**.
- Air trapping is seen, suggesting an underlying small airway disease.

UNCLASSIFIABLE IDIOPATHIC INTERSTITIAL PNEUMONIAS

- Refers to those cases where final diagnosis is not achieved even after multidisciplinary approach.
- Either there is inadequate clinical, radiologic or pathological data, or there is significant discordance **(Figs 5.20A and B)**.
- Lack of histopathologic correlation accounts for 50% of such cases. This is because of high-risk associated with surgical lung biopsy.
- Prior therapy may result in substantial changes in radiological/pathological picture, making diagnosis difficult.
- Presence of mixed patterns also leads to diagnostic dilemma.
- Management in such cases is based on most likely pattern and clinical course.

CONCLUSION

Standard scanning technique, unambiguous lexicon and structured reporting system are important components of systematic approach to interpreting of



Figs 5.20A and B: Unclassifiable IIP. Elderly male patient (53 years nonsmoker) presented with dyspnea in 2013. (A) Axial HRCT image shows reticular abnormalities and traction bronchiectasis with lower lobe predominance. Abnormalities are localized to subpleural and peribronchial regions. Radiological diagnosis of NSIP pattern was made; (B) Follow-up HRCT image after 2 years shows development of large cysts, honeycombing and pleural thickening. This patient needs histological correlation for definitive diagnosis

chest CT in patients suspected to have IIP. After diagnosis of idiopathic ILD, the next step is to determine whether it is chronic fibrosing type and then evaluate whether it fits into the criteria for UIP. Some entities may remain unclassifiable and the diagnosis in such cases may be formed after assessing the disease progression in follow-up. Mixed patterns of lung injury and presence of superimposed complications may make the task of radiologist difficult. HP, drug toxicity and connective tissue diseases are important differentials and may produce similar patterns of injury. A multidisciplinary approach is helpful in most of the cases in deciding optimal management.

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2. Travis WD, Costabel U, Hansell DM, King TE, Lynch DA, Nicholson AG, et al. An official American Thoracic Society/European Respiratory Society statement: Update of the international multidisciplinary classification of the idiopathic interstitial pneumonias. Am J Respir Crit Care Med. 2013;188(6):733-48.

6

Idiopathic Interstitial Pneumonias: Clinical Diagnosis and Management Issues

Karan Madan

INTRODUCTION

- Idiopathic interstitial pneumonias (IIPs) are usually characterized by a diffuse pattern of involvement of the lung parenchyma and as the name signifies, the underlying cause is not clear. Pulmonary interstitium is involved and the degree of inflammation or the fibrotic component varies depending on the different subtypes.
- The initial landmark classification of the major IIPs was provided in the year 2002 by a collaborative statement by the American Thoracic Society (ATS) and the European Respiratory Society (ERS).[1] IIPs were classified into seven subcategories and diagnostic criteria were proposed. This classification enabled a standardized nomenclature for this group of lung diseases. Histological diagnosis using a lung biopsy that was considered the gold standard for diagnosis of these ILDs was replaced and an integrated approach collating the opinions from the clinician, radiologist and the pathologist as part of a multidisciplinary discussion (MDD) was conceived.
- In a subsequent revised statement in 2013, most of the major entities were the same and a few new entities were added.[2]

The categories of IIPs recognized in the updated classification include:

Major idiopathic interstitial pneumonias
- Idiopathic pulmonary fibrosis (IPF)
- Idiopathic nonspecific interstitial pneumonia
- Respiratory bronchiolitis–interstitial lung disease
- Desquamative interstitial pneumonia
- Cryptogenic organizing pneumonia
- Acute interstitial pneumonia.

Rare idiopathic interstitial pneumonias
- Idiopathic lymphoid interstitial pneumonia
- Idiopathic pleuroparenchymal fibroelastosis.

Unclassifiable idiopathic interstitial pneumonias

- Clinical presentation and symptoms, smoking history, history of significant exposures, pulmonary function tests, serum autoimmune immunological markers should always be taken into consideration. A close communication between the clinician, radiologist and pathologist is essential.
- Before labeling a patient to be having an IIP, known causes of ILD should be excluded especially focusing on history of exposure to drugs which are associated with ILD causation or collagen vascular diseases or inhalational exposures.

SPECIFIC IDIOPATHIC INTERSTITIAL PNEUMONIA ENTITIES

Idiopathic Pulmonary Fibrosis

Guidelines focusing on the diagnostic and management perspectives of idiopathic pulmonary fibrosis (IPF) were published in the year 2011.[3] These were developed in collaboration between ATS/ERS/Japanese Respiratory Society (JRS) and Latin American Thoracic Association (ALAT).

Clinical Presentation and Diagnostic Issues

- IPF is chronic interstitial lung disease which is characterized by progressive pulmonary parenchymal fibrosis. The condition has a predilection for affecting the older adults and extrapulmonary symptoms are minimal to absent. Pathological hallmark is the histopathological pattern of usual interstitial pneumonia (UIP). UIP pattern is also the radiological hallmark.
- A possible diagnosis of IPF should always be considered in an adult patient with unexplained exertional breathlessness that is chronic. The common clinical findings include dyspnea and cough and physical signs include bibasilar velcro crackles and digital clubbing.
- Patients in the sixth or seventh decade are most commonly affected.
- Majority are current or past smokers and there is a male predominance of affliction.
- The diagnosis of IPF requires exclusion of other causes of ILD with a known etiology along with visualization of HRCT UIP pattern in patients who are not undergoing a surgical lung biopsy. A combination of biopsy and CT features in combination should be considered in patients who undergo lung biopsy.
- The natural history of IPF is unpredictable but most of the patients have a gradually worsening course. Acute flares and rapid worsening can ensue on the background of a relatively stable course.
- Progressive disease is characterized by worsening dyspnea, declining lung volumes on spirometry and radiological progression of pulmonary fibrosis.
- Definite UIP features on HRCT are associated with a shorter patient survival.
- Comorbidities that may be commonly observed include pulmonary hypertension, sleep apnea, COPD and gastroesophageal reflux.

Management

No pharmacological agent has shown to be of definitive proven benefit in this condition. Recent evidence indicates that anti-fibrotic agents like Pirfenidone and tyrosine kinase inhibitor Nintedanib may provide clinically meaningful improvement in some patients. Corticosteroids alone or in combination with immunosuppressive/immunomodulators should not be used in patients with IPF.

- The definitive line of management in patients with advanced disease remains lung transplantation.
- Long-term oxygen should be definitely considered in patients with advanced disease and resting hypoxemia.

Idiopathic Nonspecific Interstitial Pneumonia

Clinical Presentation and Diagnostic Issues

- Nonspecific interstitial pneumonia (NSIP) pattern of involvement occurs in a number of situations like association with connective tissue diseases, hypersensitivity pneumonitis and drug reactions. When an underlying cause or association is not obvious, the condition is labeled as idiopathic NSIP.
- Idiopathic NSIP is now accepted as a distinct clinical condition with features that differentiate the condition from other ILDs.[4]
- MDD is especially important to establish the diagnosis of idiopathic NSIP
- The most important clinical characteristics that distinguish NSIP from UIP are subacute rather than insidious onset of symptoms, associated fever in about one-third of patients, lack of a strong male predominance, predominance in never smokers, relative absence of clubbing in NSIP, and increased frequency of features and serology suggestive of connective tissue disease.
- Pulmonary function test (spirometry) demonstrates restrictive physiology.
- On histopathological examination, the involvement is usually uniform and varying degrees of inflammation and areas of fibrosis are seen.
- Long-term prognosis is favorable and most patients improve with treatment.

Management

- In patients with NSIP and underlying collagen vascular disease, if the clinical symptomatology is minimal and lung function impairment is mild, the therapies used for the management of the primary underlying CTD may be sufficient and patients can be kept under close observation with periodic lung function assessment.
- Patients with mild disease may improve or stabilize without therapy. A close watch should be kept on the progression of symptoms and lung function assessments may be performed every three months and if progression occurs, therapy should be immediately instituted. Patients with severe

symptoms and previously patients undergoing observation if they worsen should be treated with glucocorticoids, with or without combination of other drugs like azathioprine, cyclophosphamide, and cyclosporine.
- Pneumocystis prophylaxis should be considered in patients receiving moderate to high dose steroids or immunosuppressive drugs.
- Response to treatment is indicated by improvement in symptoms, improvement in radiographic involvement and pulmonary function test parameters.
- Patients with progressive disease and progression to end-stage fibrosis are candidates for lung transplantation.

Respiratory Bronchiolitis–Interstitial Lung Disease

Clinical Presentation and Diagnostic Issues

- Respiratory bronchiolitis (RB) was defined in 1974 as a distinct entity in smokers wherein bronchiolar accumulation of macrophages occurs.
- Respiratory bronchiolitis interstitial lung disease (RB-ILD) is defined as histopathological demonstration of RB like changes as the only demonstrable abnormality in an active or reformed smoker patient and clinicoradiological profile consistent with ILD.
- RB-ILD and DIP are uncommon forms of IIP. Many features of these conditions are similar and the overall long-term prognosis is favorable.[1]
- Clinical and examination findings are not specific.
- Patients are usually in the fourth or fifth decades of life. Most patients are active smokers. There is a male predominance of affliction.
- Spirometry demonstrates a restrictive or a mixed pattern
- Bronchoscopy may be performed with Bronchoalveolar lavage (BAL) and Transbronchial lung biopsy (TBLB) in patients where other diagnoses are being considered.
- Surgical lung biopsy is not always necessary and in presence of clinico-radiological features suggestive of RB-ILD, a trial of smoking cessation may precede lung biopsy. When there is a diagnostic uncertainty and patients fail to improve following a trial of smoking cessation lung biopsy may be considered.
- Symptomatic and physiological improvement occur only a minority of patients but overall prognosis is favorable and progression to end-stage fibrotic ILD is rare.

Management

- Smoking cessation is the cornerstone of management and up to two-thirds of patients improve
- Patients may have coexistent COPD which may benefit from specific bronchodilator medications.

- For patients who have progressive RB-ILD despite smoking cessation, glucocorticoid therapy and other immunosuppressive agents (e.g. azathioprine) are sometimes used, but data supporting their efficacy are conflicting.

Desquamative Interstitial Pneumonia

Clinical Presentation and Diagnostic Issues

- Desquamative interstitial pneumonia (DIP) usually affects cigarette smokers in their fourth or fifth decades of life. DIP is an uncommon IIP. A small percentage of cases are associated with connective tissue diseases.
- Spirometry demonstrates restrictive defect and diffusing capacity for carbon monoxide is reduced.
- Clinical symptoms are similar with RB-ILD and mostly smokers are affected.

Management

- Smoking cessation is the cornerstone of management.
- Oral glucocorticoid therapy may be considered when symptomatic progression occurs despite smoking cessation.
- Some patients might require treatment with immunosuppressive agents and the overall prognosis is excellent.

Acute Interstitial Pneumonia

Clinical Presentation and Management

- Acute interstitial pneumonia (AIP) presents with an abrupt onset of symptoms and rapidly progressive respiratory failure usually ensues. Diffuse alveolar opacities on chest radiographs are seen.
- This condition is also known as Hamman-Rich syndrome and accelerated interstitial pneumonia. The presentation may be considered akin to idiopathic ARDS.
- Management includes supportive mechanical ventilation and glucocorticoid treatment. Extracorporeal membrane oxygenation (ECMO) may be required if oxygenation fails to improve with conventional mechanical ventilatory strategies.
- Majority of patients die of respiratory failure.

Lymphoid Interstitial Pneumonia

Clinical Presentation and Management

Lymphoid interstitial pneumonia (LIP) is an uncommon entity characterized by infiltration of the pulmonary interstitium and alveolar spaces by lymphocytes and plasma cells.

- The condition may be associated with entities like HIV, rheumatic disease (like Sjögren's syndrome, rheumatoid arthritis, systemic lupus erythematosus) and immunodeficiencies.
- Nearly 25% of reported cases of LIP may be associated with Sjögren's syndrome.
- Most patients are symptomatic. Clinical features are nonspecific.
- Clinical examination findings are nonspecific and usually crackles are audible. Spirometry demonstrates a restrictive ventilatory defect.
- Clinicoradiographic features and BAL fluid analysis findings are nonspecific and a surgical lung biopsy is usually required for definitive diagnosis.
- Once the diagnosis of LIP is ascertained, an attempt should be made to identify the specific underlying cause.

Management

- Most patients with idiopathic LIP have significant respiratory symptoms and/or evidence of physiological impairment at the time of diagnosis. Mildly symptomatic patients may be observed, but most patients require treatment.
- Systemic glucocorticoids with or without additional immunosuppressive agents are the treatment of choice.
- The prognosis is poorly understood.

Cryptogenic Organizing Pneumonia

Clinical Presentation

- Disease onset is typically in the fifth or sixth decades of life. There is no gender predilection.
- Most patients have a subacute onset of symptoms and the overall presentation may closely resemble community-acquired pneumonia with a preceding flu-like illness.
- Spirometry usually shows a restrictive pattern with reduction in diffusion capacity.
- BAL and TBLB may be performed to exclude alternative diagnoses and surgical lung biopsy is usually preferred for confirmation of a definitive diagnosis.

Management

- Asymptomatic or patients with minimal symptoms may be carefully observed with monitoring of lung functions and chest radiographs.
- Glucocorticoids are the cornerstone of management and most patients have excellent response to treatment.
- For patients with severe involvement and respiratory failure, intravenous methylprednisolone may be considered.

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Section 3

Connective Tissue Disorders/Immune-mediated Disorders

- Connective Tissue Disorders and Related Interstitial Lung Diseases: Imaging
- Connective Tissue Disease-related Interstitial Lung Diseases: Clinical, Diagnostic and Management Issues
- Vasculitis-related Interstitial Lung Diseases: Imaging
- Vasculitis-related Interstitial Lung Diseases: Clinical Diagnostic and Management Issues
- Sarcoidosis

7

Connective Tissue Disorders and Related Interstitial Lung Diseases: Imaging

Surabhi Vyas

Connective tissue diseases or collagen vascular diseases are a group of multisystem disorders with possible autoimmune etiology and variable pulmonary complications. The pulmonary manifestations can be due to the disease process per se or as a result of various medical therapies and secondary infections, which complicate the imaging picture.

The conditions included in this broad term are:

- Rheumatoid arthritis
- Progressive systemic sclerosis
- Systemic lupus erythematosus
- Polymyositis, dermatomyositis
- Sjögren syndrome
- Mixed connective tissue disease.

These conditions will be discussed individually later in the chapter.

IMAGING MODALITIES

- *Radiography:* Chest radiograph remains usually the first and most common initial investigation to rule out pulmonary involvement. Radiograph may be normal in early involvement due to its low sensitivity or may show increased interstitial markings, shaggy mediastinal margins and reduced lung volume in more established cases.
  - It may also show areas of consolidation, fibrosis as well as pleural and pericardial effusion.
  - A normal chest radiograph however does not rule out lung involvement and further evaluation in the form of high resolution CT (HRCT) is often required.
- *Echocardiography:* Echocardiography is a commonly performed investigation in cardiac status evaluation in such patients. Its relatively low expense and availability makes it an attractive modality. However it is an operator dependent modality and interobserver variability is often present.

- *Computed tomography:* HRCT remains the mainstay for evaluation of connective tissue disease (CTD) related interstitial lung disease (ILD) and many studies have shown positive correlation between HRCT and histologic findings. HRCT is the imaging modality of choice to look for patterns and extent of ILD.[1] A contrast enhanced study may be required for assessment of enlarged nodes in the setting of secondary infections or in vascular complications where a CT pulmonary angiography is more useful.[2]
- *Magnetic Resonance Imaging (MRI):* Although previously, the role of MRI in thoracic disorders was limited to evaluation of chest wall and mediastinal pathologies, it also helps in evaluation and gradation of pulmonary hypertension as a complication of various CTD.[3] Cardiac MRI with respiratory and ECG gating is now regarded as a reference modality for right ventricular functional assessment. Moreover contrast enhanced MR angiography, perfusion MRI and blood oxygenation level-dependent (BOLD) imaging has been found useful in assessment of lung circulation and perfusion.

SYSTEMIC SCLEROSIS

Introduction

- Systemic sclerosis is an autoimmune inflammatory condition characterized by excessive collagen deposition in the skin and connective tissue.
- Prevalence range from 60 to 250 per million, with male : female ratio of 1:3–6.
- The cutaneous manifestation of systemic sclerosis include variably progressive skin thickening usually in the fingers and hands and face.
- Interstitial fibrosis has been reported in 20–60% of patients.[4]
- Pulmonary hypertension develop in around one fifth of patients and is an important cause of death in such patients. Morbidity may also be due to orofacial, esophageal manifestations and psychological issues associated with it.

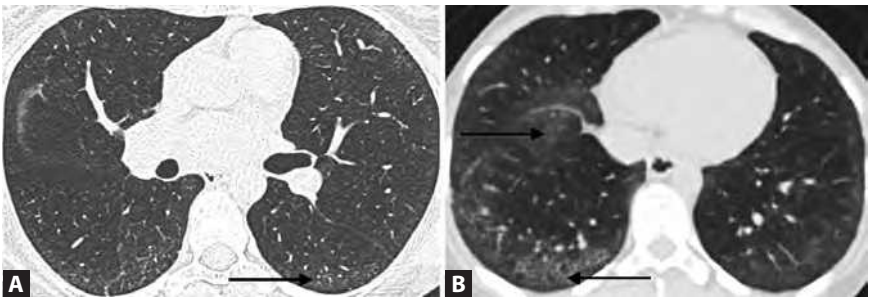
Etiopathogenesis

- Systemic sclerosis is a result of excessive collagen deposition in skin and various internal organ including lungs, heart, digestive system, kidneys, muscles, joints, and nervous system.
- Genetic, environmental, vascular and auto immunologic factors are involved in the pathogenesis. Antigens from the human leukocyte antigen (HLA) histocompatibility complex, including HLA-B8, HLA-DR5 and HLA-DR3 are also associated.
- Resultant autoimmune mechanisms lead to activation of fibroblasts, T lymphocytes, macrophages, and mast cells which secrete various cytokines causing increased collagen deposition.

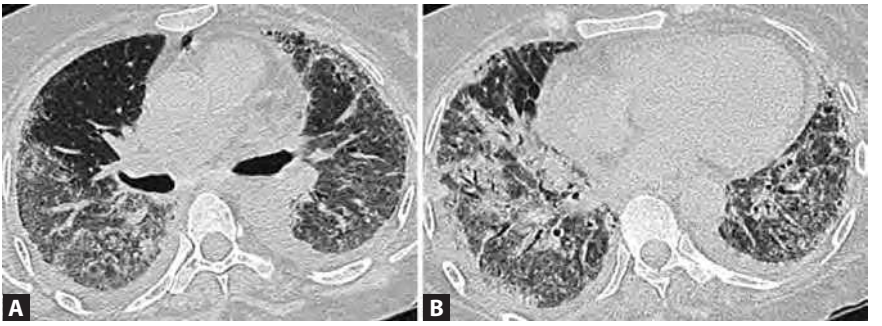
Patterns of Pulmonary Involvement

Nonspecific Interstitial Pneumonitis (NSIP)

- It is the most common pattern seen on HRCT
- In the early phase of the disease ground glass haze may be the only finding. It may represent early cellular change or early fibrosis **(Figs 7.1 and 7.2)**.



Figs 7.1A and B: Early NSIP changes in systemic sclerosis. HRCT axial image (A) showing interlobular septal thickening and ground glass haze in bilateral lower lobes posterior segments and in upper lobes, with subpleural sparing (lower arrow) and ground glass opacities better appreciated on lung window (B)



Figs 7.2A and B: Cellular NSIP in systemic sclerosis. HRCT images of a 45-year-old lady showing extensive areas of ground glass opacities and patchy air space opacities in both lungs

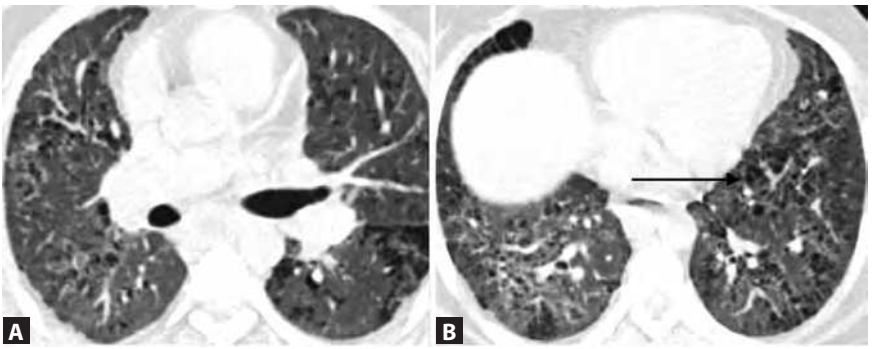
- Changes of NSIP involve the juxtapleural, posterior, and basal lungs, which initially may be seen as ground glass opacities and later as inter and intralobular interstitial thickening, traction bronchiectasis and bronchiolectasis, microcyst formation and honeycombing, the latter more commonly seen in fibrotic NSIP **(Figs 7.3 and 7.4)**.

Usual Interstitial Pneumonia (UIP)

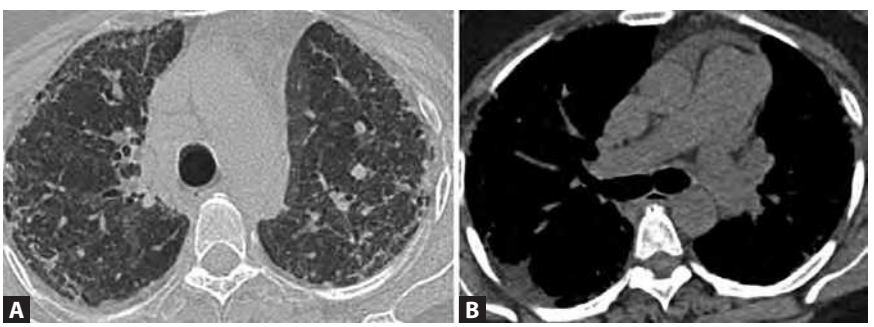
- It is the second most common pattern. Honeycomb cystic changes are seen in up to one third of patients with systemic sclerosis-ILD (SSILD).[5]
- Occasionally there is an overlap of UIP and NSIP patterns in SSILD.
- UIP pattern of ILD carries worse prognosis.

Organizing Pneumonia

- Less commonly seen in SS.



Figs 7.3A and B: NSIP with microcyst formation. Axial CT images showing extensive ground glass opacities (A) in both lungs with few areas of interlobular septal thickening and microcyst formation, arrow (B)



Figs 7.4A and B: Fibrotic NSIP in systemic sclerosis. HRCT image showing advanced areas of fibrosis, bronchiectasis with ground-glass opacities (A). Also note dilated pulmonary trunk and both pulmonary arteries (B)

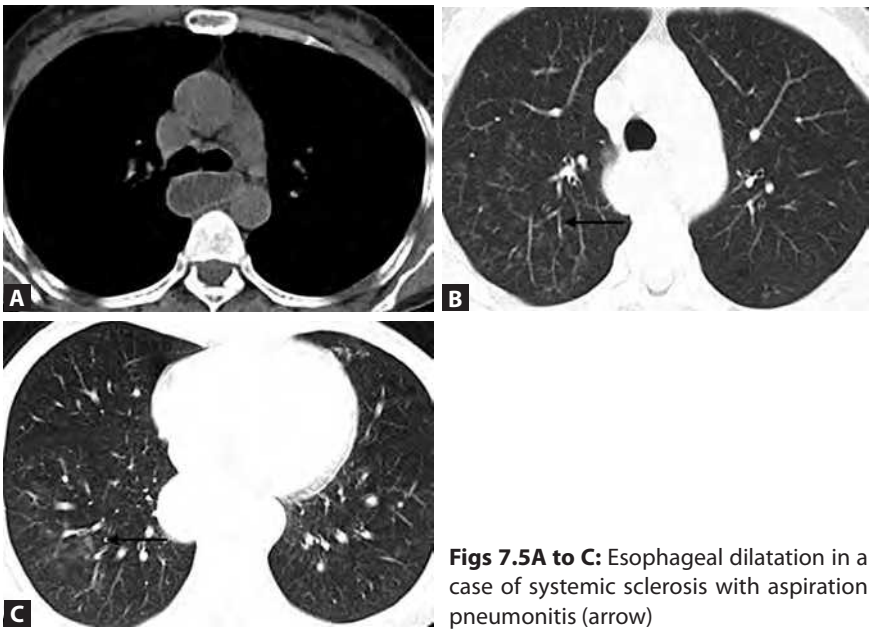
Other Changes

- Dilated patulous esophagus, aspiration pneumonitis, and pulmonary arterial dilatation, enlargement of right sided chambers in severe pulmonary hypertension and rib erosions at superior surfaces **(Figs 7.5A to C)**
- Rarely pneumothorax may be seen as complication **(Figs 7.6A and B)**

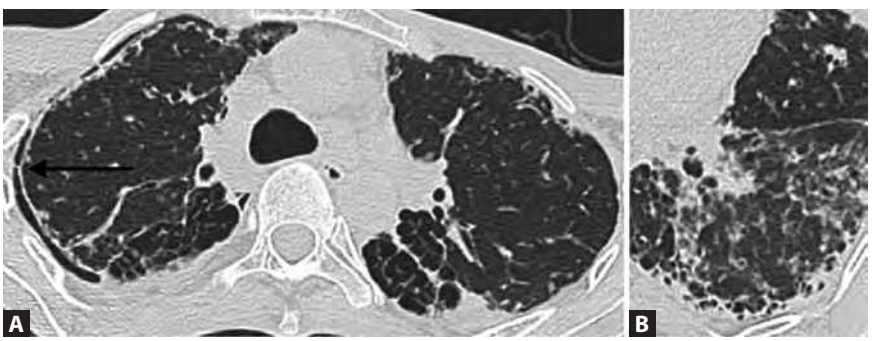
RHEUMATOID ARTHRITIS

Introduction

- Rheumatoid arthritis (RA) is the most common connective tissue disease with prevalence ranging from 0.5 to 2% in general population.
- Pulmonary parenchymal involvement is a common extra-articular complication of RA and is found in around 40% of the RA patients.[6]
- Rarely patients present with respiratory complications prior to development of osseo-articular disease.
- Severity of lung involvement does not correlate to the osseous articular disease severity.



Figs 7.5A to C: Esophageal dilatation in a case of systemic sclerosis with aspiration pneumonitis (arrow)



Figs 7.6A and B: Extensive changes of UIP in a case of systemic sclerosis with sudden onset dyspnea and pneumothorax (arrow)

- Respiratory involvement is a progressive disease and is often the cause of death in these patients. Other complications such as cardiovascular diseases and infections also contribute to the morbidity and mortality with the latter being the most common cause of death.

Structures Involved

- Pulmonary parenchyma as interstitial lung disease
- Pleura as effusion and/or thickening
- Vasculature as vasculitis and
- Airways as small airway disease.

Risk Factors for Development of Pulmonary Complications in Rheumatoid Arthritis

- Increasing age
- Long standing disease
- Male gender, male: female ratio 2:1
- Smoking—increased risk with more than 25 number of pack years
- High levels of RA factor and anti CCP antibodies.
- Genetic factors such as HLA B40, HLA DRB1 and Alpha 1 antitrypsin positivity.

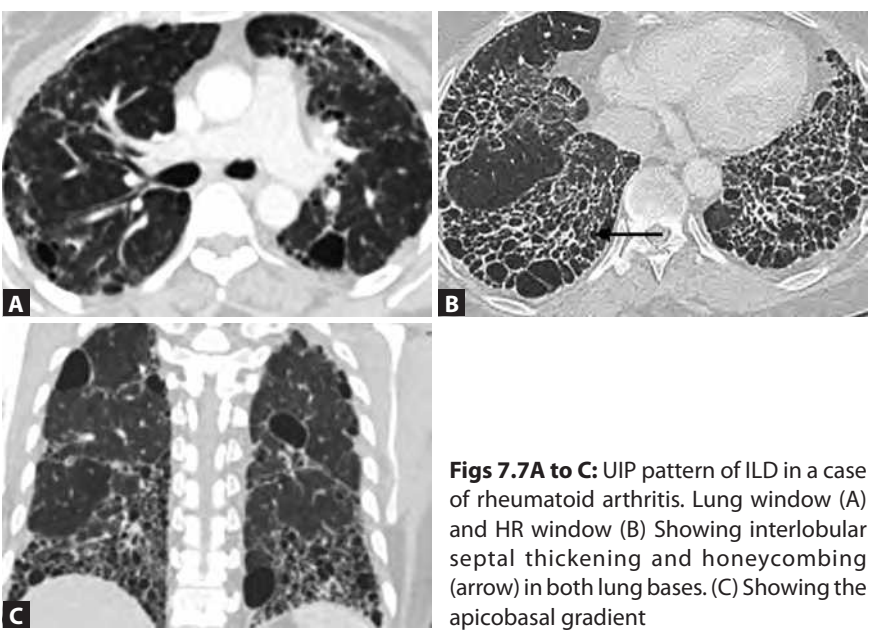
Etiopathogenesis

- Genetic and environmental factors lead to increased citrullination of proteins in the lung which lead to enhanced autoimmune response.
- Consequent injury to the alveolar epithelium leads to change in the cytokine milieu and release of inflammatory mediators. These lead to proliferation of fibroblasts among other responses.

INTERSTITIAL LUNG DISEASE

Types

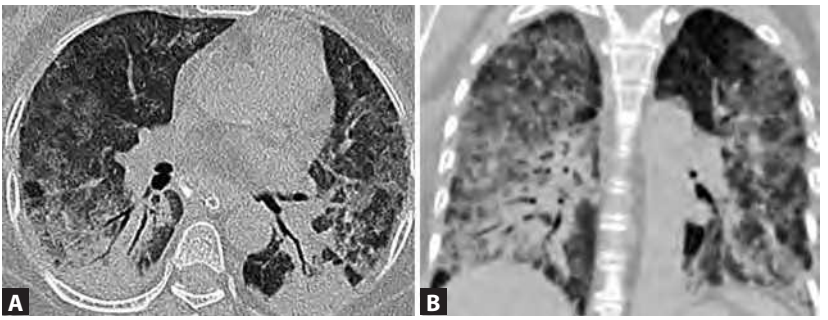
- *Usual interstitial pneumonia (UIP):* Most common pattern with frequency around 40–66%, characterized by basal predominant fibrosis, interlobular septal thickening and honeycombing **(Figs 7.7A to C)**. With long-standing disease there is volume loss and traction bronchiectasis. Ground-glass



Figs 7.7A to C: UIP pattern of ILD in a case of rheumatoid arthritis. Lung window (A) and HR window (B) Showing interlobular septal thickening and honeycombing (arrow) in both lung bases. (C) Showing the apicobasal gradient

opacities are relatively uncommon. The prognosis is worse in UIP than NSIP. However there is no difference in the survival of RA-UIP patients from idiopathic UIP.

- *Nonspecific interstitial pneumonia (NSIP):* Second most common with frequency of around 11–32% and characterized by variable ground-glass opacities and typical subpleural sparing. Honeycombing is relatively rare in NSIP. Areas of interlobular septal thickening and fibrosis may be seen in fibrotic stages.
- *Acute interstitial pneumonia (AIP/Hamman Rich syndrome):* It is a rapidly progressing condition appearing as bilateral asymmetrical patchy alveolar and ground-glass opacities. The condition carries high mortality and is similar to ARDS in presentation and radiological appearance. Lung histology reveals diffuse alveolar damage (DAD) pattern.
- *Organizing pneumonia:* It is present in around 4–6% patients of RA and is seen on HRCT as multifocal peripheral areas of consolidation with or without air bronchogram, which may show fibrotic changes with time.
  - It may be differentiated from idiopathic chronic eosinophilic pneumonia on the basis of peripheral eosinophilia in the later and from bronchoalveolar carcinoma by presence of nodules and nonresponse to steroids in the later.
- *Overlap syndrome:* In a small percentage of patients, overlap syndrome may occur.
- *Combined pulmonary fibrosis and emphysema:* It is seen radiologically as centrilobular or paraseptal emphysema with lower lobe fibrosis. These patients are at an increased risk of pulmonary hypertension.
- Lymphocytic interstitial pneumonia (LIP), desquamative interstitial pneumonia (DIP), constrictive bronchiolitis (mosaic attenuation and air trapping on imaging) and follicular bronchiolitis (centrilobular nodules and branching shadows)patterns are rarely seen in RA and are more frequently seen in other CTDs than RA.
- Drug induced ILD-methotrexate is the most commonly prescribed disease-modifying antirheumatic drug (DMARD) in the treatment of rheumatoid arthritis, however it is also a known causative agent for interstitial lung disease. The common patterns of pulmonary involvement are organizing pneumonia, acute hypersensitivity pneumonitis, and accelerated appearance of pulmonary nodules. The most common pattern is that of hypersensitivity pneumonitis which show patchy ground glass opacities, air space consolidation and centrilobular nodules.
  - Other immunomodulators like anti-TNF agents (etanercept, infliximab and adalimumab) are also shown to have been associated with ILD in combination with methotrexate.
  - Out of all these agents, Methotrexate has been shown to have a greater association with incidence of pulmonary involvement varying from 2-8 %.
  - The imaging differentiation of methotrexate induced lung changes and RA ILD is often difficult **(Figs 7.8A and B)**. However the temporal relationship and reversal of changes on stopping the offending drug may help differentiate the two.



Figs 7.8A and B: Interstitial pneumonia in rheumatoid arthritis resulting from methotrexate with sudden worsening. Multiple patchy peripheral areas of ground glass opacities and air space consolidation in both lungs

Other respiratory system manifestations include:

Pleural Involvement

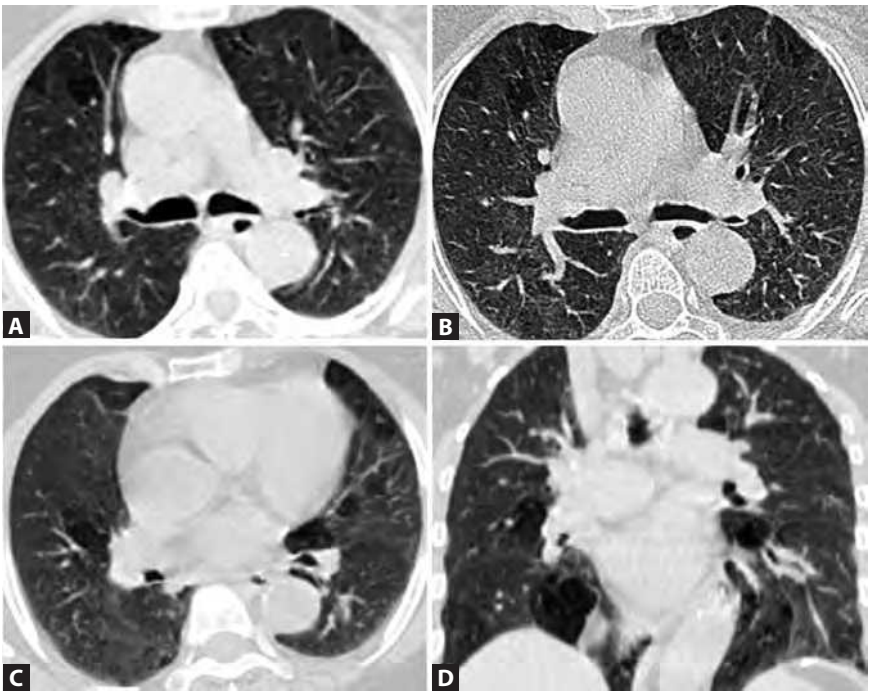
- Pleural disease is a frequent manifestation of RA, so that effusions may be seen in 50–70% of patients.
- Effusion is more commonly unilateral and may be present bilaterally and is more frequently in patients having co-existent rheumatoid nodules.
- The effusion is aseptic and may show cholesterol crystals, resulting in "pseudochylous" pleural fluid in chronic cases.
- Rheumatoid factor is also present in the pleural fluid often in higher values than serum.
- Other pleural manifestations include pneumothorax, and rarely bronchopleural fistula.

Airways

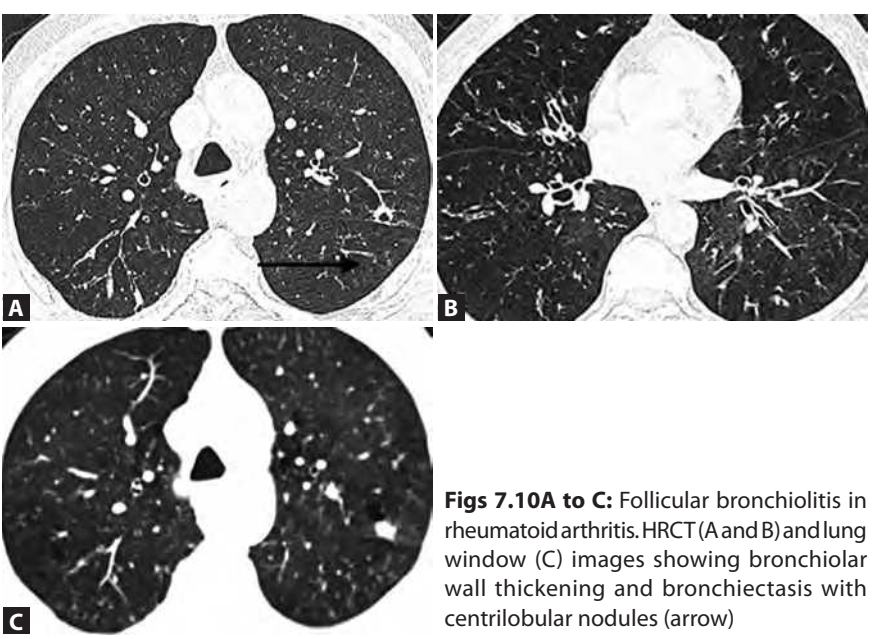
- Airway involvement may be seen in 30–60% of patients and may be due to affection of small and large airways.
- Large airway disease may be due to involvement of the cricoarytenoid joints leading to cricoarytenoid arthritis, causing upper airway compromise or due to bronchiectasis.
- Small airway disease may be manifested as follicular bronchiolitis and obliterative or constrictive bronchiolitis. It affects more distal airways measuring 2 mm or less and may be caused by the disease per se or due to repeated secondary infections **(Figs 7.9 and 7.10)**.

Nodules

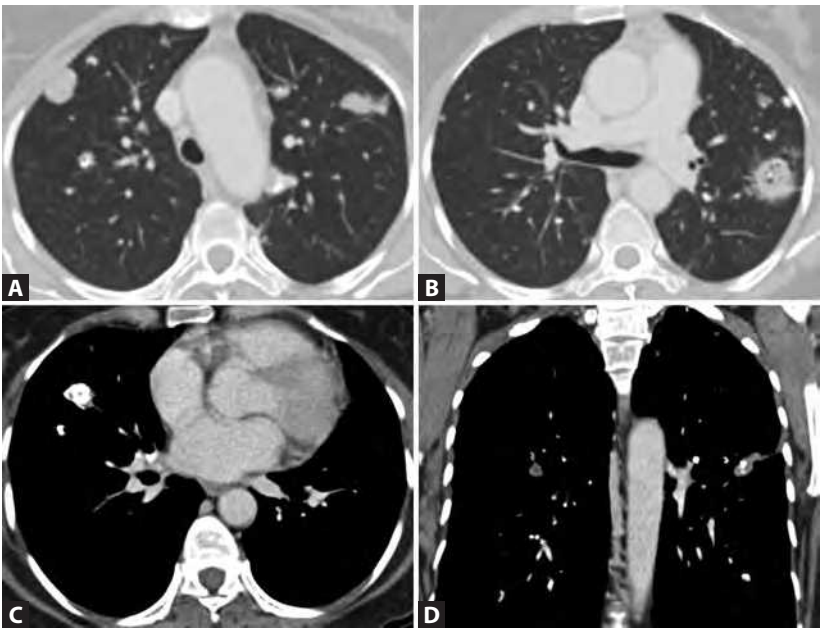
- Rheumatoid nodules are necrobiotic nodules in predominantly pleural and subpleural location and may show cavitation **(Figs 7.11A to D)**.
- These nodules may disappear and reappear in the course of the disease and need to be differentiated from malignancy, infections and septic emboli.



Figs 7.9A to D: Obliterative bronchiolitis in rheumatoid arthritis – HRCT (A and B) and lung window (C and D) CT images showing areas of air trapping



Figs 7.10A to C: Follicular bronchiolitis in rheumatoid arthritis. HRCT (A and B) and lung window (C) images showing bronchiolar wall thickening and bronchiectasis with centrilobular nodules (arrow)



Figs 7.11A to D: Rheumatoid nodules. Lung window (A and B) and mediastinal window (C and D) images of a known case of rheumatoid arthritis showing scattered calcified and noncalcified nodules in both lungs

- Caplan syndrome is a combination of seropositive rheumatoid arthritis and pneumoconiosis with a characteristic pattern of fibrosis. These are seen on imaging as variable sized nodules in upper lobes and peripheral lungs which may calcify, cavitate or remain unchanged with background changes of pulmonary fibrosis. The syndrome was originally described in coal workers but was later found in a variety of pneumoconiosis.

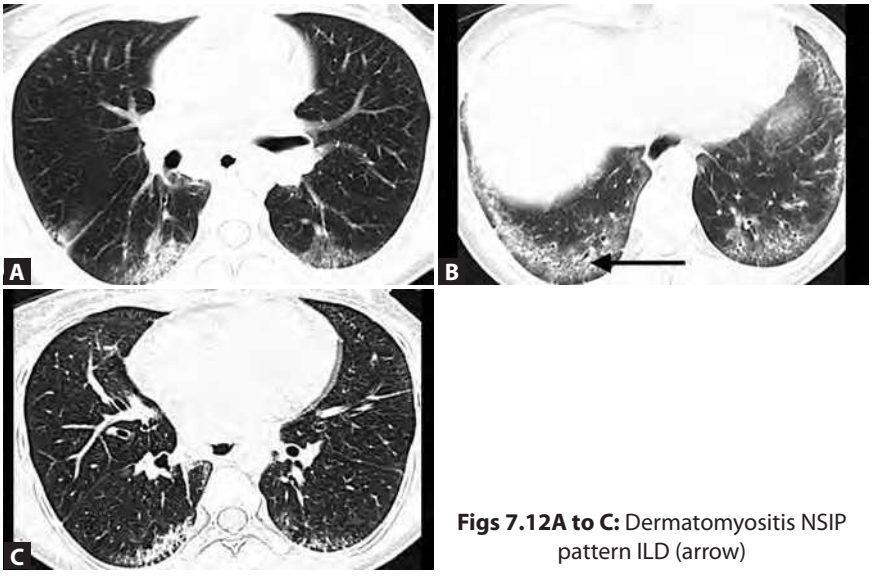
Vascular Disease

- Rheumatoid vasculitis involves small and middle calibre vessels and carries significant morbidity, however pulmonary vasculitic involvement is not so common.
- Pulmonary hypertension is also less commonly associated with RA than other connective tissue disorders like systemic lupus erythematosus (SLE) and systemic sclerosis.

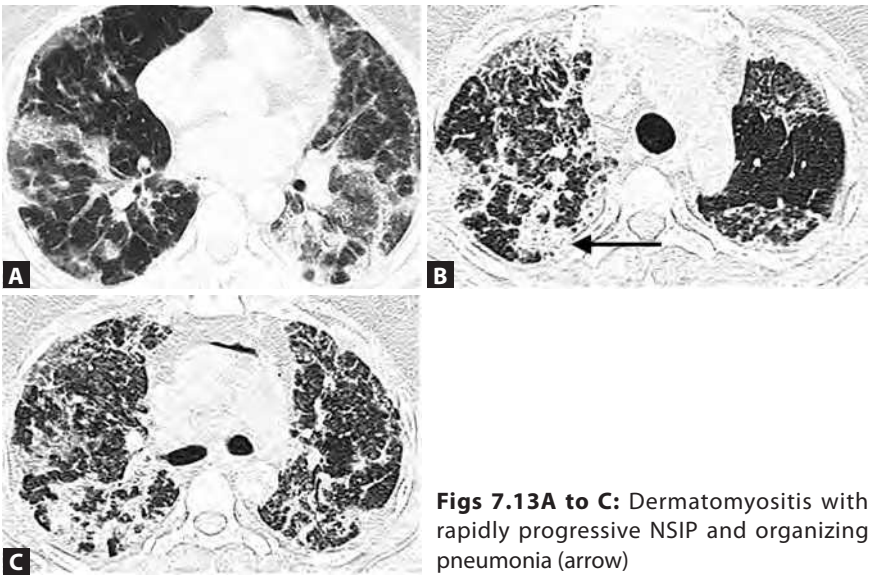
POLYMYOSITIS AND DERMATOMYOSITIS

- *Synonym*: Idiopathic inflammatory myopathies (IIM)
- Pulmonary involvement seen in 40% patients, and lung is the most common organ involved in these patients other than muscles and skin.
- Pulmonary complications include aspiration pneumonia, pulmonary hypoventilation and ILD.

- *Infections* are the most common pulmonary complications seen in up to 26% patients. Predisposing factors being dysphagia, thoracic muscle weakness, aspiration and immunosuppression due to therapy.
- *Pulmonary hypoventilation* due to muscle weakness resulting in dyspnea is uncommon (< 5%); and causing respiratory failure is rare. No apparent parenchymal abnormality is seen on imaging while diaphragmatic dysfunction may be demonstrated on USG/fluoroscopy.[3]
- *ILD*
  - *ILD* can develop prior to overt manifestations of polymyositis and dermatomyositis (PM-DM), present concurrently or occur subsequently during the course of the diseases.
  - Common patterns are NSIP followed by UIP and organizing pneumonia **(Figs 7.12 and 7.13)**.
  - Less common patterns include lymphoid interstitial pneumonia (LIP) and acute fibrinous and organizing pneumonia (AFOP).
  - Diffuse alveolar damage (DAD) and diffuse alveolar hemorrhage can also occur, especially in those with DM.
  - Most patients follow a chronic course, while an acute presentation/rapid deterioration is less common
  - Patients with DM are more likely to have acute and rapidly progressing forms of ILD including DAD.
  - In those with PM; presence of joint involvement, an older age, positive anti-Jo-1 antibodies, concomitant anti-Ro/SSA autoantibodies with antisynthetase antibody are reported as poor prognostic indicators for pulmonary involvement.
  - Long-term outcomes do not necessarily correlate with HRCT patterns.



Figs 7.12A to C: Dermatomyositis NSIP pattern ILD (arrow)



Figs 7.13A to C: Dermatomyositis with rapidly progressive NSIP and organizing pneumonia (arrow)

- *Pneumothorax and pneumomediastinum* can occur and occur as a consequence of rupture of alveoli/blebs and is especially seen in those with rapidly progressing ILD.
- *Pulmonary hypertension* is uncommon.
- *Drug induced DLD* due to cyclophosphamide and methotrexate may also be seen.

Cardiac causes though infrequent can also be responsible for dyspnea in PM-DM. These include congestive heart failure due to left ventricular dysfunction, coronary artery disease, arrhythmias and myocarditis/cardiomyopathy.

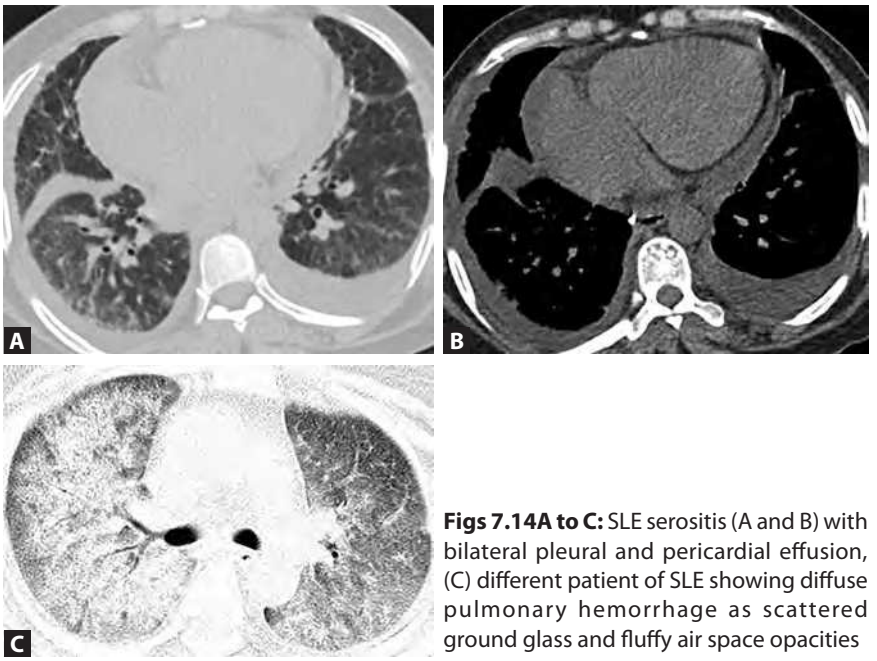
SYSTEMIC LUPUS ERYTHEMATOSUS

- Systemic autoimmune disease involving multiple organs in the body.

Pulmonary Manifestations

- Pulmonary involvement in >50% patients[2]
- Involvement of lungs, its vasculature, pleura and diaphragm.
  Acute:
  - Infections (common)
  - Pulmonary hemorrhage (common)
  - Pulmonary edema
  - Pulmonary embolism [common with antiphospholipid antibody (APLA) syndrome]
  - Lupus pneumonitis (less common).

  Subacute:
  - Pleural effusion, thickening and fibrosis (most common) **(Figs 7.14A to C)**
  - Interstitial lung disease (rare).



Figs 7.14A to C: SLE serositis (A and B) with bilateral pleural and pericardial effusion, (C) different patient of SLE showing diffuse pulmonary hemorrhage as scattered ground glass and fluffy air space opacities

Chronic:
- Pulmonary fibrosis (rare)
- Atypical findings such as cavitation and airway involvement as bronchiolitis (rare).

- *Infections:* These patients are more prone to infections, including community acquired pneumonia, atypical pneumonia and TB. Atypical pneumonia can present as diffuse lung disease.
- *Diffuse pulmonary hemorrhage (DPH)*—seen on imaging as bilateral ground glass opacities and consolidation.
- *Lupus pneumonitis*—seen in about 12% patients, and seen as multifocal consolidation, basal predominance
- *Pulmonary edema:* Renal involvement occurs in up to 50% patients with SLE. Hence, pulmonary edema occurring consequent to renal involvement is not uncommon.
- *Fibrosing interstitial lung disease:* UIP/NSIP pattern is seen in < 3% of cases. May progress to chronic fibrosis. Bronchiolitis is rare.
- *Diaphragmatic involvement and "shrinking lung syndrome"*
  - 25% patients have respiratory muscle dysfunction but "Shrinking lung syndrome", rare, 0.6%[2]
  - Consider when dyspnea is out of proportion to the severity of chest radiographic abnormalities
  - Progressive loss of lung volume with dyspnea and pleuritic chest pain
  - *Pulmonary function tests (PFTs):* Restrictive pattern.
  - *Chext X-ray (CXR):* Elevated hemidiaphragm (unilateral/bilateral)
  - Linear basal atelectasis, ill-defined juxta-diaphragmatic opacity.

MIXED CONNECTIVE TISSUE DISEASE

- Condition with combined features of SLE, RA, systemic sclerosis, polymyositis or dermatomyositis
- Biochemically high titer of antinuclear antibodies (ANCA), and ribonucleoproteins antibodies
- Male: Female ratio nearly 1:10
- Thoracic manifestations in 20–80% patients.

Interstitial Lung Disease

- Usually of NSIP pattern with ground glass opacities in the lower lobes predominantly with subpleural sparing
- Less commonly UIP characterized by interlobular septal thickening and honeycombing in lower lung fields with an apico-basal gradient
- LIP pattern less frequently seen.

Vascular Manifestations

- Pulmonary arterial hypertension is seen in up to 50% of patients and is cause of morbidity and mortality
- Vasculitis
- Thromboembolism.

Pleural Manifestations

- Pleural effusion
- Pleural thickening.

Other Findings

- Esophageal dysmotility
- Esophageal dilation.

SJÖGREN SYNDROME

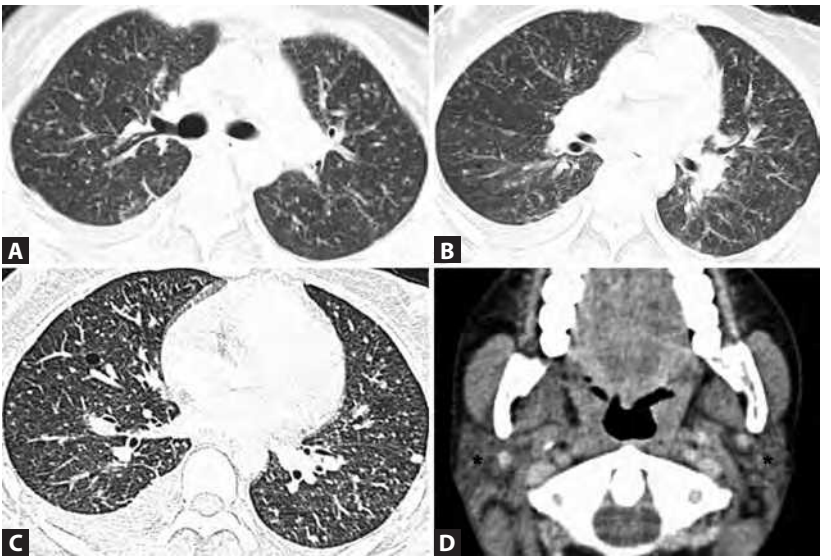
- Autoimmune disease characterized by dry eyes and dry mouth
- Histologically there is T lymphocyte infiltration of various glandular and extraglandular structures.

Interstitial Lung Disease

- ILD more common with primary Sjogren
- Most common pattern NSIP
- LIP less common but more specific **(Figs 7.15A to D)**
- Other patterns UIP, COP, amyloidosis.

Airways

- Bronchiectasis
- Bronchial wall thickening
- Bronchiolitis (follicular, constrictive)
- Air trapping.



Figs 7.15A to D: LIP pattern Sjögren syndrome. Lung window (A and B) and HR (C) images showing few random and centrilobular nodules with parenchymal cysts (arrow). (D) section through upper neck of the same patient showing enlarged heterogeneous bilateral parotids (*) with fatty infiltration

Pleural Manifestations

- Pleural effusion and thickening are uncommon.

Other Features

- Increased risk for lymphoma, usually B cell non-Hodgkin lymphoma
- Mostly arise from glandular tissue and less commonly from gut and lung.

The major findings of specific entities are summarized in **Table 7.1**.

Table 7.1: Summary of HRCT findings of various conditions of connective tissue diseases (CTD)

Type of CTD	*Most common HRCT pattern*	*Less common HRCT patterns*
Systemic sclerosis	NSIP	UIP, OP
MCTD	NSIP	UIP, OP
RA	UIP	SAD, NSIP, LIP, OP
PM-DM	NSIP/OP	DAD
SLE	Diffuse alveolar hemorrhage, serositis, infections	NSIP, DAD
Sjögren syndrome	NSIP, LIP, SAD	UIP, DAD

Abbreviations: NSIP, nonspecific interstitial pneumonia; UIP, usual interstitial pneumonia; OP, organizing pneumonia; LIP, lymphocytic interstitial pneumonia; DAD, diffuse alveolar damage; SAD, small airway disease.

Causes of Acute Deterioration in Patients of CTD Associated ILD

- Progression of ILD
- Fluid overload (more with renal involvement)
- Diffuse alveolar hemorrhage is more with vasculitis.

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8

Connective Tissue Disease-related Interstitial Lung Diseases: Clinical, Diagnostic and Management Issues

Ghan Shyam Pangtey, Uma Kumar

INTRODUCTION

- The connective tissue diseases (CTD) are the group of heterogeneous disorders also known as collagen vascular disease (CVD).
- The common CTD includes systemic sclerosis (scleroderma), rheumatoid arthritis, Sjögren syndrome, systemic lupus erythematosus, polymyositis-dermatomyositis, mixed connective tissue disease (MCTD) and undifferentiated CTD.
- These CTD may be associated with lung involvement in the form of pleuro-parenchymal or interstitial lung disease (ILD).
- In some instances, the ILD in CTD may develop months or years before the primary diagnosis of any rheumatologic illness or may develop many years after the fully differentiated autoimmune disease.
- The diagnosis of CTD-associated ILD (CTD-ILD) is challenging due to occurrence of a variety of interstitial pneumonia patterns.
- The CTD-ILD is most commonly seen in systemic sclerosis (SSc) and is least common in systemic lupus erythematosus (SLE).
- An acute presentation of ILD is more common in SLE and polymyositis/dermatomyositis (PM/DM).
- Infections and drug-induced pulmonary toxicity are also important cause to be considered in these patients, as these patients are often on immunosuppressive medications for management of underlying CTD.
- A multidisciplinary diagnostic approach should be considered for the optimum management of CTD-ILD, which should include an experienced radiologist, pulmonologist and a rheumatologist.

SCLERODERMA

- Systemic sclerosis (SSc) is an acquired autoimmune chronic connective tissue disease with vascular damage, and tissue fibrosis.
- Systemic sclerosis (SSc) has the highest association with ILD among the CTDs and up to half of SSc patient may develop ILD at some stage of their illness.

- The SSc disease can be divided into early stage (2–5 years), chronic stable stage and late fibrotic stage according to initial symptom duration.
- The initial 2–5 years from the development of first non-Raynaud's disease sign or symptom (such as joint pain and swelling, reflux disease, digital ulcer, etc.) is considered to be early stage of SSc, and during this phase, there may be progressive fibrosis of skin and internal organs, thus giving the window of opportunity to monitor and treat SSc in early stage with aggressive disease modifying agents before irreversible tissue fibrosis and organ failure sets in.

Clinical Features

- The SSc can be divided into three types based on pattern of skin involvement, as well as clinical and laboratory features:
  1. *Diffuse cutaneous SSc (DcSSc):* It is characterized by extensive skin induration, starting in the fingers and progressing proximally to limbs and the trunk. There is an association with early interstitial lung disease and acute renal involvement also.
  2. *Limited cutaneous SSc (LcSSc)*: In these patients, skin involvement is limited to the extremities and face, trunk remains unaffected. Raynaud's phenomenon may precede other manifestations of SSc by years. The presence of calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia, in some LcSSc patients, is termed the Calcinosis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia (*CREST) syndrome* **(Fig. 8.1)**
     Visceral organ involvement in LcSSc has slow progression, and pulmonary arterial hypertension (PAH), ILD, hypothyroidism, and primary biliary cirrhosis are usually late complications.
  3. *SSc sine scleroderma*: in some patients, Raynaud's phenomenon and other characteristic features occur in the absence of skin thickening. This syndrome is termed *SSc sine scleroderma*.



Fig. 8.1: Scleroderma patient with CREST syndrome



Fig. 8.2: Digital gangrene in a patient with scleroderma

- The SSc associated ILD is more common in women in the age group 30–55 years.
- Patients with SSc presents with skin thickening (sclerodactyly), Raynaud's disease, digital ulcer-gangrene **(Fig. 8.2)**, gastroesophageal reflux, chronic cough and dyspnea.
- Patients with SSc-ILD usually complains of exertional breathlessness (with/without non-productive cough) and on auscultation bilateral basilar inspiratory crackles are heard.
- When SSc presents with ILD as an initial manifestation, a careful clinical evaluation to highlight presence of mucocutaneous telangiectasias, Raynaud's phenomenon and abnormal nail fold capillaries should be carried out.

Laboratory Evaluation

- Anti-nuclear antibody (ANA) is positive in most patients with scleroderma and the most common pattern found is speckled pattern
- The autoantibody specific for SSc includes Anti-Scl-70, Anti-centromere, Anti-RNA topoisomerase III, and few SSc-ILD patients may not demonstrate any specific antibody.
- Anti-Scl-70 positivity is more commonly seen in dcSSc and is associated poor prognosis and development of ILD; the antibody anti-centromere and anti-Scl-70 are usually mutually exclusive and do not occur together in the same patient.
- Pulmonary function testing is suggestive of restrictive pattern along with decreased forced vital capacity (FVC) and decreased diffusion capacity for carbon monoxide (DL_{CO}) **(Table 8.1)**.
- Chest HRCT in SSc-ILD patients commonly presents with NSIP pattern with bilateral ground-glass opacities, most notable in lower lobes after long-

Table 8.1: Restrictive pattern of pulmonary function test with reduced DLCO in a case of scleroderma with ILD

Spirometry	*Predicted value*	*Observed value*	*% Predicted*
FVC (L)	2.75	1.52*	55%
FEV1 (L)	2.17	1.14*	52%
FEV1/FVC (%)	77	75	97
PEF 25-75 (L/S)	2.13	0.85	39
DIFFUSION			
DLCO UNC	20.32	10.16	50
DLCO CORR	20.32	9.73	47
Hb: 14.8			

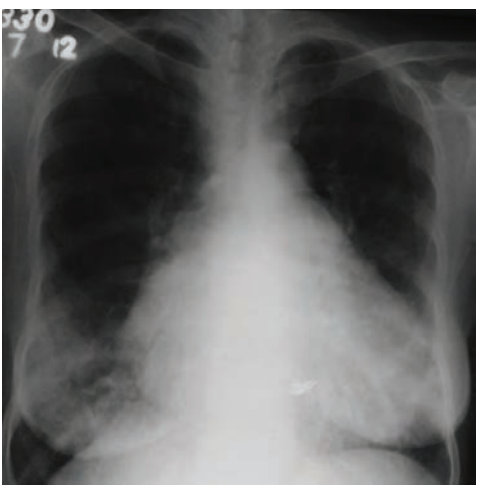


Fig. 8.3: Chest radiograph of systemic sclerosis with NSIP pattern of ILD

standing disease, lower lobe traction bronchiectasis and fibrotic changes may become manifest and may be seen in a chest radiograph also (**Fig. 8.3**).

- The second common pattern is UIP pattern, i.e. reticular abnormalities and honeycombing principally in a subpleural and basal distribution.

Diagnosis

- The diagnosis of systemic sclerosis is mainly a clinical diagnosis. It is defined by the 2013 ACR-EULAR criteria for diagnosis of SSc.
- Diagnosis of SSc ILD is made on the basis of clinical and radiological features only, the lung biopsies are not routinely recommended for evaluation of SSc-ILD.
- Open or thoracoscopic lung biopsy is required for confirmation only in the event of unusual or atypical clinical and/or radiological features.

Histopathological evaluation of lungs in SSc-associated ILD may unveil any one of the following patterns:

1. Fibrotic NSIP (interstitial fibrosis and modest inflammatory cell infiltrates)
2. Cellular NSIP (interstitial lymphocytic infiltrates without fibrosis) or
3. UIP pattern (regions of dense fibrosis and fibroblastic foci).

- Non-NSIP, non-UIP patterns are quite rare in SSc-ILD. The acute/subacute presentation of DAD is also rare.
- Goh et al. have proposed a staging system based on assessment of disease severity. Based on this method, extent of disease on CT and/or lung function can help in identifying patients with worse survival. Classification of patients into extensive or limited disease depends upon CT extent above or below 20% respectively.
- If this categorization is uncertain, then the distinction amongst extensive and limited disease can be made by assessment of FVC lower or more than 70% predicted respectively.
- In concurrence to this system, the possibility of future progression and adverse outcome was found to be dependent upon radiological disease progression, functional impairment and decline, early disease onset and a relatively early onset of pulmonary symptoms from the initial diagnosis.

Treatment

- Early systemic sclerosis associated ILD (<2 years) with poor prognostic markers such as anti-Scl 70 positivity, significant lung involvement on HRCT chest or deranged PFT (FVC<70%) should be treated with immunosuppressive drugs.
- The evidence at present is only modest improvement pulmonary function and skin thickening and the drug most commonly recommended is injectable cyclophosphamide (600 mg/m^2) for 6–12 months followed by maintenance therapy with mycophenolate or azathioprine for next 3–4 years.
- The intravenous cyclophosphamide is preferred over oral cyclophosphamide (2 mg/kg) due to lower incidence of bone marrow toxicity, severe infections and gonadal failure.
- Mycophenolate mofetil (MMF), being less toxic, is a promising alternative to azathioprine and cyclophosphamide.
- Low dose oral corticosteroids (prednisolone 10 mg OD) is preferred over high dose corticosteroids, i.e. prednisolone > 10 mg should be avoided because of risk of scleroderma renal crisis.
- Recurrent microaspiration due to chronic gastroesophageal reflux disease is significant contributors to advancement of lung disease in SSc-ILD and there is some evidence that patients with SSc-ILD, may benefit with anti-reflux medications (proton pump inhibitors plus domperidone).

MIXED CONNECTIVE TISSUE DISEASE

- This heterogeneous group of patients has overlapping phenotypic features of SLE, scleroderma, rheumatoid arthritis and inflammatory muscle disease.

- The characteristic hallmark of mixed connective tissue disease (MCTD) is very high titers of U1-Ribonucleoprotein (U1-RNP) antibodies.
- The pleuropulmonary complications, especially ILD can be present in 20–80% patients.
- Overall prognosis is better as these patients rarely have renal involvement as compared to SLE.

Clinical Presentation

- The disease is more common in female and the age of onset is 4th to 5th decade.
- The patient may present with any of the symptoms of primary disease such as Raynaud's phenomenon, sclerodactyly, non-deforming arthritis and proximal muscle weakness.
- Raynaud's phenomenon and proximal muscle weakness are seen in > 2/3rd of patients with MCTD.
- Severe CNS and renal involvement are rare.
- Pulmonary involvement in the form pulmonary hypertension and ILD are common.
- Most common cause of mortality in MCTD is due to severe progressive pulmonary hypertension.

Diagnosis

The MCTD should be suspected in any patient with combination of features of scleroderma with arthritis and proximal muscle weakness due polymyositis and with very high U1RNP antibodies titers (>1000 IU).

Laboratory Investigations

- Pulmonary involvement is very common in MCTD up to 20–80% patients have some pleural or parenchymal disease.
- Interstitial lung disease on radiology has been found in 20–60% patients in different series and NSIP/or UIP pattern are the most common presentation.
- Diffuse alveolar hemorrhage (DAH) may develop in very few unfortunate MCTD patients.
- Pulmonary function test reveals restrictive pattern and reduced DLCO suggestive of ILD and this type of physiological finding can be seen in up to 80% of MCTD patients.
- Echocardiography with Doppler and sometimes right ventricular catheterization may be required to confirm pulmonary hypertension.

Treatment

- The management depends upon specific abnormality present in individual patients.
- ILD associated with MCTD can be treated with immunosuppressive drugs (cyclophosphamide and azathioprine) similar to scleroderma patient, as the type and pattern of ILD are similar.

- Patients with moderate-severe pulmonary hypertension should be treated with calcium-channel blockers (nefedipine retard) and with severe cases, intravenous prostacyclin or oral phosphodiesterase inhibitors can be used.

RHEUMATOID ARTHRITIS

- Rheumatoid arthritis (RA) is a chronic inflammatory disease with predominant involvement of symmetrical small and large joints of body **(Fig. 8.4)**. The etiology is not known.
- The inflammatory arthritis may lead to joint erosion, damage and disability with progressive inflammation and disease.
- Extra-articular manifestations including fatigue, polyneuropathy, Vasculitis, reticuloendothelial, pericardial and lung involvement can also occur in RA.
- Pleuropulmonary abnormalities are one of the common extra-articular manifestations of rheumatoid arthritis (RA).
- In some studies, RA-associated ILD has been reported to occur in 20–40% of the patient either after few year of disease, and in few cases, before RA diagnosis.
- The respiratory involvement due to opportunistic infections and drugs causing hypersensitivity pneumonitis are also commonly seen in RA.
- One of the significant risk factor for ILD is smoking.
- Usual interstitial pneumonia (UIP) is the most frequent histopathological pattern in RA followed by NSIP (fibrotic > cellular).
- Various studies have highlighted a worse survival for UIP pattern in rheumatoid arthritis associated ILD (RA-ILD), which was not much different from that of IPF.
- A rare presentation of RA-ILD may be in form of diffuse alveolar damage (DAD) in a normal lung or in a previously undiagnosed ILD. This pattern is recognized on HRCT as widespread ground-glass opacities associated with



Fig. 8.4: Rheumatoid arthritis with active synovitis at the wrist, metacarpophalangeal and PIP joints

areas of dependent consolidation, very similar to that seen in opportunistic infections or in congestive heart failure.
- Organizing pneumonia with or without ILD, DIP or LIP are some of the other histopathological patterns seen infrequently in RA-ILD.

Clinical Presentation

- Rheumatoid arthritis (RA) is characterized by symmetric, peripheral polyarthritis.
- It often results in joint damage and physical disability. RA affects approximately 1% of the adult population worldwide with female-to-male ratio of 2–3:1.
- The patients of RA usually present between 25 to 55 years of age and often complain of early morning stiffness lasting for more than one hour and decreasing with physical activity. Other symptoms include fatigue, low-grade fever, muscle and joint pain, and joint stiffness. Joints appear swollen, red in color and are tender.
- The earliest involved joints are typically the small joints of hand and feet, MCP, PIP and MTP.
- Being a systematic disease, it also results in extra-articular manifestations including lung involvement, pericarditis, peripheral neuropathy, vasculitis and hematologic abnormalities.

Diagnosis

- In the absence of a single confirmatory test for the definitive diagnosis of RA; clinical history, physical examination and laboratory tests are used to diagnose RA.
- The 1987 ACR criteria for diagnosis of RA have been recently replaced by the new ACR-EULAR 2010 Rheumatoid arthritis classification criteria, which is increasingly being used for classifying the RA patients, the new criteria is good for classifying the early inflammatory arthritis into rheumatoid arthritis and non-rheumatoid arthritis.

Laboratory Features

- Serum Rheumatoid factor (RF) may be found positive in 75-80% of RA patients. It is not very specific for RA diagnosis and may be found in other autoimmune disease like Primary Sjögren syndrome, SLE, as well as in non-autoimmune diseases such as chronic hepatitis B and hepatitis C infections.
- RF may also be found positive in up to 5% of normal populations although in low titers.
- Anti-citrullinated cyclic peptide antibody has similar sensitivity to RF but with more specificity (up to 90–95%) and may be useful in diagnosis of early inflammatory arthritis.
- Inflammatory markers ESR and CRP are usually raised in active RA.
- Radiograph of hands may show osteopenia around the joints in early active disease and erosions may be present in long-standing RA.

- Chest radiograph is usually adviced before starting disease-modifying antirheumatic drugs (DMARDs) to rule out pulmonary tuberculosis, but it may show evidence of early asymptomatic ILD in few RA patients.

Treatment

- There is paucity of randomized controlled trials for management of RA-ILD and therefore the treatment usually is empirical.
- Organizing pneumonia, due to its possible reversibility, may be treated with high dose oral or intravenous methylprednisolone.
- Immunosuppressive medications (cyclophosphamide, azathioprine or mycophenolate mofetil) may be used as steroid-sparing agents or may be utilized in patients not responding to steroids alone.
- Oral prednisolone along with azathioprine or mycophenolate may be used in UIP/fibrotic NSIP patterns, although this may require further validation with help of appropriate studies.
- Though there is no evidence as yet, intravenous cyclophosphamide may be a rational choice for rapidly progressive fibrotic NSIP or UIP.
- As DAD presenting as acute respiratory failure portends poor survival, high-dose intravenous methylprednisolone with or without intravenous cyclophosphamide may be utilized in form of an aggressive therapy.
- Salvage rituximab therapy may be reserved for severe and/or unresponsive forms of disease.

ANTIRHEUMATIC DRUG-INDUCED INTERSTITIAL LUNG DISEASE

- In literature almost all major anti-rheumatic drugs used in RA treatment has been implicated in lung injury and fibrosis.
- It becomes difficult at times to differentiate the clinical and radiological findings is due to primarily DMARDs or due to primary disease RA associated ILD.
- The only clues may be the "onset of ILD" and its improvement in relation to the stoppage of the culprit drug.
- If ILD was present before the starting of any specific DMARDs, then that specific anti-rheumatic drug will be unlikely to be the cause of ILD and can be tried to at least treat RA.
- However, delayed reactions and progression of ILD despite stoppage of drugs is also a rare possibility.
- Bronchoalveolar lavage (BAL) and microbiological sampling may be required to differentiate drug induced ILD from infectious causes of pulmonary involvement.
- Although, histopathology samples (obtained after trans-bronchial or surgical lung biopsy) may be helpful in identifying infectious causes, they are not useful to differentiate between drugs induced ILD and RA-ILD.
- Methotrexate, the anchor drug for RA may lead to lung injury in 0.3–11.6% of the patients. Frequently, methotrexate leads to a subacute presentation, within a few weeks of its use, with fever, cough and breathlessness.

- Radiological assessment in such a clinical scenario portrays non-specific findings such as parenchymal infiltrates.
- Whereas, BAL evaluation may reveal lymphocytosis with or without eosinophilia (may be associated with peripheral eosinophilia), histopathology may highlight a cellular interstitial pneumonitis along with granulomas.

Treatment

- This presentation may be managed with systemic corticosteroid treatment and stoppage of methotrexate, which usually has a good response.
- At the other extreme, a rapidly progressive, life-threatening presentation may also occur due to methotrexate. Such an acute presentation may be difficult to differentiate from RA associated acute lung injury, as the histopathological pattern picked up in both is usually diffuse alveolar damage (DAD) pattern. Presentation in the form of OP and fibrotic NSIP has also been described in literature.
- Methotrexate-induced lung injury usually carries a good prognosis. However, patients with pre-existing ILD may have continued progression of interstitial fibrosis. For this reason, American College of Rheumatology (ACR) recommends to avoid using methotrexate in RA-ILD patients.
- Leflunomide, either used alone or along with methotrexate, has been associated with acute lung injury/DAD-like pattern in about 1% of the cases.
- The pulmonary affliction is more common within first 20 weeks of treatment with leflunomide, occurrence of secondary alveolar proteinosis and vasculitis associated with acute pulmonary hemorrhage has also been reported. There is rapid improvement on stopping of the drug along with use of high-dose corticosteroids.
- Sulfasalazine-induced pulmonary toxicity usually presents as cough and/or fever. The pattern observed is mostly pulmonary eosinophilia. Organizing pneumonia and UIP with/without fibrosis are the other forms, which have been reported.
- Most of the patients improve on discontinuation of the drug with or without a short course of corticosteroid therapy. However, deaths have been reported secondary to respiratory failure due to non-stoppage of sulfasalazine therapy.
- Gold salts which were used in past as anti-rheumatic agents were associated pulmonary reactions in up to 1% of RA patients with histopathological patterns of DAD, OP and fibrotic NSIP.
- The anti-rheumatic biological agents have also been reported to be associated with lung injury. The TNF antagonists may lead to rapid development of respiratory failure especially in older patients, and in those with pre-existing ILD. This class of drugs may also lead to non-infectious granulomatous lung disease.
- Rituximab-associated ILD has been reported to be around < 0.03% patients in a study done on cancer patients. It may present as macronodular organizing pneumonia, which may occur weeks to months after the last

infusion. Most of these patients showed improvement on discontinuation of therapy with or without corticosteroid therapy.

POLYMYOSITIS-DERMATOMYOSITIS

- Polymyositis and dermatomyositis are group of acquired autoimmune inflammatory muscle disease associated with proximal muscle weakness.
- Proximal muscle weakness may accompany or preceded by skin involvement in patients of dermatomyositis, the skin involvement presents in the form of heliopropes rash over eyelid, Gottron's papule **(Fig. 8.5)** over knuckles and erythematous rash over body.
- Polymyositis/dermatomyositis (PM/DM) patients may develop ILD in up to 20–50% of cases. It may present before or after the development of muscle weakness.
- A high index of suspicion is often required to pick up subtle presentations of ILD in presence of PM/DM. The patients may be asymptomatic or may present either acutely or insidiously.
- The most common presentation of PM/DM-ILD is in the form of an insidious onset and the radiological and histopathological findings are consistent with fibrotic NSIP, organizing pneumonia and UIP.
- The acute presentation includes a rapid development of breathlessness along with pulmonary infiltrates within a span of weeks to months. The prognosis is not good and this often leads to respiratory failure. This form of presentation is associated with DAD type of histopathological pattern, which may have some overlap with OP with or without NSIP.
- Bronchoalveolar lavage along with a transbronchial biopsy may be required to rule out an opportunistic infection considering that most of these patients are on immunosuppressive medicine.
- However, lung biopsy is usually not favored as part of the usual work-up of PM/DM-ILD.



Fig. 8.5: Violaceous papule over knuckles (Gottron's) papule of patient with dermatomyositis

- Interstitial lung disease has been identified as part of *antisynthetase syndrome*, which also includes polymyositis, arthritis, Raynaud's phenomenon, fever, 'mechanic's hands' (cracking of lateral aspects of fingers) and antisynthetase antibodies.

Laboratory Tests

- Antisynthetase antibodies include anti-Jo 1 (most common), anti PL-7, anti PL-12, anti-OJ, anti-EJ, anti-KS and anti-Wa.
- Markers predicting PM/DM-ILD have been identified to be KL-6, SP-D and CK-19.
- Lung function tests in pure ILD are suggestive of restrictive pattern along with reduced DL_{CO}, which may be complicated to interpret, if there is accompanying respiratory muscle weakness.
- Reduced maximal inspiratory pressures are characteristic of respiratory muscle involvement.
- Since risk of associated malignancy (including lung cancer) is high in dermatomyositis and polymyositis, the screening for occult neoplasm with routine chest and abdominal imaging is commonly advised.

Treatment

- Due to the paucity of enough randomized controlled trials, no definite recommendations regarding management of PM/DM-ILD are in place.
- High dose oral prednisolone is used in treating myositis associated ILD, high dose intravenous methylprednisolone is utilized for rapidly progressive disease.
- Intravenous cyclophosphamide may be used along with methylprednisolone, if severity of disease is more.
- In patients with significant disease, it may be worthwhile to use immunosuppressant's (azathioprine, mycophenolate) for their steroid sparing effect.
- Rituximab (anti-CD-20) therapy has been utilized in patients who fail traditional immunosuppression. However, further studies are required to validate its efficacy.
- Respiratory muscle involvement has been targeted with immunosuppressive therapies with/without intravenous immunoglobulins.

SYSTEMIC LUPUS ERYTHEMATOSUS

- Systemic lupus erythematosus (SLE) is a multisystem disease associated with antibodies against nuclear antigen (ANA)
- The mechanism of tissue injury is appeared to be secondary to immune complex formation and complement fixation.
- Acute respiratory illness may develop suddenly in patients with SLE but the most common cause in these situations is secondary to infection, especially if it is associated with fever, productive cough and new pulmonary infiltrates.

- In rare situations, patients with SLE may present with sudden onset diffuse alveolar hemorrhage (DAH) leading to sudden onset dyspnea cough and hemoptysis.
- Diffuse alveolar hemorrhage usually appears in the setting of known SLE and in most instances, it is the rule rather than exception. In 60–90% of DAH, there may be involvement of kidney also (glomerulonephritis)
- The treatment of DAH in SLE should be done with high dose methylprednisolone along with cyclophosphamide or azathioprine. Plasmapheresis and immunoglobulin therapy has been used considering SLE an immune complex disease but it is not proven to have any benefit.
- In a recently published study, 38% of patient were found to have finding suggestive of ILD on high resolution CT in spite of normal chest radiographs.
- The various forms of ILD from UIP, LIP, NSIP and bronchiolitis obliterans all have been documented in SLE.
- In some patients with UIP pattern of ILD, there is past history of acute lupus pneumonitis.
- Pulmonary function test reveals restrictive pattern with reduced DLCO.
- Response to immunosuppressant and steroid is more common in patients with NSIP or Organizing pneumonia pattern of ILD instead of UIP pattern.

SJÖGREN'S SYNDROME

- *Sjögren syndrome* is characterized by lymphocytic infiltration of the exocrine organs and has an autoimmune basis. Sjögren syndrome may present as primary disease or as secondary Sjögren syndrome when it is not associated with any other rheumatic condition.
- The prevalence of Sjögren syndrome is 0.5–2%, and it is more common in females.

Clinical Manifestations

- Sjögren syndrome typically presents as dry eyes, dry mouth and swollen parotid glands.
- Up to one-third of patients may have extraglandular manifestations in the form of joint, skin (vasculitis), pulmonary (ILD), neurologic (neuropathy), hematological (lymphoma), or renal (nephritis) involvement.
- Respiratory involvement in the form ILD is seen in up to 25% of Sjögren's syndrome (SjS).
- However, SjS associated ILD (SjS-ILD) is mostly mild, and clinically significant disease is seen in only a minority of patients.

Laboratory and Histopathology

- Lung function tests may reveal restrictive, obstructive or mixed patterns due to frequent airway abnormalities in SjS-ILD.

- Restrictive lung function tests and decreased DL_{CO} is seen in 17–37.5% of SjS patients.
- Bronchoalveolar lavage is often suggestive of subclinical lymphocytic and neutrophilic alveolitis, which is associated with progression of disease.
- The most common histopathological pattern observed in SjS-ILD is NSIP, although other patterns such as OP, LIP or UIP are also observed.
- The higher frequency of LIP in previous studies could be attributable to either reclassification of LIP into cellular NSIP in some cases or due to low-grade lymphoma in some of the others.
- The HRCT findings are mostly reflective of NSIP or LIP pattern.
- Researchers have compared HRCT findings of LIP and pulmonary lymphoma. Whereas findings such as consolidation, large nodules > 10 mm or pleural effusion were associated with lymphoma.
- LIP pattern more commonly associated with cysts as compared to lymphoma. Centrilobular and subpleural nodules also occasionally indicative of lymphocytic bronchiolitis.
- Ground-glass attenuation, small nodules and hilar-mediastinal lymphadenopathy were associated with both lymphoma and LIP.

Diagnosis

- The Revised International American-European classification criteria 2002 are commonly used for diagnosis of SjS.
- Surgical biopsy is not indicated in radiological NSIP patterns, it should always be considered in non-NSIP patterns, especially with suspicion of lymphoma.
- It may be sometimes difficult to differentiate malignant lymphoma from a LIP pattern, without a surgical biopsy.

Treatment

- There are no clinically controlled trials to guide therapy. Corticosteroids are used as first line therapy, with immunosuppressive therapy reserved as steroid sparing agents and for patients not responding to steroids.
- Prognosis is usually good, however, a small proportion of patients with UIP pattern of ILD may show progressive worsening.

CLINICAL APPROACH

- Clinical approach includes looking for extrathoracic features of CTD in any patient diagnosed with ILD, each patient should be subjected to autoimmune markers (autoantibodies), radiology (HRCT) and sometimes histopathology to detect an underlying CTD**.**
- Younger age, female sex, Raynaud's phenomenon, symmetric joint swelling or stiffness, synovitis, digital edema, 'mechanic hands', palmar telangiectasia, Gottron's papule, higher titers of antinuclear antibody (ANA), elevated muscle enzymes are the most common features of underlying CTD.

- A thoracic high resolution computed tomography (HRCT) imaging depicts the extent, distribution and pattern of lesions in interstitial lung disease along with extraparenchymal abnormalities.
- A lung biopsy in CTD-ILD is often not opted for, as it does not influence treatment decisions in presence of typical HRCT findings strongly suggestive of underlying CTD.
- It should be performed in cases atypical presentation and to rule out more treatable cause like hypersensitivity pneumonitis or drug-induced injury.
- When performed, histopathological features suggestive of underlying CTD are presence of multi-compartment involvement, lymphoid aggregates, germinal centers, increased perivascular collagen, lymphoplasmacytic inflammation, eosinophil infiltration or pleural inflammation.
- CTD-ILD can have UIP or non-UIP pattern on histology or HRCT. Although UIP pattern have poor prognosis compared to non-UIP pattern but as compared to UIP-IPF (nonconnective tissue disease), the subset of CTD-UIP patients have slightly better prognosis.
- At present time with the available evidence, the CTD-ILD patient with non-UIP pattern and significant disease should be treated with immunosuppressive treatment (steroid and cytotoxic drugs), while patients with UIP pattern should be enrolled in experimental clinical trials and lung transplantation protocol similar to IPF.

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9

Vasculitis-related Interstitial Lung Diseases: Imaging

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TERMINOLOGY

- Refers to inflammatory disorders (noninfectious) that primarily affect the pulmonary blood vessels ranging from the alveolar capillaries to the main pulmonary artery.
- Multisystem disorders
- Other systems involved include—kidney (glomerulonephritis), paranasal sinuses (rhinosinusitis)
- *Pathology:* Inflammation in the walls of the vessels leading to its destruction and also necrosis of the adjoining lung. The inflammation may be acute or chronic, cellular or acellular.
- The inflammation in the wall may be predominantly granulomatous, eosinophilic, neutrophilic or lymphoplasmacytic.

CLASSIFICATION

- Heterogeneous group of disorders
- This chapter focuses on those entities which can present as diffuse lung diseases.
- *Small vessel vasculitis*
  - Antineutrophil cytoplasmic antibody (ANCA) associated vasculitis, which includes microscopic polyangiitis, granulomatosis with polyangiitis, formerly called Wegener's granulomatosis and Eosinophilic granulomatosis with polyangiitis (EGPA), formerly called Churg-Strauss syndrome
- *Vasculitis associated with collagen vascular disease*
  - Systemic lupus erythematosus (SLE)
  - Rheumatoid arthritis

- *Miscellaneous*
  - Goodpasture's syndrome
  - Behçet disease
  - Drug related.

For a comprehensive classification, refer to Chapter 10.

IMAGING MODALITIES

Chest Radiograph

- Primary modality for diagnosis and follow-up
- The International Labor Office (ILO), Geneva–an agency of the United Nations has published guidelines on the Classification of chest X-ray (CXRs) in those with occupational inhaled dust exposures
- These have been published since 1950, with latest update being in 2011
- The purpose being to provide a standardized format for interpreting and describing CXRs in pneumoconiosis and reduce interobserver variability
- The published guidelines comprise of a series of standard chest radiographs (22 in number) with detailed footnotes to codify the spectrum of abnormalities seen.

Thin-section Computed Tomography

- Despite the use of CT in most DLDs as the primary modality and its obvious superiority to CXRs; it has not replaced CXRs in evaluation of occupational lung disease.
- This is due to additional cost and radiation burden involved in epidemiological surveillance.
- It is used as a secondary modality in those with clinicoradiographic discordance.
- A low dose CT (LDCT) with exposures of <1.5 mSv is now advocated for screening for carcinoma lung in those with occupational exposure; and also for screening for pneumoconiosis.
- A standard classification system for scoring extent of disease on CT is also now available.
- It is called the "International classification of occupational and environmental respiratory diseases (ICOERD)"
- Similar to the ILO system for chest radiograph reporting, this is a semi-quantitative reporting system.

MAGNETIC RESONANCE IMAGING

It may aid in differentiation of progressive massive fibrosis (PMF) from carcinoma lung.

18^F FDG PET-CT

The FDG PET shows areas of increased uptake in the vessel wall and pulmonary parenchyma due to active vasculitis, which is difficult to differentiate from malignant lesion uptake.

Patterns of Pulmonary Involvement

Pulmonary involvement may be in various forms:

- Pulmonary arterial aneurysms
- Pulmonary arterial stenosis
- Diffuse focal pulmonary hemorrhage
- Focal lung opacities (nodules/consolidation)
- Bronchiolitis
- Pulmonary arterial hypertension

This chapter focusses on pulmonary parenchymal involvement by various vasculitic disorders.

Diffuse Alveolar Hemorrhage

- Diffuse alveolar hemorrage (DAH) may result from pulmonary capillaritis or may occur without associated capillaritis (*See* Chapter 10)
- Imaging findings are similar regardless of etiology
- *Chest radiograph:* Bilateral diffuse air space opacities with relative sparing of the apices.
- *High-resolution computed tomography (HRCT)* acute phase—bilateral ground glass opacities/consolidation, lobular appearance more in dependent areas **(Fig. 9.1)**
  Intermediate phase: After two or three days the intralobular and interlobular septal thickening superimposes on the ground glass opacities giving "crazy-paving" appearance. The septal thickening is smooth.
  Chronic phase: Ground glass opacities resolve. Recurrent episodes result in interstitial fibrosis.
  In the phase between the acute episodes, ill-defined, uniformly distributed centrilobular nodules may persist.

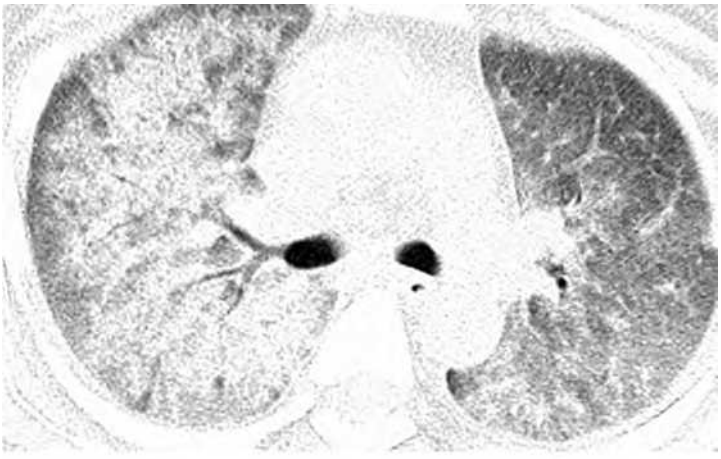
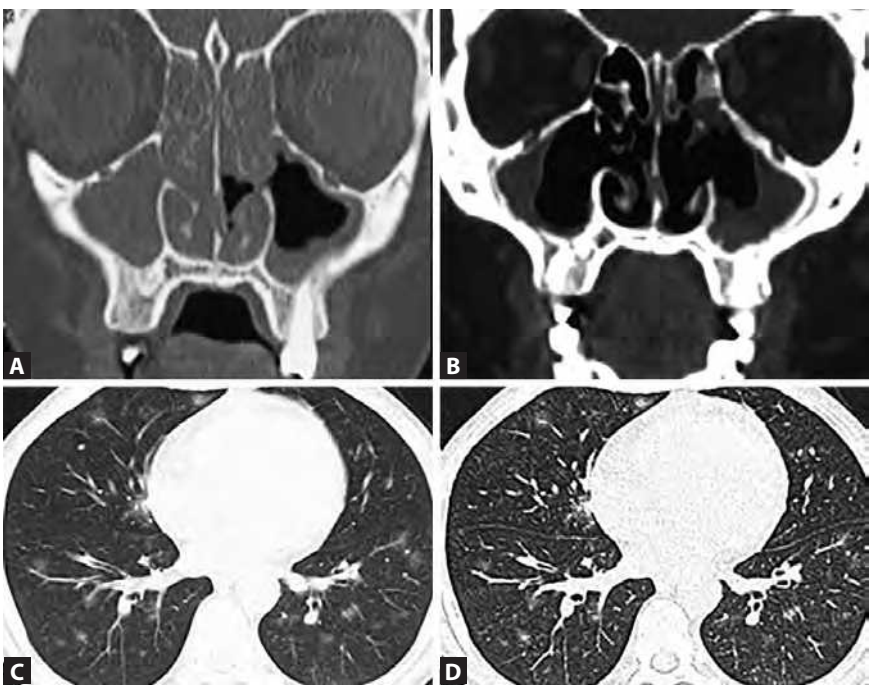


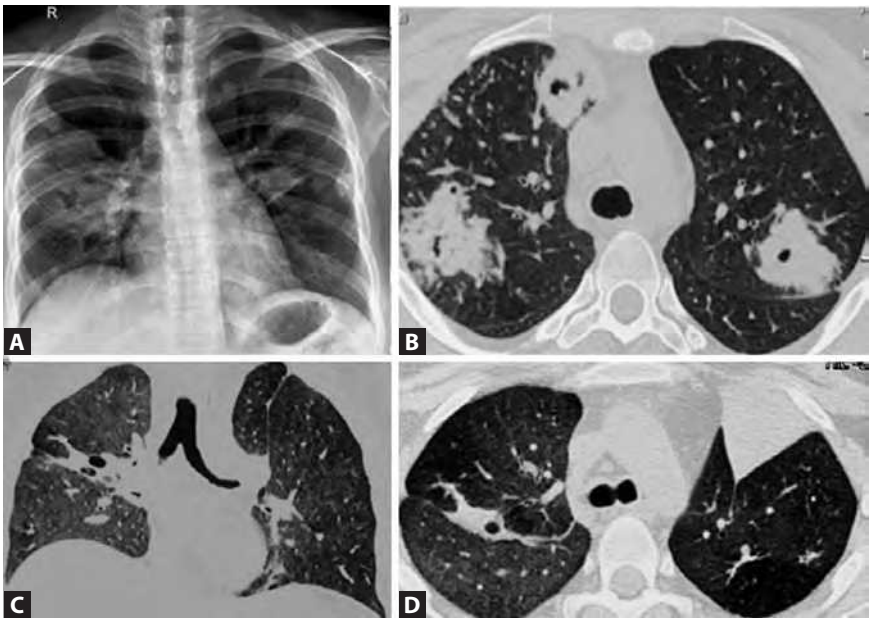
Fig. 9.1: Diffuse alveolar hemorrhage (DAH) in a patient of SLE. HRCT axial image showing bilateral ground-glass opacities (right more than left), which appear denser in the dependent parts—acute DAH

GRANULOMATOSIS WITH POLYANGIITIS

- *Previously called*: Wegener *granulomatosis*.
- *Pathology:* Necrotizing, small vessel granulomatous vasculitis.
- *Clinical manifestations:* Classical triad of upper respiratory tract involvement, lower respiratory tract lesions and glomerulonephritis (*See* Chapter 10) **(Figs 9.2A and B)**.
- *Age of onset*: 3rd to 5th decades
- Cytoplasmic ANCA is positive in more than 90% patients
- *Pulmonary manifestations:*
  - Pulmonary nodules/masses (with or without cavitation), most common finding, seen in up to 90% patients, frequently multiple and bilateral **(Figs 9.2C and D)**
    Distribution is often subpleural or peribronchovascular.
  - *Bronchial wall thickening*: Seen in segmental or subsegmental bronchi in about 70% cases, appearing as bronchovascular thickening on CXR.
  - Large airway involvement, in up to 30% patients with subglottic trachea being the most frequent site of involvement, with wall thickening initially and subsequent stenosis.



Figs 9.2A to D: Sinonasal and pulmonary involvement of granulomatosis with polyangiitis (GPA). Coronal reconstruction CT image of paranasal sinuses (A) showing opacification of the bilateral ethmoid and right maxillary sinuses with mucosal thickening in the left maxillary sinus, thinning of ethmoid septae. On follow-up imaging (B) on treatment there is partial reduction in the maxillary and ethmoid thickening; (C and D) Pulmonary nodules in GPA. Ground-glass nodules scattered in both lungs on lung window and HR window in early changes of vasculitis



Figs 9.3A to D: Consolidation with cavitation in GPA. (A) Chest radiograph showing multiple areas of consolidation with cavity formation, better demonstrated on CT (B), showing marked resolution in the follow-up CT images (C and D) which show fibrotic areas and decreased size of consolidation and cavitation areas

- Consolidation, ground-glass opacification (GGO) (in 25–50%): Multifocal with random/patchy distribution. These represent areas of DAH resulting from necrotizing capillaritis **(Figs 9.3A to D)**
- Centrilobular nodules, tree-in-bud opacities (in about 10% cases)—likely due to bronchiolar inflammation, usually occur in association with other findings.

- *Follow-up imaging:* On treatment, pulmonary lesions show good response with little or no residual fibrosis.
  - Lesions which resolve without any residue—GGO lesions, about 50% of nodules/masses.
  - Some scarring seen on resolution of remaining focal lesions, as well as airway involvement.
- *Differential diagnosis:* Necrotizing pneumonia with cavitation, secondary infection in pre-existing cavities, cavitary metastases.
  - Secondary infection can occur within the cavitary nodules of vasculitis; and this complication is difficult to differentiate from disease progression. This complication results in increase of the size of pre-existing consolidation/mass with areas of enlarging necrosis or new air-fluid levels. Rarely pneumothorax may also result **(Fig. 9.4)**.

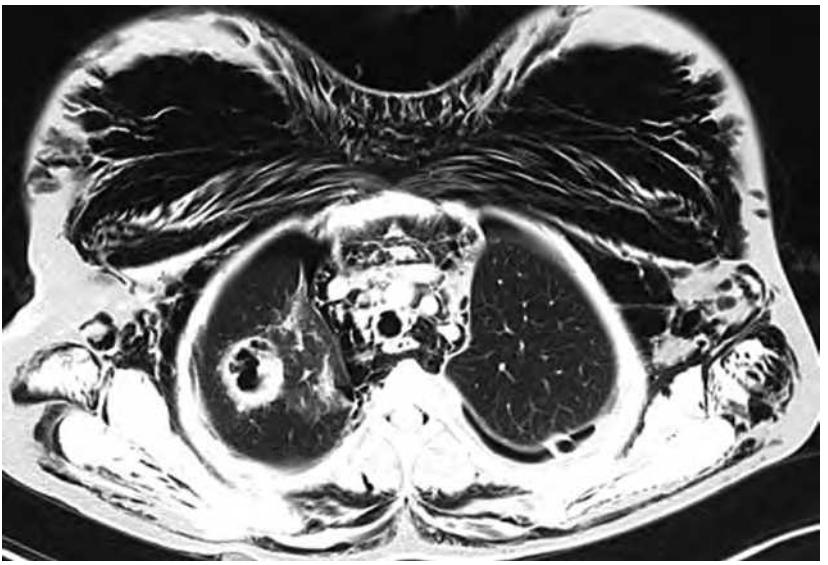
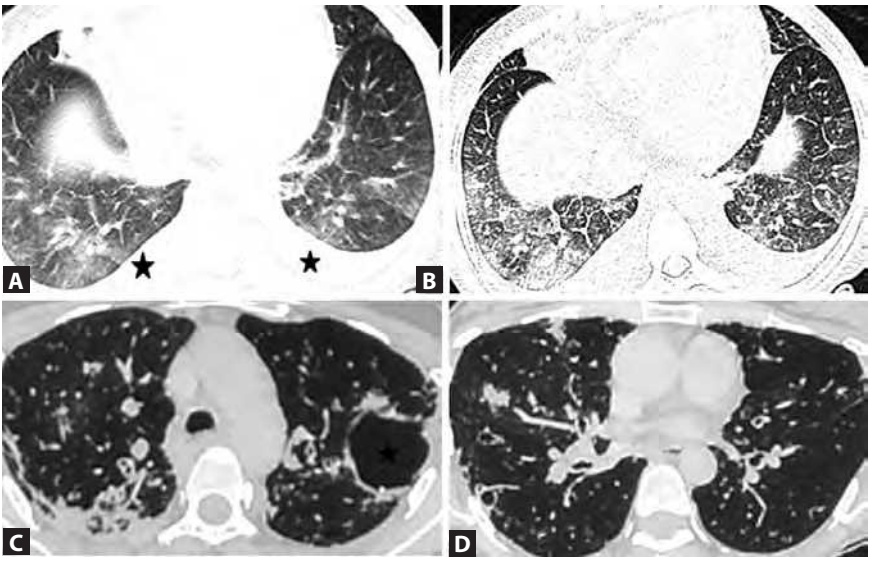


Fig. 9.4: Extensive subcutaneous emphysema and bilateral pneumothorax in a patient of GPA with cavitating nodule. Lung window CT, showing extensive subcutaneous emphysema, bilateral pneumothorax and pneumomediastinum as a complication

EOSINOPHILIC GRANULOMATOSIS WITH POLYANGIITIS

- Previously called Churg-Strauss syndrome (CSS)
- *Clinical presentation* distinct from ANCA associated granulomatous vasculitis and microscopic polyangiitis (*See* Chapter 10)
  - Eosinophilic granulomatosis with polyangiitis (EGPA) presents with asthma and eosinophilia
  - Cardiac involvement is frequently seen. Peripheral nerve involvement is also seen
  - Pulmonary hemorrhage is less common, as is glomerulonephritis.
- *Pathology:* Eosinophilic infiltration of bronchial walls (bronchitis) and of alveolar spaces (pneumonia) along with other macrophages. Necrotizing vasculitis involving arterioles and venules with eosinophilic infiltration of vessel walls with or without granuloma formation.
- *Imaging findings:* Pulmonary involvement seen in up to 50% patients[1]
  - *Parenchymal opacities*: GGO which are bilateral, patchy in distribution with no zonal predilection. These opacities are mostly transient **(Figs 9.5A and B)**
  - Nodules (without cavitation) or reticular opacities may also be seen.
  - *Small airway diseases pattern*: Centrilobular nodules, bronchial wall thickening, bronchial dilatation, GGO and mosaic attenuation **(Figs 9.5C and D)**
  - *Associated findings*: Pleural effusion, cardiac wall thickening.

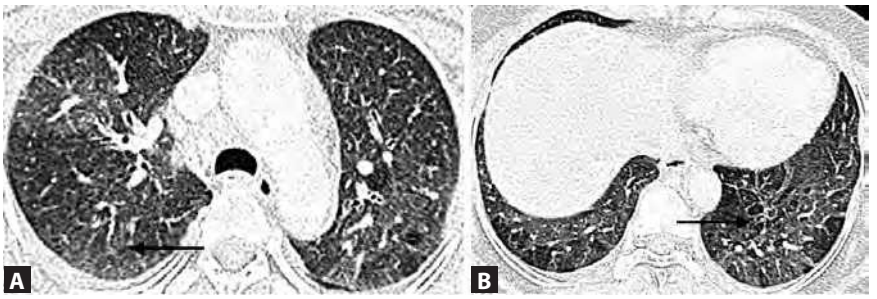


Figs 9.5A to D: Pulmonary parenchymal changes in EGPA. Lung window (A) and HR window (B) CT image showing patchy areas of ground glass opacities, asymmetric in distribution. Also to be noted is bilateral pleural effusion(*) in (A). (C and D) Different patient with EGPA showing small airway disease pattern with bronchial wall thickening, bronchiectasis and nodule formation. Note a thin walled cavity in left upper lobe in (C) (*)

- *Differential diagnosis:* Primarily other eosinophilic lung diseases—simple pulmonary eosinophilia, chronic eosinophilic pneumonia. These can be difficult to differentiate even histologically
  - Due to the small airway diseases pattern, the differentials include subacute hypersensitivity pneumonitis, respiratory bronchiolitis; and other vasculitis—SLE, microscopic polyangiitis.

Microscopic Polyangiitis

- Small vessel systemic vasculitis
- *Pathology*: Necrotizing but not granulomatous, thickened alveolar walls with neutrophilic infiltration of capillary walls.
- *Clinical presentation:* Common cause of pulmonary renal syndrome with pulmonary involvement seen in up to 50% cases[2] (*See* Chapter 10).
- *Imaging findings:* Findings of diffuse alveolar hemorrhage seen.
- *CXR*: Bilateral GGO/normal.
- *CT*: Thin section CT should be performed if clinical suspicion of DAH or rapidly deteriorating renal function, even in those with normal CXRs.
  - May reveal GGOs when CXR appears normal
  - In those with repeated hemorrhages—superimposed interstitial fibrosis seen **(Figs 9.6A and B)**
  - Interstitial fibrosis though less common is a poor prognostic indicator.



Figs 9.6A and B: Microscopic polyangiitis HRCT images showing areas of ground-glass opacification with superimposed interstitial fibrosis changes (arrows) likely due to repeated hemorrhages

- *Differential diagnosis:* ANCA associated granulomatous vasculitis, Goodpasture's syndrome, SLE.

Goodpasture's Syndrome

- A pulmonary renal syndrome characterized by glomerulonephritis and diffuse alveolar hemorrhage.
- *Pathology:* There is deposition of IgG antibody on the basement membrane of the glomeruli and alveoli. Immunofluorescence demonstrates linear deposition of the antibody. Intra-alveolar hemorrhage and hemosiderin laden macrophages seen.
- *Imaging findings:* Diffuse alveolar hemorrhage is the main finding which resolves within days. There may be features of interstitial thickening.

Behçet Disease

- Disease of young men with unknown etiology
- *Pathology*: Vessel wall thickening and perivascular cellular infiltration
- *Thoracic manifestations* are a result of pulmonary artery aneurysms, thrombosis, infarction and hemorrhage.
- *Chest radiograph and CT* may show lung mass due to an aneurysm. CT is a better modality to demonstrate the origin, size and complications of the aneurysm in the form of rupture and hemorrhage, which can be seen as areas of ground glass opacities.
- Thrombotic occlusion of the small vessels result in focal areas of peripheral infarcts and mosaic perfusion
- Aneurysms may resolve after immunosuppression therapy.

The pulmonary imaging features of various vasculitis are summarized in **Table 9.1**.

Table 9.1: Pulmonary imaging features of various vasculitis on HRCT

Vasculitic conditions	*Predominant pulmonary parenchymal involvement pattern*	*Secondary pulmonary parenchymal involvement*
Granulomatosis with polyangiitis (GPA)/ Wegener's granulomatosis	Pulmonary nodules—subpleural, peribronchovascular distribution	Bronchial wall thickening, ground-glass opacities, consolidation, centrilobular nodules
Microscopic polyangiitis	Ground-glass opacities (GGO) due to hemorrhage	Interstitial fibrosis
Eosinophilic granulomatosis with polyangiitis (EGPA)/Churg-Strauss syndrome	GGO-bilateral, patchy nodules (without cavitation) reticular opacities may also be seen	Centrilobular nodules bronchial wall thickening, bronchial dilatation mosaic attenuation
Behçet disease	GGO due to hemorrhage	Interstitial thickening
Goodpastures syndrome	GGO due to hemorrhage	Interstitial fibrosis

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10

Vasculitis-related Interstitial Lung Diseases: Clinical Diagnostic and Management Issues

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INTRODUCTION

The systemic vasculitis is a heterogeneous group of disorders which are associated with the common feature of blood vessel wall inflammation and necrosis involving arteries, veins or capillaries.

- The various systemic vasculitic disorders involve blood vessels according to their size, but at times they may have overlapping features also.
- The classification of vasculitis has undergone various modifications since the publication of 1990 American College of Rheumatology criteria for primary vasculitis.
- The latest classification and nomenclature criteria the *Revised 2012 International Chapel Hill Consensus Conference (CHCC 2012)* is well accepted and presently in use. The classification is based on vessel size, antineutrophil cytoplasmic antibody (ANCA) as well as histology **(Table 10.1)**.
- Pulmonary involvement can be seen in a large number of vasculitis due to the extensive anastomotic network, dual blood supply, and direct exposure to inhalational agent over alveolar epithelium and response to immunological injuries.
- Any systemic vasculitis can affect lung, the site and extent depend upon the type of vasculitis.
- The ANCA-associated small vessel vasculitis (AAV) which includes microscopic polyangiitis (MPA), granulomatosis with polyangiitis (GPA) and eosinophilic granulomatosis with polyangiitis (EGPA) are the most common vasculitis associated with ILD and will be discussed in the following sections.
- In the revised CHCC 2012 classification the antiglomerular basement membrane (anti-GBM) disease is also classified as immune complex mediated small vessel vasculitis and it may be associated with diffuse alveolar hemorrhage (DAH) secondary to pulmonary capillaritis.

DIFFUSE ALVEOLAR HEMORRHAGE

- Diffuse alveolar hemorrhage (DAH) is an acute rheumatologic emergency and if not diagnosed in time can lead to high mortality within a short period of time.
- The pulmonary capillaritis secondary to systemic vasculitis can lead to massive pulmonary hemorrhage.

Table 10.1: Revised 2012 International Chapel Hill Consensus Conference Nomenclature of Vasculitides

Large vessel vasculitis

- Giant cell arteritis
- Takayasu arteritis

Medium vessel vasculitis

- Polyarteritis nodosa (PAN)
- Kawasaki disease (KD)

Small vessel vasculitis

- ANCA associated
  - Microscopic polyangiitis (MPA)
  - Granulomatosis with polyangiitis (GPA)
  - Eosinophilic granulomatosis with polyangiitis (EGPA)
- *Immune complex mediated*
  - Anti-GBM disease
  - Cryoglobulinemic vasculitis (CV)
  - IgA Vasculitis (Henoch-Schönlein) (IgAV)
  - Hypocomplementemic urticarial vasculitis (Anti-C1q vasculitis)

Variable vessel vasculitis (VVV)

- Behçet's disease (BD)
- Cogan's syndrome (CS)

Single organ vasculitis (SOV)

- Cutaneous leukocytoclastic angiitis
- Cutaneous arteritis
- Primary CNS vasculitis
- Isolated aortitis
- Others

Vasculitis associated with systemic disease

- SLE, RA, sarcoid

Vasculitis associated with probable etiology

- HCV, HBV, drugs, cancer, syphilis

Table 10.2: Causes of diffuse alveolar hemorrhage

Bland hemorrhage without pulmonary capillaritis

- Coagulation disorder
- Pulmonary hypertension
- Severe mitral stenosis
- Sepsis
- Antiphospholipid syndrome (APLA)
- Chemotherapeutic drugs
- Radiation
- Idiopathic pulmonary hemosiderosis
- Others

Hemorrhage secondary to pulmonary capillaritis

- Microscopic polyangiitis (MPA)
- Granulomatosis with polyangiitis (GPA)
- Eosinophilic granulomatosis with polyangiitis (EGPA)
- Isolated pauci-immune pulmonary capillaritis
- Anti-GBM disease
- IgA Vasculitis (IgAV)
- Connective tissue diseases (SLE, RA, APLA, inflammatory muscle diseases)
- Behçet's disease
- *Miscellaneous*: Infections, drugs, toxins, post-transplant

- The DAH can also occur without capillaritis, also called bland hemorrhage. The various causes of bland hemorrhage are coagulopathy, mitral stenosis, pulmonary hypertension, sepsis and idiopathic pulmonary hemosiderosis **(Table 10.2)**.

Clinical Features

- Massive hemoptysis is the most common feature of DAH although in some cases hemoptysis may be scanty to absent.
- Hemoptysis is usually associated with dyspnea and easy fatigability due to anemia. In severe cases, it may lead to hypoxic respiratory failure secondary to alveolar drowning due to hemorrhage.

Diagnosis

- A high index of suspicion should be kept for vasculitis in any known patient of rheumatologic illness presenting with sudden hemoptysis, dyspnea, falling hemoglobin with radiological alveolar infiltrates.
- Alveolar hemorrhage can be acute or subacute and sometimes recurrent also.
- Radiologic pattern can be variable from unilateral patchy alveolar infiltrate in case of mild DAH to diffuse pattern/complete consolidation in severe cases **(Fig. 10.1)**.



Fig. 10.1: Chest radiograph in a patient with diffuse alveolar hemorrhage due to microscopic polyangiitis (MPA) secondary to pulmonary capillaritis

- Computed tomography is helpful in small bleed, where chest radiograph may be normal, it may also show nodules with or without cavitation in cases of GPA (Wegener's).
- Fiberoptic bronchoscopy with bronchoalveolar lavage (BAL) may be helpful in diagnosis of DAH in patients without hemoptysis. BAL fluid with >5% hemosiderin-laden macrophages suggest recent alveolar hemorrhage. This can be confirmed by Prussian blue staining of centrifuged BAL fluids.
- Bronchoscopy lavage can also exclude infectious cause of DAH like tubercular or bacterial infections.
- Once DAH is diagnosed clinically and radiologically, patient should be further evaluated for the cause of hemorrhage (bland vs capillaritis).
- Autoantibody (ANCA, ANA, RF, anti-GBM, C3, C4) can be helpful in making appropriate diagnosis.
- Renal function (urea, creatinine) and urine evaluation can be extremely useful in DAH associated with MPA, GPA and anti-GBM disease.
- In other cases histologic confirmation may be required through open lung biopsy or video assisted Thoracoscopic biopsy before starting toxic chemotherapeutic drugs.

GRANULOMATOSIS WITH POLYANGIITIS

- Granulomatosis with Polyangiitis (GPA) was previously known as Wegener's granulomatosis, the name was withdrawn recently after Nazi association of Friedrich Wegener was discovered.
- GPA usually involves small vessels of the upper and lower respiratory tract; and kidneys.

- The term "Limited" GPA-Wegener's is used when the disease spares kidneys and is limited to upper or lower respiratory tract or eyes.
- It is associated with raised protienase 3/antineutrophil cytoplasmic antibody (PR3-c ANCA) and usually have recurrent/relapsing course.

Clinical Features

- Equally affect both sexes with nearly 1:1 ratio.
- Mean age of diagnosis is 50 years.
- *Constitutional symptoms*: Fever, weight loss, migratory arthritis.
- *Nose, sinuses and ears:* Nose is most commonly involved. The typical symptoms are persistent rhinorrhea, epistaxis, severe nasal obstruction, nasal crusting and serosanguinous discharge. Cartilaginous inflammation may lead to perforation and collapse of nasal septum leading to saddle nose deformity **(Fig. 10.2)**. Conductive and sensorineural deafness may also be present.
- *Eyes:* Orbital pseudotumor in retrobulbar space lead to proptosis. There may be scleritis, photophobia and pain in the eyes.
- *Mouth*: Strawberry gums, tongue ulcers, palatal perforation.
- *Pulmonary involvement*: It may be seen in up to 80% of patients, often it is asymptomatic and lung involvement may be detected on chest imaging only. Patients may present with cough, hemoptysis, dyspnea and pleuritis.
- *Renal involvement*: Renal disease is quite common in patients with GPA. It presents as rapidly proliferative glomerulonephritis (RPGN) with hematuria, RBC casts, non-nephrotic range proteinuria and is untreated may lead to end stage renal disease.

Investigations

- Highly raised acute phase reactants (ESR, C-reactive protein).
- Active urinary sediments on urine analysis.
- *C-ANCA* with *antiproteinase-3* positivity in around 70% of cases.
- The ANCA is best demonstrated by a combination of indirect immunofluorescence test (IIF) using normal peripheral blood neutrophils as

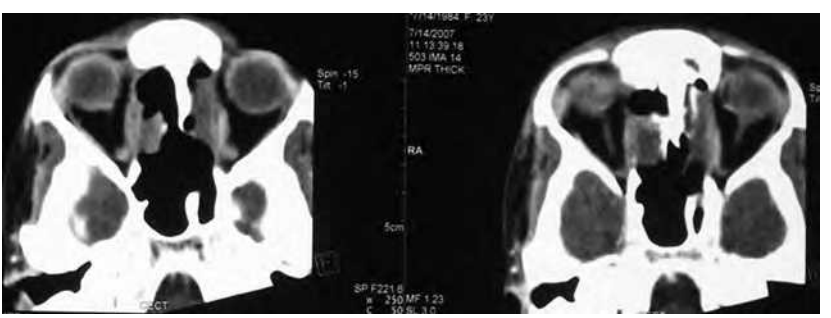


Fig. 10.2: CECT Head showing maxillary sinus involvement and septal destruction in a case of GPA

substrate, by an enzyme-linked immunosorbent assay (ELISA) for proteinase 3 (PR3) and myeloperoxidase MPO. By using combination of these tests, false positive ANCA (by IIF) due to multiorgan disorders can be eliminated.
- HRCT chest in a typical case may show multiple lung nodules (often cavitatory), pulmonary infiltrates and fibrosis. Radiologically, presence of ground glass opacities with lung nodules of more than 3 cm size indicate disease activity (*See* Chapter 7 for detailed discussion).
- Biopsy of the affected tissue shows granulomatous inflammation, vasculitis and necrosis.

EOSINOPHILIC GRANULOMATOSIS WITH POLYANGIITIS

- This disease was previously known as *Churg-Strauss syndrome*.
- This is the least common form of ANCA associated vasculitis and is associated with asthma and peripheral eosinophilia.
- Eosinophil rich necrotizing granulomatous inflammation of respiratory tract and necrotizing vasculitis of small to medium sized vessels are the hallmark on histopathology.

Clinical Features

- Mean age of presentation is around 40 years with no gender predominance.
- There are 3 phases
  1. Prodromal—allergic rhinitis, asthma
  2. Eosinophilic—peripheral eosinophilia, tissue infiltration by eosinophils.
  3. Vasculitic—systemic necrotizing vasculitis.
- *Constitutional symptoms*: Fever, malaise, arthralgia, myalgia, weight loss.
- *Sinonasal involvement*: Atopy, allergic rhinitis, sinusitis, nasal polyps, epistaxis, rhinorrhea, purulent nasal discharge, crusting.
- *Pulmonary involvement*: Asthma is present quite early and is poorly controlled on inhaled glucocorticoids. EGPA should be suspected in patients with late onset asthma. Hemoptysis is not a common feature of EGPA.
- *Peripheral nervous system involvement*: Presents with features of mononeuritis multiplex or polyneuropathy.
- *Skin involvement*: Maculopapular rash, nodular skin lesions over the extensor surface (Churg-Strauss nodules).
- *Heart involvement*: The most common cardiac manifestation is congestive heart failure and may be cause of death in EGPA patients. Others include coronary arteritis and valvular abnormalities.
- *Renal involvement*: It is less common. Interstitial nephritis may be present.

Investigations

- Peripheral eosinophilia, increased IgE level, raised ESR, C-reactive protein.
- MPO-ANCA is positive in around 50% of patients with perinuclear staining pattern (p-ANCA).
- Pulmonary function tests suggestive of obstructive airway disease.

- Pulmonary infiltrate, pleural effusion and DAH are rare in EGPA.
- Diagnostic confirmation needs clinical setting of asthma and histological confirmation of vasculitis by biopsy of lung, skin or affected nerve.

MICROSCOPIC POLYANGIITIS

- The MPA is a necrotizing vasculitis of small vessels (arterioles, capillaries and venules) of unclear etiology without granulomatous inflammation.
- Associated with raised serum myeloperoxidase/antineutrophil cytoplasmic antibody (MPO-ANCA).
- The disease usually have monophasic phase (non-recurrent) compared to relapsing remitting course of GPA.

Clinical Features

- More common in males and whites in around 6th decade of life.
- *Constitutional symptoms*: Fever, malaise, weight loss are commonly present during active disease.
- *Renal involvement*: Renal manifestations present in around 80–100% of MPA patients ranging from active urinary sediments to end stage renal disease. The most common presentation is RPGN which presents as hematuria, dysmorphic RBCs, non-nephrotic range proteinuria and rapidly rising urea and creatinine (days-week).
- *Pulmonary involvement*: Present in around 25–55% of patients as dyspnea, cough and hemoptysis. 15–25% of patients can develop DAH which may be life-threatening and present with massive hemoptysis, anemia and acute respiratory failure. Other pulmonary manifestations include interstitial lung disease, pleural effusion, pleuritis, and pulmonary infiltrates.
- *Skin involvement*: 40–50% of patients have dermal manifestations such as palpable purpura (most common), cutaneous nodules, urticaria, ulcers and livedo reticularis.
- *Neurologic involvement*: Manifestations are seen in 50–60% of patients. Peripheral neuropathy with mononeuritis multiplex and distal symmetrical polyneuropathy are common presentations.
- *Gastrointestinal involvement*: Most common GI symptom is abdominal pain. Other manifestations include gastrointestinal bleeding, bowel ischemia, perforation and colonic ulceration.

Investigations

- The acute phase reactants such as ESR and C-reactive proteins are elevated in active disease.
- P-ANCA with anti-MPO is positive in around 60–70% of patients.
- Routine urine analysis for active urinary sediments is usually positive in patients with renal involvement or with pulmonary renal syndrome (dysmorphic RBC plus proteinuria).
- Chest radiographs show fleeting focal infiltrates often bilaterally to massive lung hemorrhages **(Fig. 10.3)**.



Fig. 10.3: Chest radiograph of a patient with MPA showing alveolar infiltrate with lower lobe predominance

- CECT chest may be better able to differentiate between interstitial lung disease, alveolar hemorrhage or infection, rarely pleural effusion may also be seen.
- Biopsy of the affected tissues confirms small vessel necrotizing vasculitis without any granulomas.
- Renal biopsy in patients with renal involvement reveals focal segmental glomerulonephritis (FSGN) with crescents.

GOODPASTURE'S SYNDROME

- Goodpasture's syndrome is also known as antiglomerular basement membrane (anti-GBM) disease.
- It is associated with pulmonary involvement in the form of DAH and renal involvement in the form of rapidly proliferative glomerulonephritis (RPGN).
- The antigen target is type IV collagen, which is major component of the basement membrane.
- This combination of lung-kidney involvement is also known as pulmonary renal syndrome (PRS).
- The most common cause of PRS is ANCA associated vasculitis (AAV) followed by Goodpasture syndrome, which may account for 20–30% of PRS. Other causes of PRS include SLE, idiopathic pulmonary hemosiderosis and microscopic polyangiitis.

Clinical Features

- Most common presentation is with progressive dyspnea, hypoxemia and alveolar infiltrate on chest radiograph.
- Hemoptysis may also be present in up to 80–94% patients.
- Glomerulonephritis is cardinal feature and frequently seen and presents as acute nephritic syndrome (microscopic hematuria, red cell casts and proteinuria).

- In up to one-third of patients only rapidly progressive glomerulonephritis (RPGN) is the presentation without any pulmonary DAH, while in up to 10% of patients only lung involvement may be seen without renal involvement.
- Azotemia at presentation may be seen in up to half to 2/3rd patients, which if not immediately treated may lead to severe life threating renal injury leading to end stage renal disease (ESRD) or death.

Diagnosis

- Anti-GBM antibody with ELISA are highly sensitive (>95%) and specific (97%) for diagnosis of Goodpasture's syndrome and also helpful in monitoring the response to treatment.
- A percutaneous kidney biopsy with immunofluorescence can also confirm the diagnosis which shows immunofluorescence pattern of IgG and complement (C3) deposits along the GBM.
- Lung biopsy is usually not required if diagnosis is established with antibodies and kidney biopsy.

Management

- The mainstay of treatment is institution of early plasmapheresis (8–10 cycles may be required) accompanied with systemic steroid and cyclophosphamide pulse for initial 2 weeks.
- Diffuse alveolar hemorrhage with hemoptysis may also respond to plasmapheresis treatment.
- Patient who develop oliguria, requires acute dialysis, serum creatinine > 5 mg/dL, kidney biopsy with >50% crescents/necrosis and advanced fibrosis have bad prognosis and respond poorly to treatment.
- Patients on renal support for ESRD can be undertaken for kidney transplantation after 6 months when serum antibodies are undetectable.

BEHÇET'S DISEASE

- Behçet's disease is a chronic, multisystem, variable vessel vasculitis which may involve small, medium or large vessels.
- The disease is characterized by recurrent attacks of mucocutaneous, eye, musculoskeletal, vascular, CNS and gastrointestinal symptoms.
- The disease is commonly suspected in subjects with recurrent nonhealing oral or genital ulcers and sometime may also be associated with uveitis thus making the Behçet's classic 'triad.'
- The hallmark of the disease is "Pathergy test", a nonspecific hyper-reactivity to skin prick by needle and development of papule/pustule.
- The lung involvement is due to pulmonary vascular involvement in the form of arterial aneurysms, vascular thrombosis, pulmonary infarcts and pulmonary hemorrhage.
- Massive hemoptysis or bleeding presenting as DAH can be rarely encountered in Behçet's disease due to erosion of bronchial artery wall, rupture of aneurysm or formation of bronchial wall-arterial fistulae.

Diagnosis

- The diagnosis is based on clinical manifestations as there is no diagnostic test for Behçet's disease. The International Study Group (ISG) 1990 classification criteria is widely used with sensitivity and specificity of >90%.
- The diagnosis requires recurrent oral ulcers and at least 2 genital ulcers, erythema nodosum such as lesions, folliculitis, uveitis and positive Pathergy test.
- Diagnosis of lung involvement in Behçet's requires clinical correlation and high clinical suspicion. Imaging of lung with MRI/CT angiography may demonstrate aneurysms, but sometimes lung biopsy may also be required.

Management of Behçet's Disease

- The treatment is according to organ involvement and age of the patient.
- The minor organ involvement in the form of mucocutaneous or musculoskeletal involvement can be successfully treated with colchicine and thalidomide and in refractory cases corticosteroid, azathioprine or methotrexate may be used.
- In young patients with major organ involvement in the form of uveitis, vascular, lung, central nervous system, and gastrointestinal tract involvement, high dose corticosteroid and other immunosuppressive agents (azathioprine, methotrexate, cyclophosphamide and anti-TNF inhibitors) may be required for disease control.

MANAGEMENT OF VASCULITIS ASSOCIATED WITH INTERSTITIAL LUNG DISEASE

- Systemic vasculitis are heterogeneous group of potentially fatal diseases with high morbidity and mortality.
- Once a diagnosis of systemic vasculitis is confirmed by clinical, radiologic and histologic evaluation, they should be aggressively treated with immunosuppressive agents.
- The vasculitis with or without ILD are similarly treated although it is not clear whether AAV (ANCA associated vasculitis) with ILD has worse prognosis than AAV without ILD.
- Few small studies suggests AAV with ILD have worse prognosis but others have not shown such relationship.
- Acute vasculitis with imminent threat to vital organs or life, needs emergency treatment with any one or more of the immunomodulating agents, which can be lifesaving: pulse methylprednisolone, plasmapheresis, intravenous immunoglobulin (IVIG), surgery/interventional vascular procedure (vascular embolization, etc.) for acute bleed, and/or renal replacement therapy in case of acute kidney injury.
- Cytotoxic and immunosuppressive drugs (cyclophosphamide, methotrexate, azathioprine, mycophenolate mofetil (MMF), tacrolimus, cyclosporin) and Anti-B cell therapy (rituximab) are the backbone of vasculitis therapy.

- Intravenous cyclophosphamide is given at dose of 15 mg/kg every month for 4–6 months as *induction therapy* and this is usually followed by azathioprine, MMF, methotrexate for next 2–3 years as *maintenance therapy*.
- CD-20 inhibitor-rituximab is equally effective as compared to intravenous cyclophosphamide in induction of remission in patients.
- Rituximab is given at a dose of 375 mg/m^2 every week for 4 weeks for induction therapy and can be followed by 6 monthly infusion for maintenance therapy. It is an expensive drug as compared to cyclophosphamide with advantage of decreased adverse effects like hemorrhagic cystitis, cytopenia and ovarian failure in females.

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11

Sarcoidosis

Priyanka Naranje

INTRODUCTION

- Sarcoidosis is an immune-mediated multisystem disorder of yet unidentified etiology
- It is characterized by presence of noncaseous epithelioid cell granulomas in the affected organs
- Disease associated morbidity and mortality is predominantly attributed to thoracic involvement which is the most common system to be involved
- Twenty percent of pulmonary sarcoidosis leads to chronic lung disease progressing to pulmonary fibrosis[1]
- There is a wide variation in incidence and prevalence due to many factors such as environmental exposure, genetic factors, race, ethnicity, age, and sex as well as surveillance methods
- It can affect any gender and age-group; however it is seen more often in females less than 40 years of age.

CLINICAL FEATURES

- Respiratory symptoms—dyspnea, cough, bronchial hyper-reactivity
- Constitutional symptoms—fatigue (most important), weight loss, night sweats
- Extrapulmonary symptoms—erythema nodosum, lupus pernio, visual disturbances, headache, arrhythmias, fever, polyarthritis (part of Lofgren's syndrome), Heerfordt syndrome (fever, uveitis, parotiditis)
- Variable clinical courses—stable, relapsing-remitting, progressive, pulmonary fibrosis (25% cases).[1]

LABORATORY INVESTIGATIONS

Among the wide gamut of available investigations, the recommended basic investigations in the initial evaluation of suspected sarcoidosis patients are as follows:[2]

- Pulmonary function tests
  - Parenchymal involvement—restrictive pattern, decreased lung volumes
  - Airway involvement—obstructive pattern (less common)
- Serum angiotensin converting enzyme
  - Used as a measure of disease activity
  - Supportive measure for diagnosis
  - Elevation more than twice the normal value seen in 60–80% cases
- Serum calcium—for hypercalcemia
- Urine for hypercalciuria
- Tuberculin skin test—to exclude tuberculosis
- Histopathology of mediastinal/hilar/peripheral lymph nodes, skin or conjunctiva—for confirmation of granulomatous inflammation.

IMAGING MODALITIES

- Chest radiography (CXR) and computed tomography (CT), especially high resolution CT (HRCT) are imaging modalities of choice
- Indications for CT—timely diagnosis, staging, follow-up, rule out other treatable/curable diseases, narrow differential diagnosis, for activity assessment[3] and detection of complications
- Magnetic resonance imaging (MRI)—evolving role as radiation-free modality for serial imaging follow-up and assessment of pulmonary fibrosis
- Positron emission tomography (PET) CT—to identify occult sites of involvement, suitable sites for biopsy and activity assessment in chronic symptomatic patient.

IMAGING CLASSIFICATION/STAGING SYSTEM OF SARCOIDOSIS

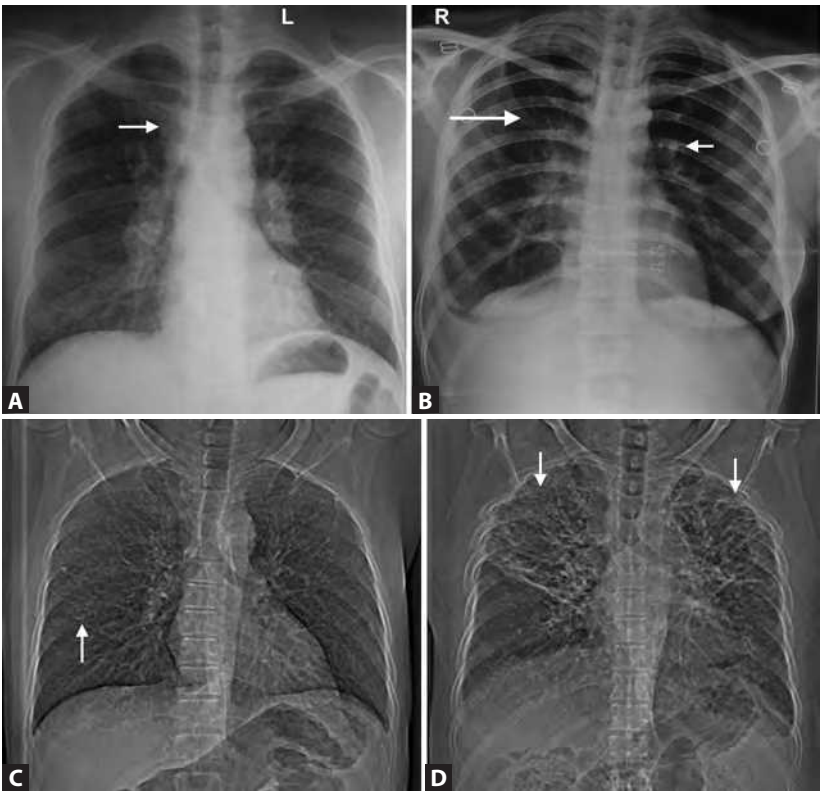
The Siltzbach classification system of sarcoidosis developed more than 40 years back in 1967, is based on CXR. It is still carries a strong prognostic value.[4] Although the disease usually follows a stable course, the chances of spontaneous remission declines with advancing stage of the disease. However, this staging system does not correlate well with the pulmonary function tests.

Stage 0—Normal chest radiograph
Stage 1—Lymphadenopathy only **(Fig. 11.1A)**
Stage 2—Lymphadenopathy and parenchymal lung disease **(Fig. 11.1B)**
Stage 3—Parenchymal lung disease only **(Fig. 11.1C)**
Stage 4—Pulmonary fibrosis **(Fig. 11.1D)**.

RADIOLOGIC FEATURES OF SARCOIDOSIS IN THORACIC INVOLVEMENT

Lymph Nodal Involvement

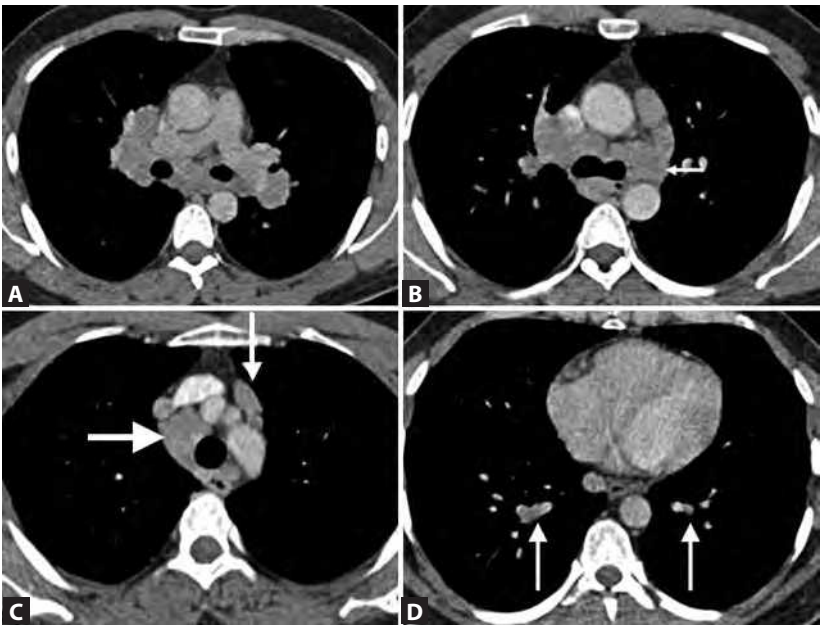
- Enlarged bilaterally symmetric hilar and right paratracheal lymphadenopathy (most commonly seen and most typical pattern) **(Figs 11.1A and 11.2A)**
- Lymph node enlargement at other mediastinal stations with or without hilar lymph nodal involvement—at the left paratracheal, subcarinal,



Figs 11.1A to D: Typical radiographic staging patterns of sarcoidosis. (A) Plain radiograph showing enlarged bilaterally symmetric hilar and right paratracheal lymphadenopathy (arrow) (most typical pattern of sarcoidosis) representing Siltzbach stage 1; (B) Plain radiograph showing hilar lymphadenopathy (small arrow) and pulmonary parenchymal nodules (big arrow) representing stage 2; (C) CT topogram showing multiple nodules (arrow) in bilateral lung fields without significant hilar lymphadenopathy—stage 3; (D) CT topogram showing bilateral upper zones fibrotic changes (arrows) suggesting stage 4

aortopulmonary window, prevascular regions and bronchopulmonary groups may also be seen **(Figs 11.2B to D)**

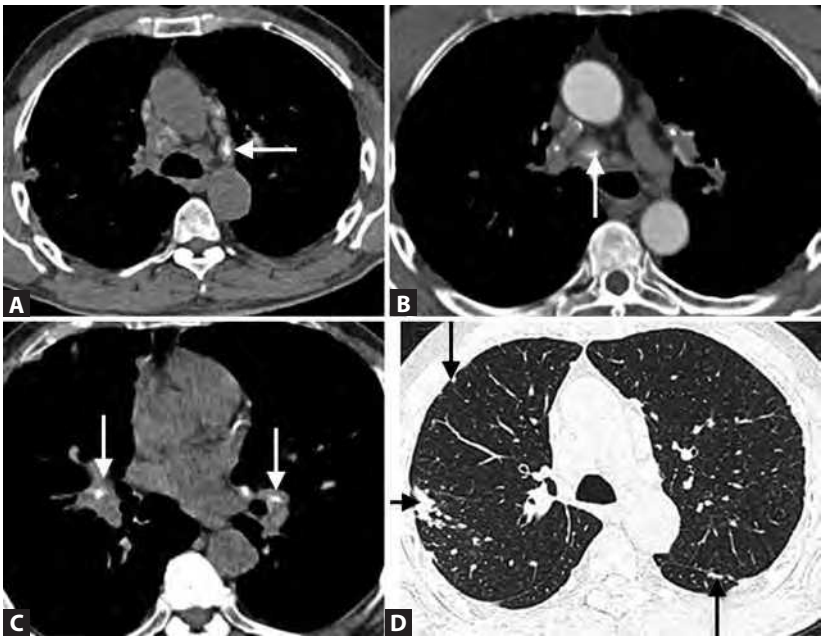
- 'Asymptomatic' bilateral hilar lymph node enlargement—points towards sarcoidosis over other common conditions such as tuberculosis or malignancy
- Occasionally, lymph nodes are asymmetric or seen in rare locations such as along the internal mammary group, retrocrural or paravertebral regions
- The calcification in enlarged nodes (20%)—amorphous, hazy (icing sugar type), punctate, popcorn like, or eggshell-like patterns, represent long standing disease **(Figs 11.3A to D)**.



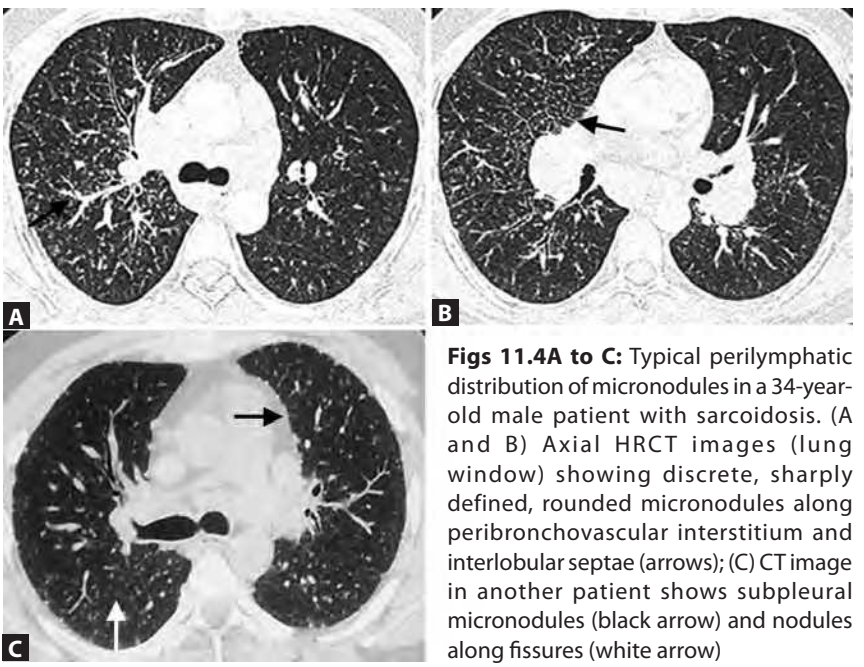
Figs 11.2A to D: Typical (A, C, D) and atypical (B) radiologic findings of lymphadenopathy in a patient with sarcoidosis. (A) Contrast-enhanced CT axial image showing multiple homogeneous non-necrotic enlarged lymph nodes in bilateral hilar regions; (B and C) Enlarged nodes at other mediastinal stations—aortopulmonary window (small arrow), right paratracheal location (thick arrow) and prevascular regions (vertical arrow); (D) Bronchopulmonary group of nodes (arrows) in a typical case of sarcoidosis

Lung Parenchymal Involvement

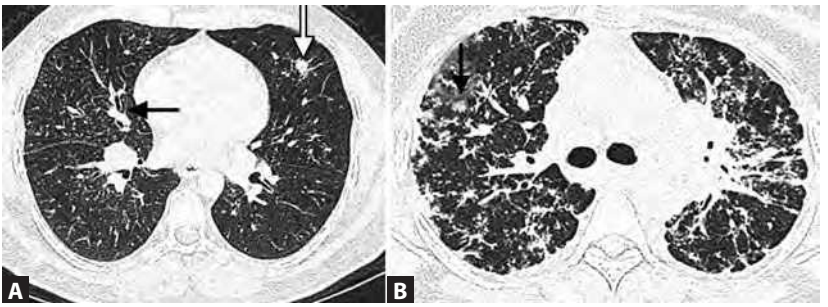
- *Micronodules*—representing granulomas
  - A perilymphatic distribution of micronodules (most common pattern 75–90%[1])
  - Small (2–4 mm in diameter), discrete, sharply defined, rounded nodules
  - Along peribronchovascular interstitium, subpleural regions, interlobular septa and fissures **(Figs 11.4A to C)**
  - Usually bilaterally symmetrical
  - Mostly in the upper and middle lobes.
- *Macronodules and masses:* About 15–25%, representing coalescence of granulomas
  - Ill-defined irregular opacities
  - 1–4 cm in diameter
  - Multiple and bilateral
  - Perihilar or peripheral regions **(Figs 11.5A and B)**
  - Air bronchograms may be seen
  - Cavitation rare (in <3% of patients with parenchymal opacities)



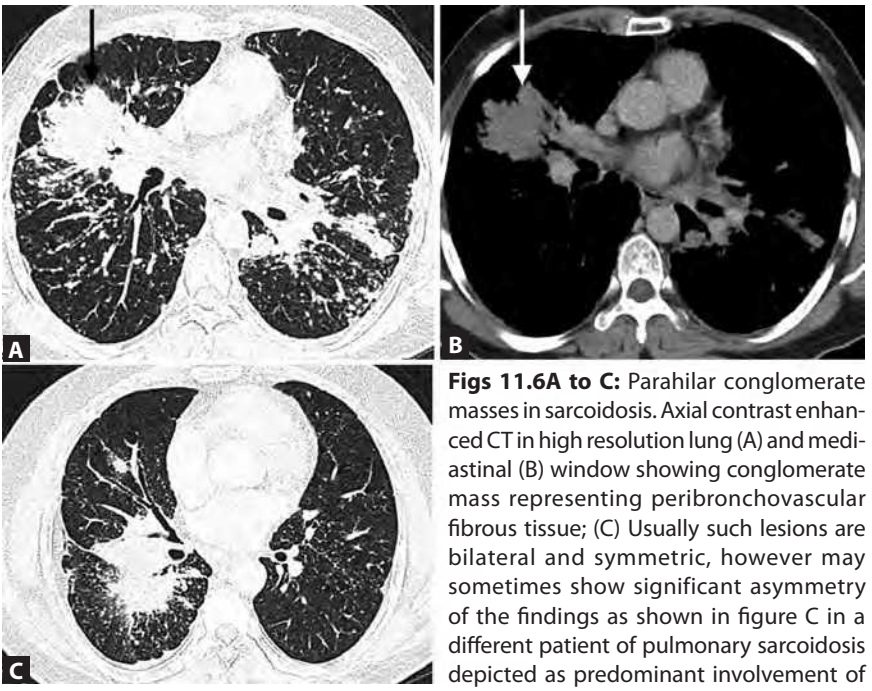
Figs 11.3A to D: (A to C) Pattern of calcification in sarcoidosis. Axial unenhanced (A and C) and contrast enhanced (B) CT scan showing calcification within the lymph nodes (amorphous/icing sugar type) (arrows) in AP window, precarinal and bilateral hilar regions in a longstanding case of sarcoidosis; (D) Lung window shows presence of subpleural nodules, fissural nodules and a macronodule (black arrows) in the same patient



Figs 11.4A to C: Typical perilymphatic distribution of micronodules in a 34-year-old male patient with sarcoidosis. (A and B) Axial HRCT images (lung window) showing discrete, sharply defined, rounded micronodules along peribronchovascular interstitium and interlobular septae (arrows); (C) CT image in another patient shows subpleural micronodules (black arrow) and nodules along fissures (white arrow)

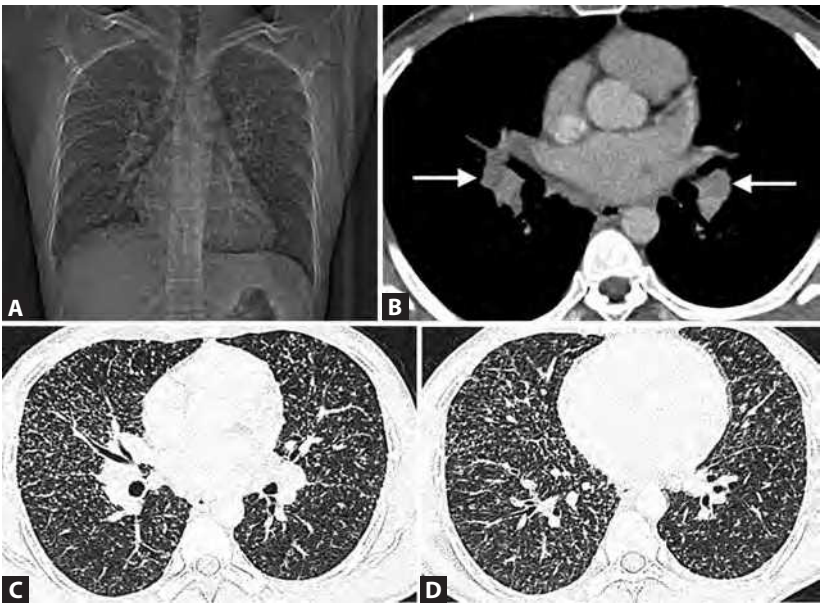


Figs 11.5A and B: Macronodules in sarcoidosis. (A) Axial HRCT image (lung window) depicting one cluster of nodules in the periphery of the left upper lobe (white arrow) which has coalesced to form a conglomerate lesion (macronodule). Note multiple discrete peribronchovascular nodules in bilateral lungs (black arrow); (B) Axial HRCT image in another patient showing multiple bilateral irregular macronodules representing coalescence of granulomas (arrow)



Figs 11.6A to C: Parahilar conglomerate masses in sarcoidosis. Axial contrast enhanced CT in high resolution lung (A) and mediastinal (B) window showing conglomerate mass representing peribronchovascular fibrous tissue; (C) Usually such lesions are bilateral and symmetric, however may sometimes show significant asymmetry of the findings as shown in figure C in a different patient of pulmonary sarcoidosis depicted as predominant involvement of right lung as compared to left

- Conglomerate masses consisting of peribronchovascular fibrous tissue (mimics PMF when predominantly in upper lobes) **(Figs 11.6A to C)**
- Usually symmetrical, however can be asymmetric **(Fig. 11.6C)**
- Multiple well-defined rounded macronodules (mimics metastasis)
- Miliary pattern—rare in sarcoidosis (<1% of cases) (think of other differential such as tuberculosis, metastasis or pneumoconiosis) **(Figs 11.7A to D)**.

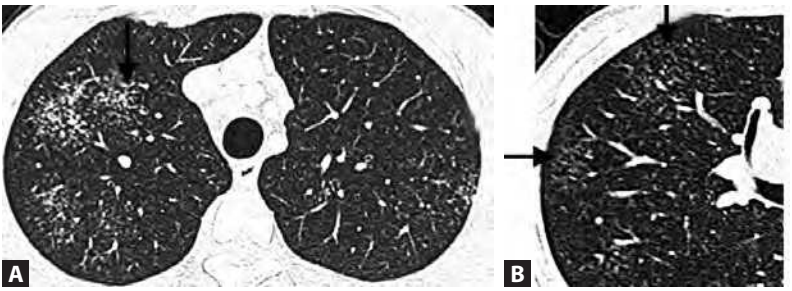


Figs 11.7A to D: Miliary nodules in sarcoidosis (atypical manifestation). (A) CT topogram showing multiple diffuse tiny nodules in bilateral lung fields; (B) Axial contrast enhanced CT scan shows bilateral non-necrotic hilar lymph nodes (arrows); (C and D) HRCT lung window reveals innumerable tiny micronodules in a random distribution

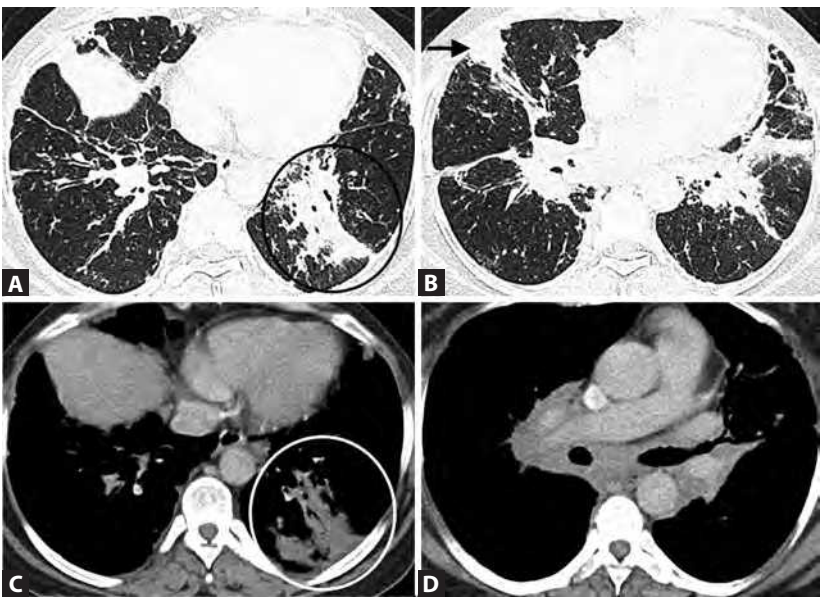


Fig. 11.8: Axial high-resolution CT scan (lung window) depicting the macronodules with multiple fine small nodules around these large nodules (arrows). This appearance suggests sarcoid galaxy sign

- *Galaxy sign:* Suggested by presence of multiple small satellite nodules surrounding the periphery of masses **(Fig. 11.8)**.
- *Cluster sign:*
  - Multiple perilymphatic micronodules in focal, rounded clusters **(Figs 11.9A and B)**



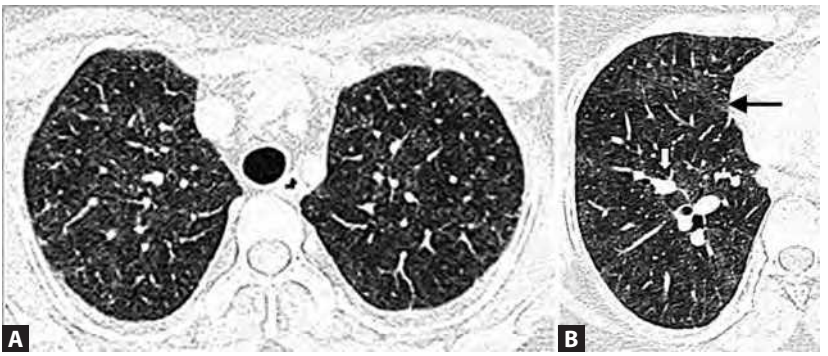
Figs 11.9A and B: Cluster sign in sarcoidosis. Axial HRCT scan showing multiple micronodules in focal, rounded clusters (arrows) representing noncoalescent granulomas



Figs 11.10A to D: Alveolar sarcoid pattern. (A and B) Axial HRCT scan (lung window) and (C) (mediastinal window) shows alveolar consolidation in left lower lobe (circle) and subpleural alveolar opacities in right middle lobe (black arrow) suggesting alveolar sarcoid pattern in a case of pulmonary sarcoidosis; (D) Mediastinal window shows typical bilateral hilar and subcarinal enlarged lymph nodes in the same patient

- Represents noncaseous, noncoalescent granulomas without fibrosis
- Found in peripheral, subpleural regions of the upper and middle zones of the lungs.

- *Patchy alveolar/airspace consolidation/alveolar sarcoid pattern:* About 10–20% **(Figs 11.10A to D)**
  - Consolidation
  - Having ill-defined, blurred margins with or without air bronchograms
  - Bilateral and symmetric
  - In peribronchovascular regions



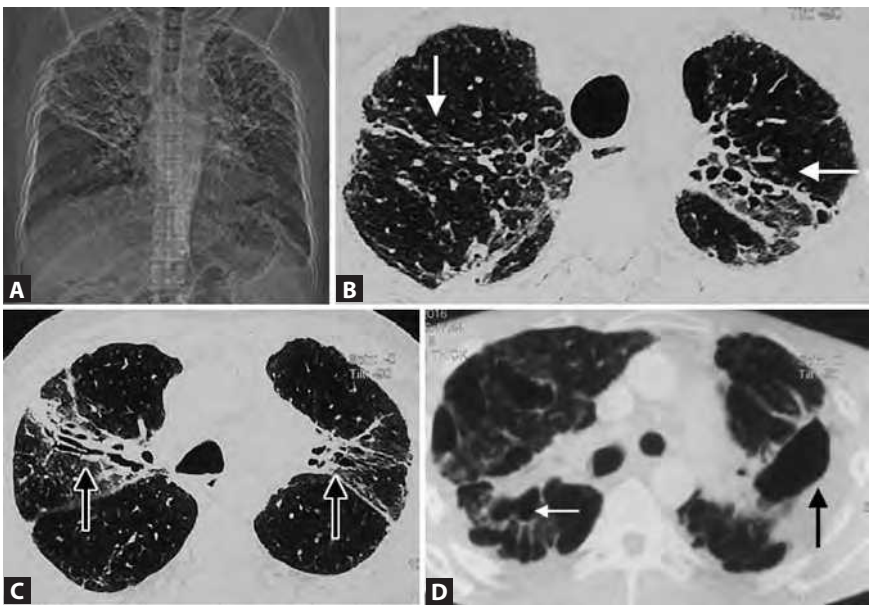
Figs 11.11A and B: (A) Axial HRCT scan shows patchy ground-glass opacities in both lungs in a case of pulmonary sarcoidosis; (B) Magnified axial HRCT image at a different level in same patient shows GGO accompanied by other findings of peribronchovascular (white arrow) and fissural nodules (black arrow) in right lung

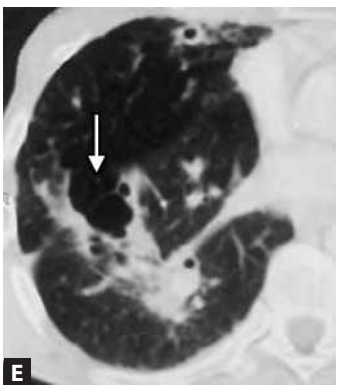
- Predominantly middle and upper lung zones
- Surrounding micronodules along the periphery of consolidation **(Figs 11.10A to D)**.

- *Patchy ground-glass opacities (GGOs):* About 40% of parenchymal involvement
  - Usually patchy distribution
  - Extensive GGOs less common
  - Ground-glass pattern is produced by multiple confluent micronodules **(Figs 11.11A and B)**
  - Always accompanied by other findings.
- *Interlobular and intralobular septal thickening:*
  - Subpleural
  - Upper and middle lung zones
  - Bilateral
  - Irregular **(Fig. 11.12)** or linear
  - Differential diagnosis—lymphangitic carcinomatosis.
- *Fibrotic changes:*
  - Upper and mid-zones predominantly **(Figs 11.13A and B)**
  - Follow the large airways in a perihilar distribution (fairly common and characteristic) **(Fig. 11.13C)**.
  - Patchy
  - Volume loss
  - Reticulations
  - Traction bronchiectasis
  - Architectural distortion (displacement of fissures and bronchovascular bundles)
  - Fibrotic cysts, paracicatricial emphysema and bullae **(Figs 11.13D and E)**
  - Honeycomb-like cysts, which are seen in subpleural regions of the middle and upper lung zones, sparing the lung bases



Fig. 11.12: Interlobular septal thickening in magnified axial high-resolution CT scan of right lung shows hilar lymph nodal enlargement and nodularity and thickening of interlobular septa (black arrow) and fissures (white arrow), in a case of pulmonary sarcoidosis





Figs 11.13A to E: Fibrotic changes in sarcoidosis. (A) CT topogram in a patient with stage 4 sarcoidosis shows typical bilateral upper and mid zone fibrotic changes; (B and C) Axial high-resolution CT scan in a different patient of stage 4 sarcoidosis shows fibrosis and traction bronchiectasis typically in bilateral perihilar regions (black arrows) and predominantly in upper lobes (white arrows); (D) Fibrotic cysts (white arrow) and cavities (black arrow) in another patient representing advanced fibrocystic stage of sarcoidosis; (E) Paracicatricial emphysema (arrow) seen adjacent to fibrotic areas

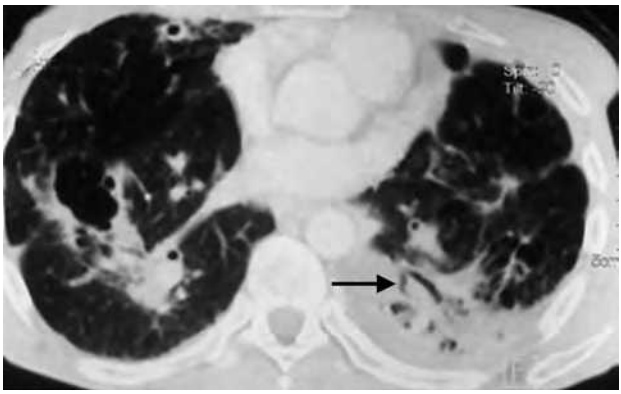


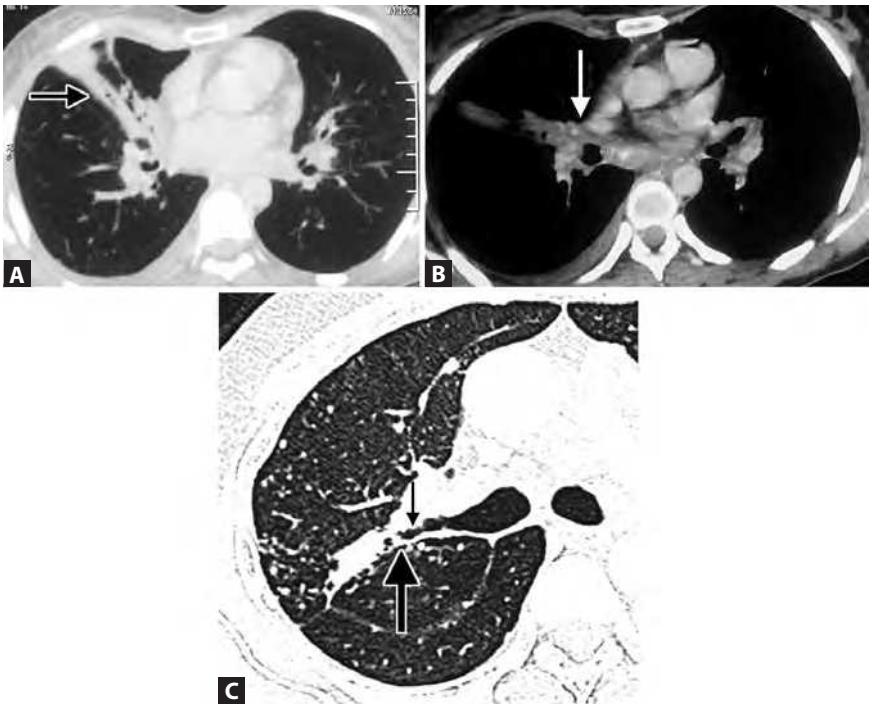
Fig. 11.14: Axial CT image lung window obtained in a patient with stage 4 sarcoidosis depicts a mycetoma (aspergilloma) within a cystic space in the left lower lobe (arrow)

- Pulmonary arterial hypertension—suggested by dilated main pulmonary artery (PA), enlarged right PA and left PA, right ventricular enlargement, and attenuation of peripheral vessels.

- *Cavitation of parenchymal lesions:*
  - Rare finding
  - Ten percent in end-stage disease and most often occurs in areas of advanced fibrosis **(Fig. 11.13D)**
  - Primary cavitary sarcoidosis—<0.8% in acinar or nodular active disease
- *Mycetoma:*
  - Complication of stage 4 fibrocystic sarcoidosis
  - Develop in pre-existing cysts and bullae **(Fig. 11.14)**
  - Usually in the upper lobes.

Airway Involvement

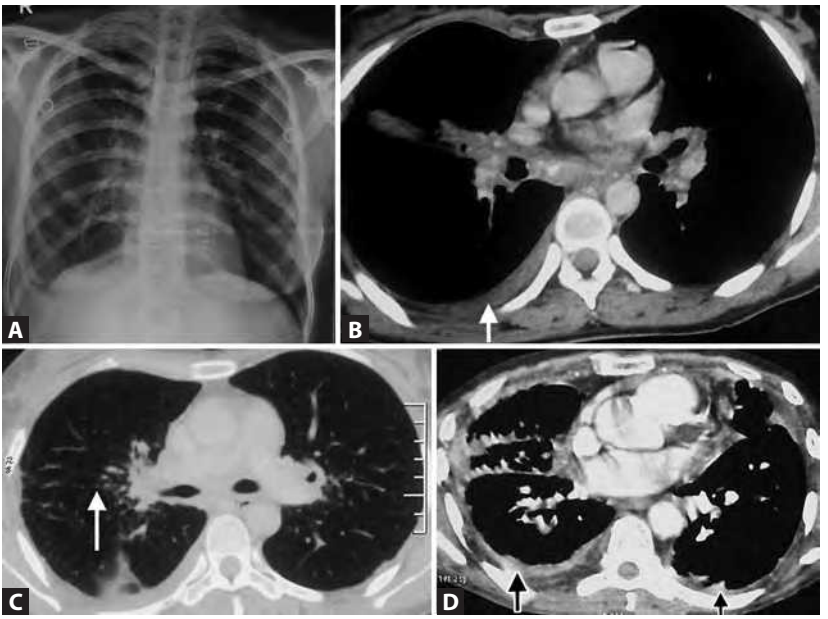
- Mosaic attenuation pattern
  - Can represent either small airways disease (most often), patchy interstitial disease or vascular disease
  - Do inspiratory-expiratory CT.
- Air trapping
- Atelectasis **(Figs 11.15A and B)**
  - Endobronchial granulomas or enlarged peribronchial lymph nodes causing obstruction of lobar or segmental bronchi
  - Common in right middle lobe due to anatomically having small caliber and acute angulation from bronchus intermedius.
- Tracheobronchial abnormalities
  - Bronchial stenosis—secondary to granulomas within the bronchial wall **(Fig. 11.15C)**, extrinsic nodal compression, or end-stage parenchymal disease distorting the major bronchi
  - Tracheal or major bronchial wall thickening can be seen.



Figs 11.15A to C: Airway involvement in sarcoidosis. (A) Axial CECT lung window image in a known case of sarcoidosis shows segmental atelectasis in right middle lobe (black arrow) likely due to compression on right middle bole bronchus by enlarged right hilar lymph nodes (white arrow) as seen on mediastinal window (B); (C) Magnified axial HRCT image in a different patient with pulmonary parenchymal sarcoidosis depicts irregularity due to nodules along the walls of right upper lobe bronchus (thin arrow) and an endobronchial rounded nodule (thick arrow) suggesting bronchial involvement by sarcoid granulomas. Rest of the lung shows subpleural and fissural micronodules

Pleural Involvement

- Rare, only 1–4% of patients
- Exudative or transudative pleural effusion, usually transient **(Figs 11.16A to C)**
- Hemorrhagic or chylous pleural effusion
- Pneumothorax
- Pleural thickening **(Fig. 11.16D)**
- Rarely, pleural calcification
- Plaque like pulmonary opacities which are contiguous with the pleura, generally in the upper and middle lung zones.



Figs 11.16A to D: (A) Pleural involvement in sarcoidosis. Chest radiograph in a 32-year-old female patient shows bilateral hilar prominence and multiple perihilar nodules typical of stage 2 sarcoidosis. Blunting of right costophrenic angle is seen suggestive of pleural effusion; (B) Axial contrast-enhanced CT scan obtained at the subcarinal level in the same patient as (A) shows right sided pleural effusion (arrow) and hilar and mediastinal lymphadenopathy; (C) Lung window images of this patient shows multiple peribronchovascular nodules in right lung (arrow); (D) Another patient with pulmonary sarcoidosis shows predominant bilateral pleural thickening (arrows) with minimal fluid

TYPICAL AND ATYPICAL RADIOLOGICAL FEATURES FOR SARCOIDOSIS

It is critical to identify both the typical as well as atypical radiologic appearances of the disease in order to achieve timely diagnosis. The various manifestations described here are given in **Table 11.1**.

ACTIVITY ASSESSMENT FOR SARCOIDOSIS

High-resolution CT may be particularly helpful for distinguishing active inflammation from irreversible fibrosis in selected patients with stage 2 or 3 sarcoidosis **(Table 11.2)**.

COMMON DIFFERENTIAL DIAGNOSIS

- Conditions that closely mimic sarcoidosis on imaging include tuberculosis, lymphangitis carcinomatosa, silicosis or coal worker pneumoconiosis and

Table 11.1: Typical and atypical radiological findings of pulmonary sarcoidosis

Attribute	*Typical radiologic findings as a predominant radiologic feature of the disease*	*Atypical radiologic findings as a predominant radiologic feature of the disease*
Lymphadenopathy	Symmetric Bilateral Hilar, right paratracheal and bronchopulmonary groups Non-necrotic	Asymmetric Unilateral Prevascular, subcarinal, precarinal without hilar lymph nodes Retrocrural, internal mammary, paravertebral regions
Nodules	Micronodules (2–5 mm) Macronodules (>5 mm) Peribronchovascular, subpleural, along interlobular septa	Miliary nodules—random distribution Solitary nodule Cluster sign Galaxy sign
Consolidation	Bilateral perihilar regions Radiating from the hilum toward the periphery Noncavitatory	Peripheral consolidation Alveolar sarcoid pattern Conglomerate masses Cavitation
Fibrotic changes	Reticular opacities, traction bronchiectasis, architectural distortion, volume loss	Cysts, bullae, emphysema, honeycomb-like cystic changes
Distribution of abnormalities	Upper and mid zones predominance Axial predominance	Lower lobe predominance
Septal thickening	Isolated mild interlobular thickening	Extensive interlobular and/or intralobular septal thickening in upper and mid zones
Ground-glass-opacities (GGO)	—	Patchy or extensive
Airway involvement	—	Mosaic attenuation pattern, tracheobronchial wall thickening or stenosis Atelectasis
Pleural involvement	—	Effusion, pleural thickening, calcification, chylothorax, pneumothorax, hemothorax, Pleural plaque-like thickening

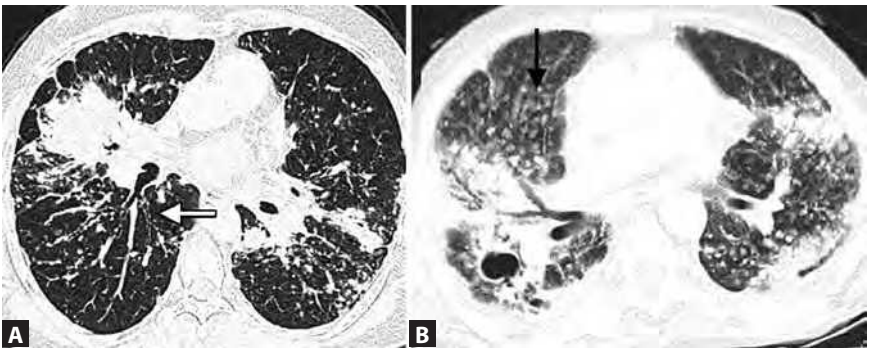
LIP. **Table 11.3** summarizes the imaging-based differentiating points for sarcoidosis and silicosis **(Figs 11.17A and B)**. **Table 11.4** summarizes the main differences between sarcoidosis and tuberculosis **(Figs 11.18A to D)**.

Table 11.2: Radiologic findings to suggest disease activity versus fibrotic/inactive disease

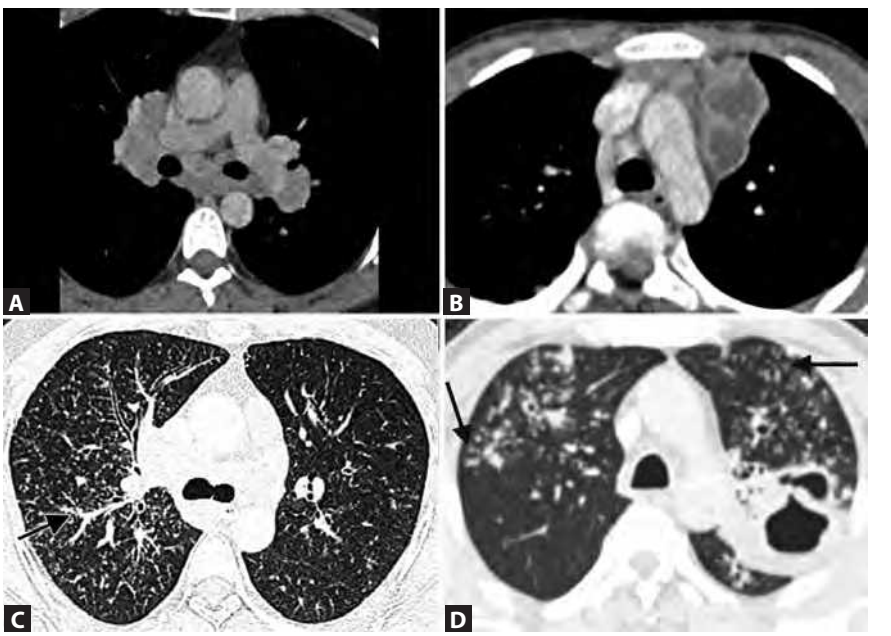
Active disease	*Fibrotic sequelae/Inactive disease*
Perilymphatic nodules Micronodules, macronodules	Traction bronchiectasis, bronchiolectasis
Parenchymal consolidation—confluent alveolar opacities	Reticular opacities
Ground-glass-opacity	Architectural distortion
Interlobular septal thickening	Cysts, bullae, paracicatricial emphysema
Pleural effusion	Volume loss, tracheal displacement
Tracheal wall/ major bronchial wall thickening	Cavity with mycetoma
	Secondary pulmonary arterial hypertension

Table 11.3: Sarcoidosis versus silicosis based on common radiological attributes in absence of relevant history of exposure

Attribute	*Sarcoidosis*	*Silicosis*
Eggshell-like nodal calcifications	Uncommon	More frequent
"Icing sugar" type of calcification within nodes	Common	Not seen
Conglomerate masses	Typically bilateral hilar/ perihilar	More typically posteriorly situated within the upper lobes and not in hilar location
		The long-axis of the mass tend to be parallel to the chest wall
Nodules	Peribronchovascular, septal distribution	Centrilobular and subpleural in location
	Less well-defined margins	More well-defined nodules
	Usually bilaterally symmetric, but asymmetric distribution of nodules is common	Bilaterally symmetric and uniformly distributed
		Posterior predominance
	Axial and anterior predominance	



Figs 11.17A and B: Sarcoidosis versus silicosis. (A) Axial HRCT scan in a patient with sarcoidosis. Note the micronodules in peribronchovascular distribution (arrow) with bilateral parahilar conglomerate masses; (B) Axial CT image in a patient with silicosis shows larger discrete nodules in predominant centrilobular distribution with PMF in bilateral lungs with cavitation—a feature rare in sarcoidosis (Ref to Table 11.3)



Figs 11.18A to D: Sarcoidosis versus tuberculosis. (A) Typical bilateral hilar non-necrotic nodes in a case of sarcoidosis versus; (B) Necrotic nodes in a case of tuberculosis showing peripheral rim enhancement; (C) Typical perilymphatic nodules (arrow) in a case of sarcoidosis versus. (D) Centrilobular nodules in tree-in-bud pattern (arrows) in a case of tuberculosis. Also note the presence of cavitatory lesion in left upper lobe, feature of tuberculosis (Ref to Table 11.4)

Table 11.4: Sarcoidosis versus tuberculosis

Attribute	*Sarcoidosis*	*Tuberculosis*
Lymph nodes	Bilateral hilar Nonnecrotic Homogeneous Noncoalescent Hazy or egg shell calcification	Unilateral Necrotic Heterogeneous Conglomeration common Densely calcified
Micronodules	Perilymphatic distribution	Centrilobular distribution Tree-in-bud pattern seen
Cavitation in nodules or masses	Rare	Common
Consolidation	Peribronchovascular region	More common in upper lobes or RML, lingula, superior segment any LL with ipsilateral hilar lymph nodes
Miliary pattern	Rare	Common
Pleural involvement/ empyema	Rare	Common

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Section 4

Occupation/Exposure-related Diffuse Lung Diseases

- Occupational/Environmental Exposure-related Lung Diseases: Imaging
- Organic Dust Exposure and Hypersensitivity Pneumonitis: Imaging
- Occupational/Exposure-related DLDs: Clinical Diagnosis and Management Issues

12

Occupational/Environmental Exposure-related Lung Diseases: Imaging

Ashu Seith Bhalla, Manisha Jana

TERMINOLOGY

- Refers to broad group of pulmonary disorders caused by inhalation of dust, chemicals or proteins.
- Severity depends on the nature of the inhaled substance; as well as intensity and duration of exposure.
- Exposure is often occupational. However, can also be environmentally acquired in communities residing in neighborhood of such industries—indirect exposure.
- Clinical and radiographic screening programs are essential in workers exposed to known fibrogenic substances.
- Exposure history may not however be available upfront, hence recognition of radiographic patterns essential.
- Radiologist may be the first to suggest diagnosis, and then look for exposure history.
- This is especially true in cases of indirect (environmental) exposure.

CLASSIFICATION

- *Pneumoconiosis*: Those resulting from inhalation of mineral (norganic) dusts
  - May be fibrogenic/nonfibrogenic
  - *Fibrogenic*
    - Silicosis, Coal worker's pneumoconiosis (CWP), asbestos-related pleural and parenchymal disease
    - *Less common*: Berylliosis, talcosis, calcinosis, hard metal pneumoconiosis
  - *Nonfibrogenic*: Siderosis (iron), barium, tin
  - May also be classified as being immunologically mediated/due to direct exposure (this classification is used especially in cases of organic dust exposure).

- *Immunologically mediated*: Hypersensitivity pneumonitis, asthma
- *Direct irritants*: Irritant gases, vapor, smoke.

IMAGING MODALITIES

Chest Radiograph (Figs 12.1 to 12.3)

- Primary modality for diagnosis and follow-up
- The International Labour Office (ILO), Geneva—an agency of the United Nations has published guidelines on the Classification of Chest X-ray (CXRs) in those with occupational inhaled dust exposures.
- These have been published since 1950, with latest update being in 2011.[1]

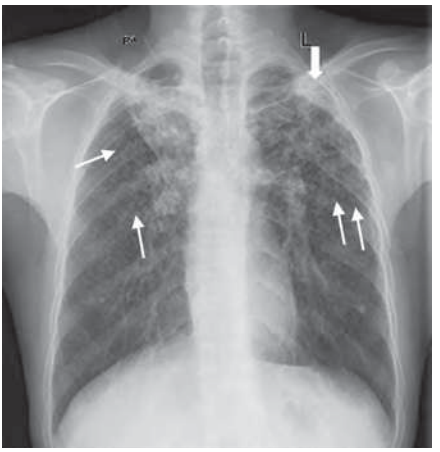


Fig. 12.1: Small opacities (arrows) and pleural thickening (block arrow) in both upper zones in silicosis

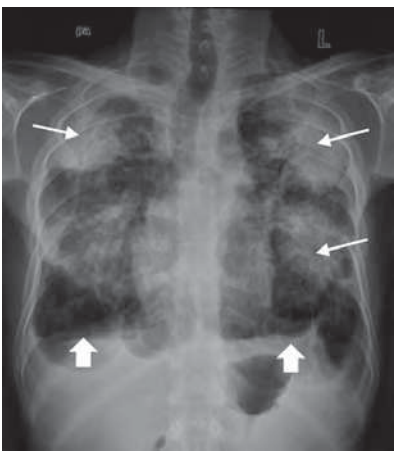


Fig. 12.2: Large opacities (arrows) in silicosis. The domes of diaphragm are flattened secondary to hyperinflation of the lower lung lobes (block arrows)



Fig. 12.3: Eggshell calcification of supraclavicular, mediastinal and hilar nodes in silicosis. Note the fibrotic parenchymal changes in both upper zones

- The purpose of these guidelines is to provide a standardized format for interpreting and describing CXRs in pneumoconiosis and reduce inter-observer variability.
- Also, National Institute for Occupational Safety and Health (NIOSH)—a United States Federal Agency; runs a "B" reader program which trains and certifies physicians for reporting radiographs according to this classification system.

Basic Methodology

- The published guidelines comprise of a series of standard chest radiographs (22 in number) with detailed footnotes to codify the spectrum of abnormalities seen.
- These include only standard PA (posteroanterior) CXRs
- The recent version has digital CXRs unlike the analog radiographs in the previous versions.
- The reader compares the radiograph of the "subject" with the "standard radiographs" provided.
- It deals with parenchymal and pleural abnormalities.
- The abnormalities are classified into different types based on shape and size.
- Further these are described based on the "profusion"/severity of the abnormalities.
- A structured reporting format referred to as the "Roentgenographic Interpretation Form" is available on which the findings need to be recorded.
- A simplified synopsis of this classification system is given below:

Terminology

Quality assessment

- Systematic reporting requires that the technical quality be reported initially as: (1) Good, (2) Acceptable: with no technical defect, (3) Acceptable: some technical defect but still adequate, (4) Unacceptable.

- Quality defects include those related to exposure (over- or under), positioning, artifacts and under inflation.

Parenchymal abnormalities

- Classified as small or large opacities.

Small opacities (Fig. 12.1)

- Small opacities are further classified based on shape and size as:
  - *Small, rounded opacities*: p (up to about 1.5 mm), q (about 1.5–3 mm), or r (3–10 mm).
  - *Small, irregular opacities, based on width:* s (up to about 1.5 mm), t (about 1.5–3 mm), u (3–10 mm).
- *Distribution*: Mentally the reader is asked to divide each lung into even sized three zones: Upper, mid and lower and presence of small opacities in these zones recorded.
- *Profusion*: The "concentration" of abnormalities is described on a four point scale (0,1,2,3) with "0" being absent and 3 being "most profuse". The profusion is to be compared to the standard radiographs available. A further minor category is assigned and the final profusion reported in the form of two numbers, e.g. 1/1; with the first number representing the major category and the second the minor category.

Large opacities (Fig. 12.2)

- Refers to opacities more than 1 cm in size
- *Further subclassified as:*
  Category A: One or more in number, combined largest dimension not more than about 50 mm
  Category B: Combined largest dimension more than 50 mm but less than approximate equivalent area of right upper zone
  Category C: Combined largest dimension more than the equivalent area of right upper zone.

Pleural abnormalities

- Pleural abnormalities need to be detailed regarding the type, and location
- *Type*: Pleural plaques or diffuse pleural thickening
- *Location*: Along the chest wall/the diaphragm/others
- *Pleural calcification*: Present/absent
- *Width*: Of the "in profile" thickening seen along the edge of the chest wall and reported as being: a (3–5 mm); b (5–10 mm) or c (more than 10 mm).
- *Extent*: Combined distance along the chest wall measured and classified as: Up to 1/4 of lateral chest wall; 1/4 to 1/2 or more than 1/2 of lateral chest wall.

Other abnormalities

- There are 29 other abnormalities included which also need to be reported
- These are enlisted in alphabetic order and commenting on all is obligatory
- Examples: Atherosclerotic aorta (aa), bullae (bu), cancer (ca), cor pulmonale (cp), round atelectasis (ra), tuberculosis (tb).

Thin-section CT

- Despite the use of CT in most DLDs as the primary modality and its obvious superiority to CXRs; it has not replaced CXRs in evaluation of occupational lung disease.
- This is due to additional cost and radiation burden involved in epidemiological surveillance.
- It is used as a secondary modality in those with clinicoradiographic discordance/severe functional impairment.
- A low dose CT (LDCT) with exposures of <1.5 mSv is now advocated for screening for carcinoma lung in those with occupational exposure; and also for screening for pneumoconiosis.
- A standard classification system for scoring extent of disease on CT is also now available.[2]
- It is called the "International Classification of Occupational and Environmental Respiratory Diseases (ICOERD)"
- Similar to the ILO system for chest radiograph reporting, this is a semi-quantitative reporting system.
- This classification describes the parenchymal abnormalities as small opacities, large opacities, ground glass, honeycombing, and emphysema.
- The shape of the focal opacities is further detailed as regular, irregular or linear
- Also, the location and profusion of the abnormalities is detailed.
- Severity of involvement is graded from 0 to 3 (four grades). This grade is to be given separately for each region of bilateral lungs.
- *The regions are divided as:* Upper, middle, bottom left and bottom right.
- Pleural abnormalities are to be detailed too, and expressed separately for the visceral and parietal pleura.
- These are further detailed by extent, pleural thickness and calcification.

MAGNETIC RESONANCE IMAGING

May aid in differentiation of progressive massive fibrosis (PMF) from carcinoma (ca) lung.

18^F FDG PET-CT

Investigated for the same (Ca lung vs PMF) but role not defined.

PNEUMOCONIOSES

Fibrosing Forms

Silicosis

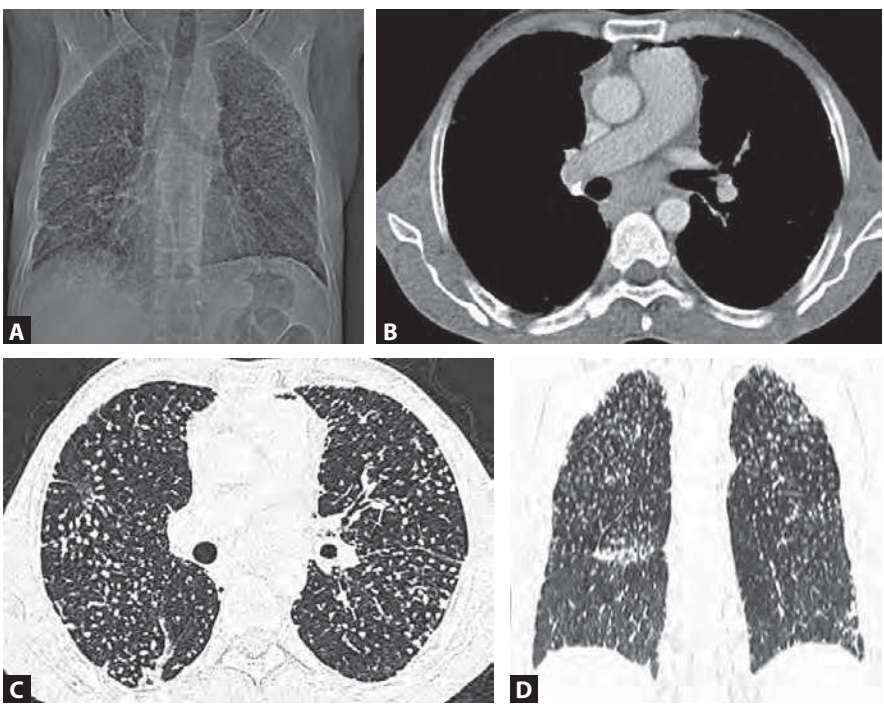
- *Causative mineral*: "Respirable silica"—crystalline silicon dioxide, fine particles
- Include quartz, tridymite, cristobalite.
- *Common industries*: Mining, construction, sandblasting of denim, ceramics, glass workers

- *Clinicoradiological forms*: Acute or classic. Classic further divided into simple/complicated.
- Classic/chronic silicosis is by far the more common form.

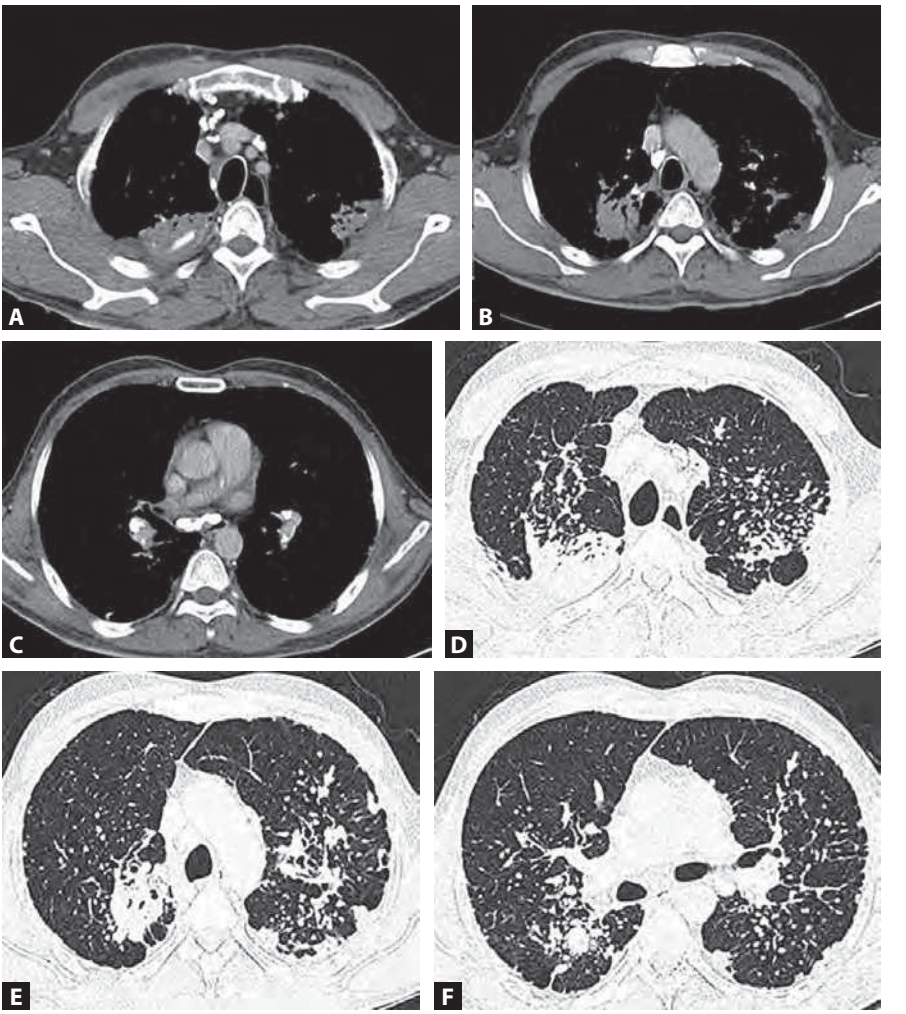
Classic/Chronic Silicosis (Figs 12.4 and 12.5)

Simple form

- Characterized by presence of multiple nodules
- *Pathology*: Gray or black colored nodules which have collagen in center with a periphery of macrophages containing particles. On polarized light microscopy, birefringent silica crystals may be seen.
- *CXR*: Multiple, well-defined, relatively uniform nodules measuring 1–10 mm with predominantly upper zones distribution. Calcification seen in 10–20%.[3]
- *High-resolution computed tomography (HRCT)*: Nodules—small 2–5 mm
  - Distribution appears more diffuse on CT but more numerous in upper zones and posterior parts
  - May be calcified
  - Perilymphatic (subpleural, paraseptal) and centrilobular. Subpleural nodules appear round/triangular, may coalesce forming "pseudo-plaques"



Figs 12.4A to D: Silicosis. Chest radiograph (A) shows diffuse perilymphatic nodules. Axial CECT chest soft tissue window (B) image reveals mediastinal and right hilar lymph nodes with calcification. HRCT lung (C and D) images show-well defined perilymphatic nodules in both lungs



Figs 12.5A to F: Chronic silicosis. Axial CECT chest soft tissue window (A to C) and lung window (D to F) reveal calcified mediastinal and hilar lymph nodes, bilateral upper lobe fibrotic changes with pleural thickening, and perilymphatic nodules (arrows). Note the paracicatricial emphysema in the upper lobes

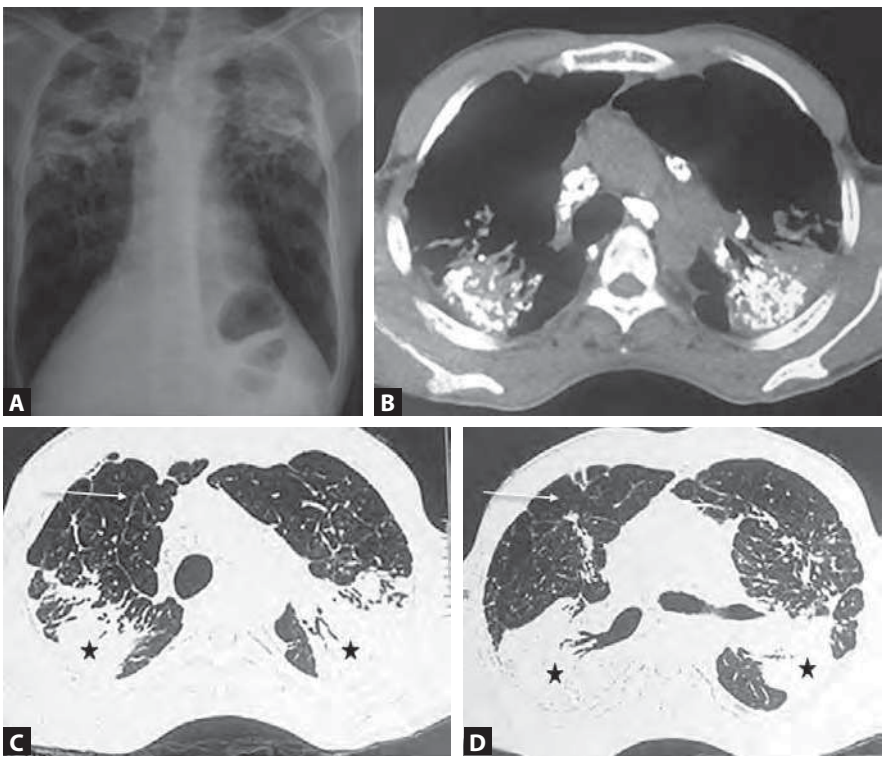
- Nodes–enlargement with or without calcification, classical pattern—"egg-shell"
- Bronchial wall thickening maybe seen.

- *Differential diagnosis*: Coal worker's pneumoconiosis, sarcoidosis, lymphangitis carcinomatosis
  - Besides evaluation of clinical history, following radiological features help in differentiation
  - *Sarcoidosis*: More central clustering of nodules (peribronchovascular), fissural or septal irregularity, reticular opacities, intervening normal lung areas.
  - *Silicosis:* Distribution of nodules more diffuse and symmetrical

- CWP nodules ill-defined compared to silicosis
- *Lymphangitis carcinomatosis*: Beaded irregular septae, reticular opacities.

Complicated form

- *Synonym*: Progressive massive fibrosis **(Figs 12.6A to D)**
- Results from conglomeration of smaller nodules
- *Pathology*: Similar to nodules, multiple central foci of mature collagen surrounded by rim of macrophages. Central necrosis may be seen.
- *CXR*: Conglomerate masses—more than 1 cm in size
  - Bilateral, symmetrical, more in mid or upper lung zones.
  - *Margins*: Irregular/ill-defined, calcification may be seen.
- *CT*: Distributed in upper lobes (posterior parts), lower lobes (apical segments)
  - Lateral interface of the fibrotic masses is typically parallel to the lateral chest wall.



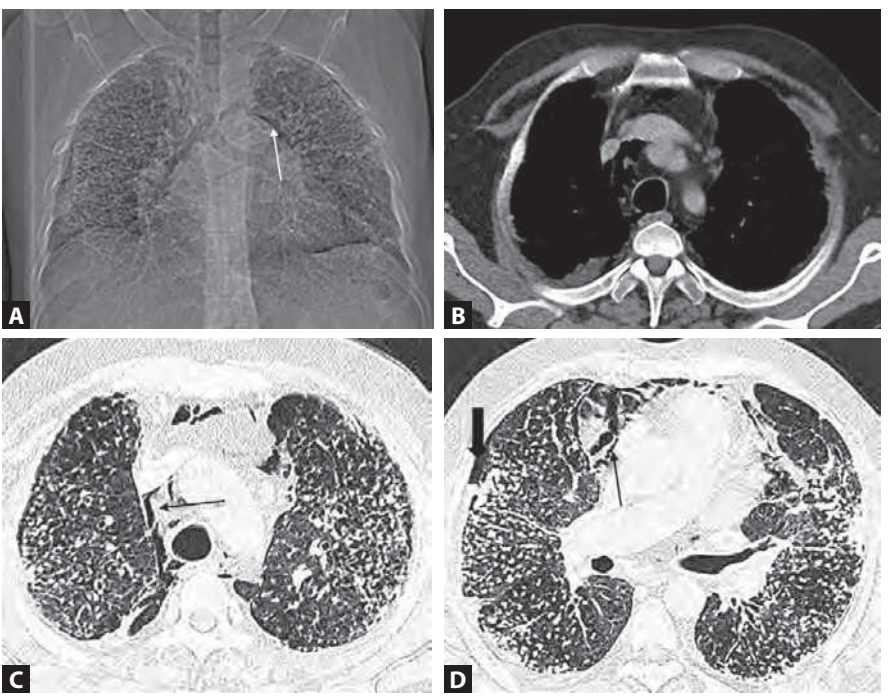
Figs 12.6A to D: Silicosis with progressive massive fibrosis (PMF). Chest radiograph (A) shows bilateral upper zone fibrotic masses with 'pulled up hila; and resultant lower zone hyperinflation. Axial CECT chest soft tissue window (B) image reveals mediastinal lymph nodes with eggshell calcification; and bilateral upper lobe calcified mass-like lesions. CECT lung window (C and D) images show bilateral upper lobe fibrotic masses (asterisk) with a lenticular shape and convergence towards hila. *Note* the paracicatricial emphysema in the surrounding lungs (arrows)

- Surrounding emphysema, air trapping
- Adjoining pleural thickening
- With progression–migrate towards hila with worsening paracicatricial emphysema
- May show central cavitation.

- Can show uptake on *18FFDG PET-CT*. Differentiation from Ca lung difficult
- *Differential diagnosis*: CWP, post-tubercular sequelae (especially in endemic areas), sarcoidosis.

Uncommon patterns (Figs 12.7A to D)
- Usual interstitial pneumonia (UIP) pattern (in < 10%)[4]
- Unexplained pleural effusions—uncommon 11%[4]

Acute silicosis
- *Synonym*: silicoproteinosis
- Acute/subacute form, rare
- After sudden, large exposure. Reported in sandblasting, including sandblasting of denim
- Results in alveolar filling with proteinaceous material



Figs 12.7A to D: Silicosis with pneumothorax and pneumomediastinum. Chest radiograph (A) reveals diffuse well-defined nodules in both lung fields; with presence of pneumomediastinum (arrow). CECT chest axial soft tissue window (B) and HR sections (C and D) show presence od pneumomediastinum (arrow), pneumothorax (block arrow) and perilymphatic nodules

- *Pathology*: Air spaces filled with acid Schiff positive material—alveolar lipoproteinosis
- *Imaging:* Consolidation—characteristic, more in posterior segments of lungs
  - Crazy paving pattern also reported.
  - Centrilobular nodules.

Accelerated Silicosis

Seen when there is exposure to large amounts of silica within a short duration (4–8 years).

Radiographic findings progress much faster than in simple silicosis with faster development of fibrosis.

Complications

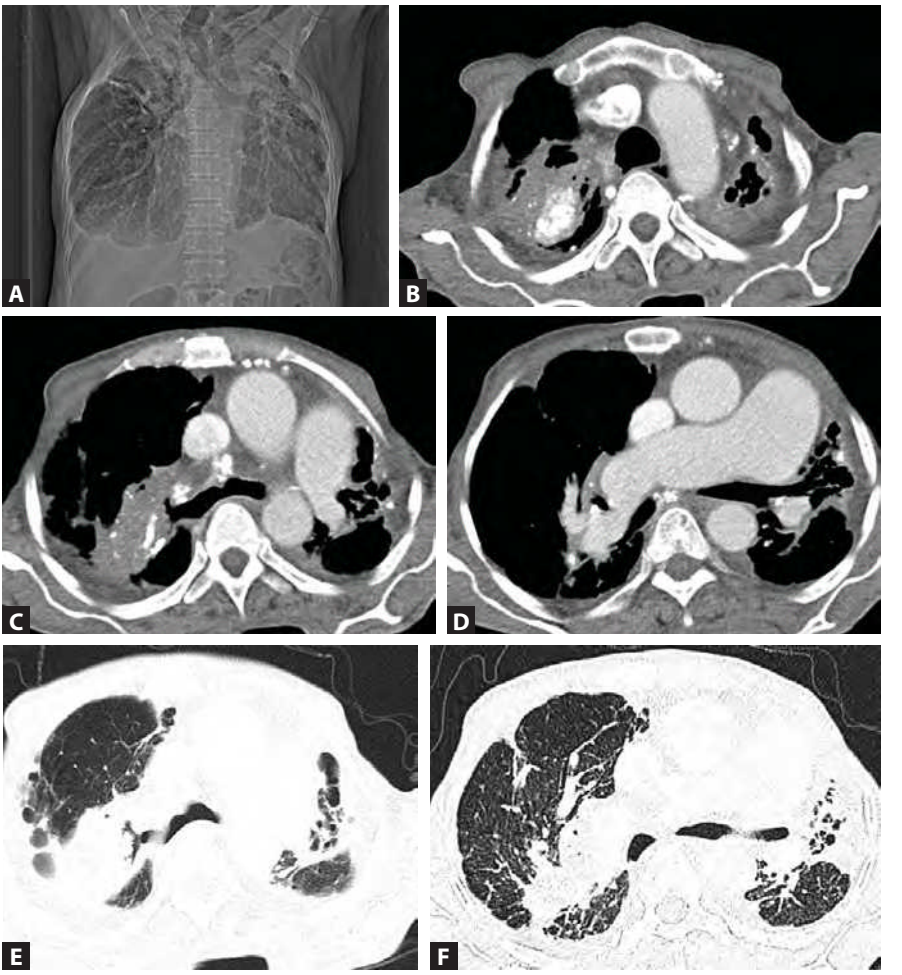
Tuberculosis, carcinoma lung.

Silicotuberculosis (Figs 12.8 and 12.9)
- Silicosis significantly increases risk of tuberculosis
- Suspect in case of cavitation, asymmetric consolidation or rapid progression
- Cavitation is taken as the strongest indicator
- D/D: Cavitation can also result from ischemic changes within an area of PMF
- Worsens prognosis.

Carcinoma lung
- Increased risk of Ca lung, although lesser than in asbestosis
- Difficult to differentiate from PMF
- *MRI*: Ca lung shows T2 hyperintensity, while PMF appears hypointense on both T1 and T2 weighted sequences (Vs muscle). On postcontrast images, PMF shows peripheral enhancement.[3]
- *18^F FDG PET-CT*: Both Ca lung and PMF may show uptake.

Coal Worker's Pneumoconiosis

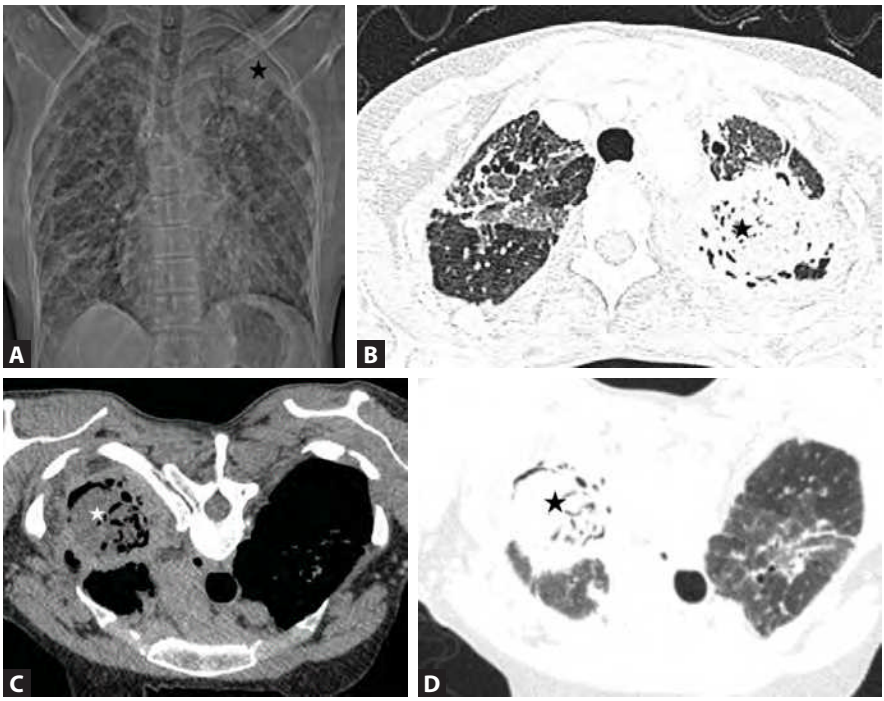
- "Real risk" amongst coal miners.
- Exposure to coal mine dust. Coal dust often a mixture with components of silica and kaolin. Coal worker's pneumoconiosis (CWP) typically refers to pneumoconiosis resulting from exposure to washed coal, devoid of silica.
- *Pathology:* Coal macules—1–5 mm nodules with anthracotic pigments with no fibrosis, these are centerd around the bronchioles in the center of the pulmonary lobule.
  - *Progressive massive fibrosis*: Fibrotic masses measuring more than 1 cm
- Though pathology different, imaging findings resemble silicosis.
- CWP similarly has "simple" and "complicated" forms.
- Also, increased risk of tuberculosis and chronic obstructive pulmonary disease (COPD).
- Increased mortality in those with complications, and patients with more severe PMF.



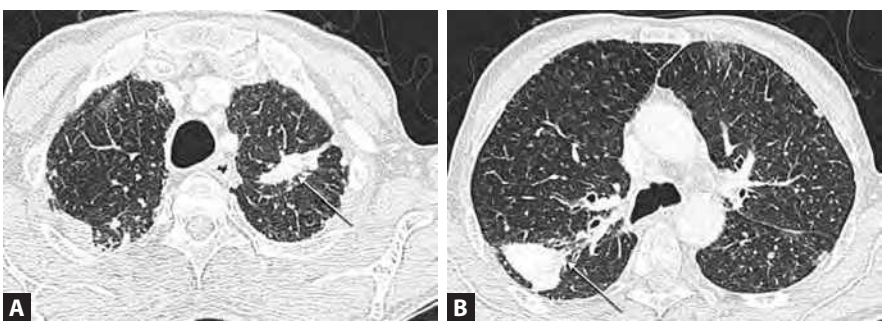
Figs 12.8A to F: (A) Silicotuberculosis. CT scout film reveals extensive fibrosis involving both lung fields, extensive bronchiectasis in the upper zones with apical pleural thickening; and lower zone hyperinflation. CT thorax soft tissue (B to D) and lung window (E and F) reveal bilateral upper lobe fibrobronchiectatic changes and parenchymal calcification

Simple form

- *CXR*: Nodular pattern dominant, occasionally reticulonodular.
  - *Nodules*: Small 1–5 mm, granular with ill defined margins.
  - Bilateral diffuse, more in upper zones.
  - Calcification in 10–20%.[3]
- *CT* **(Figs 12.10A and B)**
  - *Distribution of nodules*: Perilymphatic, less frequently-centrilobular.
  - Coalescence of sub-pleural nodules gives a plaque like appearance.
  - Calcification in the nodules seen in up to 30%, may be a "central dot" like appearance.[3]



Figs 12.9A to D: Silicotuberculosis with aspergilloma. CT scout film reveals extensive fibrosis involving both lung fields, and a soft tissue mass in left upper zone (asterisk). CT thorax supine (B) and prone (C and D) reveal a cavity with low attenuation intracavitary soft tissue in the right upper lobe, having a nondependent air crescent



Figs 12.10A and B: Coal worker's pneumoconiosis. CT axial images reveal small perilymphatic nodules in both upper lobes, with bilateral upper lobe fibrotic masses (arrows)

- Hilar/mediastinal nodal enlargement
- Bronchial wall thickening may be seen

- *Differential diagnosis*: Silicosis **(Table 12.1)**.

Complicated form

- *Pathology*: Pigmented macrophages and free pigment with interspersed haphazardly placed collagen.

Table 12.1: Differentiating features of CWP and silicosis on imaging

	CWP	*Silicosis*
Nodule size and margin	Smaller, less distinct margins	Larger
Calcification in nodules	Central dot-like	More diffuse
Lymph node calcification	Less common	More common, 'egg shell' appearance

- *Imaging findings*: Large (>1 cm) conglomerate masses similar to silicosis
- Margins may be irregular/well-defined.
- *Vs silicosis*: Incidence is less, appearance similar.

Diffuse lung fibrosis
- Uncommon pattern (seen in 10–40%)[4,5] is diffuse lung fibrosis, wherein irregular opacities are seen in the lower lung zones.
- This form is associated with more lung function impairment than "PMF"
- On CT, this reticular pattern may resemble UIP/NSIP with honeycombing.
- Pneumoconiotic nodules may also be present
- Higher incidence of carcinoma lung, preferentially seen in the areas of fibrosis.

Anthracofibrosis is discussed in Chapter no 13.

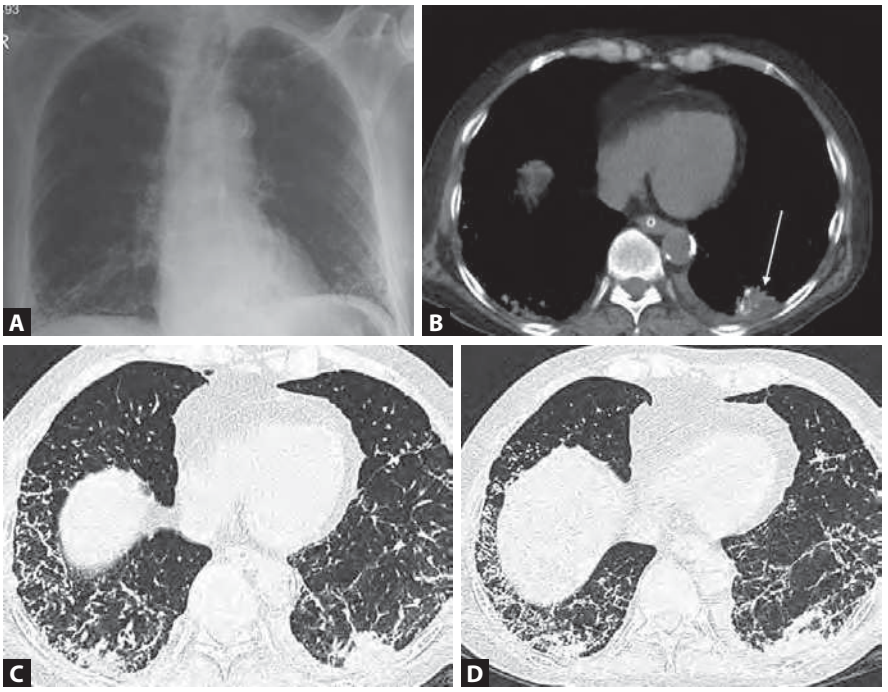
Asbestos-related Lung and Pleural Disease (Figs 12.11 to 12.14)

- Seen in asbestos workers as well as those receiving environmental exposure in the community.
- Vermiculite, tremolite
- *Industries*: Asbestos mining and milling, ship demolition
- Restrictive pulmonary function defect seen
- *Carcinogenic*: Predisposes to mesothelioma and lung carcinoma.

Lung disease
- Also called asbestosis
- Subpleural fibrosis
- *CT*: Interlobular septal thickening, curvilinear subpleural opacities, parenchymal bands **(Figs 12.11 and 12.12)**
- Traction bronchiectasis seen, honeycombing-less frequent
- Centrilobular nodules, ground-glass opacities may also be present
- *Differential diagnosis*: Idiopathic interstitial pneumonia (IIP) [usual interstitial pneumonia (UIP), fibrotic nonspecific interstitial pneumonia (NSIP)] **(Table 12.2)**.

Table 12.2: Differentiating features of asbestosis and IIP on imaging

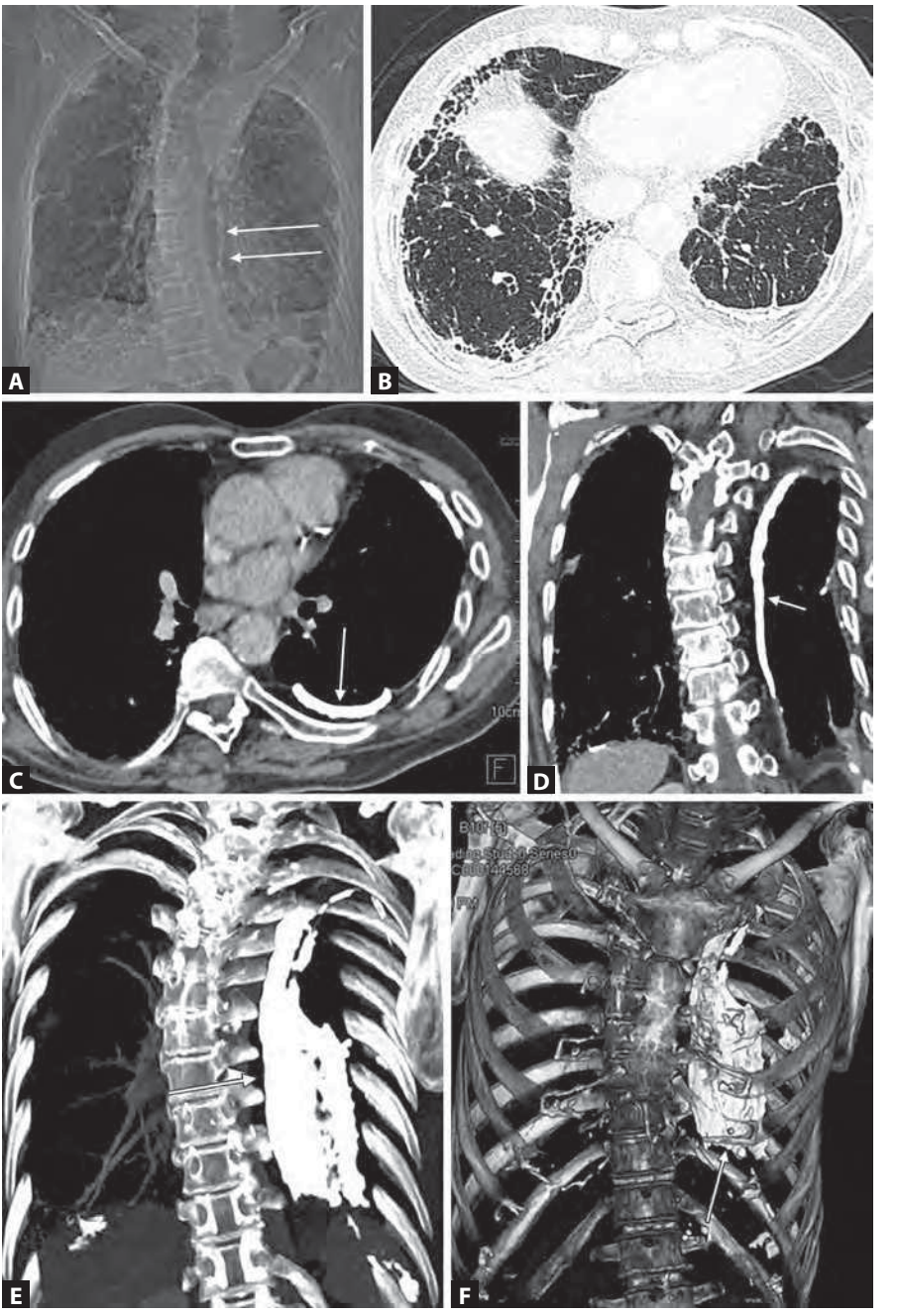
	Asbestosis	*IIP*
GGO	Not seen	May be present
Centriacinar nodules	Not seen	May be present
Distribution of fibrosis	Basal	Basal and subpleural
Pleural calcification	May be present	Uncommon



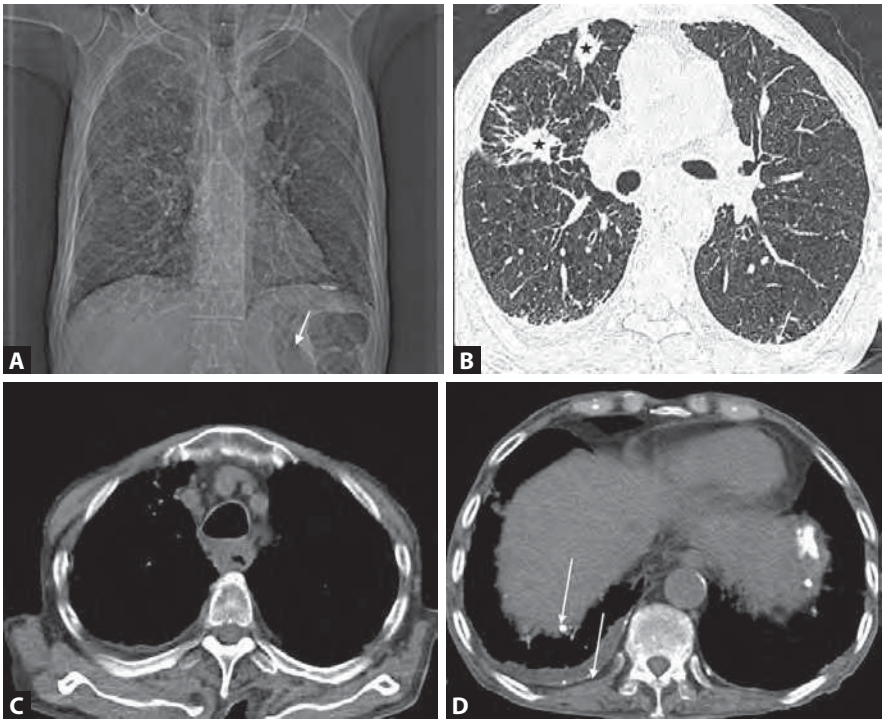
Figs 12.11A to D: Asbestosis. The chest radiograph (A) shows honeycombing in both lung bases. CECT chest axial soft tissue window (B) and HRCT lung window (C and D) reveal pleural calcification and a rounded pleural based mass (arrow). Also note the interlobular septal thickening in the basal segments of both lungs

Pleural disease

- *Pleural effusion*
  - Earliest manifestation but infrequently seen
  - Arise 10–15 years following the commencement of exposure
  - Unilateral/bilateral
  - Pleural fluid-exudative
  - Resolves with pleural thickening (affecting visceral pleura)
- *Pleural plaques*
  - Refers to focal irregular areas of pleural thickening, mostly involving the parietal pleura **(Figs 12.12A to F)**
  - Arise 20–30 years following the commencement of exposure[4]
  - The most common finding.
  - Along posterior or lateral chest wall (6th to 10th ribs) and along central diaphragm.
  - Apices and costophrenic angles are relatively spared.
- However costophrenic angles are involved in those with *diffuse pleural thickening.*



Figs 12.12A to F: Asbestosis with extensive pleural calcification. CT scout film (A) reveals reticular shadows both lung fields; basal pleural thickening (arrows) and left hemithoracic volume loss. HRCT lung (B) shows extensive interlobular septal thickening with basal and peripheral preponderance (arrows). Axial soft tissue window images (C), Coronal MPR (D), MIP (E) and VRT (F) images reveal sheet like mediastinal pleural thickening (arrow)



Figs 12.13A to D: Asbestosis with development of carcinoma esophagus. CT scout film (A) reveals reticular shadows both lung fields; and basal pleural thickening (arrow). HRCT lung (B) shows extensive architectural distortion and speculated mass like fibrotic lesions in right upper lobe (asterisk). Axial soft tissue window images (C and D) reveal costal pleural thickening and calcification (arrow). The upper thoracic esophagus shows concentric esophageal thickening abutting the posterior wall of trachea

- *Round atelectasis*
  - Often associated with pleural thickening
  - *Synonyms*: Asbestos pseudotumor, Blesolvsky's sign
  - Subpleural rounded/oval mass-like lesion
  - Vessels converge towards the lesion- "Comet-tail sign"
- *Differential diagnosis*: IIPs, CTD related ILD, infective, lymphangitis carcinomatosis
- *Mesothelioma*
  - Thickened, nodular pleura measuring more than 1 cm
  - Circumferential with mediastinal pleura involvement
  - Unilateral pleural effusion.

Chronic Beryllium Disease

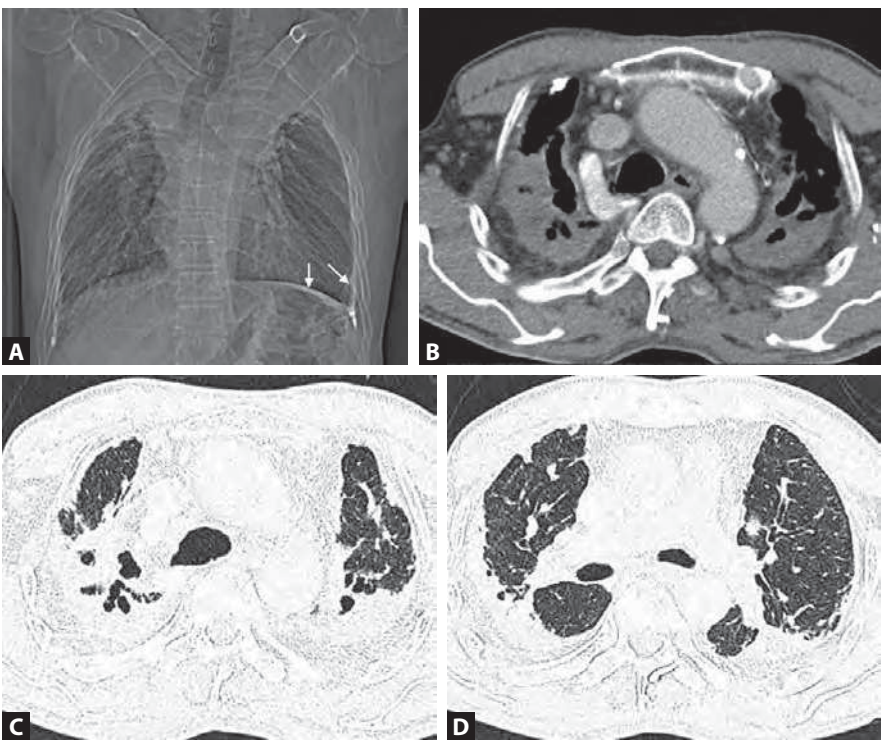
- *Synonyms*: Beryllium related lung disease, berylliosis
- *Industries*: Ceramics, aerospace industry, nuclear reactors, dentistry
- Granulomatous hypersensitivity reaction
- Acute/chronic forms

- Acute form is rare nowadays due to strict safety regulations.
- Severity not always proportional to the intensity or duration of the exposure.
- Imaging findings are similar to sarcoidosis, may be indistinguishable on histopathology also
- *Nodules*: Peribronchovascular
- Bronchial wall thickening
- Interlobular septal thickening
- Ground glass opacities
- Conglomerate masses
- Lymph nodes—less common, seen in about 25%.[4]

Less Common Pneumoconiosis

Talcosis

- Inhalation of hydrated magnesium silicate
- *Industries*: Leather, ceramic, paint, cosmetic
- Also results from intravenous drug abuse of recreational drugs
- *Imaging findings:* Nodules and conglomerate masses **(Figs 12.14A to D)**.



Figs 12.14A to D: Talcoasbestosis. The CT scout film (A) shows bilateral apical pleural thickening and lower lobe hyperinflation; with left sided basal pleural calcification (arrow) reaching the costophrenic angle. CT soft tissue window (B) and lung window HRCT images (C and D) reveal extensive apical pleural thickening and fibro-bronchiectatic changes

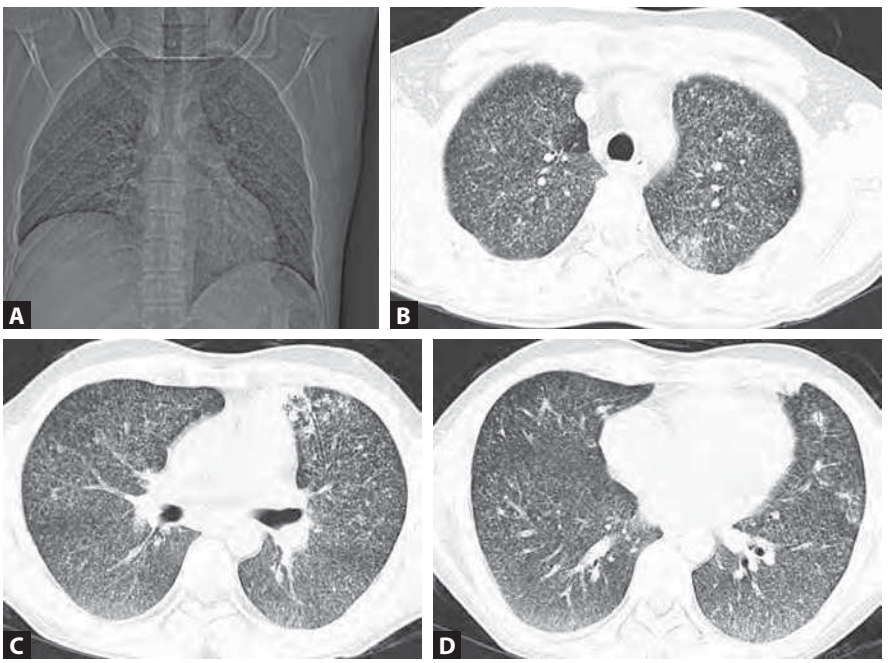
- *Nodules:* Centrilobular, perilymphatic
- Conglomerate, large, fibrotic masses with high attenuation foci due to talc deposition.

Calcicosis

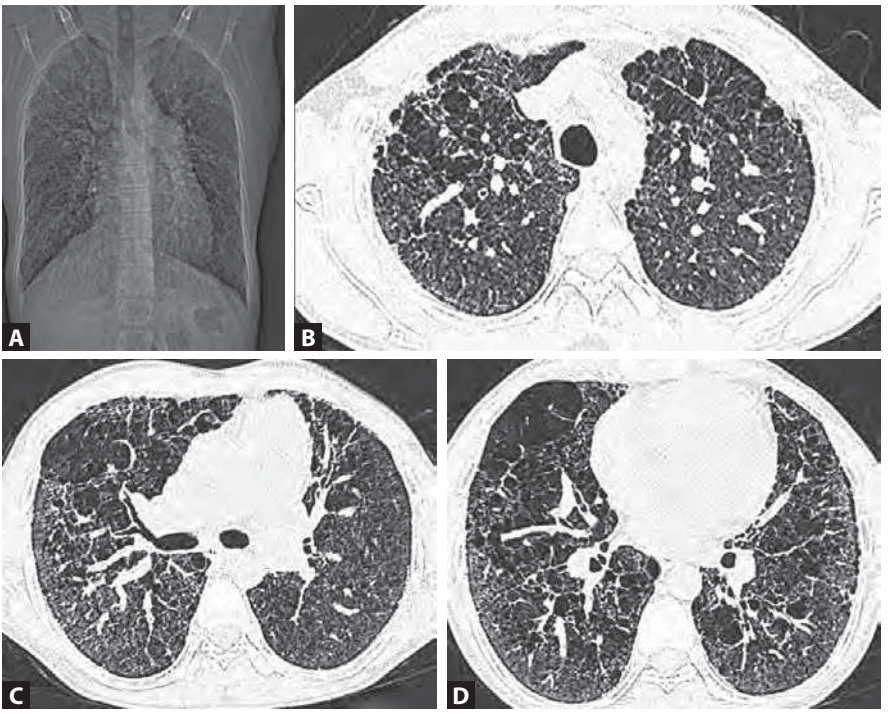
- Limestone dust inhalation –mixed dust with calcium carbonate as its main component, plain limestone is not fibrogenic
- CT features not well described
- Small nodules reported.

Hard Metal Pneumoconiosis (also see chapter 13)

- *Earlier called*: Giant cell interstitial pneumonia
- *Industries*: Hard metal **(Figs 12.15 and 12.16)**
- Inhalation of Tungsten carbide, diamond dust and cobalt
  - Aeronautical, automobile, machinery industries, jewelry and polishing industries
- *Early manifestations*: Airway diseases- asthma, obliterative bronchiolitis
- *Late manifestations*: Interstitial fibrosis
- *CT*: Tiny nodules, ground-glass opacity, interstitial fibrosis
- Lower lobe dominance.



Figs 12.15A to D: Subacute onset of dyspnea in an automobile tool manufacturing industry worker. The chest radiograph (A) and the HRCT chest images (B to D) reveal diffuse centriacinar nodules, and ground-glass opacities



Figs 12.16A to D: Chronic occupational exposure in a diamond polisher. Chest radiograph reveals (A) extensive cystic changes with hyperinflation in both the lungs. HRCT (B to D) revealonterlobular septal thickening and multiple cystic changes with peripheral and subpleural preponderance

Siderosis (Figs 12.17A to D)

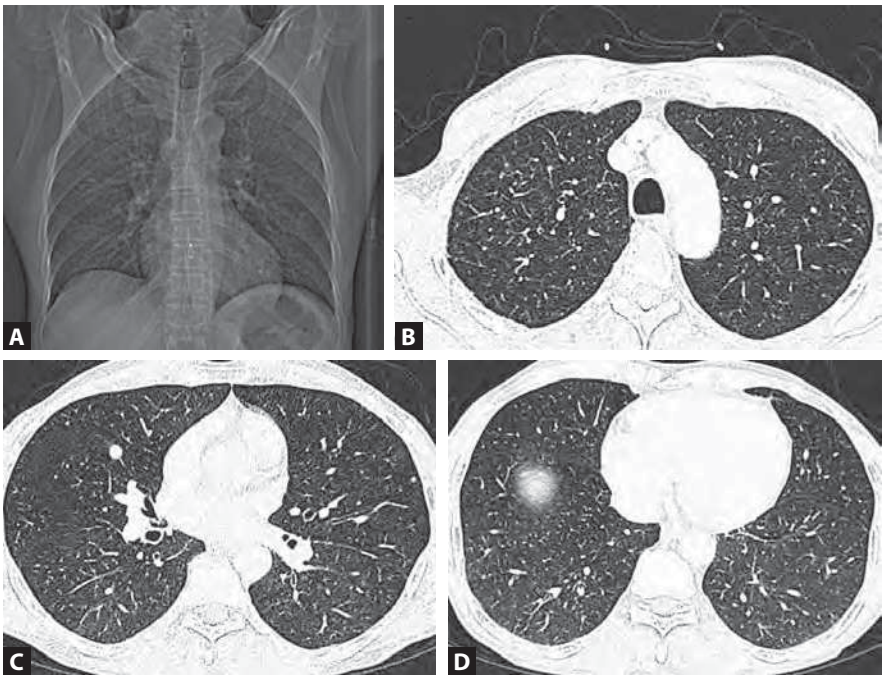
- Iron ore mining, foundry workers, silver polishing
- More common in electric arc and oxyacetylene welders
- *CT*: Nodules-centrilobular, GGO
- Mainly, nonfibrotic form but interstitial fibrosis has been reported in welders.

Kaolin Dust Exposure (Figs 12.18A to D)

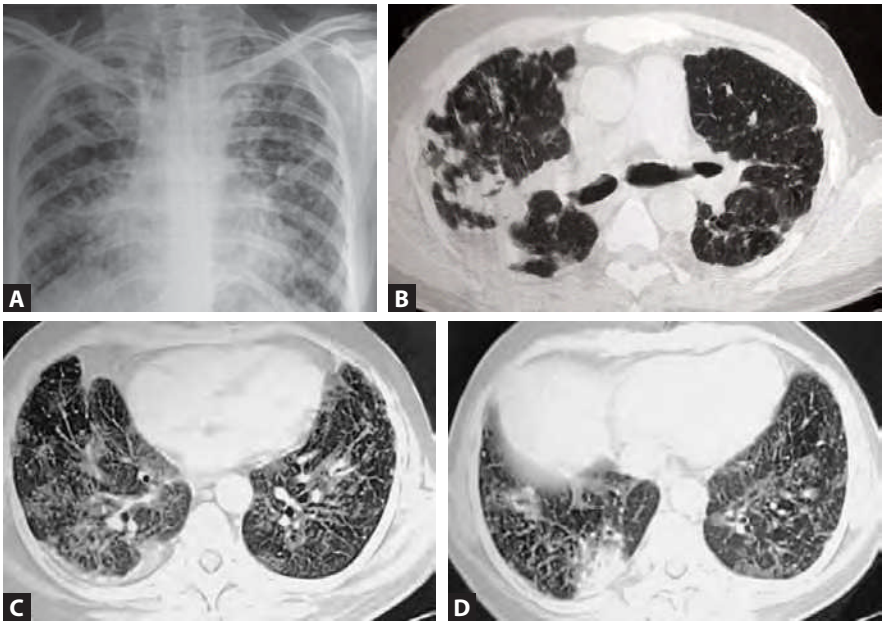
- Brick factory workers exposed
- Mixture of quartz, mica and other constituents
- Results in interstitial fibrosis.

Aluminum Dust Pneumoconiosis (Figs 12.19A to D)

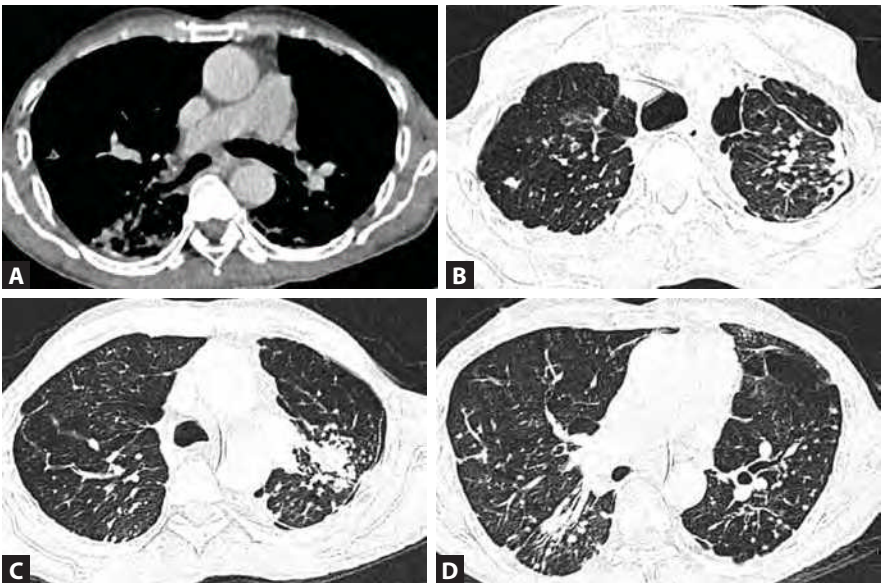
- *Industries:* Aluminum arc-welding, aluminum production and processing
- CT—Ground-glass opacities with DIP or PAP pattern
- Interstitial fibrosis—resembling UIP pattern
- *Nodules*: Centrilobular, upper lobes dominant (D/D silicosis)



Figs 12.17A to D: Siderosis in an electrical arc welder worker. The chest radiograph (A) appears normal; but the HRCT chest images (B to D) reveal diffuse ground-glass opacity in both the lungs; with centriacinar nodules



Figs 12.18A to D: Occupational exposure to kaolin (brick factory). The chest radiograph (A) and the HRCT chest images (B to D) reveal septal thickening and fibrosis in the upper lobes; and perilymphatic nodules in the lower lobes



Figs 12.19A to D: Aluminum exposure. The HRCT images reveal bilateral upper lobe fibrosis with traction bronchiectasis and perilymphatic nodules

CONCLUSION

- A complete coverage of the entire list of pneumoconiosis is beyond the scope of the chapter; here the commoner entities have been covered.
- It is important to remember that in these patients with nonoccupational, environment related exposure, radiologist may be the first to suggest the diagnosis; and a history of exposure may not be available at the outset.
- This is because patients themselves are not aware of the exposure.

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13

Organic Dust Exposure and Hypersensitivity Pneumonitis: Imaging

Manisha Jana, Ashu Seith Bhalla

INTRODUCTION

- Organic dust and chemical antigens are responsible for a wide variety of occupational lung diseases.
- The mechanism of injury may be direct or immune mediated.
- The immunologically mediated entities include hypersensitivity pneumonitis and occupational asthma.
- A complete list of the reported offending agents/industries or occupations is beyond the scope of this chapter
- More frequently encountered conditions are discussed below.

IMMUNOLOGICALLY-MEDIATED ENTITIES

Hypersensitivity Pneumonitis

- *Synonym:* Extrinsic allergic alveolitis.
- Group of disorders resulting from repeated aerosol exposures and sensitization to a variety of inhaled aerosols (organic as well as chemical). Inorganic antigens responsible are usually low molecular weight particles.
- *Pathology:* Granulomatous disorder involving lung interstitium as well as terminal airways.
- Both type III and type IV hypersensitivity reactions are responsible.
- Diagnosis based on combination of clinical and radiographic features, along with history of exposure to offending antigen.
- *Work related exposures causing hypersensitivity pneumonitis:* Organic dust in farming, isocyanates in paint, contaminated metal working fluids, plastics, avian (bird droppings), farmer's lung.
- Only a minority of people exposed to these antigens develop HP.
- Also, in several patients the offending antigen cannot be identified through clinical history.
- Hence, diagnosis can be challenging.

- Classically clinico-radiographically presentation divided into acute, subacute and chronic.
- Several patients however have an insidious onset of the disease.[1,2]

Imaging Findings

Do not always correlate with the stage of the clinical presentation–features of acute and subacute overlap. Also these findings may be superimposed on the chronic phase.

Chest Radiograph

- Often normal
- Ill-defined nodular opacities (<5 mm); relative sparing of apices and bases
- Airspace opacities, ground-glass opacity (GGO)/infrequently consolidation, patchy/diffuse **(Figs 13.1 and 13.2)**.



Fig. 13.1: Acute hypersensitivity pneumonitis on chest radiograph. There is diffuse ground glass opacity in both lung fields



Fig. 13.2: Acute hypersensitivity pneumonitis on CT scout film. There are multiple ill-defined airspace nodules in both lung fields

- *Distribution*: Variable
- *Chronic HP*: Reticular pattern–fine, with or without honeycombing, volume loss
- Upper lobe predominant **(Figs 13.3 and 13.4)**

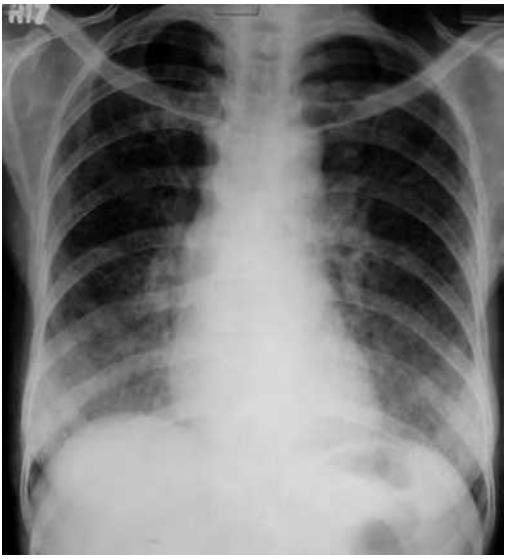


Fig. 13.3: Subacute hypersensitivity pneumonitis on chest radiograph. There is diffuse ground-glass opacity and reticular shadows in both lung fields lower zones

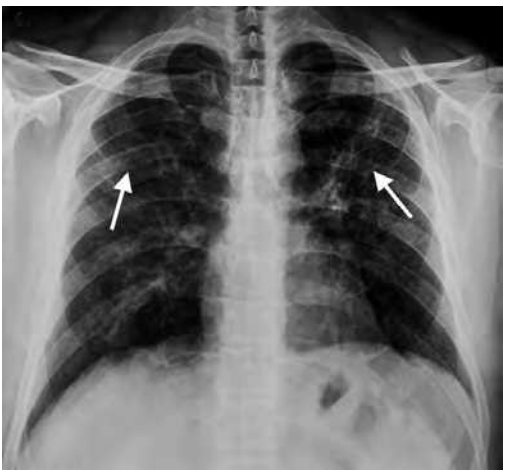


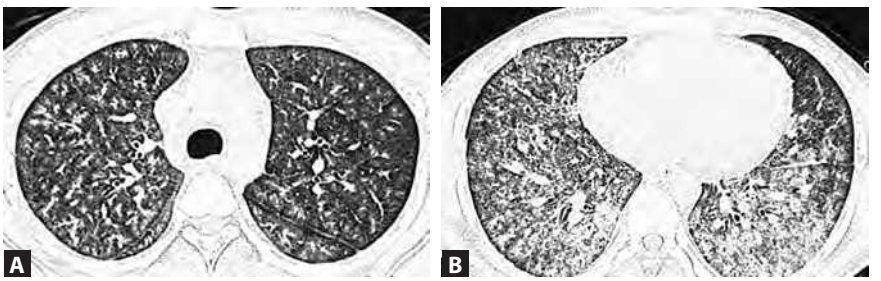
Fig. 13.4: Chronic hypersensitivity pneumonitis on chest radiograph, manifested by presence of reticular opacities (arrows) with upper lobe predominance

High-resolution Computed Tomography (HRCT)

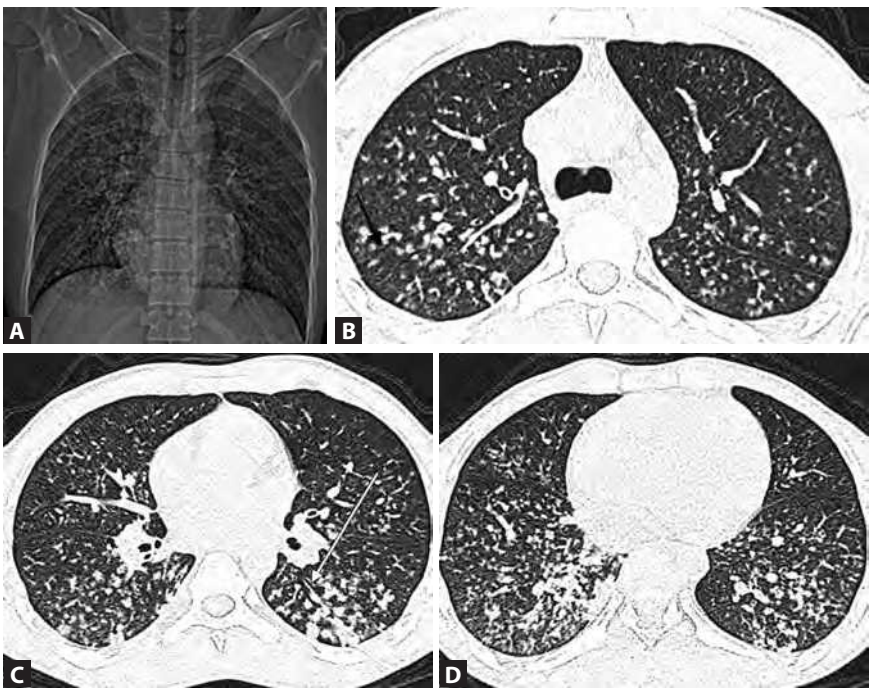
Significantly superior to chest radiograph.

Acute phase (Figs 13.5 and 13.6)

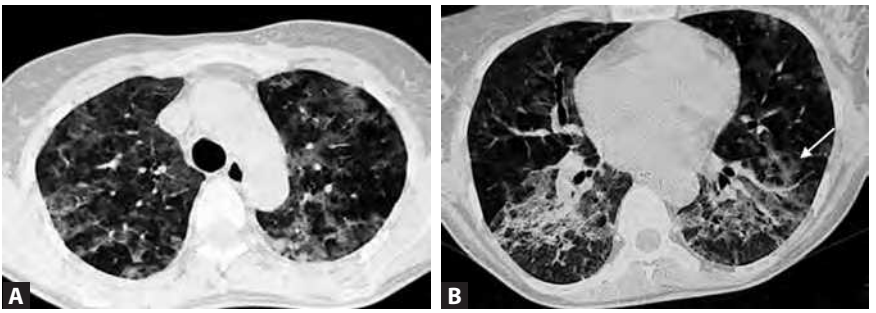
- *Poorly defined nodules:* Centrilobular, <5 mm size, often ground glass nodules
- Diffuse ground glass opacities (GGO), may be patchy, more in midzones/along bronchovascular bundles
- Mid/lower lobe dominance with sparing of lung bases.



Figs 13.5A and B: Acute HP, manifested as diffuse ill-defined centriacinar nodules in both lungs; having a lower lobe and dependent area preponderance



Figs 13.6A to D: Extrinsic allergic alveolitis in mosquito Repellent ('All out') Poisoning. The CT scout film (A) reveals diffuse ill defined airspace in both lung fields. HRCT chest (B to D) reveal acinar nodules (arrow) in both lungs; confluent at places; and peribronchial thickening (long arrow)



Figs 13.7A and B: Subcute HP. The HRCT images (A and B) show classical 'head cheese' pattern. This pattern results from a combination of ground glass opacities, air trapping and interspersed normal lung

Subacute phase (Figs 13.7A and B)

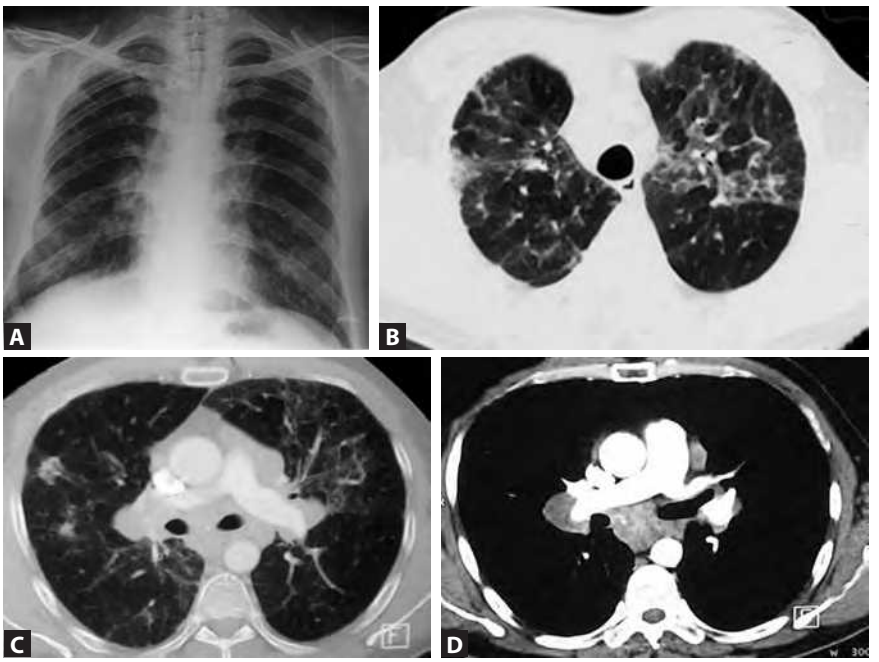
- *Poorly defined nodules:* Centrilobular
- *GGOs*: Diffuse/patchy (DD: Cryptogenic organizing pneumonia in peribronchovascular distribution).
- Bronchiolar wall thickening
- Areas of air trapping
- Lobular areas of low attenuation—"head-cheese sign". Head-cheese sign results from combination of GGOs, areas of air trapping and intervening normal lung
- *Consolidation:* Less common
- *Cysts*: Seen in few (<10%), thin walled
- Mediastinal adenopathy (about 50%)
- DD: Respiratory bronchiolitis-interstitial lung disease (RB-ILD).

Chronic phase (Figs 13.8 and 13.9)

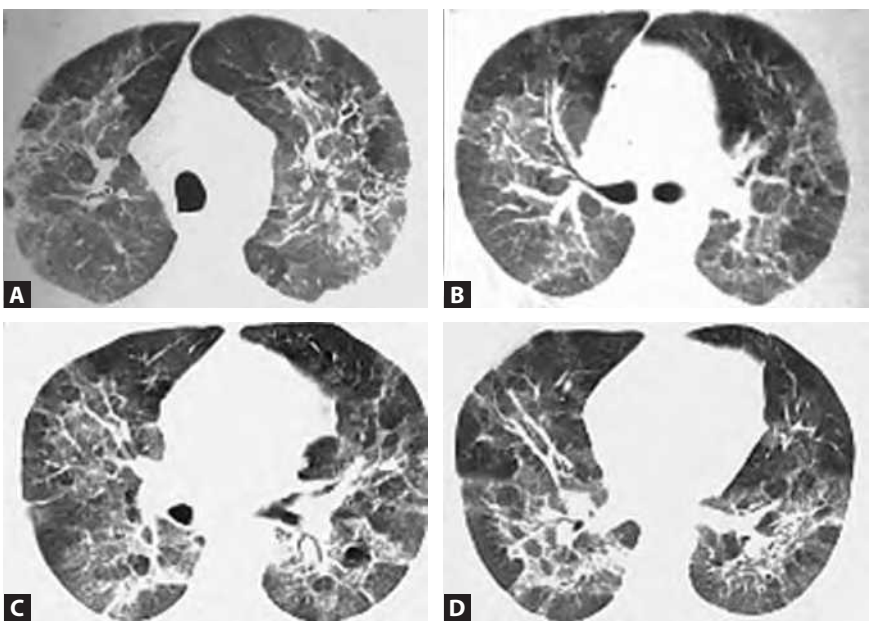
- *Lung fibrosis:* Reticular opacities, architectural distortion, bronchiectasis and bronchiolectasis, honeycombing, volume loss.
- *Honeycombing:* Subpleural and midzone; when in bases resembles UIP.
- Presence of fibrosis is a poor prognostic indicator with increased mortality.
- Emphysema, cysts—in farmer's lung or bird fancier's lung
- Fibrosis has upper lobe predominance
- Pulmonary artery hypertension
- DD:
  - Usual interstitial pneumonia (UIP) in those with honeycombing
  - Nonspecific interstitial pneumonia (NSIP) in those with GGOs, reticulation, traction bronchiectasis.

Occupational Asthma

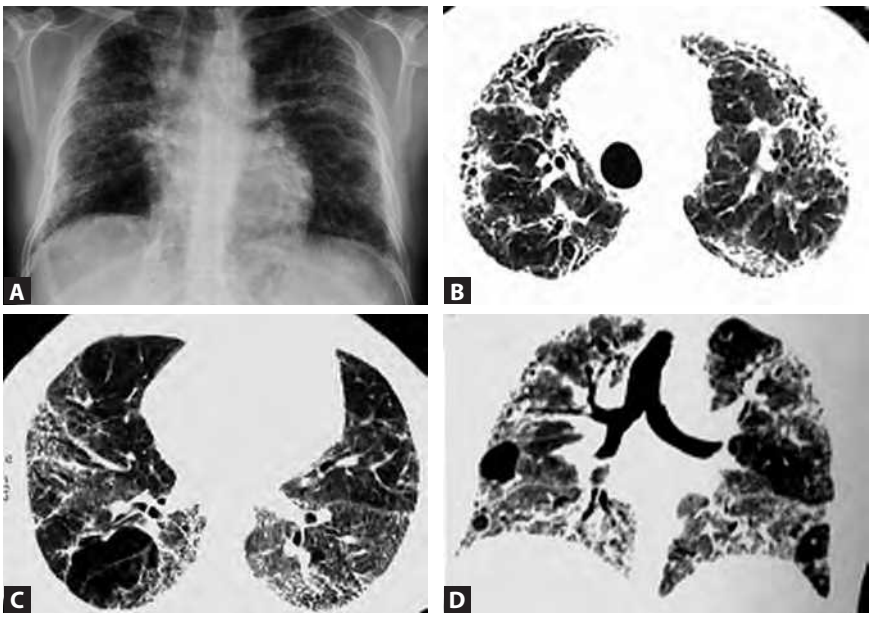
- Two different terms included in the description of 'work related asthma'.
- 'Work exacerbated asthma' refers to patients who have pre-existing asthma which is worsened due to nonspecific work-related stimuli.
- 'Occupational asthma' results from exposure to aerosols/dusts/vapours encountered in the work environment.



Figs 13.8A to D: Chronic HP. The chest radiograph shows fine reticular shadows at both lung bases (A). HRCT images (B and C) show interlobular septal thickening and GGO. Classical 'head cheese' pattern. This pattern results from a combination of ground glass opacities, air trapping and interspersed normal lung. Axial soft tissue window image (D) reveals mediastinal and right hilar lymphadenopathy



Figs 13.9A to D: Chronic HP. HRCT images show interlobular septal thickening, fibrosis and air trapping with a central and upper lobe predominance



Figs 13.10A to D: Farmer's lung. Chest radiograph (A) shows reticular shadows in peripheral subpleural distribution. HRCT images (B to D) show interlobular septal thickening, fibrosis, traction bronchiectasis and honeycombing in bilateral peripheral subpleural distribution

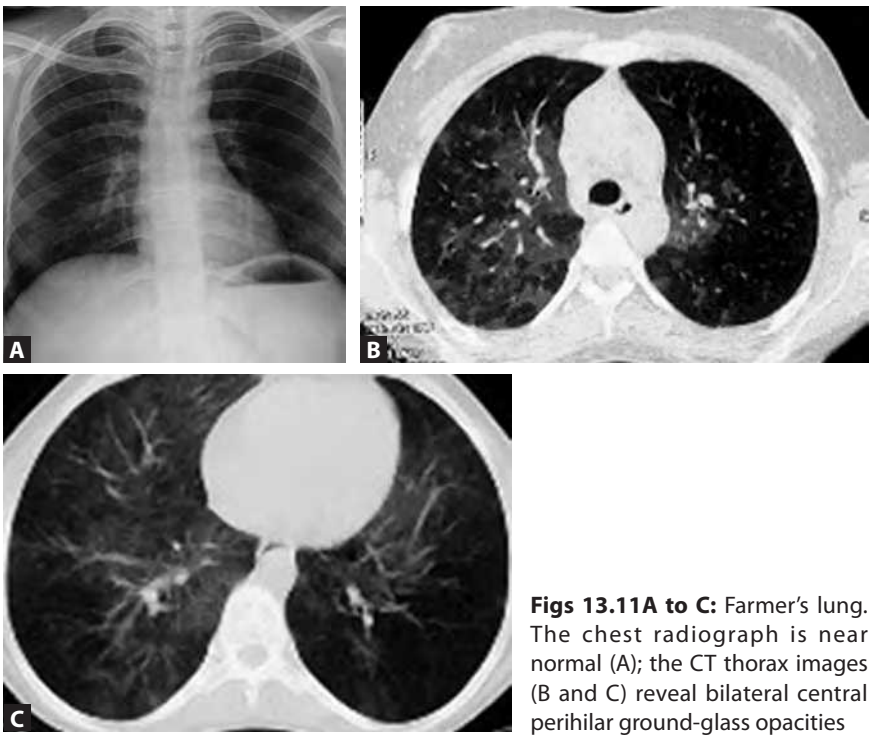
- Thought to be the *most common form of occupational lung disease*
- *Occupational asthma is further divided into two types*: Immunological or nonimmunological
- *Immunological:* IgE mediated, latency period is present
- *Nonimmunological:* Due to direct irritation, no latency period.
- Hundreds of agents implicated in a wide range of occupations.
- *Occupations:* Spray paints, farming, hairdressing or even nurses.
- *Agents:* High/low molecular weight including proteins, latex, isocyanides.
- A comprehensive list of these are beyond the purview of this chapter.
- *Imaging:* Bronchial wall thickening, hyperinflation and air trapping, peribronchial cuffing, atelectasis.

Other forms of HP include farmer's lung, bird fancier's lung, hot tub lung, humidifier lung, malt worker's lung, cheese worker's lung, mushroom worker's lung and several more **(Figs 13.10 and 13.11)**.

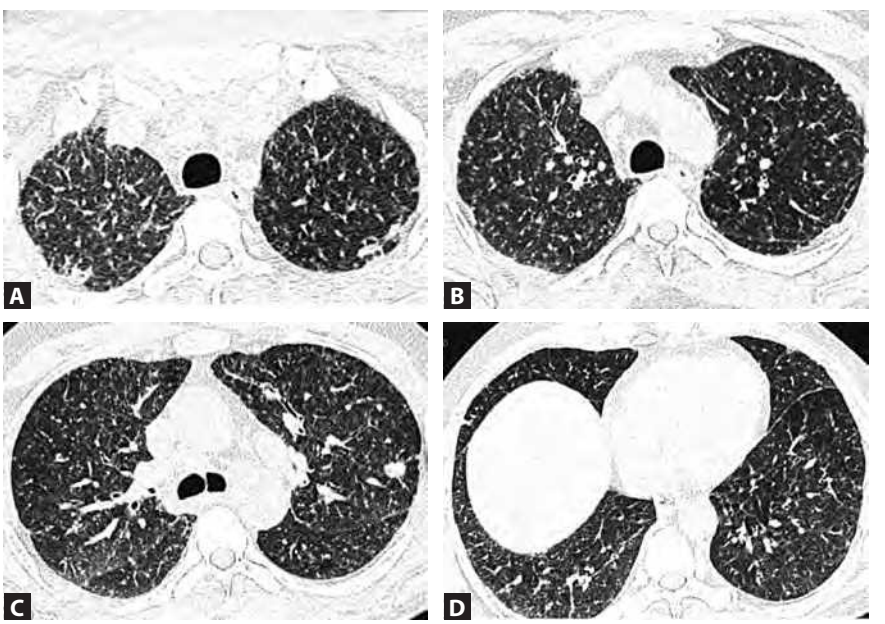
DIRECT EXPOSURE RELATED

Organic Dust Toxic Syndrome

- *Exposure:* Organic dust
- Acute febrile illness following exposure without evidence of HP
- *Examples:* Cotton fever (byssinosis) **(Figs 13.12A to D)**, humidifier fever (office goers), grain fever
- *Imaging findings:* GGOs, centrilobular nodules, lower zone dominance.



Figs 13.11A to C: Farmer's lung. The chest radiograph is near normal (A); the CT thorax images (B and C) reveal bilateral central perihilar ground-glass opacities



Figs 13.12A to D: Byssinosis. HRCT images show diffuse centrilobular nodules scattered over both lungs, and interspersed septal thickening at the apices

Chemical Pneumonitis/Airway Disease

- *Exposure:* Acute exposure to inhalable noxious chemical substances, may be organic or inorganic
- *Organic:* Organophosphates, polymer fumes, smoke.
- *Inorganic:* Hydrogen sulfide, sulphur dioxide, nitrogen oxide
- *CT in acute phase:* Centrilobular nodules, GGOs. Pneumonitis when exposed to high doses.
- Bronchiolitis obliterans like picture may develop weeks to months following the acute exposure.
- *CT in chronic phase:* Air trapping/mosaic perfusion, bronchiolectasis.

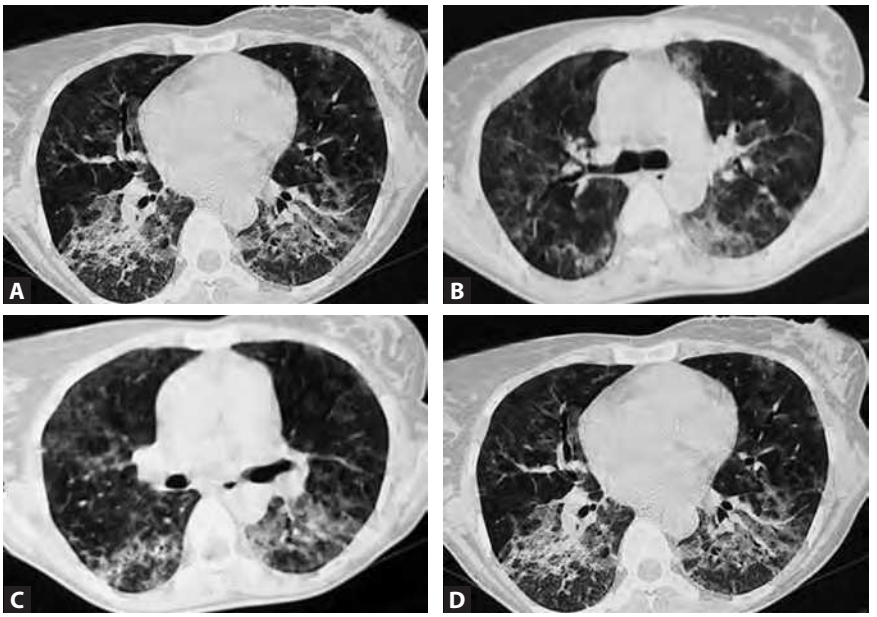
Biomass Fuel Exposure

- Still a concern in developing countries
- It is estimated that up to half of world's population still uses biomass as a source of energy for cooking or heating
- Biomass fuel is made of various combinations of wood, dung cakes, straw, shrubs, kerosene and coal.
- On burning, biomass emits toxic fumes which are composed of carbon monoxide, small solid particles, hydrocarbons and formaldehyde
- These fumes are especially harmful when burnt in small, confined spaces. There are several names given to these disorders depending on the type of exposure, e.g. hut lung, Gujjar lung.
- Chronic exposure to these fuels can lead to a variety of respiratory illnesses
- These include respiratory infections, chronic obstructive pulmonary disease (COPD), asthma, anthracosis and lung carcinoma.
- Imaging features **(Figs 13.13 and 13.14)** change depending on the predominant pattern of involvement and include
  - Hyperinflation, mosaic attenuation, GGO
  - Bronchial wall thickening and cuffing
  - Centrilobular nodules (changes of SAD).

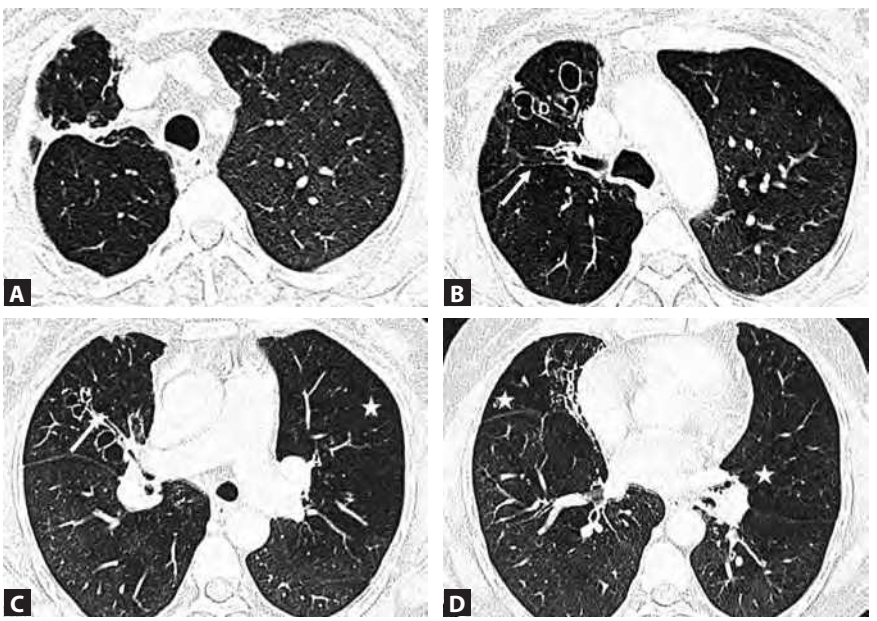
See also Chapter no. 15.

Anthracofibrosis[3]

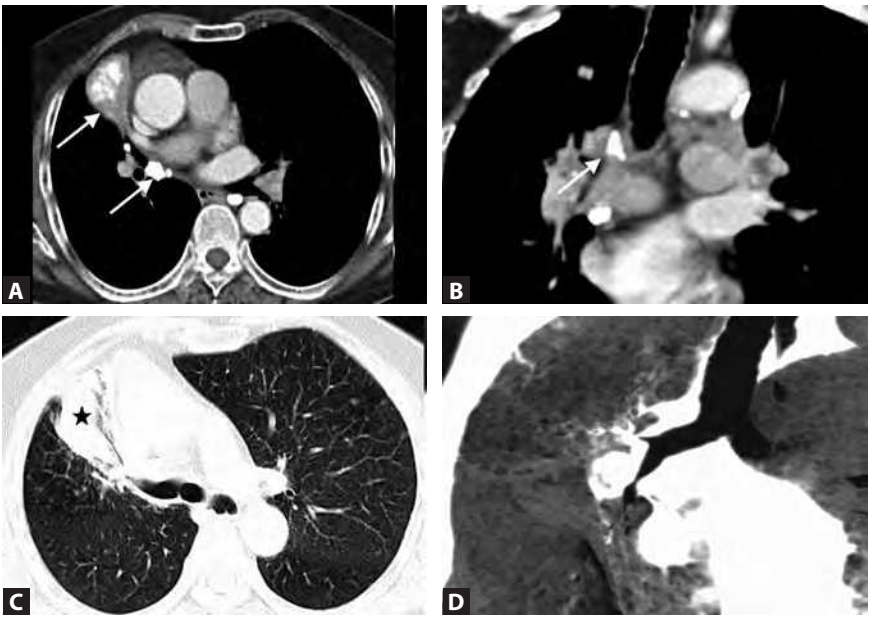
- Recently recognized entity, over last decade or so
- Anthracosis refers to black pigment deposition seen in the bronchial mucosa on bronchoscopy
- It results from deposition of carbon, silica or other air pollutants
- When resulting in fibrosis (bronchial stenosis) it is referred to as anthracofibrosis
- The term is used when there is no occupational exposure to silica/coal
- It has been reported in middle aged/ elderly adults especially females
- This gender predilection is because of its association with biomass fuel exposure



Figs 13.13A to D: Acute exposure to *chullah* smoke. CT images reveal classical 'head cheese' pattern



Figs 13.14A to D: Chronic exposure to *chullah* smoke. CT images reveal upper lobe fibrosis, traction bronchiectasis (arrows) and lower lobe compensatory emphysematous changes (asterisk)



Figs 13.15A to D: Anthracofibrosis. The CT imaging findings include calcified mediastinal nodes and a pulmonary mass with calcification (arrows in A and B); right middle lobe bronchial narrowing and RML collapse (C, D) (asterisk). Tubercular bronchial stricture is an imaging differential for this entity

- It is often associated with active TB/sequelae of TB
- CT findings are predominantly of bronchial stenosis, peribronchial thickening and calcified nodes **(Figs 13.15A to D)**
- However lung changes are also reported in this condition
- Pulmonary changes include atelectasis/collapse (most commonly right middle lobe), conglomerate masses as in progressive massive fibrosis (PMF), and also multiple nodules
- DD: Silicosis, tuberculosis, carcinoma lung
- The nodes of anthracosis may show uptake on 18FDG PET CT further confusing the diagnosis
- There is however no occupational exposure to coal/silica.

Giant Cell Interstitial Pneumonitis (also *See* Chapter 12)

- GIP is characteristic finding on pathology of hard metal lung
- *Hard metal:* A compound made of tungsten carbide, cobalt
- Workers of tools composed of hard metals
- *Exposure:* Causes obliterative bronchiolitis
- *Pathology*: Constrictive bronchiolitis
- *Imaging findings*: Chest radiograph: Normal/hyperinflated lung fields

- *HRCT:* Air trapping/mosaic attenuation
  - Bronchial wall thickening, bronchiectasis
  - Volume loss of upper lobes
  - Subpleural nodules.

Table 13.1 enumerates some of the organic dust exposures, both immunologically and non-immunologically mediated.

Table 13.1: Organic dust exposures and their clinic radiopathologic features

Disease	*Occupation/agent*	*Clinic radiopathologic findings*
Farmer's lung	Agricultural products (e.g. hay dust, mold spores, thermophilic actinomycetes, aspergillus)	Hypersensitivity pneumonitis (type III pattern) *Imaging*: Similar to HP
Bird fancier's lung/ pigeon breeder's lung	Bird droppings/feathers Avian proteins	Similar to HP
Byssinosis/brown lung disease/Monday fever	Cotton dust, especially in closed environment Effect direct/due to gram-negative bacteria Raw cotton, flax, hemp	Direct irritant Symptoms most marked on first day of work week Imaging not well described Ill defined nodules
Biomass exposure	Biomass smoke	Direct irritant COPD (emphysema) May show GGO, mosaic attenuation, peribronchovascular thickening, fibrosis Pulmonary hypertension
Flavor Worker's lung	2–3 butanedione, a diacetyl used in flavoring of microwave popcorn	*Pathology*: Constrictive bronchiolitis *Imaging findings—chest radiograph*: normal/ hyperinflated lung fields *HRCT*: Air trapping/mosaic attenuation bronchial wall thickening, bronchiectasis (volume loss of upper lobes, subpleural nodules)
Indium	Television manufacturing Used in the coatings	Pulmonary alveolar proteinosis type picture Crazy paving pattern *Less common*: Nodules (CLN or perilymphatic) Subpleural emphysematous blebs

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14

Occupational/Exposure-related DLDs: Clinical Diagnosis and Management Issues

Anant Mohan

INTRODUCTION

- Evolving field due to newer industrial/environmental exposures and recognition of diseases emerging out of exposure.
- Occupational interstitial lung diseases (ILDs) comprise four main clinical entities[1-3]*: Hypersensitivity pneumonitis, pneumoconiosis, other immune-mediated granulomatous conditions such as beryllium associated ILD, and diffuse interstitial fibrosis.*
- Safety regulations variable across countries.
- Need for regular screening programs in those persons exposures with known high risk of lung injury, fibrotic/carcinogenic.
- Incidence of common exposures encountered will vary depending on area of practice
- *Common occupational agents include*: Silica, asbestos, coal mine dust, beryllium, organic dust as in farming activities, toxic chemical fumes (e.g. nitrogen dioxide and phosgene), smoke from combustion, and bird droppings.
- Silicosis and coal workers pneumoconiosis are the most prevalent occupational ILDs.
- *Common nonoccupational/environmental exposures*: Pets, bird feathers/droppings, dust from furnishings, humidifiers, indoor hot tubs, and fungi.

CLINICAL PRESENTATION

- Variable latency periods, usually years in the case of pneumoconioses
- Pulmonary manifestations are predominant
- Extrapulmonary manifestations seen occasionally in some forms, e.g. beryllium disease or silica induced renal or autoimmune disease
- Occasionally may involves the airways and present with asthma or bronchitis symptoms.

Initial assessment: Four criteria to be satisfied:
1. Establishing evidence of structural lesion consistent with the interstitial process (e.g. fibrosis) by chest radiograph and/or HRCT scan.
2. To elicit an agent—disease relationship.
3. Evidence of exposure to an agent known to cause occupational ILD, including a dose deemed sufficient to cause the disease.
4. Exclusion of possible alternative diagnoses.

HISTORY

- Comprehensive exposure history is critical to diagnosis.
- Occupational exposure history needs to be detailed—nature of exposure, duration, temporal relationship to onset of complaints.
- Owing to long latencies, all previous occupations/exposures should be sought, keeping in view the possibility of recall bias
- Establishing causal relationship important not only for management, but also for medicolegal (compensation) implications.
- Difficult to identify offending agent in cases of nonoccupational environmental exposures, since exposures rarely occur in isolation.
- Knowledge of common regional exposures and local practices important in such cases.
- Presence of rheumatological/connective tissue diseases/autoimmune/inflammatory bowel disease make the diagnosis of occupational/environmental ILD less likely, although concomitant diseases may occur
- History of cigarette smoking, including passive smoke exposure should always be elicited.
- Pneumoconioses typically become clinically apparent over a period of years
- Hypersensitivity pneumonitis may present acutely within a few weeks or may be delayed up to several months.
- Most common symptoms are dyspnea, cough, chest tightness and wheezing.
- Hemoptysis/weight loss in those with superadded infections/malignancy.

PHYSICAL EXAMINATION

- Physical findings may be entirely normal or nonspecific and often appear late.
- Assessment of respiratory status (e.g. rate, depth, use of accessory muscles, nasal flaring).
- *In those with interstitial fibrosis*: Basal inspiratory crackles on auscultation, digital clubbing (asbestosis, IPF).
- Presence of wheezing and/or end-inspiratory rhonchi on auscultation signifies airway disease.
- In later stages—features of right heart failure (raised JVP, loud P2, pedal edema, pleural effusions)
- Assessment for rheumatological/autoimmune disorders, e.g. erythema nodosum (Sarcoidosis), joints/skin/eye for connective tissue diseases.

LABORATORY INVESTIGATIONS

Serological tests are important supportive evidence in excluding other close differentials such as autoimmune diseases, e.g. sarcoidosis.

PULMONARY FUNCTION TESTS/SPIROMETRY

- Essential part of work-up.
- Aid in diagnosis, management decisions and follow-up (For details *See* Chapter 4)
- Additional testing for diffusion capacity of carbon monoxide (DLCO) is desirable
- Based on standard interpretations of the forced vital capacity (FVC), forced expiratory volume in one second (FEV1), and FEV1/FVC ratio, a diagnosis of restrictive, obstructive or mixed defects may be obtained depending on the type of occupational lung disease
- Spirometry in isolation is not specific for any type of ILD
- Correlation with CT picture enhances utility and diagnostic accuracy.

SPUTUM EXAMINATION, BRONCHOSCOPY, BRONCHOALVEOLAR LAVAGE, TRANSBRONCHIAL LUNG BIOPSY, LUNG BIOPSY

- Sputum microscopy may help identify ferruginous bodies seen in asbestosis
- Bronchoalveolar lavage (BAL) is seldom required, especially in those with known history of exposure.
- When indicated, a targeted BAL based on high-resolution computed tomography (HRCT) segment localization should be performed.
- BAL fluid must be subjected to cellular counts. A predominant inflammatory cellular pattern frequently narrows the differential diagnosis of ILD.
- BAL may be useful for asbestosis and transbronchial lung biopsy (TBLB) useful in those with granulomatous lung reaction, e.g. hypersensitivity pneumonitis, berylliosis.
- Pathologic examination of lung tissue may at times be required in atypical settings, particularly to exclude treatable nonoccupational disorders or malignancy.
- Surgical biopsy indicated occasionally in those with unresolved diagnosis.

SPECIFIC ENTITIES

Silicosis

- Most commonly occurs after a long latent period of 10–40 years
- Occasionally may present acutely in case of larger exposures
- Gradually progressive dyspnea and productive cough
- PFT reveals a mixed pattern of obstruction and restriction with a reduced diffusion capacity
- Increased risk for developing active tuberculosis, COPD, and lung cancer

- Lung biopsy usually not required; if done, reveals round, hyalinized nodules known as a "silicotic nodules"
- No therapy, including steroids, has any proven benefit
- Exposure should be removed/discontinued
- Screen patients with tuberculin test; evaluate further for treatment if reading is >10 mm.

Asbestosis

- Latent period from exposure to symptoms ranges from 15–40 years
- Exertional dyspnea, dry cough are predominant symptoms
- Physical examination reveals bibasilar crackles and occasionally clubbing
- Evidence of cor pulmonale occurs in advanced disease
- *PFT*: Restrictive pattern; impaired DLCO
- Lung biopsy rarely required if compatible history, radiographic and spirometric findings are present
- Disease progresses in 20–40% cases depending on severity and cumulative exposure
- No pharmacotherapy is effective
- Management includes removal from exposure and supportive care, including.
  - Pneumococcal and influenza vaccinations
  - Treatment of intercurring respiratory infections
  - Supplemental oxygen when required
  - Diuretics for cor pulmonale
  - Pulmonary rehabilitation.

Chronic Beryllium Disease

- Associated with exposure to occupations involving electronics, defense machinery, automotives, nuclear weapons, metal recycling, etc.
- Highly variable latent period from 2 months to more than 40 years
- May be asymptomatic or presents with exertional dyspnea, dry cough and constitutional symptoms
- Digital clubbing and cor pulmonale occur in advanced stages
- Increased propensity to develop lung cancer
- Similar to sarcoidosis on radiology and lung biopsy histopathology
- *PFT:* May be normal or show any form of restrictive/obstructive/mixed defect
- *Supportive lab evidence*: Sensitivity to beryllium demonstrated in blood/BAL fluid shown at blood—beryllium lymphocyte proliferation test (BeLPT)
- Corticosteroids form the mainstay of treatment.

Hypersensitivity Pneumonitis/Extrinsic Allergic Alveolitis

Definition

A syndrome caused by an exaggerated immune response to the inhalation of a variety of antigenic particles found in the environment.

Diagnostic Criteria[4,5]

1. Hypersensitivity pneumonitis (HP) is a pulmonary disease which may or may not be accompanied by systemic manifestations (e.g. fever and weight loss)
2. Caused by the inhalation of an antigen to which the individual is sensitized and hyperresponsive
3. Defined by exposure to a given antigen, sensitization to this antigen, and the presence of clinical symptoms.

Clinical Features

Acute

- Results from intense exposure to the inciting agent
- Presents with flu-like symptoms
- Tends to improve if antigen is avoided.

Subacute

- Continues over weeks to a few months
- Results from prolonged low-level antigen exposure
- Insidious fever, weight loss, fatigue
- Generally, a progressive condition.

Chronic

- Insidious course over months to years
- Slowly progressive cough, exertional dyspnea, fatigue, and weight loss
- May progress to end—stage fibrosis and cor pulmonale.

Identifying the Offending Agent

A variety of agents are implicated and are broadly divided into five main groups: Bacteria (e.g. thermoactinomyces vulgaris causing farmers lung), mycobacteria (e.g. *Mycobacterium avium* intracellulare causing hot tub lung), fungi (e.g. *Aspergillus* sp. causing mushroom workers lung), animal proteins (e.g. bird droppings causing pigeon breeders lung), and chemicals (e.g. diisocyanates causing chemical workers lung).

Laboratory Investigations

- *PFT*: Usually restrictive pattern during symptomatic episodes
- *Serum precipitins*: IgG antibodies against a potential agent
- Inhalational challenge of the patient to the suspected culprit agent
- *Bronchoalveolar lavage*: Most sensitive tool to detect alveolitis; lymphocyte count > 20% and up to 50% of total cell count
- *Transbronchial/surgical lung biopsy*: Demonstrate typical histopathological findings such as granulomas and cellular infiltrate, especially in subacute and chronic HP.

Management

- Avoidance of exposure of the suspected agent is most essential
- Indications for corticosteroids:
  - Acute symptomatic episodes
  - Progressive subacute or chronic HP
  - May not affect long-term outcome
- *Lung transplantation*: For chronic advanced HP progressing to pulmonary fibrosis.

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Section 5

Cystic Lung Diseases

- Chronic Obstructive Airway Disease and Asthma: Imaging
- Chronic Obstructive Airway Disease and Asthma: Clinical Diagnosis and Management Issues
- Other Cystic Lung Diseases: Imaging

15

Chronic Obstructive Airway Disease and Asthma: Imaging

Chandrashekhara SH, Mukesh Kumar Yadav

INTRODUCTION

- According to Global Initiative for Chronic Obstructive Lung Disease (GOLD), chronic obstructive pulmonary disease (COPD) is 'disease characterized by airflow limitation that is not fully reversible'
- COPD caused by chronic inhalation of noxious particles resulting in airway remodeling and emphysema
- *COPD reflects a complex syndrome comprising of following diseases*:
  - Pulmonary emphysema
  - Chronic bronchitis
  - Small airways disease
  - Nonpulmonary causes leading to COPD—cardiac disease, neurologic-cognitive dysfunction, and musculoskeletal disorders
- Recognizing the type of COPD is important as it influence management. Chronic bronchitis need medical management and emphysema require surgery.
- Computed tomography (CT) has been used to differentiate between airway-predominant and emphysema-predominant COPD
- *Pulmonary function tests:* FEV_1: FVC ratio should be <0.70 for COPD
- Many types of classification and patterns of COPD has been described.

CLINICAL PHENOTYPES

Several clinical phenotypes have been described for COPD, which are as follows:

- Emphysema predominant
- Airways predominant
  - Small airways predominant
  - Large airways predominant
- Mixed.

CLASSIFICATION OF CHRONIC OBSTRUCTIVE PULMONARY DISEASE

- Three COPD phenotypes were identified based on morphologic CT changes and clinical features of COPD (based on the presence or absence of apparent emphysema and bronchial wall thickening).
  1. Phenotype A—no or minimal emphysema with or without bronchial wall thickening:
  2. Phenotype E—emphysema without bronchial wall thickening
  3. Phenotype M—emphysema with bronchial wall thickening. It shows best response to bronchodilators.

CLINICAL FEATURES

- Symptoms—dyspnea on exertion, wheezing, productive cough and hemoptysis
- Emphysema patients are known as "pink puffers". Chronic bronchitis patients are called as "blue bloaters,"(as cyanosis related to hypoxemia)
- *Cor pulmonale:* Right heart failure caused secondary to COPD.

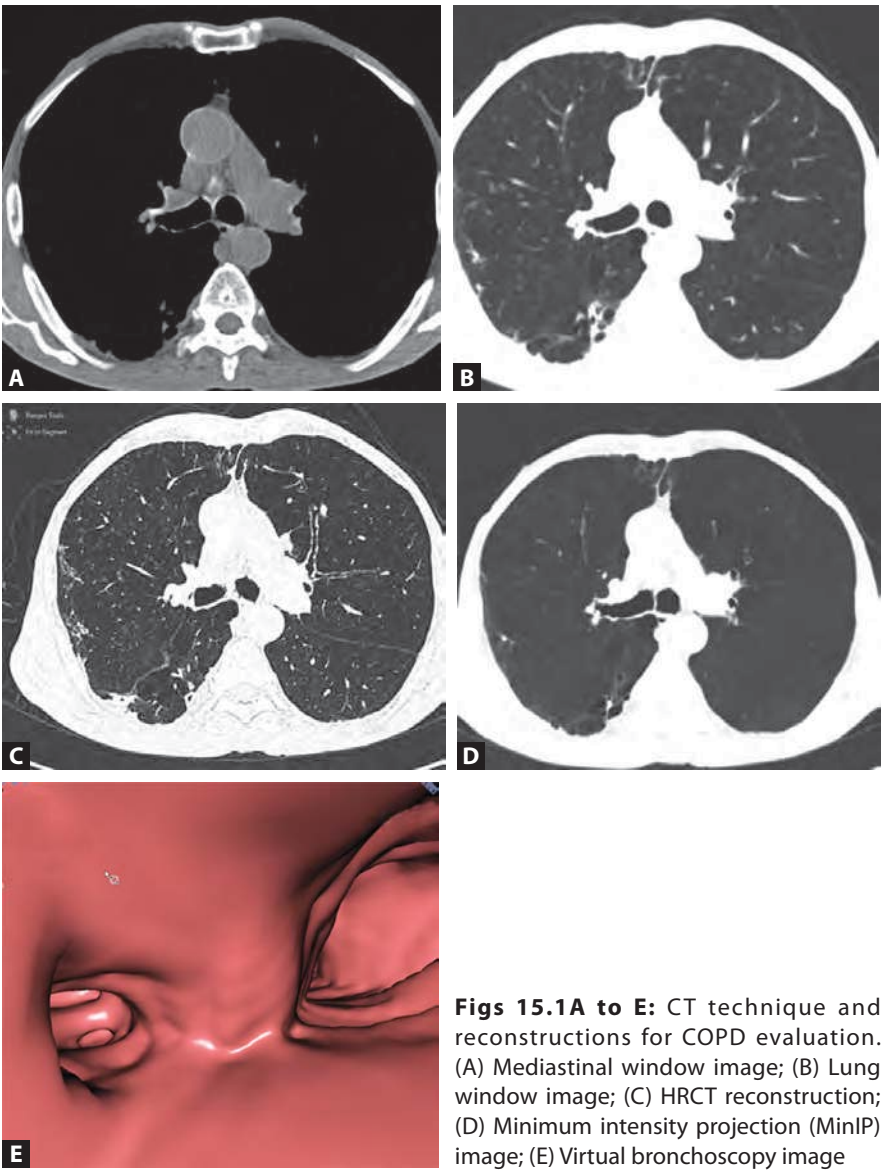
RADIOGRAPHIC FEATURES

Plain Radiograph

- Chest radiograph findings are nonspecific, which may show increased bronchovascular markings; sometimes cardiomegaly may be seen
- Emphysema seen as lung hyperinflation, flattened hemidiaphragms, small heart and bullous changes
- Lateral chest radiograph may show "barrel-shaped chest" with widened anterior-posterior diameter
- "Saber-sheath Trachea" sign - marked coronal narrowing of the intrathoracic trachea (on PA view) with sagittal widening (on lateral view).

CT Protocol and CT Features

- Images should be acquired with high signal-to-noise ratio
- Thin sections (0.5–1 mm reconstructions) are to be required for proper evaluation
- Noncontrast-enhanced volumetric CT acquisition is a standard technique
- Two separate sets of reconstructions are essential for CT assessment
- CT images should be viewed at window level settings which are appropriate for lung assessment (window level:700, window width:1500). A narrower window (1000 or 800) may be useful for detecting or excluding early emphysema. Mediastinal window and virtual bronchoscopy images may also be helpful in the disease evaluation **(Figs 15.1A to E)**.
- Acquisition at full inspiration is required for quantification of pulmonary emphysema
- Appropriate breathing instructions during the scanning are of paramount importance



Figs 15.1A to E: CT technique and reconstructions for COPD evaluation. (A) Mediastinal window image; (B) Lung window image; (C) HRCT reconstruction; (D) Minimum intensity projection (MinIP) image; (E) Virtual bronchoscopy image

- Sometimes image noise may resemble emphysema
- Reducing the dose might be appropriate for CT quantification of pulmonary emphysema
- Iterative reconstruction (IR) techniques help in reducing image noise in chest imaging. But they alter the quantitative assessment of emphysema and air trapping

- Expiratory/inspiratory ratio of mean lung density (E/I-ratio$_{MLD}$)—preferred method for assessment of air trapping
- Minimum intensity projection (MinIP) technique—help in assessing mild emphysema and airways
- In COPD patients, the number of small airways are reduced on CT
- Narrowing and loss of terminal bronchioles subsequently lead to emphysematous destruction in COPD
- Significant wall thickening of large airways is seen in chronic bronchitis patients
- Small airway obstruction is assessed by evaluation the air trapping: retention of air in the lung distal to an obstruction that can be evaluated on expiratory CT scans
- COPD patients show excessive expiratory tracheal collapse
- COPD patients demonstrate pulmonary artery hypertension; show narrowing and significant reduction of small pulmonary arteries.

Assessing Disease Severity

- Visual qualitative and semiquantitative assessment—subjective bias, less accurate
- Visual quantitative assessment time-consuming
- Quantitative image analysis—more direct and objective method; based on digital data
- Quantitative HRCT helps in objectively assessing the severity of disease activity.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE PATTERNS ON COMPUTED TOMOGRAPHY

The various patterns of COPD are described on CT as follows:*

- Emphysema (more than 6% of pixels <-950 HU by quantitative CT)
  - Centrilobular emphysema
    - Trace centrilobular emphysema (involving <0.5% lung zones, upper lobe predominance)
    - Mild centrilobular emphysema (involving 0.5–5% lung zones upper lobe predominance)
    - Confluent centrilobular emphysema (involving more than 5% lung zones, upper lobe predominant, diffuse)
    - Advanced destructive emphysema (Diffuse, lower lobe predominant)
  - Panlobular emphysema (A1AT deficiency-related emphysema; lower lobe predominance)
  - Paraseptal emphysema
    - Mild paraseptal emphysema (small <1 cm cyst like lucencies)
    - Substantial paraseptal emphysema (large> 1cm cyst like lucencies/ bullae)

- Airway disease (less than 6% of pixels <-950 HU by quantitative CT)
  - Bronchial disease—thickening of segmental and subsegmental bronchial walls
  - Small airway disease
- *Associated features*:
  - Large airway disease—tracheobronchomalacia, saber-sheath trachea, tracheobronchial outpouching/diverticula
  - Interstitial lung abnormality
  - Pulmonary artery hypertension
  - Bronchiectasis.

  **Modified from CT-Definable Subtypes of Chronic Obstructive Pulmonary Disease: A Statement of the Fleischner Society.*

EMPHYSEMA

- Emphysema is caused by permanent dilatation of the airspace distal to the terminal bronchioles with associated destruction of the alveolar wall
- There are three types, i.e. centrilobular, panlobular and paraseptal emphysema **(Fig. 15.2)**
- In patients with COPD, destruction of the lung parenchyma leads to the loss of alveolar attachments to the small airways and a resultant decrease in elastic recoil, which causes small airway collapse during expiration
- CT pick-up the early-stage disease before airflow obstruction or alteration in diffusing capacity
- The extents of emphysema do not always correlate with the severity of airflow limitation
- The relative low-attenuation area (i.e. the area with attenuation below a specific threshold value) measured at expiratory CT correlates more closely with the severity of airway obstruction than does that measured at inspiratory CT.

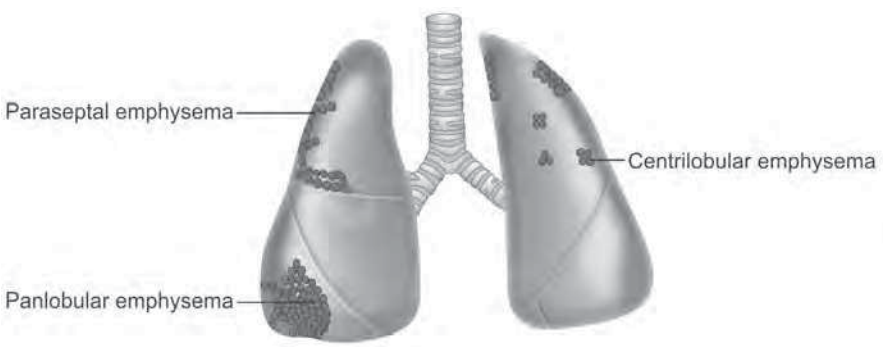


Fig. 15.2: Schematic diagram showing types of emphysema

CT Protocol

- The scans should be acquired at end inspiration
- Expiratory scans are not of much value in the evaluation of emphysema
- Minimum-intensity projection (MinIP) images increase sensitivity for detecting mild disease.

CT Assessment of Emphysema

- CT quantification analysis more accurate at grading extent of disease compared to subjective visual grading
- Attenuation values with cut off –950 HU represent emphysema
- Usually lower lobe emphysema lead to significant functional impairment
- Central emphysema affects more diffusion lung capacity than peripheral emphysema.

Types of Emphysema

Centrilobular Emphysema

- Most common type
- It usually affects lobules around the central respiratory bronchioles
- *Location:* Typically seen in the upper lung zone adjacent to the pleura and interlobar fissures
- It is frequently associated with spontaneous pneumothorax in young adults
- It is strongly associated with smoking.

Imaging

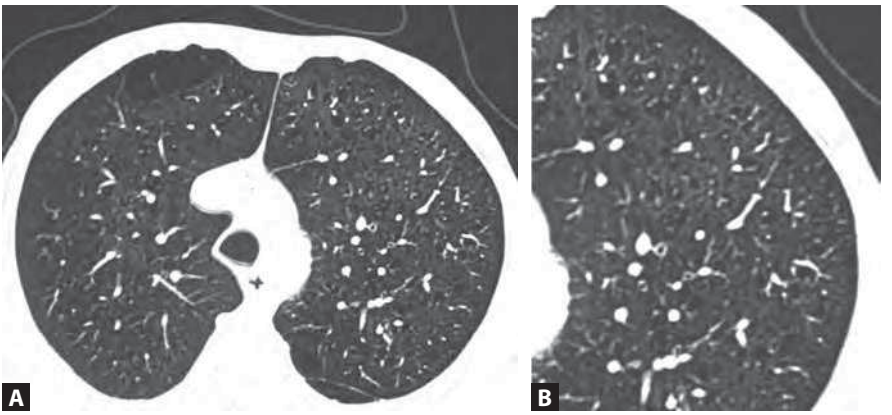
Chest Radiograph

Chest radiography is may be normal as it is insensitive to early disease:

- In advanced disease lung may show heterogeneous density with distortion of vascular markings
- Remaining vessels may show increased branching angle
- Associated hyperinflation with flattening of hemidiaphragm may be seen
- Increased AP diameter of thorax, widened retrosternal space with narrow-shaped heart may be seen.

HRCT

- Well-defined round lucencies in centrilobular portion of secondary pulmonary lobule on HRCT **(Figs 15.3A and B)**.
- Anatomical borders of secondary pulmonary lobule are preserved
- Well-defined margins between normal and emphysematous lungs result in heterogeneous attenuation
- *Location:* Upper lung zone predominant, lung apices, lower lobe superior segments
- CT is more sensitive than chest radiography and detect early disease
- Low-attenuation areas located in center of secondary pulmonary lobule
- Borders of secondary pulmonary lobule are preserved



Figs 15.3A and B: CT in centrilobular emphysema. Axial HRCT lung window images showing focal areas of decreased attenuation in the center of secondary lobules

- Pulmonary artery hypertension seen as enlarged central pulmonary arteries, peripheral arterial pruning.

Associated Abnormalities

- Chronic bronchitis
- Hyperinflation
- Respiratory bronchiolitis
- Secondary pulmonary hypertension.

Differential Diagnosis

- Panlobular emphysema
- Paraseptal emphysema
- Constrictive bronchiolitis
- Langerhans cell histiocytosis
- Cystic lung disease
- Asthma.

Panlobular Emphysema

- Uniformly affects the whole of secondary lobule
- Seen as generalized diffuse decrease in CT attenuation, mainly in the lower lobe
- The vessels in affected lung are decreased in number and caliber
- Associated with α_1-antitrypsin deficiency, it also may be seen in advanced smoking-related emphysema and emphysema related to intravenous drug abuse
- Affects the whole secondary lobule
- It is seen in α_1-antitrypsin deficiency and smokers with advanced emphysema
- Sometimes it becomes difficult to appreciate mild to moderate disease
- Associated with more severe small airway obstruction.

Imaging

Chests Radiograph

- Very insensitive
- May appear normal
- Advanced disease seen as hyperinflation, flattened diaphragm, increased retrosternal space with narrow heart.

HRCT

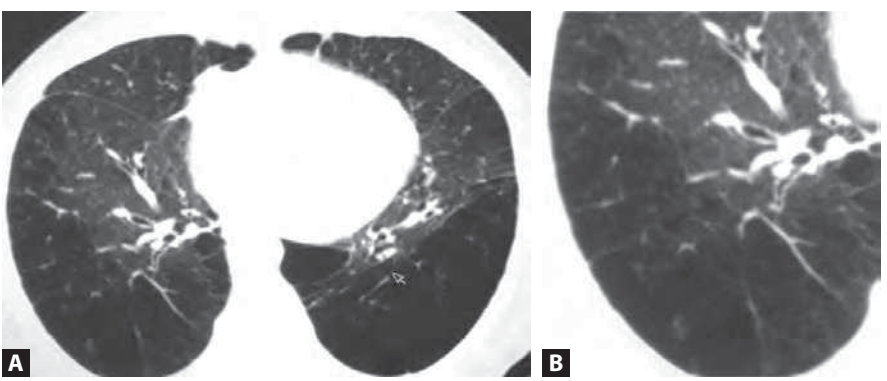
- Best evaluate and assess of PLE
- Show ill-defined areas of low attenuation and a paucity of vessels
- Sometimes it is difficult to distinguish between normal lung and PLE
- Homogeneously distributed in lungs with lower lobe predominance **(Figs 15.4A and B)**
- Vascular distortion and disruption, a paucity of vascular markings
- Hyperinflation
- Flat hemidiaphragms
- Increased AP diameter of thorax
- Increased retrosternal/retrocardiac spaces
- Associated bronchiectasis may be seen.

Associated Abnormalities

- Recurrent infection
- Chronic bronchitis
- Secondary pulmonary hypertension.

Differential Diagnosis

- Paraseptal emphysema
- Centrilobular emphysema
- Cystic lung disease
- Asthma.



Figs 15.4A and B: CT in panlobular emphysema. Axial CT lung window sections revealing the diffusely randomly distributed focal areas of decreased attenuation in the lungs

Paraseptal Emphysema

- Characterized by permanently enlarged distal acinus with destruction of alveolar ducts and sacs; destruction pattern located in the periphery of the lung adjacent to the pleura or along the interlobular septa
- Single or multiple bullae circumscribed, well demarcated thin-walled dilated airspace measuring > 1 cm; subpleural in location
- It may occasionally occur as an isolated finding
- Association with tall and thin body habitus—postulated gravitational forces with increased negative pressure at lung apices as cause
- Higher incidence in smokers, IV drug users
- Associated with Marfan and Ehlers-Danlos syndromes.

Imaging

Chest Radiograph

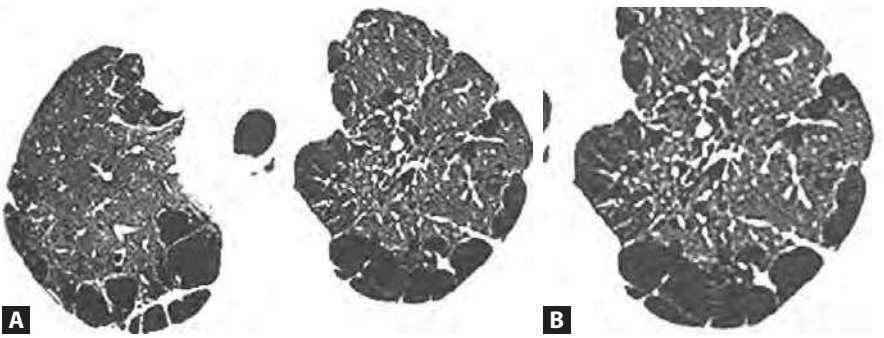
- Radiography may be normal
- Peripheral lung lucencies with thin walls.

HRCT

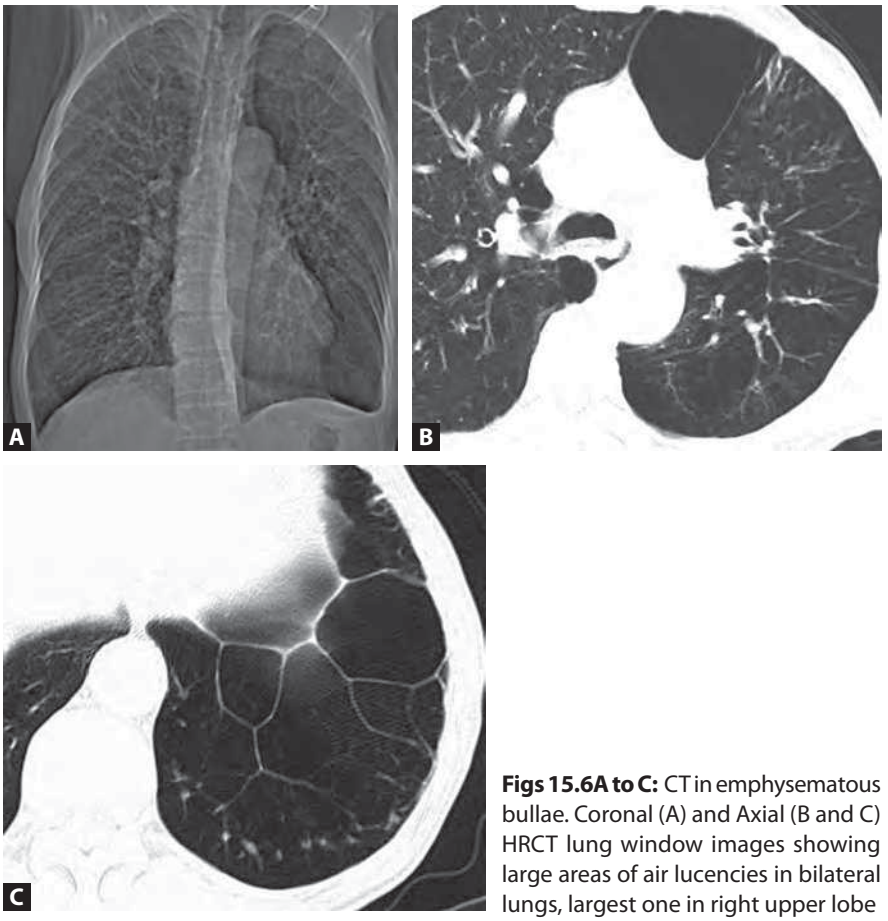
- Small subpleural lucencies arranged as single tier of "cystic" spaces **(Figs 15.5A and B)**
- *Location:* Upper lung zone predominant
- Subpleural and peribronchovascular arcades of cystic spaces
- Cystic spaces separated by intact interlobular septa
- Large emphysematous bullae may be seen **(Figs 15.6A to C)**
- Apical bullae may cause spontaneous pneumothoraces
- Minimum-intensity projection (MinIP) images may improve detection
- Quantitative data may be obtained from CT densitometry software .

Differential Diagnosis

- Cystic lung disease
- Centrilobular emphysema



Figs 15.5A and B: CT in paraseptal emphysema. Axial HRCT lung window sections of different patients showing peripherally located cysts along the pleura and fissures



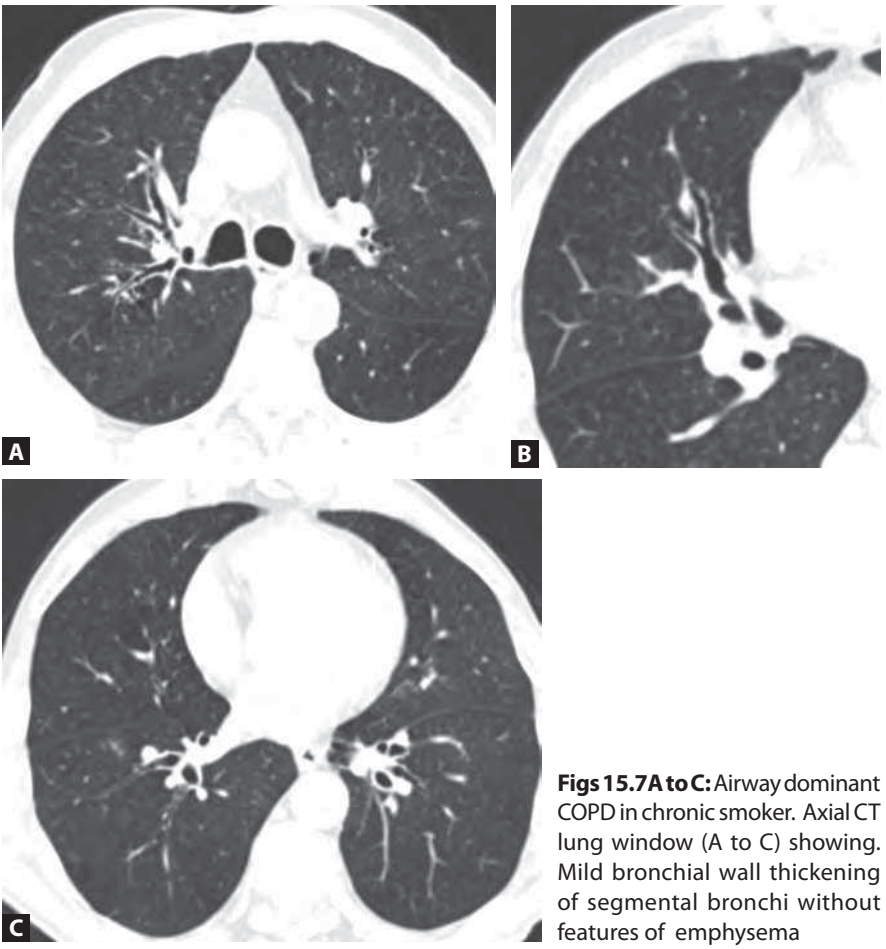
Figs 15.6A to C: CT in emphysematous bullae. Coronal (A) and Axial (B and C) HRCT lung window images showing large areas of air lucencies in bilateral lungs, largest one in right upper lobe

- Panlobular emphysema
- Honeycomb lung (Pulmonary fibrosis).

AIRWAY DISEASE

Bronchial Wall Thickening

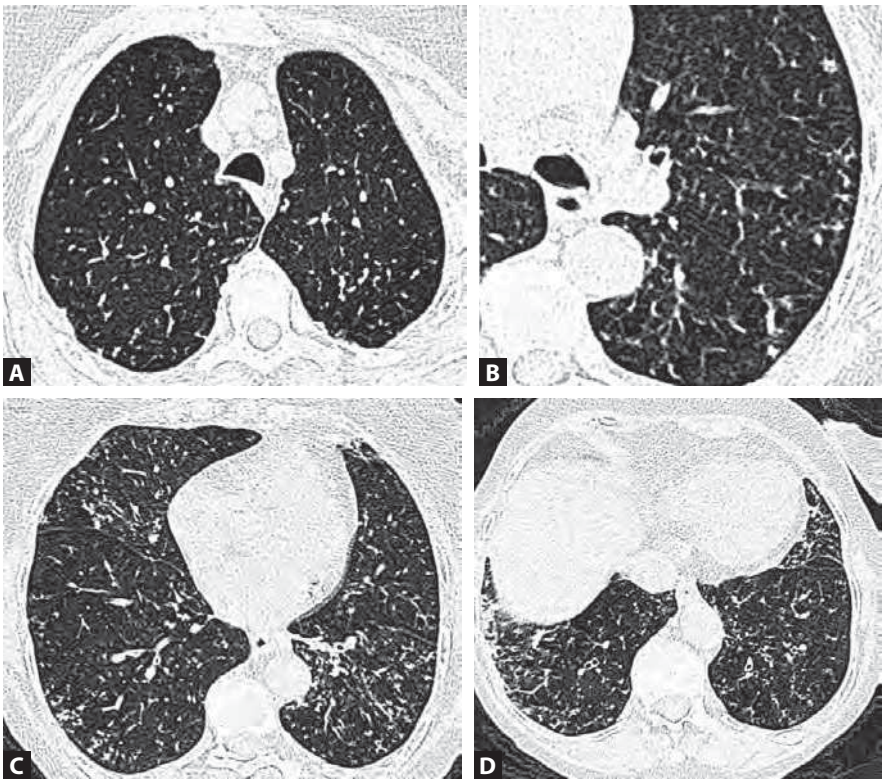
- Commonly seen in chronic cigarette smokers, those with chronic bronchitis
- CT show bronchial wall thickness more than adjacent vessel **(Figs 15.7A to C)**
- Chest radiograph in chronic bronchitis may be nonspecific. It may show bronchial wall thickening with increased bronchovascular markings
- Quantitative CT may provide an estimate of small airway remodeling
- In COPD, bronchial wall thickening may predict the abnormalities of FEV and risk of acute exacerbation.



Figs 15.7A to C: Airway dominant COPD in chronic smoker. Axial CT lung window (A to C) showing. Mild bronchial wall thickening of segmental bronchi without features of emphysema

Small Airways Disease

- Small airway disease may be seen in COPD
- *Direct signs on CT:* Tubular, branching linear structures representing diseased small airways
- *Indirect signs on CT:* Lung changes secondary to small airway disease, which include air trapping, atelectasis, centrilobular emphysema, and air-space nodules
- *Types of small airway disease:*
  - Inflammatory small airway disease
    - It occurs due to inflammation around the small airways in COPD patients
    - CT show clusters of centrilobular nodules of ground-glass attenuation **(Figs 15.8A to D)**
  - *Obstructive small airway disease*: Seen as gas trapping at expiratory CT.



Figs 15.8A to D: CT in small airway disease. Axial HRCT scan (A to D) show multiple centrilobular nodules in bilateral lungs. Mild bronchiectasis and fine fibrosis seen at places

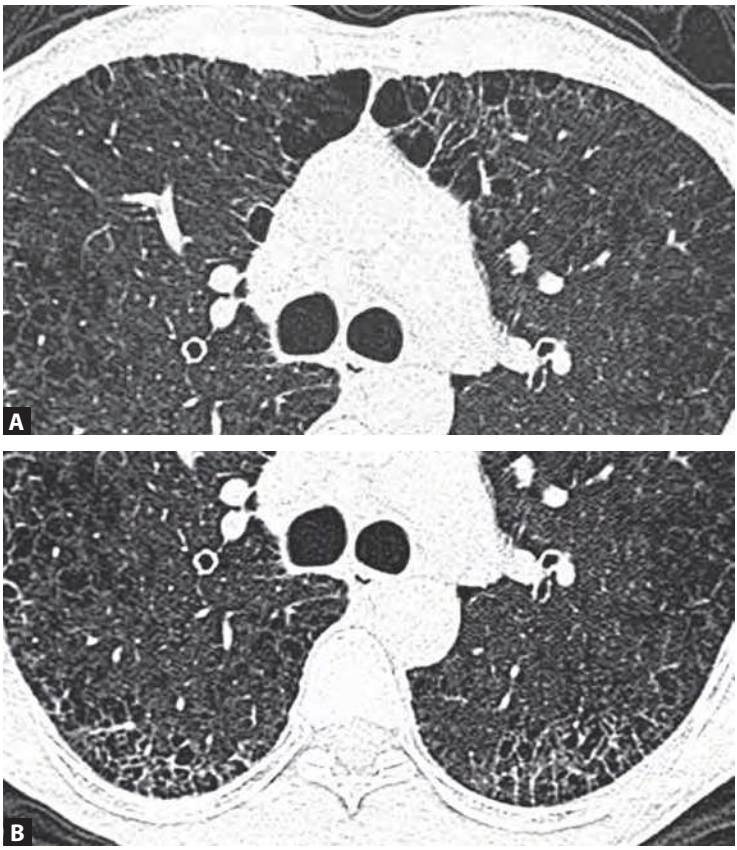
ASSOCIATED FEATURES

Interstitial Lung Abnormalities

- Cigarette smokers may show abnormalities compatible with infiltrative lung disease, including ground-glass opacities and reticular abnormalities
- There may be occurrence of emphysema and pulmonary fibrosis, especially in smokers, which constitute combined pulmonary fibrosis and emphysema (CPFE)
- CPFE show upper-lobe emphysema and lower-lobe fibrosis/interstitial thickening **(Figs 15.9A and B)**.
- Interstitial abnormalities may show various combinations of imaging features such as respiratory bronchiolitis, fibrosis and interstitial fibrosis/thickening.

Pulmonary Vascular Disease

- Pulmonary hypertension may occur as complication of COPD
- Chronic hypoxia and acidosis in COPD cause vascular contraction leading to pulmonary vascular resistance resulting in pulmonary artery hypertension

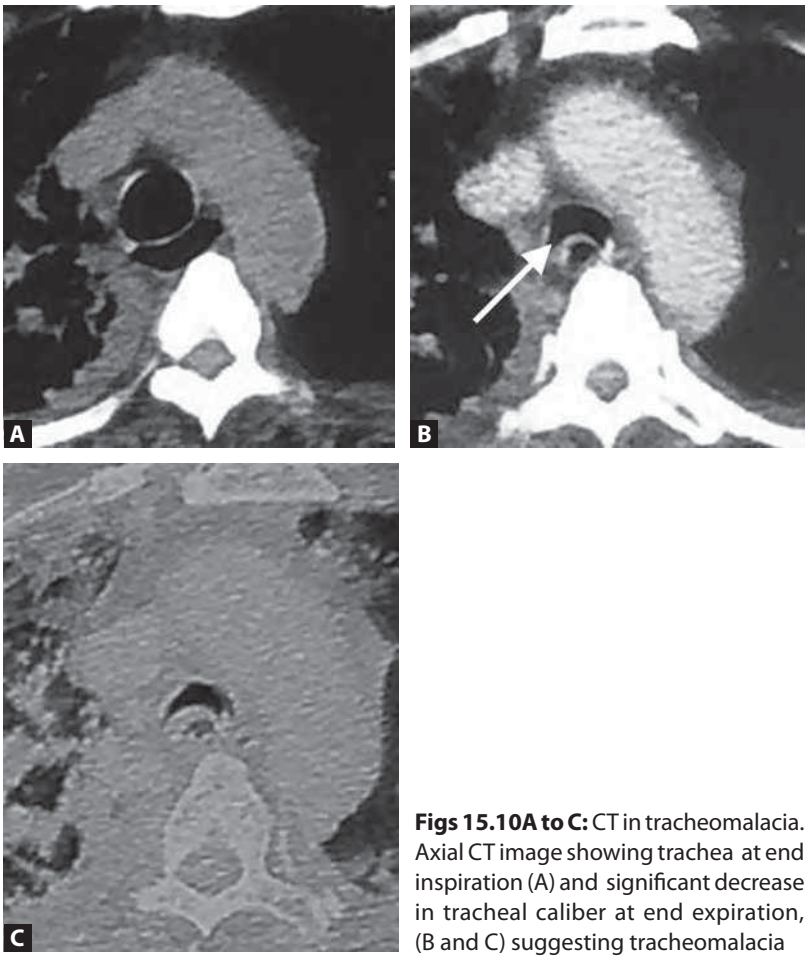


Figs 15.9A and B: CT in combined pulmonary fibrosis and emphysema. Axial HRCT image show upper lobe paraseptal emphysema (A) and lower lobe basal fibrosis and interstitial thickening (B)

- CT show significant dilatation of pulmonary artery in comparison to aorta with diameter ratio more than 1.

Abnormalities of the Trachea and Central Bronchi

- *Tracheobronchomalacia*
  - It is caused due to focal or diffuse tracheal or bronchial wall weakness
  - CT show excessive reduction in the tracheal lumen by more than 80% at expiratory scan **(Figs 15.10A to C)**.
- *Saber-sheath trachea*
  - AP (sagittal) diameter of intrathoracic trachea is more than lateral (coronal) diameter by ratio of 2:1 **(Figs 15.11A to C)**
  - Associated with advanced stage of COPD
  - Extrathoracic trachea is normal
  - Inward bowing of lateral walls of trachea, which become worse on expiration scans may be noted



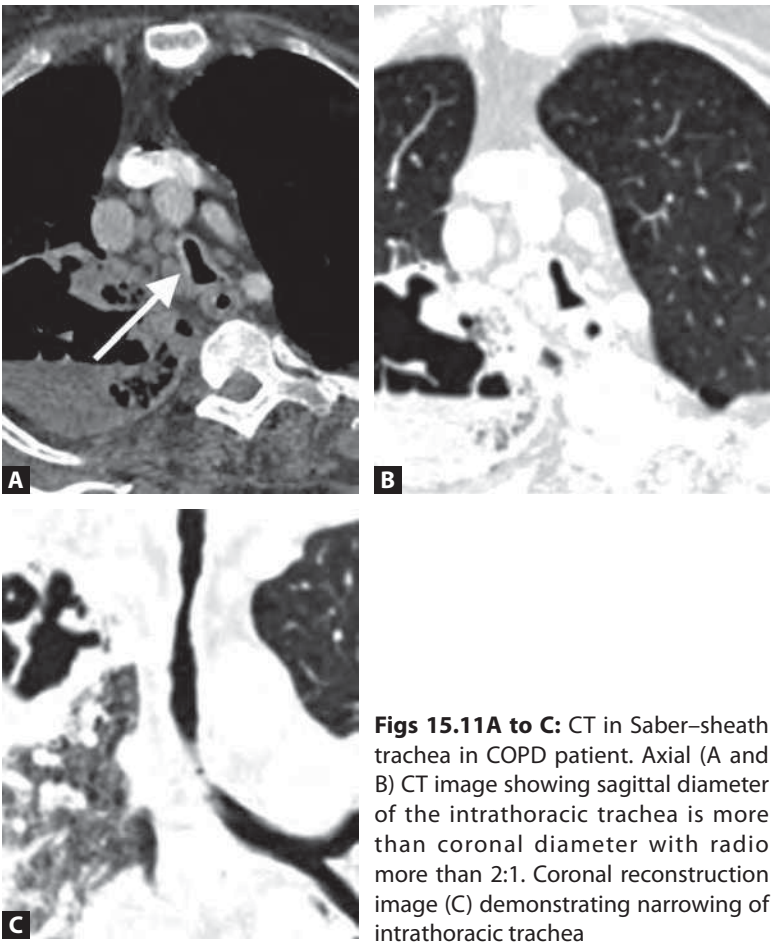
Figs 15.10A to C: CT in tracheomalacia. Axial CT image showing trachea at end inspiration (A) and significant decrease in tracheal caliber at end expiration, (B and C) suggesting tracheomalacia

- Associated tracheal wall thickening, ossification of tracheal rings may be seen.

- *Tracheal diverticula or outpouching*
  - COPD patients may show tracheal diverticula, which occur due to increased intraluminal pressure associated with chronic cough
  - CT show small air filled outpouchings arising from trachea **(Figs 15.12A to C)**.
  - Increased number of diverticula may be seen in chronic cigarette smoking and chronic cough
  - Sometimes it may mimic pneumomediastinum.

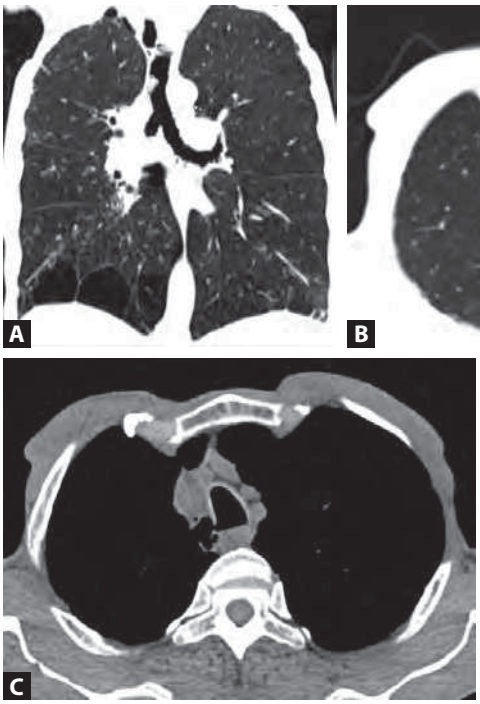
Bronchiectasis

- Bronchiectasis is localized bronchial dilatation
- It may occur in patients with chronic bronchitis and COPD patients **(Figs 15.13A and B)**.

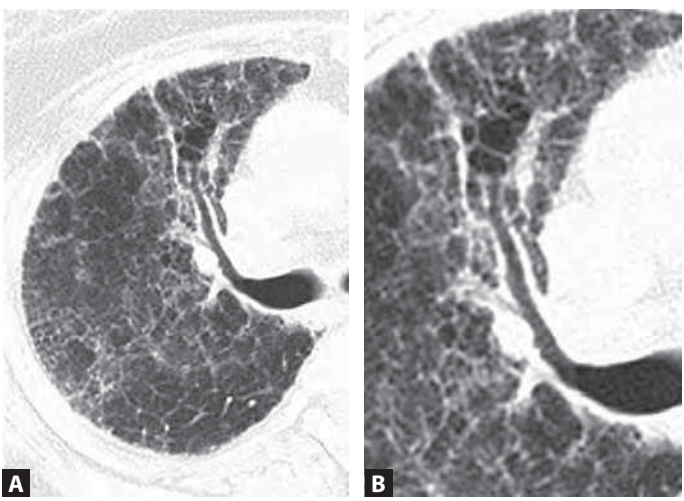


Figs 15.11A to C: CT in Saber–sheath trachea in COPD patient. Axial (A and B) CT image showing sagittal diameter of the intrathoracic trachea is more than coronal diameter with radio more than 2:1. Coronal reconstruction image (C) demonstrating narrowing of intrathoracic trachea

- *The bronchiectasis is usually has following features*:
  - Bronchial dilatation (signet-ring sign)
  - Bronchial wall thickening
  - Lack of normal tapering with visibility of airways in the peripheral lung
  - Mucus retention in the bronchial lumen
  - Associated atelectasis and sometimes air trapping.
- 'Signet-ring sign'—seen in axial section of a dilated bronchus (ring) with its accompanying small artery (signet).
- Bronchiectasis caused by primary airway disease should be differentiated from traction bronchiectasis as a result of fibrosis
- Associated bronchial wall thickening with areas of air trapping seen on expiratory CT scans.



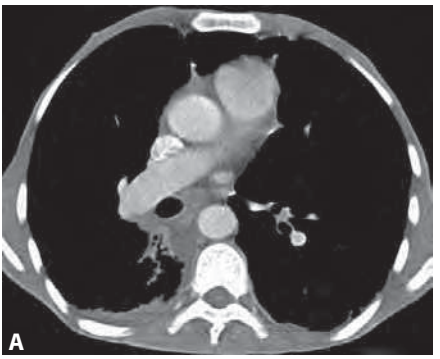
Figs 15.12A to C: CT in tracheal diverticulum. Coronal (A) CT lung window, axial (B) lung window, axial (C) mediastinal window show irregular outpouching from right side of trachea. Note is made of emphysematous changes in bilateral lungs



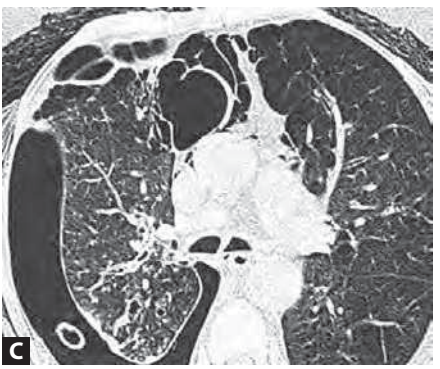
Figs 15.13A and B: CT showing bronchiectasis in COPD patient. Axial HRCT (A and B) showing mild segmental bronchiectasis in the background of COPD changes

Complications of Chronic Obstructive Pulmonary Disease (Figs 15.14A to C)

- Pulmonary artery hypertension
- Secondary infection







Figs 15.14A to C: Complications in COPD. (A) Pulmonary artery hypertension seen as dilated main pulmonary arteries in comparison to ascending aorta diameter; (B) Secondary infections in other patient seen in the form of consolidation with cavity formation in the background of COPD changes; (C) Pneumothorax caused by the rupture of emphysematous bullae

- Pneumothorax
  - In case of co-existence of COPD and malignancy, obtaining adequate biopsy/FNAC sample (under CT guidance) for histopathological examination become challenge because increased risk of pneumothorax.
- Cor pulmonale.

TREATMENT OF CHRONIC OBSTRUCTIVE PULMONARY DISEASE

- Smoking cessation
- Bronchodilators
- Lung volume reduction surgery
- Candidates with heterogeneous emphysema (usually upper lobe predominant)
- Endobronchial valve placement in patients with advanced emphysema shown to modestly improve pulmonary function, exercise tolerance, and symptoms
- 1-way valves which allow air to exit but not enter treated bronchus
- Placed bronchoscopically
- Complications include hemoptysis, pneumonia, and COPD exacerbation
- Lung transplantation: Unfortunately too few organs for those in need.

ASTHMA

- Reversible reactive small airway obstruction
- Occurs in 5% of adults, 10% of children
- Mosaic attenuation less frequent than in chronic bronchitis (CB)
- Severe asthma may be indistinguishable from CB
- Bronchial wall thickening, bronchial dilation, and mucus plugging may be present in severe cases
- Typically reversible airway obstruction, chronic airway inflammation, and nonspecific airway hyper-reactivity.

Etiology

Various agents may precipitate the asthma which may include animals (pet hair or dander), dust, weather changes, air or food chemicals, exercise, mold, pollen, respiratory infections (e.g. common cold), emotional stress, tobacco smoke, medications (e.g. aspirin).

Associated Abnormalities

- Allergic bronchopulmonary aspergillosis
- Bronchocentric granulomatosis
- Chronic eosinophilic pneumonia
- Churg-Strauss syndrome.

Clinical Features

- Cough, shortness of breath, wheezing, and chest discomfort
- On exposure to allergen, the clinical symptoms worsen.

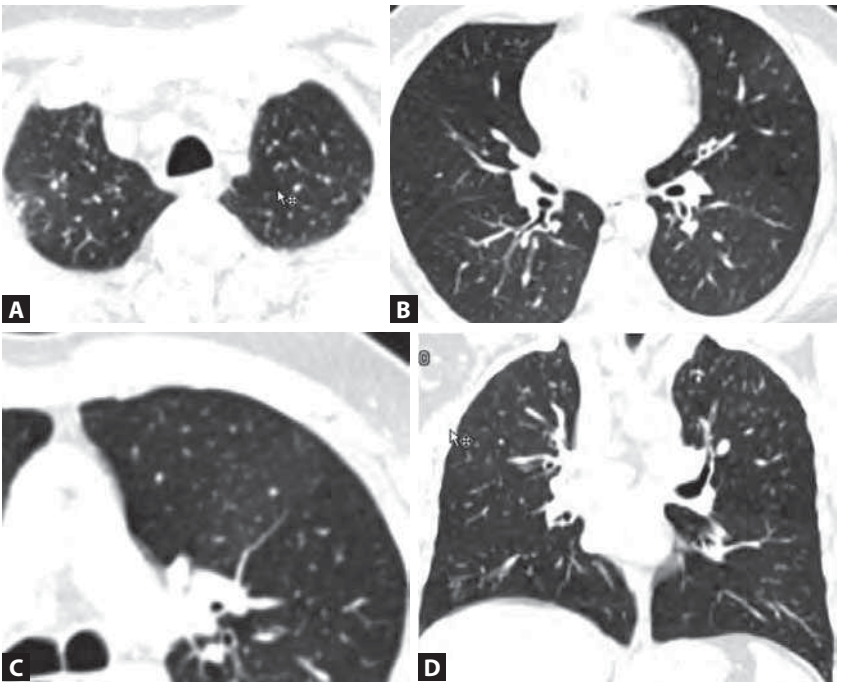
Imaging

Chest Radiograph

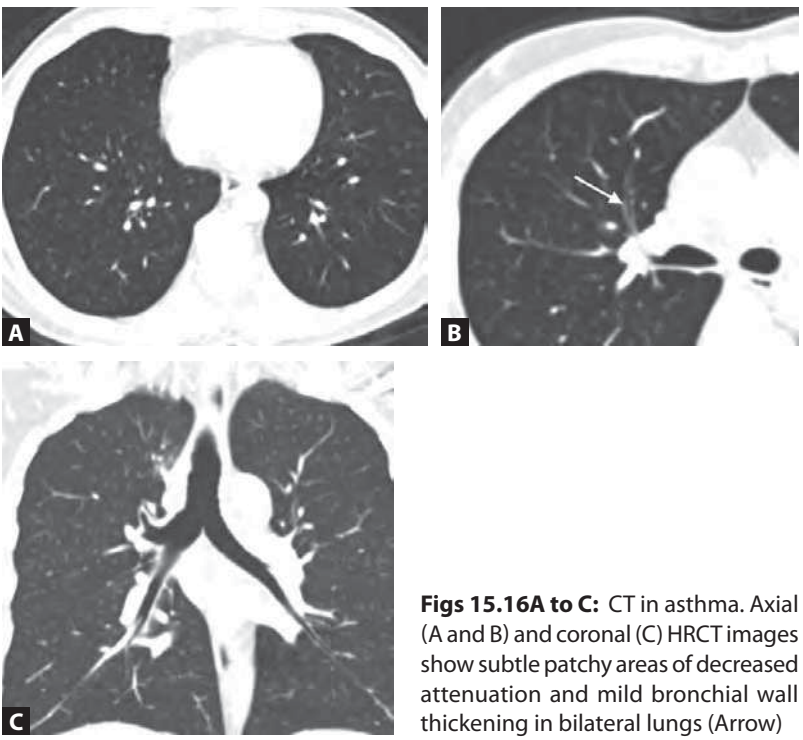
- Most common finding is bronchial wall thickening or peribronchial cuffing
- *Hyperinflation:* Transient or fixed
- *Complications may include*:
  - Atelectasis which is caused by mucoid impaction; right middle lobe commonly affected
  - Pneumothorax and pneumomediastinum may be seen
  - Pneumopericardium, pneumoretroperitoneum
  - Rarely pneumorrhachis and subdural air
  - Pneumonia.

Computed Tomography

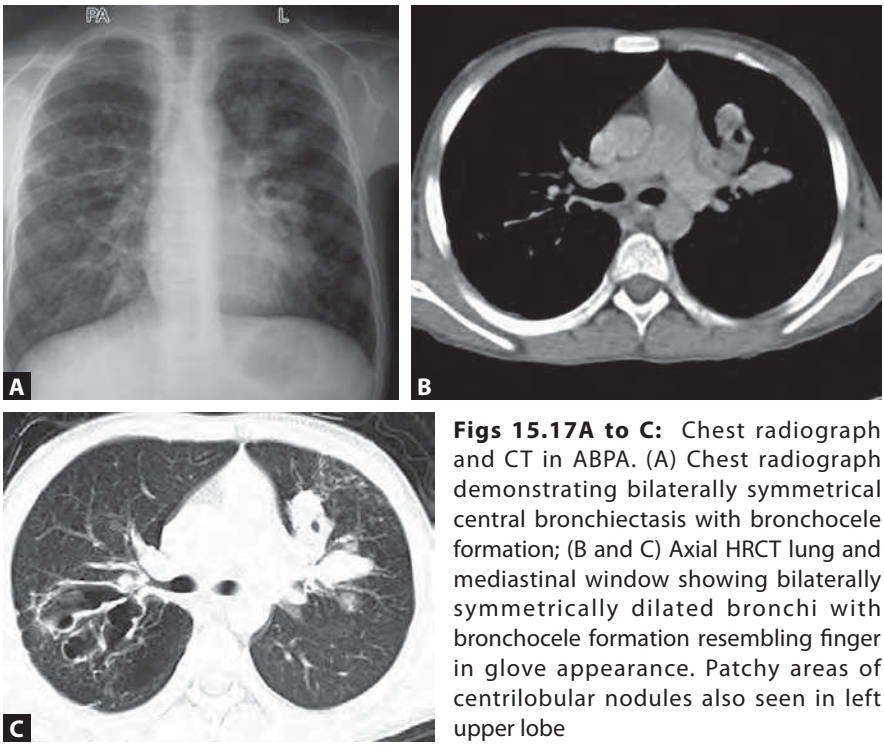
- Helps in assessing extent and severity of airway wall thickness/luminal narrowing
- ↑ thickness correlates with severity of disease
- Bronchiolitis
- Mosaic attenuation during inspiration



Figs 15.15A to D: CT in asthma. Axial (A to C) and coronal (D) HRCT images show subtle patchy areas of decreased attenuation and vascularity in bilateral lungs



Figs 15.16A to C: CT in asthma. Axial (A and B) and coronal (C) HRCT images show subtle patchy areas of decreased attenuation and mild bronchial wall thickening in bilateral lungs (Arrow)



Figs 15.17A to C: Chest radiograph and CT in ABPA. (A) Chest radiograph demonstrating bilaterally symmetrical central bronchiectasis with bronchocele formation; (B and C) Axial HRCT lung and mediastinal window showing bilaterally symmetrically dilated bronchi with bronchocele formation resembling finger in glove appearance. Patchy areas of centrilobular nodules also seen in left upper lobe

- Expiratory air-trapping **(Figs 15.15 and 15.16)**
- Small centrilobular nodules may be seen
- Associated bronchiectasis may be seen, which is usually cylindrical bronchiectasis
- *Allergic bronchopulmonary aspergillosis*:
  - It is usually caused be hypersensitivity reaction to *Aspergillus fumigatus* in the airways
  - It occurs in patients with asthma or cystic fibrosis
  - CT show central bronchiectasis, bronchocele with mucoid impaction fingure in glove appearance, and atelectasis **(Figs 15.17A to C)**.

Treatment

- Avoid exposure to precipitating allergens
- Treatment of concomitant infection
- Corticosteroids and augmentation of immunosuppression in transplant recipients
- Extracorporeal photopheresis.

CONCLUSION

- Chest radiographs are primary screening modality

- HRCT is the workhorse for evaluation of COPD
- Following recommended protocols, CT acquisition techniques and reconstruction parameters are very much are essential for proper assessment of COPD.

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16

Chronic Obstructive Airway Disease and Asthma: Clinical Diagnosis and Management Issues

Saurabh Mittal, Randeep Guleria

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Definition

- Chronic obstructive pulmonary disease (COPD) is a common, preventable lung disorder characterized by progressive, poorly reversible airflow limitation often with systemic manifestations, in response to tobacco smoke and/or other harmful inhalational exposures.[1]
- The natural history of COPD includes intermittent worsening of symptoms and lung function, called as exacerbations.

Incidence

- The 'Indian Study of Asthma, Respiratory Symptoms and Chronic Bronchitis' (INSEARCH) study of 85,105 men and 84,470 women from 12 urban and 11 rural sites reported the prevalence of chronic bronchitis to be 3.49% (4.29% in males and 2.7% in females).

Risk Factors

- Smoking is known to be the most important risk factor for development of COPD.
- *Bidi* and other indigenous forms of tobacco smoking are at least as harmful as cigarette smoking.
- Passive exposure to cigarette smoke (also known as environmental tobacco smoke or (ETS) may also contribute to respiratory symptoms and development of COPD.
- Exposure to biomass fuels like crop residues or woods or animal dung is also considered a major cause of the causation of COPD especially in low income countries.

- Outdoor air pollution mainly from emission of pollutants from motor vehicles and industries is also considered a risk factor for COPD.
- The most studied genetic risk factor is deficiency of alpha-1 antitrypsin, a serine protease inhibitor.
- Asthma is also a risk factor for development of COPD.

Pathogenesis

- Chronic obstructive pulmonary disease is characterized by increased numbers of neutrophils, macrophages, and T lymphocytes (CD8 more than CD4) in the lungs.
- These inflammatory cells release a variety of cytokines and mediators which results in the pathophysiological abnormalities—mucous hypersecretion (causing a chronic productive cough) and ciliary dysfunction leading to airflow obstruction and hyperinflation.

Clinical Features

- Early stages of COPD may be asymptomatic.
- Persistent cough and sputum production is the most common symptom.
- Shortness of breath with intermittent worsening is classical feature of COPD.
- Fatigue and loss of muscle mass occur commonly in severe COPD.
- Physical examination may be normal in early stages. Advanced COPD patients have hyperinflated chest (also called as barrel chest) along with obliterated cardiac dullness and downward displacement of the liver dullness.
- They have prolonged expiratory phase of breath sounds. In emphysema, breath sounds are uniformly diminished in intensity and may be accompanied with rhonchi.
- Development of complications such as corpulmonale may manifest as signs of pulmonary artery hypertension and right ventricular enlargement and/ or failure [loud P2, parasternal heave and raised jugular venous pressure (JVP)].
- Many patients have associated comorbidities including heart failure, coronary artery disease, osteoporosis and musculoskeletal abnormalities which should be looked for.

Phenotypes

- Phenotype is a single or combination of patient features that helps in differentiating between individuals with COPD and is usually related to clinically meaningful outcomes.[2]
- Phenotype is defined by symptoms, radiology, physiology and biomarkers.
- Lately, there has been a lot of discussion on COPD phenotypes and many have been proposed based on different features.[3]
- Infrequent exacerbator (<2 exacerbations per year) have few exacerbations and usually need only long acting bronchodilator therapy. They may have either chronic bronchitis or emphysema phenotype.

- Frequent exacerbators (≥2 exacerbations per year) with emphysema is another important phenotype. It is associated with poor quality of life, a rapid decline in lung function and a higher mortality. They usually need bronchodilators and inhaled steroids.
- Frequent exacerbators (≥2 exacerbations per year) with chronic bronchitis have chronic cough with expectoration >3 months in a year for 2 consecutive years. This group benefits most with anti-inflammatory therapy with phosphodiesterase 4 inhibitor (roflumilast) and macrolides.
- Asthma COPD overlap syndrome (ACOS) is a recognized phenotype associated with an incompletely reversible obstruction of airflow along with symptoms of increased reversibility of the obstruction or if the diagnosis of COPD is established in a patient with a history of previously diagnosed asthma (before 40 years of age).
- ACOS is an entity which basically has features of both asthma and COPD. It is associated with more exacerbations, rapid decline of lung function, increased healthcare utilization and poor outcome.
- Alpha-1 antitrypsin deficiency predisposes an individual to COPD and liver cirrhosis. It is recognized as a specific phenotype that may respond to augmentation therapy.
- Upper lobe predominant emphysema phenotype is associated with severe airflow limitation and may respond to lung volume reduction strategies.

Diagnosis and Patient Assessment

- Spirometry is considered gold standard for the diagnosis of COPD.
- A post-bronchodilator forced expiratory volume in first second (FEV_1)/forced vital capacity (FVC) below < 0.7 may be used as the cut-off for defining airflow obstruction.
- A global assessment of COPD patient includes assessment of severity, patient's symptoms, exacerbation risk (high-risk if ≥ exacerbations in last year) and presence of comorbidities.
- As per Global Initiative for Chronic Obstructive Lung Disease (GOLD) guidelines, severity of airflow limitation is divided in 4 grades—Grade 1 FEV_1 >80% of predicted, Grade 2 FEV_1 ≥ 50% to 80%, Grade 3 FEV_1 ≥30% to 50% and Grade 4 FEV_1 < 30% predicted.
- Patient's symptoms are evaluated by Modified British Medical Research Council (mMRC) scale or by COPD Assessment Test (CAT) score.
- The GOLD combined assessment is done based on these and patient is classified into one of the four classes as shown in **Table 16.1**.

Other Investigations

- Chest X-ray may be normal in early disease. It helps in excluding alternate diagnosis. Chest X-ray features of COPD include hyperinflation and lung hyperlucency **(Fig. 16.1)**.
- High resolution CT chest helps in identification of involved areas and is most useful for assessment for lung volume reduction surgery.

Table 16.1: Combined COPD assessment

Patient	*Characteristic*	*Exacerbations per year*	*CAT*	*mMRC*
A	Low-risk Less symptoms	≤ 1	< 10	0–1
B	Low-risk More symptoms	≤ 1	≥ 10	≥ 2
C	High-risk Less symptoms	≥ 2	< 10	0–1
D	High-risk More symptoms	≥ 2	≥ 10	≥ 2

Abbreviations: COPD, chronic obstructive pulmonary disease; CAT, COPD assessment test; mMRC, modified British Medical Research Council



Fig. 16.1: Posteroanterior chest radiograph in a 62-year-old male with COPD having history of smoking (smoking index 400). Spirometry revealed FEV_1/FVC ratio 68%, FVC 79% of predicted and FEV_1 46% of predicted (GOLD group 3)

- Pulse oximetry and arterial blood gas helps in assessment of oxygenation and decide for long-term oxygen therapy.
- Six-minute walk test and cardiopulmonary exercise testing is also useful for functional assessment.

Prevention

- Smoking cessation is most important intervention for prevention of COPD
- Use of safe fuels and reduction of second hand smoke are few other ways to prevent COPD.

Treatment

- Smoking cessation
- Bronchodilator medications are cornerstone of the symptomatic management of COPD.
- They are prescribed on an as-needed or on a regular basis to prevent or reduce symptoms.
- The main bronchodilator treatments include-short acting β-agonist (SABA, e.g. Salbutamol), long acting β-agonist (LABA, e.g. Salmeterol, Formoterol, Vilanterol, Indacaterol), short acting muscarinic antagonist (SAMA e.g. Ipratropium bromide), long acting muscarinic antagonist (LAMA e.g. Tiotropium, Glycopyrronium), theophylline or combination therapy.
- The choice of treatment depends on the availability of medications and each patient's individual response in terms of symptom relief and side effects.
- Group A is treated with short acting β-agonist (SABA) or short acting muscarinic antagonist (SAMA) as and when required.
- For group B, LAMA therapy is first line of management.
- In group C, LAMA alone or in combination with LABA can be used.
- For group D, ICS with LABA with or without LAMA is preferred drug therapy.
- Theophylline can be used as add-on therapies.
- In patients with moderate to severe COPD with recurrent exacerbations, roflumilast, a phosphodiesterase inhibitor is useful in reducing risk of exacerbations.
- All patients should be vaccinated with pneumococcal vaccine and annual influenza vaccination is also recommended.
- Pulmonary rehabilitation is integral part of COPD management and a dedicated rehabilitation program is desirable for all patients.

BRONCHIAL ASTHMA

Definition

Asthma is defined a chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness and cough that is often reversible either spontaneously or with treatment.[4]

Incidence

- The burden of bronchial asthma is immense.
- Indian study on epidemiology of asthma, respiratory symptoms and chronic bronchitis (INSEARCH) study estimated that around 18 million patients are suffering from the disease with overall prevalence rate of about 2.05% among those aged >15 years.

Risk Factors

- Several risk factors are postulated though no causal effect is established.
- Asthma is more common in younger population.

- History of atopy or family history of asthma, eczema or atopy is a non-modifiable risk factor for asthma
- Exposure to environmental allergens, second hand smoke, viral infections in childhood and obesity are some of the other risk factors.

Pathogenesis and Pathology

- Hyper-reactive airways and airway inflammation is central to the pathogenesis of bronchial asthma.
- Eosinophilic phenotype has predominant eosionophilic inflammation and has high exhaled breath nitric oxide levels while neutrophilic phenotype is late onset and less responsive to steroid therapy.
- Small airway inflammation occurs in almost all patients of bronchial asthma and Curschmann's spirals (mucous plugs) are seen on pathology in patients with acute severe asthma.
- Some degree of bronchiectasis is also commonly seen in long standing asthma and usually does not require specific therapy.
- In patients with co-existent allergic bronchopulmonary aspergillosis (ABPA) the bronchiectasis is usually central.

Clinical Features and Diagnosis

- Bronchial asthma can usually be diagnosed by detailed history and physical examination.
- Common features include cough, wheezing, shortness of breath and chest tightness.
- The symptoms are usually episodic, variable in severity and worse during night and early morning.
- A history of improvement in symptoms with therapy (bronchodilators or inhaled corticosteroids) is also very helpful in making a diagnosis of bronchial asthma.
- Spirometry may be normal while patient is asymptomatic.
- Bronchodilator reversibility on spirometry is tested with inhaled SABA or SAMA and is said to be present if there is an increase in FEV_1 by 12% and 200 mL.
- Diurnal peak flow rate variability of more than 10% is also suggestive of reversible airway obstruction associated with bronchial asthma.
- Other causes of these symptoms such as COPD, bronchiectasis, gastro-esophageal reflux disease and foreign body aspiration should be excluded by clinical examination or appropriate investigations.
- Disease may be complicated by sensitization to *Aspergillus* and may progress to development of allergic bronchopulmonary aspergillosis (ABPA).

Investigations

- Chest radiograph is usually normal. Long standing disease may lead to some degree of bronchiectasis and hyperinflation due to air trapping **(Fig. 16.2)**.



Fig. 16.2: Posteroanterior chest radiograph in 42-year-old lady with long standing asthma. Spirometry revealed FEV_1/FVC of 72%, FVC 60% of predicted and FEV_1 32% of predicted. She had positive bronchodilator reversibility response. Lung volume measurement with body plethysmography revealed Total lung capacity at 170% of predicted and residual volume at 250% of predicted suggesting marked air trapping

- Absolute eosinophil count blood eosinophila suggests eosinophilic inflammation.
- *Aspergillus* specific IgE is helpful in screening these patients for ABPA and then for diagnosis of ABPA total Serum IgE levels are usually performed.
- High-resolution computed tomography (HRCT) is required in a minority of patients not responding to therapy or to assess degree of bronchiectasis.

Management

- Exposure prevention in very important step but many a times a specific trigger for acute worsening is not identified.
- The disease can be sufficiently controled with appropriate therapy and most patients are able to live a normal and symptom free life
- Pharmacotherapeutic agents in asthma can be broadly divided into two groups: relievers and controlers.
- Relievers include SABA or SAMA and are to be used on as and when needed basis.
- Patients requiring frequent relievers usually need controller medications.
- Controller medications include inhaled corticosteroids (ICS), LABA, leukotriene antagonists and theophylline.
- Initially low dose ICS are used and based upon response stepwise treatment with LABA, LAMA, theophylline or oral corticosteroids can be initiated.

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17

Other Cystic Lung Diseases: Imaging

Chandrashekhara SH, Mukesh Kumar Yadav

LUNG CYST

- Lung cyst is defined as well-defined air containing parenchymal space surrounded by thin wall (which is <3 mm in thickness)
- Cystic lung diseases include many complex diseases.
- The thickness of the wall differentiates cyst from cavity (where wall thickness is often more than 4 mm).

CLASSIFICATION OF CYSTIC LUNG DISEASES*

- *Neoplastic causes:* Pulmonary Langerhans cell histiocytosis, Erdheim-Chester disease, lymphangioleiomyomatosis, other malignancies like sarcomas, adenocarcinomas, pleuropulmonary blastoma.
- *Genetic/developmental/congenital causes:* Birt-Hogg-Dubé syndrome, Ehlers-Danlos syndrome, neurofibromatosis, congenital pulmonary airway malformation, bronchopulmonary dysplasia.
- *Cystic lung disease associated with lymphoproliferative disorders:* Lymphocytic interstitial pneumonia, follicular bronchiolitis, Sjögren syndrome, amyloidosis light-chain deposition disease.
- *Infectious causes: Pneumocystis jiroveci*, staphylococcal pneumonia, recurrent respiratory papillomatosis, other unusual infections like fungal diseases.
- *Associated with interstitial lung diseases:* Hypersensitivity pneumonitis, desquamative interstitial pneumonia
- *Smoking related cystic lung diseases:* Pulmonary Langerhans cell histiocytosis, desquamative interstitial pneumonia, respiratory bronchiolitis
- *Other/miscellaneous causes:* Post-traumatic pseudocysts, hyper-IgE syndrome
- *Mimickers of cystic lung diseases:* Emphysema, bulla, Bronchiectasis, honeycombing, cavity

**Modified from cystic lung diseases:* Algorithmic approach. 2016;150(4):945-65.

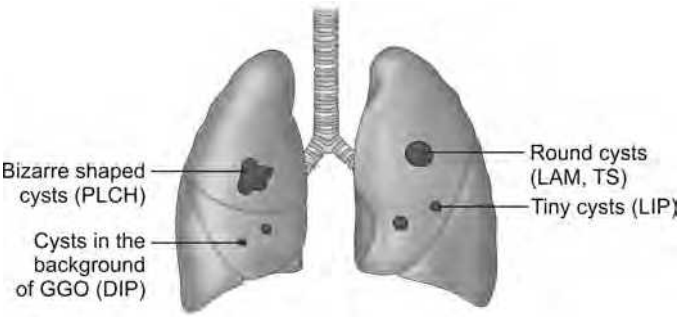


Fig. 17.1: Cystic lung diseases

We will discuss the following important cystic lung diseases in detail in this chapter ***(Fig. 17.1)***

- Pulmonary Langerhans' cell histiocytosis
- Lymphangioleiomyomatosis
- Tuberous sclerosis
- Birt-Hogg-Dubé (BHD) syndrome
- Lymphocytic interstitial pneumonia (LIP)
- Mimickers of cystic lung diseases.

Lung cysts may be seen in the background ground glass opacities (e.g. DIP, LIP). The lung cysts may occur as result of chronic fibrosing stage of ILD.

SPECIFIC ENTITIES

Pulmonary Langerhans' Cell Histiocytosis

- *LCH include 3 subtypes of disease:* Letterer-Siwe disease, Hand-Schüller-Christian disease and eosinophilic granuloma.
- *Hand-Schüller-Christian disease:* Occur in young adults and adolescents with involvement of lung, bone, and pituitary gland (associated with diabetes insipidus).
- *Letterer-Siwe disease:* Occurs in infants with multiorgan involvement, has relatively poor prognosis.
- *Eosinophilic granuloma:* Usually involve single organ.
- Pulmonary Langerhans' cell histiocytosis (PLCH) (eosinophilic granuloma) is preferred term for isolated pulmonary involvement; occurs most commonly in young adults.

Clinical Features

- Dyspnea, nonproductive cough.
- May present with fatigue, chest pain, weight loss or spontaneous pneumothorax.
- No sex predilection; 20–40 years of age.
- Many of the patients will have history of cigarette smoking.

Pulmonary Function Tests

- Carbon monoxide diffusing capacity is decreased.
- Total lung capacity is preserved.

Diagnosis

- Transbronchial lung biopsy
- Bronchoalveolar lavage: > 5% CD1a(+) Langerhans cells.
- Sometimes, open lung biopsy may be required.

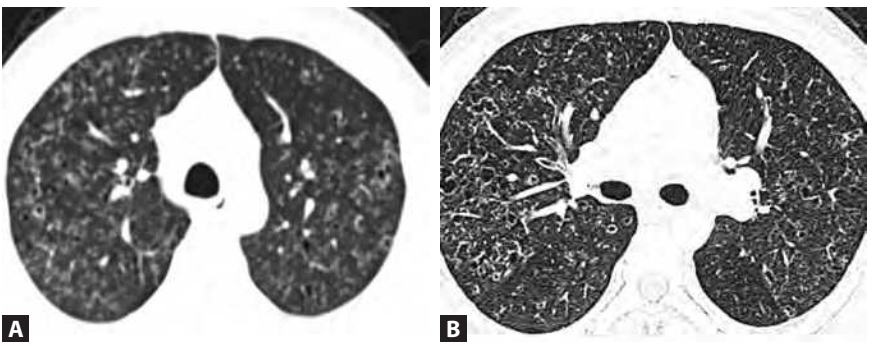
Imaging

Chest Radiograph

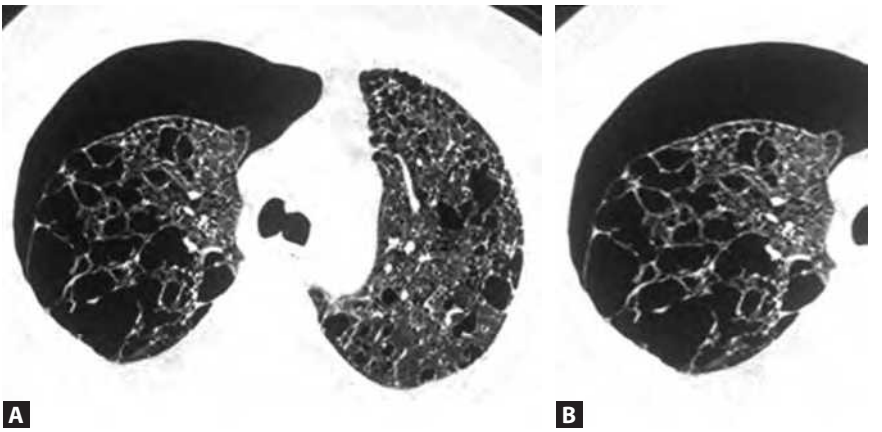
- Bilateral diffuse symmetrical reticulonudolar opacities, which are predominantly distributed in upper zones.
- Lung bases are usually spared.
- Small nodules may be seen in early disease; cavity formation occur in the later coarse of the disease.
- Lung volumes are preserved, or may show hyperinflation.

Computed Tomography

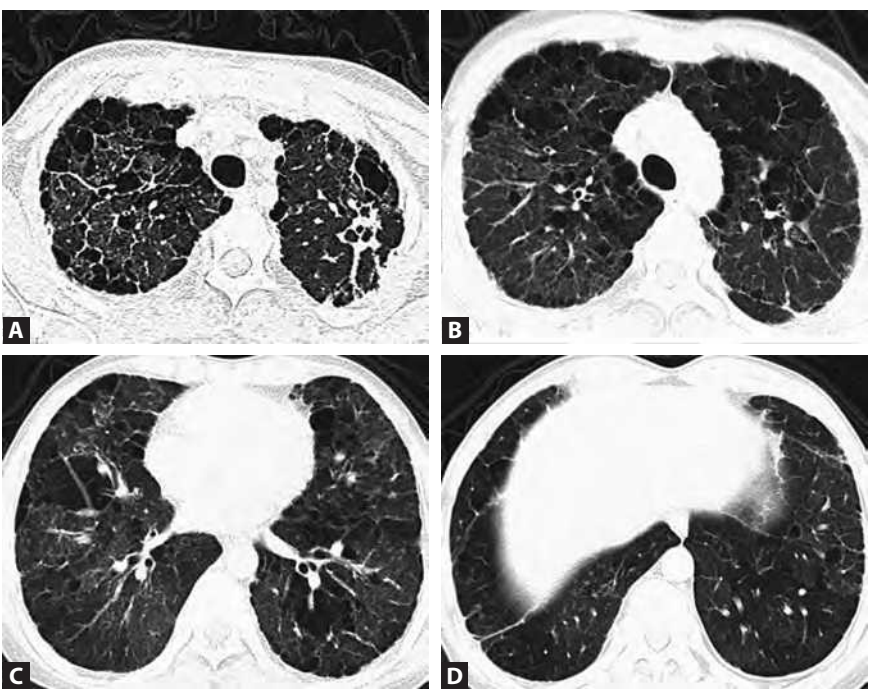
- HRCT—imaging modality of choice for evaluation of pulomanry LCH.
- CT findings depend on the stage of the disease **(Figs 17.2 and 17.3)**.
- *Early disease:* Nodular pattern dominates, which may get cavitated. These nodules are small (1–10 mm in size), irregular shaped, distributed in centrilobular, peribronchial, peribronchiolar location.
- *Later stage of disease:* Cystic pattern dominates. These cysts are irregular, small (1–3 cm) and sometimes bizarre shaped.
- Nodules and cysts usually seen in upper/mid lung zone **(Figs 17.4A to D)**.
- Basal zones near costophrenic angle are typically spared.
- Intervening lung parenchyma appears nodular.
- Lung volume is relatively preserved, which differentiate it from IPF.



Figs 17.2A and B: CT in pulmonary Langerhans cell histiocytosis. Axial lung (A) and HRCT (B) window showing small nodules with few small cysts in bilateral lungs



Figs 17.3A and B: CT in pulmonary Langerhans cell histiocytosis with pneumothorax. Axial HRCT lung window (A and B) showing multiple small cystic lesions with right sided pneumothorax (likely due to secondary rupture of cysts)



Figs 17.4A to D: CT in pulmonary Langerhans cell histiocytosis. Axial HRCT lung window (A and B) showing multiple small bizarre shaped cysts more in upper and middle lobes, relatively sparing lower lobes and CP angle regions

- Associated ground-glass haze, reticular interstitial thickening, irregular bronchovascular structures may be seen.
- Secondary spontaneous pneumothorax (in 10–15%) seen

- *Other features of LCH:* Bony lytic lesions
- Mediastinal/hilar lymphadenopathy usually not seen in PLCH
- Pulmonary artery hypertension may be observed in 33% of cases.
- When LCH is confined to the lungs, the prognosis is good.

Differential Diagnosis

Lymphangioleiomyomatosis

- Pulmonary cysts are uniformly distributed.
- Nodules are unusual.
- Plural effusion/chylothorax may be seen (which is unusual in LCH).

Pneumocystitis Jiroveci Pneumonia (PJP)

- Cysts are usually associated with ground-glass opacities
- Pneumatoceles may be seen.
- Associated pneumothorax may be seen.

Sarcoidosis

- End-stage of sarcoidosis may demonstrate cysts, honeycombing
- Associated with hilar/mediastinal lymphadenopathy
- Nodules are typically distributed in perilymphatic location with associated interstitial thickening.

Bullous Emphysema

- Pulmonary cysts will have indistinct walls with associated signet ring sign.

Treatment

- Initial stage of disease—stop smoking
- Progressive disease—corticosteroids
- Advanced disease may require lung transplantation.

Lymphangioleiomyomatosis

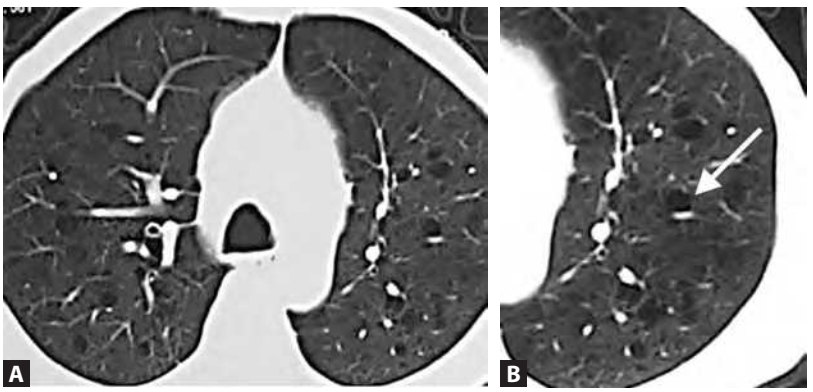
- It is a disease which is characterized by proliferation of peribronchovascular smooth muscle causing air trapping and subsequent development of cysts.
- Rare and affects perominantly females, age:30–35 years.
- May be seen in postmenopausal and those who are on hormonal replacement therapy
- LAM show many similarities to tuberous sclerosis; small percentage of cases associated with TSC. It occurs in 30–40% of tuberous sclerosis patients.
- Sporadic LAM occur exclusively in women.
- As disease progress, pulmonary function tests gets worsened.
- Majority of patients die within 10 years of the onset of clinical features.

Chest Radiograph

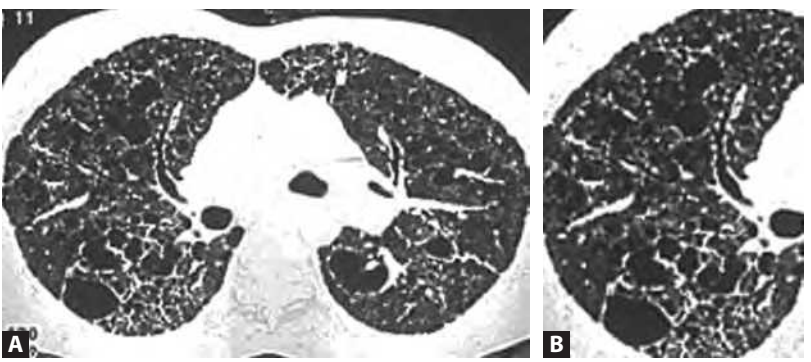
- Early disease may show normal radiograph.
- Bilateral reticular opacities
- Lung volumes are preserved.

CT Findings

- *HRCT:* Best modality to assess cysts and disease pattern
- Diffuse parenchymal cysts are usually thin walled (less than 2 mm in size), uniform in size, randomly distributed without zonal predominance **(Figs 17.5A and B)**.
- Sporadic LAM show thin-walled cysts surrounded by normal intervening parenchyma; tuberous slcerosis—IAM show diffuse small nodular lesions and thin-walled cysts.
- Whole of lung parenchyma get replaced by cysts in advanced disease.
- Occasionally interlobular interstitial thickening and ground-glass opacities may be seen **(Figs 17.6A and B)**.
- Chylous pleural effusions may be seen.
- Rupture of cysts may cause pneumothorax, which may occur either one side or both sided; recurrence of pneumothorax is quite common.



Figs 17.5A and B: CT in LAM. Axial HRCT lung window (A and B) showing multiple small randomly distributed well-defined cysts in bilateral lungs



Figs 17.6A and B: CT in known case of LAM. Axial HRCT lung window (A and B) showing multiple small clustered well-defined cysts in bilateral lungs. Associated interstitial thickening and fibrosis seen at places

- May be associated with hilar and mediastinal lymphadenopathy and pericardial effusion

Differential Diagnosis

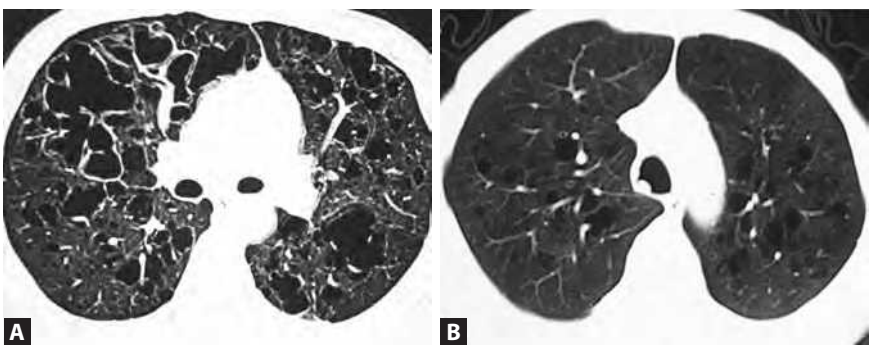
- The differntiating features of PLCH and LAM are given in **Table 17.1**.
- Idiopathic pulmonary fibrosis
- Lymphocytic interstitial pneumonia
- Centrilobular emphysema.

Staging, Grading and Classification

- Performed using LAM histologic scoring system (LHS)

Table 17.1: Radiological difference between PLCH and LAM **(Figs 17.7A and B)**

	PLCH	*LAM*
Age	20–40 years M=F	30–35 years Female
Cysts	Cysts are irregular and are of varying shapes such as bilobed, cloverleaf shaped or bizarre shaped. Nodules may be seen in the intervening lung parenchyma between the cysts	Cysts are round and uniform May show features similar to tuberous sclerosis
Location	Upper and mid zones, relatively sparing lower zones and CP angles	No zonal predominance Uniformaly distributed
Lung volume	Preserved	Preserved
Intervening lungs	Nodular with mild architectural distortion	Usually normal
Other findings	Pneumothorax	Chylous effusion Pneumothorax



Figs 17.7A and B: CT showing difference between PLCH and LAM. (A) Axial CT image of PLCH showing multiple bizzare shaped cysts, with upper and middle lobe predominance. (B) Axial CT image of LAM showing round cysts, distributed diffusely in bilateral lungs. Intervening lung parenchyma is normal

- Total percentage of parenchyma affected by cysts and LAM cells
- Strong correlation with overall patient survival.

Treatment

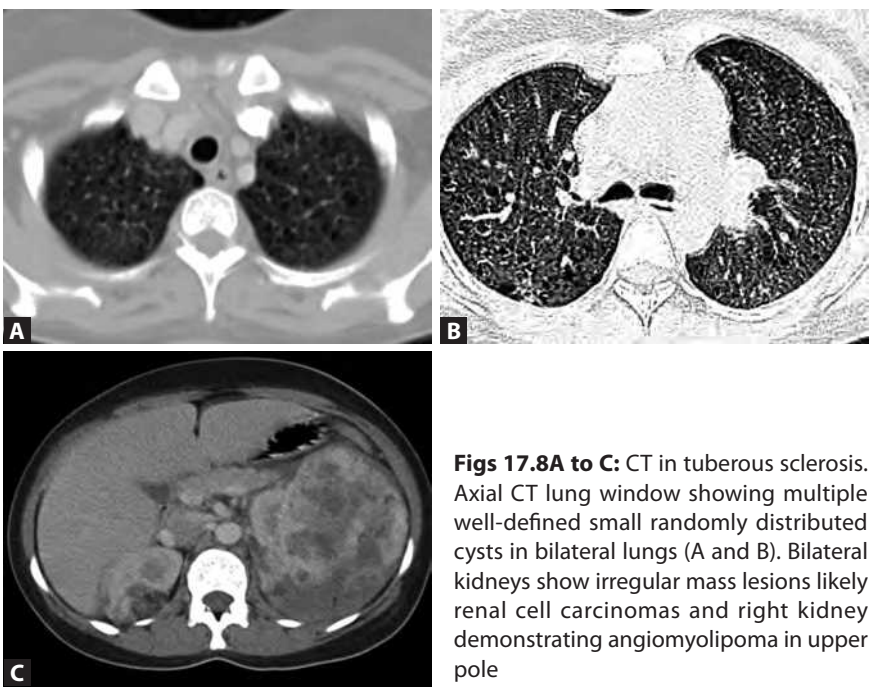
- Pneumothorax should be treated by chest tube drainage or by aspiration
- Recurrence of chylous pleural effusions may require drainage, pleurodesis, or pleurectomy
- Advanced disease may require lung transplantation.

Tuberous Sclerosis

- Autosomal-dominant disorder
- Female show more pulmonary involvement; age: 30–40 years.
- LAM occurs in one-third of patients with tuberous sclerosis and is less severe than sporadic LAM.

Imaging

- HRCT findings of tuberous sclerosis similar to those of LAM **(Figs 17.8A to C)**.
- Chylous pleural effusions usually not seen in tuberous sclerosis as compared to LAM.



Figs 17.8A to C: CT in tuberous sclerosis. Axial CT lung window showing multiple well-defined small randomly distributed cysts in bilateral lungs (A and B). Bilateral kidneys show irregular mass lesions likely renal cell carcinomas and right kidney demonstrating angiomyolipoma in upper pole

- Multiple uniformly distributed cysts may be seen
- Few small nodules (Multifocal micronodular pneumocyte hyperplasia) may seen.

Birt-Hogg-Dubé Syndrome

- Autosomal dominant disease
- Syndrome comprise of multiple lung cysts, cutaneous fibrofolliculomas, spontaneous pneumothorax, and renal cancer.
- Multisystemic disease caused by a deletion mutation on the *FLCN* gene (chromosome 17).
- It consists of
  - Multiple lung cysts and secondary spontaneous pneumothoraces.
  - CT show multiple small cysts are located in lower zones, may be septated when they become large.
  - The cysts usually abut or encase proximal lower pulmonary veins **(Figs 17.9A and B)**
  - Multiple bilateral renal tumors (renal cell cancer and oncocytoma); renal cell cancers occur in > 25% of Birt-Hogg-Dubé Syndrome (BHD) syndrome.
  - Cutaneous manifestations (angiofibromas, perifollicular fibromas, acrochordons, fibrofolliculomas, etc.)
  - It is be noted that not all patients with pulmonary manifestations of BHD have renal or skin findings.

Lymphoid Interstitial Pneumonia

- LIP is due to diffuse polyclonal lymphocytic infiltration of alveolar septa
- LIP is part of lymphoproliferative disorders
- High-risk of lymphoma in patients with Sjögren syndrome
- *Age of presentation*: 50–60 years



Figs 17.9A and B: CT in Birt-Hogg-Dubé syndrome. Axial CT lung window showing multiple well-defined cystic lesions in bilateral lungs, which are closely abutting pulmonary vessels (A and B). Right sided pneumothorax (B) seen, which is caused secondary to rupture of lung cysts

Associated Abnormalities

Most patients have autoimmune disease like Sjögren syndrome or immunodeficiency diseases.

Clinical Features

- Cough, dyspnea
- Fatigue
- Other symptoms related to underlying or predisposing diseases such as Sjögren syndrome, immunodeficiency
- Recurrent infections.

Diagnosis

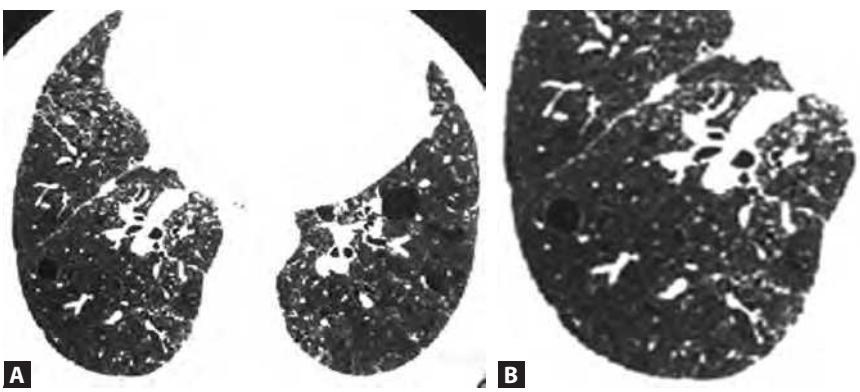
- Bronchoalveolar lavage may show lymphocytosis
- Definitive diagnosis usually requires surgical lung biopsy.

Chest Radiograph

- Multiple small cysts and ground-glass opacities are seen, mainly in basal zones.

Computed Tomography

- *HRCT:* Imaging modality of choice for evaluation of LIP.
- Thin walled cysts (< 3 cm), few poorly defined nodules are typical findings **(Figs 17.10A and B)**.
- The cysts may gradually increase in size. The deep parenchymal cysts are seen due to partial obstruction of bronchiole by lymphocyte infiltration.



Figs 17.10A and B: CT in LIP. Axial CT lung window (A and B) showing multiple perivascular cysts located in bilateral lower lobes. Bilateral lungs demonstrating tiny nodules, ground-glass opacities and interstitial thickening

- Diffuse or patchy ground-glass opacities, interstitial thickening, tiny nodules consolidation may be seen
- Mediastinal lymphadenopathy may be seen.

Differential Diagnosis

- Lymphangioleiomyomatosis (LAM)
- Hypersensitivity pneumonitis
- Nonspecific interstitial pneumonia (NSIP)
- Pulmonary langerhans cell histiocytosis (PLCH).

Treatment

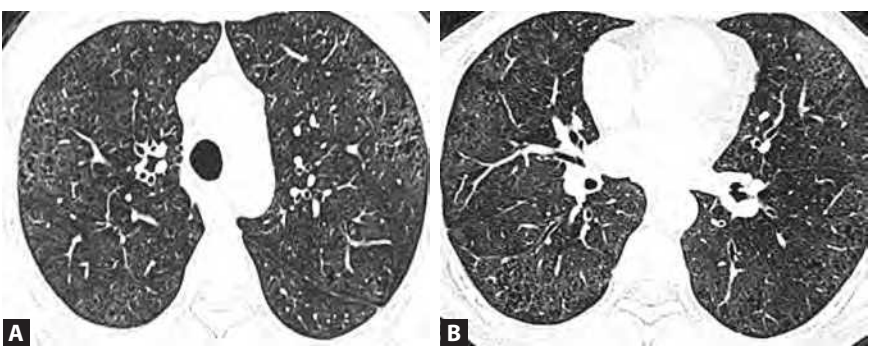
- Steroids, cytotoxic drugs
- LIP may progress with fibrosis. Sometimes, it may evolve into lymphoma.

Desquamative Interstitial Pneumonia

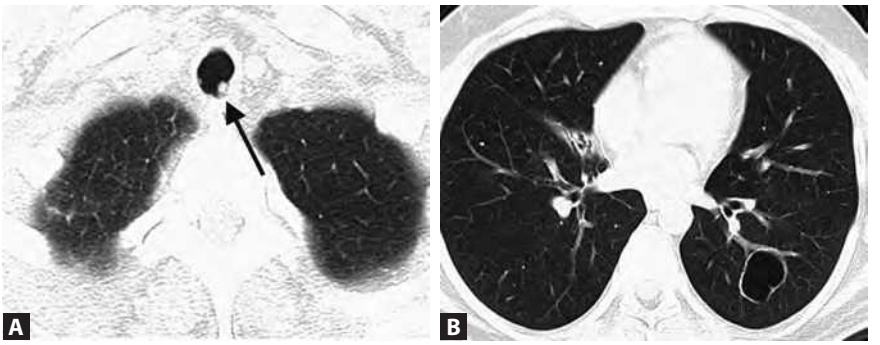
- More in men than in women
- 90% of cases occur in smokers.
- CT show small cysts within areas of ground-glass opacity **(Figs 17.11A and B)**
- Cysts are usually discrete and have imperceptible walls.

Tracheobronchial Papillomatosis

- Common benign tumor of laryngotracheal region
- Associated with human papilloma virus infection
- It show small nodular lesions in larynx and trachea on CT **(Figs 17.12A and B)**.
- The lungs may show randomly distributed cysts with thin or thick walls with nodular lesions.



Figs 17.11A and B: CT in DIP. Axial CT lung window (A and B) showing multiple small cysts in bilateral lungs. Patchy areas of ground glass opacities and interstitial thickening seen in bilateral lungs



Figs 17.12A and B: CT in tracheobronchial papillomatosis. Axial CT lung window show small intraluminal nodules in trachea (A) and well-defined thick walled cyst in left lower lobe (B)

Table 17.2: Differentiating important cystic lung diseases

Disease	*Imaging*	*Distribution*	*Additional findings*
PLCH	Nodules with cysts (Irregular bizarre shaped cysts)	Predominantly upper zones, spares basal zones	Lung volume normal or hyperinflated
LAM	Thin walled cysts (uniform round cyst)	Random, diffuse	Chylous effusion
Tuberous scleorsis	Thin walled cysts	Randomly distributed	Angiomyolipomas of kidneys and liver; other featurs of TS
LIP	Well-defined cysts	Predominantly in basal zones	Ground glass haze

Incidental Lung Cysts

- The cysts may be detected incidental
- They may or may not be significant
- They sometime represent early manifestation of disease or residual manifestation of previous disease.

Differentiating Cystic Lung Diseases

- The differentiating various cystic lung diseases is based on identifying the disease pattern, distribution of cysts and other associated findings as described in **Table 17.2**.

MIMICKERS OF CYSTIC LUNG DISEASE

The following conditions mimic the cystic lung diseases ***(Figs 17.13 and 17.14)***:

- Honeycombing
- Cystic bronchiectasis

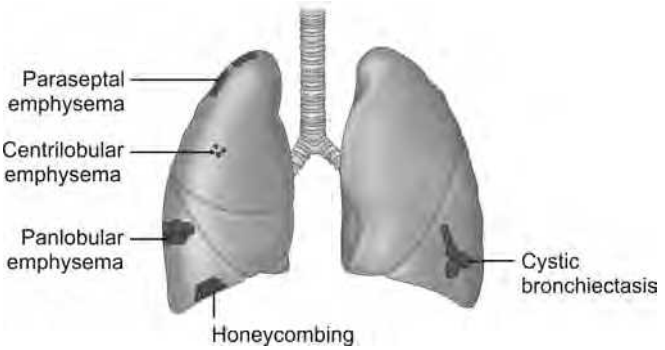
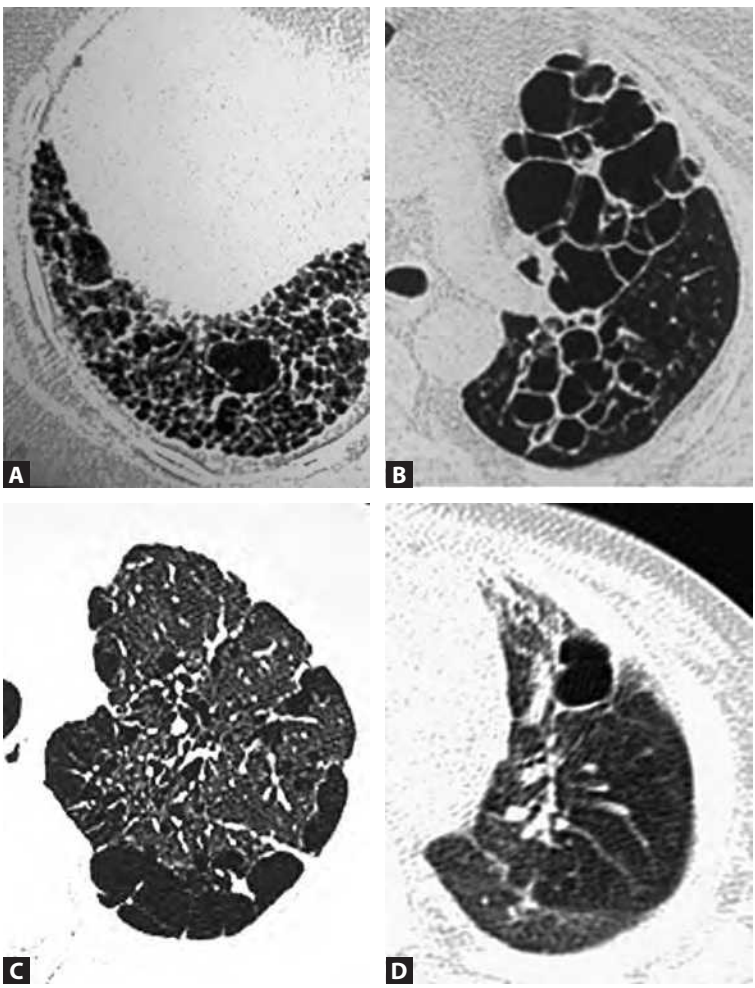


Fig. 17.13: Mimickers of cystic lung diseases



Figs 17.14A to D: Mimickers of cystic lung diseases. (A) Cysts/honeycombing in ILD; (B) Cystic bronchiectasis; (C) Paraseptal emphysema (D) Postinfective sequelae

- Centrilobular emphysema
- Paraseptal emphysema
- Cysts in endstage fibrosis
- Bullae
- Cavity
- Postinfective sequelae.

Honeycombing

- Honeycombing indicates complete destruction of acinar and bronchiolar parenchyma which are replaced by subpleural, peripheral tiny cysts, arranged in multiple tiers with associated interstitial thickening. The cysts have shared thick walls. Honeycombing is typically seen in basilar zones.
- It indicates end-stage pulmonary fibrosis
- It is a feature of diffuse pulmonary fibrosis (IPF).
- IPF accounts for > 60% of UIP.
- Lung cysts should be differentiated from honeycombing **(Figs 17.15A to C)**.

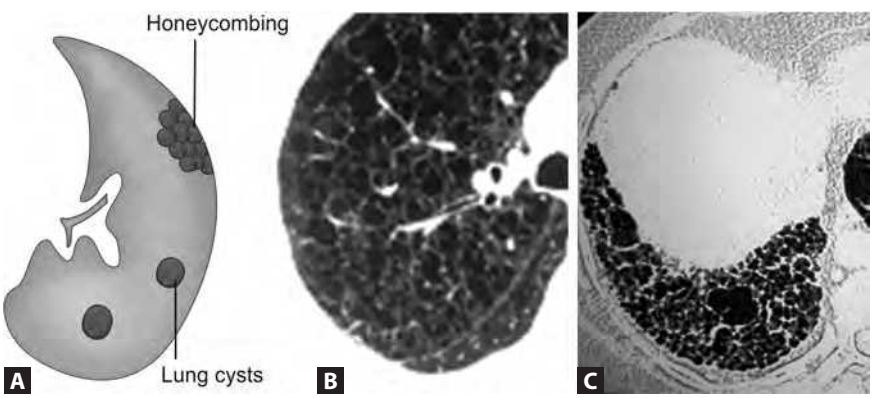
Imaging

Chest Radiograph

- Honeycombing is seen as peripheral, subpleural, basilar tiny cysts with associated reticular opacities
- Lung volume loss may be seen.

Computed Tomography

- Honeycombing seen as multiple tiny cysts, which are located peripherally in subpleural location, predominantly in basal zones.



Figs 17.15A to C: Difference between cyst and honeycombing. (A) Cyst and honeycombing (B) coronal CT image showing thin walled cyst in right lung (C) axial HRCT image showing honeycombing having shared thick walls forming layers, predominantly located in subpleural and peripheral location

- These cysts have thin walls and are of different sizes varying from 2–20 mm.
- *Associated findings:* Inter-or intralobular interstitial thickening, traction bronchiectasis and parenchymal architectural distortion
- Honeycombing is a feature of IPF, which are seen in 80–90% cases. these honeycombing cysts helps in differentiating IPF from other ILD, e.g., DIP, NSIP.
- On follow-up scans, these honeycomb cysts may increase in size.

Differential Diagnoses

- Paraseptal emphysema and paracicatrical emphysema
- Idiopathic pulmonary fibrosis (UIP pattern)
- Fibrosing nonspecific interstitial pneumonia
- Fibrotic sarcoidosis
- Chronic hypersensitivity pneumonitis.

Cystic Bronchiectasis

- Cystic bronchiectasis may resemble cystic lung diseases.
- It is identified by following the continuity of the cysts with bronchial tree **(*see* Fig. 17.7)**
- Presence of the signet ring sign is clinching point in differentiation.

Associated diseases: Tracheobronchomegaly, Williams-Campbell syndrome, and cystic fibrosis.

- Tracheobronchomegaly usually involves trachea and bronchi
- Williams-Campbell syndrome typically involve the bronchi distal to the third order branching,
- Cystic fibrosis usually diffuse with upper lung predominance.

Emphysema

- Cysts have imperceptible wall
- In emphysema, vessels will be coursing through cysts, in contrast to other cystic lung disease, where vessels run along the edges or separate from the cysts.
- Paraseptal emphysema will have upper lobe predominance with peripheral/subpleural distribution **(*see* Fig. 17.8)**.

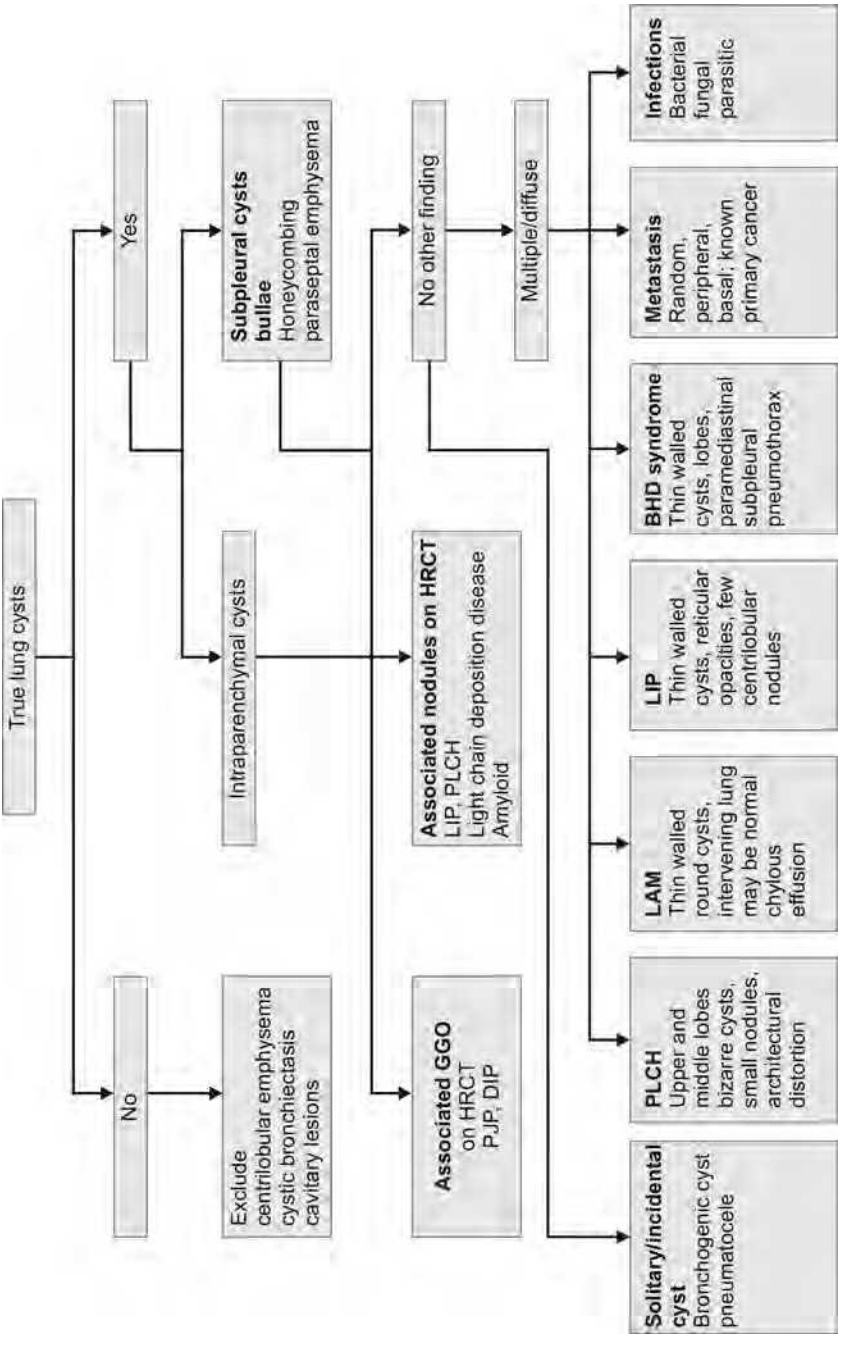
Mimickers of Cystic Lung Disease

The imaging features of mimickers of cystic lung diseases are described in **Table 17.3**.

Diagnostic Approach to Cystic Lung Diseases*

The diagnostic approach to cystic lung disease is described in **Flow chart 17.1**.

* Modified from Cystic Lung Diseases: Algorithmic Approach. 2016;150(4):945-65.



Flow chart 17.1: Diagnostic algorithm for evaluating cystic lung diseases

Table 17.3: Features of mimickers of cystic lung disease

Conditions	*Imaging*	*Distribution*	*Additional findings*
Honey-combing	Multilayered honeycomb cysts	Subpleural, peripheral, basal zones	Intra- and interlobular interstitial thickening, ground glass haze
Cystic bronchiectasis	Cysts continuous with bronchial structures	Focal or diffuse, depending on etiology	Signet ring sign
Centrilobular emphysema	Cystic air spaces without wall	Upper lobe predominance	Vessels coursing through the cyst
Paraseptal emphysema	Cystic air spaces without wall	Subpleural, more in upper lobes	Cysts adjacent to pleural surface
Bullae	Round focal lucency or decreased attenuation >1 cm in size	Focal or diffuse	Adjacent lungs show emphysematous changes
Cavity	Often thick walled (> 4 mm), irregular shaped gas-filled space/ lucency within pulmonary consolidation or mass, or nodule	Distribution depending on the etiology.	Consolidation or mass seen

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Section 6

Pediatric Diffuse Lung Diseases

- Classification and Diseases Common in Infancy
- Diffuse Lung Diseases in Older Children: Imaging
- Pediatric DLDs: Clinical Diagnosis and Management Issues

18

Classification and Diseases Common in Infancy

Manisha Jana, Ashu Seith Bhalla

INTRODUCTION

- Pediatric interstitial lung diseases (ILDs) are rare disorders of diverse etiology
- Characterized by impaired gas exchange, and diffuse changes on imaging
- Entities significantly different compared to adults
- Although they are less common in children than in adults, it is more common in infants and young children than in older children
- As these may involve not only the pulmonary interstitium but also the airways or alveoli the term diffuse lung diseases (DLDs) is more appropriate and desirable over ILD
- To establish a childhood ILD syndrome (chILD); common causes of DLDs in infancy should be ruled out (e.g. immunodeficiency, bronchopulmonary dysplasia, aspiration, primary ciliary dyskinesia)
- At least three out of the four criteria should be fulfilled for diagnosis
  1. Respiratory symptoms like cough, exercise intolerance or labored breathing
  2. Respiratory signs
  3. Hypoxemia
  4. Diffuse lung abnormalities (CT or CXR).

CLASSIFICATION

Classifying pediatric DLDs has always remained a challenge; and various ways of description exist in the literature. The new system of classification **(Table 18.1)** of DLDs describes them into three broad categories:

1. *Disorders more prevalent in infancy*
2. *Disorders not specific to infancy*
3. *Unclassifiable*

The disorders more prevalent in infancy are also diverse in etiology; and a classification of the same is provided in **Table 18.1**. Of note is that the age limit has been considered as below two years of age.

Table 18.1: New classification of childhood diffuse lung diseases (disorders more prevalent in infancy)[1]

Diffuse developmental disorders	• Acinar dysplasia • Congenital alveolar dysplasia • Alveolar capillary dysplasia with misalignment of pulmonary veins
Alveolar growth abnormalities	• Prenatal conditions—secondary pulmonary hypoplasia • Postnatal conditions—chronic neonatal lung disease – *Prematurity-related chronic lung disease [bronchopulmonary dysplasia (BPD)]* – *Term infants with chronic lung disease* • Associated with chromosomal abnormalities – *Trisomy 21* – *Others* • Associated with congenital heart disease in chromosomally normal infants
Surfactant dysfunction disorders and related abnormalities	• Surfactant dysfunction disorders – *SpB genetic mutations* – *SpC genetic mutations* – *ABCA3 genetic mutations* – *Congenital GMCSF receptor deficiency* – *TTF1 genetic mutations* – *Others with histology consistent with surfactant dysfunction disorder but without a recognized genetic disorder* • Lysinuric protein intolerance (PAP histological pattern)
Specific conditions of unknown or poorly understood etiology	• Neuroendocrine cell hyperplasia of infancy (NEHI) • Pulmonary interstitial glycogenosis (either *primary or associated with other pulmonary conditions*)

Disorders not specific to infancy can be classified into various groups (discussed in Chapter 19).

A large number of pediatric DLDs remain unclassified; either because of inadequate sampling or nondiagnostic biopsies; or an end stage disease.

Radiologic-Histological Patterns

- Non-specific interstitial pneumonia (NSIP), LIP and DIP are seen in children
- Underlying etiology and pathogenesis of NSIP, LIP and DIP is different.
- UIP (Usual interstitial pneumonia) as seen in adults and causing idiopathic pulmonary fibrosis is not seen in children
- AIP is also exceedingly rare.
- DIP patterns—noted in genetic surfactant disorders
- LIP pattern—noted in—immunodeficiency syndromes

- NSIP pattern—noted in underlying autoimmune disease, hypersensitivity pneumonitis, surfactant protein deficiency
- Alveolar proteinosis—found in genetic surfactant disorders (Surfactant protein B deficiency).

DISORDERS MORE PREVALENT IN INFANCY

Diffuse Developmental Disorders

- Include *acinar dysplasia, congenital alveolar dysplasia (CAD) and alveolar capillary dysplasia with misalignment of pulmonary veins (ACD/MPV)*
- All entities fatal in infancy
- Present usually within 48 hours after birth
- Lack specific radiographic abnormalities
- CAD and ACD/MPV have high association with congenital cardiovascular, gastrointestinal and genitourinary abnormalities
- *CXR*: Normal to low lung volume, diffuse hazy opacity resembling surfactant deficiency disorders
- Prone to develop pulmonary artery hypertension early in life.

Alveolar Growth Abnormalities

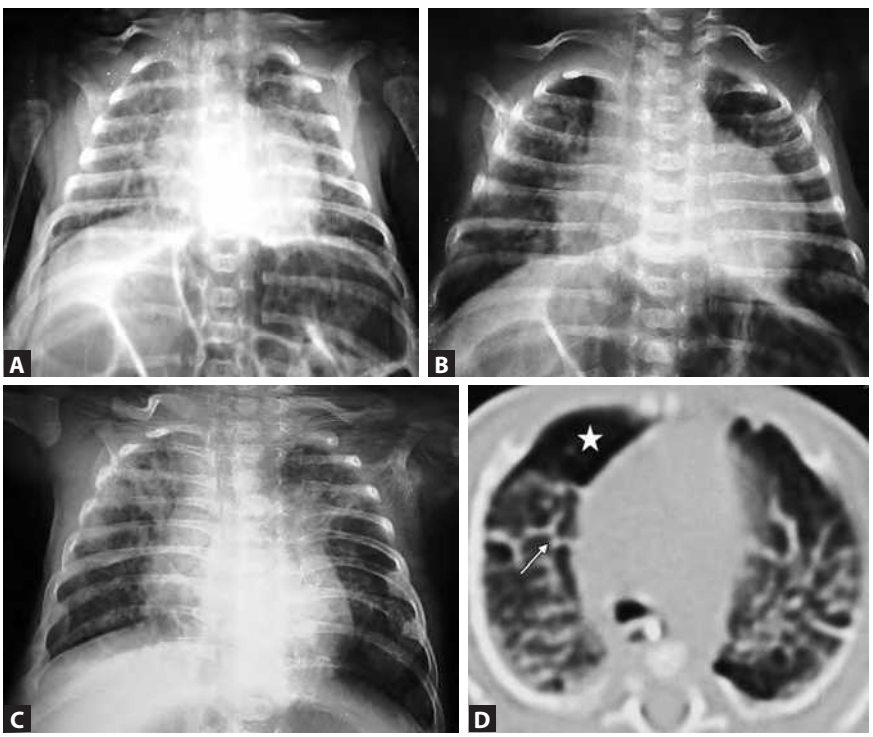
- The most common cause of DLD in infants
- Lung pathology occurs as a result of superimposed events, either prenatal or postnatally
- *Prenatal condition leading to pulmonary hypoplasia*: Polyhydramnios, congenital diaphragmatic hernia
- *Postnatal insult*: Bronchopulmonary dysplasia (BPD)
- Clinical presentation variable
- Imaging findings variable, few entities having specific imaging features (ex-BPD, Trisomy 21)
- *BPD*:
  - Affects premature infants, present with chronic lung disease and oxygen dependence
  - *Imaging*: Coarse reticular opacities, cystic lucencies **(Figs 18.1 and 18.2A to F)**.
- *Trisomy 21*: Peripheral subpleural cysts
- DLD associated with congenital heart disease can present with mosaic attenuation due to ground-glass opacities and altered lung vascularity **(Figs 18.3A to D)**.

Genetic Surfactant Dysfunction Disorders and Related Diseases

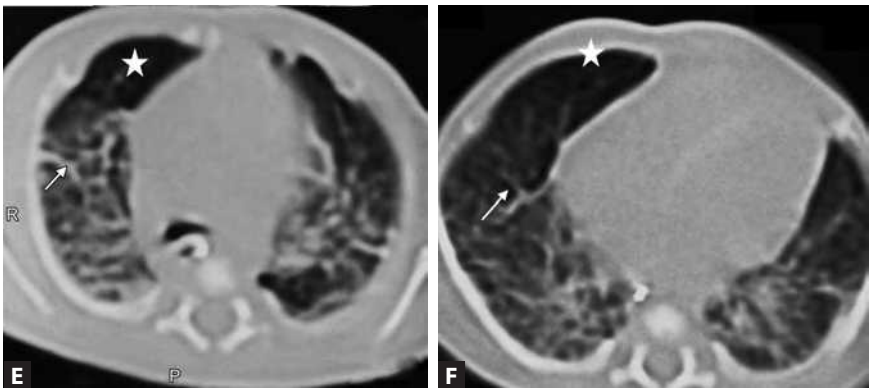
- *Inborn errors of surfactant metabolism*: A group of rare disorders resulting from genetic mutations involving surfactant proteins.
- Clinical-radiographic and histologic pattern of presentation vary according to the type of gene mutation.



Fig. 18.1: Bronchopulmonary dysplasia (BPD). The chest radiograph shows hyperinflated lung fields and streaky 'bubbly' appearance owing to hyperinflation, cystic changes and fibrosis

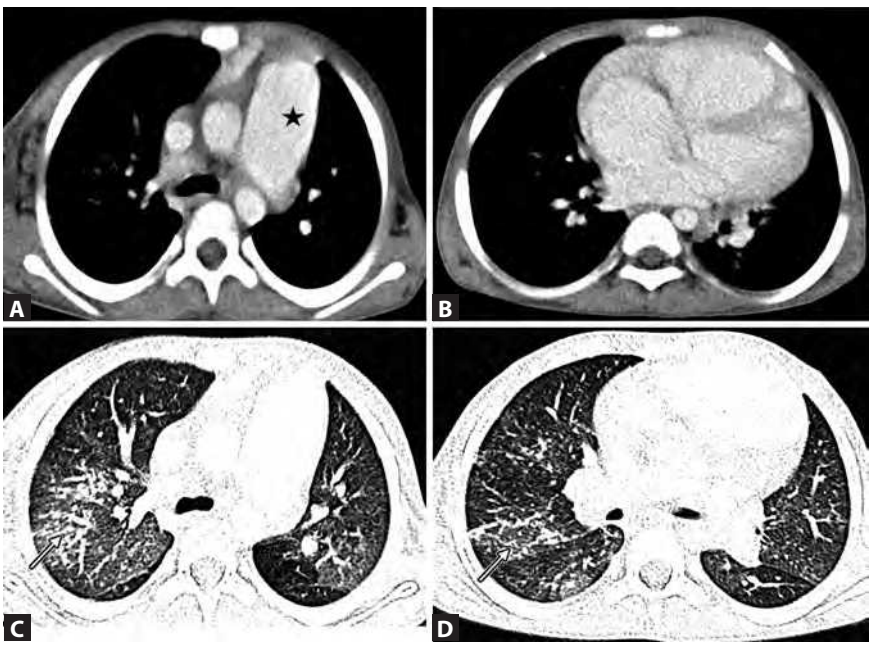


Figs 18.2A to D

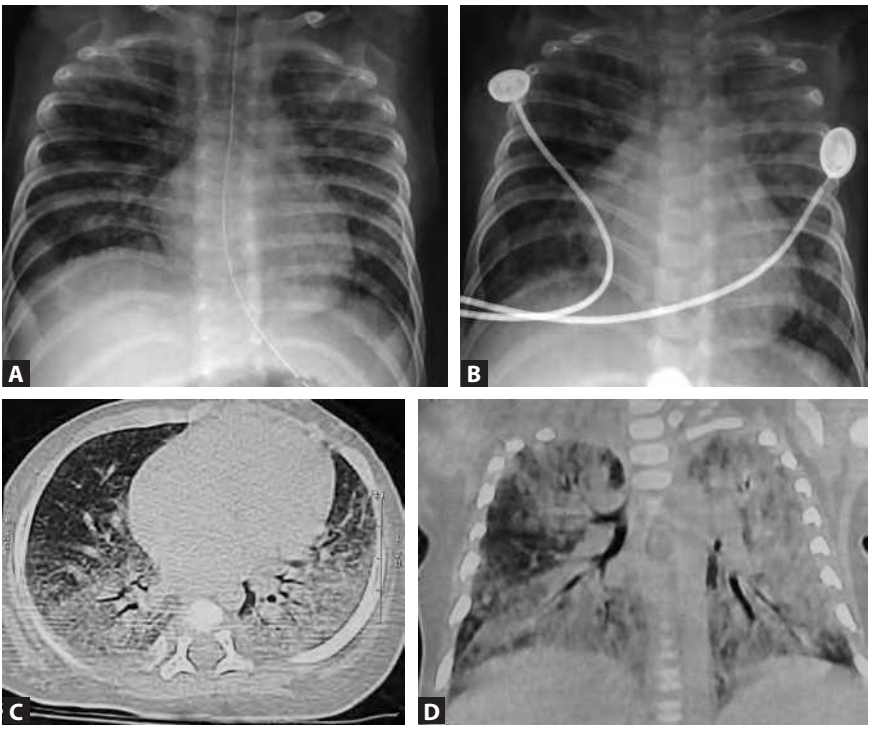


Figs 18.2E and F

Figs 18.2A to F: Radiologic evolution of BPD on chest radiograph in a neonate with prolonged NICU stay. Serial chest radiographs (A to C) show an initial small volume lung with reticular shadows; which gradually changed to hyperinflated lungs having bubbly appearance. Axial CECT lung window images; (D to F) show irregular septal thickening (arrow); and areas of air trapping (asterisk)



Figs 18.3A to D: ASD with pulmonary artery hypertension. Axial CECT soft tissue window images (A and B) show cardiomegaly and dilated main pulmonary artery (asterisk). HRCT sections (C and D) reveal ground glass opacities in both lungs (arrows); and septal thickening



Figs 18.4A to D: Surfactant protein ABCA 3 deficiency. The chest radiographs (A and B) of a 5-month-old infant show patchy areas of ground glass opacity in both lungs. CT thorax images (C and D) reveal multifocal ground glass opacities, predominantly in dependent locations

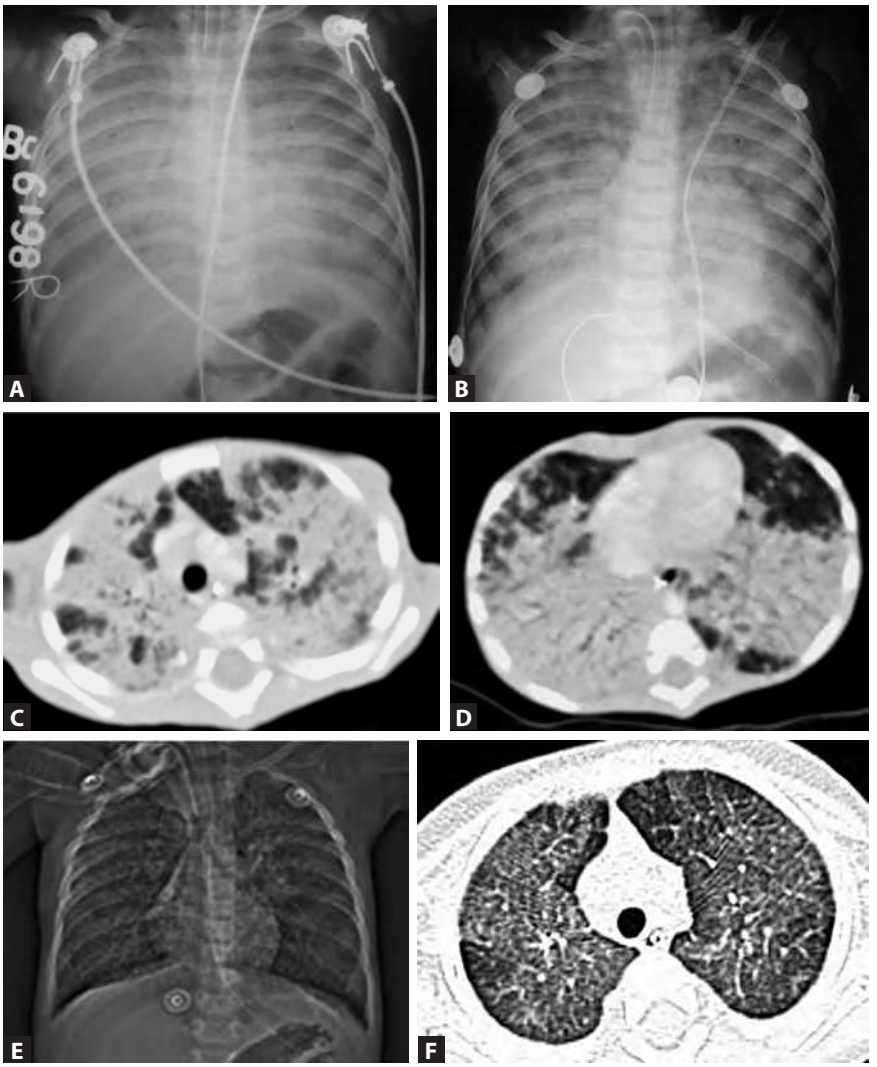
- *Radiologic patterns **(Figs 18.4 to 18.8)** in infants include*:
  - Pulmonary alveolar proteinosis (PAP)
  - Chronic pneumonitis of infancy (CPI)
  - Desquamative interstitial pneumonia (DIP)
- *Radiologic patterns in older children and adolescents*: NSIP and UIP.

Surfactant Protein B Deficiency

- Presents in neonatal age group
- Fatal, presents with neonatal respiratory distress and respiratory failure
- Chest radiographic findings resemble respiratory distress of prematurity
- *HRCT pattern*: PAP.

Surfactant Protein ABCA3 Deficiency

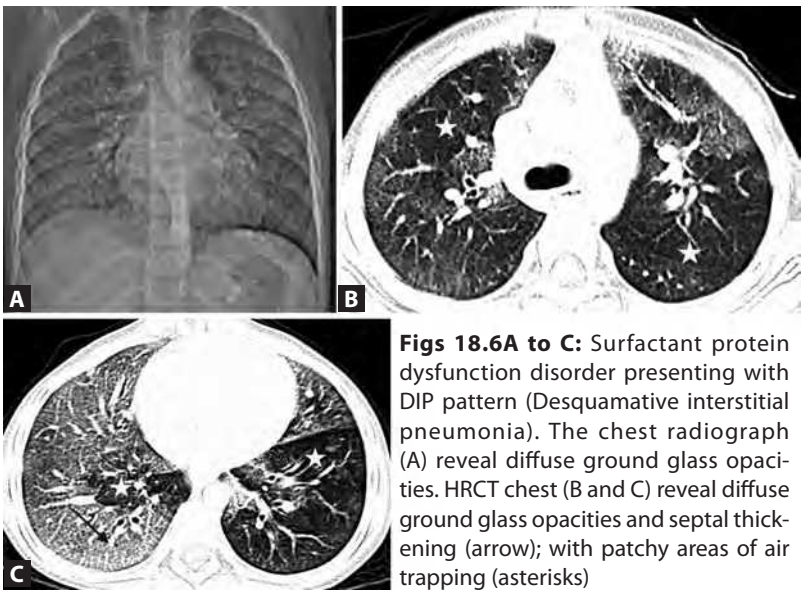
- Age of presentation—neonates, infants or older children
- Imaging pattern—PAP or DIP (neonates and young infants)
- NSIP pattern with areas of lipoid pneumonia like picture (older children).



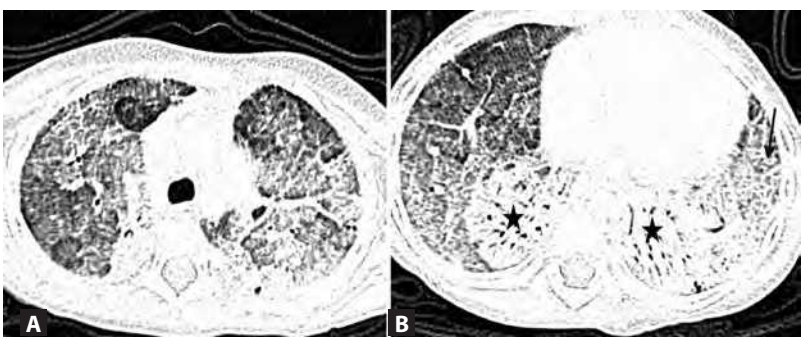
Figs 18.5A to F: Surfactant protein dysfunction disorder presenting as dense consolidation in both lungs. The chest radiographs (A and B) of a 10-month-old infant 'whiteout lung'. CT thorax images (C and D) reveal multifocal dense consolidations. A follow-up radiograph (E) and HRCT (F) after three months reveal significant resolution of the lesions; leaving behind ground glass opacities and mild septal thickening

Surfactant Protein C Deficiency

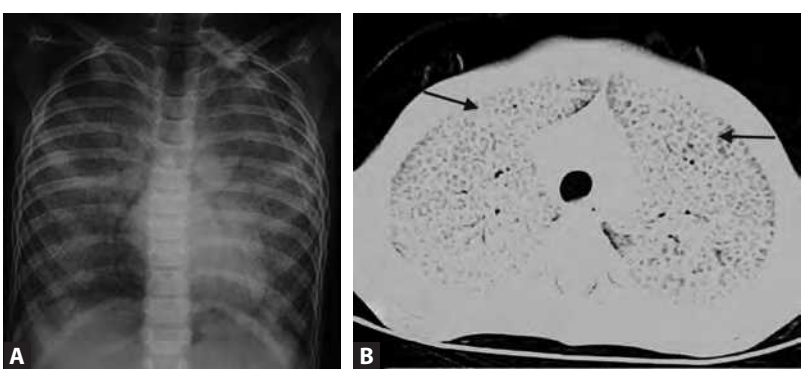
- Age of presentation—Older infants, children, adults
- Imaging pattern—CPI in older infants
- Chronic interstitial pneumonia and pulmonary fibrosis in older children, adults.



Figs 18.6A to C: Surfactant protein dysfunction disorder presenting with DIP pattern (Desquamative interstitial pneumonia). The chest radiograph (A) reveal diffuse ground glass opacities. HRCT chest (B and C) reveal diffuse ground glass opacities and septal thickening (arrow); with patchy areas of air trapping (asterisks)



Figs 18.7A and B: Surfactant dysfunction disorder presenting as 'crazy paving' pattern (arrow) and areas of fibrosis (asterisks) in both lower lobes



Figs 18.8A and B: Pulmonary alveolar proteinosis in a 1-year-old infant. The chest radiograph (A) shows a 'whiteout lung'; and the axial lung window CT image (B) also reveal dense GGO, with superimposed septal thickening (arrows)

Specific Conditions (Unknown or Poorly Understood Etiology)

Neuroendocrine Cell Hyperplasia of Infancy-NEHI[2]

- Histopathological entity which correlates with syndrome of persistent tachypnea of infancy
- Term infants with presenting symptoms of tachypnea, hypoxia, crackles and retractions; not responding to steroids or bronchodilators
- Pathogenesis is not clear-genetic/acquired.
- *Histopathology*: Lung biopsy almost normal on routine H and E stain/non-specific, mild airway wall abnormalities. Immunohistochemistry: increased neuroendocrine cells in airway wall.
- *Chest radiographs*: Hyperinflation.
- *HRCT*:
  - Patchy areas of ground glass opacities, most marked in the central right middle lobe and lingula
  - Peripheral air trapping **(Figs 18.9A to D)**



Figs 18.9A to D: Neuroendocrine cell hyperplasia of infancy (NEHI). The CT scout film (A) reveal mild hyperinflation of the lung fields. Axial CECT lung (window images) (B to D) show characteristic central perihilar ground glass opacities (arrows) and peripheral lucent areas due to air trapping (asterisk)

Cellular Interstitial Pneumonitis/Pulmonary Interstitial Glycogenosis (PIG)[3]

- Also known as infantile cellular interstitial pneumonia—ICIP or cellular interstitial pneumonitis of infancy. It is different from cellular pneumonitis of infancy (CPI).
- Probably a reactive condition of developing lung
- Often seen in association with other pathologies such as meconium aspiration and other congenital anomalies.
- *Presentation*: Tachypnea since birth.
- Good prognosis with improvement seen over time.
- *Histopathology*: Resembles CIP with increased cellularity of the interstitium. Electron microscopy shows primitive interstitial cells with abundant cytoplasmic glycogen. *Lacks inflammation (unlike UIP, DIP, LIP); dysmaturity of cells seen*
- *Chest radiographs*: Nonspecific with diffuse lung infiltrates.
  - Diffuse fine reticular opacities with hyperinflation
  - Early stages resemble hyaline membrane disease, with a hyperinflated lung
  - Later, may resemble BPD.
- *HRCT*: Very few case reports **(Figs 18.10A to D)**
  - *Early stage*: GGOs
  - *Later*: Interlobular septal thickening/fibrosis interspersed with areas of emphysematous changes
  - Worst at the bases.

Chronic Pneumonitis of Infancy

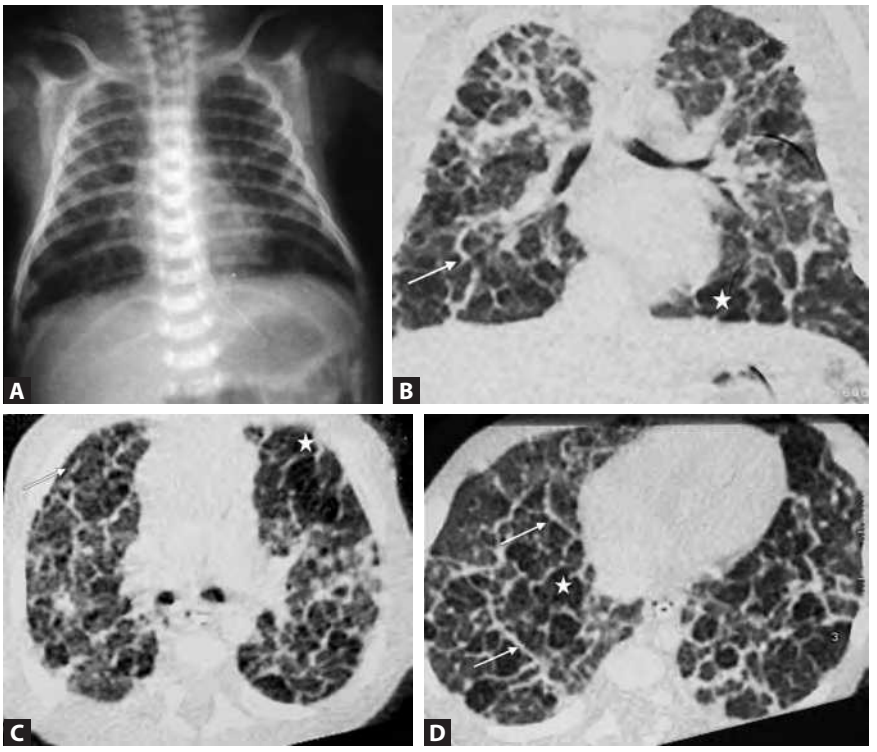
- Not included in the new classification system
- Infants with evidence of severe lung disease. Has a high mortality rate. Some cases maybe associated with genetic surfactant abnormalities.
- *Histopathology*: Alveolar exudates with numerous macrophages, alveolar septal thickening, pneumocyte hyperplasia.
- *Chest radiographs*: Extensive ground-glass opacities, areas of fibrosis, volume loss or hyperinflation.

DISORDERS WHICH CAN PRESENT AS SEVERE NEONATAL DIFFUSE LUNG DISEASES[1]

- Diffuse developmental disorders
- PIG
- Surfactant protein B deficiency (homozygous SFTPB mutations)
- *ABCA3* gene mutations TTF-1 (NKX2.1) mutations
- Pulmonary hemorrhage syndromes
- Pulmonary lymphangiectasia.

Pulmonary Lymphatic Disorders

- Diverse imaging manifestation



Figs 18.10A to D: Pulmonary interstitial glycogenosis. The chest radiograph (A) of a 6-month-old infant shows a pattern which resembles BPD. Neuroendocrine cell hyperplasia of infancy (NEHI). CT chest coronal (B) and axial (C and D) images show smooth interlobular septal thickening (arrow) and patchy air trapping (asterisk)

- Disorders of this group reflect pulmonary manifestations of abnormal lymphatic system development and/or overgrowth
- Entities included
  - Thoracic lymphangioma (can rarely be intrapulmonary)
  - Lymphangiohemangioma
  - Lymphangioleiomyomatosis
  - Lymphangiomatosis
  - Lymphangiectasia
  - Kaposi's sarcoma
  - Kaposiform lymphangiomatosis
  - Lymphatic dysplasia.

Lymphangioleiomyomatosis (LAM)

- Abnormal smooth muscle and lymphatic proliferation.
- Sporadic form of LAM is commoner than syndromic association with tuberous sclerosis.

- Sporadic form is a disease seen almost exclusively in women of child bearing age.
- Even in syndromic LAM, females are more commonly affected than male; although males can be affected too.
- Even in syndromic LAM, the older children or adolescents are more commonly affected than very small children and infants.
- In proven cases of tuberous sclerosis, about 30–40% patients can have LAM.
- Patients of tuberous sclerosis with LAM may not have the classical imaging triad of tuberous sclerosis (epilepsy, mental retardation and adenoma sebaceum).
- The pulmonary imaging findings of sporadic and syndromic LAM are similar.
- *Imaging*: Thin walled cysts uniformly distributed in lungs, and chylous pleural effusion.
- *DD*: LCH (cysts are more irregular, bizzare shaped; intervening lung often abnormal, systemic manifestations commonly associated with pediatric LCH).

Pulmonary Lymphangiectasia[4]

- Congenital anomaly of abnormal lymphatics development
- High mortality; 50% of affected are stillborn[5]
- Those who survive present in early infancy/neonatal period with respiratory distress and chylous pleural effusion
- *Associations*: Down's syndrome, Noonan syndrome, congenital cardiac anomalies
- *Pathology*: Abnormal dilation of the axial and peripheral lymphatics (peribronchial, septal, subpleural)
- *Imaging*:
  - Smooth interlobular septal thickening
  - Pleural effusion
  - Marked pleural thickening
  - Mediastinal soft tissue thickening
- *MRI*: T2W hyperintense septal thickening in the axial and peripheral interlobular septae
- *Imaging differential*: Lymphangiomatosis (presents late, clinical features different).

Lymphangiomatosis

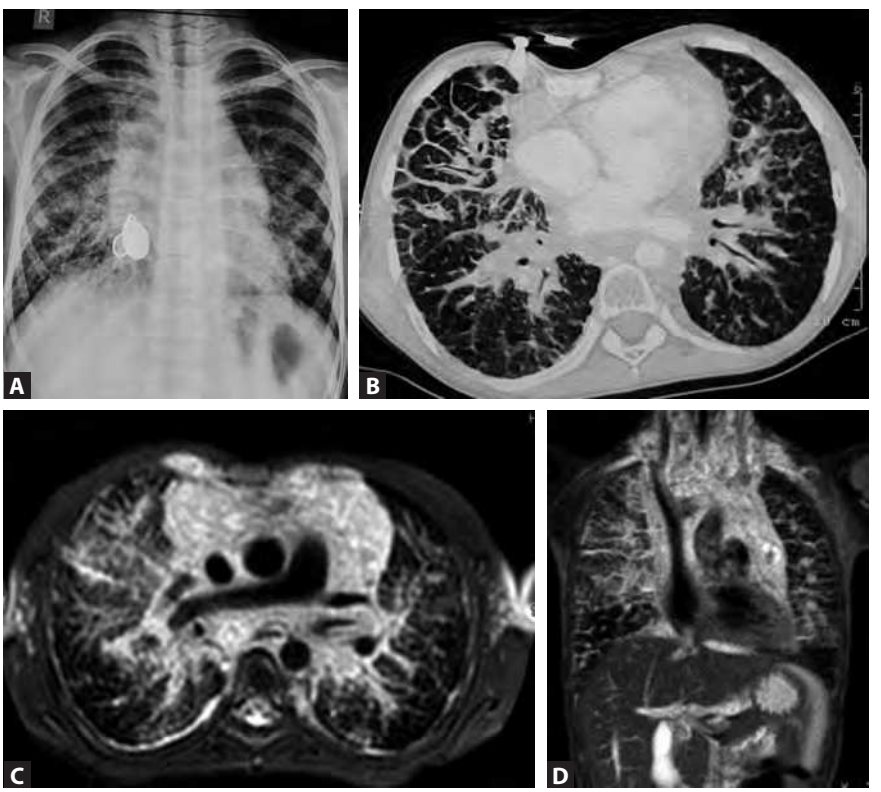
- Proliferation and dilation lymphatics; *multisystem disorder*
- Congenital; but presents later in childhood
- *Pulmonary involvement*: Diffuse bilateral smooth interlobular and peribronchial septal thickening (septal thickening can sometimes be nodular)
- Young children presents with worse clinical features and progressive dyspnea

- *Imaging*: Septal thickening, visceral and parietal pleural thickening; mediastinal soft tissue without mass effect; little or no lymphadenopathy.

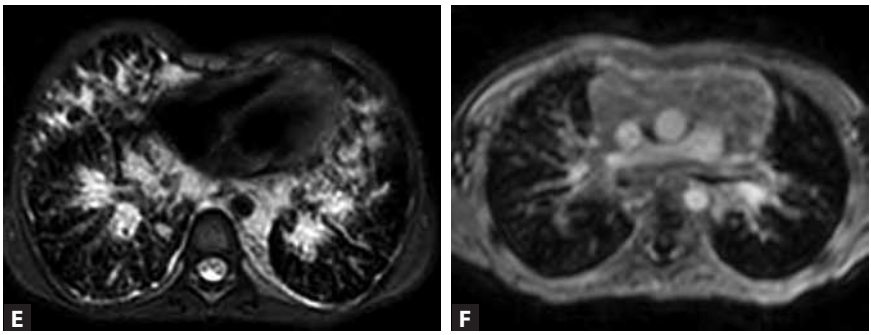
Kaposiform Lymphangiomatosis

- Rare, newly described entity[6] with overlapping features with generalized lymphangiomatosis
- Progressive disorder with foci of 'kaposiform' spindle cell proliferation in the endothelium of lymphatics on histopathology
- *Presentation usually in childhood*, with respiratory difficulty, bleeding disorders; cutaneous hemanio/lymphangioma rare
- Thrombocytopenia, Kasabach-Merritt syndrome associated
- *Other organs affected*: Spleen, bones
- Often refractory to therapy
- *Imaging*: Enhancing mediastinal/peribronchial soft tissue in lymphatic distribution
  - Hemorrhagic effusion[7] **(Figs 18.11A to F)**.

DLDs of known cause: DLD secondary to systemic diseases, aspiration syndromes, eosinophilic pneumonia, and immunocompromised host are discussed in Chapter 19.



Figs 18.11A to D



Figs 18.11E and F

Figs 18.11A to F: Pulmonary lymphatic disorder. Kaposiform lymphangiomatosis in a 8-year-male having insidious onset dyspnea and bicytopenia. The chest radiograph (A) shows reticular opacities in both lungs; with central perihilar predominance. The CT (B) reveals smooth interlobular septal thickening; which appear very hyperintense on T2W MR images (C to F) and show no contrast enhancement. The hyperintensity on T2W images suggests a diagnosis of dilation of lymphatics in the interlobular septae

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19

Diffuse Lung Diseases in Older Children: Imaging

Ashu Seith Bhalla, Manisha Jana

INTRODUCTION

- Diffuse lung diseases (DLDs) in children are characterized by impaired gas exchange, and diffuse changes on imaging
- Entities significantly different compared to adults
- The disease spectrum is different from those causing DLDs in infancy
- Less prevalent than DLDs in infancy
- DLDs in children are seldom idiopathic, usually having an underlying etiology.

CLASSIFICATION

Diffuse lung diseases (DLDs) in older children can be classified in various ways. The new classification of DLDs which are not specific to infancy has been described in **Table 19.1**.

The following disorders are covered in detail elsewhere in the book.

Infections (*See* Chapters 23 and 28); Hypersensitivity pneumonitis (*See* Chapter 13); Toxic inhalation (*See* Chapters 13 and 24); Malignancies (*See* Chapter 23); Opportunistic infections (*See* Chapters 23 and 24); Immune mediated (*See* Chapters 7 and 9).

SPECIFIC ENTITIES

Infections

Bacterial Pneumonia

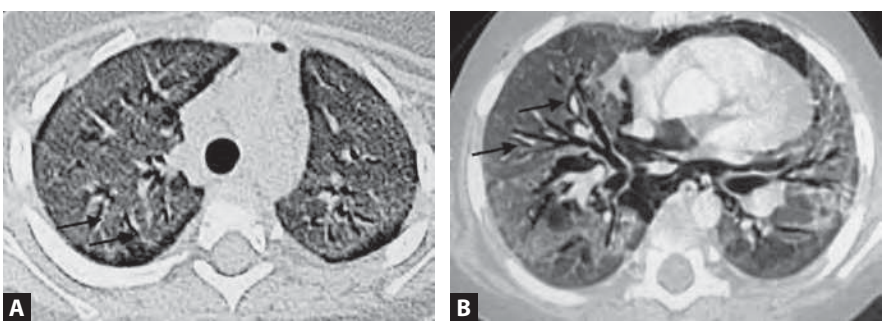
- Presents with lobar consolidation
- Interstitial pattern not commonly seen.

Pneumocystis Pneumonia

- Affects immunocompromised host
- Caused by *Pneumocystis jirovecii*
- *Presentation:* Respiratory distress, hypoxia

Table 19.1: Diffuse lung disorders not specific to infancy[1]

• Disorders of the normal host	• Infectious and postinfectious processes • Disorders related to environmental agents – Hypersensitivity pneumonitis – Toxic fume inhalation • Aspiration syndromes • Eosinophilic pneumonia
• Disorders related to systemic disease processes	• Immune-related disorders • Storage disorders • Sarcoidosis • Langerhans cell histiocytosis • Malignant infiltrates
• Disorders of the immunocompromised host	• Opportunistic infection • Disorders related to therapeutic intervention • Disorders related to transplantation and rejection • Diffuse alveolar damage of unknown etiology
• Disorders masquerading as interstitial disease	• Arterial hypertensive vasculopathy • Congestive vasculopathy, including veno-occlusive disease • Lymphatic disorders • Congestive changes related to cardiac dysfunction

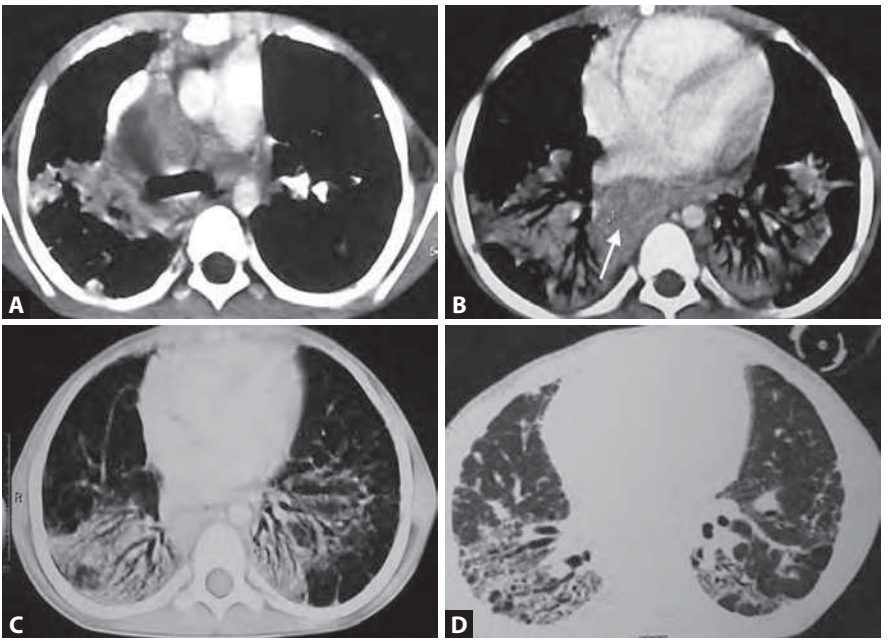


Figs 19.1A and B: Pneumocystis pneumonia with PIE. Axial CT images of lung reveal diffuse ground-glass opacity (A and B). Note the 'black bronchus sign'; described as prominence of the bronchial air in the background of GGO (arrows in A). PIE is seen as extension of air lucencies surrounding the pulmonary vessels (arrows in B)

- *Computer tomography (CT):* Perihilar or diffuse ground-glass opacity (GGO), consolidation, pneumatocele **(Figs 19.1A and B)**
- *Complications:* Pulmonary interstitial emphysema (PIE), pneumothorax, pneumomediastinum.

Tuberculosis

- Interstitial lung disease (ILD) unusual imaging feature of tuberculosis
- Mediastinal fibrosis secondary to tuberculosis can cause lymphatic obstruction, giving rise to smooth septal thickening on high-resolution computed tomography (HRCT).



Figs 19.2A to D: Interstitial lung disease in non-Hodgkin lymphoma. Axial CECT of chest soft tissue window (A and B) and lung window (C) reveal enlarged homogeneous lymph nodes in right paratracheal, right hilar and subcarinal locations (arrow); with high-density consolidation extending along the peribronchovascular interstitium contiguous with the lymph nodes. Follow-up axial CT of the lungs (D) after six months show development of interstitial fibrosis with traction bronchiectasis

Malignancy

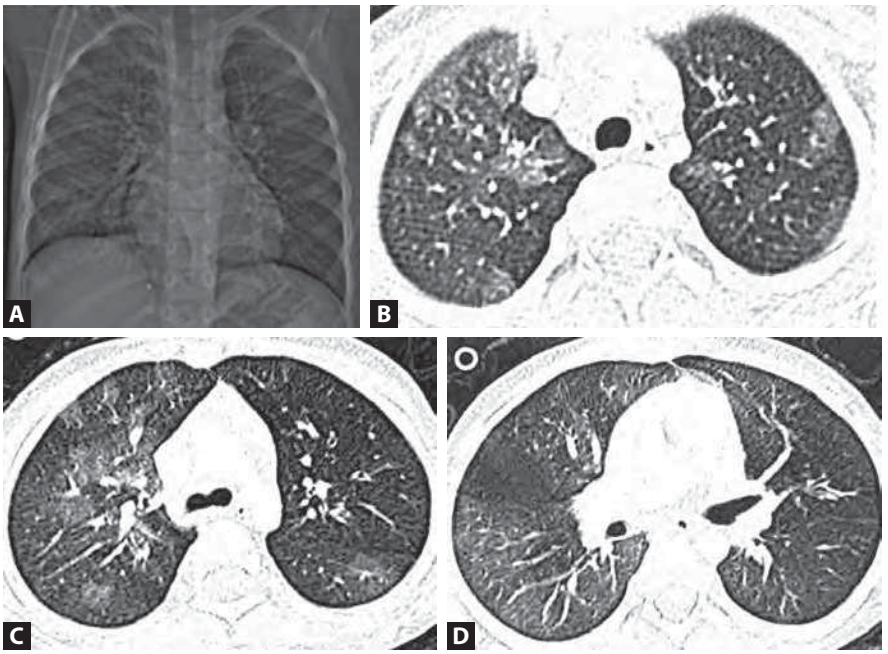
- Malignant mediastinal lymphadenopathy can give rise to lymphatic obstruction and smooth septal thickening on HRCT **(Figs 19.2A to D)**
- Lung interstitium involvement in Hodgkin's lymphoma usually secondary to mediastinal and hilar lymphadenopathy and contiguous peribronchovascular extension
- Primary lung interstitium involvement commoner in NHL than Hodgkin's lymphoma and more in relapse rather that primary diagnosis.

Hypersensitivity Pneumonitis (also *See* Chapter 13)

- Similar to adults, hypersensitivity pneumonitis (HP) in children is related to environmental exposure to antigens
- Common exposures in children are often related to bird or fungal antigens, unlike occupational exposure in adults **(Figs 19.3A to D)**
- Chronic HP shows interstitial fibrosis.

Aspiration Syndromes

- *Gastroesophageal reflux-related DLDs:* Lower lobes, dependent areas
- Consolidation and septal thickening

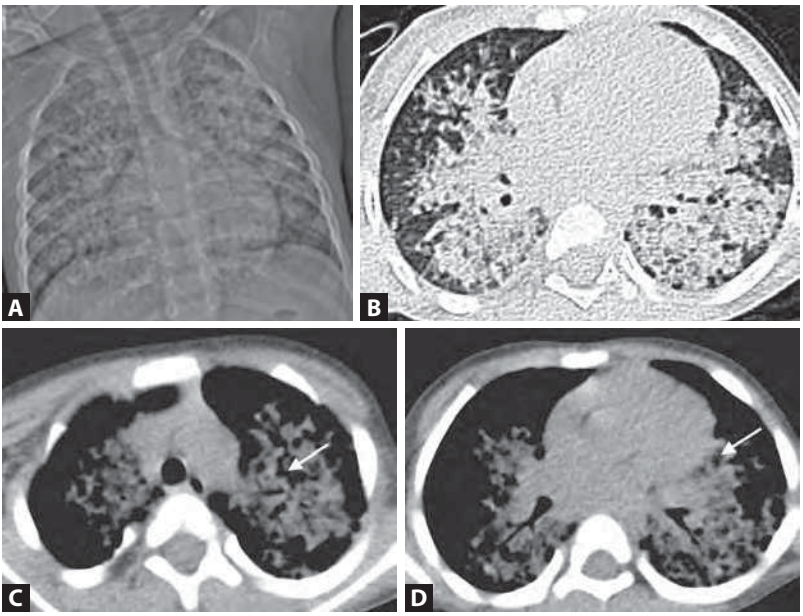


Figs 19.3A to D: Hypersensitivity pneumonitis in a 3-year male. The CT scout image (A) is near normal; whereas the CT images (B to D) reveal patchy GGO with interspersed lobular geographic areas of air trapping, giving rise to 'head cheese sign'

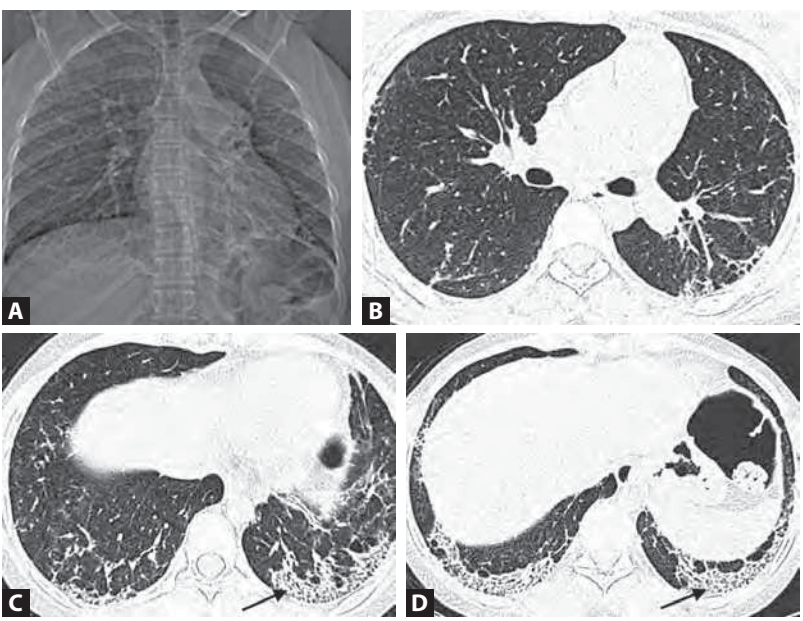
- Fibrosis in chronic stage
- *Barium swallow, scintigraphy:* Demonstrate reflux
- *Exogenous lipoid pneumonia:* Aspiration of oily substances (mineral oils used in constipation) **(Figs 19.4A to D)**
  - Vs Endogenous lipoid pneumonia—lipid accumulation in alveoli in obstructive pneumonitis
  - Acute/chronic
  - Chronic inflammation, lipid laden macrophages
  - Bilateral consolidation/GGO with low density (fat) areas within
  - "Crazy-paving pattern".

Immune-mediated/Collagen Vascular Disorders

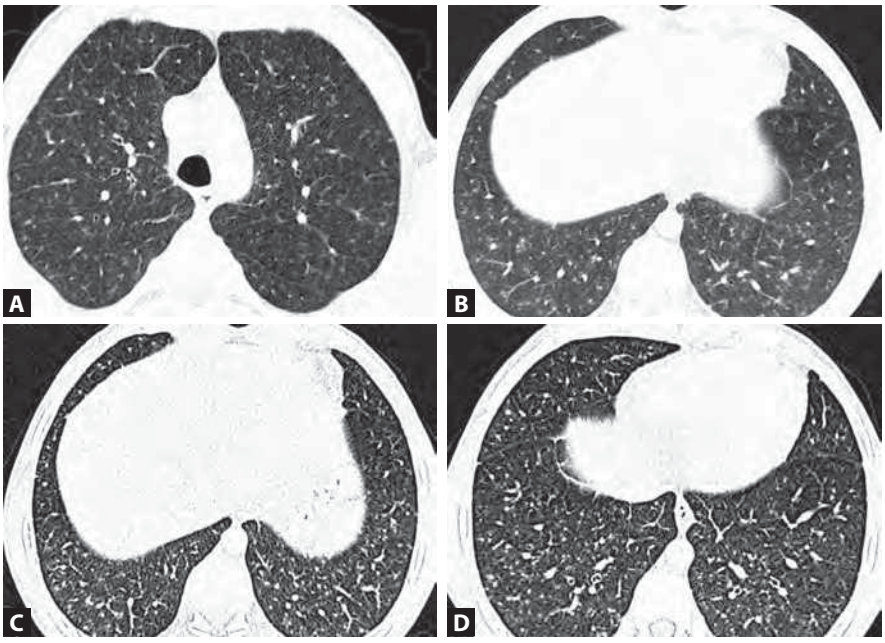
- Common histologic/radiological patterns include follicular bronchiolitis, nonspecific interstitial pneumonia or lymphocytic interstitial pneumonia **(Table 19.2)** (NSIP or LIP) **(Figs 19.5 and 19.6)**
- Pleural effusion, thickening or fibrosis may also be seen
- Underlying disorders include progressive systemic sclerosis (PSS), dermatomyositis, juvenile idiopathic arthritis and Sjögren syndrome. The frequency of the pattern differs according to the underlying disease (also *See* Chapters 7 and 9).



Figs 19.4A to D: Exogenous lipoid pneumonia in a one year male child who presented with cough, dyspnea and low grade fever for 2 weeks; not responding to antibiotics. CT scout film (A) and axial HRCT image (B) revealed bilateral dense perihilar consolidations. CT axial soft tissue window images (C and D) show low density areas (arrows) within the consolidations, characteristic for lipoid pneumonia



Figs 19.5A to D: NSIP pattern of DLD in a 17-year-old girl with SLE. CT scout film (A) and axial HRCT images (B to D) revealed bilateral peripheral subpleural areas of ground-glass opacity, fibrosis and septal thickening. There is no traction bronchiectasis, although microcystic honeycombing is seen (arrows)



Figs 19.6A to D: Follicular bronchiolitis pattern of DLD in a 12-year-old boy with SLE. CT lung window images (A, B) show ill marginated centrilobular nodules scattered over both lungs. The nodules are well-appreciated on HRCT images (C and D). Compared to centrilobular nodules seen in tuberculosis, they appear smaller

Table 19.2: ILD associated with collagen vascular disorders

Collagen vascular disorder	*ILD pattern*
PSS	NSIP, interstitial fibrosis, pulmonary artery hypertension
Dermatomyositis	LIP, interstitial fibrosis
Juvenile idiopathic arthritis	LIP, chronic bronchiolitis
Sjögren syndrome	LIP, chronic bronchiolitis
Wegener's granulomatosis	Nodules, consolidation

Abbreviations: PSS, progressive systemic sclerosis; ILD, interstitial lung disease; NSIP, nonspecific interstitial pneumonia; LIP, lymphocytic interstitial pneumonia

Follicular Bronchitis/Bronchiolitis (Figs 19.6A to D)

- Infants presenting with tachypnea, fine crackles, and chronic cough by the age 6 weeks
- Good prognosis, patients improving gradually over several years
- *Histopathology:* Follicular lymphocytic infiltration involving the bronchial walls
- Imaging appearance is similar to neuroendocrine cell hyperplasia of infancy (NEHI), but biopsy findings are different as airway inflammation is not

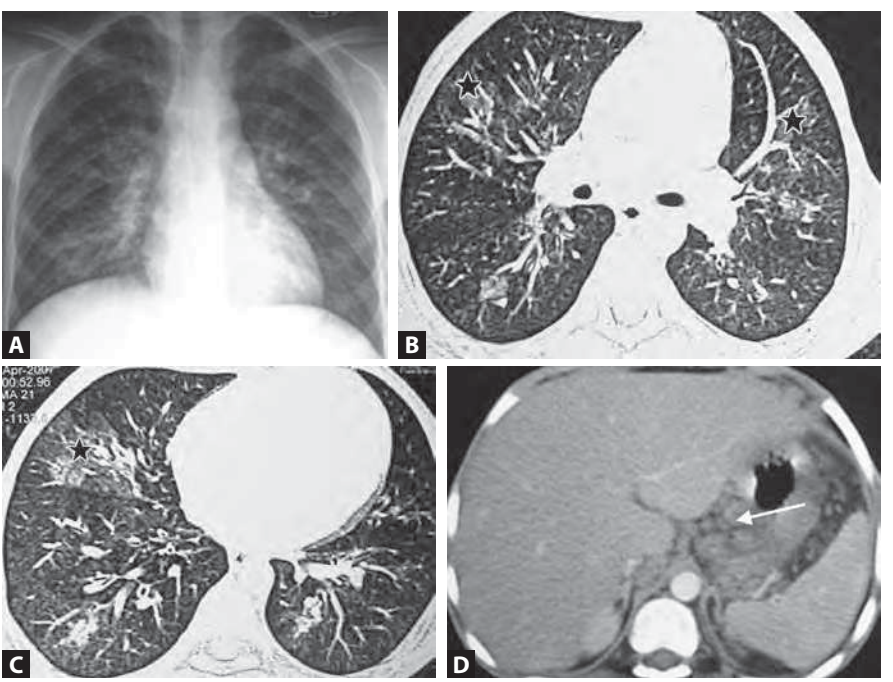
a prominent finding in NEHI. Also neuroendocrine cells are not seen in follicular bronchiolitis
- Chest radiographs, CT—Similar to NEHI.

Lymphocytic Interstitial Pneumonia

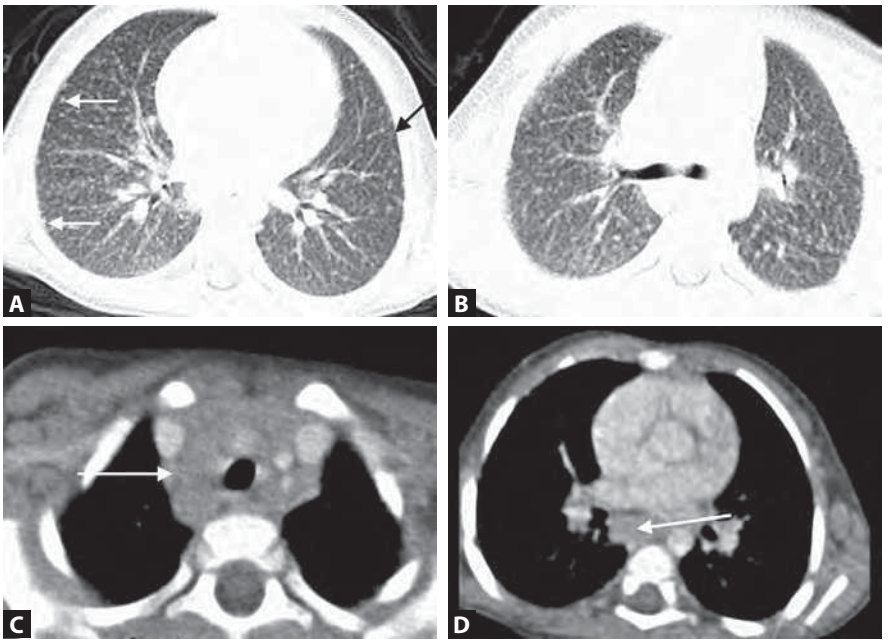
- Lymphocyte infiltrative disorders have a continuum of morphologic findings—follicular bronchitis/bronchiolitis (localized) to LIP (more diffuse)
- LIP associated with EBV and HIV infection
- HIV in children—30% acquire LIP (2nd–3rd year)
- *HRCT findings:* Nodules, GGO, peribronchovascular cysts, lymphadenopathy **(Figs 19.7 and 19.8)**.

Storage Disorders

- Rare disorders
- Storage disorders can result in a wide gamut of respiratory involvement which includes airway involvement (large and small), aspiration, ventilatory dysfunction, interstitial lung diseases and resultant pulmonary hypertension[2]
- Among the disorders causing ILD, lysosomal storage disorders (Gaucher disease, Niemann-Pick disease) are the commoner ones



Figs 19.7A to D: LIP in a 13-year-old male child with HIV infection. The chest radiograph (A) shows perihilar ground-glass opacities. HRCT chest (B and C) show peribronchovascular ground-glass opacities (asterisks) and bronchial wall thickening. CT of the upper abdomen (D) show retroperitoneal lymphadenopathy (arrow)



Figs 19.8A to D: LIP in a 6-year-old male child with HIV infection. The CT chest lung window images (A and B) show diffuse perilymphatic nodules (arrows). CT chest axial soft tissue window images (C and D) reveal mediastinal lymphadenopathy (arrows)

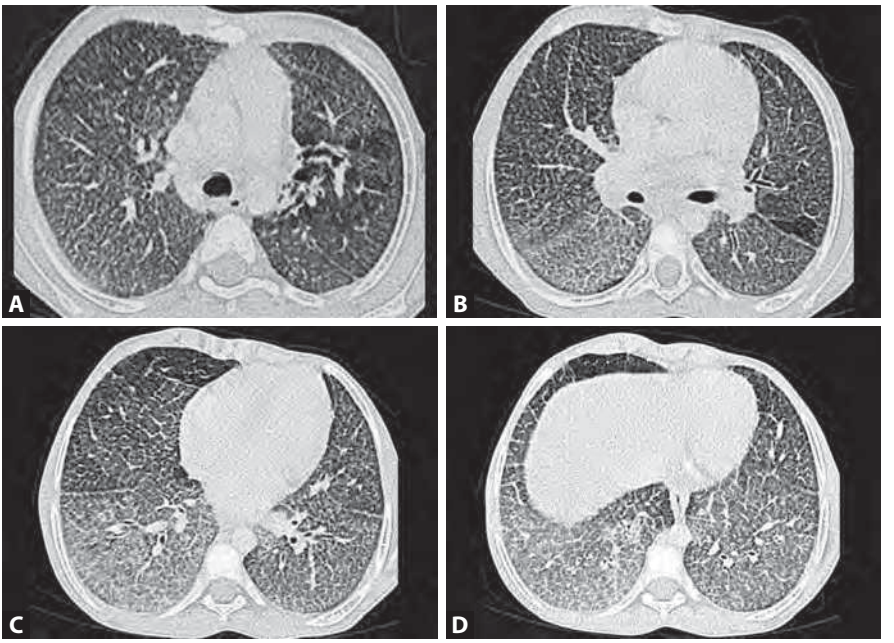
- Chest radiograph shows a reticular or less commonly reticulonodular pattern.

Gaucher Disease

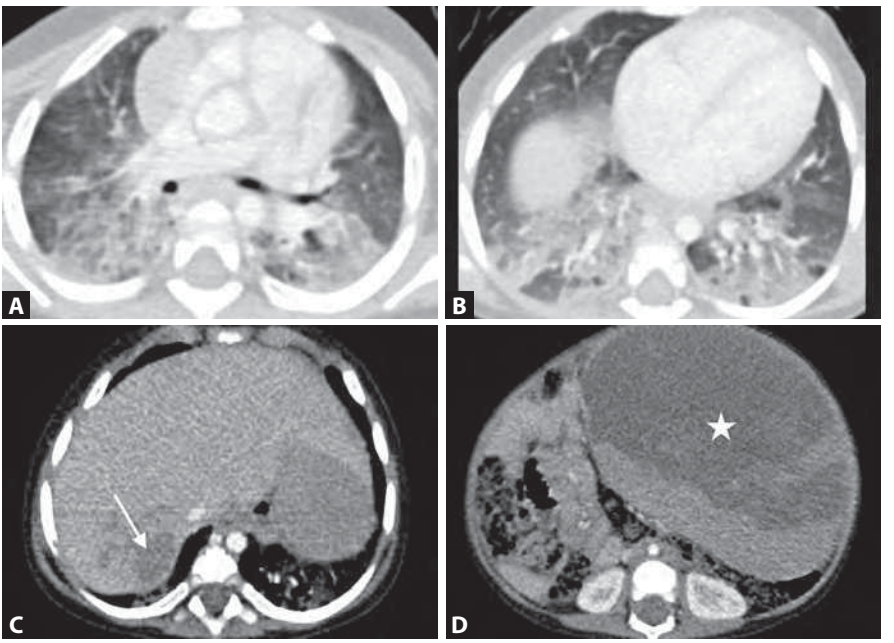
- The most common lysosomal storage disorder
- Deficiency of enzyme glucocerebrosidase
- Lipid-laden macrophages (Gaucher cells) collect in several organs including liver, spleen, bone marrow and less frequently in lungs
- HRCT findings **(Figs 19.9 and 19.10)** include septal thickening—Gaucher cells infiltrating interstitium
  - Desquamative interstitial pneumonia (DIP pattern)—alveolar filling by Gaucher cells
  - Pulmonary hypertension (involvement of capillary walls)[3]
  - Hepatopulmonary syndrome in long standing cases—dilation of subpleural pulmonary vessels before opening of arteriovenous shunt[4]
- *Management:* Enzyme therapy, substrate reduction therapy.

Niemann-Pick Disease

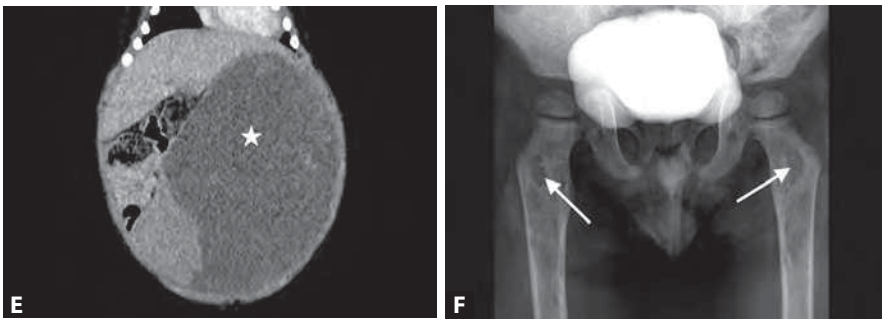
- Deficiency/absence of sphingomyelinase
- Niemann-Pick cells (NP cells) (lipid-laden macrophages) deposit in various organs



Figs 19.9A to D: Pulmonary manifestation in a 10-year-old child with Gaucher's disease. HRCT chest images reveal smooth interlobular septal thickening superimposed on the background of GGO, giving rise to 'crazy paving' appearance



Figs 19.10A to D



Figs 19.10E and F

Figs 19.10A to F: . Gaucher's disease in a 2-year-old male child; with hepatosplenomegaly and respiratory distress. CT chest (A and B) reveal bilateral diffuse ground-glass opacities and minimal interlobular septal thickening. CT abdomen axial and coronal reformatted images (C to E) reveal hypodense focal liver lesion (arrow) and a large splenic infarct (asterisk). Radiograph of the pelvis and proximal femora (F) reveals multiple geographic lytic lesions (arrows) in both femora; suggestive of Gaucher cell deposits

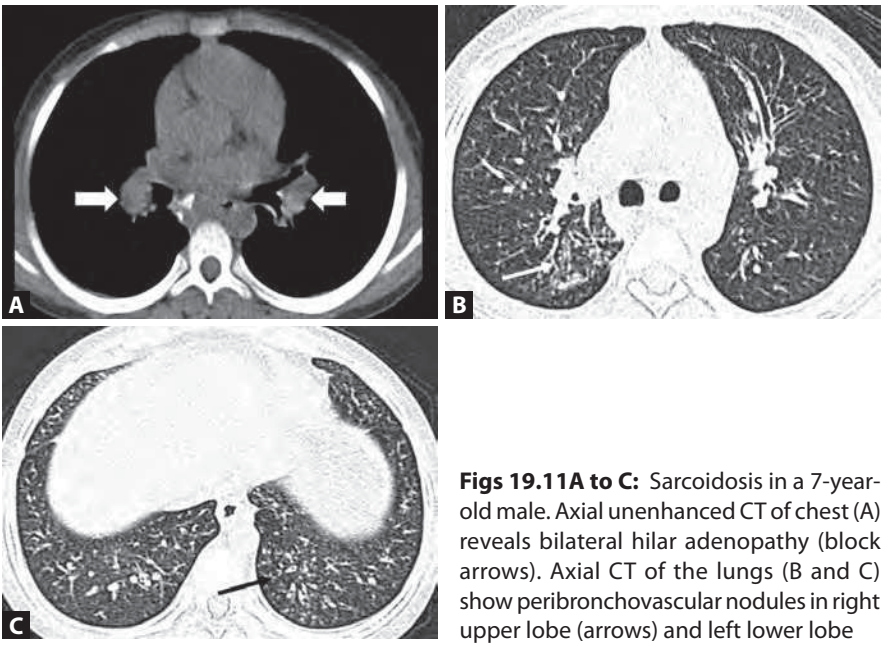
- In lungs, NP cells are deposited in interstitium of alveolar walls[5]
- *Bronchoalveolar lavage (BAL)/lung biopsy:* Characteristic NP cells seen
- *HRCT:* Manifestation of lipoid pneumonia as NP cells cause partial filling of alveoli
  - Ground-glass opacity in upper zones
  - Interlobular septal thickening (lung bases)
  - Background lung architecture is preserved
- *Management:* Supportive, lung lavage, hematopoietic stem cell transplantation.

Sarcoidosis

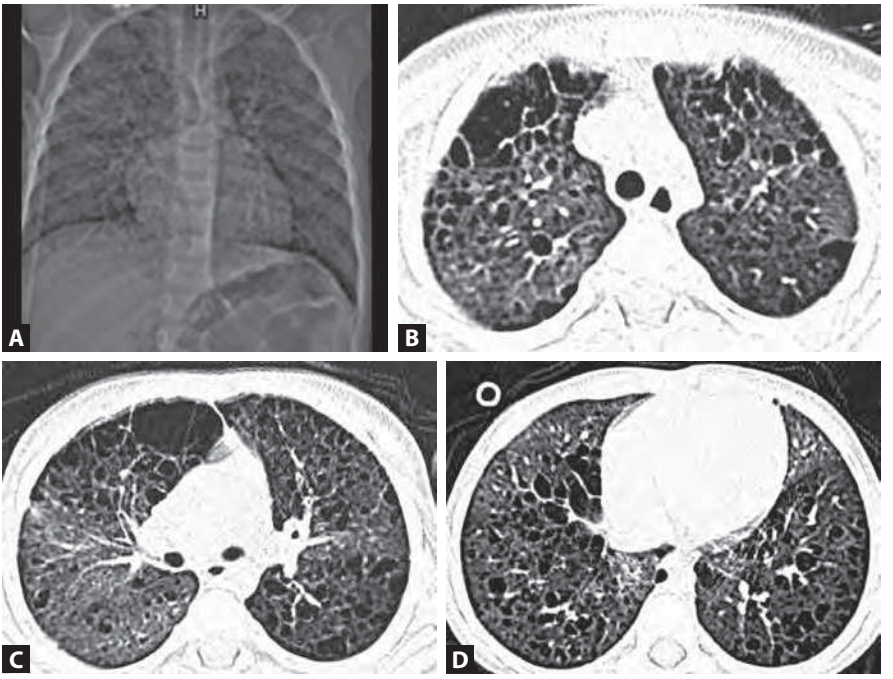
- *Age group:* Late childhood and adolescents
- *Imaging features:* Similar to adult sarcoidosis; hilar and mediastinal-adenopathy, peribronchovascular nodules, irregular septal thickening **(Figs 19.11A to C)**
- Alveolar sarcoid shows consolidation on CT
- *Advanced stage:* Interstitial fibrosis.

Langerhans Cell Histiocytosis (Figs 19.12 to 19.15)

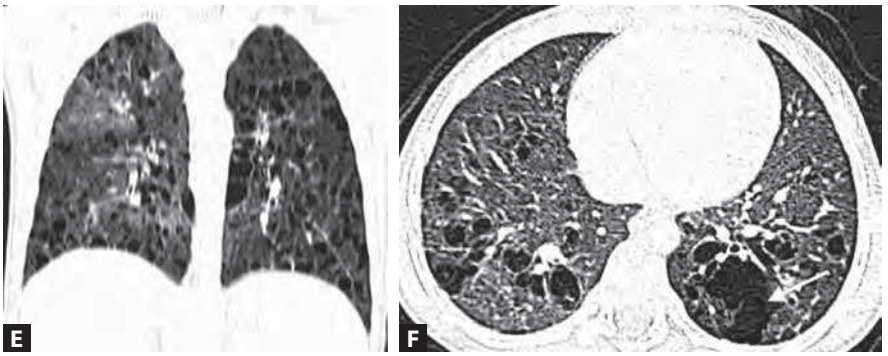
- Pulmonary Langerhans cell histiocytosis (LCH) seen approximately in 10% of children affected with multisystem disease
- Majority of childhood pulmonary LCH have systemic involvement, unlike adult pulmonary LCH
- Systemic features include ear discharge, mastoiditis, skeletal involvement, hepatobiliary involvement, maculopapular rashes, exophthalmos, diabetes insipidus secondary to pituitary stalk involvement, etc.



Figs 19.11A to C: Sarcoidosis in a 7-year-old male. Axial unenhanced CT of chest (A) reveals bilateral hilar adenopathy (block arrows). Axial CT of the lungs (B and C) show peribronchovascular nodules in right upper lobe (arrows) and left lower lobe

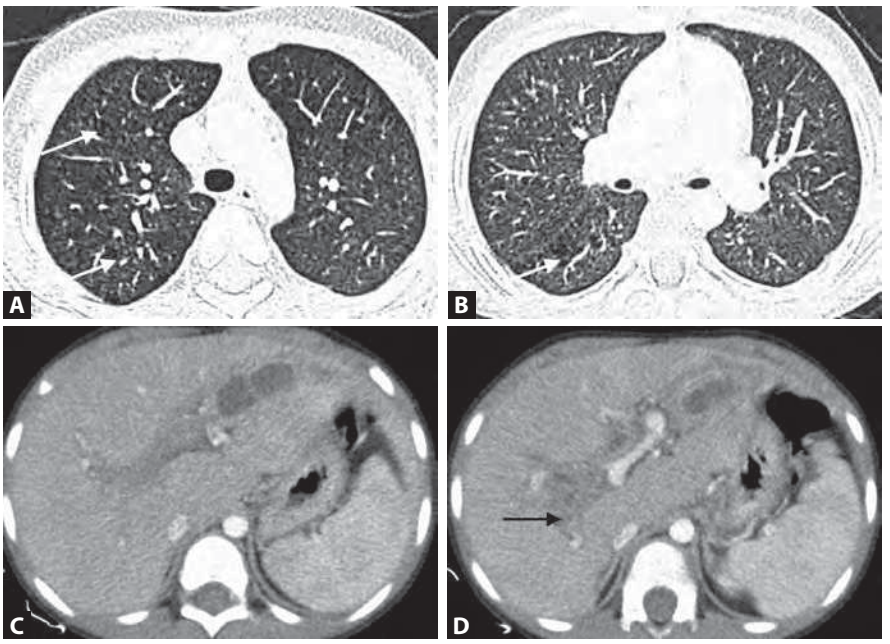


Figs 19.12A to D

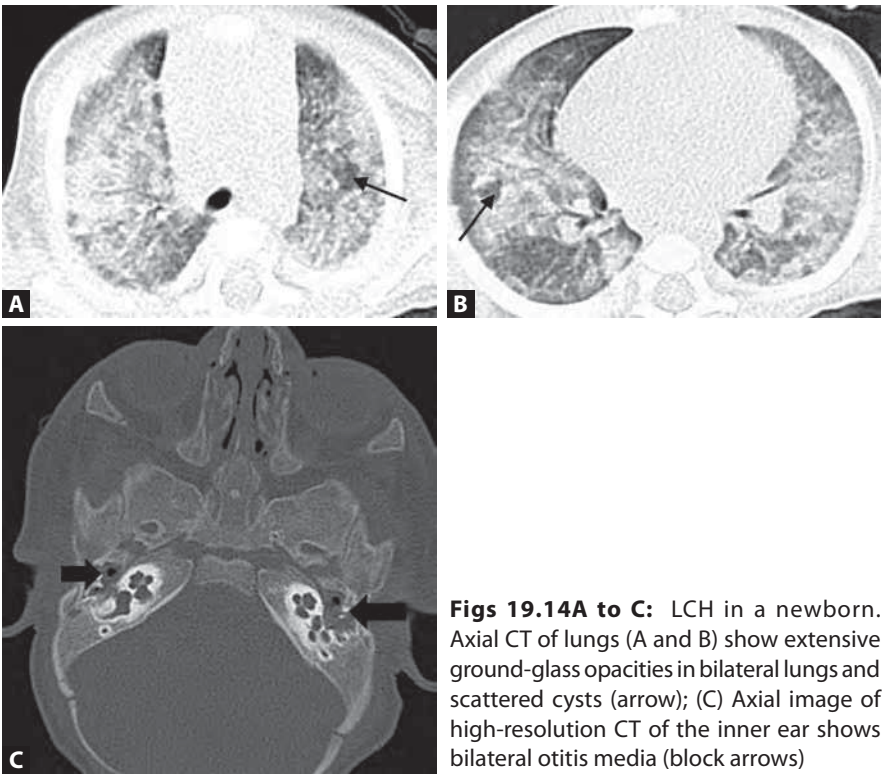


Figs 19.12E and F

Figs 19.12A to F: LCH in a 3-year-old boy. CT scout film (A) shows reticular opacities with cystic changes in both lungs. Axial HRCT of lung (B to D) and coronal reformatted image (E) show multiple irregular thin walled cysts involving both the lungs without any zonal predominance. Follow-up CT (F) after 1 year reveals progression of the cysts (arrow)



Figs 19.13A to D: Multisystem LCH in a 4-year-old old boy having cholestatic jaundice. Axial HRCT of lung (A and B) show multiple thin walled cysts involving both the lungs (arrows). CT of the abdomen (C and D) show periportal hypodensity (arrow) and biliary dilation, suggestive of hepatobiliary involvement in LCH



Figs 19.14A to C: LCH in a newborn. Axial CT of lungs (A and B) show extensive ground-glass opacities in bilateral lungs and scattered cysts (arrow); (C) Axial image of high-resolution CT of the inner ear shows bilateral otitis media (block arrows)

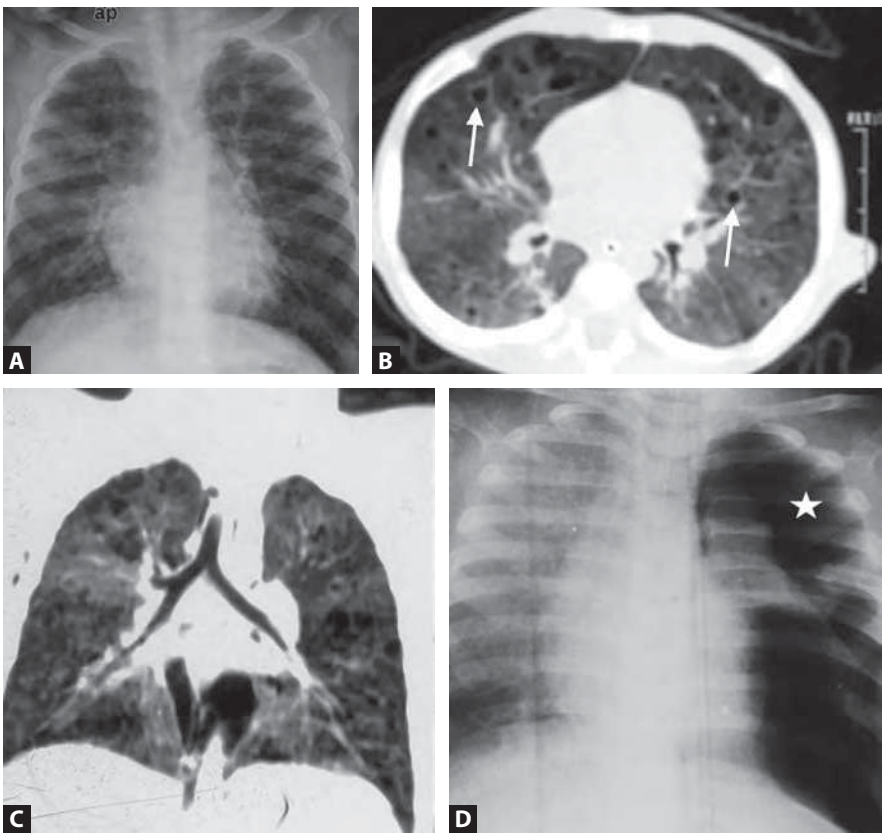
- Hepatobiliary involvement—periportal hypodensity (echogenic on US), focal hepatic lesions, sclerosing cholangitis
- Contrary to adult LCH, childhood pulmonary LCH has no upper lobe predilection
- Not associated with smoking
- CT findings—irregular nodules which tend to cavitate and form thin walled irregular cysts, interstitial fibrosis
- Pneumothorax is a frequent complication.

Diffuse Lung Disease Secondary to Cardiovascular Surgeries

- May accompany surgery for congenital heart diseases
- Imaging findings may mimic localized mediastinal fibrosis.

End-stage/Unclassified Interstitial Lung Disease

- A large number of childhood ILD remain unclassified (either end stage ILD, or inconclusive biopsy)



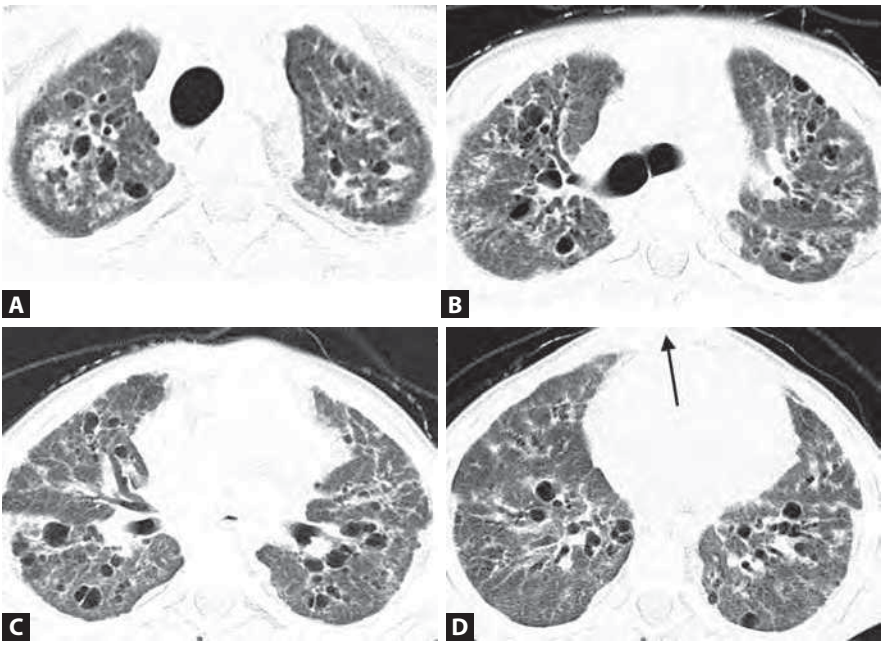
Figs 19.15A to D: Pneumothorax as a complication of pulmonary LCH. Chest radiograph (A) and CT (B and C) reveal multiple thin walled lung cysts (arrows) scattered bilaterally; with accompanying increased lung volume. A repeat radiograph done on an episode of worsening of dyspnea (D) revealed left sided pneumothorax (asterisk)

- *Causes:* End-stage ILD with extensive fibrosis, insufficient biopsy sample, improper area selection for sampling
- *Imaging features:* Fibrosis, traction bronchiectasis, volume loss **(Figs 19.16 and 19.17).**

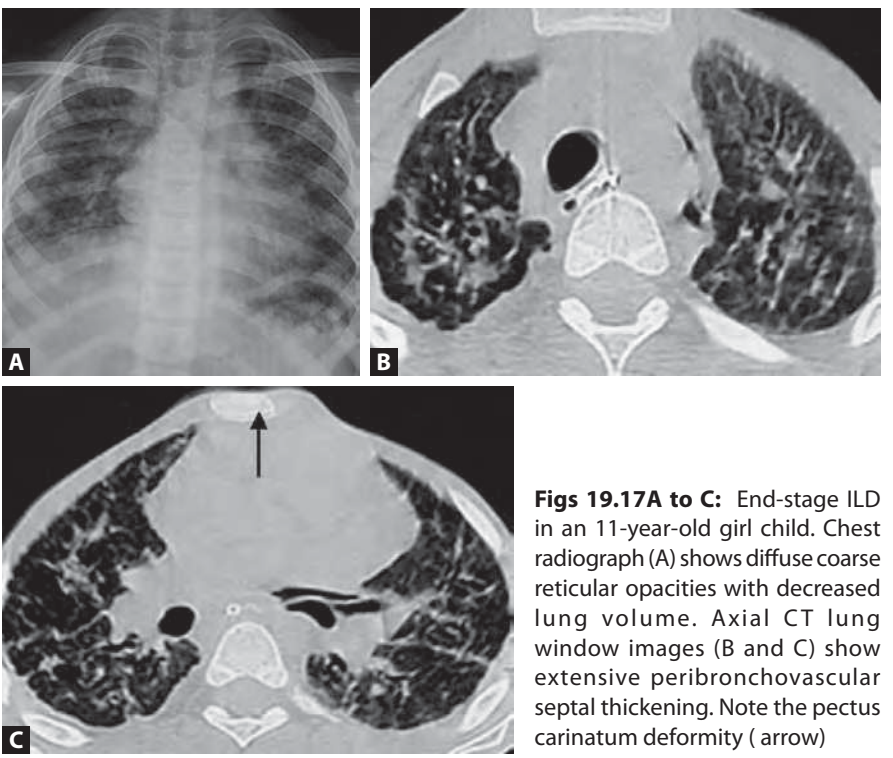
Miscellaneous

Diffuse Pulmonary Hemorrhage Disorders (alsoSee Chapter 21)

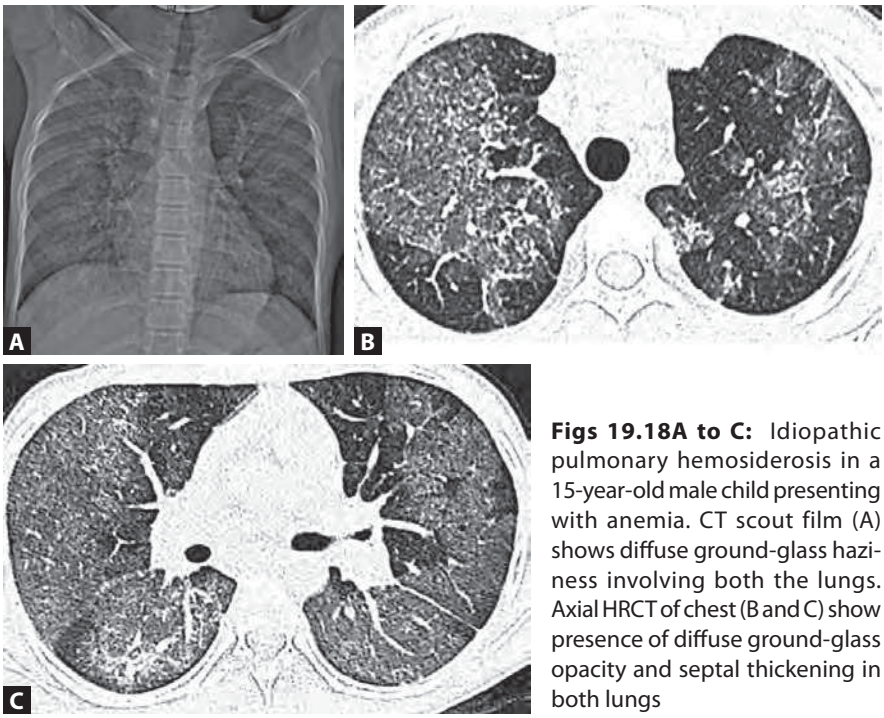
- Inflammatory disruption of interstitial capillaries with or without capillaritis
- Idiopathic form known as idiopathic pulmonary hemosiderosis (IPH) **(Figs 19.18A to C)**
- Cause of diffuse pulmonary hemorrhage in children
- Other common cause of diffuse pulmonary hemorrahge in children—Goodpasture syndrome
- *Age:* Usually under 10 years; both sex equally affected



Figs 19.16A to D: End-stage ILD in a 10-year-old male child. Axial CT images of chest lung window show extensive interstitial septal thickening, cyst and traction bronchiectasis. Note the pectus carinatum deformity (arrow)



Figs 19.17A to C: End-stage ILD in an 11-year-old girl child. Chest radiograph (A) shows diffuse coarse reticular opacities with decreased lung volume. Axial CT lung window images (B and C) show extensive peribronchovascular septal thickening. Note the pectus carinatum deformity (arrow)



Figs 19.18A to C: Idiopathic pulmonary hemosiderosis in a 15-year-old male child presenting with anemia. CT scout film (A) shows diffuse ground-glass haziness involving both the lungs. Axial HRCT of chest (B and C) show presence of diffuse ground-glass opacity and septal thickening in both lungs

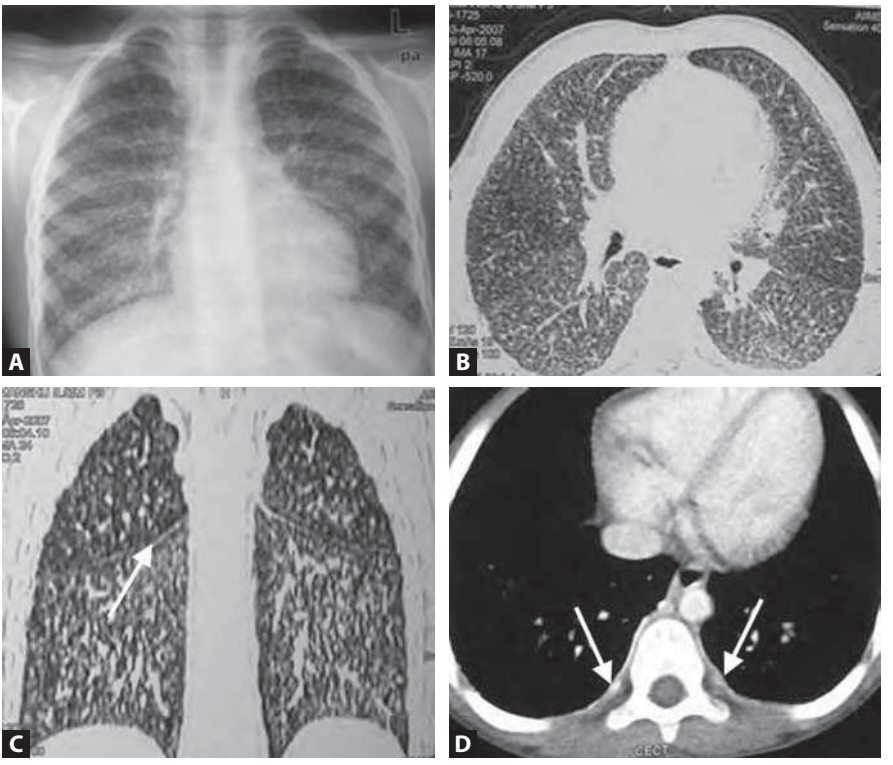
- *Etiology:* Unknown; autoimmune theory and sensitivity to cow milk protein proposed
- *Presentation:* Cough, dyspnea, fever, pallor, hematemesis/hemoptysis
- *Acute phase:* Acinar nodules/consolidation, GGO
- *Chronic phase:* Interstitial thickening and fibrosis
- *Acute hemorrhage:* Imaging findings revert to normal within 2 weeks.

Pulmonary Alveolar Microlithiasis

- Rare disease of unknown etiology
- Deposition of calcium phosphate microliths in the alveoli
- May be inherited as autosomal recessive trait
- *Early imaging findings:* Multiple small high density alveolar nodules **(Figs 19.19A to D)**
- *Advanced stage:* Extensive nodules coalescing to give rise to high density consolidation, obscuring the heart borders on X-ray, widespread nodules with sparing of subpleural lung giving rise to a subpleural black line on HRCT.

PATTERN APPROACH

Although the DLD spectrum in childhood is very complex; adopting a simplified pattern approach in HRCT may help in reaching a reasonable set of differential diagnoses **(Table 19.3)**.



Figs 19.19A to D: Pulmonary alveolar microlithiasis. Chest radiograph frontal view (A) shows extensive high density nodules involving both lungs. Axial CT of chest lung window (B), and coronal reformatted image (C) show extensive acinar nodules and nodular thickening of left sided major fissure (arrow). (D) Axial CT soft tissue window shows the hyperdense pleural due to conglomeration of the subpleural nodules

Table 19.3: Pattern approach towards differential diagnosis of childhood DLDs on HRCT

GGO	Acute/subacute presentation	Infections, pulmonary edema/ARDS, hypersensitivity pneumonitis, DAH
	Chronic presentation	Errors of surfactant metabolism, storage disorders, NSIP
Septal thickening/ reticular pattern	NSIP/UIP pattern	Connective tissue diseases, errors of surfactant metabolism
	Interlobular septal thickening	Venous/lymphatic edema
	Upper lobe fibrosis	Sarcoidosis
Nodules	Random	Miliary-infective High density-alveolar microlithiasis
	Centrilobular	Infective (TB), bronchiolitis
	Perilymphatic	Sarcoidosis, LIP

Contd…

Contd…

Cysts	Peribronchovascular/ perilymphatic	LIP
	Random	LCH, sequel of infection (PJP)
	Subpleural/basal	Fibrotic ILD/NSIP

Abbreviations: ARDS, acute respiratory distress syndrome; DAH, diffuse alveolar hemorrhage; GGO, ground-glass opacity; ILD, interstitial lung disease; LCH, Langerhans cell histiocytosis; LIP, lymphocytic interstitial pneumonia; NSIP, nonspecific interstitial pneumonia; PJP, *Pneumocystis Jirovecii* pneumonia; UIP, usual interstitial pneumonia

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20

Pediatric DLDs: Clinical Diagnosis and Management Issues

Rakesh Lodha, Sushil K Kabra

Childhood interstitial lung diseases (ILD) are a heterogeneous collection of disorders characterized by abnormal gas exchange due to altered structure of the interstitial region of the lung. As many entities affect distal bronchioles and alveolar spaces, the term "diffuse lung disease (DLD)" more accurate.

The exact prevalence is unknown; studies from UK and Ireland have reported prevalence of 3.6 per million children.[1] In a study from Delhi, using 10 years' retrospective data, 90 of 2017 children in a pediatric pulmonology service were diagnosed to have ILD.[2]

CLASSIFICATION AND CAUSES

Many of the conditions that cause ILD in children and adults are common; there are a few distinct causes in infants and children. There are different ways to classify the ILDs in children. Some disorders primarily affect the lungs while others may be associated with systemic diseases. Disorders seen more often in infancy are classified separately as diffuse developmental disorders, alveolar growth abnormalities, specific conditions of uncertain cause and surfactant dysfunction disorders[3] (*See* Chapter 18, Table 18.1 for detailed classification). Disorders not specific to infants are classified into various groups such as disorders of the normal host, disorders related to systemic disease processes, disorders of the immune-compromised host and disorders masquerading as ILD (*See* Chapter 19, Table 19.1).

Table 20.1 shows a broad classification based on the pathologic processes.

In addition to causes listed in previous classifications, there may be conditions that do not clearly fit into specific category (i.e. end-stage disease, inadequate or nondiagnostic biopsy specimen).

The following conditions should not be included in the list of causes of interstitial lung diseases: Cystic fibrosis, congenital or acquired immunodeficiency, congenital heart disease, bronchopulmonary dysplasia, pulmonary infection, primary ciliary dyskinesia presenting with newborn respiratory distress, and recurrent aspirations.

Table 20.1: Interstitial lung diseases in children

	Main pathologic abnormality	*Examples*	*Common symptoms/signs/ lab test abnormalities*
1.	Increased collagen in interstitium	Systemic lupus erythematosus Scleroderma Juvenile dermatomyositis Juvenile idiopathic arthritis	Reduced compliance Restrictive pattern Hypoxemia
2.	Vascular problems	Scleroderma Capillary alveolar dysplasia	Hypoxemia Pulmonary hypertension Reduced compliance Restrictive pulmonary function tests
3.	Increased cellular infiltrates in lungs	Pulmonary hemorrhage Langerhans cell histiocytosis Repeated aspirations Viral/ bacterial pneumonia	Hypoxemia Restrictive lung disease Systemic manifestations
4.	Exudates in alveoli	Pulmonary alveolar proteinosis Hypersensitivity pneumonitis Surfactant protein deficiency Toxic environmental exposure	Hypoxemia Restrictive pattern Marked breathlessness
5.	Acute alveolitis	Hypersensitivity pneumonitis Toxic exposure	Acute onset breathlessness Hypoxemia

In a study performed at our center, the common causes of ILD in children were: Langerhans cell histiocytosis, desquamative interstitial pneumonitis, systemic lupus erythematosus, idiopathic pulmonary hemosiderosis, hypersensitivity pneumonitis, pulmonary alveolar proteinosis, and pulmonary microlithiasis.[2]

PATHOPHYSIOLOGY

The mechanisms of lung abnormalities in ILD is not well-understood in many of the conditions. The proposed mechanisms are: response to lung injury similar to other organs; inflammation (alveolitis) followed by fibrosis. The various cells- lymphocytes, neutrophils, macrophages, fibroblasts play important roles.

CLINICAL PRESENTATION

The common clinical features in children with ILD include: Cough, exercise intolerance, tachypnea, cyanosis, crepitations on auscultation, failure to thrive. In addition, one should take history for frequent respiratory infections, exposure to dust or drugs, tiring during feeding, unexplained fever, features of systemic involvement – joint examination, rashes, lymphadenopathy, other systems.

When should Interstitial Lung Disease be Suspected?

In the neonatal period, ILD/DLD should be suspected if there is unexplained respiratory distress in a term baby. Interstitial lung disease (ILD) should be suspected in the first 2 years of life in a child with unexplained fast breathing, failure to thrive, dry cough in absence of respiratory tract infection. In children older than 2 years, ILD should be suspected in presence of unexplained respiratory symptoms/signs; failure to thrive; hypoxemia; diffuse radiological changes or unexplained exercise-induced breathlessness.

DIAGNOSIS

In a child suspected to have ILD, careful history and physical examination should be performed. A neonate or infant with DLD is regarded as having "chILD syndrome" if *at least three* of the following four criteria are present: respiratory symptoms (cough, rapid and/or difficult breathing, or exercise intolerance), respiratory signs (tachypnea, adventitious sounds, retractions), digital clubbing, failure to thrive, or respiratory failure), hypoxemia; and diffuse abnormalities on a chest radiograph (CXR) or computed tomography (CT) scan. The above approach is sensitive but not specific for ILD and has not been has studied prospectively.

There is a need for structured investigations as the presentation of ILD can be nonspecific. Hypoxemia may be documented at rest or on exercise. The chest radiograph may be normal or show nonspecific changes; this is rarely specific. The spirometry/Infant pulmonary function testing may show restrictive pattern and reduced diffusion capacity of the lungs. Echocardiography may reveal evidence of pulmonary arterial hypertension and rule out structural heart diseases. The following laboratory tests may be considered in the initial work-up:

- *Collagen vascular disorders:* Antinuclear antibody (ANA), antineutrophil cytoplasmic antibody (ANCA), rheumatoid factor (RF).
- *Hypersensitivity pneumonitis:* Serum precipitins
- *Infectious disease:* Cultures, serology, smear
- *Immunodeficiency:* Immunoglobulin assays, T-cell and B-cell subset quantification, serology for HIV infection
- Studies for gastroesophageal reflux, chronic aspiration
- Anti GBM antibody

In infants, various genetic tests may be performed **(Table 20.2)**.

Bronchoscopy and bronchoalveolar lavage (BAL) can be diagnostic in:

- Pulmonary hemorrhage syndromes
- Pulmonary alveolar proteinosis
- Eosinophilic lung disease

One can rule out hypersensitivity pneumonitis if BAL is normal. The target site for BAL can be guided by the imaging and airway inspection.

In children, particularly infants, transbronchial biopsy is not recommended as the tissue sample obtained is usually too small for diagnosis.

Table 20.2: Genetic tests in infants suspected to have ILD

	Age group	*Genetic tests*
1.	Newborns; severe disease	Mutations in genes: *SFTPB*, *SFTPC*, *ABCA3*
2.	Newborns with ILD, hypothyroidism, hypotonia	*NKX2.1* mutations/ deletions
3.	Infants with ILD; initial studies do not provide a diagnosis	*SFTPC*, *ABCA3* mutations
4.	Infants; alveolar proteinosis	*CSF2RA*, *CSF2RB* gene analysis

Reference no. 3.

Lung biopsy can provide a definitive diagnosis. However, the need and timing of the procedure is controversial. If a child's lung condition has progressed to a point where the child needs supplemental oxygen, one may consider a biopsy. A practical approach is to consider lung biopsy only if treatment is likely to change as a result. Preferably, the biopsy should be performed before initiating steroids. The biopsy site is based on recent CT chest findings; it is preferably taken from 2 sites by VATS/minithoracotomy. An adequate biopsy should be of 10 × 10 × 10 mm size.

MANAGEMENT

As most of the childhood ILDs do not have a specific treatment, supportive care is the mainstay:

- Nutrition
- Immunizations
- Supplemental oxygen for hypoxemia
- Supervised exercise (older children)
- Limiting exposure to smoke/environmental irritants
- Aggressive treatment of intercurrent infections.

Specific treatment may be available for some of the conditions:

- Antimicrobials for infections
- Avoidance of offending antigen in hypersensitivity pneumonitis
- Whole lung lavage/GM-CSF for pulmonary alveolar proteinosis

Corticosteroids are the mainstay of treatment; prednisolone is usually administered at 1–2 mg/kg/d. Pulse methyl prednisolone 30 mg/kg/d for 3–5 days every 3 months may be considered. With steroids, a favorable response is observed in 40–65%. Hydroxychloroquine is preferred in conditions like desquamative interstitial pneumonitis (DIP); the usual dose is 5–10 mg/kg/d. In children, immunosuppressive therapy with azathioprine, cyclophosphamide, cyclosporine, or methotrexate may be considered.

The European protocols for management of children with ILD include: pulse methylprednisolone, oral prednisolone, hydroxychloroquine and azithromycin.[4] The treatment guidelines are similar for children who are mechanically ventilated or close to being ventilated versus those who are not

mechanically ventilated or close to being ventilated; the difference is that the time period to assess response is shorter for the former group.[4]

Some of the newer therapies include:

- *Interferon gamma:* Antiproliferative, antifibrotic, anti-infective
- N-acetylcysteine (NAC)
- Combination-steroids+ NAC+ Azathioprine
- Bosentan-endothelin receptor antagonist

Lung transplantation including living donor lobar lung transplantation is an option.

The children should be monitored periodically:

- Respiratory rate and heart rate
- Weight, height
- SpO_2
  - Awake
  - Overnight while sleeping
  - On exercise
- Echocardiography for pulmonary artery hypertension
- Chest X-ray at 6 and 12 months
- Spirometry at each visit

The suggested frequency is monthly for 3 months; then at 6, 12 months.

PROGNOSIS

The prognosis is variable and depends on cause:

- *ILD of infancy:* NEHI, Interstitial glycogenosis good
- *Surfactant protein deficiency:* Poor
- Severity score at diagnosis is an important predictor of outcome

In our study: 3 out of 90 died and risk factor for poor outcome was high ILD score at time of diagnosis.[2]

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Section 7

Acute Diffuse Lung Diseases

- Acute Diffuse Lung Diseases: Imaging
- Clinical and Management Issues

21

Acute Diffuse Lung Diseases: Imaging

Ashu Seith Bhalla, Manisha Jana

INTRODUCTION

- Many diffuse diseases, both infectious and noninfectious, can have a severe, acute presentation.
- These can mimic acute respiratory distress syndrome (ARDS) clinically and radiologically.

CAUSES

Common Causes

- Acute respiratory distress syndrome (ARDS)
- Pulmonary edema
- Acute interstitial pneumonia
- Acute exacerbation of idiopathic pulmonary fibrosis (AEx-IPF)
- Infections
- Hypersensitivity pneumonitis
- Diffuse pulmonary hemorrhage.

Less Common Causes

- Acute cryptogenic organizing pneumonia
- Nonspecific interstitial pneumonia (NSIP)
- Acute eosinophilic pneumonia
- Connective tissue disease (CTD) related interstitial lung diseases (ILDs)
- Drug-induced diffuse lung diseases (DLDs)
- *Rare-Sarcoidosis:* Lofgren's syndrome.

IMAGING MODALITIES

Chest Radiograph

- Primary modality for initial evaluation and follow-up
- In most differentials reveals bilateral air space opacifications

- Pleural effusions, lung/lobar collapse may also be seen
- Assessment of width of vascular pedicle useful as it correlates with intravascular volume status and pulmonary artery occlusion pressure. Especially useful if intravascular pressure measurements not available.

Computed Tomography

- Higher sensitivity than chest radiograph (CXR) for delineation of abnormalities, aids in prognostication and differentiating amongst various causes
- However due to disadvantages of cost and radiation exposure; and primarily the risk and logistics of transporting a sick patient to the CT suite, it is not imperative for management
- Qualitative/quantitative assessment
- *Quantitative:* Volumetric assessment of areas of consolidation/ground glass opacity (GGO) (also see chapter in general section). Larger volumes of involvement correlate with poorer prognosis.

Ultrasound

- Thoracic ultrasound increasing being used in emergency and ICU settings as a supplement to clinical examination and CXR
- Increasing evidence in literature of its additive value
- May help in reducing the number of CT scans ordered
- Performed at bedside
- Limitations—obese patients, subcutaneous emphysema, learning curve.

Approaches

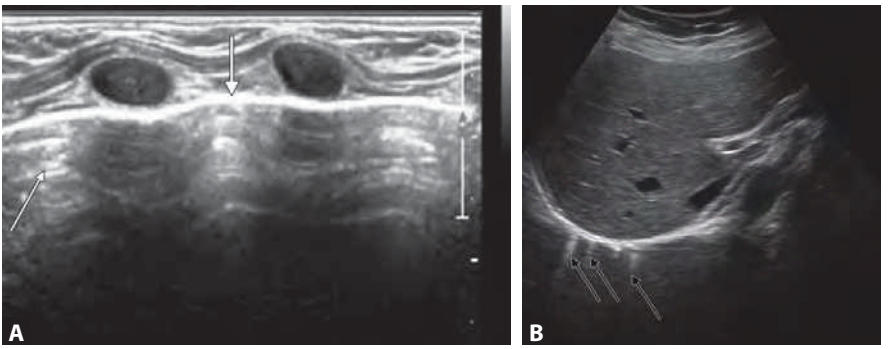
- Guided by a prior radiograph to focus on probable area of abnormality
- Trans-sternal, parasternal, and intercostal approaches—for the lung, pleura, anterior mediastinum
- Sector transducers useful in sub-xiphoid and trans diaphragmatic approaches—juxtaphrenic paravertebral lesions
- Suprasternal and supraclavicular approaches—upper mediastinum and lung apex.

Normal Chest Sonography ***(Figs 21.1A and B)***

- *A-lines:* Normal; horizontal lines parallel to the pleural surface
- Normal lung not visible under the A line
- *B-lines:* Vertical lines intersecting the A lines, 'comet-tail artifacts'
  - 3–4 can be normally seen.
  - If more in number, indicate in alveolar or septal infiltrates (ex-pulmonary edema/ARDS).

Consolidation

- Airless lung looks similar to echotexture of the liver and spleen: 'Tissue like configuration'/hepatization
- *'Shred sign'*: Interrupted appearance (shredded appearance) of the pleural line due to subpleural area of consolidation



Figs 21.1A and B: USG chest using a linear transducer (A) showing A-lines (arrow) as horizontal lines parallel to the echogenic pleural line (block arrow). USG chest through subxiphoid approach (B) shows normal appearance of B-lines (arrows)

- Multiple bright dot-like and branching linear structures—*sonographic air bronchogram*
- *Consolidation:* Lung volume is not decreased and bronchi show normal branching pattern
- *Atelectasis:* Decreased lung volume and of the involved lung can be crowded
- *Passive atelectasis:* Air bronchogram present as long as bronchi are patent
- *Mucoid impaction*: Bronchial tree is filled with fluid (replacing normal air), *sonographic fluid or mucus bronchogram.*

SPECIFIC ENTITIES

Acute Respiratory Distress Syndrome

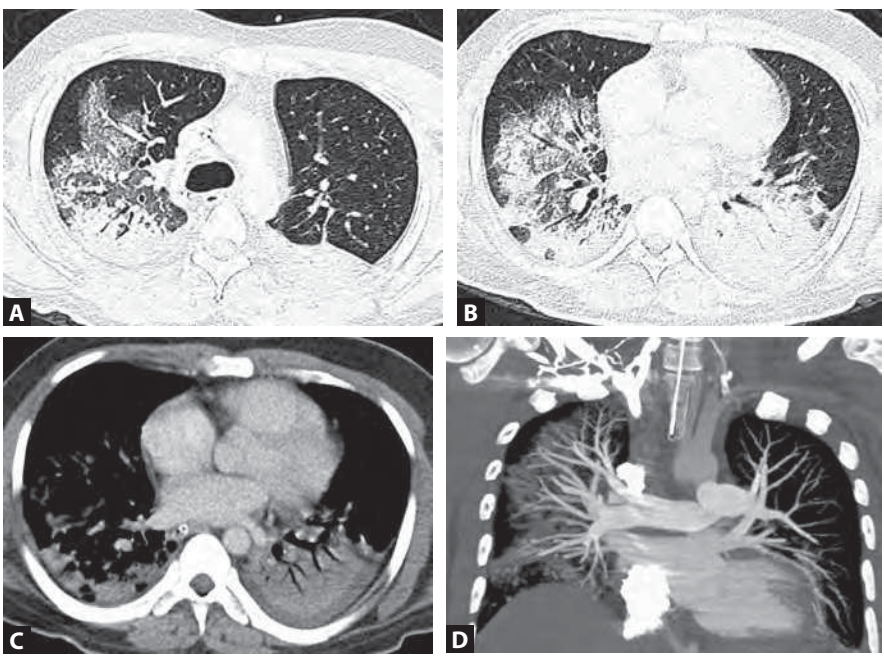
- Diffuse alveolar damage resulting from variety of insults, causing diffuse alveolar damage and requiring mechanical ventilation (also see clinical section)[1]
- *Berlin definition: To standardize terminology used by intensivists and radiologists*[2]
- *Pathology*: Diffuse alveolar damage, increased capillary and alveolar permeability, lead to edema (interstitial and alveolar), followed by a phase of fibrosis.
  - Three phases, can overlap.
  - *Acute (exudative) phase:* Initial seven days following insult; characterized by alveolar edema, interstitial inflammation, hemorrhage and hyaline membrane formation.
  - *Subacute (proliferative) phase:* Proliferation of fibroblasts in the interstitium also in alveoli; proliferation of type II pneumocytes in alveoli. Conversion of alveolar exudates into cellular granulation tissue.
  - *Chronic (fibrotic) phase:* Two weeks or more following injury. Diffuse parenchymal fibrosis with collagen deposition, bullae.

Chest Radiograph

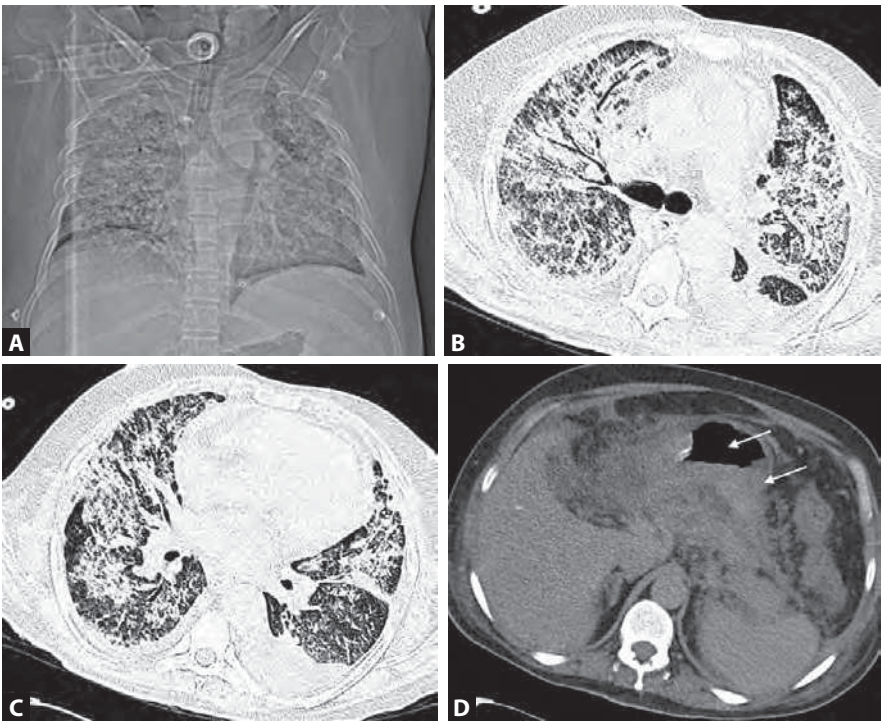
- Chest radiograph (CXR) can be normal in the first 24 hours—stage of early alveolar edema
- Over the next 2–3 days—extensive, bilateral air space opacification/consolidation (whiteout lung)
- Homogeneous/patchy multifocal
- Picture similar regardless of the etiology
- *Radiographic criteria for diagnosis of ARDS:* Bilateral opacities (consistent with pulmonary edema), findings not explainable by pleural effusion, lung collapse or nodules.

High-resolution Computed Tomography

- CT is not an essential part of diagnostic criteria for ARDS, but when available implications of findings same as seen on CXR.
- Acute phase **(Figs 21.2A to D)**
  - Normal initially, abnormalities appear after 12 hours.
  - *Ground glass opacities:* Dominant finding, diffuse or patchy (geographic distribution).
  - *Consolidation:* Initially patchy, increase later in the exudative phase and become more homogeneous.



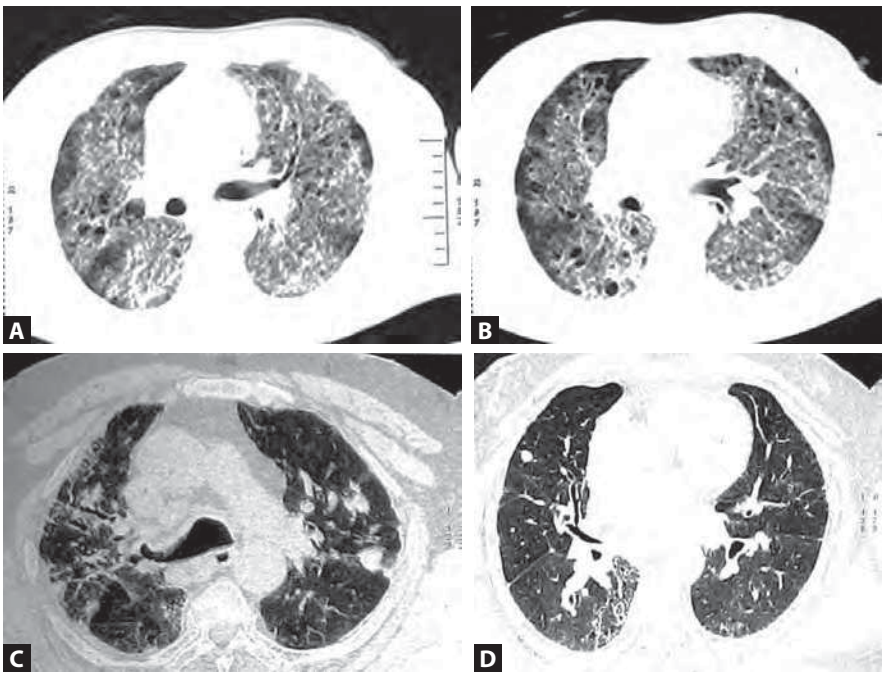
Figs 21.2A to D: ARDS in a young male with septicemia. HRCT sections (A and B) revealed the central perihilar distribution of consolidation and ground glass opacities. Soft tissue window axial (C) and MIP image (D) show the bilateral dependent and perihilar predominance of disease



Figs 21.3A to D: ARDS in subacute phase in a young male with acute pancreatitis. CT scout film (A) reveals whiteout lung with reticular shadows. HRCT sections (B and C) show the central perihilar distribution of consolidation, with superimposed septal thickening and ground glass opacities. Soft tissue window axial (D) image shows bulky pancreas with peripancreatic fat stranding (arrows)

- Apicobasal gradient may be seen, with more opacification seen in lung bases.
- Areas of atelectasis may also be seen.
- Lung morphology determines response to therapy.

- Subacute phase **(Figs 21.3A to D)**
  - Increasing consolidation in early phase, followed by appearance of interstitial fibrosis
  - Interlobular septal thickening superimposed on GGO—"crazy paving pattern"
  - Bronchiectasis, bronchiolectasis
  - Severity of changes helps prognosticate
  - Those with more severe changes liking to require more prolonged mechanical ventilation, with subsequent risk of its attendant complications.
- Chronic phase **(Figs 21.4A to D)**
  - Interstitial fibrosis increases
  - Coarse reticular fibrosis



Figs 21.4A to D: ARDS in subacute and chronic phase. HRCT images in subacute phase (A and B) reveal ground-glass opacities with superimposed septal thickening; having a central perihilar preponderance. HRCT repeated five months later (C and D) reveal fibrosis, traction bronchiectasis in a subpleural distribution

- Architectural distortion
- Bronchiectasis, bronchiolectasis with distortion of airways
- Volume loss
- Bullae
- Severity of changes more in those with prolonged positive pressure ventilation
- Morphological changes may persist in survivors even when imaged months after the acute episode.

Ultrasonography

- Aids in differentiation from hydrostatic pulmonary edema
- Heterogeneous distribution of abnormalities
- *B lines:* Well separated and discrete/coalescent
- Areas of consolidation—more in posterior regions, areas of sparing seen.

Hydrostatic Pulmonary Edema

- Cardiac failure or fluid overload
- Pulmonary edema essentially refers to accumulation of fluid in the extravascular compartment of the lung

- *Pathology:* The mechanism includes hydrostatic edema (as in fluid overload/left ventricular failure); diffuse alveolar damage (DAD) and edema due to increased permeability but no DAD (e.g. high altitude pulmonary edema, transfusion related acute lung injury)
- *Two pathological phases:* Interstitial edma and alveolar filling
- Imaging **(Figs 21.5 and 21.6)**.
  Chest Radiograph
  - Primary modality for diagnosis and follow up
  - *Early changes:* Enlargement of upper lobe pulmonary veins (due to diversion of blood flow)
  - Airspace opacities with central/parahilar (batwing) distribution
  - Kerley lines
  - Pleural effusion.

  Computed Tomography
  - HRCT not performed for diagnosis of hydrostatic pulmonary edema
  - However recognition of features necessary to allow differentiation from other causes
  - HRCT findings include:
    - *Ground glass opacities:* Diffuse/patchy and parahilar. Gravitational predominance may be seen
    - *Vs increased permeability edema; which has more subpleural and peripheral distribution*
    - In early cases centrilobular ground glass nodules may be seen, while areas of consolidation may be present in areas of more alveolar filling
    - Smooth interlobular septal thickening. In conjunction with GGO, characteristic of hydrostatic edema. Focal nodularity if seen reflects dilated septal veins; often prominent in lung bases

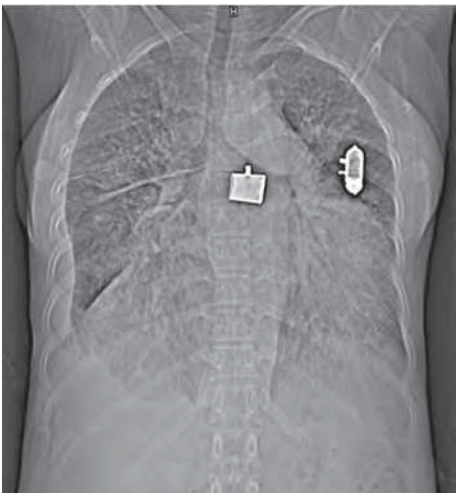
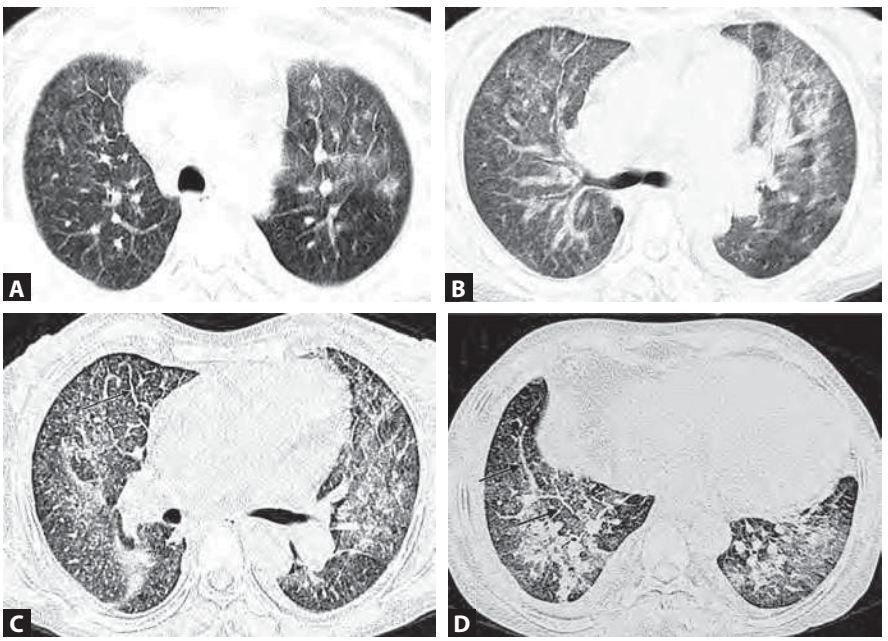


Fig. 21.5: Hydrostatic pulmonary edema. CT scout film reveals bilateral perihilar reticular shadows and pleural effusion on right side



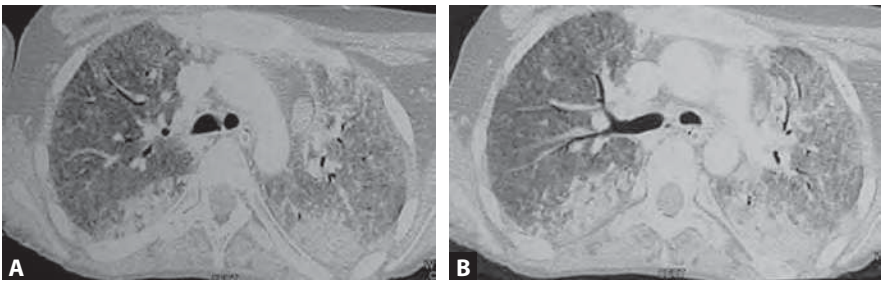
Figs 21.6A to D: Hydrostatic pulmonary edema. CT lung window images (A and B) and HRCT (C and D) images show central distribution of ground-glass opacities and consolidation; with smooth interlobular septal thickening (arrows)

- Peribronchovascular interstitial thickening—appears as bronchial wall thickening or prominence of pulmonary vessels
- *Vascular prominence:* In cardiogenic pulmonary edema, pulmonary arteries and veins appear dilated. These include central vessels, intrapulmonary, centrilobular vessels as well as septal veins.

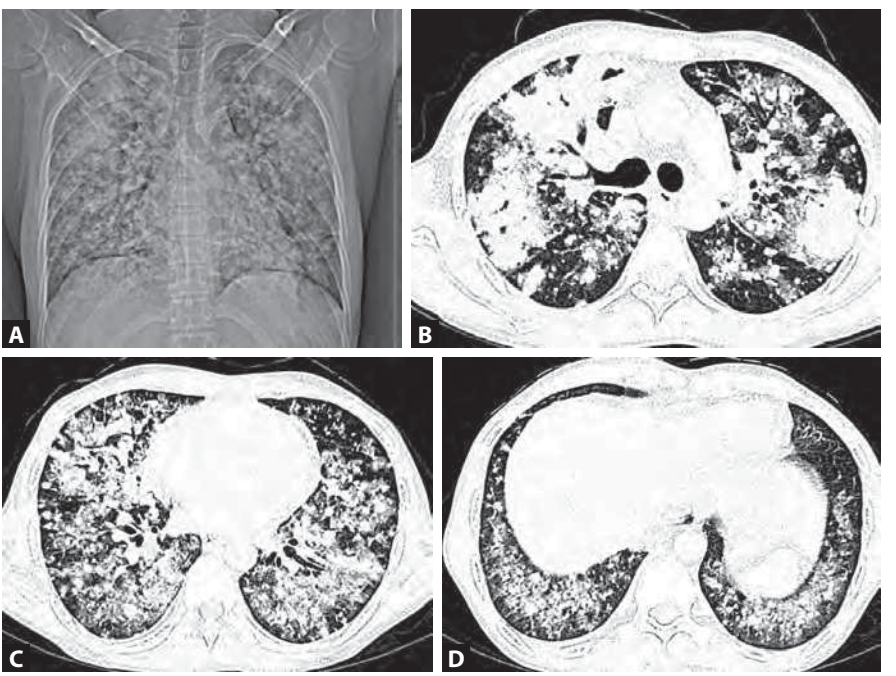
Pleural effusion: Transudative pleural effusion, unilateral or bilateral with smooth fissural thickening.

Acute Interstitial Pneumonia

- Synonym/earlier called Hamman-Rich syndrome
- Pathology similar to ARDS, i.e. diffuse alveolar damage
- *However idiopathic:* In patients who were otherwise healthy, with no known predisposing cause
- Dyspnea and rapidly progressing respiratory failure
- Often preceded by a flu like illness with symptoms such as fever and arthralgias
- Poor prognosis
- HRCT features similar to ARDS and vary according to the stage of the illness **(Figs 21.7 and 21.8)**
- *Vs ARDS:* Acute interstitial pneumonia (AIP) more likely to have predominantly lower zones, symmetrical distribution of GGOs and consolidation **(Table 21.1)**



Figs 21.7A and B: Acute interstitial pneumonia. HRCT chest (A and B) show diffuse ground glass opacities in both lungs; and consolidation in bilateral dependent areas



Figs 21.8A to D: Acute interstitial pneumonia in a patient with Crohn's disease. CT scout image (A) shows diffuse consolidation in both lung fields. HRCT chest (B to D) show airspace nodules and consolidation with a central predominance

- With progression the areas of involvement become more extensive
- Presence of bronchiectasis and bronchiolectasis within the areas of increased attenuation are bad prognostic indicators, as these imply onset of fibrosis
- Residual abnormality in survivors—mild reticulation.

Acute Exacerbation of Idiopathic Pulmonary Fibrosis (AEx-IPF)

- IPF is a fibrotic ILD of unknown etiology with typically a chronic, progressive course.

Table 21.1: Differentiating features of ARDS, hydrostatic pulmonary edema, AIP and DAH on imaging

	Acute respiratory distress syndrome (ARDS)	*Hydrostatic pulmonary edema*	*Acute interstitial pneumonia (AIP)*	*Diffuse alveolar hemorrhage (DAH)*
Distribution of radiographic abnormality	Diffuse or patchy, geographical distribution	Central, perihilar, dependent areas, 'batwing pattern'	Lower zone predominance, symmetrical	Central and dependent areas
Smooth interlobular septal thickening	Not common in acute phase	Commonly seen	Not common	Seen in intermediate phases
Pleural effusion	Not an essential finding	Commonly seen	Not common	Not common

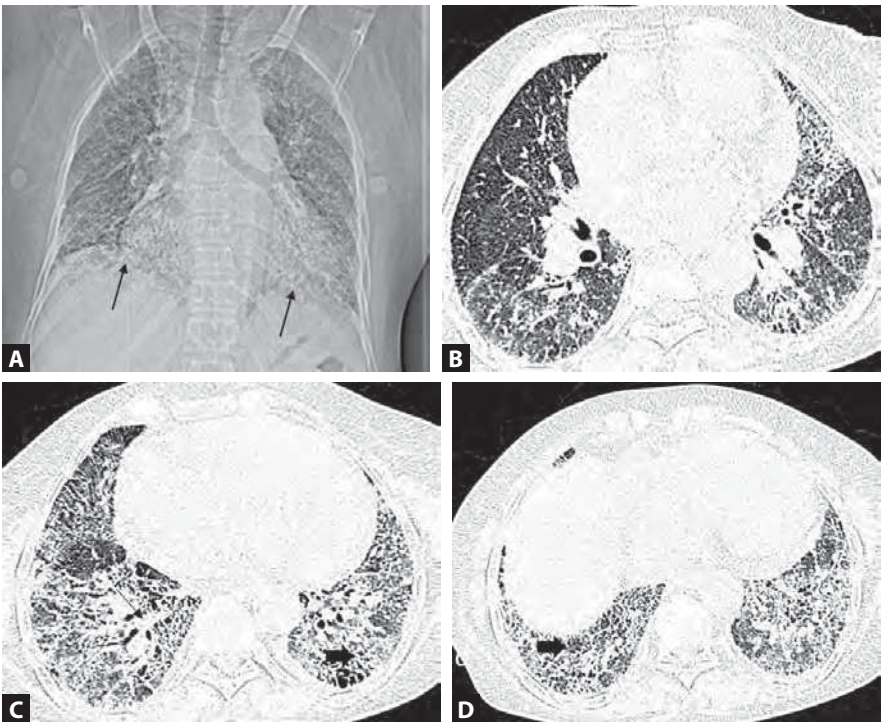
- There can however be episodes of acute exacerbation of pulmonary symptoms in these patients.
- Causes of acute exacerbation include: Infection, acute exacerbation of the ILD, pulmonary embolism, pneumothorax or pulmonary edema.
- AEx-IPF is diagnosed after excluding the other known causes of severe worsening.[3]
- *Pathology:* DAD or organizing pneumonia (OP) superimposed on underlying UIP.
- *HRCT:* Bilateral ground glass opacities/consolidations. Distribution is diffuse or multifocal in those with DAD; and peripheral in OP. Underlying UIP pattern seen **(Figs 21.9A to D)**.
- Prognosis is worse in those with diffuse/multifocal pattern, with poorer survival rates.
- Onset of traction bronchiectasis and bronchiolectasis also indicates worse prognosis, as in all causes of DAD.

Infections

- Infections can also result acute/subacute interstitial pneumonia
- *Causative agents:* Influenza virus, *Pneumocystis jirovecii*.

Hypersensitivity Pneumonitis

- Acute Phase
- *CXR:* Non-specific findings/normal
  - May show diffuse GGO with ill defined nodules
- *HRCT*: Seldom performed as symptoms improve rapidly.
  - However, severe cases may present with acute respiratory failure due to DAD
  - In such cases, HRCT picture is similar to exudative phase of DAD as detailed in ARDS



Figs 21.9A to D: Acute exacerbation of IPF in a 50-year-old lady. CT scout image (A) shows reticular opacities and honeycombing at bilateral basal areas (arrows). HRCT chest (B to D) show diffuse ground-glass opacities; with a background of interlobular septal thickening, traction bronchiectasis (arrow) and honeycombing (block arrow) having peripheral subpleural and basal predominance

- Bilateral GGO seen, background of chronic changes may be seen
- Consolidation, however, is unusual and only seen if there is superimposed infection or areas of organizing pneumonia.

- *Differential diagnosis:* Difficult to differentiate from other causes of bilateral GGO. Subacute phase on the other hand has a more specific findings with centrilobular nodules and lobular areas of air trapping.

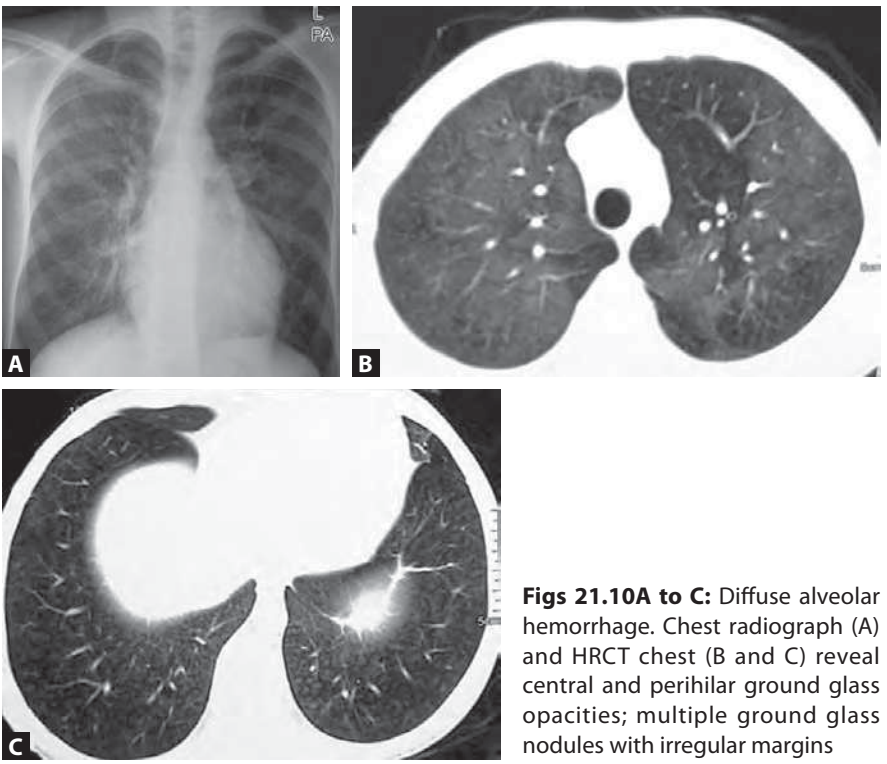
Diffuse Alveolar Hemorrhage (Also see chapter 9)

- *Synonym:* Diffuse pulmonary hemorrhage (DPH).
- Numerous causes (also see chapter on pulmonary vasculitis).
- Variable presentation—with/without hemoptysis.
- *Pathology:* Three different patterns have been described—associated with pulmonary capillaritis, bland alveolar hemorrhage with no vasculitis and DAD related.
- *HRCT:* Bilateral GGO/consolidation **(Figs 21.10 and 21.11)**; lobular appearance more in dependent areas **(Figs 21.12A to D)**.
- In most cause, the pulmonary changes are reversible, except in those with DAD.

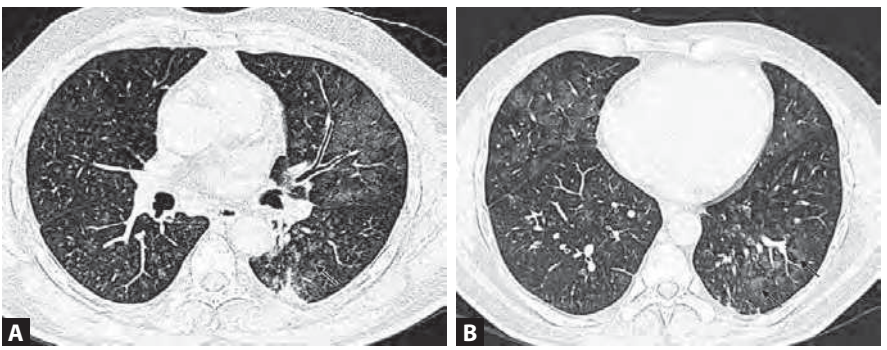
LESS COMMON CAUSES

Cryptogenic Organizing Pneumonia

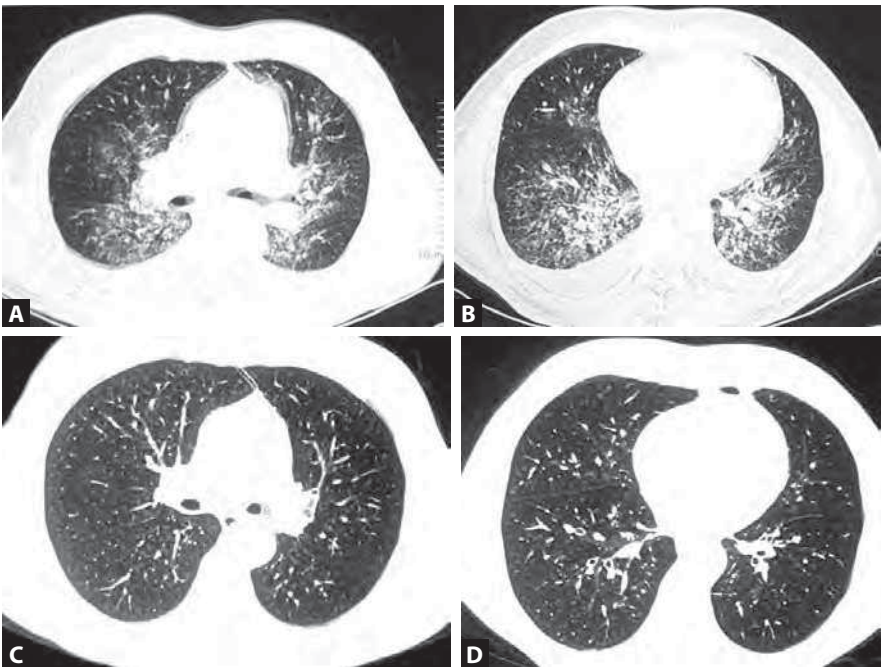
- Also see chapter 5
- Usually cryptogenic organizing pneumonia has a subacute presentation,[3] with cough and dyspnea, occasionally acute onset



Figs 21.10A to C: Diffuse alveolar hemorrhage. Chest radiograph (A) and HRCT chest (B and C) reveal central and perihilar ground glass opacities; multiple ground glass nodules with irregular margins



Figs 21.11A and B: Diffuse alveolar hemorrhage in a male with repeated episodes of hemoptysis and anemia. HRCT chest images reveal diffuse ground glass opacities; multiple ground glass nodules with irregular margins (arrows)



Figs 21.12A to D: Temporal evolution in diffuse alveolar hemorrhage. HRCT chest images at presentation (A and B) reveal diffuse ground glass nodules in both lungs. Repeat CT performed 7 days later (C and D) reveal near complete clearance of nodules

- *CT:* GGO, consolidation which is patchy in distribution—peripheral, peribronchovascular; may be migratory over time
- Most patients recover with corticosteroids, though relapse common
- Some may show a progressive course with worsening fibrosis
- Occasionally acute, severe form seen which progresses to respiratory failure with high mortality.

Acute Fibrinous Organizing Pneumonia

- Also see chapter 5
- Classified as a rare histological idiopathic interstitial pneumonia (IIP)
- Idiopathic or associated with CTD, HP or drug reaction.
- Clinical presentation has two subtypes
- Subacute form
- Rapidly progressive, acute, fulminant form
- *CT:* Consolidation that is predominantly basal in distribution.

Nonspecific Interstitial Pneumonia

- Occasionally cases with the histological pattern of NSIP can have an acute/subacute presentation; the classical presentation being of an idiopathic chronic fibrotic disease (also see chapter 5)

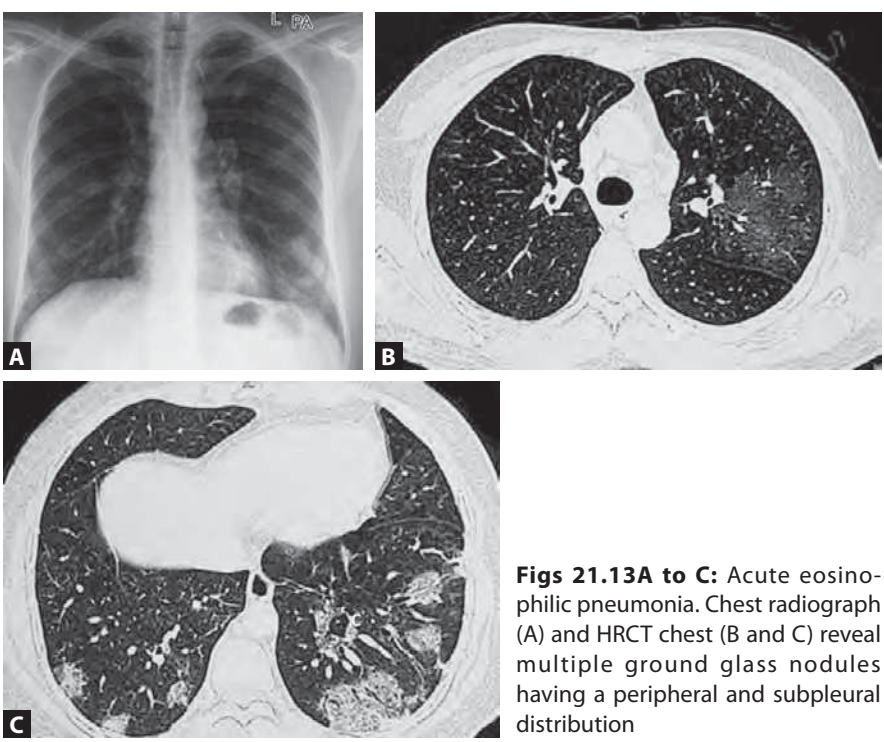
- Amongst the idiopathic IIPs presently acutely COP/NSIP have a better prognosis than DAD superimpose on UIP; and hence the distinction is important.[3]

Connective Tissue Disease-related Interstitial Lung Diseases

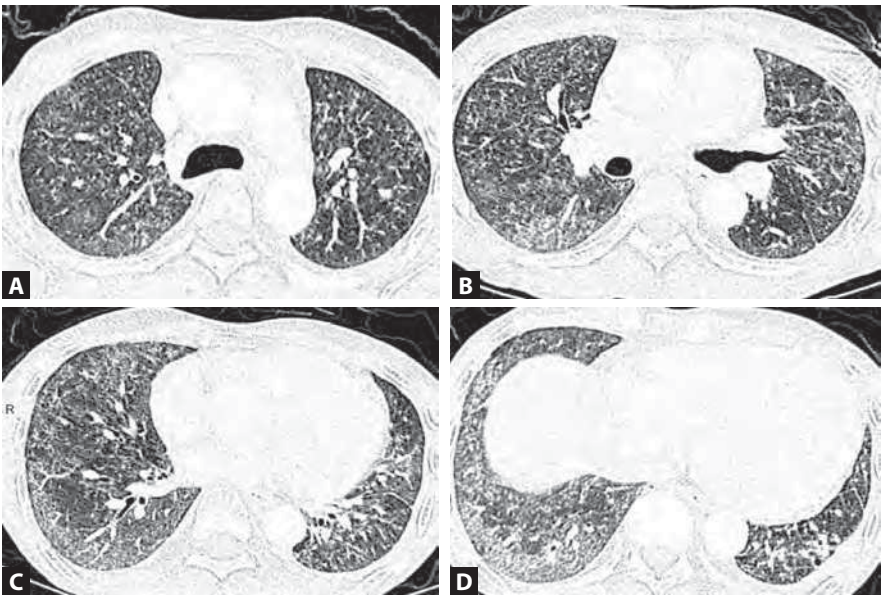
- Amongst the CTDs, pulmonary involvement in dermatomyositis/polymyositis can have an acute/subacute presentation
- In addition diseasese with chronic ILDs such as rheumatoid arthritis can have an acute exacerbation of the pulmonary disease
- Prognosis is poor in both the clinical situations.

Acute Eosinophilic Pneumonia

- Form of eosinophilic lung disease with infiltration of lung parenchyma by eosinophils
- *Pathology:* Infiltration of interstitium by eosinophils, with interspersed areas of DAD
- Short duration illness usually following dust exposure, drug related or commencement of smoking. However, often the precipitating event is not evident
- Few require mechanical ventilation
- Peripheral eosinophilia usually absent, but BAL eosinophils are elevated
- *CXR:* Bilateral, diffuse GGO **(Figs 21.13A to C)** with superimposed Kerley lines
- Good response to corticosteroids.



Figs 21.13A to C: Acute eosinophilic pneumonia. Chest radiograph (A) and HRCT chest (B and C) reveal multiple ground glass nodules having a peripheral and subpleural distribution



Figs 21.14A to D: Tacrolimus related acute interstitial pneumonia in a renal transplant recipient. HRCT images (A to D) reveal diffuse centrilobular nodules, ground glass opacities with minimal interlobular septal thickening: NSIP pattern

Drug-induced Diffuse Lung Diseases (*See* chapter 23)

- Drug-induced DLDs typically occur weeks/months after exposure; and often has an acute/subacute presentation **(Figs 21.14A to D)**
- Besides withdrawal of the responsible drug, corticosteroids may be required in the severe cases.

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22

Clinical and Management Issues

Pawan Tiwari, GC Khilnani

TERMINOLOGY

- Acute onset of respiratory symptoms, in conjunction with diffuse bilateral pulmonary infiltrates can result from a variety of causes.
- While the milder forms may only have shortness of breath or chest tightness; severe forms can lead to respiratory failure.
- Hypoxemic respiratory failure is most common; hypercarbia may be an indicator of advanced disease or respiratory muscle fatigue.

EVALUATION

- History of pre-existing connective tissue disorder or interstitial lung disease
- Immune status and use of immunosuppressant drugs or drugs known to cause lung toxicity
- History of occupational exposure and risk factors for hypersensitivity pneumonitis
- Clinical features suggesting acute infective process (e.g. antecedent upper respiratory tract infection, concurrent viral illness, history of contact with patient with active tuberculosis).

PHYSICAL EXAMINATION

- Nonspecific
- *General physical examination:*
  - Assessment of hemodynamic stability and sensorium
  - Presence of tachypnea (>30/min), tachycardia (>110/min); use of accessory muscles of respiration.
  - Pandigital clubbing; may point towards pre-existing disease.
  - Pedal edema, elevated jugular venous pressure, tender hepatomegaly, or other signs of fluid overload or cor pulmonale
  - Assessment for mucocutaneous manifestations of connective tissue disorders

- *Chest examination:* Inspiratory crackles on auscultation, inspiratory squeaks in patients with bronchiolitis, or traction bronchiectasis associated with pulmonary fibrosis.
- *Cardiac examination:* S_3 gallop to rule out cardiogenic pulmonary edema; murmurs to rule out valvular diseases. Loud and palpable P_2, findings of tricuspid regurgitation may indicate associated pulmonary hypertension.

ANCILLARY INVESTIGATIONS

Important for diagnosis and monitoring of patients.

PULSE OXIMETRY

- Provides non-invasive measurement of peripheral arterial oxygen saturation (SpO_2) using spectrophotometry
- May be used for continuous non-invasive titration of oxygenation and prevention of hypoxemia
- May not be reliable in patients with shock and reduced peripheral tissue perfusion
- Cannot measure oxygen tension or ventilation.

ARTERIAL BLOOD GAS ANALYSIS

- Common sites for arterial blood gas (ABG) sampling include radial, brachial or occasionally axillary artery in the upper limb, and femoral or dorsalis pedis in the lower limb. Radial artery remains the most commonly used site.
- Taking sterile precautions, two fingers of one hand should be used to palpate the artery while holding the needle in the other hand. Both fingers should be proximal to the desired puncture site. The artery should be punctured with the needle at a 30–45° angle (radial, brachial, axillary, dorsalis pedis) or at a 90° angle (femoral artery) relative to the skin. Around 2–3 mL of blood should be removed.
- Normal values are: pH 7.35–7.45, $PaCO_2$ 35–45 mm Hg (4.7–6 kPa), HCO_3 24 (22–26) mEq/L. For PaO_2, generally resting PaO_2 >80 mm Hg (10.7 kPa) and SaO_2 >95% are considered normal, unless the new values are substantially different than prior values.
- In acute hypoxemic respiratory failure, patients commonly have respiratory alkalosis with hypoxemia. Hypercarbia and respiratory acidosis are features of advanced disease or respiratory muscle fatigue. Patients with shock, lactic acidosis, renal involvement or acute kidney injury may have concomitant metabolic acidosis.

INTRAVASCULAR PRESSURE MEASUREMENTS

- May be utilized in patients with shock, or those with acute respiratory failure, on ventilatory support. May also be used to take frequent samples for ABG analysis. Should not be placed routinely in all patients, and should be removed once the patients improve.

PULMONARY ARTERY OCCLUSION PRESSURE

Not routinely used. These may be used in cases of pulmonary edema when diagnosis is not clear using transthoracic 2D echocardiography. It can also be used for the diagnosis of pulmonary hypertension and response to vasodilator therapy.

MONITORING OXYGENATION

- *SpO_2*: Noninvasive measure of oxyhemoglobin in peripheral tissue
- *PaO_2*: Partial pressure of oxygen in arterial blood; measures the portion of O_2 dissolved in plasma
- *FiO_2*: Fractionated inspired O_2; on the basis of oxygen delivery devices or ventilatory settings
- In general, SpO_2 of > 90% implies a PaO_2 of 60 mm Hg or more.
- PaO_2/FiO_2 ratio is used to classify the severity of hypoxemic respiratory failure and to monitor improvement.

TRACHEAL ASPIRATE, BRONCHOSCOPY, BRONCHOALVEOLAR LAVAGE

- Used for diagnosis of specific disorders (like acute eosinophilic pneumonia or diffuse alveolar hemorrhage), or as an indicator for possible etiology (lymphocytosis in hypersensitivity pneumonitis or cellular NSIP).
- Exclusion of infective etiology or malignancy.
- Bronchoalveolar lavage (BAL) may be done with supplemental oxygen or noninvasive ventilation in patients who have hypoxemic respiratory failure
- BAL samples are usually sent for cell counts, cultures (e.g. bacterial, mycobacterial, fungal, viral), enzyme immunoassay, direct immunofluorescence, and cytologic analysis (e.g. viral inclusion bodies, malignant cells).

RADIOLOGY

- Chest radiograph is the mainstay
- Risks of transporting patient to CT suite

MANAGEMENT

Oxygen

- Oxygen supplementation using nasal cannula or venturi masks may suffice for patients with mild hypoxemia.
- Patients with high FiO_2 requirement and high minute ventilation may benefit from non-rebreathing masks with reservoir.
- High flow nasal cannula provides heated humidified high flow oxygen up to FiO_2 of 1.0 with some positive end inspiratory pressure, and has shown efficacy in hypoxemic respiratory failure.

Mechanical Ventilation

- The term refers to a form of assisted ventilation where mechanical means are used to assist or replace spontaneous respiration.
- *Indications:* Poor sensorium, hemodynamic instability, unable to handle respiratory secretions, respiratory rate >30/min, unable to maintain PaO_2 >90% and FiO_2 >0.60, $PaCO_2$ >50 mmHg and pH <7.25
- *Non-invasive:* In conscious and hemodynamically stable patients, or those with "do not intubate" status. Provided through facemasks or helmet
- *Invasive:* Through endotracheal intubation
- Lung protective ventilation, with optimal FiO_2 and PEEP should be utilized.
- Adequacy of mechanical ventilation should be assessed on the basis of clinical parameters, SpO_2, ABG parameters and plateau and peak pressures.

Drugs

- High dose systemic corticosteroids, typically 500 mg to 1 g of methylprednisolone pulse for 1–3 days is given in patients with acute hypoxemic respiratory failure. Subsequently, oral prednisone (daily 0.5–1 mg/kg) is started, which is then gradually tapered.
- Other immunosuppressants may include cyclophosphamide (pulse or oral), rituximab, azathioprine or mycophenolate mofetil, according to primary disease.
- Patients should receive intravenous antibiotics including treatment for pneumocystis in suspected infective exacerbations. In cases of suspected or proven CMV or fungal pneumonia, appropriate antivirals or antifungals may be added.
- Other supportive treatment includes prevention of complications (e.g. venous thromboembolism, gastrointestinal bleeding, nosocomial pneumonia).

SPECIFIC ENTITIES

Acute Respiratory Distress Syndrome (ARDS)

- *Definition/diagnostic criteria:* Revised Berlin definition given in 2012. Following components:
  - Acute onset of respiratory distress
  - Develops within one week of the clinical insult or new/worsening respiratory symptoms
  - Chest X ray showing bilateral lung infiltrates **(Fig. 22.1)**
  - Respiratory failure which is not fully explained by cardiac failure or fluid overload
- *Categories of severity (based on Pa_2/FiO_2 ratio at PEEP or CPAP* $\geq$ 5 cm H_2O)
  - *Mild:* PaO_2/FiO_2 ratio 200 to 300 mm Hg
  - *Moderate:* PaO_2-FiO_2 ratio between 100 to < 200 mm Hg
  - *Severe:* PaO_2-FiO_2 ratio < 100 mm Hg

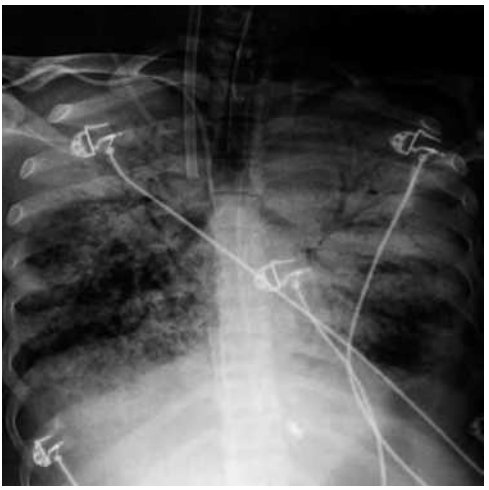


Fig. 22.1: Acute respiratory distress syndrome, with bilateral air bronchograms, on mechanical ventilation. Also seen is jugular venous access for venovenous extracorporeal membrane oxygenation

- *Causes*
  - *Pulmonary/direct injury:* Pneumonia, aspiration
  - *Extrapulmonary/indirect injury:* Sepsis, pancreatitis, major trauma, multiple transfusions.

    Pathogenic mechanisms depend on inciting insult; common features include increased permeability due to epithelial and endothelial damage, and neutrophilic infiltration in early ARDS.

  Management
  - Ventilatory management includes lung protective ventilation (i.e., low tidal volume < 6 mg/kg of calculated ideal body weight), along with adequate positive end expiratory pressure (PEEP).
  - Early recognition and management of refractory hypoxemia (defined as PaO_2 < 100 mm Hg, inability to keep plateau pressure Pplat below 30 cm H_2O despite tidal volume < 4 mL/kg of ideal body weight, occurrence of barotrauma or oxygenation index of < 30)
  - Prone positioning and early institution of extracorporeal membrane oxygenation have been shown to improve outcome.
  - Cautious use of noninvasive ventilation in mild ARDS under close observation
  - Use of newer interfaces such as helmet for NIV in ARDS has shown some benefit in recent studies.
  - Non-ventilatory management includes antimicrobials, fluid conservative strategy, optimal use of sedation and neuromuscular blockade and regular sedation free intervals.

Prognosis

- Hospital mortality of 35–46% has been reported.
- In most patients, spirometry values normalise within 6 months, though diffusion capacity (Dlco) may normalize by 5 years.
- A high proportion of survivors have cognitive dysfunction (>40%) and impaired physical function (66%) persisting after 12 months.

Pulmonary Edema

- May be cardiogenic or non-cardiogenic.
- Common causes of cardiogenic pulmonary edema include decompensation of heart failure, valvular dysfunction, ischemia, or volume overload.
- Non-cardiogenic pulmonary edema may be associated with pneumonia, sepsis, aspiration or drowning.
- Up to 10% patients with pulmonary edema may have multiple causes, e.g. ARDS with concomitant fluid overload, or acute decompensated cardiac failure with associated lung injury.
- Paroxysmal nocturnal dyspnea or orthopnea point towards a cardiac cause of cardiogenic pulmonary edema. Signs and symptoms of infection, altered sensorium, witnessed aspiration, drowning or multiple blood transfusions may indicate non-cardiogenic cause.
- Engorged neck veins, and peripheral edema may indicate fluid overload. S_3 gallop is highly specific for left ventricular dysfunction. Other specific findings include murmurs of valvular stenosis or regurgitation.
- *Laboratory testing:*
  - ECG findings, troponin levels and serum brain natriuretic peptide (BNP) levels may be useful in differentiating cardiogenic pulmonary edema.
  - Two-dimensional transthoracic echocardiogram is useful, if cause of pulmonary edema cannot be found on history, physical examination, ECG and chest X-ray.
  - Pulmonary-artery catheterization determines pulmonary artery occlusion pressure. This is considered as gold standard to establish cause of acute pulmonary edema, but is not required in most cases.
- Management includes treatment of primary cause, management of hypoxemic respiratory failure and supportive treatment. Non-invasive ventilation (using continuous positive pressure ventilation) is efficacious of management of respiratory failure in cardiogenic pulmonary edema along with diuretics and vasodilators. Non-cardiogenic pulmonary edema management should managed like ARDS.

Infections

- Infections such as pneumocystis, cytomegalovirus, influenza, atypical organisms (mycoplasma or chlamydia) or mycobacteria may have presentations similar to acute ILDs, or may cause acute exacerbations of pre-existing ILDs **(Fig. 22.2)**



Fig. 22.2: Bilateral nodular infiltrates. Transbronchial lung biopsy showed invasive aspergillosis



Fig. 22.3: Acute interstitial pneumonia. Patient required mechanical ventilation, and died after 2 weeks. Diffuse alveolar damage was demonstrated on postmortem lung biopsy

- Ruling out infectious etiology, with use of BAL should be a part of work-up of acute ILD.

Acute Interstitial Pneumonia

- Usually affects previously healthy middle-aged persons
- Not associated with smoking; no gender predilection
- Respiratory picture similar to ARDS but no known predisposing cause **(Fig. 22.3)**

- Rapid onset of symptoms with 7–14 days of prodromal illness
- Fever, cough, progressive dyspnea and respiratory failure are usual presenting complaints. Clubbing is not commonly seen.
- Needs to be differentiated from AE-IPF (older age, diagnosed IPF, pan-digital clubbing, UIP pattern on HRCT)
- Histology shows diffuse alveolar damage (DAD), either organizing or proliferative phase.
- Diagnosis is based upon clinicoradiologic features and lung biopsy showing DAD.
- If lung biopsy is not feasible, presumptive diagnosis of AIP should be made when there is presence of a compatible clinical syndrome and radiology, after ruling out alternate causes.
- Poor prognosis with >50% in hospital mortality; may progress to chronic ILD in significant proportion of patients.
- Treatment is primarily supportive
- Mechanical ventilation may be required
- High-dose systemic corticosteroids have shown benefit in small case series.
- Lung transplantation has been reported in a few cases of AIP.

Acute Exacerbation of IPF (AE-IPF)

- Causes of acute deterioration in patients with IPF include pneumonia, pulmonary embolism and acute worsening of ILD.
- Incidence 4.8–14% in placebo arms of various randomized controlled trials
- Risk factors for AE-IPF include younger age, higher body mass index, coronary artery disease, lower FVC, pulmonary hypertension, prior surgery and prior use of immunosuppressants.
- It is defined as an acute and clinically significant respiratory deterioration which is characterized by new widespread alveolar abnormality in patients with concurrent or previous diagnosis of IPF
- *Diagnostic criteria:*
  - Acute worsening or development of dyspnea typically < 1 month duration
  - CT with new bilateral ground-glass opacity with or without consolidation, which is superimposed on a background pattern consistent with usual interstitial pneumonia
  - Deterioration not fully explained by fluid overload or cardiac failure
- Lower PaO_2/FiO_2 ratio (<100), elevated lactate dehydrogenase (LDH) or Krebs von den Lungen-6 (KL-6) levels, and high ground- glass opacity or consolidation scores have been associated with high mortality in AE-IPF.
- Histopathology shows underlying fibrotic interstitial pneumonia with superimposed DAD.
- *Treatment:*
  - High-dose corticosteroids, typically methylprednisolone pulse for 3 days, followed by high-dose oral steroids. Other drugs including silvestat, cyclosporine or cyclophosphamide have shown some benefit in small case series.

  - Patients receive antimicrobials including cotrimoxazole irrespective of definitive evidence of infections, due to high mortality of AE-IPF.
  - Hypoxemic respiratory failure is managed with non-invasive or mechanical ventilation.
  - Preventive therapies include nintedanib and antacid therapies.
- *Prognosis:* Overall mortality rate of 70% have been reported. Systematic reviews have reported high one month (67%) and three month (67%) mortality.

Acute Hypersensitivity Pneumonitis

- Hypersensitivity pneumonitis (HP) is a syndrome caused by repeated exposure to various organic particles.
- Classical acute form is associated with exposure to thermophilic actinomycetes, fungi or other bacteria (e.g. farmer's lung, hot tub lung).
- All chronic HP cases need not have a preceding acute disease. Also, the terms acute and chronic only imply presentation, but not the underlying pathogenesis.
- The acute form of HP usually presents as a "flu-like" illness, and has a self-limiting course.
- Only occasionally can it have a severe presentation leading to respiratory failure.
- Often follows a sudden, intense exposure to the offending agent. History of previous such episodes may be present.
- Symptoms include fever, myalgias and sweating, apart from dyspnea and chest tightness.
- Onset of symptoms occurs usually 6–24 hours following the acute exposure
- May have associated wheezing and bronchial hyperresponsiveness.
- Acute episodes may be indistinguishable from acute viral or mycoplasma infections.
- *Physical examination:* Inspiratory crackles; short-high pitched end-inspiratory sound (also called "sqwaks").
- *PFT:* Seldom performed. Restrictive pattern commoner, though obstructive or mixed may also be seen.
- Diagnosis:
  - Based on high index of suspicion, recognition of antecedent antigen exposure, along with clinical, radiologic or histopathologic findings.
  - Elevated serum levels of specific circulating antibodies may be taken as a marker of exposure. However, these may be false negative in acute HP.
  - BAL cytology shows neutrophilia and lymphocytosis (>50%), along with CD4/CD8 ratio of less than 1.
  - Lung biopsy is less commonly performed. It demonstrates shows peribroncholar small ill-formed noncaseating granulomas. Peripheral airways may show features of proliferative bronchiolitis obliterans.
- *Differential diagnosis:* Asthma, obliterative bronchiolitis

- *Management*:
  - Avoidance of antigen exposure is the most important step. Patients who are unable or unwilling to avoid exposure to antigens may benefit from air-purifying respirators.
  - Corticosteroids may be required in severe or progressive disease.
  - Mostly self-limiting, with the symptoms resolving within hours or days.
  - Recurrent exposures may lead to obstructive lung disease (with or without emphysema, commonly centrilobular) instead of fibrosis.

Diffuse Alveolar Hemorrhage

- Diffuse alveolar hemorrhage (DAH) refers to diffuse bleeding from pulmonary arterioles, alveolar capillaries and pulmonary venules (i.e. pulmonary microcirculation) due to microvascular damage.
- Often a catastrophic clinical syndrome presenting with hemoptysis and dyspnea, which may progress to hypoxemic respiratory failure **(Fig. 22.4)**
- Extrapulmonary symptoms are those related to underlying systemic disease. *Causes:* Broadly classified as immune and non-immune causes.
  - Nonimmune causes include endobronchial tumors, arteriovenous malformations or aneurysms, hemorrhagic pneumonia, infections, mitral valve disease, congestive cardiac failure, uremia, coagulopathies or thrombocytopenia, pulmonary veno-occlusive disease or massive pulmonary embolism.
  - Autoimmune DAH is due to capillaritis or endotheliitis; common etiologies being ANCA-associated vasculitides, Goodpasture' syndrome,



Fig. 22.4: Diffuse alveolar hemorrhage. Bilateral lower zone interstitial infiltrates; patient had recurrent hemoptysis with anemia; bronchoalveolar lavage showed progressive hemorrhagic return; BAL and TBLB showed DAH

systemic lupus erythematosus (SLE) or drugs (D-penicillamine, trimellitic anhydride).
  - Other causes include immunocompromised status (post-bone marrow transplant or AIDS) or idiopathic pulmonary hemosiderosis (unknown cause, sometimes associated with celiac disease).
- *Diagnosis:*
  - BAL fluid shows blood, with progressively more blood with serial aliquots. DAH should be considered, if BAL cytology shows > 20% hemosiderin-laden macrophages (on Prussian blue staining).
  - Presumptive diagnosis of DAH can be made on the basis of compatible clinical and radiologic picture, along with serologic and BAL findings.
  - BAL can also be utilized to rule out alternate infective etiology.
  - Transbronchial lung biopsy may be used to diagnose pulmonary capillaritis, presence of granulomatous vasculitis and immunofluorescent staining.
  - Kidney biopsy can be done, if required in patients with pulmonary renal syndromes.
  - Thoracoscopic or surgical lung biopsy may be required for diagnosis, if rest of the work-up is non-diagnostic.
  - Elevated serum creatinine or presence of active sediments in urine point towards pulmonary-renal syndromes.
  - Markers of connective tissue disease such as ANA, ANCA (either antiproteinase or antimyeloperoxidase), anti-glomerular basement membrane antibodies (anti-GBM antibodies) may point towards specific diagnosis. Investigations should include antibodies against anti-tissue transglutaminase, IgA (anti-tTG), and work-up for celiac disease.
- *Management:*
  - Management of respiratory failure, hemodynamic status and correction of coagulation parameters
  - Prompt work-up for the cause and its treatment
  - High-dose corticosteroids are useful in immune-mediated causes; cyclophosphamide or rituximab have demonstrated benefit, especially in ANCA-associated vasculitis
  - Other drugs include hydroxychloroquine and azathioprine.
  - Plasmapheresis in anti-GBM disease or ANCA-associated vasculitis with severe renal insufficiency
  - Gluten-free diet, if associated celiac disease
- *Prognosis:* Untreated DAH has very high mortality, and thus requires aggressive treatment. Long-term prognosis depends on the underlying etiology.

Less Common Causes

Cryptogenic organizing pneumonia (COP) usually presents with subacute respiratory symptoms (less than 3 months) along with constitutional symptoms like fever, and may have neutrophilic leukocytosis. Organizing pneumonia (OP) may be idiopathic or secondary to autoimmune causes or infections. Diagnosis



Fig. 22.5: Cryptogenic organizing pneumonia. Bilateral multilobar infiltrates; CT showed reverse halo sign; TBLB showed organizing pneumonia

is based upon predisposing factors, Characteristic HRCT findings, exclusion of alternate etiology and histopathology **(Fig. 22.5)**. Typical histopathologic features include patchy involvement, organizing pneumonia involving alveolar ducts and alveoli; bronchiolar intraluminal polyps may also be seen. Majority of patients respond to oral corticosteroids.

Nonspecific interstitial pneumonia (NSIP), related to collagen vascular diseases, antisynthetase syndrome or idiopathic may present with acute onset, or more commonly, acute worsening with hypoxemic respiratory failure. Management remains similar to acute exacerbation of IPF.

Acute Eosinophilic pneumonia (AEP) is a rare disorder. It usually presents with fever and acute respiratory failure along with diffuse infiltrates on chest radiology **(Fig. 22.6)**; pulmonary eosinophilia is present. Causes include infections (more commonly parasitic), hypersensitivity to drugs or inhaled agents, systemic idiopathic-eosinophilic disorders or idiopathic. As presentation mimics ARDS or AIP, it is confused with these more common entities. BAL is the key to diagnosis of AEP, showing eosinophilic alveolitis along with increased neutrophils and lymphocytes. BAL can also exclude infectious etiology. TBLB or open lung biopsy, if performed, shows eosinophilic infiltration of alveoli, interstitial edema with or without organizing diffuse alveolar damage. Work-up should include ruling out infective etiology, vasculitis or drug exposure. High- dose systemic corticosteroids are frequently utilized, along with supportive treatment. With prompt management, outcome is usually favorable; relapse is uncommon.

Drug-induced ILDs, may have clinical presentation ranging from no symptoms to acute respiratory failure. Pathogenesis includes direct, dose-dependent toxicity or immune-mediated. Most reactions are immune-mediated. Diagnosis is based on high index of suspicion, exposure to known



Fig. 22.6: Acute eosinophilic pneumonia. Bilateral peripheral patchy consolidation. Biopsy showed eosinophilic interstitial infiltration with diffuse alveolar damage

offending drug and exclusion of alternate causes. Treatment includes withdrawal of offending agent and supportive treatment with or without corticosteroids.

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Section 8

Miscellaneous

- Diffuse Lung Diseases in Oncology Setting: Imaging
- Small Airway Disease: Imaging
- Imaging in Uncommon Disorders: Part I
- Imaging in Uncommon Disorders: Part II

23

Diffuse Lung Diseases in Oncology Setting: Imaging

Mukesh Kumar Yadav, Chandrashekhara SH

INTRODUCTION

- Many types of pulmonary complications can occur in cancer patients
- The main causative and contributory factors for complications are altered immunity status due to systemic disease or secondary to chemotherapy/radiotherapy or hematopoietic stem cell transplantation (HSCT)
- Of these lung diseases many involve the lungs diffusely while a few show focal patchy type of involvement
- To provide best clinical care to the patients, it is necessary to understand the complications associated with treatment as well as natural progression of the disease itself
- This chapter focuses on those pulmonary conditions which are involving lungs in diffuse manner, i.e. diffuse lung diseases
- Various diffuse lung diseases in cancer patients are:
  - Pulmonary edema
  - Pulmonary hemorrhage
  - Drug toxicity
  - Idiopathic pneumonia syndrome
  - Bronchiolitis obliterans
  - Organizing pneumonia
  - Leukostasis
  - Few infections such as *Pneumocystis jirovecii* and *Cytomegalovirus*
  - Hemophagocytic lymphohistiocytosis syndrome, etc.

Though bacterial and fungal infections, radiation pneumonitis are usually more focal in involvement; these are also briefly discussed in this chapter. In addition, bone marrow transplant complications and lymphangitic carcinomatosis are also briefly discussed.

INFECTIOUS CONDITIONS

- Majority of deaths associated with malignancies (particularly hematological malignancies) are due to infections and associated comorbidities

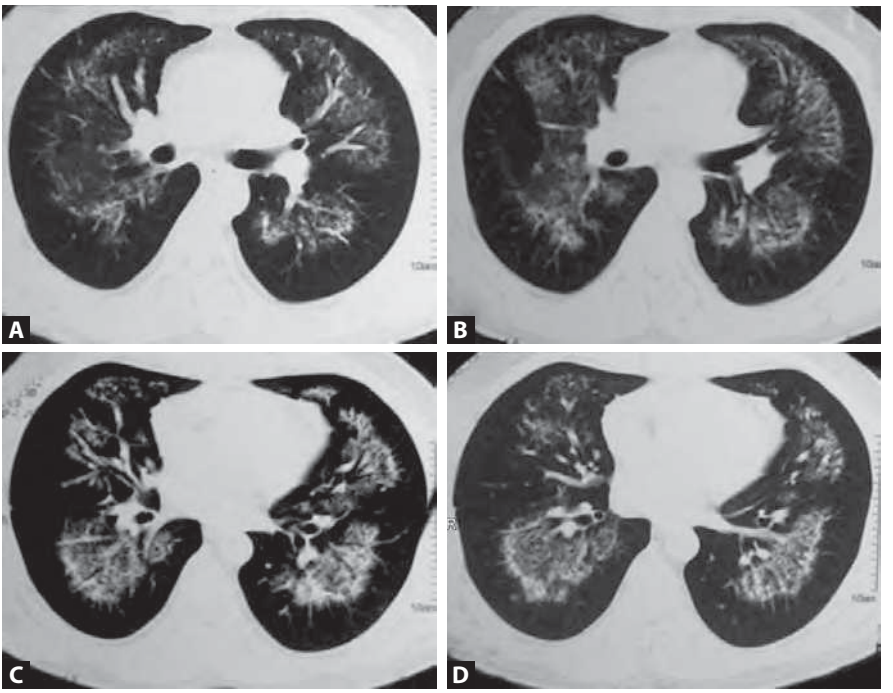
- These infections and the causative agents responsible for these tend to dominate in early phase after chemotherapy
- *Pneumocystis jirovecii* and *Cytomegalovirus* infections usually involve the lungs in diffuse manner while bacterial infections are usually focal in nature. Fungal involvement if extensive can involve the lungs diffusely.

Pneumocystis Jirovecii Pneumonia

- Pneumocystis infection occurs more frequently in those patients who receive steroids and fludarabine or stem cell transplantation in addition to intensive chemotherapy
- With the frequent use of prophylactic antibiotics the incidence has come down
- On chest radiograph **(Fig. 23.1)** typical findings are presence of bilateral perihilar opacities which are interstitial in nature
- These opacities may be confluent in nature diffusely involving entire lung fields
- It is important to note that the chest radiograph findings may be normal in as much as 40% of patients
- High-resolution computed tomography (HRCT) should be used for early diagnosis of *Pneumocystis jirovecii* pneumonia (PJP) which reveals wide spread ground-glass opacity (GGO) which are more commonly found in perihilar regions **(Figs 23.2A to D)**
- Sharp transition of involved and uninvolved pulmonary lobules may give rise to geographic pattern or mosaic pattern of GGO
- Sometimes the GGO is associated with fine interlobular and intralobular septal thickening giving rise to crazy paving appearance
- Features which are much less common are cysts (usually present in HIV positive patients), multiple nodules, consolidation and asymmetric GGO, lymphadenopathy



Fig. 23.1: Chest radiograph showing bilateral perihilar opacities



Figs 23.2A to D: *Pneumocystis jirovecii* pneumonia. Axial NCCT chest image showing wide spread ground-glass opacity in bilateral perihilar regions with sharp transition of involved and uninvolved pulmonary lobules. There is associated fine interlobular and intralobular septal thickening

- Pleural effusion is rarely present
- Main differential diagnoses include hypersensitivity pneumonitis, cytomegalovirus (CMV) pneumonia, lymphocytic interstitial pneumonia or diffuse alveolar hemorrhage **(Table 23.1)**.

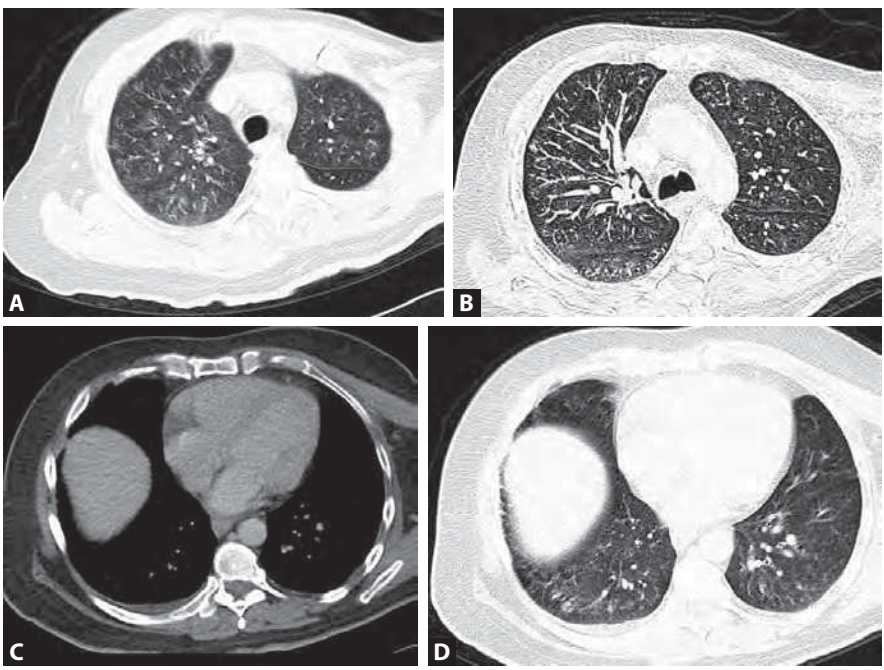
Cytomegalovirus Infection

- This condition is common in recipients of stem cell transplantation
- Although infection rates are quite high, only about one-third of these patients develop clinical pneumonia
- The cause of infection may be secondary to endogenous reactivation or exogenous infection either due to CMV seropositive marrow grafting or from transfusion of infected blood products
- Diagnosis is by detection of CMV antigen from peripheral white blood cells
- Imaging findings are nonspecific with presence of ill-defined GGO **(Figs 23.3A to D)**, nodules of various sizes and air space consolidation; and sometimes pleural effusion
- CT halo sign may also be seen in CMV infection (also seen in *Aspergillus* fungal infection) as a nodule with peripheral ground glass density

Table 23.1: Imaging pearls for differentiation (PJP vs CMV)

	PJP	*CMV*
Prevalence in hematological malignancies treated with HSCT	+	++
Clinical appearance of patient	Patient look well preserved despite severe metabolic hypoxia	Patient look severely ill
Extrapulmonary features	Retinitis, hepatitis and diarrhea are not seen	Retinitis, hepatitis and diarrhea are frequently seen
Mosaic pattern	More commonly seen	Less commonly seen
Presence of small nodules	Less common	More common
Demarcation of GGO and consolidation	More sharply defined	Less sharply defined
Dense GGO	Frequently seen dense GGO, which might confuse with consolidation	GGO less dense, never confuse with consolidation
Distribution	More commonly atypical distribution	Less commonly atypical distribution

Abbreviations: PJP, *Pneumocystis jirovecii* pneumonia; CMV, *Cytomegalovirus;* HSCT, hematopoietic stem cell transplantation; GGO, ground-glass opacity



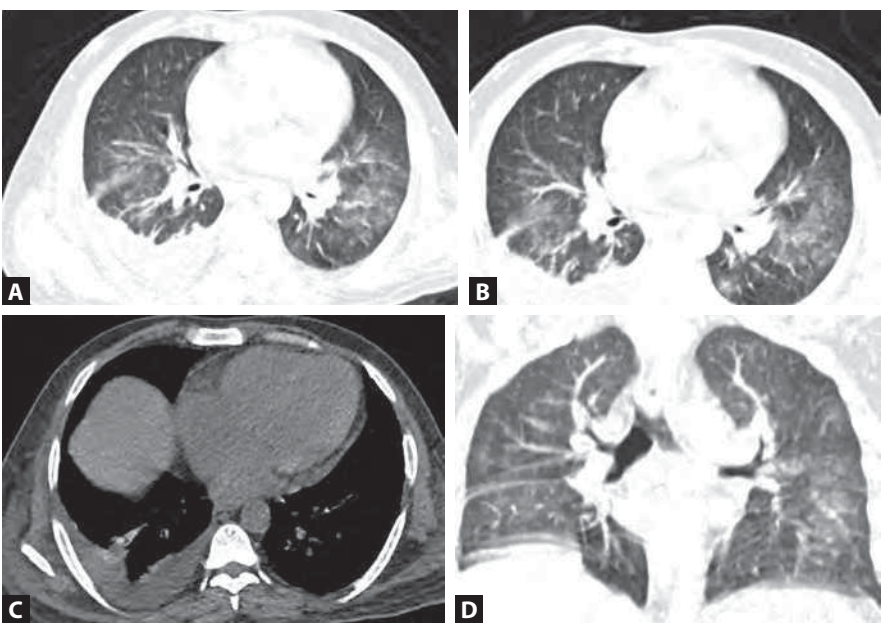
Figs 23.3A to D: *Cytomegalovirus pneumonia*. Axial NCCT chest image showing ill-defined GGO predominantly in right upper lobe, few ill-defined nodules in right upper lobe and mild pleural effusion on right side

- Other differential diagnoses are PJP and pulmonary hemorrhage which are difficult to differentiate on imaging alone
- PJP infection is most commonly associated infection. Differentiation by imaging alone is difficult, however presence of multiple centrilobular nodules indicate more towards possibility of active CMV infection **(Table 23.1)**.
- Even the timing window of susceptibility of infection after HSCT for PJP and CMV are similar (30–100 days).

NONINFECTIOUS CONDITIONS

Pulmonary Edema

- Pulmonary edema is a relatively common problem which occurs in early phase of treatment by chemotherapy or stem cell transplantation
- Common causes are graft versus host disease (GVHD), cardiac and renal derangement by intensive chemotherapeutic agents, fluid overload, damage to pulmonary vasculature due to radiation therapy or infectious agents
- Imaging findings of pulmonary edema may vary according to severity of the disease
- On chest radiograph, Kerley lines in periphery, bilateral hilar haze or prominence of pulmonary vessels may be seen
- On CT presence of bilateral perihilar ground glass density **(Figs 23.4A to D)**, air space consolidation and bilateral pleural effusion suggest pulmonary



Figs 23.4A to D: (A to D) Pulmonary edema. Axial NCCT chest lung window (A and B) and soft tissue window (C) CT images reveal smooth interlobular septal thickening in combination pleural effusion and subtle diffuse ground-glass opacity in bilateral lungs. (D) Coronal reformation showing prominent vascular bundle in upper lobes

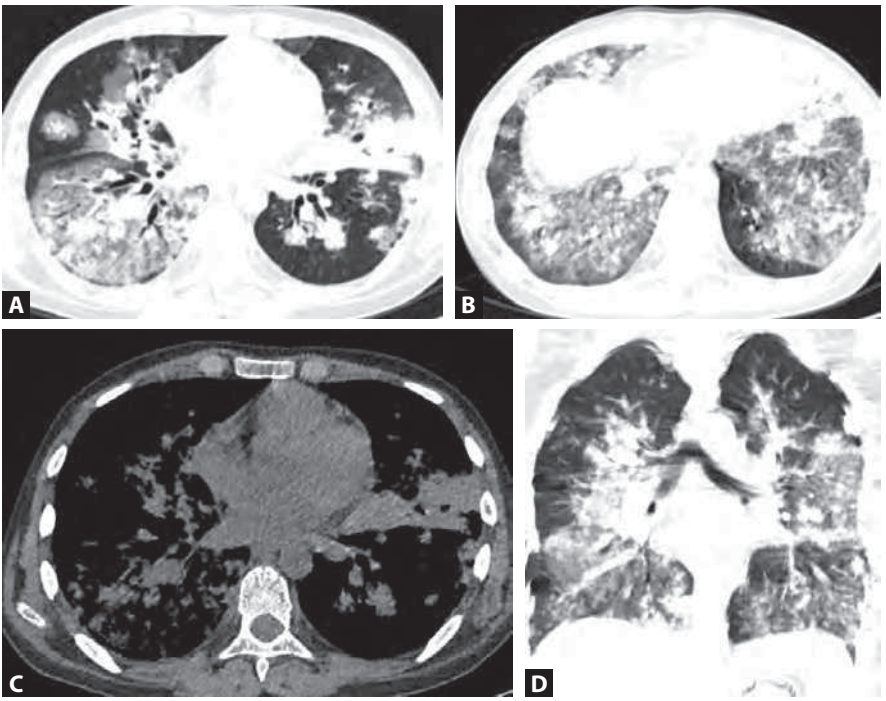
edema. In addition, smooth thickening of inter- and intralobular septae is also seen
- Rapid clearance of these abnormalities with institution of diuretic therapy also indicates this diagnosis
- Differential diagnosis includes lymphangitis carcinomatosis which is differentiated by presence of irregularity of the thickened interlobular septae in the latter. Other differential diagnosis is PJP infection which usually does not have associated pleural effusion and more commonly presents with GGO. Pulmonary hemorrhage is differentiated from pulmonary edema by presence of nodules in the former and peculiar clinical settings and deranged coagulation.

Pulmonary Hemorrhage

- It is a relatively common complication in patients with hematopoietic stem cell transplantation
- It is associated with thrombocytopenia/deranged coagulation, few infections and stem cell transplantation
- It is potential fatal complication with acute and nonspecific clinical presentation
- Hemoptysis is rare
- CT findings are extensive ground-glass opacities or consolidation in acute phase **(Figs 23.5A to D)**. Typically there is sparing of subpleural lung parenchyma. In subacute phase there is resolution usually within two weeks duration with patchy areas of ground glass opacities
- In chronic phase, there is increased reticulation, fibrosis with sometimes architectural distortion
- Bronchoalveolar lavage should be done early when there is suspicion of pulmonary hemorrhage
- It has high mortality rates
- Prompt diagnosis with early institution of steroid may decrease the mortality rates
- Differential diagnoses in early phase are pulmonary edema and PJP infection. Presence of cardiomegaly, septal thickening and thickened vascular pedicle in former and perihilar distribution in latter helps in differentiating from hemorrhage.

Drug Toxicity

- Drug-induced lung diseases (described with over 400 drugs) among cancer patients are frequently under diagnosed
- The most common amongst cytotoxic drugs which cause pulmonary toxicity are bleomycin, carmustine, busulfan and cyclophosphamide, methotrexate, gefitinib and gemcitabine
- Onset of drug induce lung disease varies from immediate period to years after initiation of drug used, but majority drugs cause it within few weeks to few months[1]



Figs 23.5A to D: Pulmonary hemorrhage. Axial NCCT image lung window (A and B) and soft tissue window (C) showing extensive ground glass-opacities and consolidation with sparing of immediate subpleural lung parenchyma. Coronal reformatted image (D) revealing significant motion artifacts due to inability of patient to hold breath

Table 23.2: Drug toxicity by common antineoplastic medications

Interstitial pneumonia	Azathioprine, bleomycin, busulfan, chlorambucil, cyclophosphamide, infliximab, melphalan, methotrexate, nitrofurantoin, paclitaxel, rituximab, sirolimus,
Hypersensitivity pneumonitis (HP)	Azathioprine, busulfan, nitrofurantoin
Organizing pneumonia	Bleomycin, cyclophosphamide, methotrexate
Granulomatous pneumonitis	Methotrexate, nitrofurantoin

- Symptoms are also variable and nonspecific, e.g. dry cough, fever and dyspnea
- Common drug-induced radiopathological entities are diffuse alveolar damage, cryptogenic organizing pneumonia and nonspecific interstitial pneumonia[1] **(Table 23.2)**
- HRCT findings of diffuse alveolar damage include multifocal or diffuse areas of ground glass densities **(Fig. 23.6)** with mid or lower lungs predominance. Later on fibrosis or honeycombing may also be seen



Fig. 23.6: Drug toxicity (Bleomycin induced). Drug-induced lung injury in a 45-year-old man who was treated with bleomycin and GCMF. Axial high-resolution CT image reveals patchy areas of consolidation with evidence of interlobular septal thickening. There is presence of fibrosis and traction bronchiectasis

- In nonspecific interstitial pneumonia secondary to drugs multifocal ground-glass density and fibrosis are the common HRCT finding, however these are quite nonspecific
- In organizing pneumonia, the main HRCT findings are of focal airspace consolidation which is frequently in subpleural and peribronchial location, ill-defined nodules or centrilobular nodules
- Other manifestations of drug toxicity may be pulmonary hemorrhage, pulmonary edema, pulmonary hypertension and veno-occlusive diseases or eosinophilic pneumonia.

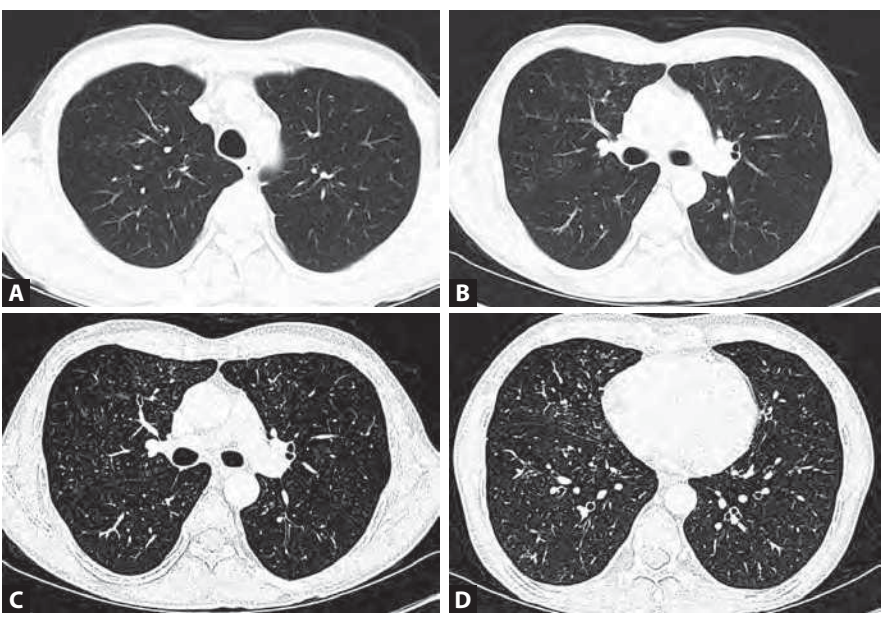
Idiopathic Pneumonia Syndrome

- It is an early phase complication with very high mortality rate
- It is defined as widespread injury to lung parenchyma after hematopoietic stem cell transfusion (HSCT) for which no other etiology is identified
- Its cause may be chemotherapy or total body radiation
- HRCT findings are similar to diffuse alveolar damage, i.e. extensive ground glass density which may be focal or diffuse.[2] These GGO are usually denser in dependent lung fields. This finding may help in differentiating this entity from PJP in which there is no gradient
- In the later phases, features of fibrosis are seen which manifest on HRCT as architectural distortion, traction bronchiectasis or honeycombing
- It is a diagnosis of exclusion.

Bronchiolitis Obliterans

- It is a relatively late complication with mean time of onset from stem cell transplantation is six months

- It is more commonly seen with allogenic HSCT as compared to autogenic HSCT
- Diagnosis is by clinical findings and pulmonary function tests, however HRCT findings contribute significantly in suggesting the diagnosis
- Main presumed causes are chronic GVHD causing injury to bronchioles resulting in fibrosis in submucosal and peribronchiolar level
- Subsequently there is progressive narrowing and eventual obliteration of bronchioles
- Prognosis is poor with no specific successful treatment of this condition which becomes fatal eventually
- Chest radiograph reveals mild to moderate pulmonary overinflation
- HRCT reveals patchy areas of mosaic attenuation **(Figs 23.7A to D)**. Low attenuation areas are due to trapped air and subsequent pulmonary oligemia in that region
- Pulmonary vessels in the involved areas are frequently smaller in caliber as compared to vessels in adjacent uninvolved areas. The adjacent lung lobules appear high attenuated due to increased blood flow in those areas.
- Frequently there is sharp demarcation of low attenuation and high attenuation areas. Bronchi appear dilated
- Expiratory scan are typically needed to reveal the air trapping

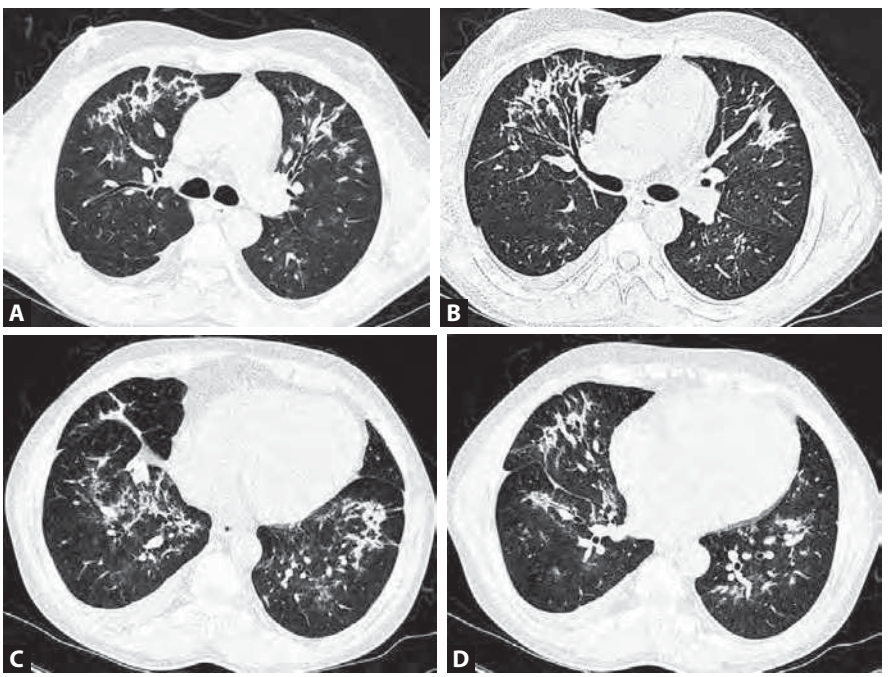


Figs 23.7A to D: Bronchiolitis obliterans. Axial NCCT images (A to D) show areas of mosaic attenuation in both lungs. There are alternating areas of decreased and increased attenuation, the caliber of vessels in hypoattenuating areas are less as compared to adjacent hyperattenuating areas. Few small nodules are also seen in addition to mildly dilated bronchioles

- Diagnosis can be established histologically by obtaining lung biopsy
- Differential diagnoses are pan lobular emphysema and pulmonary artery hypertension.

Organizing Pneumonia

- It is another late phase complication in patients with malignancy who have received HSCT.
- In this condition there is presence of edematous granulation tissue polyps in alveolar duct and bronchiolar lumen with adjacent inflammatory cellular infiltration
- The diagnosis is suggested by peribronchial distribution of the pathological changes
- Because of noncontinuous and patchy area of distribution transbronchial biopsy sometimes may not yield the diagnosis and surgical biopsy may be needed
- On HRCT typical findings are patchy areas of ground-glass densities and nodular areas of consolidation with usual distribution at peribronchial areas and in subpleural locations **(Figs 23.8A to D)**.
- The changes are usually bilateral, not conforming to segmental pattern and are seen in peripheral parts of lungs. These are more dominantly seen at lung bases. In immunocompromised patients the ground-glass density is



Figs 23.8A to D: Organizing pneumonia. Axial NCCT chest images showing patchy areas of ground-glass densities and nodular areas of consolidation bilaterally with predominant distribution at peribronchial areas and in subpleural locations

more frequently seen as compared to those patients with good immunity in which consolidation is more common finding
- Differential diagnoses are pulmonary lymphoma, low grade adenocarcinoma and sarcoidosis.

Transfusion-related Acute Lung Injury

- This is a form of severe lung injury in patients having history of/related to transfusion of blood or blood products
- It usually occurs within few hours of transfusion and patients experience severe dyspnea which is noncardiac in nature
- Chest radiograph reveals features of pulmonary edema and extensive interstitial and alveolar opacities mimicking acute respiratory distress syndrome
- Lack of volume overload and absence of cardiac disease are clinching factors in making diagnosis.

Hemophagocytic Lymphohistiocytosis

- It is a condition of disordered immune regulation in which various immune cells (macrophages, T-lymphocytes) have abnormal function
- Associated with few infectious agents (e.g. Epstein-Barr virus) as well as few malignancies
- Clinical manifestations are secondary to pancytopenia, coagulation abnormality and organomegaly
- Chest imaging shows alveolar or interstitial opacities and pulmonary edema with rapid onset and rapid resolution **(Figs 23.9 and 23.10)**
- Associated organomegaly helps in narrowing down the differential diagnosis.

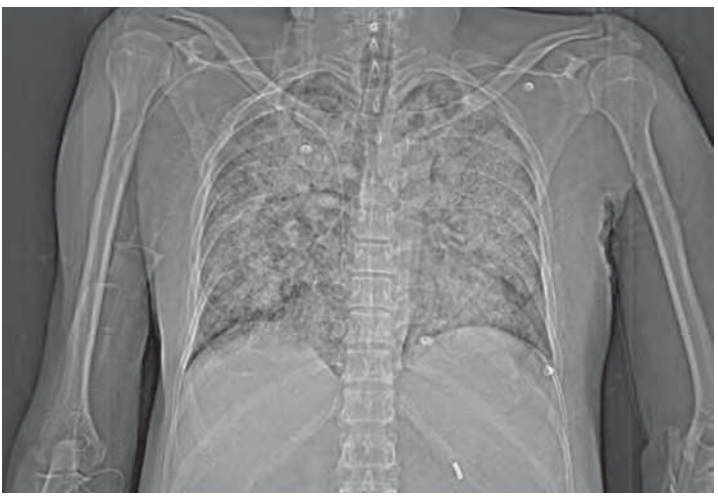
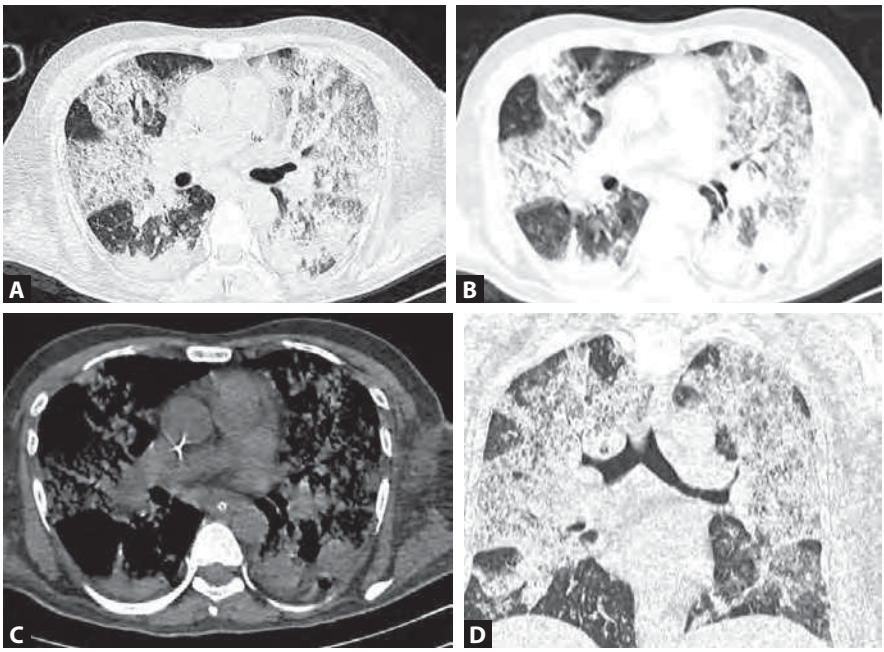


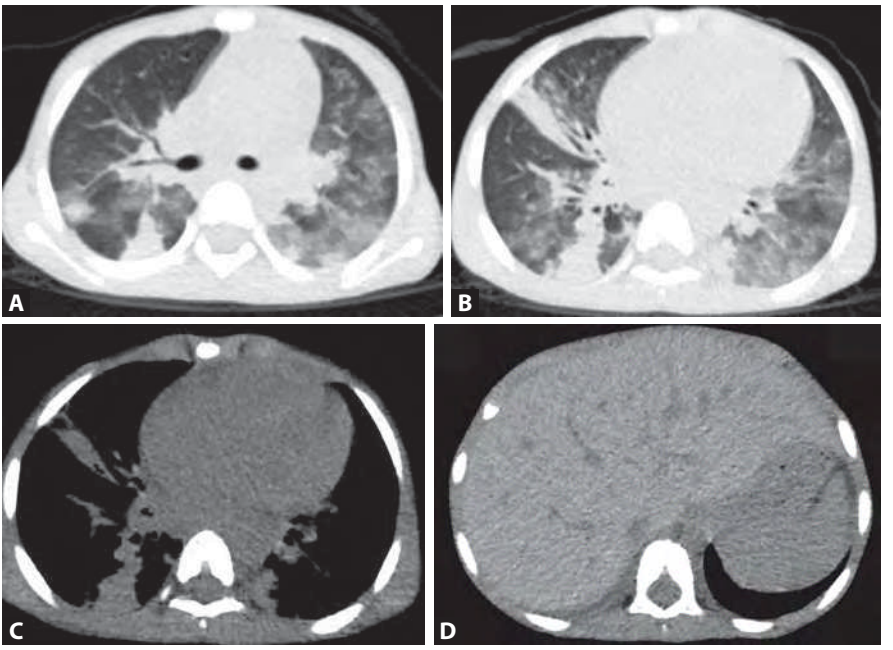
Fig. 23.9: Hemophagocytic lymphohistiocytosis. Topogram revealing extensive alveolar opacities in both lungs



Figs 23.10A to D: Axial NCCT image lung window (A and B) and soft tissue window (C), coronal reformatted image (D) showing extensive areas of consolidation and ground-glass opacities which developed in a patient with other settings consistent with hemophagocytic lymphohistiocytosis. Associated organomegaly frequently helps in narrowing down the differential diagnoses

Pulmonary Leukostasis

- This complication occurs more commonly in myelomonocytic type of acute myeloid leukemia (AML) (type M4)
- In instances when white blood cell count exceeds one lac/μL, leukemic cells start accumulating within the small vessels and clog these small vessels
- Organs commonly affected organs are lungs, heart, brain and testes
- Patients present with nonspecific symptoms such as fever, cough and dyspnea
- Radiographic findings may be normal chest radiograph even in presence of extensive symptoms or may show alveolar or interstitial opacities
- CT may reveal thickened bronchovascular bundles, ground-glass opacities which are often nonsegmental in location[3] **(Figs 23.11A to D)**
- Because of nonspecific findings—suggesting a diagnosis on the basis of CT alone is difficult. However, with a strong clinical suspicion in setting of hyperleukocytosis presence of interstitial thickening should point towards leukostasis, particularly after excluding pulmonary edema and infective etiologies.



Figs 23.11A to D: Hyperleukocytosis syndrome. In this patient with acute leukemia (peripheral WBC count 3 lac/μ L) with rapid onset dyspnea NCCT chest axial lung window images show extensive areas of ground glass opacity and peripheral wedge shaped infarcts in right lung. Lack of fever pointed away from other differential diagnosis of fungal etiology. Imaging diagnosis of leukocytosis was made in this peculiar clinical setting. Appropriate treatment for this condition resulted in rapid clearance of the widespread abnormal lung imaging finding within two days (follow-up X-ray is not shown here)

Pleuropulmonary Fibroelastosis

The idiopathic form of pleuropulmonary fibroelastosis with imaging findings has been covered elsewhere within this book, however it would be worthwhile to add that secondary form can also occur in patients of bone marrow transplantation, postradiation therapy as well as with anticancer chemotherapy.

Bone Marrow Transplant

- Hematopoietic stem cell transfusion (HSCT) has become an important treatment modality for some more aggressive forms of hematological malignancies such as acute leukemias
- The pulmonary complication incidence is different in patients who have been treated with HSCT as compared to those patients without it because of combination of immunosuppressive therapy and intensive radiation therapy in addition to the chemotherapy given in the initial period

- Pulmonary complications which occur in the patients with HSCT but are not specific to it are infections (bacterial PJP, fungal and tuberculosis), drug toxicity and pulmonary hemorrhage
- The complications which are more specific to HSCT are cytomegalovirus infection, pulmonary edema, idiopathic pneumonia syndrome, bronchiolitis obliterans (BO) and organizing pneumonitis
- Approximately half of the patients with HSCT experience complications specific to HSCT.
- Autologous HSCT entails complications related to intensive chemotherapy ± whole body radiation therapy
- Allogeneic HSCT have in addition more complications due to major immune suppression associated with it and due to GVHD
- Complications associated with allogeneic HSCT follow characteristic temporal pattern **(Table 23.3)**.[4]

Radiation Pneumonitis

- Radiation pneumonitis incidence has decreased in recent years due to less intense and decreased total radiation doses given in current regimens
- On the other hand, due to increased overall survival of these patients the late effects of radiation therapy has increased
- Postradiotherapy pulmonary side effects can be divided into two stages
- Early stage, which usually lasts for 2–3 months after radiotherapy; can range from minimal effects to extensive damage to lungs
- In the early stage, on CT diffuse ground-glass density may be seen in the exposed radiation portal with or without pleural effusion **(Figs 23.12A to D)**. Initially these are less well defined which later become more sharply defined areas conforming to the radiation portal
- In the late stage, there is progressive evolution of fibrous changes which tend to become stable in a span of 2 years **(Figs 23.13A and B)**.
- There is associated pleural thickening with hilar retractions secondary to fibrosis
- The changes are more marked and commonly seen when dose exceeds 40 Gy and when it is combined with chemotherapy (such as doxorubicin, bleomycin)
- Spontaneous pneumothorax can also be seen as a side effect of radiation therapy.

Bacterial and fungal infections do not present as diffuse lung disease but are included here for completion of pulmonary complications.

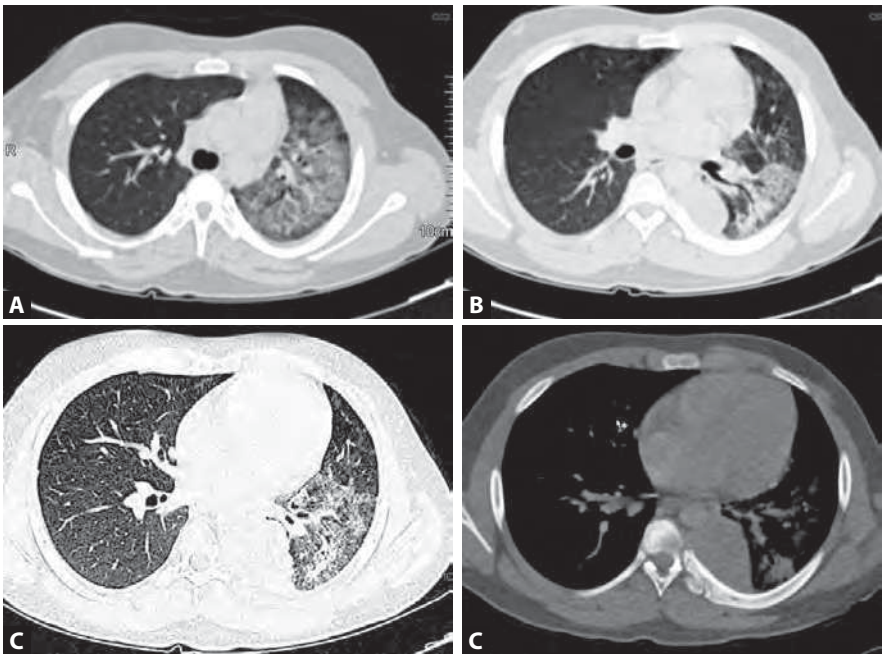
Bacterial Infections

- Subsequent to institution of chemotherapy, granulocytopenia occurs which is a dominant predisposing factor responsible for bacterial infections
- Bacterial infections are more common after hematopoietic stem cell transfusion in leukemia patients

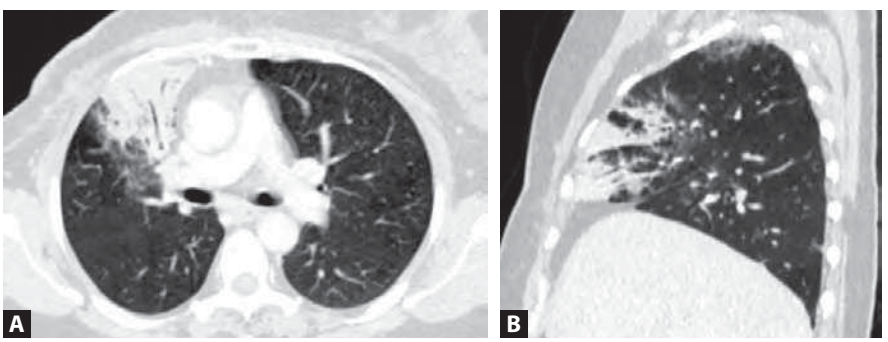
Table 23.3: Pulmonary complications in hematological malignancy patients subsequent to hematopoietic stem cell transplantation with a peculiar temporal association

	Weeks	*Prevalent disease entities*				
Onset of leukemia → Chemotherapy	-3		Leukemic infiltration			
	-2		Infection	Drug toxicity	Hemorrhage	
	-1		Infection	Drug toxicity	Hemorrhage	
HSCT → Preconditioning chemotherapy and TBI	1	Neutropenic phase (early phase)	Fungal and bacterial Infection	Hemorrhage	Edema	PJP, idiopathic pulmonary syndrome, CMV
	2					
	3					
	4	Neutropenic phase (late phase)		BO	BOOP	
	5					
	6					
	7					
	8					
	9					
	10					
	11					
	12					

Abbreviations: TBI, total body irradiations; BO, bronchiolitis obliterans; BOOP, bronchiolitis obliterans organizing pneumonia; CMV, *Cytomegalovirus*; PJP. *Pneumocystic jirovecii* pneumonia



Figs 23.12A to D: Radiation pneumonitis. Axial CECT images (A to C) lung window and (D) bone window showing extensive ground-glass opacity and patchy areas of fibrosis in entire left lung, predominantly in medial aspect, conforming to the radiation port. Note is also made of expansile lesion involving posterior end of left eighth rib with significant associated soft tissue-biopsy proven Ewing's sarcoma



Figs 23.13A and B: Another patient of CA breast with late postradiation therapy effects. Axial (A) and coronal (B) lung window images showing extensive consolidation and fibrotic changes in right middle lobe

- *Staphylococcus aureus*, coagulase negative staph and gram-negative organisms from gastrointestinal tract are main causative agents
- Various factors associated with increased incidence of bacterial infections are suboptimal function of mucociliary defence, decreased production of secretory IgA, decreased response to pneumococcal antigen and splenic dysfunction

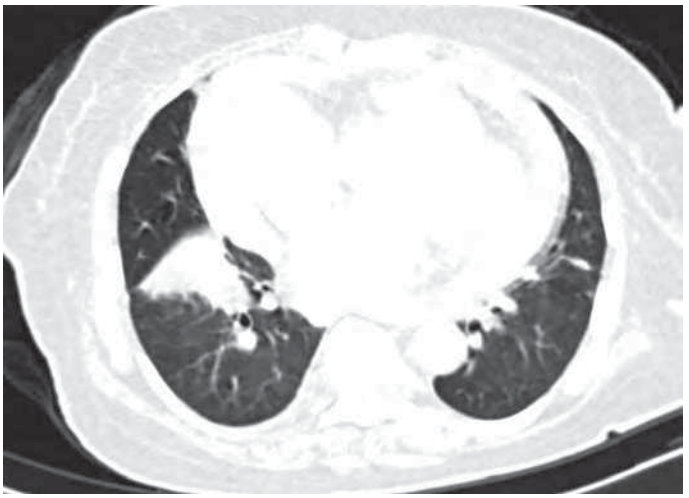
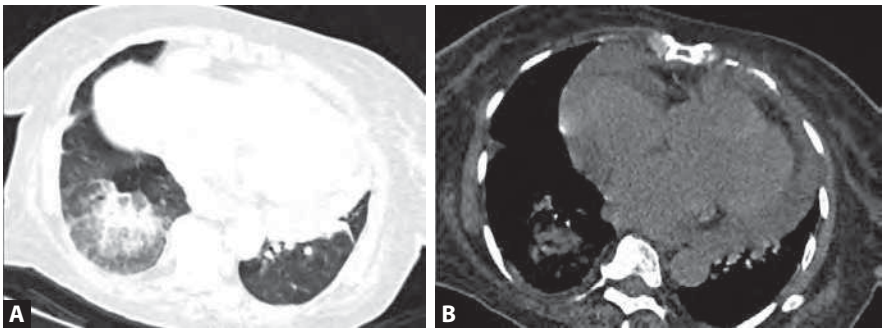


Fig. 23.14: Bacterial infection. Axial NCCT chest lung window showing segmental consolidation suggestive of bacterial infection

- On radiograph air space consolidation, ground-glass density or nodules may be seen
- Presence of ground glass opacity is more in immunocompromised patients as compared to immunocompetent patients probably due to reduced inflammatory response **(Fig. 23.14)**.

Fungal Infections

- Fungal infection is frequently seen in settings of malignancies
- An important risk factor is granulocytopenia secondary to treatment for example after chemotherapy or HSCT
- Common fungal agents are *Aspergillus* (most common), *Candida*, Mucormycosis, and *Cryptococcus*
- Generally infections because of *Apergillus* occurs in setting of low peripheral WBC count specially less than a thousand/mL.
- Out of many types of *Aspergillus* infections invasive type of pulmonary aspergillosis is the most common pattern found in these patients
- On chest radiograph single or multiple nodular opacities or consolidation commonly seen in peripheral parts of lungs is the common finding although radiograph may be normal in about 25% of patients
- Computed tomography (CT) scan is useful for making early diagnosis of invasive aspergillosis. CT halo sign—which is seen in early phase of invasive aspergillosis depicts a nodular lesion (because of central necrotic area) and peripheral ground-glass opacity (because of hemorrhagic periphery) **(Figs 23.15A and B)**. Air crescent sign; a late phase sign, occurs due to resorption of necrotic center by neutrophils, indicates that the patient is in recovery phase



Figs 23.15A and B: Axial NCCT chest lung window (A) and soft tissue window (B) images revealing patch of consolidation in right lower lobe with peripheral 'ground-glass density' suggestive of fungal etiology

- In less common invasive variety, i.e. airway invasive aspergillosis CT shows small centrilobular nodules with foci of consolidation around the bronchioles
- In *Candida* infections multiple nodules can be seen which are usually smaller than those found in aspergillosis. Other findings are consolidation or bronchopneumonia similar to bacterial infections.

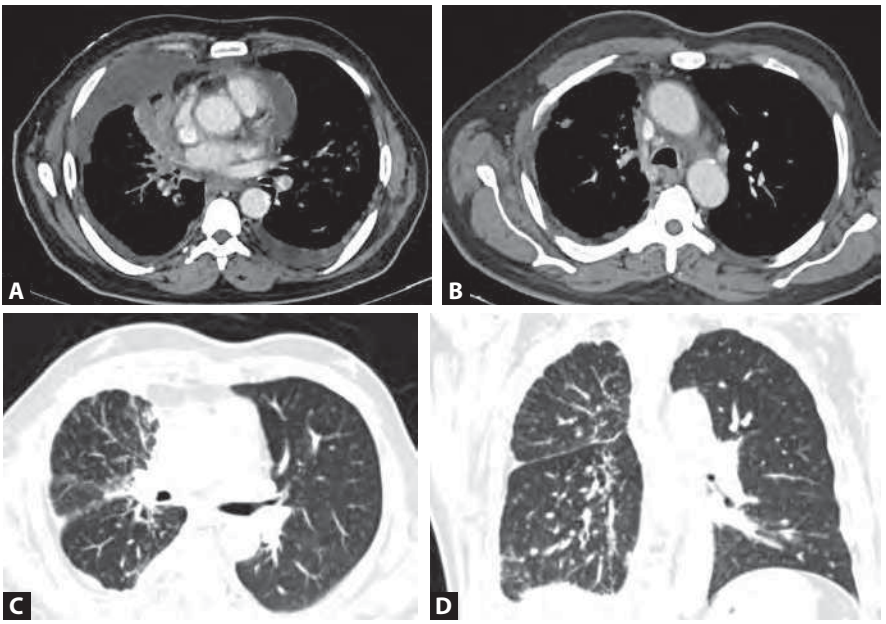
Tuberculosis

- Because of immunodeficiency associated with malignancies and associated treatments, the presentation of tuberculosis in these patients are quite atypical
- Various atypical patterns include distribution of lesions in nonsegmental pattern, presence of multiple small cavities in pulmonary lesions. Miliary pattern may also be seen. Other findings are ground-glass density peripheral to nodules
- However, as the abnormality will frequently be focal in nature and not diffusely involving the lungs, we will not discuss this topic in further details here.

MALIGNANCIES PRESENTING AS DIFFUSE LUNG DISEASES

Lymphangitic Carcinomatosis

- Lymphangitic carcinomatosis occurs when cancer cells permeate the lymphatics and spread along these lymphatics
- Chest radiograph may either be normal (30–50%) or show reticulonodular opacities which mimics interstitial edema, however it is unresponsive to diuretic therapy
- HRCT (optimal imaging modality) shows nodular, irregular beaded thickening of interlobular septa with thickening of bronchovascular bundles[5] **(Figs 23.16A to D)**
- Lung architecture is frequently preserved with presence of pleural effusion (50%) and lymphadenopathy (30%)



Figs 23.16A to D: Lymphangitic carcinomatosis. CECT chest, axial soft tissue window images (A) shows irregular hypodense mass lesion in right paracardiac location—biopsy proven CA lung. Irregular pleural thickening (B) and nodular, irregular beaded thickening of interlobular septa with thickening of bronchovascular bundles seen with preservation of lung architecture (C and D)

- Frequent differential diagnoses are pulmonary edema and idiopathic pulmonary fibrosis and sarcoidosis
- Malignancies of lung, breast, pancreas and stomach are common malignancies associated with lymphangitic carcinomatosis
- Clinically it presents with slowly progressive dyspnea and cough
- It denotes unresectable end stage disease with poor prognosis
- Differential diagnosis are pulmonary edema (shows smooth septal thickening rather than irregular, with cardiomegaly and responds to diuretic therapy) and idiopathic pulmonary fibrosis which shows smooth linear interstitial thickening, basilar subpleural predominance, honeycombing, lack of lymphadenopathy and pleural effusion.

Pulmonary Intravascular Lymphomatosis

- Pulmonary intravascular lymphomatosis is a condition wherein there is proliferation of B-cell lymphoid cells within the vascular lumen
- This is a rare condition
- Main imaging manifestations on CT images are presence of diffuse bilateral GGO.
- Less commonly consolidation and nodules may also be seen.

CONCLUSION

- Lung diseases are a frequent cause of significant morbidity and even mortality in cancer patients
- It is important to make early diagnosis for appropriate and early institution of specific therapy to alleviate patient suffering. It is important to note that although there is minimal difference in imaging findings of many of the topics discussed above, if the radiologist is aware about the clinical scenario; frequently a specific diagnosis or small differential diagnosis list can be made which has a significant bearing on the type of therapy needed for a given patient which directly results in improved patient care.

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24

Small Airway Disease: Imaging

Manisha Jana, Ashu Seith Bhalla

TERMINOLOGY

- The term 'small airways' refers to distal airways (bronchioles and their branches) that have diameters between 1 and 2 mm and no cartilage in their wall.
- Bronchioles are further divided into 'membranous' and 'respiratory' with the former being purely conducting, while respiratory bronchioles have both conducting and gas exchange (including alveoli plus alveolar ducts) function.
- Hence, the term small airway disease is used synonymously with 'bronchiolitis'.

CLINICAL CONCEPTS

- Normally the small airways make only a minor contribution to airway resistance
- It is estimated that more than 75% of small airways are obstructed when changes become apparent on pulmonary function tests (FEV1)
- Eventually patients develop airway obstruction which is not reversible with bronchodilators
- Transbronchial lung biopsy (TBLB) specimens are not sufficient for pathological diagnosis of these disorders, and surgical biopsy specimens are required for diagnosis
- Hence, the importance of early recognition of these disorders with high-resolution computed tomography (HRCT) on current generation CT scanners
- CT scan based classification aids in diagnosis of several of these disorders in conjunction with clinical background obviating the need of biopsy in many.

IMAGING MODALITIES

- Chest radiographs (CXR) have limited utility in evaluating small airway disease (SAD) as these may be normal in the early/mild involvement
- HRCT is the modality of choice
- CT signs suggestive of SAD are well described[1,2] and have been divided into direct signs and indirect signs.

Direct Signs

- Centrilobular nodules—giving 'tree-in-bud' appearance, often described as being V or Y shaped
- Ground glass opacities which are centrilobular or peribronchiolar in distribution
- Bronchiolectasis (in chronic stage).

Indirect Signs

- Air trapping (best demonstrated on expiratory scans)
- Air trapping needs to be differentiated from mosaic attenuation (*See* Chapter 1).

CLASSIFICATION

- Various basis of classification
- On the basis of clinical setting or etiology, on the basis of histology or on HRCT appearance.

On the histopathological basis, SAD is divided into primary bronchiolar disorders, interstitial lung diseases with bronchiolar component and large airway diseases with bronchiolar involvement.[3]

Primary bronchiolar disorders

- Constrictive bronchiolitis
- Follicular bronchiolitis
- Diffuse panbronchiolitis.

Interstitial lung diseases with mainly bronchiolar component

- Respiratory bronchiolitis—Interstitial lung diseases
- Cryptogenic organizing pneumonia
- Hypersensitivity pneumonitis.

Large airway diseases with bronchiolar involvement

- Asthma
- Chronic obstructive airway disease
- Bronchiectasis.

The most prevalent/commonly accepted classification is based on combined radiopathological approach

- This classification divides SAD into two major types
  1. Inflammatory/cellular bronchiolitis
  2. Constrictive/fibrotic bronchiolitis.

Chronic obstructive pulmonary disease (COPD) and asthma are covered in Chapter no 15; interstitial lung disease (ILD) with bronchiolar component are covered in Chapter no 5 and 13. This chapter will cover the primary bronchiolar disorders in detail.

Table 24.1 Summarizes the major differences between these two entities.

Table 24.1: Diagnostic pearls: Radiologic-pathologic differentiation between two types of bronchiolitis

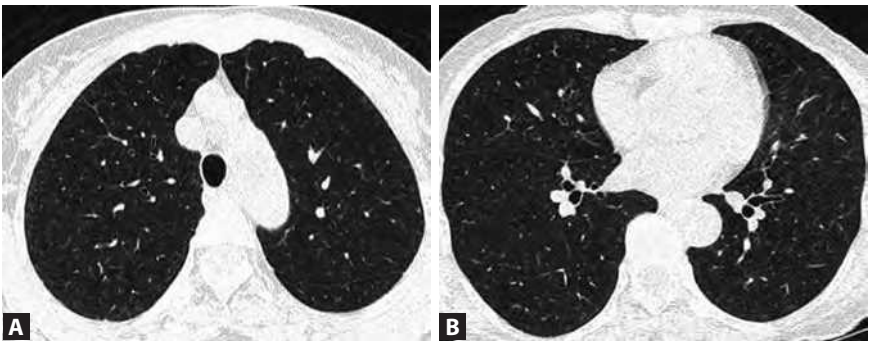
	Inflammatory/cellular bronchiolitis	*Constrictive bronchiolitis*
Pathology	Inflammatory cells (acute, chronic or a combination of these) are seen	Bronchiolar lesion resulting in narrowing of bronchiolar lumen due to collagen deposition, fibrosis and scarring of bronchiolar wall/peribronchiolar area
Etiology/ subtypes	• Infectious • Respiratory bronchiolitis • Autoimmune/Connective tissue disorders (CTD) • Hypersensitivity pneumonitis • Panbronchiolitis • Bronchiectasis associated bronchiolitis	• Idiopathic • Postinfectious • Inhalation of toxic fumes • Autoimmune/CTD • Drug related • Transplantation
Imaging	• Centrilobular nodules (CLNs) • Ground glass nodules • Bronchial wall thickening	• Air trapping (characteristic imaging feature) • Bronchiectasis/ bronchiolectasis may be seen • CLNs are rare

CELLULAR/INFLAMMATORY BRONCHIOLITIS

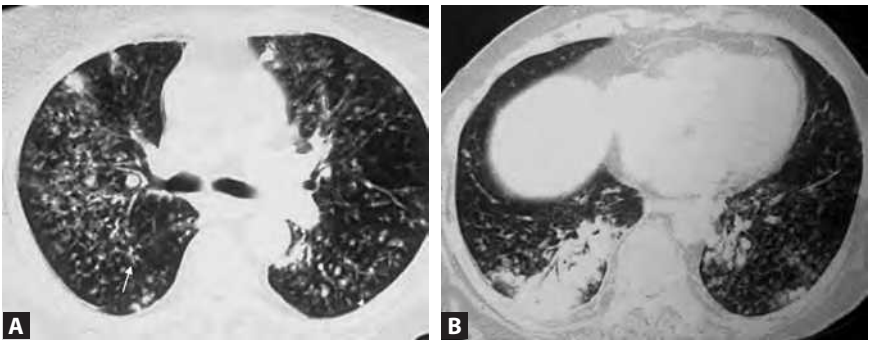
Infections

Acute

- In children, viruses are the most frequent pathogens causing acute bronchiolitis
- In adults viruses, mycoplasma or even bacterial can all cause bronchiolitis
- *Immunocompromised host*: Viral (cytomegalovirus, respiratory syncytial virus), fungal (*Aspergillus* sp) and bacteria (*Pseudomonas* sp) can all be responsible
- *HRCT:* Thickened bronchiolar walls (bronchi become visible on HRCT), well defined centrilobular nodules, 'tree-in-bud' nodules and ill-defined centrilobular nodules (CLN) **(Figs 24.1A and B)**
- Air trapping may be seen but is less frequent
- In acute bronchiolitis in children chest radiograph may be performed to exclude other diagnoses, e.g. consolidation; but CT is seldom performed



Figs 24.1A and B: Cellular bronchiolitis. HRCT chest (A and B) images reveal diffuse centrilobular nodules scattered over both lungs; accompanied by bronchiolar wall thickening



Figs 24.2A and B: Nodules and consolidation in inflammatory bronchiolitis in tuberculosis. Note the ill-defined centrilobular nodules having 'tree in bud' pattern (arrow); and area of consolidation in right lower lobe

- In late stages, bronchial wall thickening with bronchiectasis may be seen
- *Vs noninfectious bronchiolitis:* Infectious bronchiolitis has more asymmetric or patchy distribution
- Associated ground glass opacity (GGO) or areas of consolidation may be seen
- *Differential diagnosis*: Hypersensitivity pneumonitis.

Chronic

- *Organisms: Mycobacterium tuberculosis,* nontubercular mycobacteria—NTM (especially *Mycobacterium avium* intracellulare).
- Tuberculosis is associated with CLN resulting in 'tree-in-bud' appearance typically localized to a lobe or segment, although more diffuse involvement may also be seen. Associated consolidation, cavitation or necrotic lymphadenopathy aids in the diagnosis **(Figs 24.2A and B)**.

- NTM infections should be suspected when the distribution of bronchiolitis is predominantly in the middle lobe and lingular segments, especially in the elderly and postmenopausal women.
- Superadded infection/chronic colonization of pre-existing bronchiectasis is frequent. This form is difficult to differentiate from TB/NTM and is often misdiagnosed as such. The nodules in this setting can persist over long periods (as in NTM).[1]

Respiratory Bronchiolitis (RB and RB-ILD) (*See* Chapter 5)

- Respiratory bronchiolitis (RB) and respiratory bronchiolitis-associated interstitial lung disease (RB-ILD) are smoking-related disorders.
- *Pathology of RB:* Smoking-related inflammation causes accumulation of hemosiderin-laden macrophages within the respiratory bronchioles.
- *Pathology of RB-ILD:* Hemosiderin-laden macrophages accumulate in alveolar sacs in addition to respiratory bronchioles. Also, interstitial inflammation is present.
- Patients with RB are mostly asymptomatic.
- HRCT
  - *RB*: Ill-defined CLN, infrequently seen are ill-defined centrilobular GGOs.
  - *RB-ILD*: Ground glass opacities are the dominant abnormality, CLNs.
  - Air trapping in uncommon.
  - Both disorders show upper lobe predominant distribution.
- Differential diagnosis:
  - *RB*: Hypersensitivity pneumonitis (HSP), infectious bronchiolitis
  - *RB-ILD*: Desquamative interstitial pneumonia (DIP) **(Table 24.2)**, HSP, NSIP
  - *HSP*: Similar HRCT findings but seen in nonsmokers. Smokers usually do not develop HSP.

Table 24.2: Diagnostic pearls: Comparative imaging findings of RB-ILD and DIP

RB-ILD	*DIP*
Predominantly upper lung involvement	Predominantly lower and basal distribution
Ground glass opacities (GGO)	Ground-glass opacities
Centrilobular nodules seen	Centrilobular nodules absent
Cyst absent	Well-defined cysts seen in areas of GGO

Common Variable Immunodeficiency Disorders

- Common variable immunodeficiency disorders (CVID) is an immunodeficiency disorder wherein low levels of circulating immunoglobulins are seen (IgG, IgA ± IgM). Also, there are often accompanying abnormalities in T lymphocytes.
- Clinical manifestations in CVID can be divided into three broad groups: recurrent infections, autoimmune disorders and lymphoid hyperplasia (resulting in malignancies).

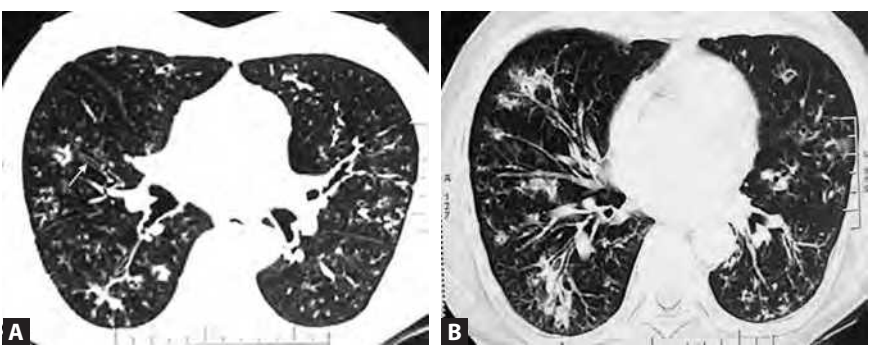
- A combination of immune dysfunction, lymphoid hyperplasia and granulocytic lesion results in a noninfectious pulmonary complication referred to as granulomatous-lymphocytic interstitial lung disease (GLILD).
- *Age:* 4th to 6th decades.
- *Clinical presentation*: Recurrent sinopulmonary infections. There is rapid worsening with development of bronchiolitis, and eventually bronchiolitis.
- Advanced cases show bronchiectasis and cyst formation.
- Distribution is diffuse with involvement of upper as well as lower lobes.
- Treatment includes prolonged course of macrolide antibiotics.
- The GLILD includes follicular bronchiolitis, lymphocytic interstitial pneumonia (LIP), granulomatous disease and organizing pneumonia.
- Follicular bronchiolitis is the characteristic pattern of SAD seen in patients of CVID.

Connective Tissue Disorders

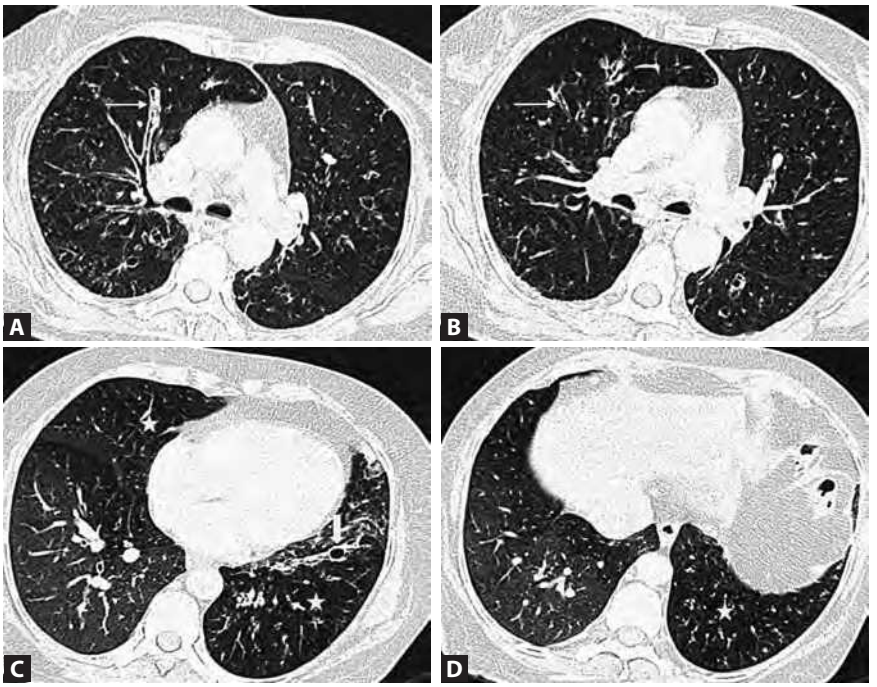
- *Pathology:* There is proliferation of lymphoid tissues around the bronchioles, which on immunohistochemistry is seen to be polyclonal
- Characteristically associated with connective tissue diseases (especially rheumatoid arthritis and Sjögren's syndrome) (*See* Chapter 7)
- SAD is seen less frequently in PSS, SLE, sarcoidosis and vasculitis
- *HRCT:* Similar to other causes of cellular bronchiolitis **(Figs 24.3A and B)**.

Inflammatory Bowel Diseases

- Inflammatory bowel disease such as ulcerative colitis and Crohn's disease may also be associated with SAD **(Figs 24.4A to D)**.
- While larger airway involvement in the form of bronchiectasis is more frequent; SAD is seen less often
- Crohn's disease is associated with cellular (granulomatous) bronchiolitis
- Ulcerative colitis is rarely associated with constrictive bronchiolitis.



Figs 24.3A and B: Inflammatory bronchiolitis in granulomatosis with polyangiitis (GPA). HRCT chest images reveal diffuse centrilobular nodules scattered over both lungs and conglomeration of these nodules at places; accompanied by bronchiolar wall thickening (arrow)



Figs 24.4A to D: HRCT findings in chronic obliterative bronchiolitis in a patient with Crohn's disease. Note the bronchiolar wall thickening (arrows), bronchiectasis (block arrow) and hyperlucent areas at lung bases because of air trapping (asterisk)

Asian Diffuse Panbronchiolitis

- Diffuse panbronchiolitis is a rare form of SAD predominantly reported from South-East and far-east Asia[2]
- *HRCT:* Centrilobular nodules with bronchiectasis/bronchiolectasis and air trapping predominantly distributed in the lower lobes
- In addition, areas of atelectasis and mucoid impaction may also be seen.

CONSTRICTIVE BRONCHIOLITIS

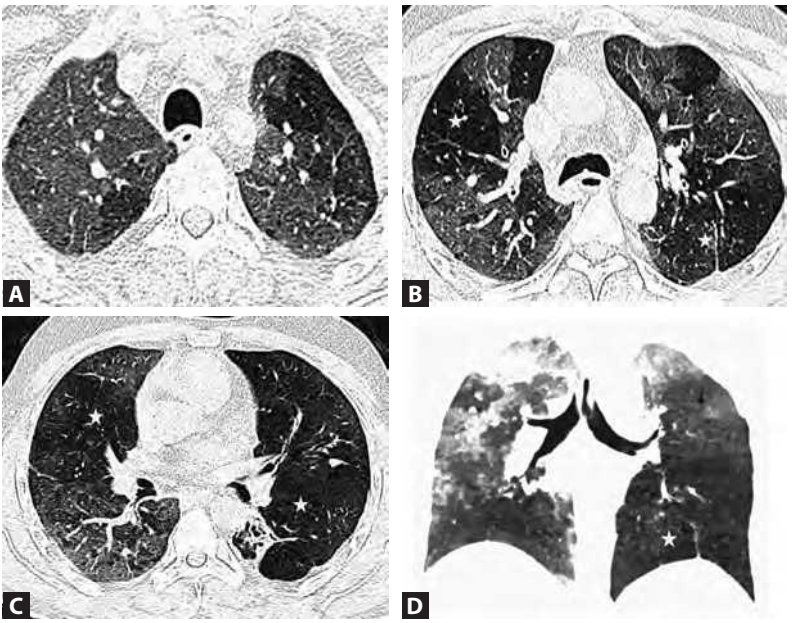
Synonym: Obliterative/fibrotic bronchiolitis, bronchiolitis obliterans.

Postinfectious

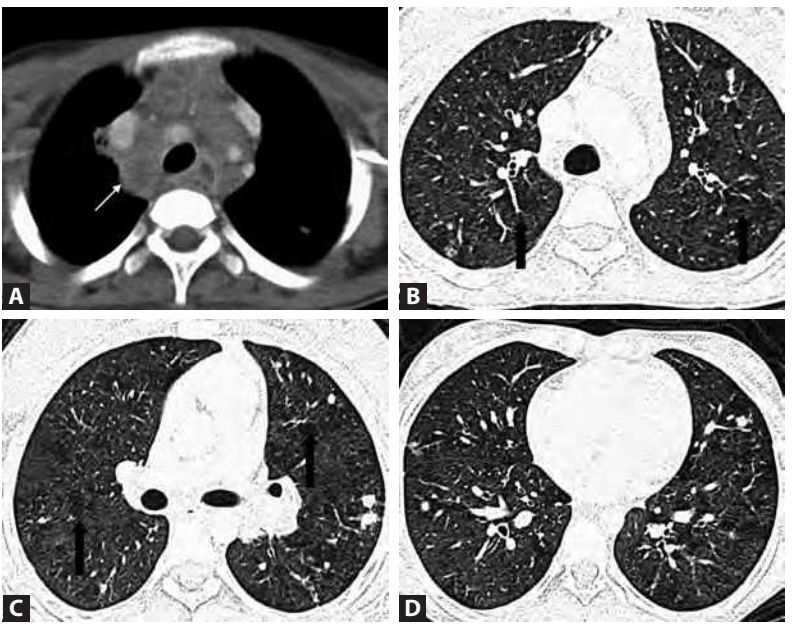
- Can occur as a consequence of infections with a variety of organisms
- Adenovirus, pertussis, tuberculosis, measles or mycoplasma
- *HRCT:* Air trapping or mosaic attenuation in a patchy distribution **(Figs 24.5A to D)**. CLN/bronchiectasis are infrequently seen **(Figs 24.6A to D)**.

McLeod/Swyer James Syndrome

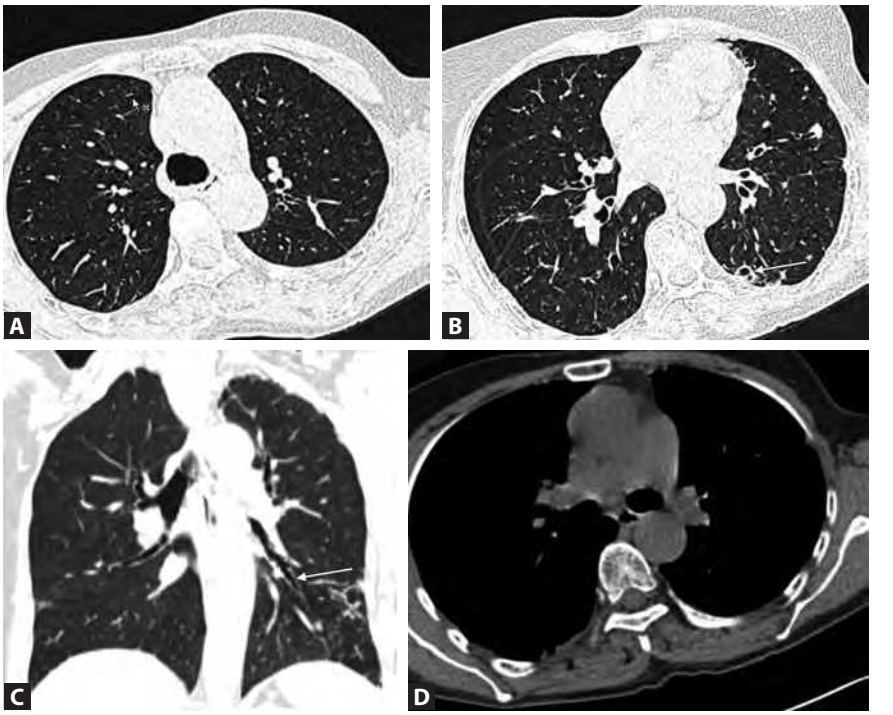
- This is a sequelae of postinfectious constrictive bronchiolitis occurring in children



Figs 24.5A to D: CT findings in chronic obliterative bronchiolitis. HRCT images (A to C) and coronal MinIP image (D) show mosaic attenuation in a patchy distribution; areas of fibrosis and air trapping more prominent in the lower lobes (asterisk)



Figs 24.6A to D: CT findings in chronic obliterative bronchiolitis secondary to tuberculosis. CECT axial image soft tissue window (A) shows necrotic mediastinal lymphadenopathy (arrow). HRCT images (B to D) show air trapping in a patchy distribution in both lungs; which earlied showed centrilobular nodules



Figs 24.7A to D: Swyer-James syndrome. HRCT images (A and B) and coronal reformatted lung window image (C) reveal a air trapping, small volume left hemithorax, areas of bronchiectasis and bronchiolectasis (arrow). Axial soft tissue window image (D) shows a small caliber left pulmonary artery

- Classically described as being unilateral, increasingly recognized that may be bilateral
- The classic description is of a small unilateral hyperlucent lung with a small pulmonary artery
- *Pathogenesis*: Infectious bronchiolitis occurring in young children often heals without sequelae or results in fibrosis and narrowing of bronchioles. This impairs subsequent alveolar development
- *Imaging*: A unilateral (or bilateral) hyperlucent lung that demonstrates air trapping
- On HRCT the air trapping is patchy and lobar/lobular in distribution. Mild bronchiectasis may also be seen. The hyperlucent appearance of the lung is due to a combination of air trapping and diminished vascularity **(Figs 24.7A to D)**
- *Differential diagnosis:* When unilateral, needs to be differentiated from pulmonary hypoplasia. Presence of air trapping suggests Swyer-James syndrome.

Toxic Fume Exposure

Exposure to noxious fumes may cause acute inflammatory response centred around the bronchiolar walls. Over a two to six weeks period subsequent fibrosis can result in obliterative bronchiolitis, e.g. nitrogen oxide exposure causing silo filler' slung, diacetyl exposure resulting in popcorn flavor manufacturer's lung (*See* Chapter 13).

Connective Tissue Disorders

- Rheumatoid arthritis and Sjögren's syndrome may be associated with both follicular and obliterative forms of bronchiolitis
- The two forms may even coexist in the same patient
- Manifestation of bronchiolitis may even precede articular manifestations
- *HRCT:* Similar to other causes of obliterative bronchiolitis.

Drug Induced

- *Rheumatoid arthritis treatment:* Follicular as well as constrictive bronchiolitis has been reported with the use of D-penicillamine and infrequently with Gold and Tiopronin
- Leads to diagnostic confusion as RA itself can also cause bronchiolitis
- Busulfan when used prior to allogeneic stem cell transplantation to suppress the immune system of the recipient.
- *HRCT:* Similar to other causes of obliterative bronchiolitis.

Transplantation

- Bronchiolitis obliterans (BO) can occur as a complication of allogeneic (not of autologous) hematopoietic stem cell transplantation (*See* Chapter 23).
- The BO is also seen in patients of lung transplantation referred to as bronchiolitis obliterans syndrome.
- The term 'bronchiolitis obliterans syndrome' refers to the clinical syndrome; while 'bronchiolitis obliterans' is used if there is confirmation on histopathology.
- Clinical presentation is of worsening dyspnea, chronic cough.
- HRCT characteristically demonstrates air trapping, and a cut off of air trapping has been used as a criteria for diagnosis.[1]
- HRCT may precede the PFT derangement in these patients.
- Advanced cases may have associated bronchial wall thickening and bronchiectasis.
- There has been interest in use of MRI for demonstration of air trapping (*See* Chapter 2).

CONCLUSION

- Following the two pattern approach in SAD helps in forming a differential diagnosis
- In an appropriate clinical setting it is even possible to make a single diagnosis.

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25

Imaging in Uncommon Disorders: Part I

Smita Manchanda

This chapter focuses on infections, pulmonary manifestations of HIV-related disease and solid organ transplant-related diseases which are interstitial lung diseases or presenting as their mimickers.

INFECTIONS

Diffuse lung involvement by infections can present as interstitial pneumonia pattern or infectious bronchiolitis pattern.[1]

Interstitial Pneumonia Pattern

- Predominantly ground-glass opacity (GGO) is seen
- *Cause:* Alveolar/Interstitial inflammatory cells and exudate
- *Organisms:*
  - Viruses, *Mycoplasma pneumoniae*, *Pneumocystis jirovecii*
  - Resolving alveolar consolidation/pneumonia.

Infectious Bronchiolitis Pattern

- Characterized by poorly defined centrilobular nodules, peribronchial consolidation, bronchial wall thickening, lobular ground glass opacity and air trapping **(Figs 25.1 and 25.2)**
- *Cause:* Peribronchial nodules enlarge and coalesce as areas of consolidation
- *Organisms:*
  - *Viruses*, *Haemophilus influenzae*, *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae.*

Viral Pneumonia

- Can be seen in both immunocompetent and immunocompromised hosts with varying organisms as the cause.[2]
  - *Immunocompetent:* Influenza virus, RSV, EBV, adenovirus



Fig. 25.1: Infectious bronchiolitis (Viral pneumonia). Chest radiograph reveals bilateral symmetrical parahilar, reticulonodular infiltrates (arrows) with peribronchial thickening and hyperaeration in adolescent male presenting with dry cough. Note that bilateral hila are also enlarged suggestive of lymphadenopathy



Fig. 25.2: Viral pneumonia. Areas of air-trapping with bronchial wall thickening, patchy consolidation and poorly defined centrilobular nodules (arrow) are seen

- *Immunocompromised hosts:* HSV, varicella-zoster, CMV, measles, adenovirus

- *Clinical features:*
  - Patients present more with systemic features than the respiratory symptoms. The course is less severe, but more prolonged.
  - Radiological manifestations are due to interstitial pneumonia and bronchiolitis



Fig. 25.3: Varicella pneumonia. Patchy ground glass opacities with few random nodules (arrow) seen scattered in bilateral lung fields

- Tracheobronchitis has no acute radiologic abnormality but may present as bronchiectasis later
- Random nodules in bilateral lung fields are a pattern characteristic of *varicella pneumonia* **(Fig. 25.3)**. However, other viruses usually present clinically and radiologically with overlapping features and it is difficult to make an etiological diagnosis on imaging alone.

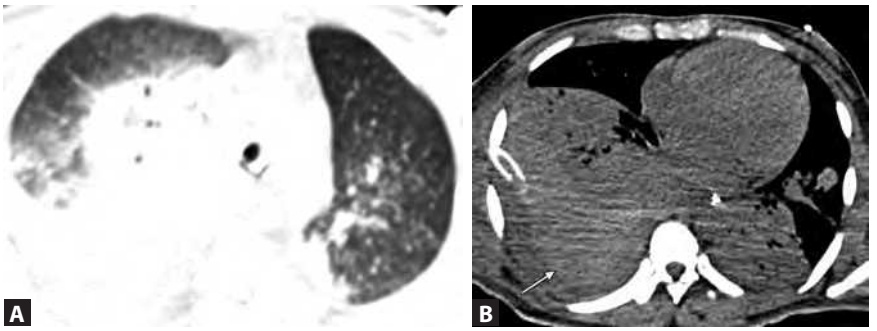
- *Chest radiograph **(Fig. 25.1)**:*
  - Poorly defined nodules
  - Patchy peribronchial ground-glass opacity
  - Air-space consolidation
  - Hyperinflation (associated bronchiolitis).
- *HRCT **(Fig. 25.2)**:*
  - Centrilobular nodules
  - Ground-glass opacity in lobular/diffuse distribution
  - Segmental consolidation
  - Air trapping.

Dengue

- *Cause:* Dengue virus (mosquito-borne disease)
- Pulmonary manifestations are uncommon
- *Increased vascular permeability:* It can manifest as pleural effusion/pulmonary edema/patchy GGO and consolidation **(Figs 25.4A and B)**
- May also be associated with diffuse pulmonary hemorrhage.

Severe Acute Respiratory Syndrome

- *Cause:* Severe acute respiratory syndrome (SARS)-associated coronavirus which causes diffuse alveolar damage and airspace edema.



Figs 25.4A and B: Dengue viral infection. Axial NCCT chest (A) showing multifocal areas of consolidation with patchy GGO in bilateral lung fields. Mediastinal window (B) reveals right pleural effusion with increased attenuation (arrow) suggestive of hemothorax in case of dengue hemorrhagic fever

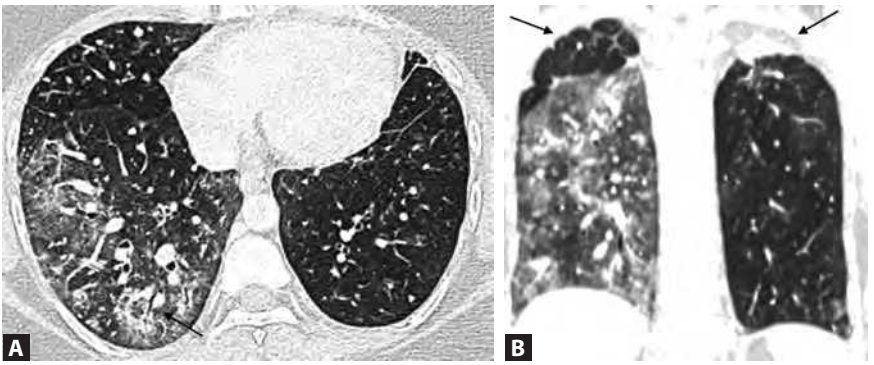
- *Clinical features:*
  - High grade fever, dry cough and dyspnea usually within 10 days after SARS exposure
- *Chest radiograph:*
  - Unifocal/multifocal consolidation
- *HRCT*:
  - *Ground-glass opacities:* Unilateral/bilateral
  - *Consolidation:* Unilateral/bilateral
  - *Crazy-paving appearance:* GGO associated with marked septal thickening.

Swine-origin Influenza A

- *Cause:* Swine-origin influenza A (H1N1) virus causing diffuse alveolar damage
- *Clinical features:*
  - Dry cough and dyspnea usually follow a prodromal period of 2–7 days
  - Usually mild disease but can be complicated by acute respiratory distress syndrome or secondary pneumonia.
- *HRCT* ***(Figs 25.5A and B)***:
  - *Ground-glass opacities:* Unilateral/bilateral
  - *Consolidation:* Peribronchovascular and subpleural.

Mycoplasma Pneumoniae

- Common cause of community acquired pneumonia in children and young adults.
- *Organism: Mycoplasma pneumoniae*
- *Clinical features:*
  - Varying from fever and productive cough to nonspecific complaints such as fatigue and myalgia



Figs 25.5A and B: Swine-origin influenza A (H1N1) virus infection. Predominantly right sided patchy ground-glass opacity with peribronchial consolidation (arrow in A) in a known case of H1N1 influenza virus infection. Fibrocalcific opacities in bilateral upper lobes (arrows in B) is sequelae of previous tubercular infection



Fig. 25.6: Mycoplasma pneumoniae. Ill-defined areas of ground-glass attenuation with nonsegmental subpleural consolidation (arrow) is seen in left lower lobe

- *Chest radiograph:*
  - Patchy consolidation
  - Diffuse reticulonodular opacities
  - Peribronchial thickening
- *HRCT* ***(Fig. 25.6)***:
  - Centrilobular nodules
  - Peribronchovascular and interlobular septal thickening
  - Lobular or segmental consolidation
  - Pleural effusion rare

***Pneumocystis jirovecii* Pneumonia**

It caused by atypical fungus *Pneumocystis jirovecii*, it is the most common cause of opportunistic infection in HIV-infected patients.

Clinical Features

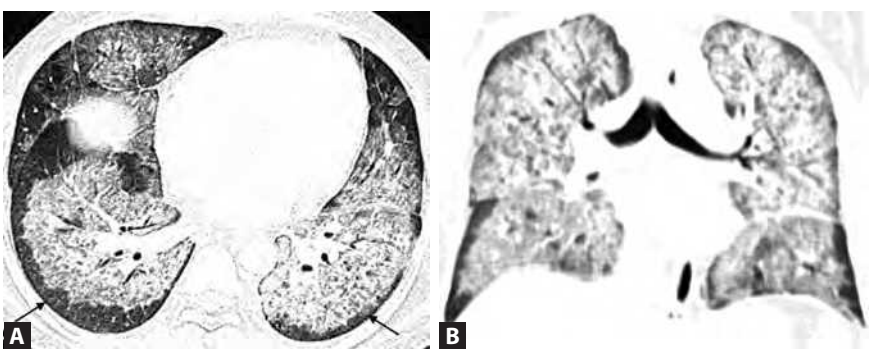
- In HIV-infected patient:
  - Subacute presentation
  - Gradual onset
  - Dry cough and dyspnea
  - Elevated serum lactate dehydrogenase levels
  - CD4 counts <100
- Patients without HIV infection:
  - Acute illness
  - Severe hypoxia
  - Respiratory failure.

Chest Radiograph

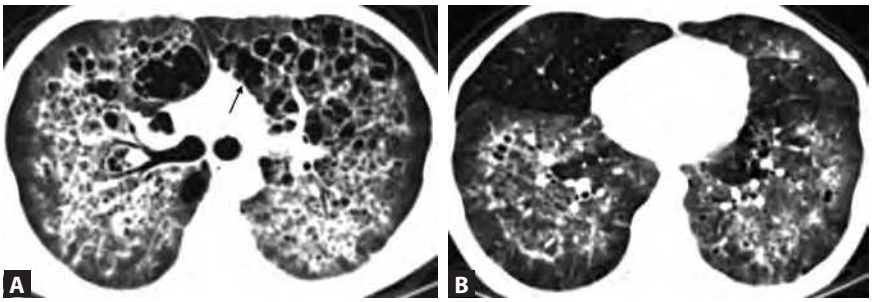
- Reticulonodular pattern
- Bilateral, symmetric
- Fine granular or poorly defined
- Can progress to consolidation.

High-resolution Computed Tomography

- Bilateral symmetric perihilar ground-glass attenuation
- Ground-glass opacity (GGO)/Consolidation with septal thickening (subacute stage) **(Figs 25.7A and B)**
- *Cystic lung disease:* Multiple, bilateral, varying size cysts **(Figs 25.8A and B)**
- Spontaneous pneumothorax.



Figs 25.7A and B: *Pneumocystis jirovecii* pneumonia. A 39-year-old male with AIDS. Axial HRCT image (A) and coronal NCCT image (B) shows extensive bilateral ground-glass opacity with superimposed interlobular septal thickening (crazy-paving pattern). Note the relative subpleural sparing (arrows in A)



Figs 25.8A and B: *Pneumocystis jirovecii* pneumonia. Bilateral perihilar ground-glass and reticular opacities seen along with numerous thin-walled cysts (arrow in A) suggestive of *Pneumocystis jirovecii* pneumonia in a retrovirus positive patient

HUMAN IMMUNODEFICIENCY VIRUS-RELATED DISEASE

Human immunodeficiency virus (HIV) is now a chronic illness with varying manifestations of pulmonary disease. These include infections, malignancies, chronic obstructive airway disease (COAD), pulmonary arterial hypertension (PAH) and immune reconstitution inflammatory syndrome (IRIS). The common manifestations are enlisted in **Table 25.1** (infections) and **Table 25.2** (non-infections) with detailed description of entities presenting as diffuse lung disease.[3,4]

Table 25.1: Infectious manifestations of HIV-related disease

Disease entity	*Etiology*	*Chest radiograph*	*HRCT*
Infections Bacterial	*Bacterial pneumonitis* Any CD4 count Usually >400	• Lobar or segmental consolidation • Multiple nodules	• Septic emboli • Cavitating nodules 'feeding vessel sign'
	Bacterial bronchiolitis	• Bronchial wall thickening • Bilateral, symmetrical, lower lobe	• Small airway disease • Ill-defined centrilobular nodules • Mosaic attenuation
	Mycobacterium tuberculosis (MTB) CD4>200: reactivation TB; CD4<200: Primary TB	• Diffuse and lower zone disease • Miliary disease **(Fig. 25.9)**	• Patchy consolidation • Lower lobe • *Lymph nodes:* Markedly enlarged and necrotic

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Disease entity	*Etiology*	*Chest radiograph*	*HRCT*
Fungal	*Pneumocystis jirovecii* ***(Figs 25.7 and 25.8)*** Most common opportunistic infection CD4 <100	• Bilateral symmetric perihilar opacities • Ground glass • Alveolar consolidation	• Bilateral symmetric perihilar ground-glass attenuation • Consolidation with septal thickening • Cystic lung disease
	Cryptococcus	• Reticular or reticulonodular infiltrates	• Reticular or reticulonodular infiltrates • Solitary or multiple nodules
	Aspergillus Common in neutropenic setting	• Mycetomas: In cavities of MTB or PJP	• Air-crescent sign • Nodules with ground-glass halo • *ABPA:* Bilateral central bronchiectasis
Viral	*Cytomegalovirus* CD4 <100 **(Figs 25.10 and 25.11)**	• Ground-glass opacities • Nodules • Perihilar and lower zone interstitial infiltrates	• Ground-glass opacity (GGO) • Nodules (varying sizes) with surrounding ground glass halo **(Fig. 25.10)** • Perihilar and lower zone interstitial thickening **(Fig. 25.11)** • Effusions • Small airway disease

Abbreviation: PJP, *Pneumocystis jirovecii* pneumonia

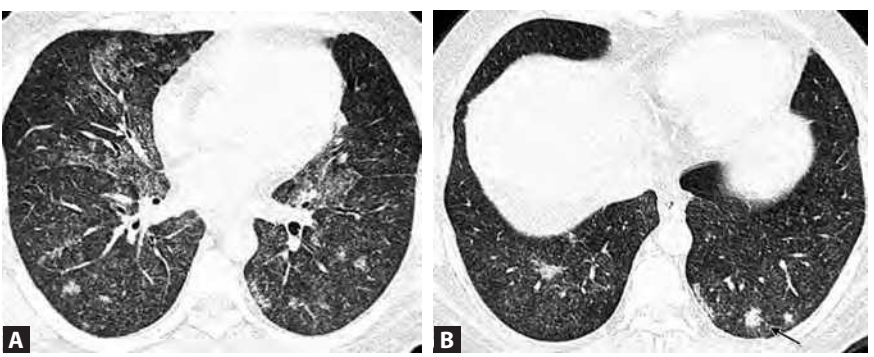


Fig. 25.9: Miliary TB. HRCT chest of a 20-year-old male with AIDS reveals multiple small (1–3 mm) noncalcified nodules scattered in bilateral lung fields suggestive of miliary TB

Table 25.2: Noninfectious manifestations of HIV-related disease

Disease entity	*Key features*	*Chest radiograph*	*HRCT*
Lymphoma CD4 counts: 200–400	• Second most common malignancy • HD more common in ART era	• Pulmonary nodules • Short doubling time	• Pulmonary nodules in NHL • Lymphadenopathy in HD
Lung carcinoma No correlation with CD4 count	• Type: Poorly differentiated or adenocarcinoma	• Similar to adenocarcinoma in non-HIV patients	• Similar to adenocarcinoma in non-HIV patients
Lymphoproliferative disorders CD4 count is usually normal	*Lymphocytic interstitial pneumonia* ***(Figs 25.12A and B)*** More frequent in children	• Ground glass nodules • Septal thickening	• Thickening of bronchovascular bundles • Centrilobular and subpleural nodules • Ground-glass opacity • Cysts
Chronic obstructive airway disease/ emphysema No correlation with CD4 count	Also in non-smokers	• Hyperinflation • Bullae • Pneumothorax	• Emphysema • Bullae • Pneumothorax • Air trapping • Cysts

Abbreviations: HD, Hodgkin's disease; NHL, non-Hodgkin lymphoma



Figs 25.10A and B: Cytomegalovirus pneumonia. A 45-year-old male with HIV infection and raised *Cytomegalovirus* (CMV) titers. Patchy GGO, poorly defined centrilobular nodules and peribronchial thickening (arrow in A) involving bilateral lung field. Few small nodules in left lower lobe are associated with areas of surrounding ground-glass halo (arrow in B)



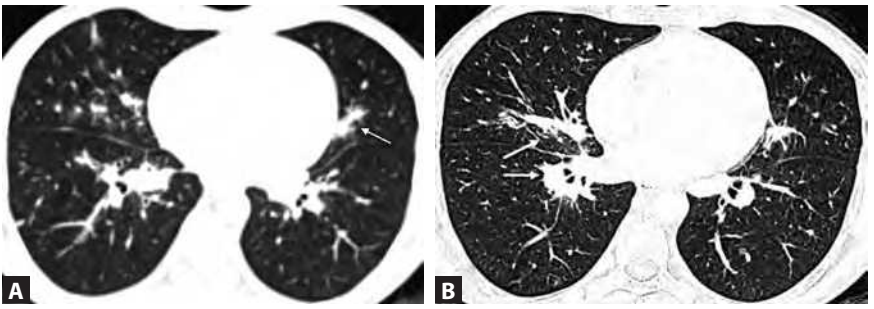
Figs 25.11A and B: Cytomegalovirus pneumonia. Patchy ground-glass opacity with poorly defined centrilobular nodules (arrow in A) in bilateral lung fields. Note the bronchial wall thickening in bilateral lower lobes (arrows in B)



Figs 25.12A and B: Lymphocytic interstitial pneumonia. CECT chest mediastinal window (A) of a 10-year-old female with HIV infection shows multiple, homogeneous mediastinal and right axillary lymph nodes. Lung window (B) reveals micronodules (1–3 mm) in a characteristic perilymphatic interstitial distribution

Kaposi Sarcoma

- Most common malignancy in HIV patients
- *Etiology:* Kaposi sarcoma (KS) herpes virus or human herpes virus 8
- Pulmonary involvement seen in up to 50%
- CD4 count < 200 cells/mm^3
- *Chest radiograph:*
  - Reticulonodular infiltrates: Bilateral, perihilar
  - Septal lines
  - Pleural effusion
- *HRCT:*
  - Thickening of bronchovascular bundles
  - Interlobular septal thickening
  - 'Flameshaped' nodules **(Figs 25.13A and B)**
  - Pleural effusions (hemorrhagic on aspiration)



Figs 25.13A and B: Kaposi sarcoma. A 40-year-old male with HIV infection. Axial lung window and HRCT images reveal flame-shaped opacities (arrow in A) and peribronchovascular thickening and consolidation (arrows in B)

Immune Reconstitution Inflammatory Syndrome

- Characterized by paradoxical worsening of symptoms with increase in CD4 counts without a progression of HIV disease.
- *Clinical diagnosis of IRIS:* When patient's condition becomes worse after an initial improvement postinitiation of ART
- Commonly seen with the following coexisting diseases:
  - *Tuberculosis-IRIS*
    - Most common form
    - Recurrence of fever, cough
    - Miliary nodules
    - Lymphadenopathy
  - *MAC-IRIS*
    - Regional lymphadenopathy with central necrosis
    - Consolidation
    - Centrilobular nodules
  - *Pneumocystis IRIS*
    - Dyspnea, cough, hypoxia
    - Diffuse ground glass opacities
    - Peribronchial consolidation and nodules
  - *Sarcoidosis IRIS*
    - Can develop/recur
    - Perilymphatic nodules
    - Hilar or mediastinal lymphadenopathy.

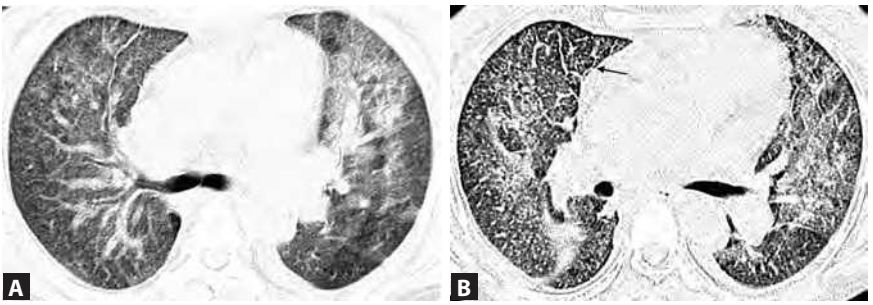
TRANSPLANT (SOLID ORGAN TRANSPLANT)-RELATED (TABLE 25.3)

Post-transplant pulmonary complications can be either infectious or non-infectious.[5]

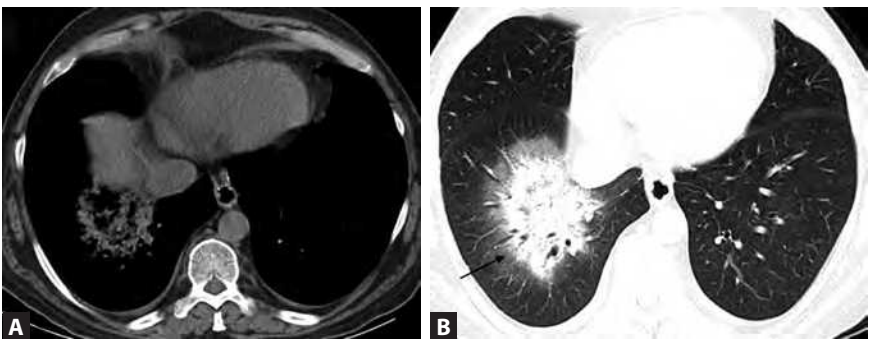
- *Infectious complications* are the same as discussed above for HIV-related disease.
- *Noninfectious complications* are further classified as acute and chronic complications.

Table 25.3: Noninfectious complications of solid organ transplant

	Noninfectious pulmonary complications			
	Liver	*Lung*	*Heart*	*Kidney*
Acute	ARDS	Reperfusion pulmonary edema	Atelectasis	Pulmonary edema
	Perioperative pleural effusions	Primary graft failure	Pulmonary edema **(Figs 25.14A and B)**	Thromboembolic events
	Pulmonary Edema	Major dehiscence of bronchial anastomosis	Mediastinitis	
	Right diaphragmatic dysfunction (phrenic nerve injury)	Acute hyperinflation of the native lung		
		Airway complications		
		Phrenic nerve injury		
Chronic	Hepatopulmonary syndrome	Post-transplant lymphoproliferative disorder	Post-transplant lymphoproliferative disorder **(Figs 25.15A and B)**	Post-transplant lymphoproliferative disorder
	Portopulmonary hypertension	Bronchiolitis obliterans **(Figs 25.16A and B)**	Immune-compressant drug induced interstitial lung disease	Pulmonary metastatic calcification
	Post-transplant lymphoproliferative disorder	Immune-compressant drug induced interstitial lung disease		
	Immune-compressant drug-induced interstitial lung disease			



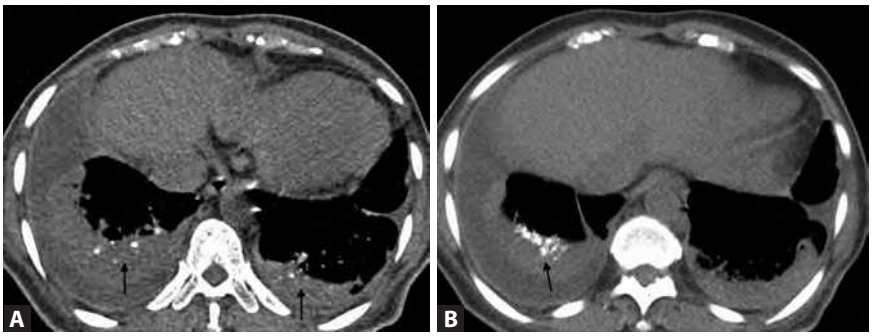
Figs 25.14A and B: Hydrostatic pulmonary edema in renal transplant recipient. CT lung window image (A) and HRCT (B) image shows central distribution of ground glass opacities and consolidation; with smooth interlobular septal thickening (arrow in B)



Figs 25.15A and B: Post-transplant lymphoproliferative disorder. NCCT chest: mediastinal (A) and lung window (B) showing focal right lower lobe consolidation with surrounding ground glass halo (arrow in B) in a biopsy proven case of post-transplant lymphoproliferative disorder



Figs 25.16A and B: Bronchiolitis obliterans. A 32-year-old female with chronic allograft nephropathy. HRCT chest reveals bilateral air trapping with bronchial wall thickening and minimal bronchiectasis in right middle lobe (arrow in A). Few ill-defined centrilobular nodules are seen in left lower lobe (arrow in B)



Figs 25.17A and B: Bilateral basal consolidation with few foci of nodular calcification (arrows in A and B) suggestive of pulmonary metastatic calcification in postrenal transplant patient

Primary Graft Failure

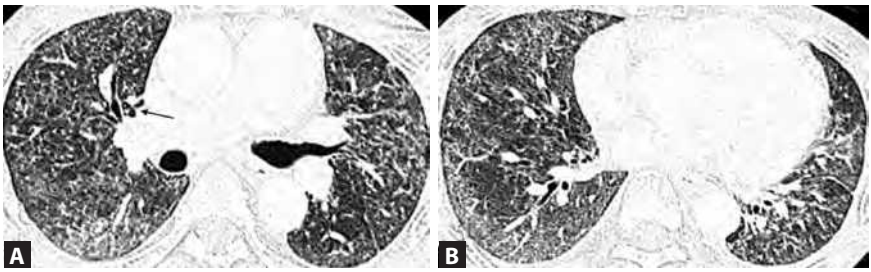
- Manifests as an acute lung injury with alveolar damage—"ischemia-reperfusion injury".
- The pattern is of diffuse alveolar damage has been described in Chapter 21.

Pulmonary Metastatic Calcification

- *Pulmonary nodules*: Single/multiple
- Alveolar consolidation
- These may appear noncalcified on chest radiograph; on CT high attenuation (greater than 100 HU) nodules/parenchymal opacities **(Figs 25.17A and B)**.

Drug-induced Lung Disease

- The use of *mammalian target of rapamycin (mTOR) inhibitors* in solid organ transplant recipients is a known cause of ILD.
- *Agents include:* Everolimus; sirolimus.
- Onset of interstitial lung disease (ILD) is variable, may present months to years later
- More common if late switch to mTORi and higher trough levels
- Recommended trough levels of everolimus: 3–8 ng/mL
- Reversible on discontinuation of the drug
- *Presentation:* Dry cough, dyspnea and easy fatigability
- *Radiographic abnormalities:*
  - Bilateral interstitial infiltrates
  - Consolidation
  - Multiple nodules
- *BAL:* Lymphocytic alveolitis
- *Tacrolimus* is a macrolide calcineurin inhibitor, which is also used in immunosuppression post-transplant. Pulmonary toxicity has been reported and can present as interstitial fibrosis **(Figs 25.18A and B)**.



Figs 25.18A and B: High-resolution computer tomography (HRCT) images of a 40-year-old male on tacrolimus, postrenal transplant reveal ground glass opacity with reticulation and minimal bronchiectasis (arrow in A) suggestive of drug induced ILD [nonspecific interstitial pneumonia (NSIP) pattern]

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26

Imaging in Uncommon Disorders: Part II

Smita Manchanda, Priyanka Naranje

This chapter shall include various entities such as pulmonary amyloidosis; eosinophilic lung disease; pulmonary alveolar proteinosis; lipoid pneumonia; interstitial lung disease (ILD) associated with inflammatory bowel disease; pulmonary veno-occlusive disease and pulmonary capillary hemangiomatosis; drug induced lung disease; age-related diffuse lung disease; IgG4-related lung disease; malignant causes of diffuse lung disease such as lymphangitis carcinomatosa, pulmonary lymphoma and invasive adenocarcinoma and vanishing lung syndrome.

PULMONARY AMYLOIDOSIS

Amyloidosis is characterized by the abnormal extracellular deposition of proteins. Thoracic involvement can be localized or a part of systemic disease.

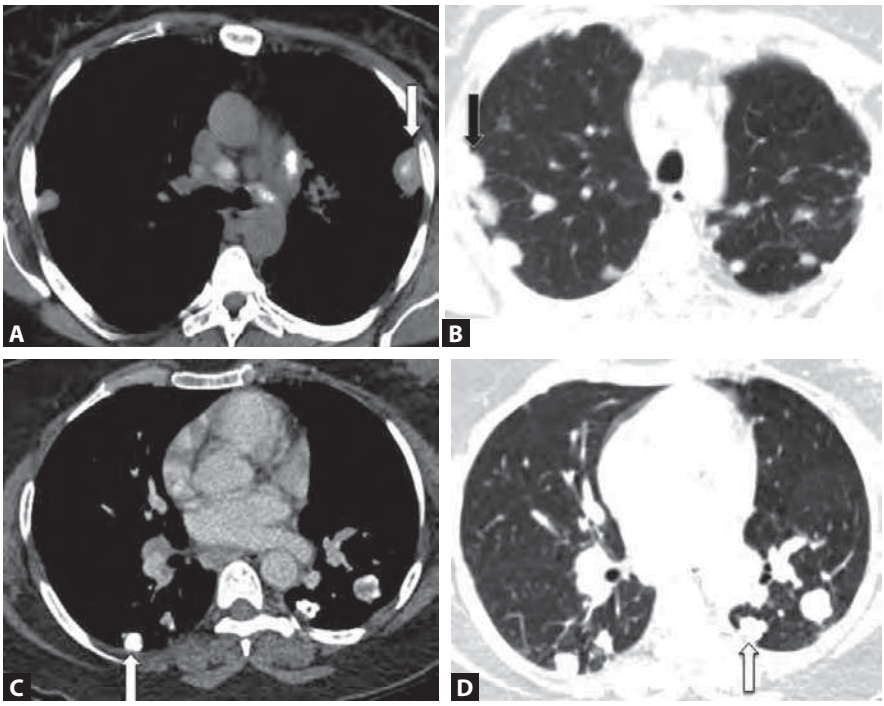
Three main patterns of pulmonary involvement are described[1]:
1. Nodular parenchymal amyloidosis
2. Diffuse alveolar septal amyloidosis
3. Airway/Tracheobronchial amyloidosis.

Nodular Parenchymal Amyloidosis

- Asymptomatic adult
- Occasional cough, dyspnea.

Chest radiograph
- Incidental abnormalities
- Solitary or multiple pulmonary nodules
- Smooth margins.

High-resolution computed tomography (HRCT) (Figs 26.1A to D)
- Solitary or multiple pulmonary nodules
- Sharp margins
- Subpleural and peripheral



Figs 26.1A to D: Pulmonary amyloidosis (nodular form). (A and B) Note the multiple predominantly subpleural well-defined nodules (arrows) and mediastinal lymphadenopathy with amorphous central calcification in a patient with pulmonary involvement in amyloidosis. (C and D) Another patient showing similar findings

- Calcification occasional
- May cavitate.

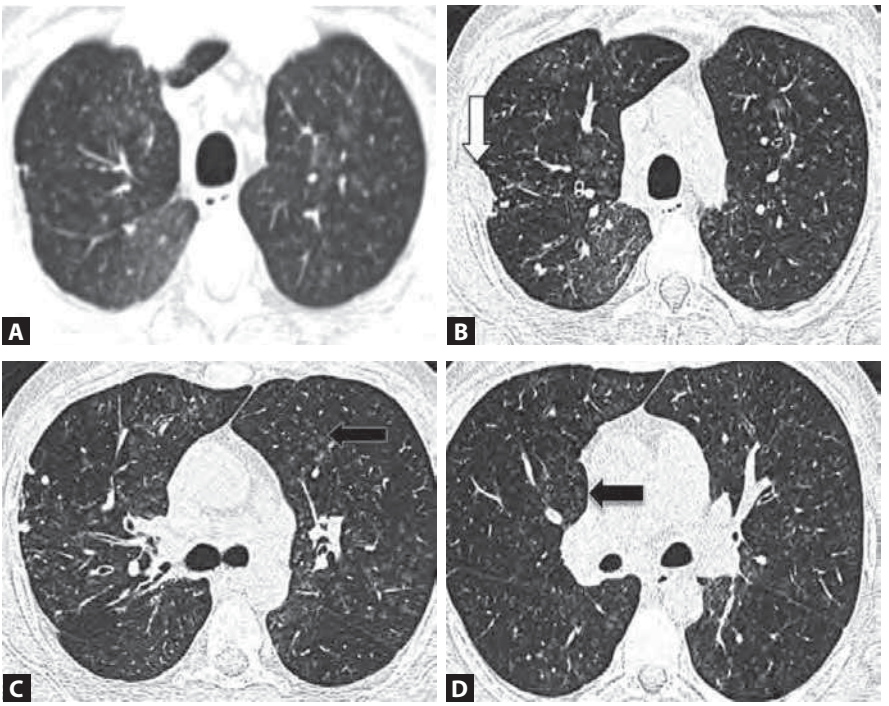
Clinical course
- *Biopsy is often required*: The deposits contain AL amyloid.
- Benign course
- No treatment required.

Diffuse Alveolar Septal Amyloidosis

Clinical features
- Associated with systemic involvement
- Cough, dysnea.

Chest radiograph
- Well-defined small nodules
- Reticular opacities.

HRCT
- Well-defined small nodules **(Figs 26.2A to D)**
- Smooth septal thickening



Figs 26.2A to D: Pulmonary amyloidosis. Note the multiple nodules (black arrows), patchy ground glass opacities in both lungs and few subpleural larger well-defined nodules (B) (white arrow) in a case of pulmonary amyloidosis

- Bilateral basal consolidations
- Punctate calcifications
- Occasional cysts
- Traction bronchiectasis.

Clinical course
- Poor prognosis
- Symptomatic, may progress to pulmonary hypertension and respiratory failure.

Airway/Tracheobronchial Amyloidosis

Clinical features
- Usually occurs in isolation
- Diffuse airway involvement occurs as submucosal plaques
- If solitary; can appear such as endobronchial neoplasm
- Wheeze, dyspnea, cough and hemoptysis.

HRCT
- Long segment narrowing
- Calcification or ossification of the airway wall

- Posterior membrane may be involved
- Postobstructive atelectasis.

Prognosis
Worse in cases of proximal airway involvement.

Mediastinal Amyloidosis

Clinical features
- Usually incidental finding
- Secondarily seen in systemic amyloidosis
- Cough, wheeze, dyspnea (if airway compression)

HRCT
- Mediastinal lymphadenopathy
- Homogeneous with punctate/eggshell calcification.

EOSINOPHILIC LUNG DISEASE

Comprises of varied disease entities, which are characterized by pulmonary manifestations and eosinophilia (peripheral or tissue eosinophilia). These have been classified as[2]:
- Eosinophilic lung diseases of unknown cause **(Table 26.1)**
- Eosinophilic lung diseases of known cause **(Table 26.2)**
- Eosinophilic vasculitis **(Table 26.3)**

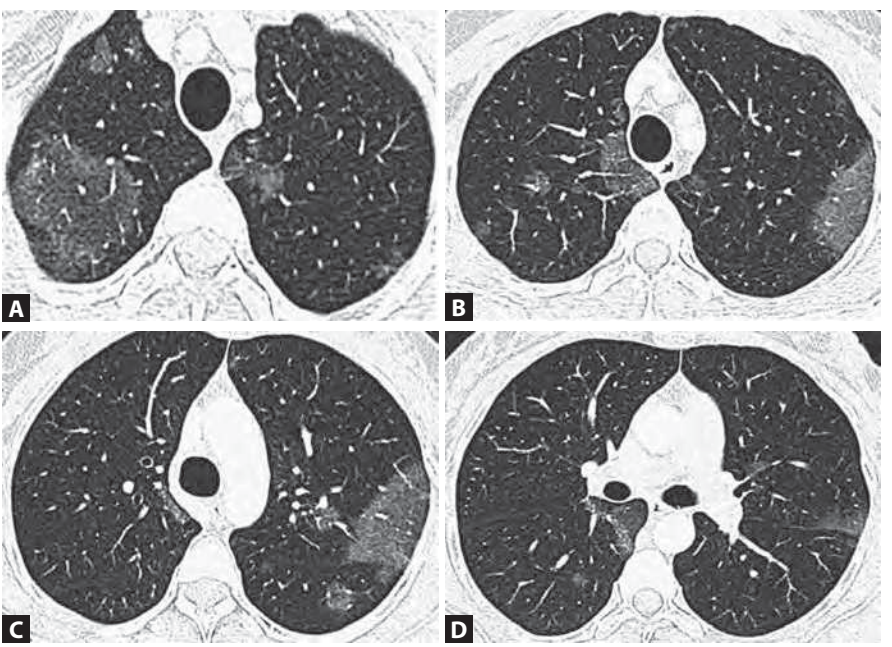
Table 26.1: Eosinophilic lung diseases of unknown cause

Disease entity	*Key feature*	*Chest radiograph*	*HRCT*
Simple pulmonary eosinophilia	• Also known as Loeffler syndrome • Spontaneous resolution	Consolidation: • Nonsegmental • Ill-defined margins • Transient • Single/multiple • Peripheral	• GGO/consolidation • Peripheral • Upper and middle lobes • Airspace nodules with perilesional GGO
Acute eosinophilic pneumonia	• Fever <5 days • > 25% eosinophils in bronchoalveolar lavage (BAL) fluid • Absence of parasitic/other infections • Immediate response to corticosteroids	• Bilateral reticular opacities • Patchy consolidation	• Bilateral patchy GGO **(Figs 26.3A to D)** • Interlobular septal thickening • Occasionally patchy consolidation/ poorly defined nodules

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Disease entity	*Key feature*	*Chest radiograph*	*HRCT*
Chronic eosinophilic pneumonia	• Insidious onset • Middle aged women • Mild eosinophilia • Serum IgE increased	• Consolidation • Nonsegmental • Peripheral • Airspace • Upper lobes	Consolidation • Nonsegmental • Peripheral • Airspace • GGO Reticulation
Idiopathic hyper-eosinophilic syndrome	• Persistent eosinophilia > 1500 cells/mm^3 • No parasitic, allergic cause • Multiorgan system dysfunction • Males • 30–40 years	• Nonspecific • Focal/diffuse • Interstitial or alveolar • Nonlobar • Pleural effusion	• Nodules • Perinodular GGO • Patchy GGO



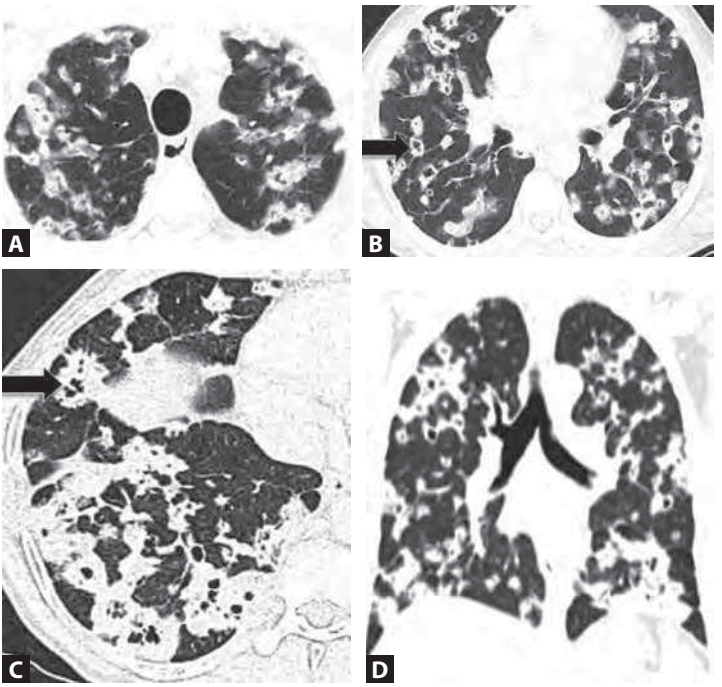
Figs 26.3A to D: Eosinophilic lung disease. Axial HRCT images showing bilateral patchy subpleural GGO predominantly in upper lobes in a case of acute eosinophilic pneumonia

Table 26.2: Eosinophilic lung diseases of known cause

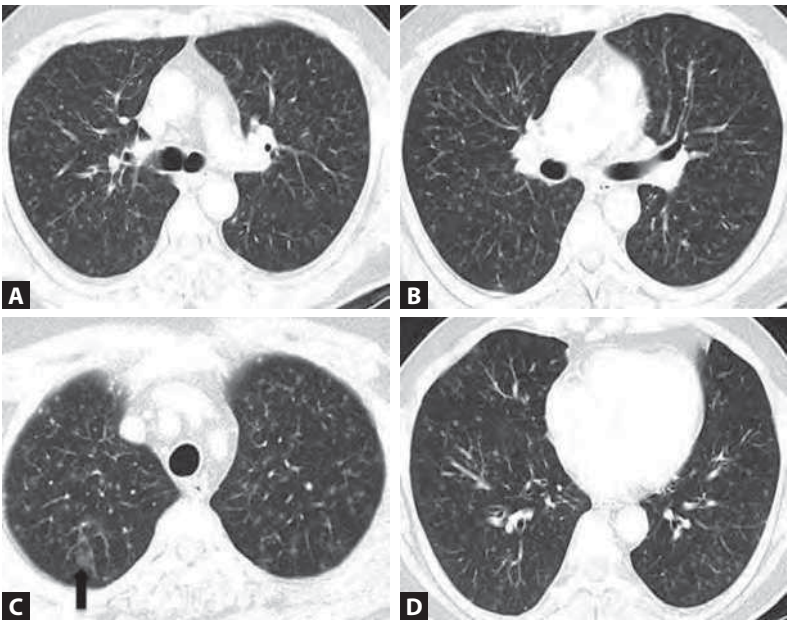
Disease entity	*Key feature*	*Chest radiograph*	*HRCT*
Allergic broncho-pulmonary aspergillosis (ABPA)	• Asthma • Peripheral blood eosinophilia • Increased serum IgE levels	• Branching linear structures • Upper lobes • Ill-defined nodules	• Bronchiectasis • Mucoid impaction • Upper lobes • Centrilobular nodules • Bronchial wall thickening
Bronchocentric granulomatosis (BG) **(Figs 26.4A to D)**	• Necrotizing granulomatous inflammation • Involves bronchial and bronchiolar epithelium	• Nodular opacities • Consolidations • Unilateral • Upper lung zones	• Focal mass • Lobar consolidation with atelectasis
Parasitic infections	• *Strongyloides stercoralis* • *Ascaris lumbricoides* • *Clonorchis sinensis* • *Schistosomiasis* • *Paragonimiasis westermani* • *Echinococcus granulosus*	• Diffuse pulmonary opacities • Single or multiple migrating nodules	• Small nodules • Larger nodules with GGO halo
	• Tropical pulmonary eosinophilia **(Figs 26.5A to D)** (*Wuchereria bancrofti* and *Brugia malayi*)	• Reticulonodular opacities • Lower lung zones	• Fine, diffuse • Reticulonodular opacities
Drug reactions	Variable (mild SPE-like syndrome/toxic AEP-like syndrome) • Aniline derivatives • L-tryptophan	• Consolidation • Hilar adenopathy • Pleural effusion • Reticulonodular densities	• GGO • Consolidation • Nodules • Reticular opacities

Table 26.3: Eosinophilic vasculitis

Disease entity	*Key feature*	*Chest radiograph*	*HRCT*
Granulomatosis [Churg-Strauss syndrome]	Diagnostic criteria (4 out of 6) • Asthma • Eosinophilia > 10% • Neuropathy • Migratory pulmonary opacities • PNS abnormalities • Extravascular eosinophils	• Bilateral consolidation • Reticulonodular opacities	• Subpleural and lobular • GGO/consolidation • Centrilobular nodules • Bronchial wall thickening • Interlobular septal thickening



Figs 26.4A to D: Bronchocentric granulomatosis. Axial (A to C) and coronal (D) HRCT images showing multiple diffuse nodular opacities in bronchocentric distribution with cavitations (arrows) in a biopsy proven case



Figs 26.5A to D: Tropical pulmonary eosinophilia. Axial HRCT images show fine, diffuse reticulonodular opacities (arrow) in both lungs in a patient belonging to endemic region and absolute eosinophil count of 15000

PULMONARY ALVEOLAR PROTEINOSIS

Characterized by the alveolar accumulation of surfactant-like proteins. It can be of three types[3]:

1. Congenital
2. *Idiopathic (primary):* Most common
3. *Secondary:* To hematological malignancies, dust/metal exposure, chemotherapy or infection and responds well to treatment of the predisposing cause.

Clinical features

- It can be asymptomatic
- Insidious onset of dyspnea, dry cough, easy fatigability
- Young males.

Chest radiograph

- Bilateral symmetrical/Patchy
- Airspace disease.

HRCT

- Bilateral involvement
- Patchy involvement with geographic/lobular sparing
- Characterized by *"crazy paving pattern":* ground glass opacity with thickened septal lines **(Figs 26.6A and B)**
- Differentials of crazy paving include pulmonary hemorrhage, pulmonary edema, radiation exposure, drug induced, inhalation disorders, bronchio-alveolar carcinoma.

Clinical relevance

- In the presence of following findings, consider superadded conditions.
- Large areas of consolidation: Infection
  - *Bulky lymphadenopathy*: Infection, malignancy.

LIPOID PNEUMONIA

Characterized by the alveolar accumulation of lipids.[4] It can be of two types depending on the source:

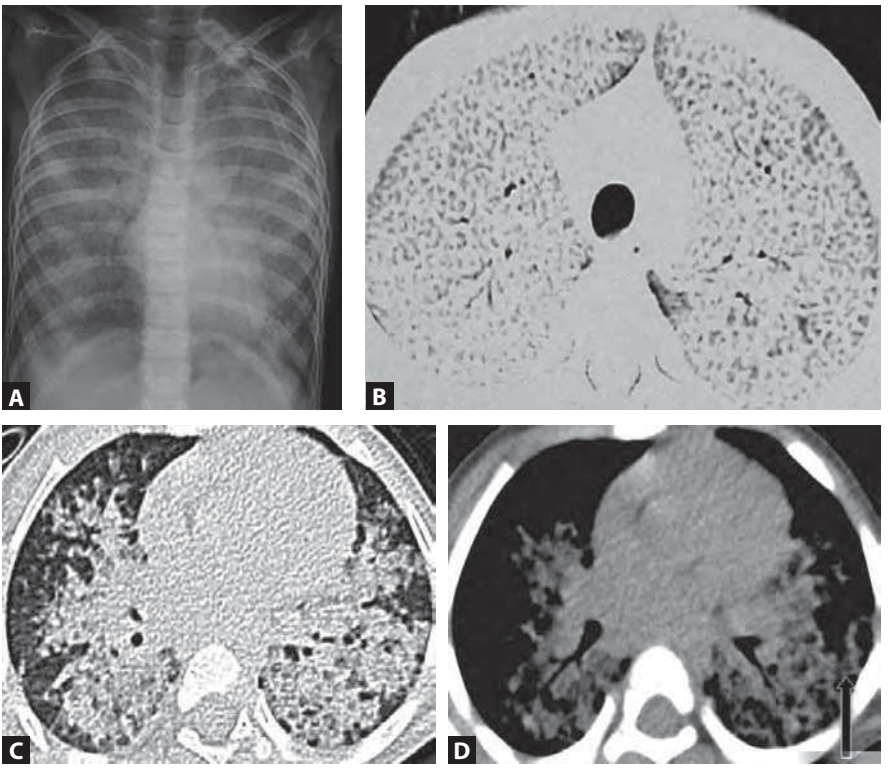
1. *Exogenous:* Inhalation/aspiration of animal fats/mineral or vegetable oils
   - *Acute*: Aspiration of large quantity of mineral oil
   - *Chronic*: Recurrent episodes of aspiration/inhalation
2. *Endogenous:* Bronchial obstruction/chronic pulmonary infection causing accumulation of fat within intra-alveolar macrophages.

Clinical features

- *Acute*: It cough, dyspnea, low-grade fever
- *Chronic*: It can be asymptomatic/chronic cough and dyspnea.

HRCT

- Acute exogenous
  - Radiological manifestations within 30 minutes to 24 hours
  - GGO/consolidation



Figs 26.6A to D: Plain radiograph (A) and axial HRCT image (B) in case of PAP showing diffuse reticular opacities on CXR and typical "crazy-paving" appearance of GGO with intervening interlobular septal thickening. Axial CT image lung window (C) and mediastinal window (D) in a patient with history of paraffin inhalation, showing consolidations in bilateral lower and middle lobes with areas of low attenuation suggesting fat (arrow); a case of exogenous lipoid pneumonia

- Bilateral segmental/lobar
- Middle and lower lobes
- *Occasionally:* Ill-defined nodules, pneumatoceles, pleural effusions

- Chronic exogenous:
  - GGO/consolidation **(Figs 26.6C and D)**
  - Segmental
  - Peribronchovascular distribution
  - Lower lobes
  - Architectural distortion/interlobular septal thickening/fibrosis
  - *Mass*: Irregular or spiculated with *presence of fat* in the mass
- Endogenous "cholesterol pneumonia"
  - Obstructive pneumonitis
  - Consolidation distal to a central obstructing lesion
  - Fat attenuation usually not seen on CT
  - Diagnosis is histopathologic.

INTERSTITIAL LUNG DISEASE ASSOCIATED WITH INFLAMMATORY BOWEL DISEASE

Pulmonary involvement in cases of inflammatory bowel disease is common but frequently unrecognized.

- The varying manifestations include[5]
  - Tracheobronchial involvement
  - Parenchymal involvement
  - Thromboembolic disease
  - Pleural disease
  - Enteric-pulmonary fistula

Tracheobronchial Involvement

- *Upper airway involvement*: Glottic/subglottic/tracheal stenosis
- Bronchial inflammation and suppuration
- *Clinical features:* Dysphonia, cough and dyspnea
- *HRCT:* Narrowing of trachea
- Circumferential tracheal wall thickening
- Bronchiectasis
- *Small airway disease*: Centrilobular ground-glass nodules, bronchiolar wall thickening and mosaic attenuation.

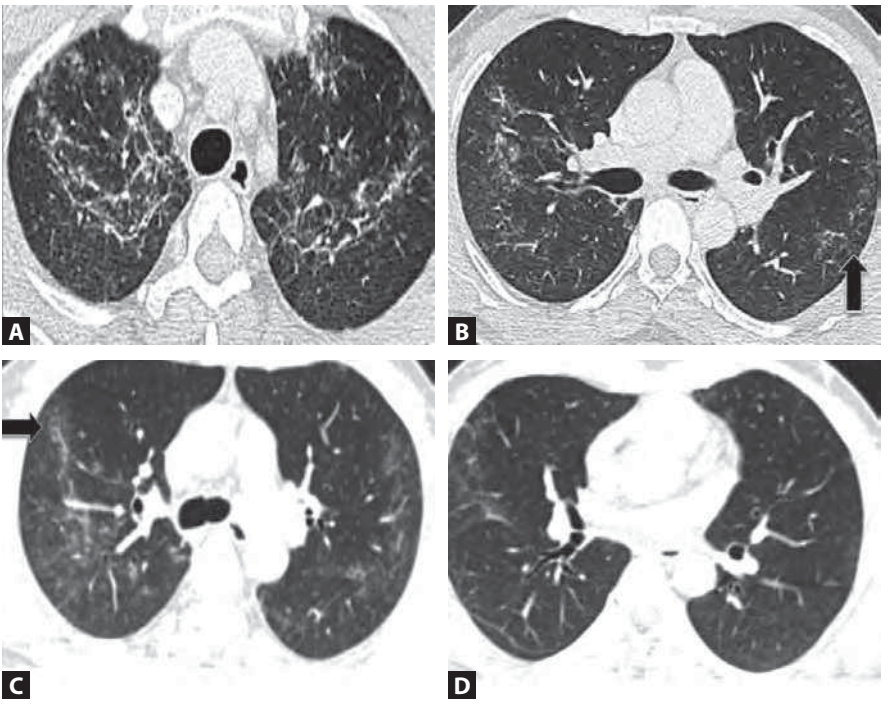
Parenchymal Involvement

- Bronchiolitis obliterans organizing pneumonia (BOOP)
  - *Clinical features:* Fever, cough and dyspnea
  - *HRCT:* Unilateral or bilateral patchy consolidation
  - Ill-defined centrilobular nodules
  - Large irregular nodules
- Eosinophilic pneumonia pattern **(Figs 26.7A to D)**
- Nonspecific interstitial pneumonitis pattern
- *Interstitial pneumonitis:* Drug induced.

PULMONARY VENO-OCCLUSIVE DISEASE AND PULMONARY CAPILLARY HEMANGIOMATOSIS

- Pulmonary veno-occlusive disease (PVOD) and pulmonary capillary hemangiomatosis (PCH) are characterized by severe pulmonary arterial hypertension with normal pulmonary artery occlusion pressure and signs of pulmonary edema.[6]
- Pulmonary hypertension results due to fibrous occlusion of pulmonary venules (PVOD) and proliferation of alveolar capillaries (PCH). Definitive diagnosis is made on surgical lung biopsy.

Clinical features: Dyspnea on exertion, chronic cough, cyanosis, syncope and chest pain.

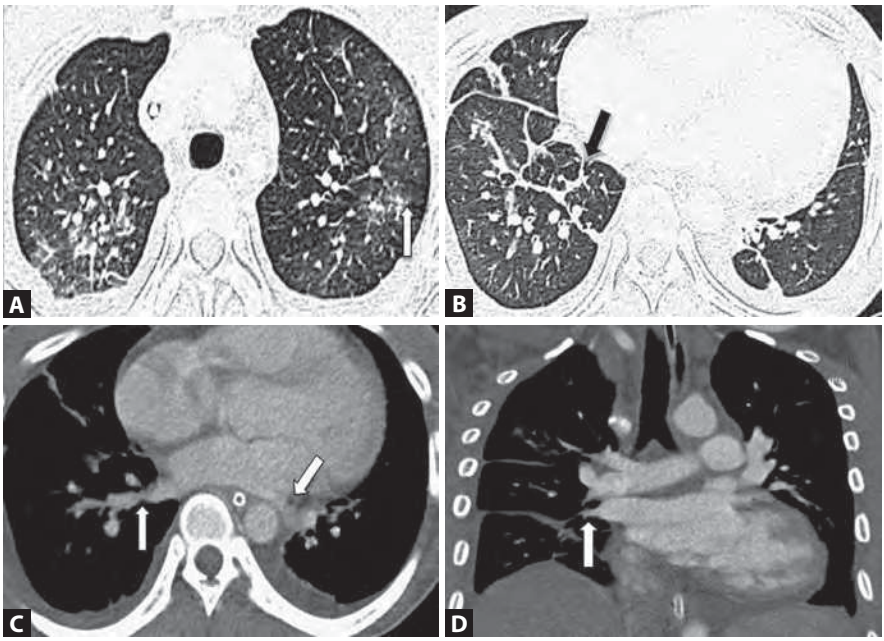


Figs 26.7A to D: ILD associated with inflammatory bowel disease. Axial HRCT images showing reticular opacities, interlobular septal thickening and GGOs (arrows) predominantly in subpleural distribution in both lungs in upper lobes showing a pattern of eosinophilic pneumonia in a case of ulcerative colitis

Chest radiograph PVOD
- Enlarged pulmonary arteries
- Normal cardiothoracic ratio
- Kerley B lines
- Pleural effusions.

Chest radiograph PCH
- Enlarged pulmonary arteries
- Bilateral basal reticulonodular opacities.

HRCT PVOD
- Dilated main pulmonary artery
- Smooth septal thickening **(Figs 26.8A to D)**
- Patchy ground-glass opacities
- *Dependent lung:* Multiple small nodules, consolidation
- Pleural effusions
- LA, LV normal size
- RA, RV may be enlarged.



Figs 26.8A to D: Axial HRCT images (A and B) show patchy GGOs (white arrow) and smooth septal thickening (black arrow) in both lungs. CT images in mediastinal window (axial C and coronal D) shows irregular narrowing in the bilateral inferior pulmonary veins (arrows) and bilateral pleural effusion. The patient had septicemia and had not developed pulmonary arterial hypertension at the time of CT

HRCT PCH

- Dilated main pulmonary artery
- Bilateral ill-defined ground-glass centrilobular nodules
- LA, LV normal size
- RA, RV may be enlarged

Clinical relevance

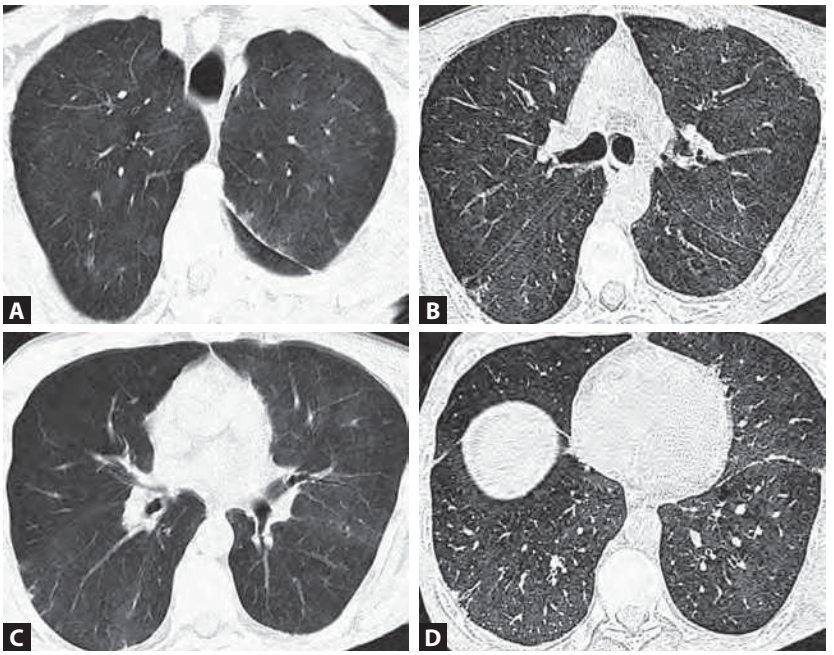
- Administration of vasodilators on a presumptive clinical diagnosis of primary pulmonary hypertension can cause pulmonary edema in case of PVOD/PCH.
- The presence of *smooth septal thickening with patchy ground-glass opacities* on HRCT, in appropriate clinical setting should alert the radiologist to suggest PVOD/PCH as the cause of pulmonary hypertension.

DRUG-INDUCED LUNG DISEASE

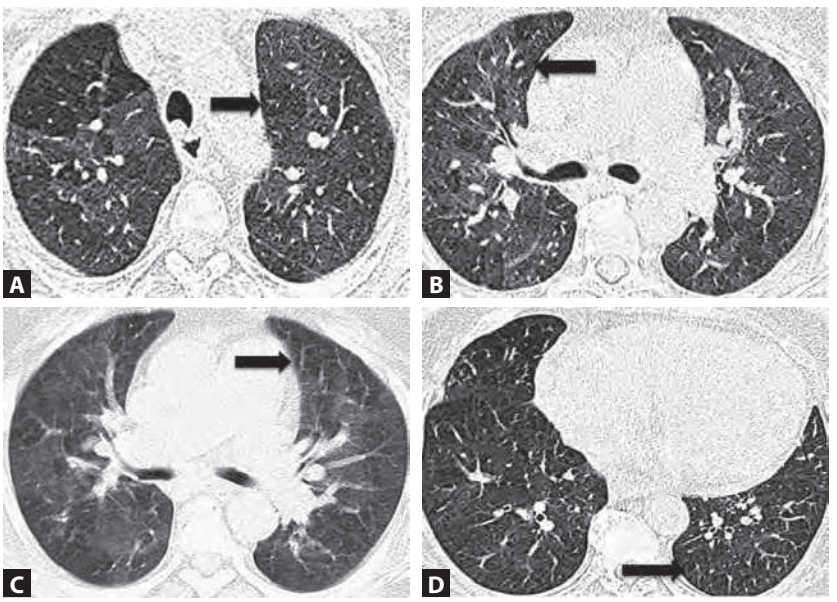
Drug-induced lung disease is on the increasing trend and early diagnosis is crucial because of the associated morbidity. The presentation is often variable and can be acute/subacute/chronic. The radiological and histological profile is also nonspecific and clinical correlation along with exclusion of other causes of pulmonary damage is essential.[7] The common HRCT manifestations along with the common drugs associated are detailed in **Table 26.4**.

Table 26.4: HRCT appearances of drug-induced lung disease

Manifestation	*Drugs associated*	*HRCT*
Diffuse alveolar damage **(Figs 26.9A to D)**	Busulfan, cyclophosphamide, carmustine (BCNU), bleomycin, paclitaxel, docetaxel, amiodarone, methotrexate, aspirin, and narcotics	Bilateral consolidation/ GGO Dependent lung regions
Acute or chronic alveolar hemorrhage	Anticoagulants, amphotericin B, amiodarone, methotrexate, nitrofurantoin, phenytoin, anticoagulants, carbamazepine, mitomycin, penicillamine, propylthiouracil, cyclophosphamide.	Bilateral GGO Intralobular linear opacities
Nonspecific interstitial pneumonia (NSIP)	Amiodarone, methotrexate, nitrofurantoin, bleomycin, hydrochlorothiazide, and carmustine (BCNU)	Patchy GGO Reticular opacities Traction bronchiectasis Bronchiolectasis Lower lung zones Subpleural/ Peribronchovascular (Nitrofurantoin)
Hypersensitivity pneumonitis	Methotrexate, cyclophosphamide, mesalamine, fluoxetine, amitriptyline, and paclitaxel	Patchy or diffuse GGO Small centrilobular nodules Lobular areas of decreased attenuation (air trapping)
Organizing pneumonia (bronchiolitis obliterans with organizing pneumonia pattern, BOOP	Amiodarone, acebutolol, minocycline, nitrofurantoin, bleomycin, gold salts, cyclophosphamide, methotrexate, penicillamine, phenytoin, carbamazepine, mesalamine, sulfasalazine, hydralazine, and interferon	Asymmetric bilateral distribution **(Figs 29.10A to D)** Consolidation: subpleural/ peribronchial GGO BOOP reaction secondary to amiodarone: Areas of consolidation with high attenuation
Eosinophilic pneumonia	Amiodarone, bleomycin, nitrofurantoin, phenytoin, beta-blockers, nonsteroidal anti-inflammatory drugs, antidepressants, hydrochlorothiazide, minocycline, sulfonamides, sulfasalazine, and mesalamine.	Consolidation/ GGO Peripheral distribution Middle and upper lung zones Focal GGO surrounded by a crescent of consolidation ('reversed halo sign')
Usual interstitial pneumonitis (UIP)	Nitrofurantoin, cyclophosphamide, chlorambucil, and pindolol	Reticular opacities Subpleural/Basal Honeycombing Traction bronchiectasis



Figs 26.9A to D: Drug-induced lung disease (Diffuse alveolar damage pattern). Axial HRCT images in a patient of psoriasis on long-term methotrexate therapy, shows diffuse GGO in both lungs suggestive of DAD pattern of the lung injury



Figs 26.10A to D: Drug-induced lung disease (bronchiolitis obliterans pattern). Axial HRCT images in a patient of carcinoma ovary received chemotherapeutic drug geftinat, shows areas of air trapping (arrows) and patchy GGO suggestive of bronchiolitis obliterans pattern of the lung injury

AGE-RELATED CHANGES

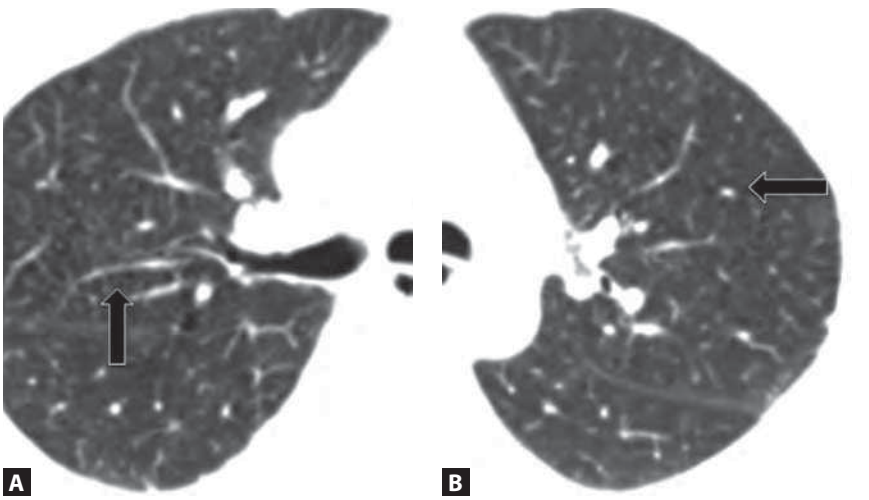
With increasing age there are physiologic and morphologic changes in the lung (decreased elastic recoil of alveoli). It is important to identify the normal appearance of 'senile lung' in the HRCT images of elderly population[8] and avoid misdiagnosis.

Clinical features
- Asymptomatic elderly population
- Also known as "Senile lung/senile hyperinflation": Distal air spaces enlarge.

Chest radiograph
- Hyperinflation
- Increased peripheral vascular markings.

HRCT
- Centrilobular emphysematous changes **(Figs 26.11A and B)**
- Subpleural reticular pattern
- Bronchial wall thickening
- Small basal atelectasis
- Progressive calcifications of the airways and the rib cage
- Loss of diaphragmatic muscle mass and other thoracic muscles.

Clinical relevance
- Senile emphysema vs chronic obstructive airway disease (COAD):
  - Cough and sputum production seen in COAD
- Age-related fibrosis vs ILD (NSIP/UIP):
  - Correlate with clinical history, previous imaging and extent
  - GGO, extensive changes and honeycombing/traction bronchiectasis not associated with age-related ILD.



Figs 26.11A and B: Age-related changes. Axial magnified CT images in lung window (A and B) shows centrilobular emphysematous changes (arrows) in a 70-year-old nonsmoker asymptomatic female as a part of senile changes of the lungs

IgG4-RELATED LUNG DISEASE[9]

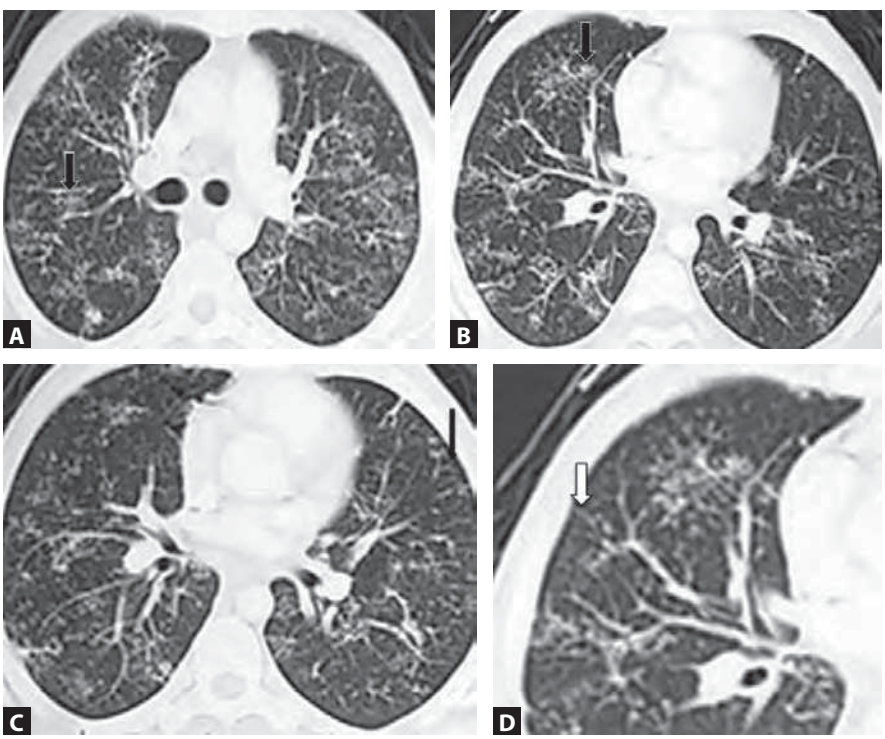
- Rare
- May occur with or without systemic involvement.
- A part of the spectrum of IgG-4-related sclerosing disease.
- Characterized by IgG4-related sclerosing inflammation along the entire intrapulmonary connective tissue.
- Diagnosis is based on histology and serum IgG4 levels.

Clinical features

- 50% asymptomatic with incidental findings on chest radiology
- Dyspnea, cough, pleuritic chest pain
- Absence of fever or other constitutional symptoms.

Inoue D, et al. (2009) described 4 patterns on HRCT chest:

1. Solid nodular type
2. Round-shaped ground-glass opacities
3. *Alveolar interstitial type:* Showing honeycombing, bronchiectasis and diffuse ground glass opacity
4. *Bronchovascular type:* Showing thickening of bronchovascular bundles and interlobular septae **(Figs 26.12A to D)**.



Figs 26.12A to D: IgG4-related lung disease. Axial CT images in lung window shows in a 40-year-old woman with IgG4-related lung disease reveal multiple small nodules in both lungs in the peribronchovascular (black arrows) and centrilobular distribution (thin arrows). Mild interlobular septal thickening is also identified (white arrow)

Other findings: Mediastinal lymphadenopathy, pleural involvement (pleural thickening).

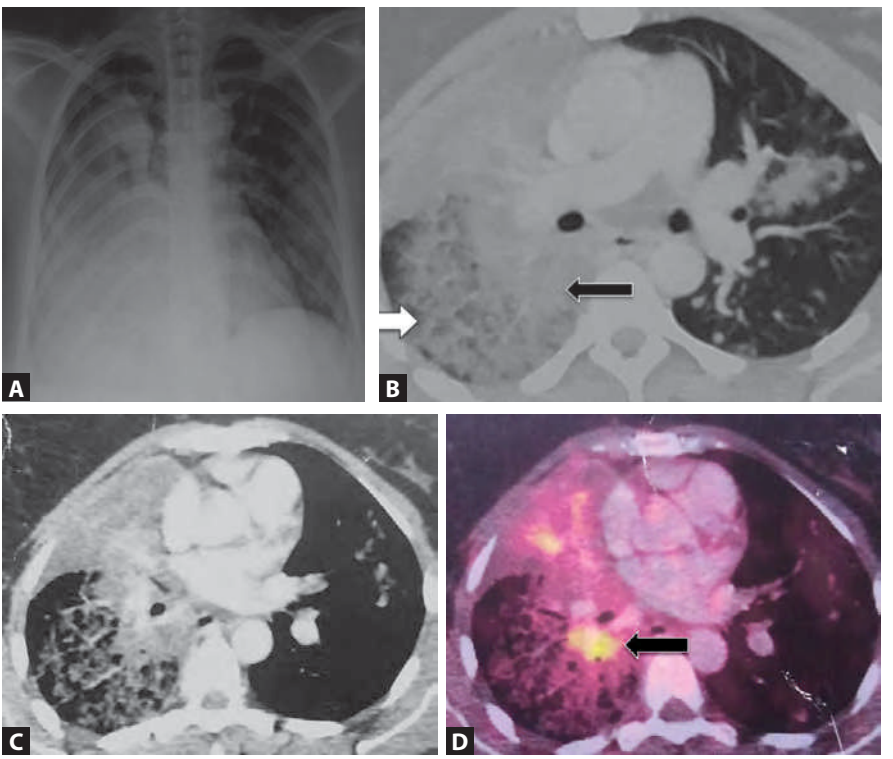
Clinical relevance
- Good response to steroid therapy
- Uncertain relationship with malignancy.

MALIGNANT CAUSES OF DIFFUSE LUNG DISEASE

Lymphangitis carcinomatosa (*See* Chapter 23).

Invasive Adenocarcinoma

New classification entities of invasive adenocarcinoma which now replaces obsolete term bronchoalveolar carcinoma (BAC) are as follows:[10]
- *Lepidic predominant adenocarcinoma:* Invasive adenocarcinoma with a predominant lepidic pattern with >5 mm invasion; formerly known as non-mucinous bronchoalveolar carcinoma **(Figs 26.13A to D)**



Figs 26.13A to D: Invasive adenocarcinoma. (A) Plain radiograph shows consolidation in mid and lower zones and airspace nodules in left mid zone; (B) Axial CT image in lung window shows dense consolidation (black arrow) in peribronchovascular region in right middle and lower lobe with adjacent ground glass opacity suggestive of lepidic growth pattern (white arrow). Left lung shows multiple metastatic nodules; (C and D) FDG PET-CT images show areas of uptake within the consolidation (arrow)

- *Invasive mucinous adenocarcinoma:* It is a variant invasive adenocarcinoma earlier known as mucinous bronchoalveolar carcinoma.

Clinical features
- Often insidious
- May be asymptomatic
- May present with bronchorrhea
- Nonresolving consolidation for weeks despite appropriate antimicrobial therapy

Chest radiograph
- Chronic unilateral airspace opacification, may show air bronchograms
- Cluster of diffuse nodules
- Extensive and widespread fluffy airspace opacities.

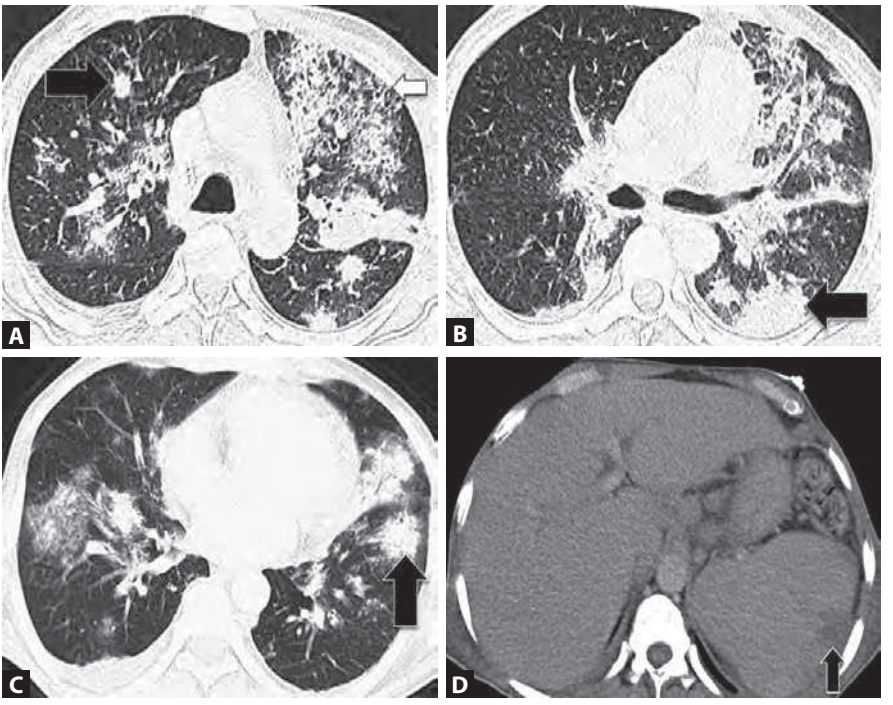
HRCT
- Solid or mostly solid nodules and masses
- Often demonstrate air bronchograms
- Lobar or multilobar distribution
- Two forms:
  - *Multinodular form:* Multifocal nodules, extensive airspace opacities with numerous air-bronchograms.
  - *Consolidative form:* Lobar consolidation, multifocal consolidation and GGO.
- Pleural effusions or significant adenopathy are uncommon.

Pulmonary Lymphoma

- Secondary lymphomatous parenchymal neoplasms (both Hodgkin (HL) and non-Hodgkin lymphomas (NHL)) are far more prevalent than primary lymphoproliferative neoplasms.
- Pulmonary parenchymal disease occurs in 24% of NHL (with or without intrathoracic lymph node enlargement)[11]
- Pulmonary parenchymal disease occurs in 38% of HL (almost always associated with intrathoracic lymph node enlargement).[11]

Chest radiograph
- Single or multiple, airspace nodules or masses
- Reticulonodular opacities.

*HRCT **(Figs 26.14A to D)***
- Similar findings in both HL and NHL
- Varied and nonspecific features
- Nodules/masses with or without air bronchograms
- Consolidations and GGOs
- Lymphangitic spread
- Endobronchial lesions
- Pleural effusion
- Lymphadenopathy (HL > NHL).



Figs 26.14A to D: Pulmonary involvement in lymphoma. (A to C) Axial HRCT images (lung window) show multiple nodules and masses (arrows) in both lungs in a case of non-Hogdkin's lymphoma. Also note interlobular septal thickening in left upper lobe (white arrow); (D) Note hepatosplenomegaly in the same patient with splenic infarct (arrow)

- Alveolar or interstitial infiltrates
- Peribronchial or perivascular thickening with or without atelectasis.

Clinical relevance

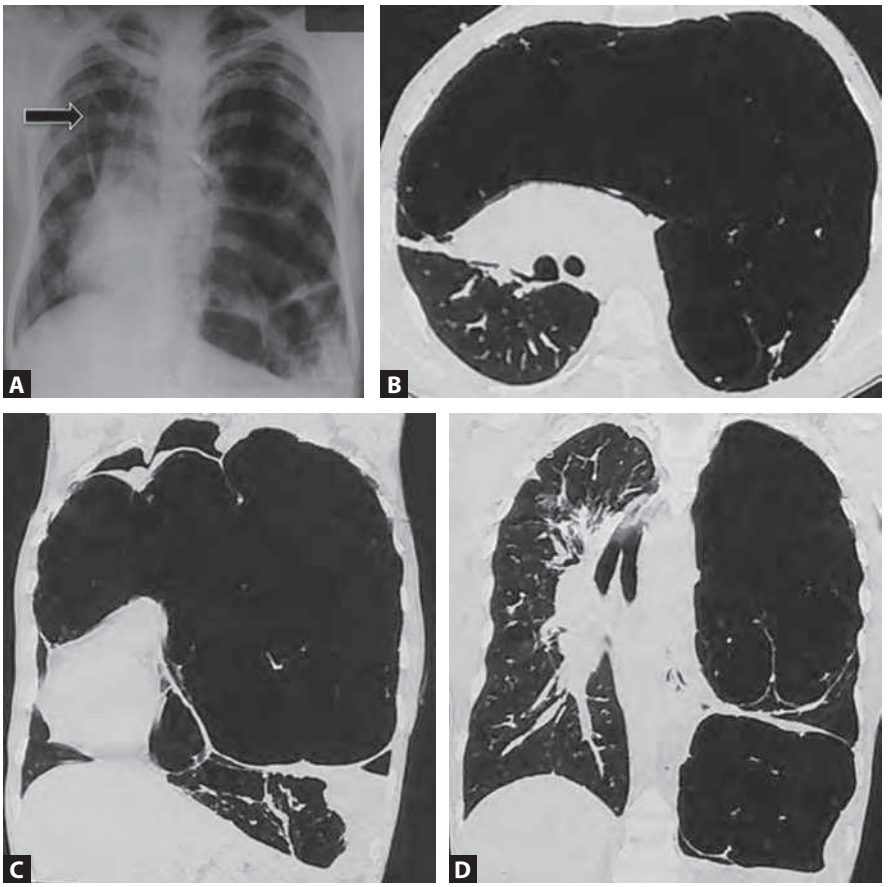
- Noncontiguous involvement of lung parenchyma in a known case of lymphoma corresponds to stage IV.

VANISHING LUNG SYNDROME

- Characterized by very large bullae which are air-filled, thin-walled (<1 mm) spaces in the lung resulting from destruction of alveolar tissue
- In this syndrome, the bullae takes up more than one-third of the occupied lung.

Clinical features

- Most common in young men, mostly in smokers[12]
- Progressive condition
- Can be asymptomatic
- Experience hypoxia, severe dyspnea and chest pain.



Figs 26.15A to D: Vanishing lung syndrome. (A) Chest radiograph shows multiple giant bullae occupying the entire left hemithorax with herniation towards right side (black arrow) and causing mediastinal shift towards right. Axial (B) and coronal CT images (C and D) show the large bullae in the left hemithorax replacing entire left lung

Chest radiograph ***(Fig. 26.15A)***

- Bullae occupy more than one-third of the affected hemithorax (vary in size 1-20 cm)
- Predominantly upper lobes involvement
- Usually asymmetric lung involvement
- Compression on adjacent structures (residual lung collapse, inversion of diaphragm or contralateral shift of the mediastinum and displacement of junction lines).

HRCT ***(Fig. 26.15B to D)***

- Paraseptal, centrilobular emphysema and subpleural bullae
- Giant bullae ranging from 1–20 cm in diameter, mostly in the range of 2–8 cm in diameter

- Asymmetrical involvement
- Extent of involvement better appreciated.

Clinical relevance

- Mistaking a large bulla for a pneumothorax and inserting a chest tube into it will lead to a pneumothorax
- Find the visceral pleural white line to identify a pneumothorax and, when in doubt, do CT
- Treatment is volume reduction surgery.

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Section 9

Approach to High-resolution Computed Tomography in Case of Interstitial Lung Disease

- Approach to Diagnosis: Distribution of Disease
- Approach to Diagnosis: Ground-glass Opacity
- Approach to Diagnosis: Linear Opacities
- Approach to Diagnosis: Nodular Pattern

27

Approach to Diagnosis: Distribution of Disease

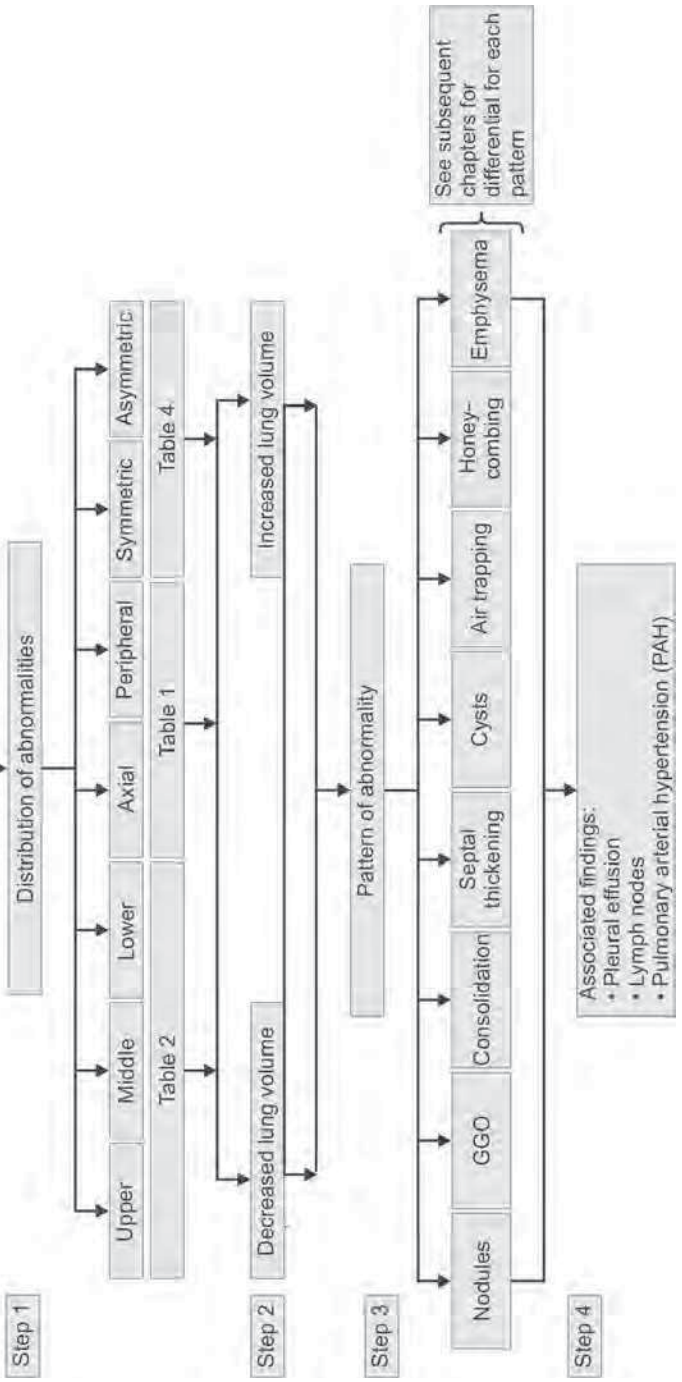
Priyanka Naranje

- Amongst the many interstitial lung diseases described in previous chapters, it is important to have a structured way of reporting them so as to transfer the clinically relevant information to the treating physician in terms of characteristic diagnostic features and subtle changes in volume and character of abnormalities on longitudinal follow-up of these patients. Such a structured reporting initiative can help in improving radiology reporting practices by provision of a clear and consistent reporting template.
- We therefore propose the following reporting format for interstitial lung disease (ILD) cases **(Appendix 27.1)** in order to provide clear diagnostic possibilities and quantify the involvement of lung parenchyma semiquantitatively based on visual interpretation.

APPROACH

- The approach to an ILD should begin with some basic background history of clinical presentation, fever, smoking; family history, drug history, occupational history, pulmonary function tests (PFTs) and relevant investigations should be noted followed by imaging characteristics on HRCT. The diagnostic approach to ILD is described in **Flow chart 27.1**.
- Apart from considering specific morphology of pulmonary abnormalities of various interstitial lung diseases, it is important to take into account the overall distribution of the lung disease on plain radiographs and HRCT. Predominant involvement of some specific lung regions is seen for many ILDs which are related to their pathophysiology.[1]
- However, it is imperative to keep in mind that these distribution patterns are quite variable and in an index case an atypical distribution should not preclude a specific diagnosis.

Flow chart 27.1: The diagnostic approach to interstitial lung disease



Such a distribution pattern of lung abnormalities can be grouped in following ways:

- Axial/central lung versus peripheral lung
- Upper lung versus lower lung
- Anterior lung versus posterior lung
- Asymmetric/unilateral versus symmetric/bilateral.

AXIAL/CENTRAL LUNG VERSUS PERIPHERAL LUNG (TABLE 27.1)

- Axial/central lung
  - Some diseases have a central, perihilar, bronchocentric, or bronchovascular distribution[2]
  - Includes sarcoidosis (most common distribution pattern[3]), silicosis, lymphangitis carcinomatosa, talcosis and airway centered interstitial fibrosis.

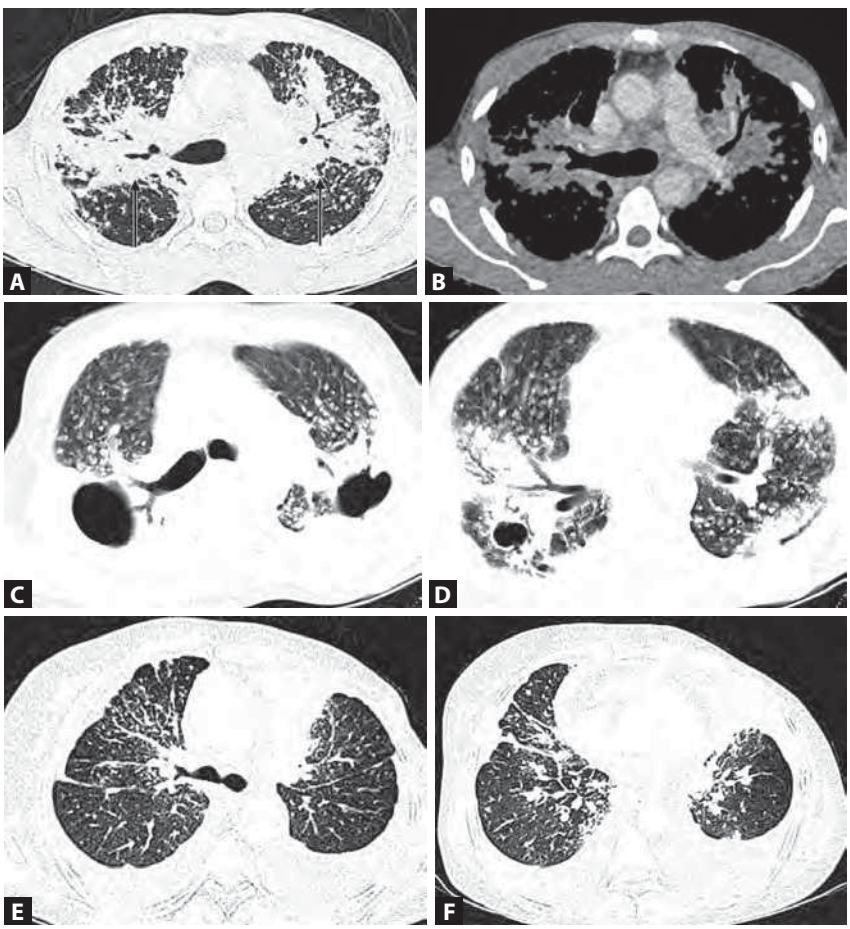
Table 27.1: Differential diagnosis of diseases according to axial/central versus peripheral lung distribution

Diseases with axial/central lung predominance		**Diseases with peripheral lung predominance**	
Central, perihilar, bronchocentric, or bronchovascular in distribution ***(Figs 27.1A to F)***		*Peripheral or subpleural parenchyma or lung cortex* ***(Figs 27.2A to F)***	
Disease entity	*Key features on HRCT*	*Disease entity*	*Key features on HRCT*
Sarcoidosis	Peribronchovascular nodules; conglomerate fibrosis with traction bronchiectasis	Usual interstitial pneumonitis	Subpleural fibrosis; honeycombing; sometimes ground-glass opacity
Silicosis	Conglomerate masses of fibrosis	Nonspecific interstitial pneumonia	Subpleural ground-glass opacity; reticulation; subpleural sparing in 50%
Lymphangitis carcinomatosa	Peribronchovascular interstitial thickening or nodules	Chronic eosinophilic pneumonia	Subpleural consolidation or ground-glass opacity
Talcosis	Conglomerate masses of fibrosis with high attenuation foci	Bronchiolitis obliterans organizing pneumonia/cryptogenic organizing pneumonia	Subpleural consolidation or ground-glass opacity
Airway-centered interstitial fibrosis (ACIF)	Peribronchovascular fibrosis and interstitial thickening Centered around the proximal bronchi No granuloma on histopathology	Acute interstitial pneumonia	Peripheral consolidation; ground-glass opacity

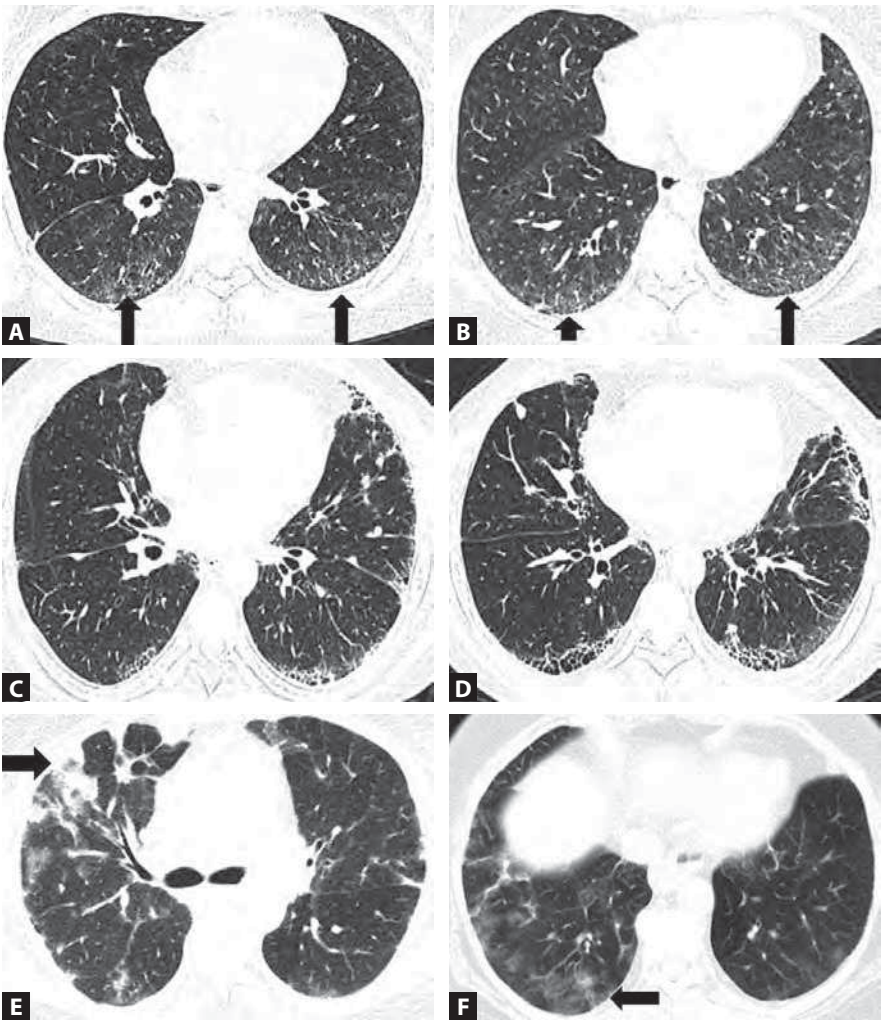
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Disease entity	*Key features on HRCT*	*Disease entity*	*Key features on HRCT*
		Desquamative interstitial pneumonia	Peripheral ground-glass opacity
		Hypersensitivity pneumonitis	Peripheral ground-glass opacity in some Generally no zonal predominance of fibrosis
		Amyloidosis	Subpleural larger nodules with central hazy calcifications
		Asbestosis	Subpleural fibrosis

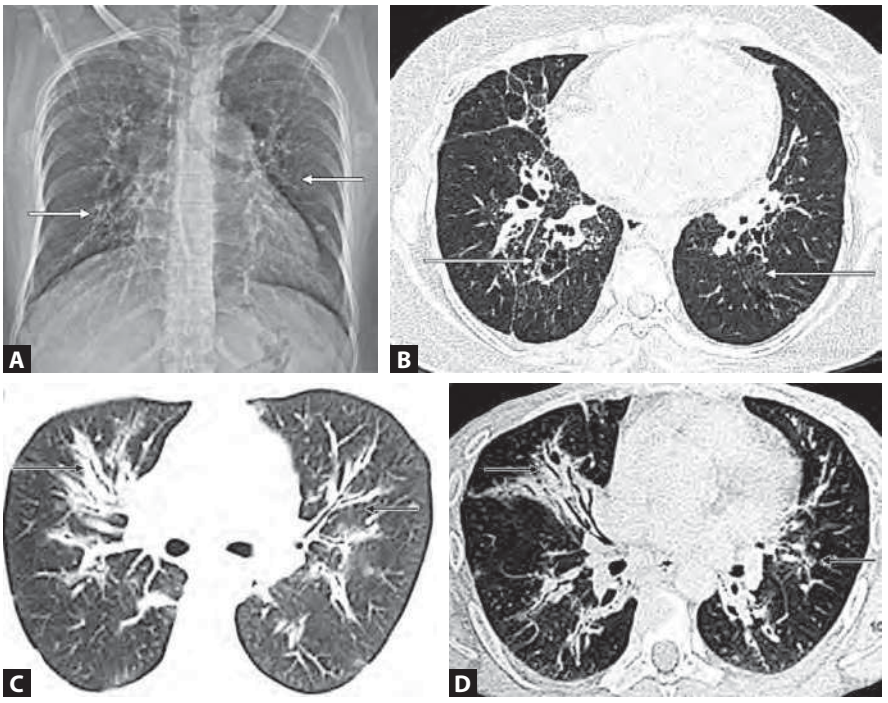


Figs 27.1A to F: Diseases showing predominant axial/central distribution pattern. (A and B) Sarcoidosis (arrows) (C and D) Silicosis. (E and F) Lymphangitis carcinomatosa in a postoperative case of left carcinoma breast



Figs 27.2A to F: Diseases showing predominant peripheral distribution pattern. (A and B) Nonspecific interstitial pneumonia (NSIP) (arrows); (C and D) Usual interstitial pneumonia (UIP); (E and F) Chronic eosinophilic pneumonia

- Peripheral lung
  - Certain diseases favor the peripheral or subpleural parenchyma, or lung cortex
  - Includes usual interstitial pneumonia (UIP), nonspecific interstitial pneumonia (NSIP), chronic eosinophilic pneumonia, asbestosis, amyloidosis and organizing pneumonia.
  - Occasionally seen in patients with hypersensitivity pneumonitis, sarcoidosis and acute interstitial pneumonia.
- The above entities are detailed in the individual chapters except ACIF which is not covered elsewhere and is detailed below.



Figs 27.3A to D: Airway-centered interstitial fibrosis (ACIF). (A) Topogram of a 40-year-old female patient shows fibrotic changes in bilateral peribronchovascular regions (arrows); (B) HRCT axial section showing fibrosis and traction bronchiectasis along bilateral peribronchovascular regions (arrows); (C and D) Another case of ACIF showing fibrotic changes along bilateral segmental bronchi/airways (arrows)

- *Airway-centered interstitial fibrosis (ACIF)*[4]
  - Recently described entity
  - Pathology—distinctive pattern of interstitial fibrosis centered and extending around the bronchioles as well as larger airways, absence of any granuloma
  - Etiology—not determined, idiopathic
  - Whether a variant of existing idiopathic interstitial pneumonia (IIP) or exists only in association with other conditions, such as hypersensitivity pneumonitis or collagen vascular disease is uncertain and hence not included in recent American Thoracic Society/European Respiratory Society (ATS/ERS) update on IIP classification.
  - Seen more common in women and mean age of onset being 46 years
  - HRCT findings—peribronchovascular fibrosis and interstitial thickening **(Figs 27.3A to D)**
  - D/D—hypersensitivity pneumonitis (history of exposure, granulomas on histopathology), gastroesophageal reflux disease (GERD) or connective tissue disease (CTD) related ILD are other differentials.

UPPER LUNG VERSUS MIDDLE VERSUS LOWER LUNG (TABLE 27.2)

Best seen on plain radiograph or coronal lung window images. Alternatively to be assessed by comparing axial HRCT scans at the levels above carina, at and those below the level of carina.

- *Upper lung:*
  - Abnormalities more severe and relatively predominant in the upper lobes
  - Diseases—Sarcoidosis, coal worker pneumoconiosis (CWP), silicosis, respiratory bronchiolitis, Centrilobular emphysema, chronic hypersensitivity pneumonitis, pulmonary Langerhan's cell histiocytosis.
- *Middle zones:*
  - Abnormalities more severe and relatively predominant in the mid zones
  - Diseases—Sarcoidosis, pulmonary Langerhan's cell histiocytosis, acute/subacute hypersensitivity pneumonitis.[5]
- *Lower lung/basal:*
  - Abnormalities more severe and relatively predominant in the lower lobes and basal segments
  - Diseases—Hematogenous or lymphangitic metastasis, UIP, CTD-related ILD, NSIP, asbestosis, lipoid pneumonia, AIP, organizing pneumonia. Acute/subacute hypersensitivity pneumonitis is more often diffuse or preponderant in the midzones and may sometimes show lower lung zones (30%) preponderance.

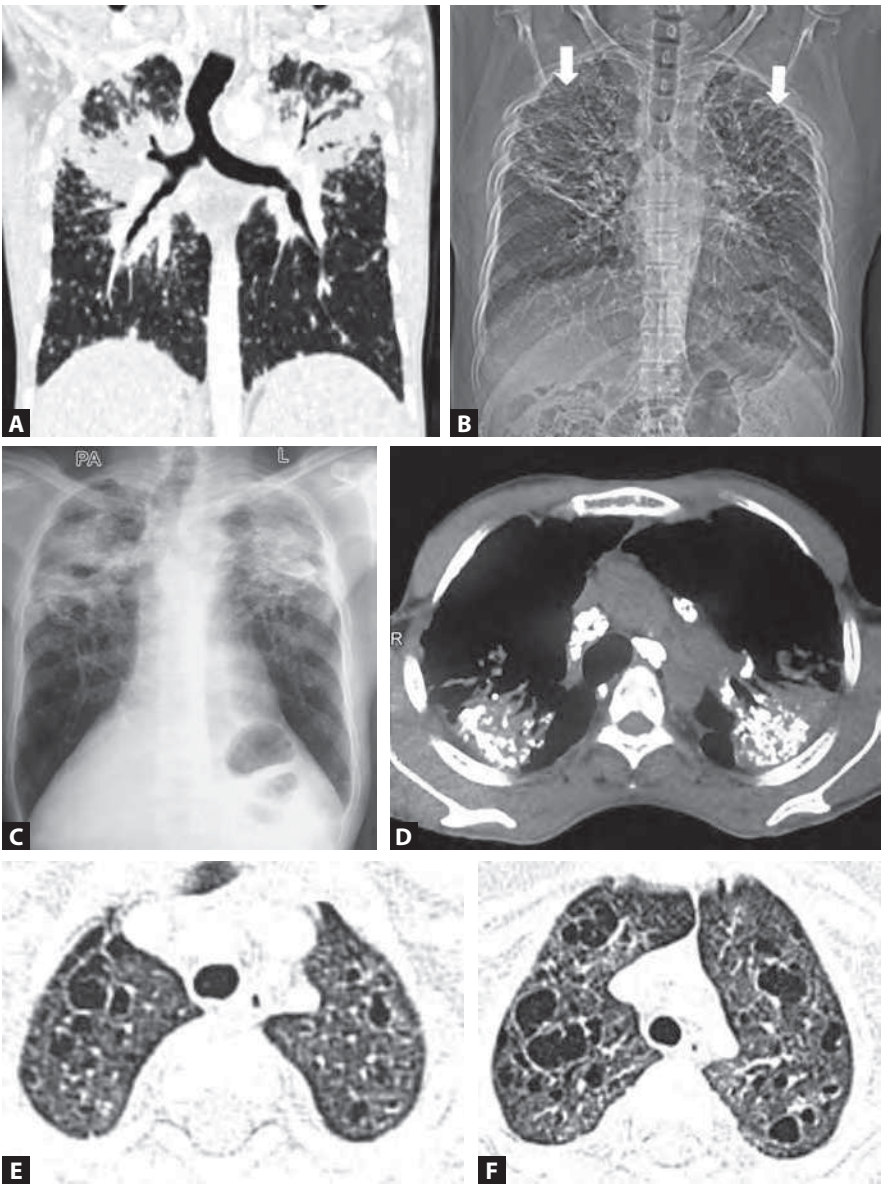
Table 27.2: Differential diagnosis of diseases according to upper lung, middle and lower/basal predominance

Diseases with upper lung predominance	
Abnormalities more severe and relatively predominant in the upper lobes and mid zones ***(Figs 27.4A to F)***	
Disease entity	*Key features on HRCT*
Sarcoidosis	Nodules Fibrosis Conglomerate masses
Langerhans histiocytosis	Cysts Nodules
CWP and silicosis	Nodules Conglomerate masses
Centrilobular emphysema	Focal lucencies
Respiratory bronchiolitis	Ground glass opacity Centrilobular nodules
Chronic hypersensitivity pneumonitis	Fibrosis Traction bronchiectasis in the upper lobes and mid zones

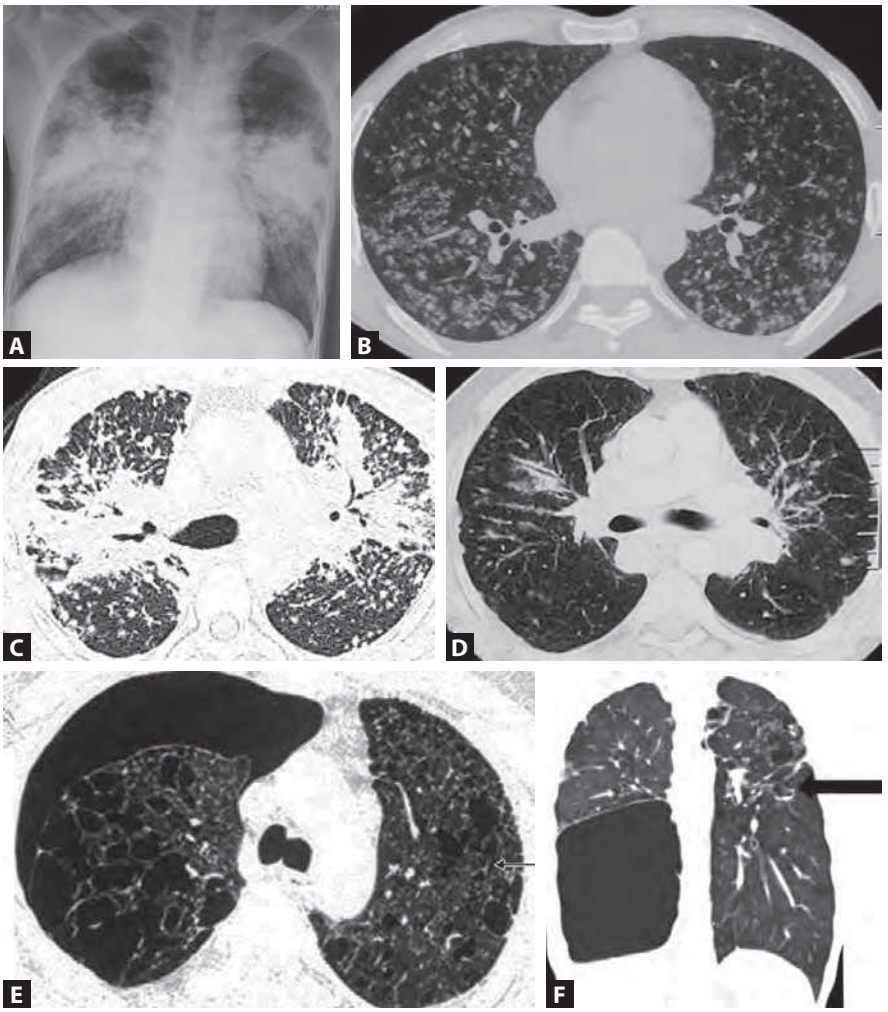
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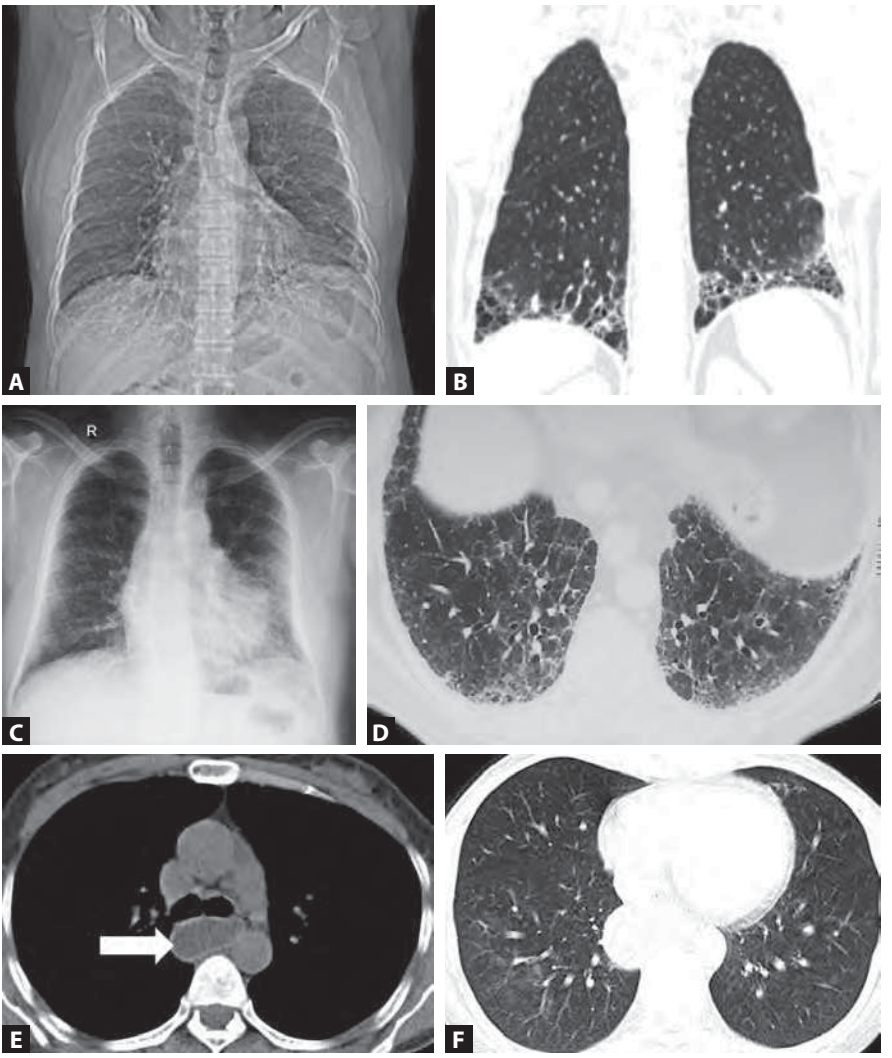
Disease entity	*Key features on HRCT*
Diseases with middle zone predominance (Figs 27.5A to F)	
Acute/Subacute Hypersensitivity pneumonitis	Diffuse or preponderant in the mid or lower lung zones GGO Centrilobular nodules
Pulmonary Langerhans' cell histiocytosis (PLCH)	Thin walled bizarre cysts Nodules Sparing the costophrenic angle
Sarcoidosis	Conglomerate parahilar masses Nodules Fibrosis
Diseases with lower/basal predominance	
Abnormalities more severe and relatively predominant in the lower lobes and basal segments ***(Figs 27.6A to F)***	
Usual interstitial pneumonitis Collagen diseases Asbestosis	Subpleural fibrosis Honeycombing Sometimes ground-glass opacity
Nonspecific interstitial pneumonia	Subpleural ground-glass opacity Reticulation
Lipoid pneumonia	Consolidation Ground glass opacity
Bronchiolitis obliterans organizing pneumonia	Subpleural consolidation Ground-glass opacity
Hematogeneous or lymphangitic metastases	Septal thickening Nodules
Acute Hypersensitivity pneumonitis	Diffuse or preponderant in the mid or lower lung zones GGO Centrilobular nodules
AIP	Symmetrical distribution of GGOs and consolidation
Aspiration pneumonitis	Ill-defined centrilobular nodules in dependent regions



Figs 27.4A to F: Diseases with upper lung preponderance of abnormalities.
(A and B) Sarcoidosis (arrows); (C and D); Silicosis (E and F) Langerhan's cell histiocytosis



Figs 27.5A to F: Mid-zone preponderance of disease. (A and B) Acute/subacute HP. Plain radiograph showing conglomeration of nodules in the mid zones of lungs. CT in the same patient done one month before depicted multiple centrilobular nodules predominantly in mid and lower lobes; (C and D) Sarcoidosis; (E and F) PLCH (in 2 different patients) showing numerous cysts in mid and upper zones (arrows)



Figs 27.6A to F: Lower lung preponderance of disease. (A and B) Usual interstitial pneumonia (UIP); (C and D) Nonspecific interstitial pneumonia (NSIP) (E and F) Aspiration pneumonitis in a case of achalasia cardia (arrow)

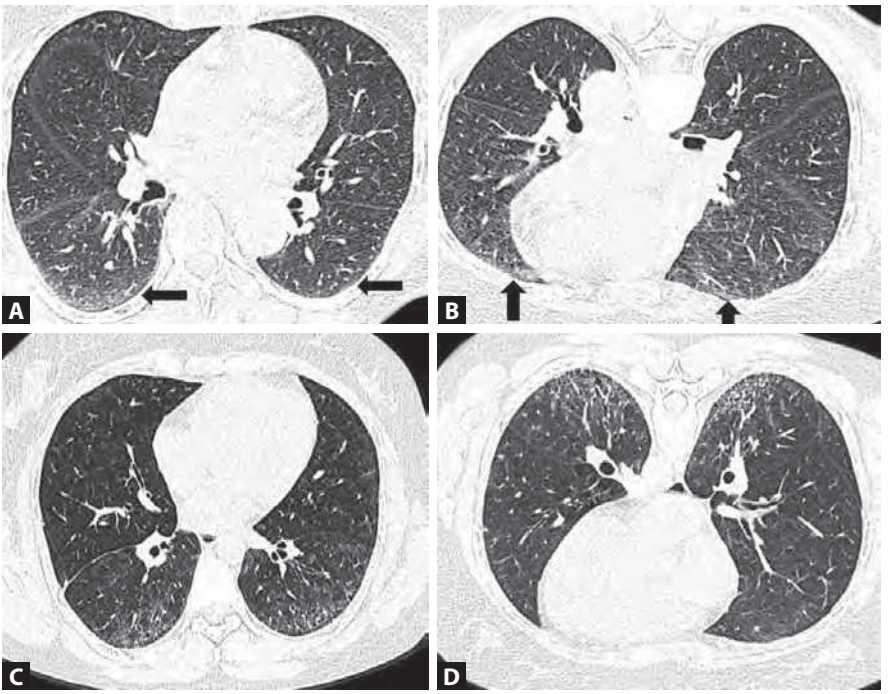
ANTERIOR LUNG VERSUS POSTERIOR LUNG (TABLE 27.3)

Value of prone scan: Confident diagnosis of early posterior lung disease as supine scan can show dependent opacities which get abolished on prone imaging whereas true septal thickening/GGO persists.

- *Diseases with posterior predominance*
  - Scleroderma, sarcoidosis, silicosis, hypersensitivity pneumonitis, UIP, NSIP, asbestosis, pulmonary edema

Table 27.3: Differential diagnosis of diseases according to anterior versus posterior lung predominance

Diseases with posterior lung predominance		**Diseases with anterior lung predominance**	
Needs supine and prone scans to confirm initial or most extensive abnormalities in posterior lung ***(Figs 27.7A to D)***		*Unusual*	
Disease entity	*Key features on HRCT*	*Disease entity*	*Key features on HRCT*
Usual interstitial pneumonia	Fibrosis		
Nonspecific interstitial pneumonia	Ground-glass opacity		
Asbestosis	Fibrosis		
Scleroderma	Fibrosis Ground-glass opacity		
Silicosis	Nodule Conglomerate masses		
Sarcoidosis	Nodules Conglomerate masses		
Pulmonary edema	Smooth septal thickening Ground-glass opacity Consolidation		
Acute respiratory distress syndrome (ARDS) AIP	Dependent and basal predominant diffuse or patchy GGO/consolidation in acute phase	Post-ARDS fibrosis	Fibrotic changes predominantly in anterior lung as subpleural fibrosis; honeycombing; traction Bronchiectasis Depends on duration of mechanical ventilation
Hypersensitivity pneumonitis	Ground-glass opacity Nodules Fibrosis		
Lipoid pneumonia	Consolidation Lucent areas of fat within Ground-glass opacity		



Figs 27.7A to D: (A) Supine HRCT scan showing GGO in posterior dependent regions of the lung; (B) Prone HRCT shows the GGOs represent dependent densities (arrows) rather than being true ILD; (C) Supine HRCT scan in another patient showing GGO in posterior dependent regions of the lung; (D) Prone HRCT of the patient shown in image C shows the GGOs represent true ILD as it did not change its position

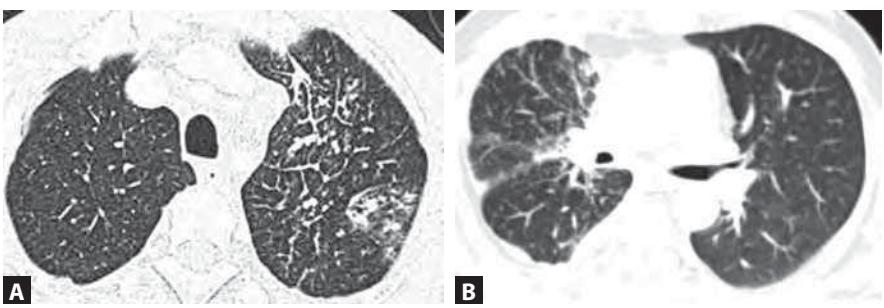
- *Diseases with anterior predominance*
  - Adult survivors of ARDS (ventral lung most prone to the effects of barotrauma).

ASYMMETRIC VERSUS SYMMETRIC (TABLE 27.4)

- *Asymmetric:*
  - Unilateral preponderance
  - Most typical of lymphangitic spread of carcinoma (50% times)[3]
  - Sarcoidosis (up to 20%)
  - Due to prior infective sequelae, some area of the lung may be fibrotic and such areas are not affected by the ILD leading to asymmetry of the findings.
- *Symmetric:*
  - Bilateral equal distribution of disease
  - Sarcoidosis, silicosis, UIP, Langerhans histiocytosis and hypersensitivity pneumonitis

Table 27.4: Differential diagnosis of diseases according to unilateral/asymmetric versus symmetric distribution

Diseases with asymmetric distribution		**Diseases with symmetric distribution**	
Typical and commonly show unilateral or asymmetric preponderance of abnormalities ***(Figs 27.8A to D)***		*More commonly bilateral symmetric However may show asymmetric distribution (less frequently)* ***(Figs 27.9A to F)***	
Disease entity	*Key features on HRCT*	*Disease entity*	*Key features on HRCT*
Lymphangitic spread of carcinoma	Peribronchovascular interstitial thickening Nodules Septal thickening	Silicosis	Nodules Conglomerate masses
Sarcoidosis	Commonly seen Peribronchovascular, subpleural and septal nodules	UIP	Honeycombing Subpleural fibrosis
Prior diseased area of the lung	Fibrotic changes Cavitation Volume loss	Langerhans histiocytosis	Cysts Nodules
		Hypersensitivity pneumonitis	GGO Centrilobular nodules
		Acute interstitial pneumonia	Variable
		Acute pulmonary edema	Smooth septal thickening
		Sarcoidosis	Commonly seen Peribronchovascular, subpleural and septal nodules

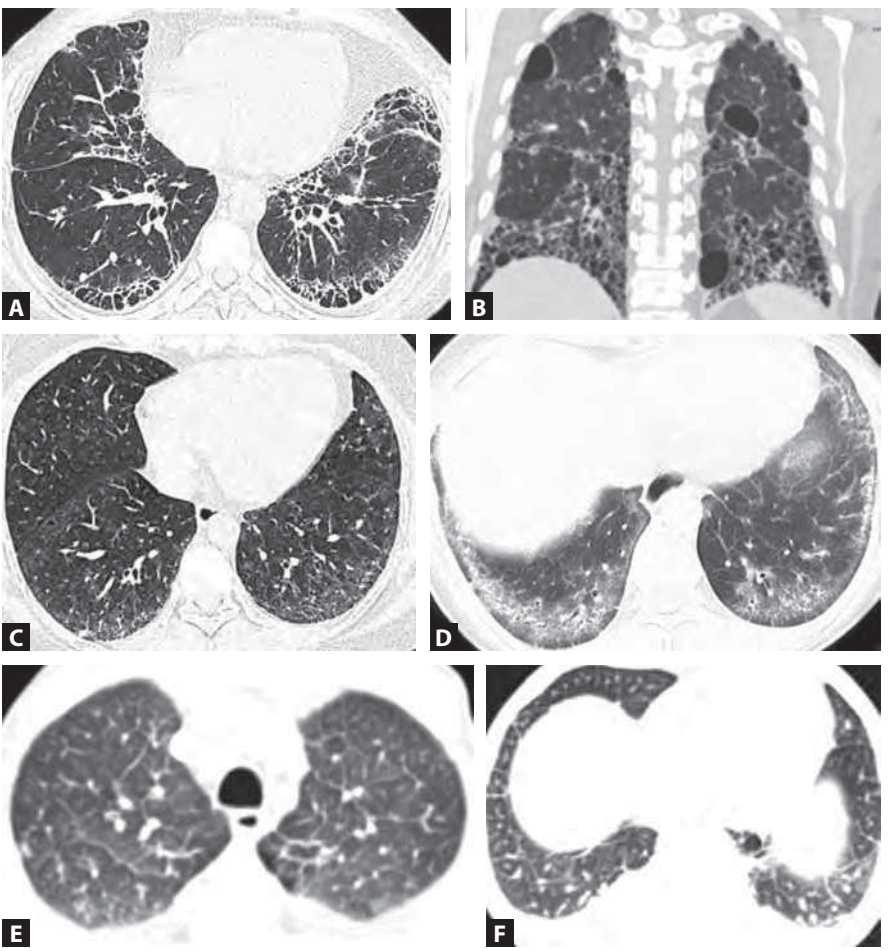


Figs 27.8A and B



Figs 27.8C and D

Figs 27.8A to D: Asymmetric lung involvement in ILDs. (A and B) Lymphangitis carcinomatosa (2 different patients); (C and D) Sarcoidosis, predominant involvement of right lung



Figs 27.9A to F: Symmetric lung involvement in ILDs. (A and B). UIP; (C and D) NSIP; (E and F) Pulmonary edema

Appendix 27.1: Proposed reporting format for interstitial lung disease and for follow-up

Findings	*Present/ Absent*	*Lobes involved*
Parenchymal abnormality:		
Ground-glass attenuation		
Consolidation		
Reticular opacities		
Septal thickening Interlobular Intralobular		
Nodules: Perilymphatic Centrilobular Random		
Microcysts		
Macrocystic honeycombing		
Architectural distortion		
Traction bronchiectasis		
Bronchiolectasis		
Bronchial wall thickening		
Air trapping/Mosaic attenuation		
Lymph nodes: Hilar Mediastinal Size Calcification Perinodal fat stranding		
Distribution at CT:		
Upper lung predominance		
Basilar predominance		
No obvious gradient		
Peripheral predominance		
Axial predominance		
Symmetric, bilateral		
Asymmetric		
Subpleural sparing		
MPA diameter		
Pleural effusion		
Pleural thickening		
Other findings		

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Conclusion: Usual interstitial pneumonia (UIP) Nonspecific interstitial pneumonia (NSIP) Sarcoidosis Respiratory bronchiolitis-associated interstitial lung disease (RB-ILD) Small airway disease Unclassified Others	Activity assessment/comparison with previous studies : Complete resolution/Stable disease/ progressive disease/advanced fibrosis Advice

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28

Approach to Diagnosis: Ground-glass Opacity

Smita Manchanda

GROUND-GLASS OPACITY

An increase in the attenuation of lung parenchyma which does not obscure the underlying vessels and bronchi.[1,2] These vessels are of normal caliber and the opacity is less dense than consolidation. The underlying cause is displacement of air and can be attributed to any of the following factors in isolation or combination:

- Partial airspace filling
- Interstitial thickening (cells/fluid/fibrosis)
- Increase in the capillary blood volume
- Partial alveolar collapse.

GROUND-GLASS NODULE

- Also known as *non-solid nodule*
- Hazy nodular increased lung attenuation
- Does not obscure underlying bronchi and vessels
- Can indicate inflammatory or premalignant/malignant pathology[3]
- Bilateral, ill-defined ground-glass nodules in peribronchial/centrilobular distribution are seen in hypersensitivity pneumonitis.

The etiological evaluation of ground-glass opacity (GGO) needs to be done in correlation with history and has been divided into acute/subacute ***(Table 28.1)*** *and chronic* ***(Table 28.2)*** *presentation.*

CRAZY PAVING PATTERN

- Pattern characterized by superimposition of linear opacities on a background of GGO

Table 28.1: Differential diagnosis of ground-glass opacity (acute/subacute) **[Figs 28.1 to 28.3]**

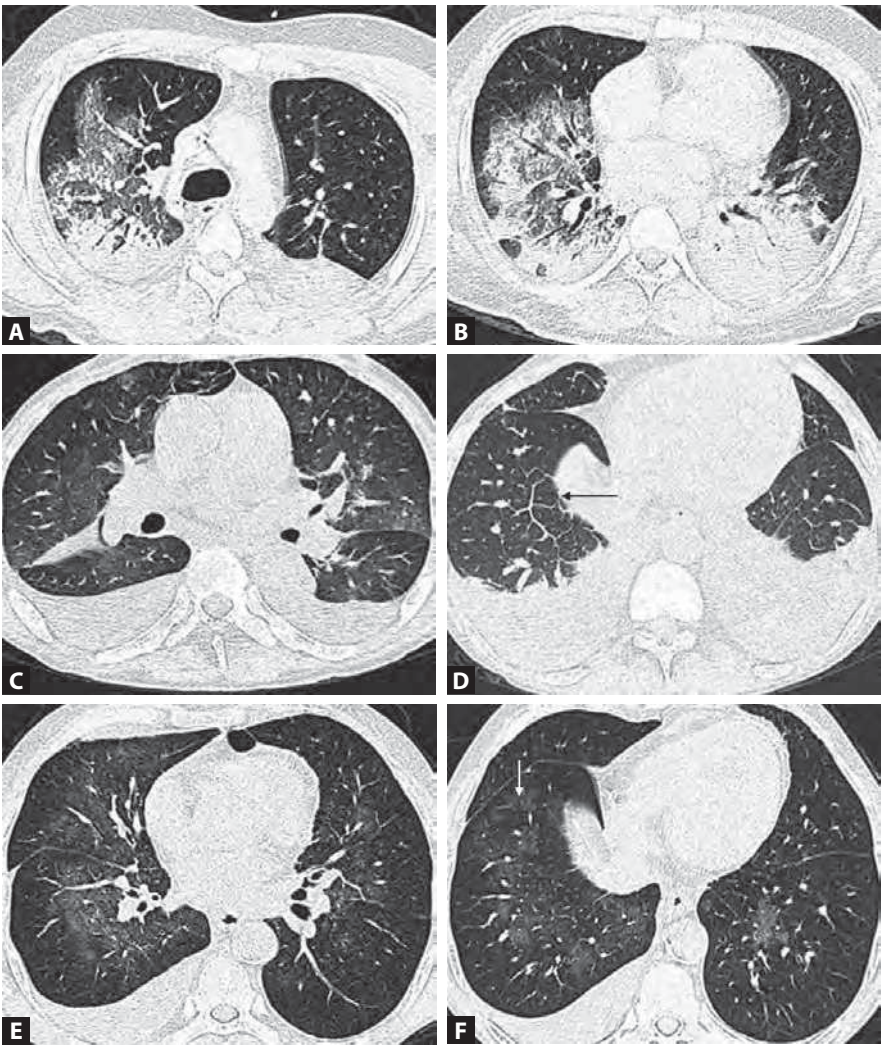
S. No.	*Disease entity*	*Clinical presentation*	*Key features on HRCT*
1.	ARDS	Rapid onset dyspnea Hypoxemia $PaO_2/FiO_2 \leq 200$ mm Hg	Diffuse or patchy GGO or consolidation (geographic distribution) Dependent and basal predominance Peripheral distribution Followed by fibrotic changes predominantly in anterior lung
2.	Pulmonary edema hydrostatic	Enlargement of upper lobe pulmonary veins Airspace opacities with central/ parahilar (batwing distribution) Kerley lines Cardiomegaly Pleural effusion	Patchy or lobular GGO Dependent or perihilar predominance Smooth peribronchovascular interstitial thickening Smooth subpleural interstitial or fissure thickening Vascular prominence Pleural effusion
3.	Diffuse pulmonary (Alveolar) hemorrhage	Recurrent hemoptysis Cough Dyspnea Iron deficiency anemia	Multifocal areas of GGO Central dependent areas Patchy consolidation *Later stage:* Interlobular septal thickening
4.	Acute interstitial pneumonia	Acute onset dyspnea following prodromal symptoms No known cause Require ventilatory support Fulfill clinical criteria of ARDS	Patchy bilateral GGO Sparing of individual lobules Dense opacification/consolidation of dependent lung Lower zone, symmetrical *Late proliferative/exudative:* Areas of increased attenuation with traction bronchiectasis/architectural distortion

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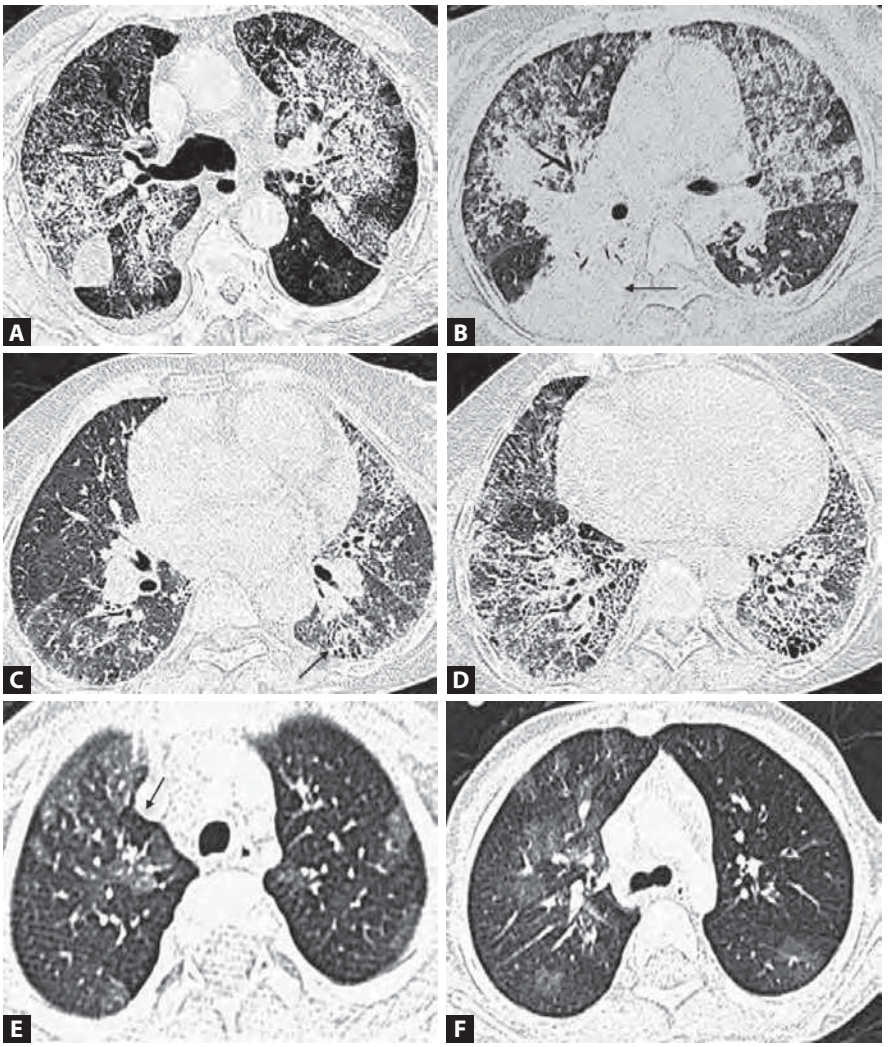
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S. No.	Disease entity	Clinical presentation	Key features on HRCT
5.	Acute exacerbation of IPF	Known case of IPF with worsening dyspnea	Patchy GGO *Distribution:* Diffuse in DAD; and peripheral in OP Reticular opacities and honeycombing Traction bronchiectasis
6.	Atypical pneumonia *Pneumocystis jirovecii* pneumonia *Viral pneumonia*	*In HIV-infected patient:* Subacute presentation Gradual onset Dry cough and dyspnea Constitutional symptoms Subacute presentation	Bilateral symmetric perihilar GGO Consolidation with septal thickening Cystic lung disease Centrilobular nodules GGO in lobular/diffuse distribution Segmental consolidation Air trapping
7.	Acute eosinophilic pneumonia	Fever <5 days > 25% eosinophils in BAL fluid Absence of parasitic/other infections immediate total response to corticosteroids	Bilateral patchy GGO Interlobular septal thickening Occasionally patchy consolidation/poorly defined nodules
8.	Hypersensitivity pneumonitis	Cough Exertional dyspnea Fever	Patchy or diffuse GGO Small centrilobular nodules Lobular areas of decreased attenuation (air trapping)

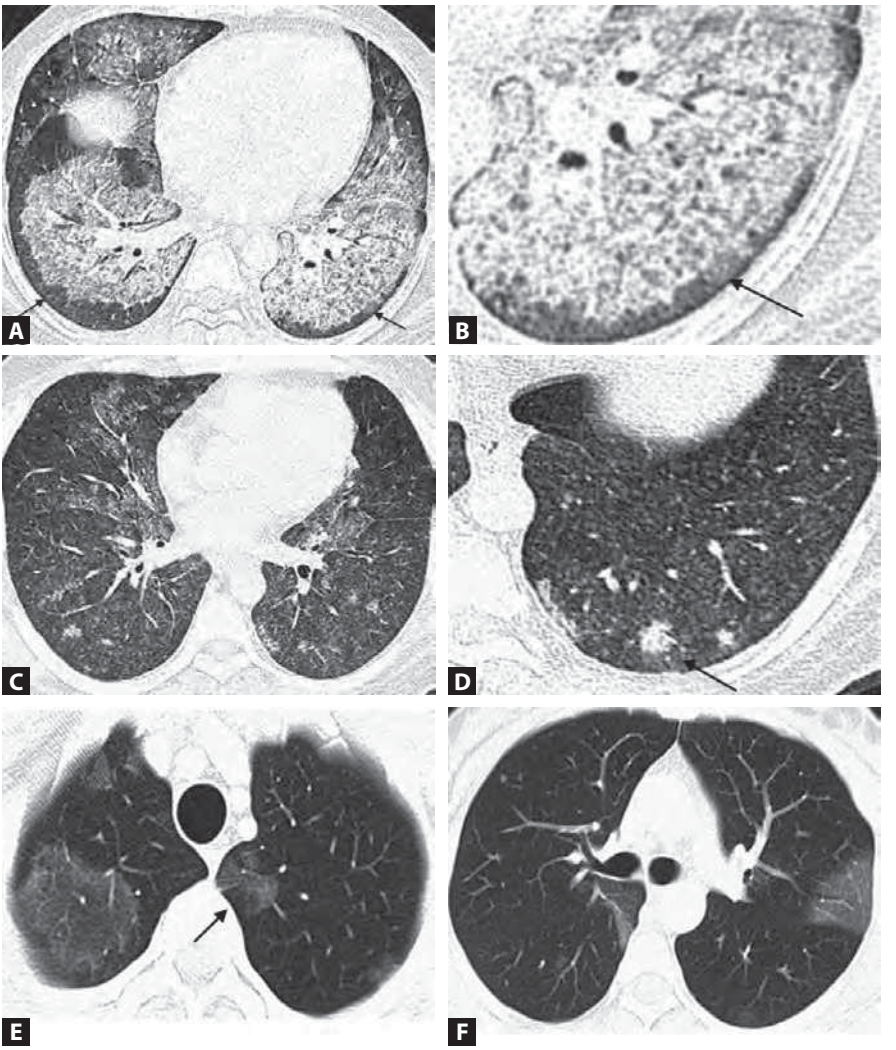
Abbreviations: HRCT, high-resolution computed tomography; ARDS, acute respiratory distress syndrome; PaO_2/FiO_2, partial pressure arterial oxygen/fraction of inspired oxygen; GGO, ground-glass opacity; IPF, ideopathic pulmonary fibrosis; HIV, human immunodeficiency virus; BAL, bronchoalveolar lavage; DAD, diffuse alveolar damage



Figs 28.1A to F: Ground-glass opacities: Acute/subacute.
(A and B) *ARDS:* Central perihilar distribution of consolidation and ground glass opacities; (C and D) *Pulmonary edema:* Central distribution of subtle ground glass opacities with smooth interlobular septal thickening (arrow) and bilateral pleural effusion; (E and F) *DAH:* Central and perihilar ground-glass opacities with poorly defined ground-glass nodules (arrow). Right pleural effusion in this case is an incidental finding



Figs 28.2A to F: Ground-glass opacities: Acute/subacute.
(A and B) Acute interstitial pneumonia*:* Diffuse B/L ground-glass opacities with consolidation in dependent areas; (C and D) Acute exacerbation of IPF. Diffuse ground-glass opacities with a background of interlobular septal thickening, traction bronchiectasis and honeycombing (arrow) in subpleural and basal locations; (E and F) Subacute hypersensitivity pneumonitis. Ground-glass opacities, air trapping (arrow) and interspersed normal lung resulting in the classical 'head cheese' pattern



Figs 28.3A to F: Ground-glass opacities: Acute/subacute.
(A and B) *Pneumocystis jirovecii* pneumonia. Extensive bilateral ground-glass opacity with relative subpleural sparing (arrows in A). Superimposed interlobular septal thickening gives the crazy-paving pattern (arrow in B); (C and D) *Viral pneumonia (CMV):* Patchy GGO, poorly defined centrilobular nodules and peribronchial thickening involving bilateral lung field. Few small nodules in left lower lobe are associated with areas of surrounding ground-glass halo (arrow in D); (E and F) *Acute eosinophilic pneumonia:* Multiple ground-glass nodules having a peripheral and subpleural distribution (arrow in E).

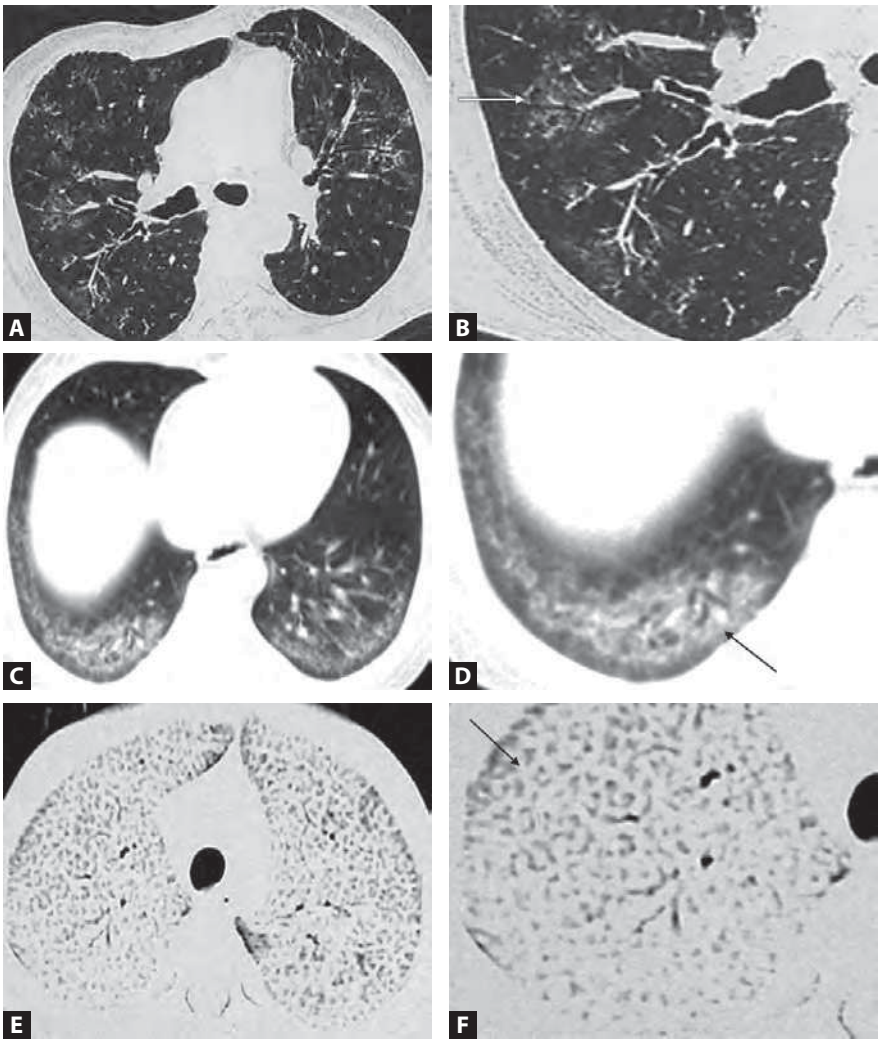
Table 28.2: Differential diagnosis of ground-glass opacity (Chronic) **(Figs 28.4 and 28.5)**

S. No.	*Disease entity*	*Clinical presentation*	*Key features on HRCT*
1.	DIP	Smoking-related dry cough, dyspnea Clubbing in 50% *PFT:* Marked reduction in diffusing capacity	Basal and peripheral subpleural GGO Occasional interlobular septal thickening Small (<2 cm), thin walled cysts in areas of GGO
2.	NSIP	Insidious onset Dyspnea, dry cough *PFT:* Restrictive pattern	GGO with reticulation Traction bronchiectasis/bronchiolectasis No honeycombing Peripheral and lower lung Subpleural sparing
3.	Pulmonary alveolar proteinosis	Alveolar accumulation of surfactant-like proteins Young males Insidious onset of dyspnea, dry cough, easy fatigability	Bilateral involvement Patchy involvement with geographic/lobular sparing Characterized by "crazy paving pattern": Ground-glass opacity with thickened septal lines
4.	Lung adenocarcinoma [as ground-glass nodules (*GGN*)]	Radiological follow-up (increase in size/attenuation/solid part) *Solitary GGN <5 mm:* No follow-up required *Pure GGN >5 mm:* Interval scan at 3 months Annual surveillance (for 3 years at least)	*Atypical adenomatous hyperplasia (AAH):* Small GGN (<5 mm) with no solid or partsolid component Less attenuation than AIS *Adenocarcinoma in situ (AIS):* GGN (5–30 mm) Greater attenuation than AAH

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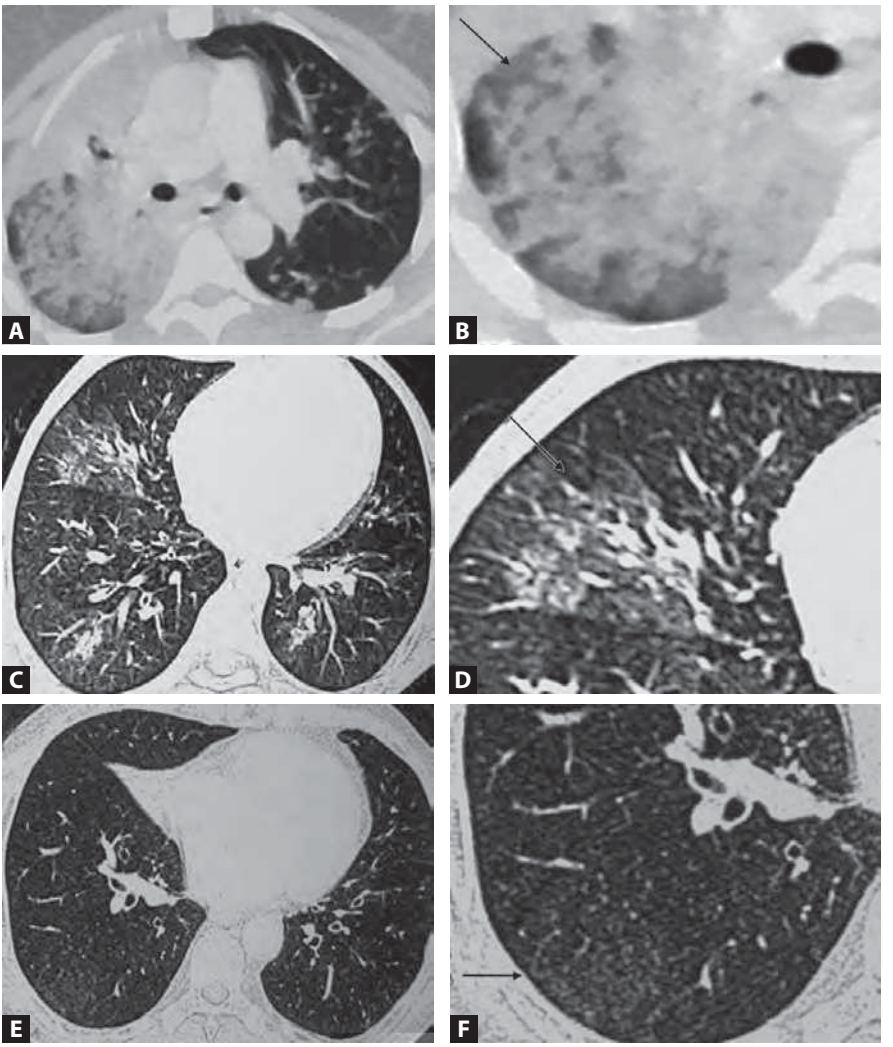
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S. No.	*Disease entity*	*Clinical presentation*	*Key features on HRCT*
		Part solid GGN: PET-CT CT-guided biopsy *Multiple GGNs <5 mm:* Surveillance CT (2 years; 4 years) *Multiple nodules >5 mm with no dominant nodule:* Interval CT at 3 months; annual CT	*Minimally invasive adenocarcinoma (MIA):* GGN with part solid component *Invasive adenocarcinoma (well differentiated):* Extensive GGO Intralesional air bronchogram Cystic lucencies No pleural retraction
5.	RB-ILD	Smoking-related Mild cough and dyspnea *PFT:* Normal or mixed pattern	Extensive centrilobular micronodules Patchy ground glass Bronchial wall thickening Minor centrilobular emphysema Predominantly upper lobe
6.	LIP	Slow onset Fifth decade *Causes:* *Children:* HIV, EBV *Adults:* Autoimmune disorders (Sjögren, SLE, RA, Hashimoto's, myasthenia gravis, pernicious anemia), HIV, Castleman's disease	GGO Perivascular cysts Reticular pattern



Figs 28.4A to F: Ground-glass opacity: Chronic.
(A and B) *DIP:* Patchy GGO with irregular linear/reticular pattern (arrow); (C and D) *NSIP:* GGO with reticular opacities in peripheral and lower lung zones with relative subpleural sparing (arrow); (E and F) *PAP:* Dense GGO with superimposed septal thickening (arrow)

- Linear pattern can be
  - Interlobular septal thickening
  - Intralobular septal thickening
  - Perilobular pattern of airspace filling
- Was initially described for the appearance of pulmonary alveolar proteinosis (PAP)
- *Differentials include*[1,2] PAP, infections (*Pneumocystis jirovecii* pneumonia), subacute stage of acute respiratory distress syndrome (ARDS), diffuse

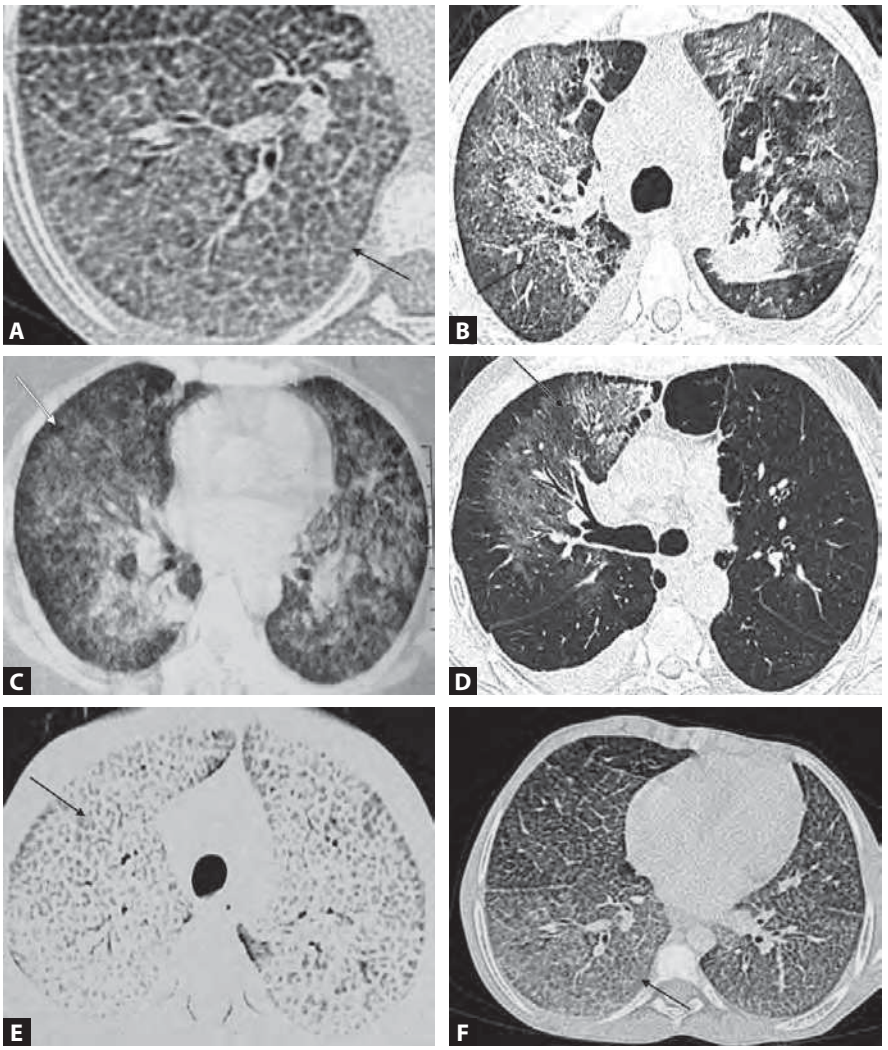


Figs 28.5A to F: Ground-glass opacities: Chronic.
(A and B) *Invasive adenocarcinoma:* Extensive GGO (arrow) with intralesional air bronchogram; (C and D) *LIP:* Peribronchovascular GGO (arrow) with bronchial wall thickening; (E and F) *RB-ILD:* Ground-glass attenuation centrilobular nodules (arrow) with bronchial wall thickening

alveolar hemorrhage, nonspecific interstitial pneumonia (NSIP), storage disorders, lipoid pneumonia **(Figs 28.6A to F)**.

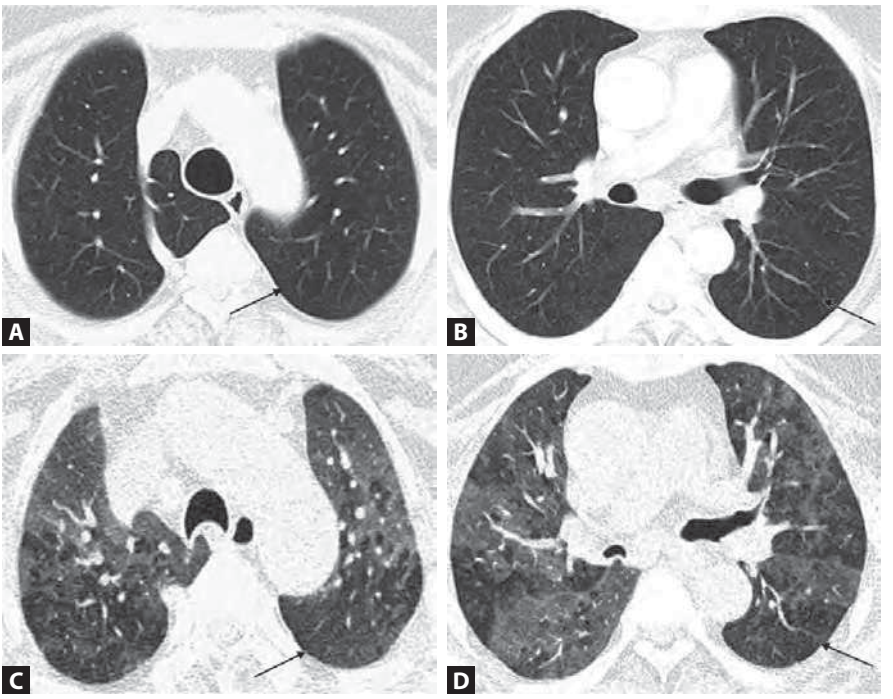
Mosaic Attenuation

- Heterogeneous appearance of differing attenuation on high-resolution computed tomography (HRCT) chest[4]



Figs 28.6A to F: Crazy paving pattern. (A) Crazy paving. Ground-glass opacities with superimposed septal thickening; (B) *Pneumocystis jirovecii* pneumonia. Bilateral perihilar GGO with areas of cystic lucencies; (C) ARDS in subacute phase. Central perihilar preponderance of GGO; (D) Diffuse alveolar hemorrhage. Crazy paving pattern in late stage of DAH, seen predominantly in the right upper lobe; (E) Pulmonary alveolar proteinosis. Diffuse symmetrical bilateral involvement; (F) Gaucher's disease. Pulmonary manifestation of a storage disorder in the form of bilateral crazy paving pattern

- *Includes previously used terms:* Mosaic perfusion and mosaic oligemia
- Etiology of mosaic attenuation:
  - *Bronchiolar:* Air trapping **(Figs 28.7A to D)**
  - *Vascular (PAH; PVH):* No air trapping **(Figs 28.8A to E)**
  - *Patchy interstitial disease:* Abnormal parenchyma

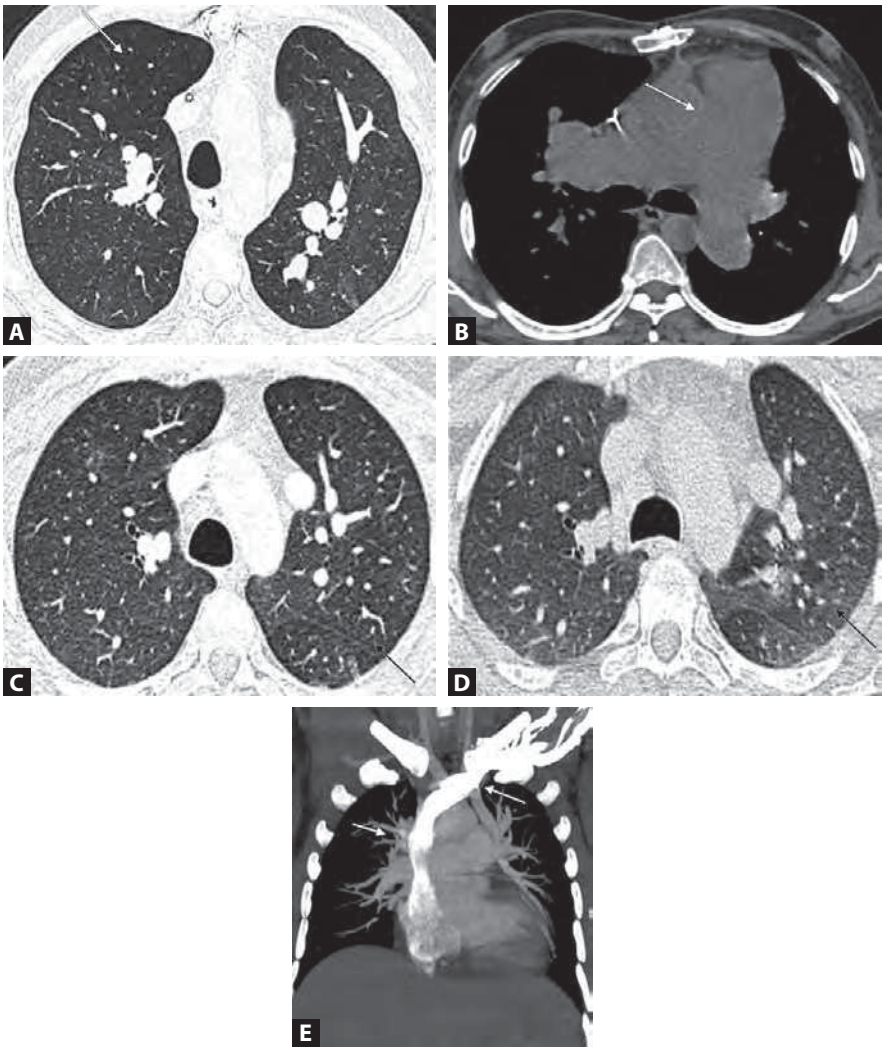


Figs 28.7A to D: Mosaic attenuation (Bronchiolar cause). (A and B) Inspiratory scan shows subtle areas of low attenuation with normal vessel size and number in the abnormal area (arrows). These areas are more evident on the expiratory scan (C and D) suggestive of air trapping

- The following points need to be assessed in evaluating heterogeneous attenuation on HRCT **(Table 28.3)**.

Table 28.3: Differential for mosaic attenuation

	Mosaic attenuation (Bronchiolar cause)	*Mosaic attenuation (Vascular cause)*	*Ground-glass opacity*
Abnormal area	Low attenuation	Low attenuation	High attenuation
Demarcation between low and high attenuation region	Sharp	Sharp	Ill-defined
Vessel size and number	Normal Occasionally decreased in region of low attenuation	Decreased peripherally in region of low attenuation (PAH) Vascular redistribution (PVH)	Normal Occasionally decreased in region of high attenuation
Air trapping	Present	Absent	Absent



Figs 28.8A to E: Mosaic attenuation (Vascular cause).
(A and B) Case 1. HRCT (A) shows subtle areas of low attenuation with decreased vessel size and number in the abnormal area (arrow). Mediastinal window (B) reveals dilated main pulmonary artery (arrow) suggestive of pulmonary arterial cause of the mosaic perfusion. (C to E) Case 2. Inspiratory scan (C) shows subtle areas of low attenuation with normal vessel size and number in the abnormal area (arrow). There is no accentuation of these areas on the expiratory scan (D). Coronal MIP image, soft tissue window (E) reveals the anomalous drainage of upper lobe pulmonary veins into the systemic circulation (arrows) suggestive of pulmonary venous cause of the mosaic pattern

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29

Approach to Diagnosis: Linear Opacities

Smita Manchanda

LINEAR OPACITIES

Linear and reticular opacities seen on high-resolution computed tomography (HRCT) chest have varied forms and underlying etiologies. Parenchymal bands, subpleural lines and irregular linear opacities are nonspecific and can be seen in fibrosis and atelectasis. The most useful is interlobular septal thickening which can be seen as smooth/nodular/irregular. Similar pattern of smooth/nodular/irregular thickening can also be seen in the peribronchovascular, perihilar and subpleural interstitium.[1]

Parenchymal Bands

- Nontapering linear or reticular opacity
- Usually 2–5 cm in length and few mm in thickness
- Is seen perpendicular to and in contact with the pleural surfaces
- *Etiology*: Atelectasis, lung fibrosis, asbestosis.

Subpleural Lines

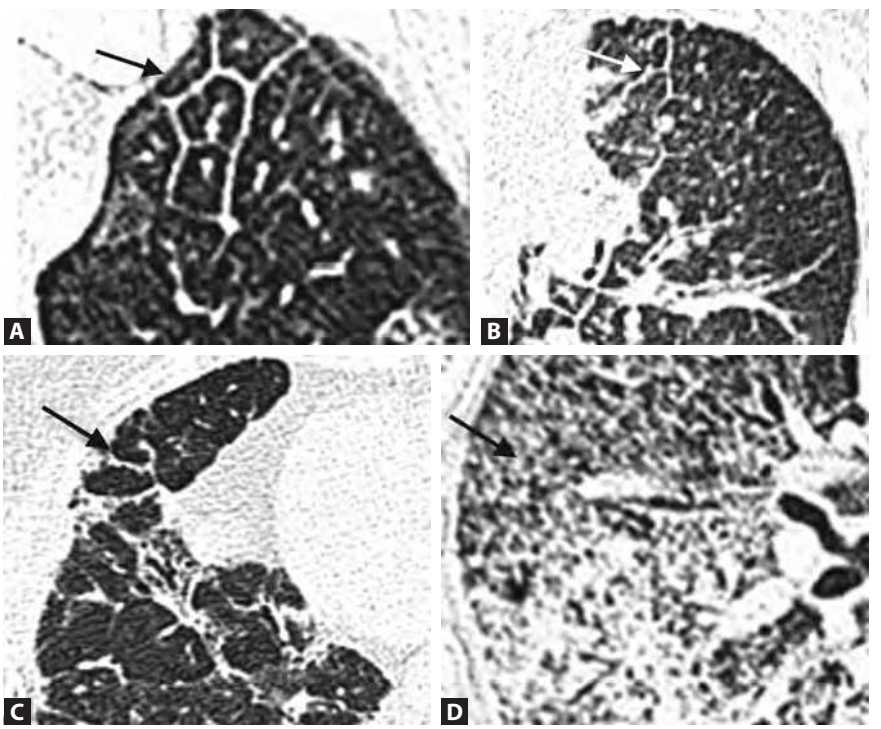
- Curvilinear opacity that parallels the pleura
- Less than 10 mm in thickness
- *Etiology*: Atelectasis, fibrosis, asbestosis, inflammation.

Irregular Linear Opacities

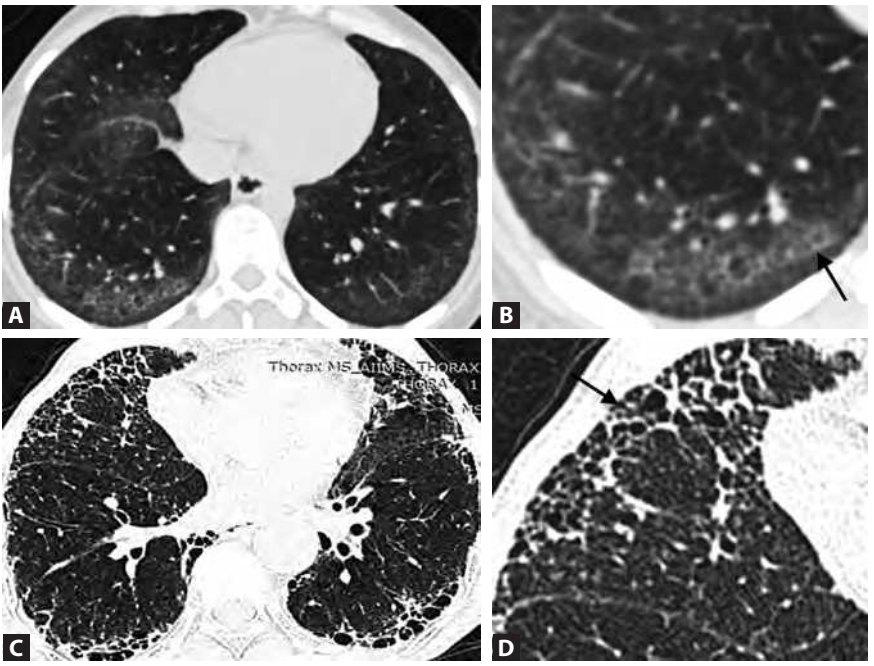
- Nonspecific linear structures that cannot be classified as a parenchymal band, subpleural line, or interlobular septa
- Usually 1–3 mm in thickness
- *Etiology*: Nonspecific, seen in fibrotic lung diseases.

Reticular Opacities (Intralobular Septal Thickening)

- Linear opacities which intersect one another at various angles and form a reticulated (netlike) pattern. Seen on HRCT as intralobular interstitial thickening **(Figs 29.1A to D)**.
- Infiltration and thickening of interstitial framework of secondary pulmonary lobule.
- Could be due to fibrosis (coarse reticulation) or inflammation.
- *Etiology*: Usual interstitial pneumonia (UIP), nonspecific interstitial pneumonia (NSIP), interstitial lung disease (ILD) with collagen vascular diseases, pulmonary edema, infections and lymphangitic carcinomatosis **(Figs 29.2A to D)**.



Figs 29.1A to D: Linear opacities. (A) Smooth interlobular septal thickening; (B) Nodular interlobular septal thickening; (C) Irregular interlobular septal thickening; (D) Intralobular septal thickening



Figs 29.2A to D: Intralobular septal thickening.
(A and B) *NSIP:* Early NSIP changes in systemic sclerosis. Intralobular septal thickening (arrow) and ground-glass haze with subpleural sparing.
(C and D) *UIP:* Reticular abnormalities with intralobular septal thickening (arrow) and honeycombing

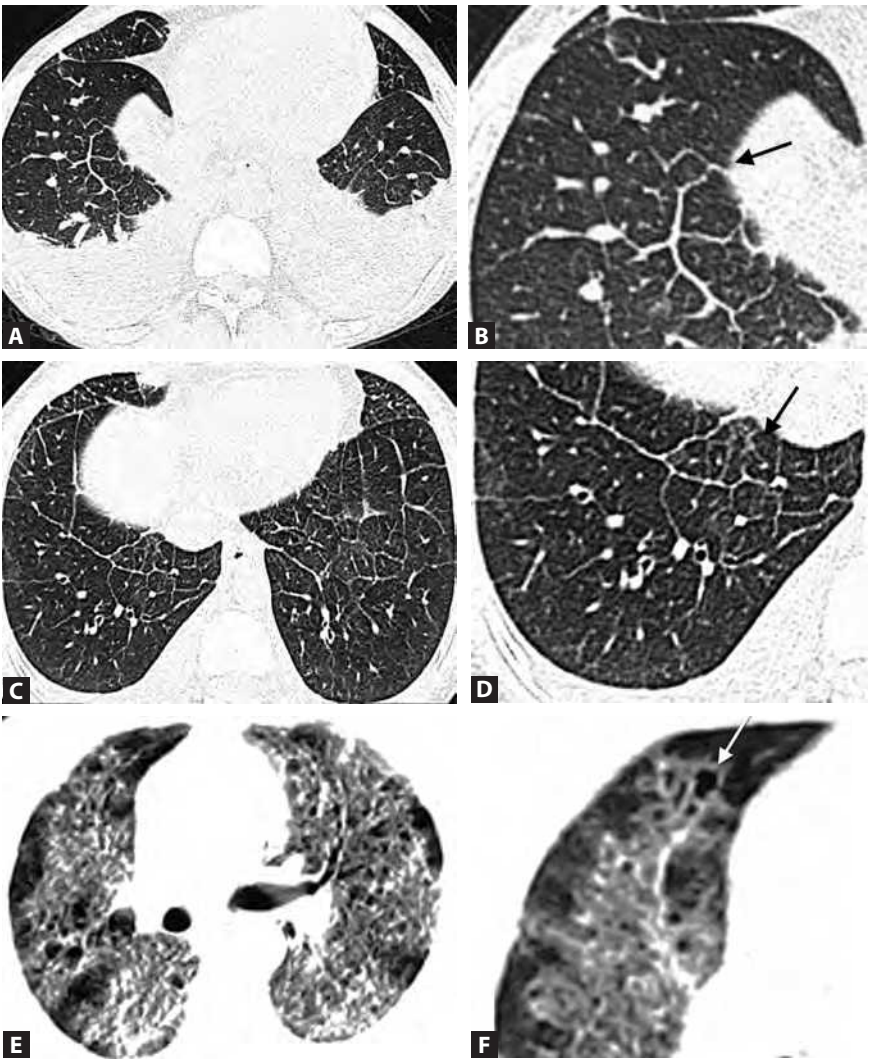
Interlobular Septal Thickening

- Interlobular septa are approximately 0.1 mm in thickness
- Few septa may normally be visualized anteriorly or along the mediastinal pleura in HRCT chest
- The presence of numerous septa or thickening of the septa indicates an underlying pathology
- Thickening can be due to *pulmonary vein dilatation*, *lymphatic infiltration* or *pulmonary interstitial infiltration*
- Septal thickening can be characterized as smooth, nodular, or irregular[1,2] **(Figs 29.1A to D)**. The differential diagnosis for the various patterns of septal thickening is detailed in **Tables 29.1 to 29.3**.

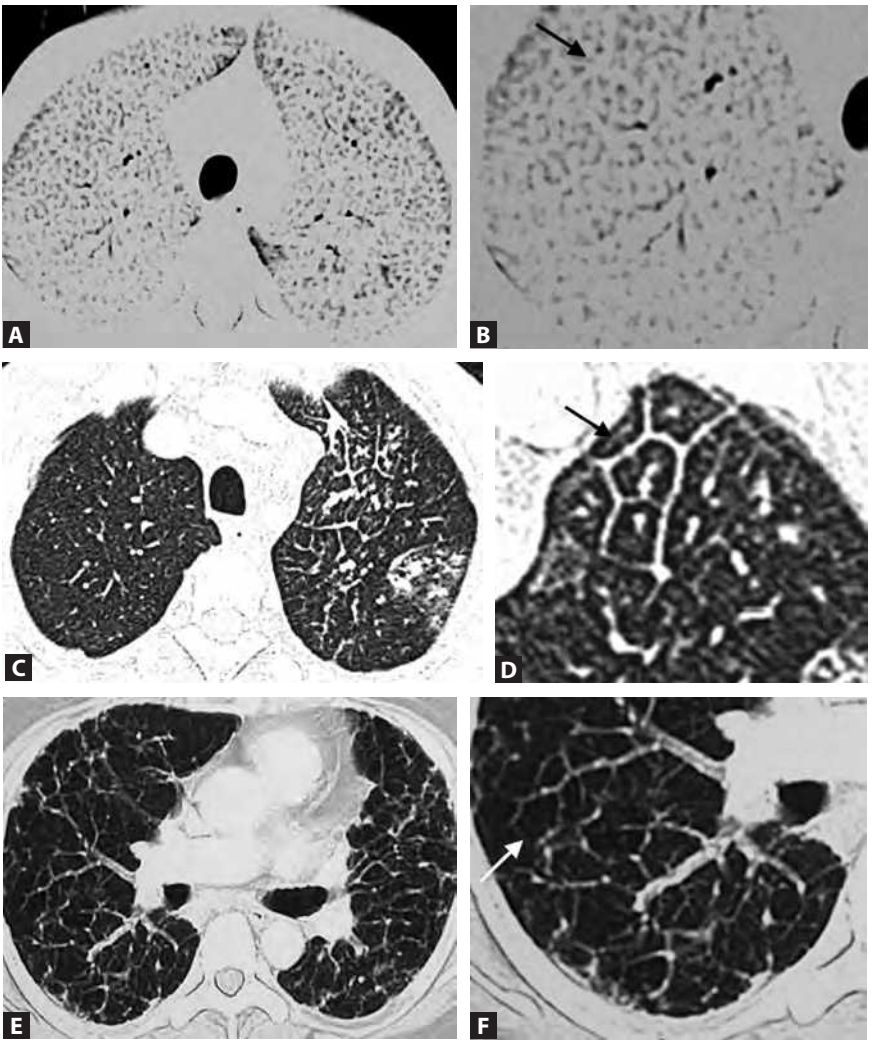
Table 29.1: Differential diagnosis of septal thickening: Smooth **(Figs 29.3 and 29.4)**

Sl. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Pulmonary edema	Hydrostatic *Cardiomegaly* Pleural effusion	Patchy or lobular GGO Peribronchovascular/subpleural interstitial/fissural thickening: Smooth *Dependent or perihilar predominance*
2.	Central venous pulmonary obstruction	Pleural effusion	Patchy or lobular GGO *Upper lobe veins enlarged*
3.	ARDS	Usually in *subacute phase*	Bilateral GGO *Central, perihilar*
4.	Pulmonary alveolar proteinosis	Young males Insidious onset	Bilateral involvement Patchy involvement with geographic/lobular sparing *"Crazy paving pattern":* Ground-glass opacity with thickened septal lines
5.	Lymphangitic carcinomatosis	Associated with interstitial thickening and *background normal lung architecture*	*Well-defined and peribronchovascular/subpleural/septal nodules* Unilateral/bilateral; Patchy Smooth/nodular interstitial thickening
6.	Pulmonary hemorrhage	*Recurrent hemoptysis* Iron deficiency anemia	*Multifocal areas of GGO* Patchy consolidation *Later stage*: Interlobular septal thickening
7.	Lymphoproliferative disease and Lymphangiectasia	LIP: Associated with dysproteinemia/ autoimmune disease/AIDS	*Nodules*: Small, well defined, subpleural Thickening of bronchovascular bundles *GGO and Cysts* Congenital lymphangiectasia: B/L thickening with effusions
8.	Infections (*Pneumocystis jirovecii* pneumonia)	In HIV-infected patient: *Subacute presentation*	Bilateral symmetric perihilar GGO Consolidation with septal thickening Cystic lung disease
9.	Diffuse alveolar septal amyloidosis	Associated with systemic involvement	Well-defined small nodules *Smooth septal thickening* Bilateral basal consolidations Punctate calcifications

Abbreviations: AIDS, acquired immune deficiency syndrome; ARDS, acute respiratory distress syndrome; GGO, ground-glass opacity; HRCT, high-resolution computed tomography; LIP, lymphocytic interstitial pneumonia



Figs 29.3A and F: Septal thickening: Smooth.
(A and B) *Pulmonary edema:* Central distribution of subtle ground-glass opacities with smooth interlobular septal thickening (arrow) and bilateral pleural effusion.
(C and D) *Pulmonary venous hypertension:* Bilateral smooth interlobular septal thickening.
(E and F) *ARDS (subacute phase):* Ground-glass opacities with superimposed smooth septal thickening in central perihilar preponderance

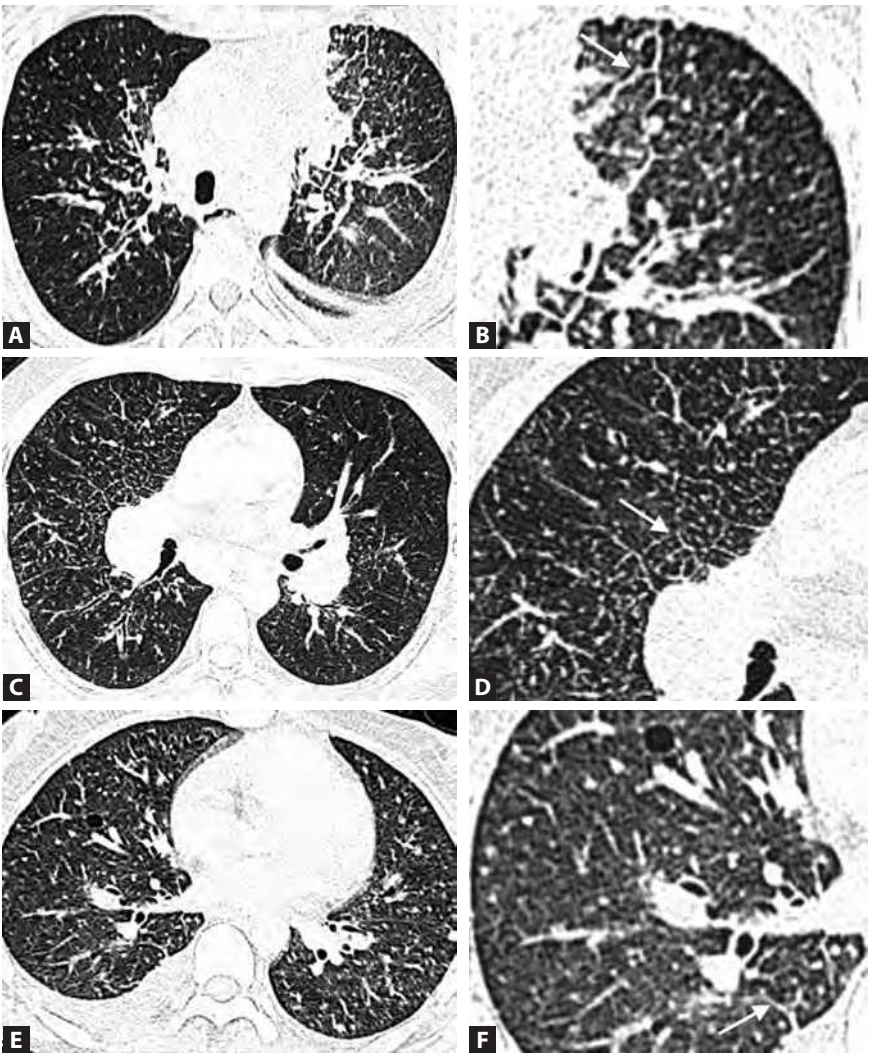


Figs 29.4A to F: Septal thickening: Smooth.
(A and B) *Pulmonary alveolar proteinosis:* Dense ground-glass opacities with superimposed septal thickening (arrow).
(C and D) *Lymphangitic spread of tumor:* Multiple nodules with interspersed smooth (arrow) and nodular septal thickening in left upper lobe. Lymphangitic carcinomatosis can cause smooth interlobular septal thickening even though nodular interlobular septal thickening is more characteristic.
(E and F) *Pulmonary lymphangiectasia:* Bilateral smooth interlobular septal thickening (arrow) with peribronchial cuffing

Table 29.2: Differential diagnosis of septal thickening: Nodular **(Figs 29.5A to F)**

Sl. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1	Lymphangitic carcinomatosis	Associated with interstitial thickening and *background normal lung architecture*	*Well-defined and peribronchovascular/subpleural/septal nodules* Unilateral/bilateral; Patchy Smooth/nodular interstitial thickening: '*beaded septa*'
2	Sarcoidosis	Respiratory and constitutional symptoms *Background fibrosis*	*Upper lobes, asymmetric* *Nodules*: Small, well-defined, subpleural, septal, perihilar, peribronchovascular Peribronchovascular interstitial thickening
3	Lymphocytic interstitial pneumonia	*LIP:* Associated with dysproteinemia/ autoimmune disease/AIDS	*Nodules*: Small, well-defined, subpleural Thickening of bronchovascular bundles *GGO and Cysts*
4	Diffuse alveolar septal amyloidosis	Associated with systemic involvement	*Well-defined small nodules* Usually smooth, can be nodular Bilateral basal consolidations Punctate calcifications
5	Lymphoproliferative disease (lymphoma/leukemia/PTLD)	Variable presentations	Multiple nodules GGO/consolidation *Peribronchovascular thickening* Lymphadenopathy

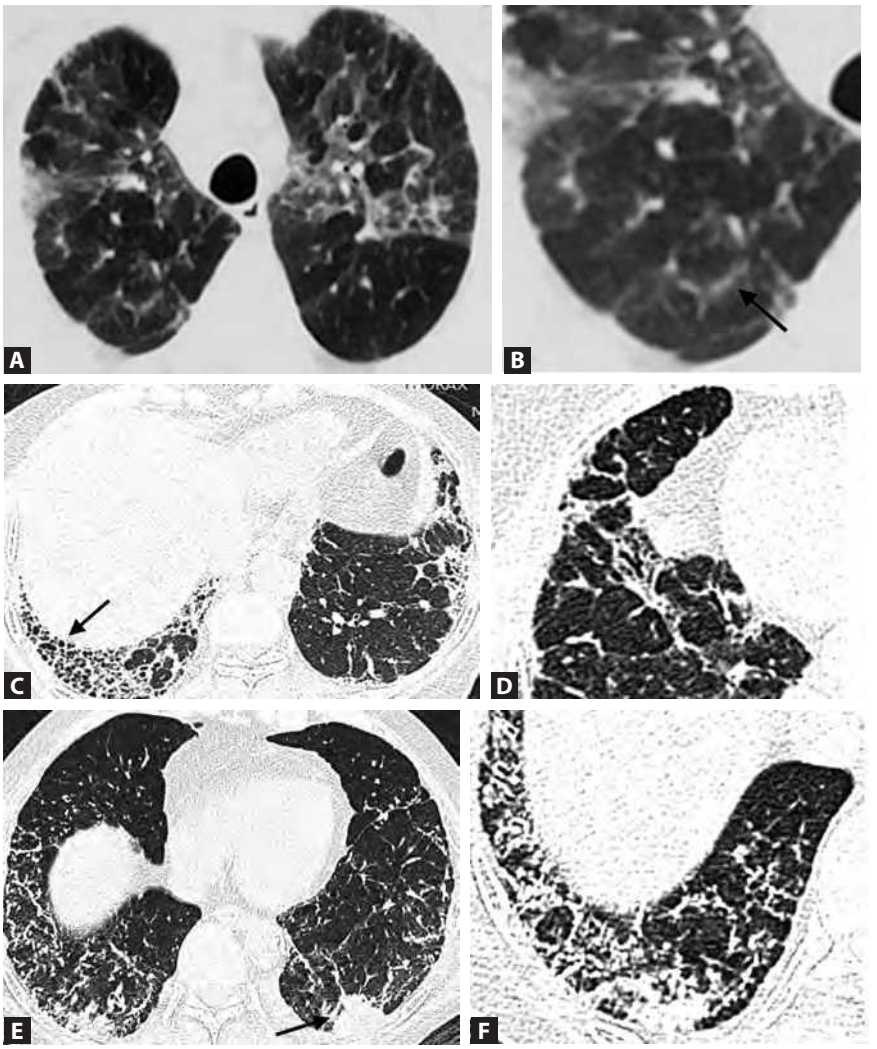
Abbreviations: AIDs, acquired immune deficiency syndrome; GGO, ground-glass opacity; HRCT, high-resolution computed tomography; LIP, lymphocytic interstitial pneumonia; PTLD, post-transplant lymphoproliferative disorder



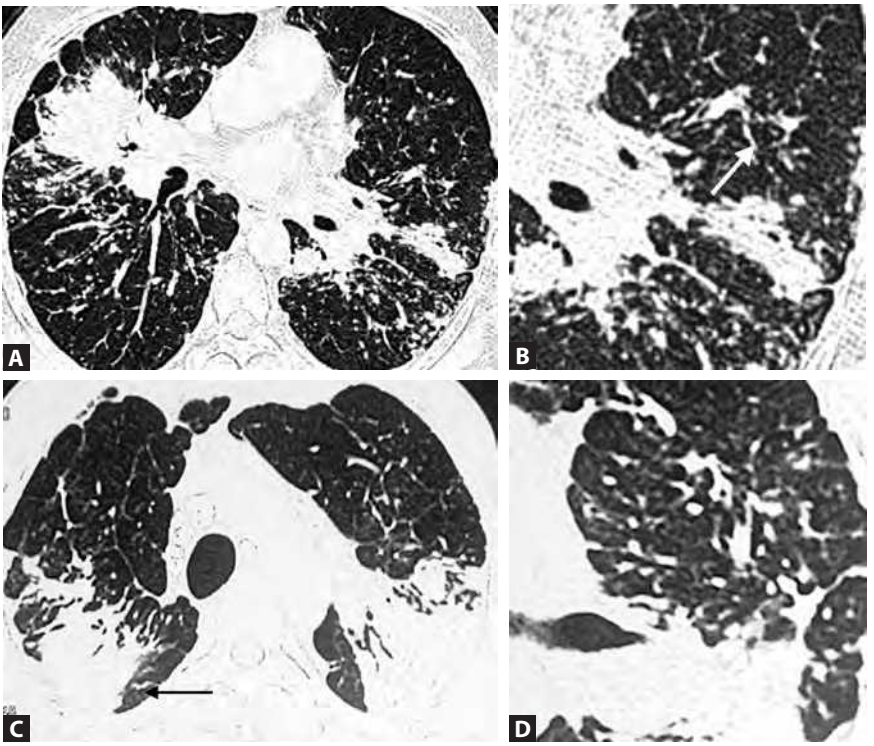
Figs 29.5A to F: Septal thickening: Nodular.
(A and B) *Lymphangitic spread of tumor:* Multiple perilymphatic nodules with interspersed beaded interlobular septa (arrow) in bilateral lung fields suggestive of lymphangitis in a known case of carcinoma lung.
(C and D) *Sarcoidosis:* Perilymphatic nodules with nodular interlobular septal thickening (arrow) and right hilar lymph nodal enlargement.
(E and F) *LIP Sjögren syndrome:* Few random and centrilobular nodules with parenchymal cysts and septal thickening (arrow)

Table 29.3: Differential diagnosis of septal thickening: Irregular **(Figs 29.6 and 29.7)**

Sl. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Chronic hypersensitivity pneumonitis	*Fibrosis: Upper lobes*	*Lung fibrosis*: Reticular opacities, architectural distortion, bronchiectasis and bronchiolectasis, honeycombing, volume loss *Honeycombing: Subpleural and midzone* Emphysema, cysts
2.	Sarcoidosis	Respiratory and constitutional symptoms Peribronchovascular interstitial thickening and lymphadenopathy Peribronchovascular clustering of nodules	*Upper lobes, asymmetric* *Nodules:* Small, well defined, subpleural, septal, perihilar, peribronchovascular *Smooth or nodular peribronchovascular interstitial thickening*
3.	Silicosis/ coal worker's pneumoconiosis	*Clinical background of dust exposure* Associated with irregular masses and fibrosis	*Diffuse and symmetrical* *Masses*: Conglomerate, irregular Fibrosis, cicatricial emphysema
4.	Asbestosis	*Subpleural fibrosis*	Subpleural dot like opacities and subpleural lines Parenchymal bands Posterior and basal predominance
5.	Usual interstitial pneumonia	*Findings of fibrosis*	Traction bronchiectasis Peripheral, subpleural, lower lung zone, posterior *Honeycombing*: *Bases*



Figs 29.6A to F: Septal thickening: Irregular.
(A and B) *Chronic hypersensitivity pneumonitis:* Interlobular septal thickening (arrow), patchy ground-glass opacities with areas of air trapping.
(C and D) *UIP:* Irregular reticular abnormalities with honeycombing (arrow) in a subpleural, posterior and lower lobe predominance.
(E and F) *Asbestosis:* Irregular interlobular septal thickening in the basal segments of both lungs with a rounded pleural based mass (arrow).



Figs 29.7A to D: Septal thickening: Irregular.
(A and B) *Sarcoidosis:* Perihilar conglomerate masses representing peribronchovascular fibrous tissue with multiple perilymphatic nodules and irregular septal thickening (arrow). (C and D) *Silicosis:* Bilateral upper lobe fibrotic masses (arrow) with a lenticular shape and convergence towards hila and surrounding paracicatricial emphysema and architectural distortion

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30

Approach to Diagnosis: Nodular Pattern

Smita Manchanda

INTRODUCTION

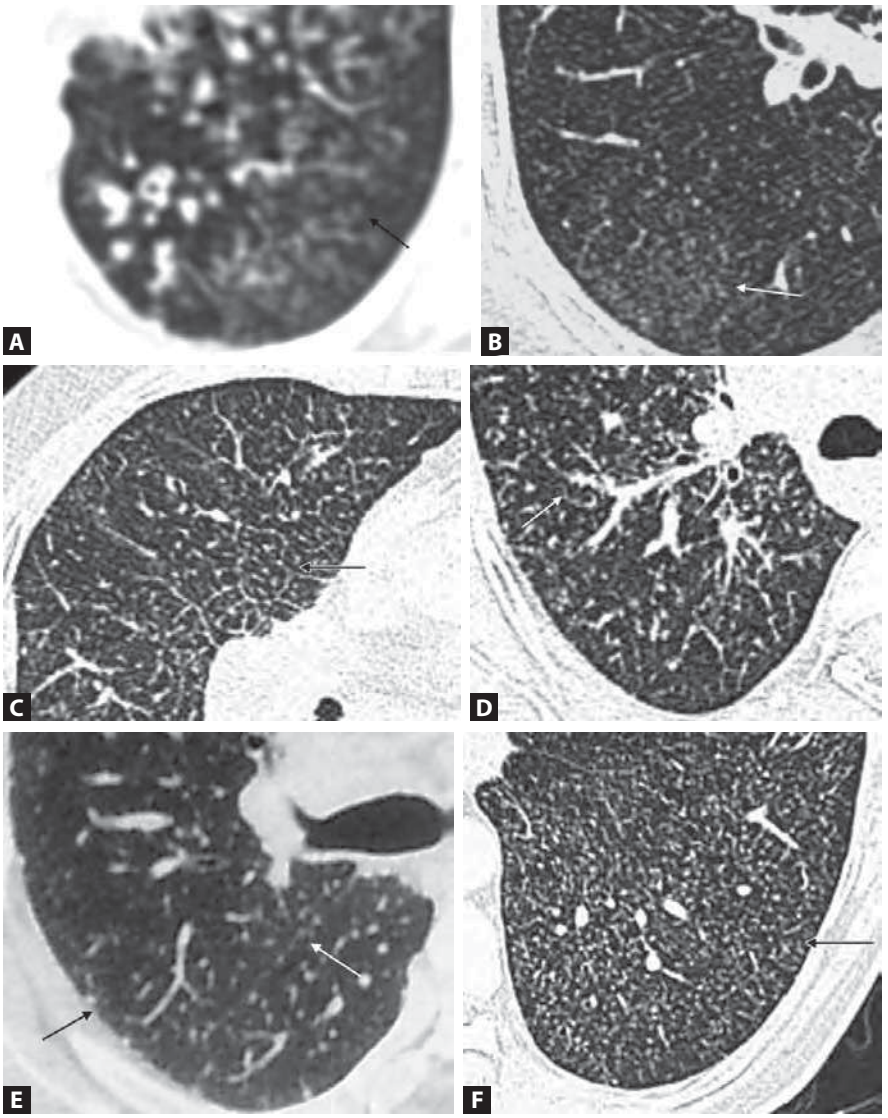
Nodule is a small, discrete, round, focal opacity within the lung (size 2–30 mm). Small nodules have a diameter less than 1 cm and the term "micronodules" is used for nodules less than 7 mm in diameter.[1]

Types of Nodules

- Based on the morphology, the nodule can be:
  - *Airspace:*
    Can be ill defined/well-defined
    Soft-tissue opacity/Ground-glass opacity
  - *Interstitial:*
    Well-defined
    Soft-tissue opacity/Ground-glass opacity
- Based on the distribution of nodules within the secondary pulmonary lobule the following three types are recognized on HRCT **(Figs 30.1A to F)**
  1. Centrilobular
  2. Perilymphatic
  3. Random.

CENTRILOBULAR NODULES

- Within the center of the secondary pulmonary lobule
- Located approximately 5–10 mm from the visceral pleural surface and are equally spaced **(Figs 30.1A and B)**
- Indicates involvement of structures in the center of the pulmonary lobule: bronchiole, artery, peribronchiolar/periarterial interstitium
- *"Tree-in-bud":* This appearance occurs because of the characteristic Y- or V-shaped branching morphology of dilated and impacted bronchi on HRCT. The bronchus may be impacted with mucous, pus, or fluid, with resultant dilation of the bronchus and associated peribronchiolar inflammation.[2]



Figs 30.1A to F: Patterns of nodules on HRCT. (A) Centrilobular nodules with tree-in-bud appearance; (B) Centrilobular nodules without tree-in-bud appearance; (C) Perilymphatic nodule: Interlobular septum; (D) Perilymphatic nodule: Peribronchovascular; (E) Perilymphatic nodule: Fissural (white arrow) and subpleural (black arrow); (F) Random nodule

Most commonly seen with pulmonary infections, the presence of "tree-in-bud" morphology also has other differentials **(Table 30.1)**

- The entities mentioned in **Table 30.1** may also present without tree in bud appearance.[3] These are also included in the differential of centrilobular nodules without tree in bud appearance (**Table 30.2A** [peribronchiolar cause] and **30.2B** [perivascular cause]).

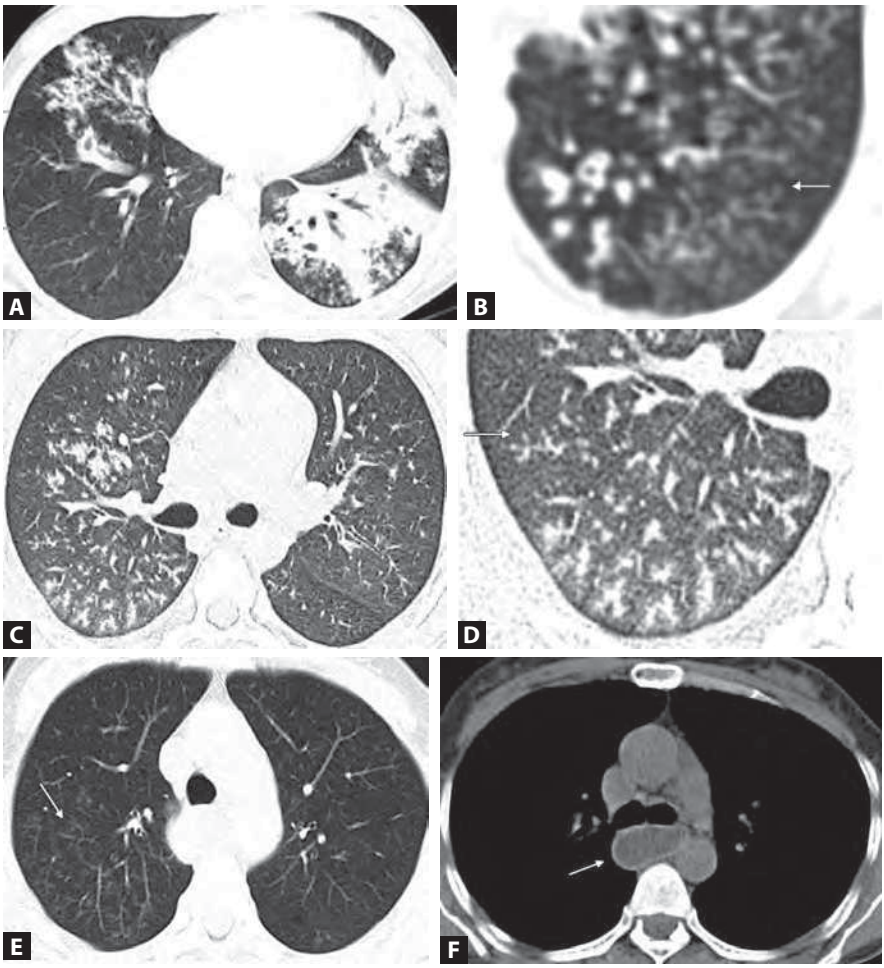
Table 30.1: Differential diagnosis of centrilobular nodules with tree-in-bud pattern **(Figs 30.2 and 30.3)**

S. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Infectious bronchiolitis	Viral, mycoplasma, *Aspergillus*, bacterial, tuberculosis	Poorly defined nodules Peribronchial consolidation Bronchial wall thickening Air trapping
2.	Aspiration	Predisposing factors: structural abnormalities of pharynx, esophagus disorders, neurologic deficits, any chronic illness	Lobar/segmental Uni- or bilateral foci of branching areas of increased attenuation
3.	Cystic fibrosis	Hereditary disease *All lobes involved*	Central bronchiectasis Bronchial wall thickening Mucous plugging Air trapping
4.	Allergic Bronchopulmonary Aspergillosis	Patients of cystic fibrosis and asthma	Central bronchiectasis *Upper lobe dominance* *High density mucous plugs* Air trapping
5.	Diffuse panbronchiolitis	Asian population Nonsmokers Idiopathic	Bronchiolectasis Bronchiectasis Diffuse/Basilar Large lung volumes Air trapping
6.	Follicular bronchiolitis	Rheumatoid arthritis, Sjögren disease, AIDS	Centrilobular nodules Ground glass nodules Bronchial wall thickening
7.	Tumor embolism	Breast, stomach, renal, prostate	Dilatation and beading of vessels Septal thickening

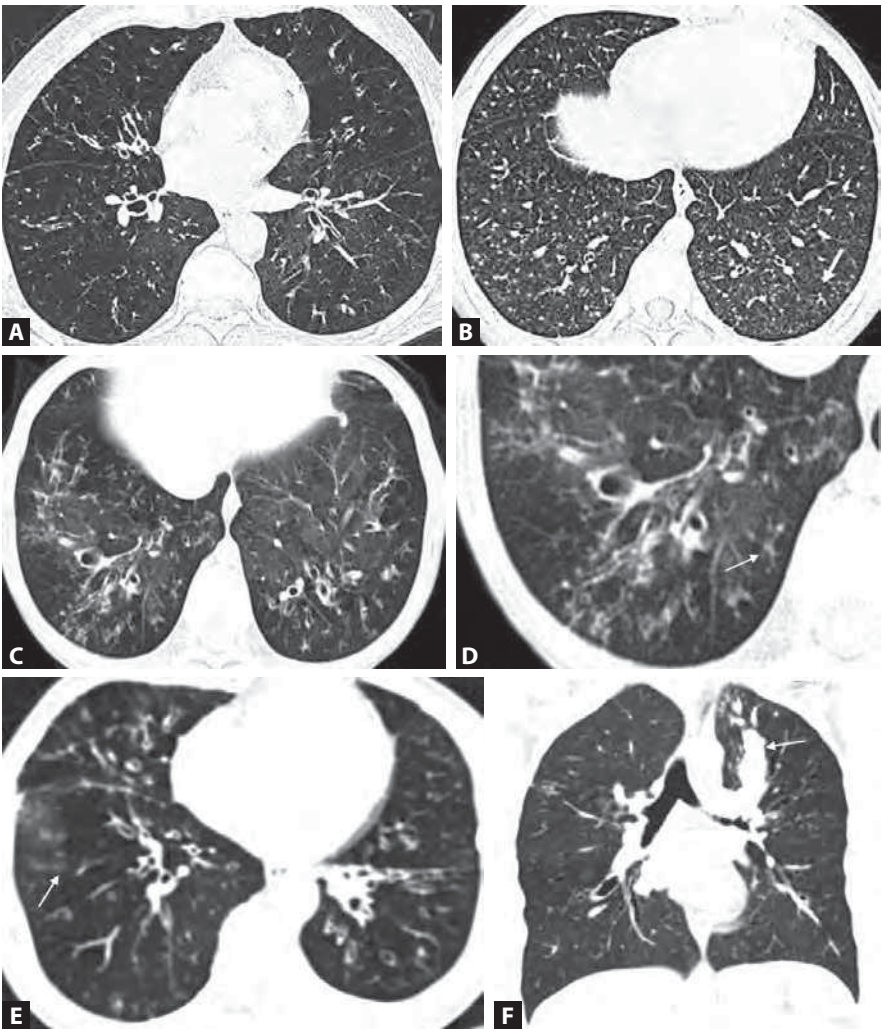
The conditions presenting with centrilobular nodules without tree-in-bud pattern may have a peribronchiolar/vascular etiology.

PERILYMPHATIC NODULES

- Seen along the pulmonary lymphatics—interlobular septa/visceral pleura/center of the secondary pulmonary lobule[3,4] **(Figs 30.1C to E)**
- Interlobular and subpleural nodules are more numerous
- Abut costal and fissural pleural surfaces
- Patchy distribution
- The differentials are discussed in **Table 30.3**



Figs 30.2A to F: Centrilobular nodules with tree-in-bud appearance.
(A and B) *Tuberculosis:* Bilateral lung fields show consolidation with centrilobular nodules, few showing the tree-in-bud pattern (arrow).
(C and D) *Viral bronchiolitis:* Centrilobular nodules and tree-in-bud opacities (arrow) with bronchial wall thickening.
(E and F) *Aspiration pneumonitis in systemic sclerosis:* Ill-defined centrilobular nodules seen in dependent lung zones (arrow in E). Note the dilated esophagus (arrow in F)



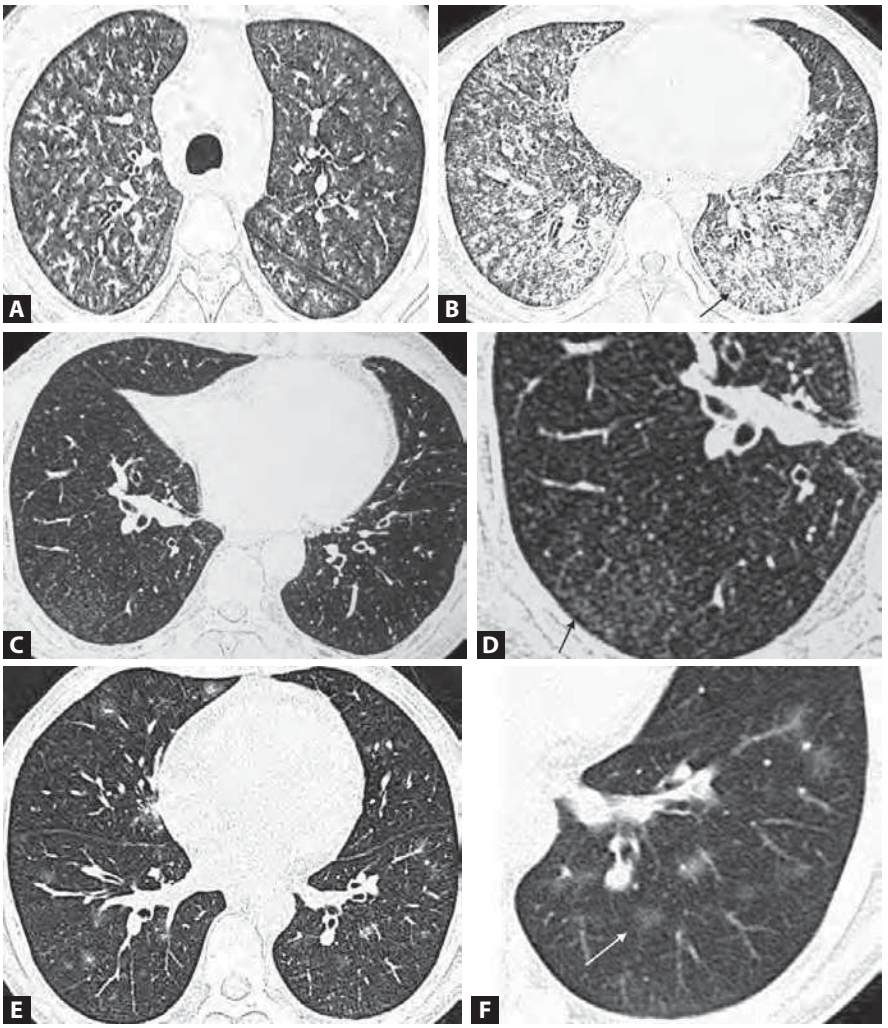
Figs 30.3A to F: Centrilobular nodules with tree-in-bud appearance.
Follicular bronchiolitis. (A) *Rheumatoid arthritis:* Bronchiolar wall thickening and bronchiolectasis with few centrilobular nodules. (B) *SLE:* Ill marginated, small centrilobular nodules (arrow) scattered over bilateral lungs.
(C and D) *Cystic fibrosis:* Dilated thick-walled bronchi, patchy GGO and centrilobular nodules with diffuse tree-in-bud pattern (arrow).
(E and F) *Allergic bronchopulmonary aspergillosis:* Bilateral central bronchiectasis with mucus plugs in the left upper lobe (arrow in F). There is bronchiolar dilatation with centrilobular nodules with tree-in-bud pattern bilaterally (arrow in E)

Table 30.2A: Differential diagnosis of centrilobular nodules without tree-in-bud pattern: Peribronchiolar cause **(Figs 30.4A to F)**

S. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Infectious bronchiolitis	Refer to **Table 30.1**	
2.	Aspiration		
3.	Cystic fibrosis		
4.	Allergic bronchopulmonary aspergillosis		
5.	Panbronchiolitis		
6.	Follicular bronchiolitis		
7.	Endobronchial neoplasms		
8.	Hypersensitivity pneumonitis (Acute/subacute)	Mid/ lower lobe dominance Head cheese sign	*Nodules*: Poorly defined, GGO nodules Head cheese sign (GGO, areas of air trapping and intervening normal lung)
9.	Respiratory bronchiolitis interstitial lung disease (RB-ILD)	Smoking related	Ill defined centrilobular nodules Patchy ground glass Bronchial wall thickening Predominantly upper lobe
10.	Early asbestosis	Later basal fibrosis and pleural calcification	Interlobular septal thickening Curvilinear subpleural opacities Parenchymal bands
11	Langerhans cell histiocytosis	Clinical background Associated with *reticular opacities and bizzare shaped cysts*	Upper and middle lung zones with sparing of costophrenic angles Reticular opacities Thin and thick walled lung cysts bizarre shapes

Table 30.2B: Differential diagnosis of centrilobular nodules without tree-in-bud pattern: Perivascular cause **(Figs 30.4A to F)**

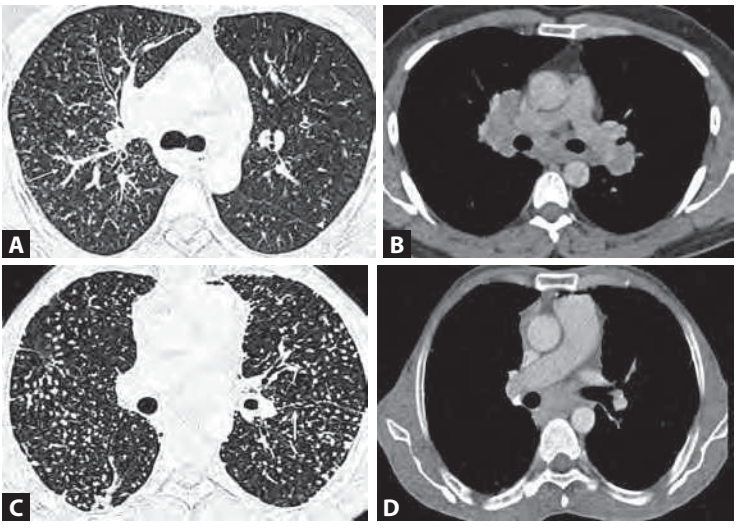
S. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Tumor thrombotic microangiopathy	Tumor cells embolize to pulmonary vessels, lead to stenosis and PAH	Ground glass centrilobular nodules Septal thickening Patchy ground-glass opacity (GGO) Mediastinal and hilar lymph nodes
2.	Vasculitis (Wegener's granulomatosis, Churg-Strauss syndrome)	Inflammatory vasculitis	Mosaic attenuation Peribronchovascular/septal thickening
3.	Pulmonary hemorrhage	Recurrent hemoptysis and iron deficiency anemia	*Multifocal areas of GGO* Patchy consolidation *Later stage*: Interlobular septal thickening
4.	Metastatic calcification	In cases of chronic renal disease/ hyperparathyroidism	*Nodules*: Multiple, calcified Patchy GGO
5.	Pulmonary edema	Hydrostatic with cardiomegaly and pleural effusion	Patchy or lobular GGO Smooth peribronchovascular/subpleural interstitial/fissure thickening *Dependent or perihilar predominance*



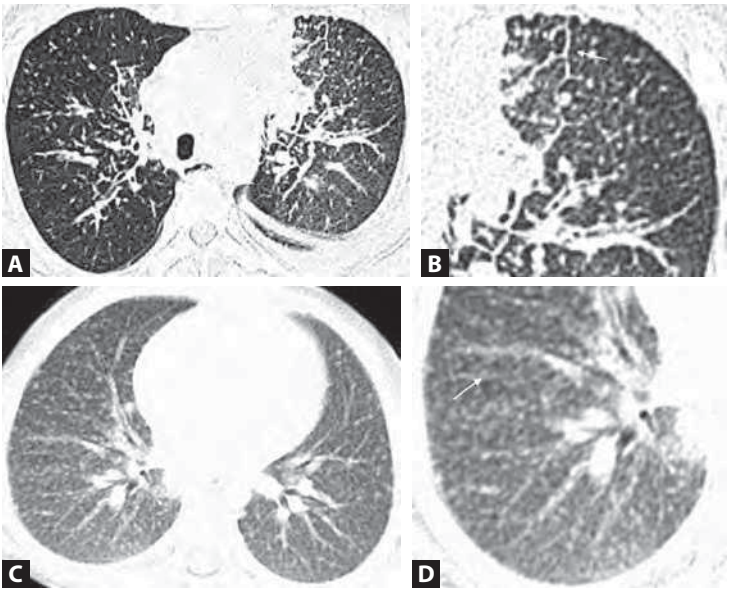
Figs 30.4A to F: Centrilobular nodules without tree-in-bud appearance.
(A and B) *Acute hypersensitivity pneumonitis:* Diffuse, ill-defined centrilobular nodules in bilateral lung fields having a lower lobe and dependent area preponderance (arrow).
(C and D) *RB-ILD:* Ground-glass attenuation centrilobular nodules (arrow) with bronchial wall thickening.
(E and F) *Granulomatosis polyangiitis:* Ground-glass nodules (arrow) scattered in both lungs in early change vasculitis

Table 30.3: Differential diagnosis of perilymphatic nodules **(Figs 30.5 and 30.6)**

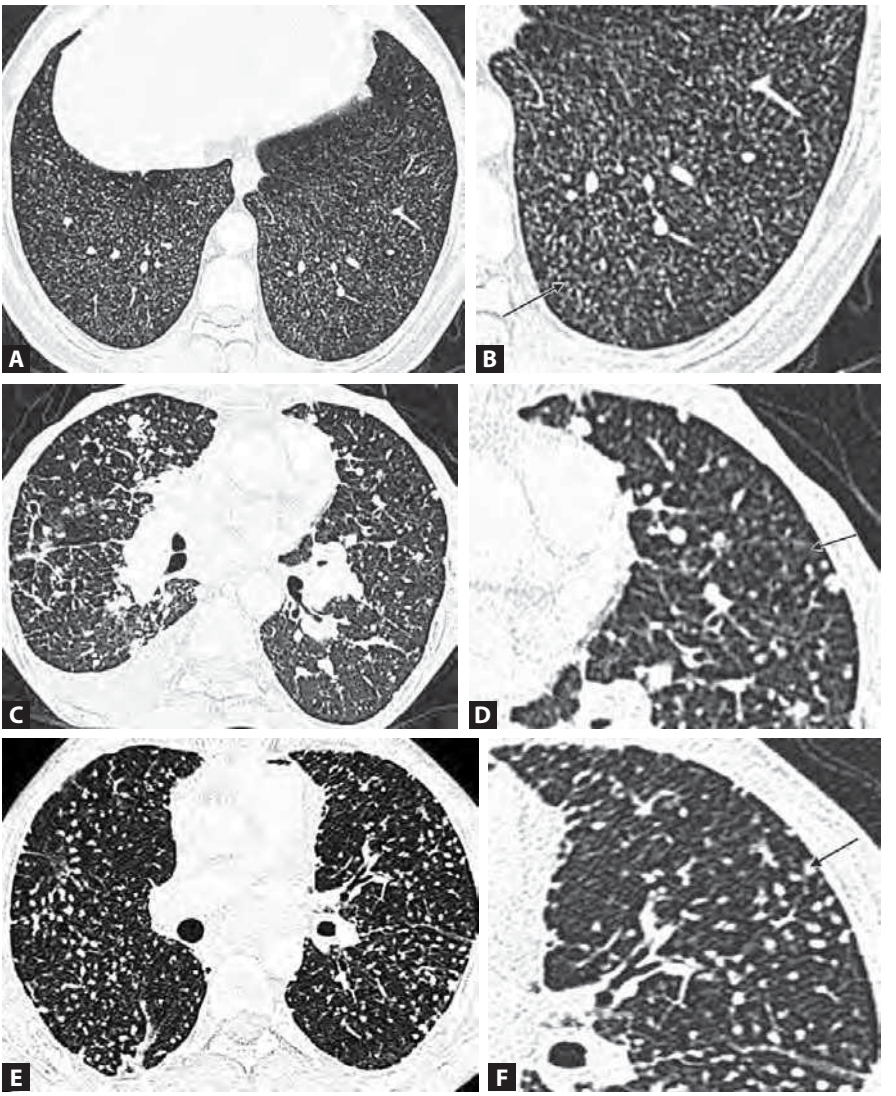
S. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Sarcoidosis	Respiratory and constitutional symtoms Peribronchovascular interstitial thickening and lymphadenopathy Peribronchovascular clustering of nodules	*Upper lobes, asymmetric* *Nodules*: Small, well-defined, subpleural, septal, perihilar, peribronchovascular *Smooth or nodular peribronchovascular interstitial thickening*
2.	Silicosis and coal workers' pneumoconiosis	Clinical background of dust exposure Associated with irregular masses and fibrosis	*Diffuse and symmetrical* *Masses*: Conglomerate, irregular Fibrosis, cicatricial emphysema
3.	Lymphangitic spread of tumor	Associated with interstitial thickening and background normal lung architecture	Well-defined and peribronchovascular/subpleural/septal nodules Unilateral/bilateral; Patchy Smooth/nodular interstitial thickening: peribronchovascular, fissural, '*beaded septa*'
4.	Diffuse alveolar septal amyloidosis	Associated with systemic involvement	Well-defined small nodules *Smooth septal thickening* Bilateral basal consolidations Punctate calcifications
5.	Lymphoproliferative disease (lymphoma, lymphocytic interstitial pneumonia)	*LIP*: Associated with dysproteinemia/ autoimmune disease/AIDS	*Nodules*: Small, well-defined, subpleural Thickening of bronchovascular bundles *GGO and Cysts*



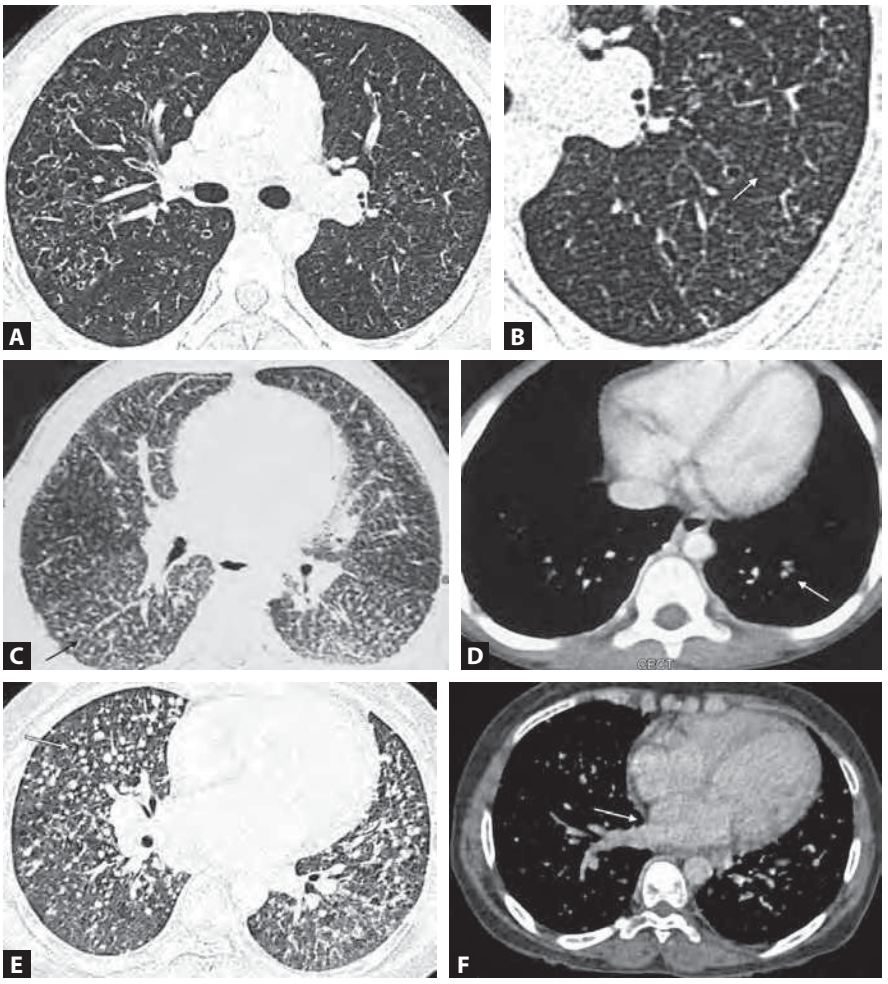
Figs 30.5A to D: Perilymphatic nodules. (A and B) *Sarcoidosis:* HRCT image (A) reveals discrete, sharply defined, micronodules along peribronchovascular interstitium (arrow in A), interlobular septae and along fissures (arrowhead in A). Note the asymmetric distribution; Mediastinal window (B) shows homogeneous enlarged bilateral hilar lymph nodes; (C and D) *Silicosis:* HRCT (C) image shows diffuse and symmetrical distribution of well- defined perilymphatic (arrow) and random nodules. Mediastinal window (D) shows mediastinal and right hilar lymph nodes with calcification



Figs 30.6A to D: Perilymphatic nodules. (A and B) *Lymphangitic spread of tumor:* Multiple perilymphatic nodules with interspersed beaded interlobular septa (arrow) in bilateral lung fields suggestive of lymphangitis in a known case of carcinoma lung; (C and D) *Lymphocytic interstitial pneumonia:* 6-year-old male child with HIV infection. Diffuse perilymphatic nodules seen in bilateral lung fields (arrow)



Figs 30.7A to F: Random nodules.
(A and B) *Miliary tuberculosis:* Multiple uniform appearing, small (1–3 mm) nodules seen diffusely throughout the lung parenchyma (arrow).
(C and D) *Miliary metastases:* Multiple varying size nodules seen scattered in bilateral lung fields suggestive of hematogeneous metastases in a case of medullary carcinoma thyroid. Note that the nodules are small and discrete but not uniform in appearance (arrow).
(E and F) *Silicosis:* Diffuse and symmetrical distribution of well-defined random (arrow) and perilymphatic nodules



Figs 30.8A to F: Random nodules.
(A and B) *Langerhans' cell histiocytosis:* Small nodules in random (arrow) and centrilobular distribution with few small cysts in bilateral lungs.
(C and D) *Alveolar microlithiasis:* Extensive symmetrical random acinar nodules with nodular thickening of left-sided major fissure (arrows). Mediastinal window (D) shows hyperdense pleura (arrow) due to conglomeration of subpleural nodules.
(E and F) *Pulmonary hemosiderosis:* High density (few calcified) soft tissue nodules (arrows) seen diffusely in bilateral lung fields. Note the mildly enlarged left atrium in this case of rheumatic heart disease with mitral regurgitation

RANDOM NODULES

- Diffuse/uniform distribution **(Fig. 30.1F)**
- Symmetrically scattered in bilateral lung fields
- The differentials are discussed in **Table 30.4**[1,3,4]

Table 30.4: Differential diagnosis of random nodules **(Figs 30.7 and 30.8)**

S. No.	*Disease entity*	*Key features to be evaluated*	*HRCT*
1.	Miliary infection	TB, fungal (Histoplasmosis), viral (Varicella)	Multiple micronodules in random pattern
2.	Miliary metastases	Thyroid cancer, renal cancer, melanoma	Hematogenous: peripheral and basal predominance/ Uniform distribution
3.	Silicosis or coal-worker's pneumoconiosis	Clinical background of dust exposure Associated with irregular masses and fibrosis	*Diffuse and symmetrical* *Masses*: Conglomerate, irregular Fibrosis, cicatricial emphysema
4.	Langerhans cell histiocytosis	Clinical background Associated with *reticular opacities and bizzare shaped cysts*	Upper and middle lung zones with sparing of costophrenic angles Reticular opacities Thin and thick walled lung cysts bizarre shapes
5.	Alveolar microlithiasis	*Small dense nodules and dense consolidation*	Patchy GGO Septal and subpleural linear calcification Consolidation
6.	Pulmonary hemosiderosis	Mitral stenosis *Multiple episodes of hemoptysis*	Densely calcified/ ossified nodules Diffuse B/L lungs

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